

**ARUNAI UNIVERSITY
TIRUVANNAMALAI**

Program Name: B.Tech – CLIMATE SCIENCE AND ENGINEERING

Program Duration: 4 years

Philosophy/Significance of the Course

In the face of increasing global challenges such as climate change, environmental degradation, and the depletion of natural resources, the **B.Tech. in Climate Science and Engineering** program has emerged as a critical educational pathway. This program equips students with the interdisciplinary knowledge and skills necessary to understand, mitigate, and adapt to the pressing issues surrounding climate change and its effects on natural and built environments.

Climate science encompasses a broad range of subjects, from atmospheric sciences and oceanography to the understanding of natural climate systems and anthropogenic influences on global warming. Engineering solutions are crucial in reducing carbon footprints, improving energy efficiency, and developing sustainable infrastructure to address the impacts of climate change. This program combines the scientific fundamentals with practical engineering applications, preparing graduates to tackle the complexities of climate-related challenges.

The increasing severity and unpredictability of extreme weather events, the rise in global temperatures, and the subsequent effects on agriculture, water resources, and biodiversity underscore the need for advanced studies in climate science and engineering. Additionally, the growing urgency for global cooperation and policy development in response to climate change makes this field highly relevant.

The program addresses the **interdisciplinary nature** of climate issues, emphasizing the need for experts who can navigate between science, technology, and policy to create sustainable solutions. As climate change continues to have widespread social, economic, and environmental consequences, professionals with expertise in both **climate science** and **engineering** are in high demand, particularly in industries such as renewable energy, urban planning, agriculture, and disaster management.

Distinctiveness and uniqueness of the program

Graduates with a degree in Climate Science and Engineering will find a broad array of career opportunities across various sectors. Some of the key career avenues include:

1. Renewable Energy Sector:

The growing demand for sustainable energy sources such as solar, wind, and bioenergy offers numerous job opportunities for graduates. Climate science experts play a key role in optimizing energy systems, developing energy-efficient solutions, and evaluating the environmental impact of energy production.

2. Environmental Consulting:

Climate science professionals are sought after by consulting firms to assess the environmental impact of projects, conduct climate risk assessments, and advise on sustainability practices. These roles often involve working with governmental and private organizations to ensure compliance with environmental regulations.

3. Government Agencies & Policy Development:

Working with governmental bodies such as the Environmental Protection Agency (EPA), United Nations Framework Convention on Climate Change (UNFCCC), or national meteorological departments, graduates can contribute to climate change research, policy-making, and regulatory frameworks.

4. Climate Change Adaptation and Mitigation:

Professionals are needed in both the public and private sectors to design and implement adaptation strategies to cope with the impacts of climate change. This includes water management, disaster preparedness, ecosystem restoration, and carbon capture and storage technologies.

5. Research and Academia:

For those interested in furthering scientific understanding, opportunities exist in research

institutes and universities, where graduates can engage in climate-related research, modelling, and development of new technologies aimed at combating climate change.

6. **Corporate Sustainability:**

Many large organizations and corporations are investing in sustainable business practices and reducing their carbon footprint. Graduates can work in sustainability roles, helping companies reduce energy consumption, waste, and carbon emissions.

7. **Disaster Risk Management:**

Climate change increases the frequency and intensity of natural disasters such as floods, hurricanes, and wildfires. Climate science and engineering graduates are essential in disaster management agencies for developing early warning systems, risk mitigation strategies, and infrastructure designed to withstand extreme weather events.

8. **International Development Organizations:**

NGOs and international development organizations are increasingly focused on climate change and environmental sustainability. Graduates can work in roles related to climate resilience, food security, and water resource management in developing countries.

Nature of Distinctiveness

The **B.Tech. Climate Science and Engineering** program stands out in its **distinctiveness** due to its interdisciplinary approach, blending climate science with engineering applications. Unlike traditional programs in environmental science or engineering, this degree provides students with a **holistic perspective**, enabling them to understand the complex interactions between climate, ecosystems, human society, and infrastructure.

Key features that make this program distinctive include:

1. **Interdisciplinary Learning:**

The program merges core subjects in **environmental science, atmospheric science, and climate physics** with essential engineering principles such as **renewable energy, sustainability engineering, and environmental management**. Students gain a comprehensive understanding of both the natural processes that drive climate and the engineering solutions that mitigate the effects.

2. **Hands-on Engineering Focus:**

The program emphasizes practical, **engineering-based solutions** to combat climate change. Students work on projects related to **green building design, sustainable transportation systems, carbon management technologies, and environmentally friendly urban planning**. This practical approach equips graduates with the skills to create innovative solutions.

3. **Global Relevance and Demand:**

Climate change is a **global challenge**, and the expertise of climate scientists and engineers is sought worldwide. The program's focus on both the scientific and engineering aspects of climate change ensures that graduates are ready to engage with international climate initiatives, as well as address local and regional environmental issues.

4. **Policy Integration:**

Students are exposed to the **policy and regulatory frameworks** that govern climate action. This is crucial in addressing the global nature of climate challenges and preparing students for careers in policy analysis, environmental consultancy, and regulatory compliance.

5. **Focus on Sustainability:**

A distinct feature of the program is its emphasis on **sustainability** at all levels. Students are trained not only to understand the scientific basis of climate change but also to design and implement **solutions that promote long-term environmental sustainability** and human welfare.

6. **Technological Innovations:**

The program encourages students to explore emerging technologies, such as **carbon capture and storage, geo engineering, and smart grid technology**, all of which are central to combating global warming and its ecosystem impacts.

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REGULATION-2025

(Choice Based Credit System)

B.Tech – Climate Science and Engineering

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

Graduates of the programme B Tech Climate Science and Engineering will

- I.** Gain knowledge and skills in Climate Science and Engineering which will enable them to have a career and professional accomplishment in the public or private sector organizations.
- II.** Become consultants on complex real life Climate Science and Engineering problems related to Infrastructure development especially housing, construction, water supply, sewerage, transport, spatial planning.
- III.** Become entrepreneurs and develop processes and technologies to meet desired infrastructure needs of society and formulate solutions that are technically sound, economically feasible, and socially acceptable.
- IV.** Perform investigation for solving Civil Engineering problems by conducting research using modern equipment and software tools.
- V.** Function in multi-disciplinary teams and advocate policies, systems, processes and equipment to support civil engineering

PROGRAM OUTCOMES (POs)

PO# Graduate Attribute

- PO1 Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2 Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, natural sciences, and engineering sciences.
- PO3 Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4 Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5 Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- PO6 The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7 Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

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- PO8 Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9 Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10 Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11 Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- PO12 Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

On successful completion of the Climate Science and Engineering Degree programme, the Graduates shall exhibit the following:

- PSO1** Knowledge of Climate Science and Engineering discipline Demonstrate in-depth knowledge of Climate Science and Engineering discipline, with an ability to evaluate, analyze and synthesize existing and new knowledge.
- PSO2** Critical analysis of Climate Science and Engineering problems and innovation Critically analyze complex Climate Science and Engineering problems apply independent judgment for synthesizing information and make innovative advances in a theoretical, practical and policy context.
- PSO3** Conceptualization and evaluation of engineering solutions to Climate Science and Engineering Issues Conceptualize and solve Climate Science and Engineering problems, evaluate potential solutions and arrive at technically feasible, economically viable and environmentally sound solutions with due consideration of health, safety, and socio cultural factors.

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**B.Tech – CLIMATE SCIENCE AND ENGINEERING (Full Time)
CURRICULUM AND SYLLABI FOR SEMESTERS I to VIII**

Semester – I

S.No	Course code	Course title	Credit			
			L	T	P	Total
1	GP2501	Moral & Value Education	2	0	0	2
2	LN2502	Professional Communication –1	3	0	0	3
3	ME2504	Workshop Practice & Technology	2	0	2	4
4	ME2501	Engineering Graphics	2	2	0	4
5	CH2513	Engineering Chemistry	3	1	1	5
6	CE2501	Engineering Mechanics	2	1	0	3
7	PH2511	Engineering Physics – I	3	0	1	4
8	MA2511	Engineering Mathematics –I	3	1	0	4

Semester – II

S.No	Course code	Course title	Credit			
			L	T	P	Total
1	LN2503	Professional Communication –II	3	0	0	3
2	ME2508	Engineering Thermodynamics	3	0	0	3
3	CS2501	Computer Programming	2	1	1	4
4	EN2501	Environmental Science & Engineering	3	0	0	3
5	MA2590	Engineering Mathematics – II	3	1	0	4
6	EE2501	Basic Electrical & Electronics Engineering	2	1	1	4
7	PH2512	Engineering Physics – II	3	0	1	4

Semester – III

S.No	Course code	Course title	Credit			
			L	T	P	Total
1	CM 2501	Climate System and Dynamics	2	0	0	2
2	MA2590	Engineering Mathematics – III	3	1	0	4
3	CS2503	Information Technology	2	0	1	3
4	CE2505	Fluid Mechanics	2	1	1	4
5	CE 2502	Surveying and leveling	2	0	2	2
6	CE2508	Strength of materials	3	0	0	3
7	CE2503	Atmospheric Science and Meteorology	2	1	1	4

Semester – IV

S.No	Course code	Course title	Credit			
			L	T	P	Total
1	CE 2535	Fundamental of Environmental Chemistry	2	0	2	4
2	CE2536	Structural Design	2	1	2	5
3	CE2522	Soil Mechanics & Foundation Engineering	3	0	0	3
4	CE2537	Building Services	3	0	0	3
5	MA2591	Computer Based Numerical & Statistical Techniques	3	1	0	4
6	CM2502	Climate Change Science	3	0	0	3
7	EN2502	Environmental Impact Assessment	3	0	0	3
8	EN2503	Global Warming and Ecosystem Impacts	3	0	0	3

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Semester – V

S.No	Course code	Course title	Credit			
			L	T	P	Total
1	BA2515	Elements of economics & principals management science	3	1	0	4
2	CE2505	Engineering Geology	2	1	0	3
3	ME2503	Heat and Mass Transfer	3	0	2	5
4	CE2522	Health Safety and Environment	3	0	2	5
5	SW2517	Land and Water Management & Engineering	4	0	0	4
6	SW2507	GIS and Remote Sensing	2	0	2	4
7	CM2503	Climate Adaptation Technologies	3	0	0	3

Semester – VI

S.No	Course code	Course title	Credit			
			L	T	P	Total
1	CE – 524	Air Pollution Control Engineering	2	1	1	4
2	CE-525	Solid Waste Engineering and Management	2	1	0	3
3	CE – 526	Wastewater Engineering and Management	4	0	0	4
4	CE-530	Industrial Waste water Treatment	3	0	0	3
5	BAM - 316	Climate Policy and Economics	3	1	0	4
6	CE-531	Green Engineering Principles	4	0	0	4
7	CE –580	Seminar -I	0	0	1	1

Semester – VII

S.No	Course code	Course title	Credit			
			L	T	P	Total
1	CE2508	Environmental Modeling & Simulation	3	0	1	4
2	CE2509	Coastal Zone Management	3	1	0	4
3	CE 2510	Noise Pollution	3	0	1	4
4	CE2511	Environmental Project management	3	1	0	4
5	CE 2512	Industrial Process and Environment pollution	3	1	0	4
6	CE 2599a	Project (Project Formulation)	0	0	2	2
7	CE 2580	Seminar - II	0	0	1	1

Semester – VIII

S.No	Course code	Course title	Credit			
			L	T	P	Total
1	CE2599b	Project (Project Execution & Report)	0	0	6	6

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GP2501 Moral & Value Education

Course Objectives

The objective of **Moral & Value Education** is to instill ethical values, integrity, and a sense of social responsibility in students. It aims to develop moral reasoning and decision-making skills to address personal and professional challenges effectively.

UNIT-I My country and my people – the Indians, being and becoming an Indian, Nationalism and Internationalism. Some life issues – Love, Sex and marriage.

UNIT-II Men and Money - value of time, meaning of work, human communication, human suffering, addiction, ecology, women's issues.

UNIT-III Understanding one's neighbor - neighbor hood groups, their structure and functions, patterns of social interaction of group dynamics. Preparation for career-choice of vocation, motivation for study and research, the present education system, curriculum and syllabus, teaching method, examination and work experience. Definition of value education, Moral and Ethics, Laws and Moral based on Ten Commandments and two great commandments.

UNIT-IV Discovery of self, Self-awareness, Growth of intellect- Man's spiritual nature, emotions, will, respect the rights of life, liberty, property, truth, reputation. Sin, Origin of sin, manifestation of sin, the results of sin, the remedy of sin, sin as an act, sin as a state, sin as a nature.

UNIT-V Consciences – as defined in Oxford dictionary and Winston Dictionary, types of consciousness (such as evil, convicted, purged, pure, weak, good, void of offence).

Course Outcomes (COs):

CO1: Understand the concepts of nationalism, internationalism, and the values associated with being an Indian, along with addressing important life issues like love, sex, and marriage.

CO2: Recognize the importance of time, work, human communication, and address societal challenges such as addiction, human suffering, ecological awareness, and women's issues.

CO3: Develop interpersonal skills, understand group dynamics, career choices, and critically analyze the education system's role in value education, morals, and ethics.

CO4: Reflect on self-awareness, spiritual growth, human rights, and the concept of sin, including its origin, manifestation, and remedies.

CO5: Understand the role of conscience, its types, and its influence on human behavior and moral decision-making.

Reference Books

1. "Ethics and Values in Social Work: An Integrated Approach for a Global World" Sarah A. Banks
2. "Moral and Spiritual Values in Education" Author: Dr. R. P. Pathak
3. "The Power of Ethics: How to Make Good Choices in a Complicated World" Susan Liautaud
4. "Moral Philosophy: A Reader" Louis P. Pojman
5. "The Moral Compass: Stories for a Life's Journey" William J. Bennett
6. "Value Education: A Manual for Teachers" Dr. B. N. Pandey

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	2	-	3	2	2	1	2	-	2
CO2	2	2	3	3	1	3	3	3	2	3	2	2
CO3	3	3	3	3	2	2	1	2	3	3	3	3
CO4	2	2	2	3	-	3	1	3	1	2	-	3
CO5	1	2	3	2	-	3	1	3	2	3	1	2

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LN2502 PROFESSIONAL COMMUNICATION –1

Course Objectives:

The objective of Professional Communication – I is to enhance students' proficiency in verbal and written communication skills. It aims to develop confidence, clarity, and professionalism in workplace and academic interactions.

Unit 1: Fundamentals of Communication

Definition and Importance of Communication

Communication Process: Sender, message, channel, receiver, feedback.

Types of Communication: Verbal (oral, written), non-verbal (body language, gestures, facial expressions).

Barriers to Communication: Physical, psychological, linguistic, and cultural barriers.

Overcoming communication barriers.

Unit 2: English Grammar and Usage

Parts of Speech, Articles, Tenses, Voice (active and passive).

Sentence structures and transformations (simple, compound, complex).

Common errors in grammar and their correction.

Punctuation and its importance in professional communication.

Unit 3: Professional Writing Skills

Writing emails: Format, tone, and etiquette.

Report writing: Structure and content.

Formal letters: Business and official communication.

Creating professional resumes and cover letters.

Unit 4: Speaking and Listening Skills

Speaking Skills: Importance of pronunciation and fluency.

Public speaking: Techniques to engage an audience.

Group discussions: Role-playing, effective participation, and moderation.

Presentation skills: Structure, visual aids, and delivery.

Listening Skills: Importance of active listening in communication.

Listening for specific information.

Paraphrasing and summarizing.

Unit 5: Non-verbal and Intercultural Communication

Non-verbal Communication: Importance of body language in professional settings.

Facial expressions, eye contact, gestures, and posture.

Intercultural Communication: Understanding cultural diversity in communication.

Overcoming cultural biases and stereotypes.

Building rapport in multicultural environments.

Course Outcomes (COs)

CO1: Understand the communication process and effectively manage communication barriers.

CO2: Demonstrate proficiency in English grammar and professional writing.

CO3: Develop strong verbal communication skills for workplace interactions.

CO4: Apply active listening techniques to enhance mutual understanding.

CO5: Recognize and adapt to cultural diversity in communication.

Reference Books

1. "Business Communication: Concepts, Cases and Applications" by P.D. Chaturvedi and Mukesh Chaturvedi.
2. "Effective Technical Communication" by M. Ashraf Rizvi.
3. "Professional Communication Skills" by A.K. Jain, Pravin S. R. Bhatia, and A.M. Sheikh.
4. "Communication Skills" by Sanjay Kumar and Pushp Lata.
5. "Developing Communication Skills" by Krishna Mohan and Meera Banerji.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1	2	1	3	3	2	3	3	1	3
CO2	3	3	2	2	1	3	2	2	2	3	1	2
CO3	3	2	3	2	2	3	2	2	3	3	2	3
CO4	2	2	2	3	2	3	2	3	3	3	1	3
CO5	2	1	1	2	3	3	3	3	3	3	2	3

ME2504 Workshop Practice & Technology

Unit 1: Introduction to Workshop Practices

Safety in workshops: Rules, precautions, and personal protective equipment (PPE).
Introduction to tools, materials, and equipment used in various shops (carpentry, fitting, welding, smithy, and machine shops).
Basic measurement techniques and instruments: Vernier caliper, micrometer, height gauge, etc.
Overview of manufacturing processes.

Unit 2: Carpentry and Fitting

Carpentry: Introduction to wood types, carpentry tools, and equipment. Basic carpentry joints and practical exercises.

Fitting: Bench work, fitting tools, and instruments. Filing, drilling, and tapping. Practical projects involving basic fitting techniques.

Unit 3: Welding and Sheet Metal Work

Welding: Fundamentals of welding, welding equipment, and types of welding (Arc, Gas, MIG/TIG). Practical exercises on simple welding joints.

Sheet Metal Work: Tools and equipment, types of sheet metal operations (cutting, bending, punching). Creation of simple objects.

Unit 4: Forging and Foundry

Forging: Basics of forging, tools, and equipment. Hands-on exercises like shaping and flattening.

Foundry: Introduction to casting processes, patterns, molding techniques, and sand preparation.

Unit 5: Machine Tools and Maintenance

Machine Tools: Introduction to lathe, milling, drilling, and grinding machines. Basic operations such as turning, facing, and drilling.

Maintenance: Preventive and breakdown maintenance of workshop equipment. Lubrication and safety practices.

Course Outcomes (COs)

CO1: Understand workshop safety and operational procedures.
CO2: Operate and handle basic tools and equipment for various workshop activities.
CO3: Perform carpentry, fitting, welding, forging, and foundry tasks.
CO4: Apply basic knowledge of machine tools for simple operations.
CO5: Develop skills for maintenance and troubleshooting of workshop machinery.

Reference Books

1. **"Workshop Technology" (Vol. I, II, III)** by S.K. Hajra Choudhury and A.K. Hajra Choudhury.
2. **"Elements of Workshop Technology"** by S.K. Hajra Choudhury, S.K. Bose, and A.K. Hajra Choudhury.
3. **"Manufacturing Technology"** by P.N. Rao.
4. **"A Course in Workshop Technology"** by B.S. Raghuwanshi.
5. **"Workshop Practice Manual"** by K.Venkata Reddy.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	1	3	3	3	2	1	1	2
CO2	3	3	2	2	2	2	1	1	3	2	2	2
CO3	3	3	3	2	3	2	1	2	3	2	2	3
CO4	3	3	3	3	3	2	1	2	3	2	2	3
CO5	3	3	3	3	2	3	3	3	2	1	3	3

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ME2501 ENGINEERING GRAPHICS

Objective:

To provide students with foundational skills in technical drawing and visualization, enabling them to effectively communicate design concepts and interpret engineering drawings.

Unit 1: Introduction to Engineering Drawing

- Importance of Engineering Graphics in engineering communication.
- Drawing instruments and their use.
- Layout of a drawing sheet and standard conventions.
- Geometrical constructions (lines, circles, polygons, tangents, etc.).
- Lettering and dimensioning practices.

Unit 2: Orthographic Projections

- **Projection Principles:** Concept of first-angle and third-angle projection.
- **Projections of Points and Lines:** Position of points, inclined lines, traces.
- **Projections of Planes:** Projection of planes parallel to reference planes, inclined to one or both planes.

Unit 3: Projections of Solids

- Types of solids: Prisms, pyramids, cylinders, cones, and spheres.
- Projections of solids with axes parallel, perpendicular, or inclined to reference planes.

Unit 4: Sections and Development of Solids

- **Sectioning of Solids:** Concept and types of sections (horizontal, vertical, inclined).
- True shape of the section.
- **Development of Surfaces:** Parallel-line and radial-line developments for solids like prisms, pyramids, cylinders, and cones.

Unit 5: Isometric Projections and Perspective Drawing

- **Isometric Projections:** Isometric scales, isometric views of simple objects.
- **Perspective Drawing:** Concept of perspective, vanishing points, and perspective views of objects.
- Introduction to CAD software for engineering drawing.

Course Outcomes (COs)

- CO1: Understand the basics of engineering graphics, tools, and standards.
CO2: Develop the ability to create and interpret orthographic projections.
CO3: Visualize and draw projections of solids in different orientations.
CO4: Apply concepts of sectioning and development for engineering applications.
CO5: Represent objects using isometric and perspective projections.

Reference Books

1. "Engineering Drawing" by N.D. Bhatt.
2. "A Textbook of Engineering Graphics" by K. Venugopal.
3. "Engineering Drawing and Graphics" by K.L. Narayana and P. Kannaiah.
4. "Machine Drawing" by P.S. Gill.
5. "Engineering Drawing with an Introduction to AutoCAD" by Dhananjay A. Jolhe.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1	2	1	1	2	1	1	2
CO2	3	3	2	2	2	1	1	1	2	2	1	3
CO3	3	3	3	2	2	2	1	1	3	2	1	3
CO4	3	3	3	2	3	2	1	2	3	2	2	3
CO5	3	3	3	3	3	2	1	2	3	2	2	3

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CH2513 ENGINEERING CHEMISTRY

Objective:

To provide students with a solid foundation in applied chemistry, enabling them to address engineering problems related to materials, environment, and technology.

Unit 1: Water Technology

Water Hardness: Types, causes, and units of hardness.

Boiler Troubles: Scale, sludge, caustic embrittlement, and boiler corrosion.

Water Treatment: Lime-soda process, zeolite process, ion-exchange process.

Desalination: Reverse osmosis and electro-dialysis.

Applications in industrial and municipal water management.

Unit 2: Electrochemistry and Corrosion

Electrochemistry: Electrochemical cells, Nernst equation, EMF, and its measurement.

Batteries: Types of batteries (primary, secondary, and fuel cells), examples (Li-ion battery, H₂-O₂ fuel cell).

Corrosion: Types of corrosion (dry and wet), galvanic and differential aeration corrosion.

Prevention of Corrosion: Coatings, sacrificial anode, and cathodic protection.

Unit 3: Polymers and Composites

Polymers: Types of polymerization (addition and condensation).

Properties and applications of polymers: Plastics, rubbers, and biodegradable polymers.

Composites: Types (metal-matrix, ceramic-matrix, and polymer-matrix).

Applications of polymers and composites in engineering.

Unit 4: Fuels and Combustion

Fuels: Classification of fuels (solid, liquid, gaseous).

Properties of fuels: Calorific value, octane and cetane numbers.

Combustion: Combustion reactions, stoichiometry, and calculation of air requirement.

Introduction to renewable fuels: Biodiesel, hydrogen fuel, and ethanol.

Unit 5: Nanomaterials and Green Chemistry

Nanomaterials: Properties, synthesis techniques (chemical and physical methods), and applications in engineering.

Green Chemistry: Principles, sustainable development, and green solvents.

Case studies on green chemical processes in industries.

Laboratory Experiments and Equipment

Laboratory Experiments:

1. Determination of total hardness of water using EDTA.
2. Conductometric titration of a strong acid and strong base.
3. Potentiometric titration of ferrous ions using KMnO₄.
4. Preparation of a polymer (e.g., urea-formaldehyde resin).
5. Study of corrosion of metals under different conditions.
6. Determination of calorific value of a fuel using a bomb calorimeter.
7. Synthesis of nanomaterials using chemical methods.

Laboratory Equipment:

- pH meter, conductometer, and potentiometer.
- Bomb calorimeter.
- UV-Vis spectrophotometer.
- Glassware: Beakers, flasks, burettes, pipettes, etc.
- Analytical balances and drying ovens.

Course Outcomes (COs)

CO1: Understand water chemistry and treatment methods for industrial and domestic use.

CO2: Analyze electrochemical and corrosion phenomena to devise preventive measures.

CO3: Gain knowledge of polymers and composites for engineering applications.

CO4: Evaluate fuel properties and understand combustion mechanisms.

CO5: Apply nanomaterial synthesis techniques and green chemistry principles for sustainable development.

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Reference Books

1. **"Engineering Chemistry"** by Jain and Jain.
2. **"A Textbook of Engineering Chemistry"** by S.S. Dara and S.S. Umare.
3. **"Engineering Chemistry"** by P.C. Jain and Monika Jain.
4. **"Principles of Physical Chemistry"** by Puri, Sharma, and Pathania.
5. **"Introduction to Nanotechnology"** by C.P. Poole and F.J. Owens.
6. **"Green Chemistry: An Introductory Text"** by Mike Lancaster.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C01	3	2	2	1	1	3	2	1	2	1	1	2
C02	3	3	2	2	1	3	3	2	2	2	1	2
C03	3	2	3	2	2	2	1	2	3	2	1	3
C04	3	3	3	2	2	3	2	2	3	2	2	3
C05	3	3	3	3	2	3	3	3	2	2	2	3

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CE2501 ENGINEERING MECHANICS

UNIT-I Force and Equilibrium

Basic concepts, Force, Moment and couple, Principle of Transmissibility, Varignon's theorem, Resultant of Force Systems concurrent and Non-concurrent coplanar Forces, Funicular polygon. Free body diagram.

UNIT-II Trusses

Plane structures, Various methods of analysis of Trusses, Method of joints, Method of sections and Graphical method.

UNIT-III Moment of Inertia

Center of gravity, centroids of Line, Area, volume and Composite Bodies, Area Moment of Inertia and Mass Moment of Inertia for plane figures and bodies including composite bodies, Product Moment of inertia, Parallel axis theorem, Principal moment of inertia.

UNIT-IV Friction

Introduction, Dry friction, Co-efficient of static friction. Friction cone, screw jack and Belt friction. Beams: Bending moment and shear force diagrams for statically determinate beams.

UNIT-V Kinematics of Rigid Bodies

Plane motion, Absolute motion, Relative motion, Translating axes and Rotating axes. Kinetics of Rigid Bodies: Plane motion, Force Mass and Acceleration, Work and energy, Impulse and momentum, Principles of energy conservation, Principle of virtual work, D'Alembert's principle and Dynamic equilibrium.

References:

Beer F.P and Johnston F.R: Mechanics for Engineers, McGraw Hill.
Meriam, J.L: Statistics, John Wiley.
Meriam, J.L: Dynamics, John Wiley.
Shames I.H: Engineering Mechanics, Prentice Hall of India.
Dayaratnam, P.: Statistics, Tata Mc Graw Hill.
Timoshenko, S. and Ypung D.: Engineering Mechanics, Mc Graw Hill.

Course Outcomes (COs):

CO1: Understand the fundamental concepts of force, moment, equilibrium, and force systems, including their graphical representations.

CO2: Analyze plane trusses using various methods, such as the method of joints, sections, and graphical methods.

CO3: Calculate the center of gravity, centroids, and moments of inertia for different geometries and composite bodies.

CO4: Apply principles of friction to practical problems, analyze friction in mechanical systems, and draw shear force and bending moment diagrams for beams.

CO5: Analyze the kinematics and kinetics of rigid bodies under planar motion using principles like energy conservation, impulse-momentum, and dynamic equilibrium.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	1	1	1	2	2	1	2
CO2	3	3	3	3	2	2	2	1	2	2	2	2
CO3	3	3	2	3	3	1	1	1	2	1	2	2
CO4	3	3	2	3	2	2	2	2	2	1	2	2
CO5	3	3	3	3	3	2	2	1	2	2	2	3

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PH2511- ENGINEERING PHYSICS – I

Objective:

To impart foundational knowledge of physics concepts with engineering applications, enabling students to apply these principles in real-world problems.

Unit 1: Oscillations and Waves

Oscillations: Simple harmonic motion (SHM), damped and forced oscillations, resonance.

Waves: Types of waves, wave equation, and superposition principle.

Ultrasonics: Generation of ultrasonic waves, applications in NDT (Non-Destructive Testing).

Unit 2: Optics

Interference: Conditions for constructive and destructive interference, Newton's rings, thin films.

Diffraction: Single-slit and double-slit diffraction, diffraction grating.

Polarization: Types of polarization, Brewster's law, applications in optical devices.

Unit 3: Quantum Mechanics

Wave-Particle Duality: De Broglie hypothesis, experimental verification (Davisson-Germer experiment).

Schrödinger Equation: Time-independent Schrödinger equation, particle in a one-dimensional box. Applications of quantum mechanics in engineering.

Unit 4: Lasers and Fiber Optics

Lasers: Principles of laser action, characteristics, types of lasers (He-Ne, semiconductor lasers), and applications.

Fiber Optics: Structure of optical fibers, types (single-mode and multimode), numerical aperture, attenuation, and applications in communication.

Unit 5: Materials Science

Crystallography: Unit cells, Bravais lattices, Miller indices.

Semiconductors: Intrinsic and extrinsic semiconductors, carrier concentration, and applications.

Superconductivity: Properties of superconductors, types, and applications in engineering.

Laboratory Experiments and Equipment

Laboratory Experiments:

1. Determination of the velocity of ultrasonic waves using a piezoelectric method.
2. Newton's rings experiment for determining the wavelength of light.
3. Diffraction grating experiment to determine the wavelength of light.
4. Measurement of numerical aperture of an optical fiber.
5. Study of I-V characteristics of a semiconductor diode.
6. Laser beam divergence and spot size measurement.
7. Determination of Planck's constant using the photoelectric effect.
8. Determination of the lattice constant using X-ray diffraction (demonstration).

Laboratory Equipment:

- Ultrasonic interferometer.
- Newton's ring apparatus.
- Diffraction grating setup.
- Optical fiber kit.
- He-Ne laser.
- Digital oscilloscope.
- X-ray diffraction apparatus (for demonstration).
- Multimeters and semiconductor diode kits.

Course Outcomes (COs)

CO1: Understand oscillatory motion and wave phenomena for engineering applications.

CO2: Apply concepts of interference, diffraction, and polarization in optical devices.

CO3: Comprehend quantum mechanics principles for modern engineering technologies.

CO4: Analyze laser and fiber optics principles for communication and measurement.

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CO5: Recognize the role of materials science in the development of advanced technologies.

Reference Books

1. **"Engineering Physics"** by R.K. Gaur and S.L. Gupta.
2. **"Concepts of Modern Physics"** by Arthur Beiser.
3. **"Optics"** by Ajoy Ghatak.
4. **"Solid State Physics"** by S.O. Pillai.
5. **"Introduction to Quantum Mechanics"** by David J. Griffiths.
6. **"Lasers and Non-linear Optics"** by B.B. Laud.
7. **"Materials Science and Engineering: An Introduction"** by William D. Callister.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	1	1	2	1	1	2	1	1	2
CO2	3	3	2	2	2	1	1	1	3	2	2	2
CO3	3	3	3	3	2	1	1	2	2	2	1	3
CO4	3	3	3	2	2	2	2	2	3	3	1	3
CO5	3	3	3	3	3	2	2	3	3	2	2	3

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MA2511 - ENGINEERING MATHEMATICS –I

Objective:

To provide students with fundamental mathematical concepts and techniques essential for engineering applications and problem-solving.

Unit 1: Differential Calculus

Successive Differentiation: n-th order derivatives, Leibnitz theorem.

Applications of Derivatives: Curvature, radius of curvature, maxima, and minima.

Partial Differentiation: Functions of multiple variables, Euler's theorem, total derivative.

Applications of Partial Differentiation: Jacobians, Taylor's and Maclaurin's series for functions of two variables.

Unit 2: Integral Calculus

Definite and Indefinite Integrals: Methods of integration (substitution, by parts, partial fractions).

Applications of Integration: Finding area, volume of solids of revolution, and length of curves.

Multiple Integrals: Double and triple integrals, change of order and change of variables.

Unit 3: Matrices and Linear Algebra

Types of Matrices: Symmetric, skew-symmetric, orthogonal, Hermitian, etc.

Rank of a Matrix: Row-reduction and echelon form.

System of Linear Equations: Consistency, Gauss elimination method, and Gauss-Jordan method.

Eigenvalues and Eigenvectors: Properties, Cayley-Hamilton theorem, diagonalization.

Unit 4: Ordinary Differential Equations (ODEs)

First-Order ODEs: Linear equations, separable variables, exact equations.

Higher-Order ODEs: Homogeneous and non-homogeneous linear differential equations with constant coefficients, method of undetermined coefficients, and variation of parameters.

Applications: Simple harmonic motion, electrical circuits, and heat conduction problems.

Unit 5: Vector Calculus

Vector Differentiation: Gradient, divergence, curl, and their physical interpretations.

Vector Integration: Line, surface, and volume integrals.

Vector Theorems: Green's theorem, Stokes' theorem, and Gauss divergence theorem (statements and simple applications).

Course Outcomes (COs)

CO1: Apply concepts of differential calculus for curve tracing, optimization, and other engineering applications.

CO2: Use integral calculus to solve problems related to areas, volumes, and engineering structures.

CO3: Solve linear algebra problems involving matrices, systems of equations, and eigenvalues/eigenvectors.

CO4: Analyze and solve ordinary differential equations for engineering scenarios.

CO5: Utilize vector calculus techniques for physical and engineering interpretations.

Reference Books

1. "Higher Engineering Mathematics" by B.S. Grewal.
2. "Advanced Engineering Mathematics" by Erwin Kreyszig.
3. "Advanced Mathematics for Engineers and Scientists" by M.D. Greenberg.
4. "A Textbook of Engineering Mathematics" by N.P. Bali and Manish Goyal.
5. "Vector Calculus" by P.C. Matthews.
6. "Introduction to Linear Algebra" by Gilbert Strang.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	1	1	1	2	1	1	2
CO2	3	3	2	2	1	1	1	1	2	1	1	2
CO3	3	3	3	2	1	1	1	1	2	1	1	2
CO4	3	3	3	3	2	1	1	1	2	1	1	2
CO5	3	3	3	3	2	1	1	1	2	1	1	3

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LN2503- PROFESSIONAL COMMUNICATION –1

Objective:

To enhance advanced communication skills, critical thinking, and professional interaction techniques, ensuring students can effectively engage in workplace communication and technical discourse.

Unit 1: Advanced Listening and Speaking Skills

Listening Skills: Active listening, barriers to listening, note-taking, and summarization.

Speaking Skills: Delivering structured oral presentations, group discussions, debates, and public speaking.

Paralinguistic Features: Voice modulation, pitch, tone, and clarity in speech.

Applications in interviews, team meetings, and client interactions.

Unit 2: Technical Writing Skills

Technical Documents: Writing proposals, project reports, abstracts, and executive summaries.

Correspondence: Emails, memos, letters, and notices in a professional context.

Technical Descriptions: Product manuals, user guides, and process documentation.

Emphasis on clarity, precision, and conciseness.

Unit 3: Reading and Analytical Thinking

Advanced Reading Skills: Critical reading, skimming, scanning, and inference.

Analysis: Evaluating arguments, identifying fallacies, and drawing logical conclusions.

Case studies and comprehension passages from technical and business contexts.

Unit 4: Professional Etiquette and Cross-Cultural Communication

Professional Conduct: Work ethics, team collaboration, and workplace behavior.

Cross-Cultural Communication: Understanding cultural diversity, avoiding stereotypes, and global communication strategies.

Importance of emotional intelligence and empathy in professional settings.

Unit 5: Soft Skills for Career Development

Interpersonal Skills: Building rapport, conflict resolution, and teamwork.

Time and Stress Management: Prioritization, multitasking, and managing deadlines.

Leadership and Negotiation Skills: Principles of leadership, decision-making, and persuasive communication.

Role of personal branding in career growth.

Course Outcomes (COs)

CO1: Demonstrate effective listening and speaking skills in professional settings.

CO2: Draft clear, concise, and well-structured technical and business documents.

CO3: Develop critical reading and analytical thinking for problem-solving and decision-making.

CO4: Exhibit professional etiquette and adapt communication strategies to diverse cultural contexts.

CO5: Apply interpersonal and soft skills for career development and workplace success.

Reference Books

1. "Effective Technical Communication" by M. Ashraf Rizvi.
2. "Communication Skills for Engineers and Scientists" by Sangeeta Sharma and Binod Mishra.
3. "Soft Skills: Enhancing Employability" by G. Ravindran.
4. "Business Communication: Building Critical Skills" by Kitty O. Locker and Stephen Kyo Kaczmarek.
5. "Developing Communication Skills" by Krishna Mohan and Meera Banerjee.
6. "The 7 Habits of Highly Effective People" by Stephen R. Covey.
7. "Emotional Intelligence" by Daniel Goleman.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	2	1	1	2	2	2	1	3	3	2	2
CO2	2	3	2	2	2	2	1	1	2	3	2	2
CO3	3	3	2	3	2	1	1	1	2	2	1	3
CO4	2	2	2	1	1	3	3	3	3	3	3	2
CO5	2	2	2	2	2	3	2	3	3	3	3	2

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ME2508 - ENGINEERING THERMODYNAMICS

Objective:

To provide students with a fundamental understanding of thermodynamic principles and their application in solving engineering problems related to energy systems..

Unit 1: Basic Concepts and Laws of Thermodynamics

Fundamentals: System, surroundings, state, process, and cycles.

Zeroth Law: Concept of temperature and thermodynamic equilibrium.

First Law: Energy conservation, internal energy, enthalpy, and applications to closed and open systems.

Second Law: Heat engines, refrigerators, Clausius and Kelvin-Planck statements, and Carnot cycle.

Unit 2: Properties of Pure Substances and Steam Tables

Properties: P-V-T behavior, phase change process, and property diagrams (P-v, T-s, h-s).

Steam Tables and Mollier Diagram: Use in determining properties of steam.

Thermodynamic Property Relations: Maxwell relations, Clapeyron equation, and property estimation.

Unit 3: Entropy and Availability

Entropy: Definition, entropy change for ideal gases and processes.

Second Law Applications: Entropy generation, increase of entropy principle.

Availability and Irreversibility: Concepts, applications to energy systems, and exergy analysis.

Unit 4: Thermodynamic Cycles

Power Cycles: Carnot, Rankine, Otto, Diesel, and Brayton cycles.

Refrigeration and Air Conditioning Cycles: Vapour compression and absorption refrigeration cycles.

Efficiency and Performance: Thermal efficiency, COP, and work ratio.

Unit 5: Ideal and Real Gases and Mixtures

Ideal Gas Laws: Equation of state, specific heat, and polytropic processes.

Real Gases: Van der Waals equation, compressibility factor.

Gas Mixtures: Dalton's and Amagat's laws, properties of mixtures of ideal gases.

Psychrometry: Humidity, specific humidity, relative humidity, and psychrometric charts.

Course Outcomes (COs)

CO1: Understand basic thermodynamic principles and laws and apply them to engineering problems.

CO2: Analyze thermodynamic properties of substances using tables, diagrams, and equations.

CO3: Evaluate entropy and availability concepts to assess energy system performance.

CO4: Apply thermodynamic cycles to power generation, refrigeration, and air conditioning systems.

CO5: Analyze the behavior of ideal and real gases and gas mixtures in engineering applications.

Reference Books

1. "Engineering Thermodynamics" by P.K. Nag.
2. "Thermodynamics: An Engineering Approach" by Yunus A. Çengel and Michael A. Boles.
3. "Fundamentals of Engineering Thermodynamics" by Moran and Shapiro.
4. "Thermodynamics" by J.P. Holman.
5. "Applied Thermodynamics for Engineering Technologists" by T.D. Eastop and A. McConkey.
6. "Engineering Thermodynamics" by Mahesh M. Rathore.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	1	1	1	2	2	1	3
CO2	3	3	3	2	2	1	1	1	1	2	1	2
CO3	3	3	2	2	1	2	1	1	1	2	1	2
CO4	3	3	3	3	2	2	1	1	1	2	1	3
CO5	3	3	2	2	2	2	1	1	1	1	1	2

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CS2501 COMPUTER PROGRAMMING

Objective:

To introduce students to the fundamentals of programming, problem-solving techniques, and the use of structured programming languages for developing efficient algorithms and programs.

Unit 1: Introduction to Programming and Problem-Solving

Basic Concepts: Components of a computer system, software, and hardware overview.

Programming Basics: Algorithms, flowcharts, and pseudocode.

Introduction to Programming Languages: Features and uses of high-level languages.

C Programming Basics: Structure of a C program, variables, data types, and operators.

Input and Output: Formatting and displaying data using printf and scanf.

Unit 2: Control Structures and Loops

Conditional Statements: if, else if, else, and switch-case.

Looping Constructs: for, while, and do-while loops.

Break and Continue Statements: Use in controlling loops.

Applications: Simple mathematical computations, pattern printing, and decision-making scenarios.

Unit 3: Arrays and Strings

Arrays: Definition, declaration, initialization, and operations on one-dimensional and two-dimensional arrays.

Strings: String handling in C, string library functions (strlen, strcpy, strcmp, etc.).

Applications: Matrix operations, string manipulation, and data sorting/searching.

Unit 4: Functions and Pointers

Functions: User-defined functions, recursion, and parameter passing.

Storage Classes: Automatic, external, static, and register.

Pointers: Introduction, pointer arithmetic, dynamic memory allocation, and pointer to arrays.

Applications: Function calls, memory management, and pointer-based operations.

Unit 5: Structures, File Handling, and Advanced Topics

Structures and Unions: Definition, declaration, and applications.

File Handling: Opening, reading, writing, and closing files.

Enumerations and Typedef: Creating user-defined types.

Introduction to Advanced Topics: Overview of Object-Oriented Programming, and basic introduction to Python or Java.

Lab Experiments

1. **Basic Programs:** Write programs to demonstrate input/output, data types, and operators.
2. **Control Structures:** Programs using if-else, switch, loops for factorial, prime numbers, etc.
3. **Arrays:** Implement operations like searching, sorting, and matrix addition/multiplication.
4. **Strings:** Programs for palindrome check, string concatenation, and substring search.
5. **Functions:** Write programs to implement user-defined functions and recursion (e.g., Fibonacci sequence).
6. **Pointers:** Demonstrate pointer arithmetic and dynamic memory allocation.
7. **Structures:** Programs to manage student records or employee details using structures.
8. **File Handling:** Write and read data from text and binary files.
9. **Project:** Develop a mini-project combining multiple topics, such as a student management system.

List of Lab Equipment

1. **Desktop Computers or Laptops:** Minimum of 30 systems with C/C++ IDE (e.g., Code::Blocks, GCC, or Turbo C++).
2. **Software:**
 - Windows/Linux Operating System.
 - GCC or Turbo C++ compiler.
 - IDEs like Visual Studio Code or Code::Blocks.
3. **Networking Setup:** For sharing resources and running collaborative programs.
4. **Projector and Whiteboard:** For demonstrations and teaching.

Course Outcomes (COs)

CO1: Develop problem-solving skills using algorithms and flowcharts.

CO2: Implement control structures and loops for decision-making and repetitive tasks.

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CO3: Apply concepts of arrays, strings, and functions to solve computational problems.

CO4: Utilize pointers and structures for efficient memory usage and data organization.

CO5: Implement file handling techniques and understand advanced programming concepts for practical applications.

Reference Books

1. **"Let Us C"** by Yashavant Kanetkar.
2. **"Programming in ANSI C"** by E. Balagurusamy.
3. **"C Programming Language"** by Brian W. Kernighan and Dennis M. Ritchie.
4. **"Problem Solving and Program Design in C"** by Jeri R. Hanly and Elliot B. Koffman.
5. **"Introduction to Computing and Programming in C"** by Reema Thareja.
6. **"Head First C"** by David Griffiths and Dawn Griffiths.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	2	1	1	1	1	2	1	2
CO2	3	3	3	2	2	1	1	1	1	2	1	2
CO3	3	3	3	3	3	2	1	1	1	2	1	3
CO4	3	3	3	3	2	2	1	1	1	2	1	3
CO5	3	3	3	3	3	2	1	1	1	2	1	3

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EN2501 ENVIRONMENTAL SCIENCE & ENGINEERING

Objective:

To develop awareness about environmental challenges, understand the impact of engineering solutions on the environment, and adopt sustainable practices in professional activities.

Unit 1: Introduction to Environment and Ecosystems

Environment Overview: Components of the environment – lithosphere, hydrosphere, atmosphere, and biosphere.

Ecosystems: Structure, function, energy flow, food chains, food webs, and ecological pyramids.

Biodiversity: Importance, hotspots, threats, and conservation strategies.

Unit 2: Natural Resources

Water Resources: Usage, conservation, and management.

Forest Resources: Deforestation, its impacts, and sustainable forest management.

Mineral Resources: Mining impacts and management.

Energy Resources: Renewable and non-renewable energy resources, energy conservation, and sustainable energy practices.

Unit 3: Environmental Pollution and Control

Air Pollution: Sources, effects, and control measures. **Water Pollution:** Causes, impacts, and treatment methods. **Soil Pollution:** Causes, effects, and remediation techniques.

Noise Pollution: Sources, impacts, and noise control measures.

Solid and Hazardous Waste Management: Types, disposal methods, and recycling.

Unit 4: Social Issues and the Environment

Sustainable Development: Principles and applications.

Climate Change: Global warming, greenhouse gases, and mitigation strategies.

Environmental Policies and Laws: Key legislations like the Environment Protection Act, Water Act, Air Act, and Wildlife Protection Act. **Environmental Impact Assessment (EIA):** Importance, procedure, and case studies.

Unit 5: Human Population and the Environment

Population Growth: Impacts on environment and resources. **Urbanization:** Environmental consequences and sustainable urban development. **Health and Environment:** Occupational health hazards, and emerging diseases due to environmental changes. **Environmental Ethics:** Role of engineers in sustainable development.

Course Outcomes (COs)

CO1: Understand the basic concepts of ecosystems, biodiversity, and their importance to the environment.

CO2: Analyze the use and conservation of natural resources and their sustainable management.

CO3: Identify the causes and effects of pollution and apply methods for mitigation and control.

CO4: Apply knowledge of environmental laws and policies to engineering solutions for sustainability.

CO5: Understand the impact of human population dynamics and ethical considerations on environmental sustainability.

Reference Books

1. "Textbook of Environmental Studies for Undergraduate Courses" by Erach Bharucha.
2. "Environmental Science: A Global Concern" by William P. Cunningham and Mary Ann Cunningham.
3. "Principles of Environmental Science: Inquiry and Applications" by William P. Cunningham and Mary Ann Cunningham.
4. "Environmental Studies" by R. Rajagopalan.
5. "Introduction to Environmental Engineering and Science" by Gilbert M. Masters and Wendell P. Ela.
6. "Environmental Chemistry" by A.K. De.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	2	2	3	3	1	1	2	2	3
CO2	2	2	1	2	2	3	3	1	1	2	2	2
CO3	3	3	2	2	2	3	3	1	1	2	2	3
CO4	2	3	3	2	2	3	3	3	2	3	2	3
CO5	2	2	1	2	2	3	3	3	2	3	3	3

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MA2590 ENGINEERING MATHEMATICS – II

Objective:

To equip students with advanced mathematical techniques and concepts, enabling them to solve complex engineering problems and develop analytical skills for modeling and analyzing systems.

Unit 1: Ordinary Differential Equations (ODEs)

First-Order ODEs: Formation, solution methods (separable, homogeneous, exact, linear).

Higher-Order ODEs: Linear differential equations with constant coefficients, complementary functions, and particular integrals.

Applications: Mechanical vibrations, electrical circuits, and growth/decay models.

Unit 2: Laplace Transforms

Definition and Properties: Linearity, first shifting theorem, and change of scale.

Inverse Laplace Transform: Methods and applications.

Convolution Theorem: Solution of differential equations using Laplace transforms.

Applications: Initial value problems and engineering systems.

Unit 3: Vector Calculus

Vector Differentiation: Gradient, divergence, curl, and their physical interpretation.

Vector Integration: Line integrals, surface integrals, and volume integrals.

Theorems: Green's, Stokes', and Gauss' divergence theorems (with proof and applications).

Unit 4: Fourier Series and Fourier Transforms

Fourier Series: Periodic functions, half-range expansions, and harmonic analysis.

Fourier Transforms: Definition, properties, and applications.

Applications: Signal processing, heat conduction, and vibration analysis.

Unit 5: Partial Differential Equations (PDEs)

Formation of PDEs: By elimination of arbitrary constants and functions.

Solution Techniques: Separation of variables, one-dimensional wave equation, heat equation, and Laplace equation.

Applications: Heat flow, wave propagation, and electrostatics.

Course Outcomes (COs)

CO1: Solve first- and higher-order ordinary differential equations and apply them to engineering systems.

CO2: Utilize Laplace transforms to simplify and solve linear differential equations.

CO3: Apply vector calculus to analyze physical phenomena in fluid mechanics, electromagnetism, and related fields.

CO4: Use Fourier series and transforms for solving problems in signal processing and thermal analysis.

CO5: Solve partial differential equations in engineering domains such as heat transfer, vibrations, and fluid flow.

Reference Books

1. "Higher Engineering Mathematics" by B.S. Grewal.
2. "Advanced Engineering Mathematics" by Erwin Kreyszig.
3. "Engineering Mathematics" by K.A. Stroud and Dexter Booth.
4. "Advanced Engineering Mathematics" by Michael D. Greenberg.
5. "Engineering Mathematics Volume II" by S.S. Sastry.
6. "Elements of Partial Differential Equations" by Ian N. Sneddon.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	2	1	1	1	1	2	1	2
CO2	3	3	3	3	2	2	1	1	1	2	1	2
CO3	3	3	3	2	2	1	2	1	1	2	1	2
CO4	3	3	3	2	3	2	1	1	1	2	1	2
CO5	3	3	3	3	2	2	2	1	1	2	2	3

EE2501 - BASIC ELECTRICAL & ELECTRONICS ENGINEERING

Objective:

To provide fundamental knowledge of electrical circuits, machines, and electronic devices essential for engineering applications.

Unit 1: Basic Electrical Circuits

Electrical Circuit Fundamentals: Ohm's Law, Kirchhoff's Laws.

Network Theorems: Superposition, Thevenin, Norton, and Maximum Power Transfer theorems.

AC Fundamentals: Sinusoidal waveforms, RMS, average values, phase, and power factor.

Analysis: Series and parallel RLC circuits, resonance in AC circuits.

Unit 2: DC Machines and Transformers

DC Generators: Construction, working principle, and types.

DC Motors: Principle, types, and characteristics.

Transformers: Single-phase transformers - construction, working principle, efficiency, and testing.

Applications: Use of DC machines and transformers in industry.

Unit 3: AC Machines

Synchronous Machines: Alternators - construction, principle, and performance.

Three-Phase Induction Motors: Construction, principle, and applications.

Single-Phase Motors: Types and uses.

Applications: Industrial and household electrical equipment.

Unit 4: Semiconductor Devices and Applications

Semiconductor Basics: PN junction diodes, Zener diodes.

Transistors: Bipolar Junction Transistor (BJT) – configurations, characteristics, and applications.

Operational Amplifiers: Basic characteristics and applications (inverting, non-inverting amplifiers).

Applications: Rectifiers, voltage regulators, amplifiers.

Unit 5: Digital Electronics and Communication

Number Systems: Binary, octal, and hexadecimal systems.

Logic Gates: AND, OR, NOT, NAND, NOR, XOR, XNOR – truth tables and applications.

Basics of Communication: Modulation and demodulation techniques, AM/FM principles.

Applications: Digital logic in computing and communication systems.

Lab Experiments

1. Electrical Circuits:

- Verification of Kirchhoff's Laws.
- Study of series and parallel resonance in RLC circuits.
- Verification of Thevenin's and Norton's theorems.

2. DC Machines and Transformers:

- Load characteristics of DC shunt and series motors.
- Open circuit and short circuit tests on single-phase transformers.

3. AC Machines:

- Regulation of a synchronous generator by the EMF method.
- Performance testing of three-phase induction motors.

4. Semiconductor Devices:

- VI characteristics of PN junction and Zener diodes.
- Input and output characteristics of BJTs.
- Basic op-amp applications (amplifiers, adders).

5. Digital Electronics and Communication:

- Implementation of basic logic gates using discrete components.
- Study of AM and FM modulation and demodulation.

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List of Lab Equipment

1. **Electrical Machines:**
 - DC generators, DC motors, transformers, alternators, induction motors.
2. **Measuring Instruments:**
 - Multimeters, wattmeters, ammeters, voltmeters, and tachometers.
3. **Power Supplies:**
 - DC power supply, variable AC source.
4. **Semiconductor Components:**
 - Diodes, transistors, op-amps.
5. **Digital Kits:**
 - Breadboards, logic gates, and ICs for basic digital experiments.
6. **Communication Kits:**
 - AM/FM modulation and demodulation trainers.

Course Outcomes (COs)

- CO1: Understand basic electrical circuit principles and analyze AC/DC circuits.
CO2: Explain the working principles and applications of DC machines and transformers.
CO3: Demonstrate the operation and applications of AC machines.
CO4: Analyze the characteristics and applications of semiconductor devices.
CO5: Comprehend the basics of digital electronics and communication systems.

Reference Books

1. **"Basic Electrical Engineering"** by D.P. Kothari and I.J. Nagrath.
2. **"Electrical Technology Vol. 1 & 2"** by B.L. Theraja and A.K. Theraja.
3. **"Electronic Devices and Circuit Theory"** by Robert Boylestad and Louis Nashelsky.
4. **"Digital Fundamentals"** by Thomas L. Floyd.
5. **"Principles of Communication Systems"** by Herbert Taub and Donald Schilling.
6. **"Basic Electrical and Electronics Engineering"** by S.K. Bhattacharya.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	2	2	1	1	1	2	1	2
CO2	3	3	3	2	2	2	2	1	1	2	1	2
CO3	3	3	3	3	2	2	2	1	1	2	1	2
CO4	3	3	2	2	3	2	1	1	1	2	1	2
CO5	3	3	2	3	3	2	1	1	1	2	1	2

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PH2512 - ENGINEERING PHYSICS – II

Objective:

To provide fundamental knowledge of electrical circuits, machines, and electronic devices essential for engineering applications.

Unit 1: Basic Electrical Circuits

Electrical Circuit Fundamentals: Ohm's Law, Kirchhoff's Laws.

Network Theorems: Superposition, Thevenin, Norton, and Maximum Power Transfer theorems.

AC Fundamentals: Sinusoidal waveforms, RMS, average values, phase, and power factor.

Analysis: Series and parallel RLC circuits, resonance in AC circuits.

Unit 2: DC Machines and Transformers

DC Generators: Construction, working principle, and types.

DC Motors: Principle, types, and characteristics.

Transformers: Single-phase transformers - construction, working principle, efficiency, and testing.

Applications: Use of DC machines and transformers in industry.

Unit 3: AC Machines

Synchronous Machines: Alternators - construction, principle, and performance.

Three-Phase Induction Motors: Construction, principle, and applications.

Single-Phase Motors: Types and uses.

Applications: Industrial and household electrical equipment.

Unit 4: Semiconductor Devices and Applications

Semiconductor Basics: PN junction diodes, Zener diodes.

Transistors: Bipolar Junction Transistor (BJT) – configurations, characteristics, and applications.

Operational Amplifiers: Basic characteristics and applications (inverting, non-inverting amplifiers).

Applications: Rectifiers, voltage regulators, amplifiers.

Unit 5: Digital Electronics and Communication

Number Systems: Binary, octal, and hexadecimal systems.

Logic Gates: AND, OR, NOT, NAND, NOR, XOR, XNOR – truth tables and applications.

Basics of Communication: Modulation and demodulation techniques, AM/FM principles.

Applications: Digital logic in computing and communication systems.

Lab Experiments

1. Electrical Circuits:

- Verification of Kirchhoff's Laws.
- Study of series and parallel resonance in RLC circuits.
- Verification of Thevenin's and Norton's theorems.

2. DC Machines and Transformers:

- Load characteristics of DC shunt and series motors.
- Open circuit and short circuit tests on single-phase transformers.

3. AC Machines:

- Regulation of a synchronous generator by the EMF method.
- Performance testing of three-phase induction motors.

4. Semiconductor Devices:

- VI characteristics of PN junction and Zener diodes.
- Input and output characteristics of BJTs.
- Basic op-amp applications (amplifiers, adders).

5. Digital Electronics and Communication:

- Implementation of basic logic gates using discrete components.
- Study of AM and FM modulation and demodulation.

List of Lab Equipment

1. Electrical Machines:

- DC generators, DC motors, transformers, alternators, induction motors.

2. Measuring Instruments:

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- Multimeters, wattmeters, ammeters, voltmeters, and tachometers.
- 3. **Power Supplies:**
 - DC power supply, variable AC source.
- 4. **Semiconductor Components:**
 - Diodes, transistors, op-amps.
- 5. **Digital Kits:**
 - Breadboards, logic gates, and ICs for basic digital experiments.
- 6. **Communication Kits:**
 - AM/FM modulation and demodulation trainers.

Course Outcomes (COs)

- CO1: Understand basic electrical circuit principles and analyze AC/DC circuits.
CO2: Explain the working principles and applications of DC machines and transformers.
CO3: Demonstrate the operation and applications of AC machines.
CO4: Analyze the characteristics and applications of semiconductor devices.
CO5: Comprehend the basics of digital electronics and communication systems.

Reference Books

1. "Basic Electrical Engineering" by D.P. Kothari and I.J. Nagrath.
2. "Electrical Technology Vol. 1 & 2" by B.L. Theraja and A.K. Theraja.
3. "Electronic Devices and Circuit Theory" by Robert Boylestad and Louis Nashelsky.
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CO3	3	3	3	3	2	2	2	1	1	2	1	2
CO4	3	3	2	2	3	2	1	1	1	2	1	2
CO5	3	3	2	3	3	2	1	1	1	2	1	2

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CM2501 CLIMATE SYSTEM AND DYNAMICS

Objective:

To provide students with a comprehensive understanding of the Earth's climate system, its components, dynamics, and the implications of climate change on the environment and society.

Unit 1: Introduction to Climate System

Overview of the Climate System: Components (atmosphere, hydrosphere, cryosphere, lithosphere, and biosphere).

Energy Balance: Solar radiation, Earth's energy budget, albedo, greenhouse effect.

Climate Zones: Koppen classification, global wind systems, and ocean currents.

Unit 2: Atmospheric Dynamics

Atmospheric Circulation: General circulation, Hadley, Ferrel, and polar cells.

Weather Systems: Cyclones, anticyclones, monsoons, and jet streams.

Climate Phenomena: El Niño, La Niña, and the Southern Oscillation (ENSO).

Weather vs. Climate: Short-term and long-term atmospheric processes.

Unit 3: Oceanic and Cryospheric Processes

Oceans in Climate System: Thermohaline circulation, upwelling, and heat transport.

Cryosphere: Ice sheets, glaciers, sea ice, and permafrost – role in climate regulation.

Sea-Level Rise: Causes, trends, and impacts.

Interactions: Ocean-atmosphere interactions and their role in climate dynamics.

Unit 4: Climate Variability and Change

Natural Climate Variability: Milankovitch cycles, volcanic activity, solar output variations.

Human-Induced Changes: Greenhouse gas emissions, deforestation, urbanization.

Climate Models: Overview, types, and role in predicting future scenarios.

Impacts of Climate Change: Biodiversity loss, extreme weather events, and socio-economic impacts.

Unit 5: Climate Policy and Mitigation

Global Frameworks: Kyoto Protocol, Paris Agreement, IPCC reports.

Mitigation Strategies: Renewable energy, carbon capture, afforestation, and geoengineering.

Adaptation Approaches: Disaster risk management, climate-resilient infrastructure.

Case Studies: National and global efforts for climate action.

Course Outcomes (COs)

CO1: Understand the components and functioning of the climate system.

CO2: Analyze atmospheric and oceanic processes driving climate dynamics.

CO3: Evaluate the role of natural and anthropogenic factors in climate variability and change.

CO4: Interpret the impacts of climate change on ecosystems and human systems.

CO5: Explore global climate policies and strategies for mitigation and adaptation.

Reference Books

1. "The Climate System" by A.M. Carleton and F.A. Walker.
2. "Atmosphere, Ocean and Climate Dynamics" by John Marshall and R. Alan Plumb.
3. "Introduction to Climate Modelling" by Thomas Stocker.
4. "Paleoclimatology: Reconstructing Climates of the Quaternary" by Raymond S. Bradley.
5. "Climate Change: The Science of Global Warming and Our Energy Future" by Edmond A. Mathez.
6. IPCC Assessment Reports (latest editions).

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
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CO3	3	3	3	3	2	3	3	1	1	2	1	3
CO4	3	3	2	3	2	3	3	1	1	2	1	3
CO5	3	3	3	3	3	3	3	1	1	3	2	3

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MA2590 ENGINEERING MATHEMATICS – III

Objective:

To enhance the analytical skills of students through advanced mathematical concepts required in various engineering domains, including linear algebra, differential equations, statistics, probability, and transforms.

Unit 1: Linear Algebra

Vector Spaces: Basis, dimension, and linear independence.

Matrix Algebra: Rank, inverse, eigenvalues, and eigenvectors.

System of Linear Equations: Solutions using Gauss elimination, LU decomposition, and matrix inversion.

Applications: Transformation in 3D space and engineering problems.

Unit 2: Partial Differential Equations (PDEs)

Introduction to PDEs: Formation and solutions of first and second-order PDEs.

Classification: Parabolic, hyperbolic, and elliptic equations.

Applications: Heat conduction equation, wave equation, and Laplace's equation.

Unit 3: Probability and Statistics

Probability: Definitions, conditional probability, and Bayes' theorem.

Random Variables: Discrete and continuous distributions, mean, variance, and standard deviation.

Statistical Methods: Correlation, regression, and curve fitting.

Applications: Reliability and quality control in engineering.

Unit 4: Fourier and Laplace Transforms

Fourier Series: Periodic functions, half-range expansions.

Fourier Transform: Properties and applications.

Laplace Transform: Properties, inverse Laplace transform, and its application to solve differential equations.

Unit 5: Numerical Methods

Root Finding: Bisection, Newton-Raphson, and secant methods.

Numerical Integration: Trapezoidal and Simpson's rules.

Numerical Solutions of Differential Equations: Euler and Runge-Kutta methods.

Applications: Engineering optimization and simulation problems.

Course Outcomes (COs)

CO1: Solve complex engineering problems using linear algebra techniques.

CO2: Formulate and solve partial differential equations for engineering systems.

CO3: Analyze data and solve engineering problems using probability and statistics.

CO4: Apply Fourier and Laplace transforms for engineering applications.

CO5: Use numerical methods to solve mathematical models in engineering.

Reference Books

1. "Advanced Engineering Mathematics" by Erwin Kreyszig.
2. "Higher Engineering Mathematics" by B.S. Grewal.
3. "Probability and Statistics for Engineers and Scientists" by Ronald E. Walpole and Raymond H. Myers.
4. "Introduction to Numerical Analysis" by Richard L. Burden and J. Douglas Faires.
5. "Linear Algebra and its Applications" by Gilbert Strang.
6. "Fourier Series and Boundary Value Problems" by James Brown and Ruel Churchill.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	1	1	1	1	1	1	1	2
CO2	3	3	3	3	2	1	1	1	1	1	1	3
CO3	3	3	3	3	2	1	1	1	1	2	1	3
CO4	3	3	3	3	1	1	1	1	1	2	1	3
CO5	3	3	3	3	2	1	1	1	1	2	1	3

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CS2503 INFORMATION TECHNOLOGY

Objective:

This course provides students with an understanding of the essential concepts and technologies in Information Technology (IT), including computer hardware, software, networking, databases, web technologies, and their applications in real-world problems. The course also introduces programming skills, database management systems, and networking concepts essential for IT professionals.

Unit 1: Introduction to Information Technology

Overview of Information Technology: Definition, scope, and the importance of IT in business, education, and industries.

Components of an IT System: Hardware (CPU, memory, storage devices), software (system software, application software), and data.

Basic Computer Architecture: Input/Output devices, computer memory, registers, ALU, CPU, and peripheral devices.

Operating Systems: Functions and types of operating systems, process management, memory management, file systems, and basic system commands.

Unit 2: Programming Fundamentals

Programming Basics: Introduction to programming languages, types of languages (procedural, object-oriented), algorithms, and flowcharts.

Data Types and Variables: Variables, constants, data types (integer, float, character, string), operators, and expressions.

Control Structures: Decision-making (if-else, switch-case), loops (for, while, do-while).

Functions and Arrays: Defining functions, recursion, arrays, and their applications.

Introduction to C/C++ Programming: Writing simple programs, basic input/output operations, and debugging.

Unit 3: Database Management Systems (DBMS)

Database Concepts: Data models (hierarchical, network, relational), database design, schemas, and instances.

Relational Model: Tables, rows, columns, keys (primary, foreign), normalization (1NF, 2NF, 3NF), relational algebra.

SQL (Structured Query Language): Introduction to SQL, SELECT queries, joins, subqueries, aggregate functions, and data manipulation (INSERT, UPDATE, DELETE).

DBMS Design: ER Model, designing and implementing databases using ER diagrams, normalization, and schema refinement.

Unit 4: Computer Networks and Internet Technologies

Basic Networking Concepts: Types of networks (LAN, WAN, MAN), OSI model, TCP/IP model, and communication protocols.

Network Topologies and Devices: Bus, star, ring, mesh topologies; routers, switches, hubs, and bridges.

IP Addressing and Subnetting: Understanding IP addresses, subnet masks, and IP classification.

Internet and Web Technologies: DNS, HTTP, HTML, CSS, JavaScript, and web browsers.

Networking Security: Introduction to firewalls, encryption, VPNs, and network security protocols.

Unit 5: Web Development and Applications

HTML & CSS: Basic HTML tags, forms, tables, and CSS for styling web pages.

Client-side Scripting: Introduction to JavaScript, event handling, and DOM manipulation.

Server-side Scripting: Introduction to PHP, ASP.NET, and JSP.

Database Connectivity: Connecting MySQL or SQL Server to web applications.

Web Security: Basic principles of web security, SQL injection, XSS, and secure coding practices.

Lab Experiments and Equipment List

1. Lab Experiments:

- Write and execute basic programs in C/C++ (data types, operators, loops, functions).
- Develop simple applications using arrays, strings, and functions in C/C++.

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- Create basic databases and perform SQL queries (SELECT, JOIN, INSERT, UPDATE, DELETE).
 - Design and develop static web pages using HTML and CSS.
 - Implement simple client-side scripting using JavaScript.
 - Build web applications with basic PHP/ASP.NET and database integration.
 - Setup and troubleshoot a small network, understanding routing, and subnetting.
2. **Equipment List:**
- Computers with C/C++ compilers (e.g., GCC, Turbo C++)
 - Web development tools (Notepad++, Sublime Text, Visual Studio Code)
 - Database Management System software (MySQL, SQL Server, PostgreSQL)
 - Server-side scripting environments (XAMPP/WAMP for PHP)
 - Networking setup (Router, Switches, Network cables, etc.)
 - Web browsers for testing (Chrome, Firefox, Edge)
 - Virtual Machines for network simulation (e.g., Cisco Packet Tracer, GNS3)

Course Outcomes (COs)

CO1: Understand the components and importance of Information Technology and its applications in different sectors.

CO2: Write simple programs using basic programming languages like C/C++ and understand programming fundamentals.

CO3: Design, implement, and manage databases using relational database systems and SQL.

CO4: Understand the fundamentals of computer networks, communication protocols, and network security.

CO5: Develop dynamic web applications using web technologies such as HTML, CSS, JavaScript, and server-side scripting.

Reference Books

1. "Introduction to Information Technology" by J. S. K. Kartha
2. "Computer Networking: A Top-Down Approach" by James Kurose and Keith Ross
3. "Database System Concepts" by Abraham Silberschatz, Henry Korth, and S. Sudarshan
4. "Web Technologies: HTML, JavaScript, PHP, Java, JSP, and MySQL" by Achyut Godbole
5. "Programming in C" by Stephen G. Kochan
6. "Head First Java" by Kathy Sierra and Bert Bates
7. "HTML & CSS: Design and Build Websites" by Jon Duckett

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	2	1	1	1	1	1	1	1	1
CO2	3	3	3	2	2	1	1	1	1	1	1	2
CO3	3	3	3	3	2	1	1	1	1	2	1	3
CO4	3	3	2	3	2	1	1	1	1	2	1	2
CO5	3	3	3	3	2	1	1	1	1	2	1	3

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CE2505 FLUID MECHANICS

Objective:

The course aims to introduce the fundamental concepts of fluid mechanics, including properties of fluids, fluid statics, fluid dynamics, flow through pipes, and open channel flow. Students will learn the mathematical analysis and application of fluid mechanics in engineering systems.

Unit 1: Introduction to Fluid Mechanics

Properties of Fluids: Density, viscosity, surface tension, compressibility.

Fluid Statics: Pressure variation in a fluid at rest, Pascal's Law, hydrostatic pressure, buoyancy, stability of floating bodies.

Manometry: Types of manometers (U-tube, inclined tube), and their applications in pressure measurement.

Fluid Kinematics: Streamlines, pathlines, streaklines, velocity field, and acceleration in fluid flow.

Types of Flow: Steady vs. unsteady, uniform vs. non-uniform, laminar vs. turbulent flow.

Unit 2: Fluid Dynamics

Continuity Equation: Derivation and application for steady flow.

Euler's Equation of Motion: Momentum balance and its applications.

Bernoulli's Equation: Derivation and practical applications in pipe flow, open channel flow, and velocity measurement.

Flow Measurement: Venturimeter, orifice meter, and rotameter.

Energy Equation: Work, energy, and power in fluid systems, including pump and turbine analysis.

Unit 3: Flow Through Pipes

Laminar and Turbulent Flow: Reynolds number, transition between laminar and turbulent flow, and types of flow in pipes.

Darcy-Weisbach Equation: Head loss due to friction in pipes, friction factor, Moody chart, and its applications.

Minor Losses: Losses due to fittings, valves, elbows, and sudden expansions.

Flow in Branched Pipes: Flow distribution in pipe networks (using Hardy Cross or equivalent methods).

Pipe Flow Design: Economic pipe diameter calculation, pump and pipe network analysis.

Unit 4: Open Channel Flow

Types of Open Channel Flow: Steady, unsteady, uniform, and non-uniform flow.

Chezy's and Manning's Formulas: Flow velocity calculation in open channels.

Specific Energy and Critical Flow: Concept of specific energy, critical depth, and flow velocity.

Flow Resistance: Roughness coefficients, and their influence on flow characteristics.

Flow in Rectangular and Trapezoidal Channels: Velocity distribution and discharge calculation.

Unit 5: Boundary Layer Theory and Flow Around Bodies

Boundary Layer Concept: Development of boundary layers, boundary layer thickness, laminar and turbulent boundary layers.

Drag and Lift: Drag coefficient, types of drag, and factors affecting drag and lift.

Flow Separation: Causes and effects of flow separation.

Applications of Boundary Layer Theory: Flow over airfoils, streamlined and bluff bodies, and analysis of flow separation.

Lab Experiments and Equipment List

Lab Experiments

1. **Verification of Bernoulli's Theorem:** Using a pipe flow setup to verify Bernoulli's equation.
2. **Flow through an Orifice:** To determine the discharge coefficient and study the flow characteristics through different orifice shapes.
3. **Measurement of Flow Rate Using Venturimeter:** To determine the flow rate in a pipe using a Venturimeter.
4. **Determination of Friction Factor in Pipe Flow:** To study head loss in laminar and turbulent flow and determine the friction factor.

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5. **Reynolds Number Experiment:** To measure the Reynolds number and classify the flow as laminar or turbulent.
6. **Flow Over Weir and Measurement of Discharge:** To determine the flow rate using different types of weirs.
7. **Flow through a Pipe Network:** To study and analyze flow distribution in a network of pipes.
8. **Boundary Layer Experiment:** To determine the boundary layer thickness and velocity distribution in the boundary layer.

Equipment List

- **Flow Measuring Devices:** Venturimeter, orifice meter, rotameter.
- **Manometers:** U-tube manometer, inclined manometer, digital manometer.
- **Pipes and Flow Setups:** Pipes of different diameters, fittings, pumps, and valves.
- **Flow Channel Apparatus:** Open channel flow setup, measuring flumes, and weirs.
- **Reynolds Number Apparatus:** Transparent pipe flow setups with seed particles to visualize flow patterns.
- **Boundary Layer Apparatus:** Wind tunnel or water channel with a flat plate for boundary layer experiments.
- **Pressure and Velocity Measurement Devices:** Pressure transducers, velocity probes, Pitot tube, and anemometers.

Course Outcomes (COs)

- CO1: Understand the fundamental properties of fluids and their behavior under different conditions.
 CO2: Apply the Bernoulli's and continuity equations to solve fluid flow problems in pipelines and open channels.
 CO3: Analyze laminar and turbulent flows and compute head losses in pipe flow using the Darcy-Weisbach equation.
 CO4: Apply Manning's and Chezy's formulas to determine the flow in open channels.
 CO5: Understand boundary layer theory and the effects of drag and lift on bodies immersed in fluid flow.

Reference Books

1. "Fluid Mechanics" by Frank M. White
2. "Introduction to Fluid Mechanics" by Robert W. Fox and Alan T. McDonald
3. "Fluid Mechanics and Hydraulic Machines" by R.K. Bansal
4. "A Textbook of Fluid Mechanics and Hydraulic Machines" by R. K. Rajput
5. "Flow in Pipes and Open Channels" by K. Subramanya
6. "Engineering Fluid Mechanics" by Donald F. Young and Bruce R. Okiishi

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	2	1	1	1	1	1	1	1	1
CO2	3	3	3	2	2	1	1	1	1	1	1	2
CO3	3	3	3	3	2	1	1	1	1	2	1	3
CO4	3	3	2	3	2	1	1	1	1	2	1	2
CO5	3	3	3	3	2	1	1	1	1	2	1	3

CE2502 -SURVEYING AND LEVELLING

Objective:

This course aims to introduce students to the principles and techniques of surveying and levelling, which are essential for civil engineering projects. It covers both theoretical aspects and practical applications related to measuring distances, angles, elevations, and preparing maps.

Unit 1: Introduction to Surveying

Surveying Fundamentals: Definition, types of surveys (land, hydrographic, topographic, etc.), importance, and principles of surveying.

Classification of Surveys: Based on instruments used, types of measurements, and the purpose (chain survey, compass survey, plane table survey, and levelling).

Surveying Instruments: Tape, chain, compass, theodolite, total station, and their uses.

Error in Surveying: Types of errors (instrumental, personal, and natural), precision, accuracy, and corrections.

Surveying Methods: Linear measurement, angular measurement, levelling, and triangulation.

Unit 2: Linear Measurement and Chain Surveying

Chain Surveying: Measurement of distances using chain and tape, ranging, and chaining of lines.

Measurement of Distances: Direct and indirect methods (tape, chain, stadia, and EDM).

Setting Out Right Angles: Using theodolite, chain, and tape for setting out right angles in the field.

Recording Field Notes: Techniques for accurate field note-taking during chain surveying, distance measurement, and compass bearings.

Errors in Linear Measurement: Effect of sag, temperature, and slope on chain measurement. Methods to minimize errors.

Unit 3: Angular Measurement and Compass Surveying

Compass Surveying: Use of compass for direction measurement, traversing, and plotting angles.

Theodolite: Parts, functions, and uses of theodolite for measuring angles, horizontal and vertical measurements.

Errors in Angular Measurement: Measurement errors, adjustment methods, and error minimization in compass and theodolite surveys.

Traversing: Open and closed traversing, method of closing the traverse, and calculation of bearings and closing error.

Ranging and Surveying of Angles: Techniques for angle measurement, angular correction, and plotting angles in a field survey.

Unit 4: Levelling and Contouring

Principles of Levelling: Definitions, terms used in levelling (line of collimation, level line, datum, benchmark, etc.).

Types of Levels: Dumpy level, tilting level, automatic level, and digital levels.

Types of Levelling Operations: Differential levelling, fly levelling, profile levelling, and reciprocal levelling.

Contouring: Methods of contouring (direct and indirect), characteristics of contour lines, and applications of contours in topographic maps.

Errors in Levelling: Instrumental and personal errors, correction for errors in levelling.

Unit 5: Modern Surveying Techniques

Total Station: Principles, types, and operation of total stations, applications in survey data collection, and reducing errors.

GPS Surveying: Introduction to GPS, its components, working principle, and applications in surveying.

Digital Levels and Laser Scanners: Working principles and applications in modern surveying.

Remote Sensing and GIS: Introduction to remote sensing technologies, GIS for mapping and spatial data analysis.

Survey Data Processing: Handling, processing, and error correction of survey data using software like AutoCAD, Surfer, and GIS tools.

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Lab Experiments and Equipment List

Lab Experiments

1. **Chain Surveying:** Measurement of distances with chain and tape, plotting and reducing the field notes.
2. **Compass Surveying:** Measurement of bearings, setting out right angles, and traversing using a compass.
3. **Levelling Operations:** Differential levelling using dumpy and automatic levels, calculation of reduced levels, and preparation of profiles.
4. **Contouring:** Preparation of contours based on levelling data.
5. **Theodolite Surveying:** Measurement of horizontal and vertical angles using a theodolite and closed traverse.
6. **Total Station Surveying:** Use of total station for distance and angle measurements, setting up the instrument, and data collection.
7. **GPS Surveying:** Collection of GPS data and processing for survey applications.
8. **GIS and Remote Sensing:** Practical application of GIS software to create maps and process survey data.

Equipment List

- **Linear Measurement Tools:** Chains, tapes (metallic and cloth), stadia rods, and distance measuring instruments.
- **Compass and Theodolites:** Prismatic compass, surveyor's compass, dumpy level, automatic level, digital level, and total station.
- **Levelling Instruments:** Dumpy level, tilting level, automatic level, laser levels.
- **GPS Receiver:** Handheld GPS devices, GNSS receivers.
- **Surveying Rods and Tripods:** Leveling rods, ranging rods, staff, and instrument tripods.
- **Field Note Books:** To record data accurately and perform calculations.
- **Computer Software:** AutoCAD, GIS software, and data processing software for survey computations.

Course Outcomes (COs)

CO1: Understand the basic principles of surveying, levelling, and their importance in civil engineering projects.

CO2: Gain proficiency in the use of traditional surveying instruments such as chains, tapes, compasses, theodolites, and levels.

CO3: Perform linear, angular, and levelling measurements with precision, and reduce and plot survey data.

CO4: Understand and apply modern surveying techniques, including the use of total stations, GPS, and GIS in surveying applications.

CO5: Gain practical experience in contouring, error corrections, and preparing field notes.

Reference Books

1. "Surveying Vol. I and II" by B.C. Punmia, Ashok Kumar Jain, Arun Kumar Jain
2. "Surveying and Levelling" by N.N. Basak
3. "Surveying" by R. Subramanian
4. "Surveying for Engineers" by A. M. Chandra
5. "Textbook of Surveying" by S. K. Duggal
6. "Surveying and Levelling" by T.P. Kanetkar, S.V. Kulkarni
7. "GPS for Land Surveyors" by Jan Van Sickle

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CO4	3	3	3	2	3	2	2	1	1	2	2	3
CO5	3	3	3	3	2	2	1	1	1	3	2	3

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CE2508 - STRENGTH OF MATERIALS

Objective:

The course on Strength of Materials (SoM) introduces students to the fundamental concepts of material strength, including stress, strain, deformation, and the behavior of materials under various loading conditions. The course aims to develop the ability to analyze and design structures subjected to loads, ensuring they can withstand different forces without failure.

Unit 1: Introduction to Strength of Materials

Basic Concepts: Stress, strain, and their types (normal, shear).

Stress-Strain Curve: Hooke's law, Young's modulus, shear modulus, and bulk modulus.

Types of Stress and Strain: Tensile, compressive, and shear stresses.

Poisson's Ratio: Relation between longitudinal strain and lateral strain.

Elasticity: Concept of elastic limit, proportional limit, and yield point.

Applications: Behavior of materials under axial loads.

Unit 2: Axial Loading

Tensile and Compressive Forces: Stress-strain relationship under axial loading.

Deformation Under Axial Load: Elongation, contraction, and deformation of bars under axial loads.

Strain Energy: Strain energy stored in elastic materials and its applications in structural design.

Factor of Safety: Determining the factor of safety under axial load.

Thermal Strain: Effect of temperature changes on elongation or contraction.

Unit 3: Bending of Beams

Bending Stress: Derivation of bending stress formula (flexure formula), moment-curvature relationship.

Neutral Axis: Location and significance of the neutral axis in bending.

Shear Stress in Beams: Distribution of shear stress across the section of the beam.

Deflection of Beams: Methods of calculating deflection in beams using double integration, moment-area, and energy methods.

Bending in Simple Beams: Analysis of simply supported, cantilever, and overhanging beams under various loading conditions.

Unit 4: Torsion and Shear in Shafts

Torsion: Concept of torsional stress, shear stress distribution in shafts.

Torsional Deformation: Angle of twist, the relationship between torque, shear modulus, and the angle of twist.

Strength of Shafts: Strength of shafts under torsion, determining the minimum diameter for a shaft based on material strength.

Solid and Hollow Shafts: Difference in torsional strength between solid and hollow shafts.

Applications: Torsion in machine elements like shafts, couplings, and axles.

Unit 5: Combined Loading and Column Buckling

Combined Loading: Superposition of axial, bending, and torsional loads.

Analysis of Combined Stress: Principal stresses and Mohr's circle for combined stress systems.

Thin and Thick Cylinders: Analysis of stresses in thin-walled and thick-walled pressure vessels.

Buckling of Columns: Euler's theory of buckling, critical load, and applications of buckling in structural design.

Slenderness Ratio: Calculation and significance of slenderness ratio in column buckling.

Course Outcomes (COs)

CO1: Understand the basic concepts of stress, strain, and material behavior under different loading conditions.

CO2: Analyze the axial, bending, and torsional stress in structural elements and determine deformations.

CO3: Calculate deflection and bending stress in beams under various loading conditions.

CO4: Understand the theory and application of column buckling and determine the stability of columns under axial loads.

CO5: Analyze and design structural components subjected to combined loading (axial, bending, and torsion).

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Reference Books

1. "Strength of Materials" by R.K. Bansal
2. "Strength of Materials" by Ferdinand P. Beer and E. Russell Johnston Jr.
3. "Mechanics of Materials" by James M. Gere
4. "Strength of Materials" by S. Ramamrutham
5. "Introduction to Solid Mechanics" by Irving H. Shames
6. "Engineering Mechanics of Solids" by E.P. Popov
7. "Mechanics of Materials" by R.C. Hibbeler

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C01	3	2	3	2	1	1	1	1	1	1	2	2
C02	3	3	3	3	2	2	1	2	1	3	2	3
C03	3	3	3	2	2	2	1	1	1	3	2	3
C04	3	3	3	2	3	2	1	1	1	2	2	3
C05	3	3	3	3	2	3	2	2	1	3	3	3

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CE2503 ATMOSPHERIC SCIENCE AND METEOROLOGY

Objective:

The course in Atmospheric Science and Meteorology aims to provide students with an understanding of the physical processes governing the atmosphere, the principles of meteorology, weather phenomena, and the tools and techniques used in atmospheric science. It also covers the application of meteorological principles to environmental and engineering problems.

Unit 1: Introduction to Atmospheric Science

Structure of the Atmosphere: Layers of the atmosphere, composition, and characteristics of each layer.

Pressure and Temperature Distribution: Variation of pressure and temperature with altitude, ideal gas law.

Atmospheric Circulation: The role of the Coriolis effect, jet streams, and global wind systems (trade winds, westerlies, and polar easterlies).

Hydrological Cycle: Evaporation, condensation, precipitation, and runoff.

Atmospheric Stability: Definition of stable, unstable, and neutral conditions, lapse rate, and stability indices.

Unit 2: Weather Systems and Phenomena

Weather Elements: Temperature, humidity, pressure, wind, cloud cover, and precipitation.

Weather Fronts: Cold, warm, stationary, and occluded fronts and their associated weather conditions.

Cyclones and Anticyclones: Formation, characteristics, and impact of cyclones (both tropical and extratropical) and anticyclones.

Thunderstorms and Tornadoes: Development of thunderstorms, tornado formation, and related severe weather phenomena.

Monsoon Systems: Seasonal winds, impact on weather and climate, and monsoon regions.

Unit 3: Meteorological Instruments and Data Collection

Basic Instruments: Thermometer, barometer, anemometer, hygrometer, and rain gauge.

Advanced Instruments: Radiosonde, weather radar, weather satellites, and Doppler radar.

Weather Stations and Networks: Operation and role of meteorological stations, the process of data collection, and reporting systems.

Remote Sensing in Meteorology: Applications of remote sensing and satellite-based data in weather forecasting and environmental monitoring.

Data Interpretation: Reading and interpreting meteorological charts, pressure patterns, isobars, and weather maps.

Unit 4: Atmospheric Dynamics and Thermodynamics

Thermodynamics of the Atmosphere: First and second laws of thermodynamics applied to atmospheric processes.

Vertical Motion: Air parcels and vertical movement of air, convection, advection, and their effects on weather.

Wind Systems and Jet Streams: Global and local wind patterns, the role of jet streams in weather systems, and their influence on weather forecasting.

Atmospheric Waves: Sound waves, gravity waves, and their role in weather phenomena.

Radiation and Energy Transfer: Solar radiation, Earth's radiation balance, and heat transfer processes in the atmosphere.

Unit 5: Climate Change and Environmental Impacts

Climate vs. Weather: Understanding the difference between short-term weather and long-term climate.

Greenhouse Effect: Role of greenhouse gases in global warming, carbon cycle, and anthropogenic climate change.

Global Warming and its Effects: Impacts on weather patterns, sea level rise, and global ecosystems.

Weather Forecasting Models: Numerical weather prediction models, the role of supercomputers in weather forecasting.

Mitigation and Adaptation Strategies: Sustainable practices, policy frameworks, and adaptation strategies for climate change.

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Lab Experiments and Equipment List

Lab Experiments

1. **Measurement of Atmospheric Pressure:** Using a barometer to measure and record atmospheric pressure at different times and locations.
2. **Temperature and Humidity Measurement:** Using thermometers and hygrometers to measure temperature and relative humidity.
3. **Rainfall Measurement:** Using a rain gauge to measure precipitation and studying rainfall patterns over time.
4. **Wind Speed and Direction:** Using an anemometer and wind vane to measure wind speed and direction at different locations.
5. **Cloud Classification:** Identifying and classifying clouds based on their appearance and the weather conditions.
6. **Observation of Weather Fronts:** Studying the changes in weather patterns associated with the passage of cold, warm, and occluded fronts.
7. **Use of Weather Radar Data:** Interpreting weather radar images to track storms and other weather phenomena.
8. **Simulation of Atmospheric Processes:** Laboratory models for demonstrating atmospheric circulation, thermal convection, and weather systems.

Equipment List

- **Thermometers:** For measuring air temperature.
- **Barometers:** For measuring atmospheric pressure.
- **Anemometers:** For measuring wind speed.
- **Hygrometers:** For measuring humidity.
- **Rain Gauges:** For measuring precipitation.
- **Wind Vanes:** For measuring wind direction.
- **Cloud Chamber:** For studying cloud formation.
- **Radiosondes:** For measuring temperature, pressure, and humidity at different altitudes.
- **Weather Satellites:** For remote sensing of atmospheric conditions.
- **Weather Radar Systems:** For monitoring storm movements and precipitation.
- **Weather Station:** A comprehensive setup for collecting meteorological data (e.g., temperature, humidity, pressure, wind speed, and rainfall).

Course Outcomes (COs)

- CO1: Understand the fundamental principles governing atmospheric science and meteorology.
CO2: Analyze and interpret various weather phenomena such as cyclones, fronts, and thunderstorms.
CO3: Gain hands-on experience in using meteorological instruments and data collection techniques.
CO4: Understand the thermodynamic processes and dynamics of the atmosphere.
CO5: Evaluate the impact of climate change on weather patterns, ecosystems, and global weather systems.

Reference Books

1. **"Meteorology: Understanding the Atmosphere"** by Steven A. Ackerman and John A. Knox
2. **"Atmospheric Science: An Introductory Survey"** by John M. Wallace and Peter V. Hobbs
3. **"Introduction to Atmospheric Chemistry"** by Daniel J. Jacob
4. **"Meteorology for Scientists and Engineers"** by A. M. Blyth and A. R. H. Baugh
5. **"Fundamentals of Atmospheric Physics"** by Murry L. Salby
6. **"Principles of Meteorology"** by J. R. Houghton
7. **"The Physics of Climate"** by Thomas R. Karl and Kevin E. Trenberth

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2	1	1	1	1	1	2	2	1
CO2	3	3	3	3	2	1	1	2	2	3	2	3
CO3	3	3	3	2	3	2	1	1	1	2	3	3
CO4	3	3	3	3	3	2	1	2	2	3	2	3
CO5	3	3	3	3	3	2	2	2	1	3	3	3

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CE2535 -FUNDAMENTAL OF ENVIRONMENTAL CHEMISTRY

Objective:

The course on Environmental Chemistry provides students with an understanding of chemical processes in the environment and their impact on environmental pollution and sustainability. The course will introduce students to key concepts in atmospheric, aquatic, and soil chemistry, and focus on the chemical dynamics of natural systems and human-made environments.

Unit 1: Introduction to Environmental Chemistry

Definition and Scope of Environmental Chemistry: Basic principles, key concepts, and interdisciplinary nature.

Environmental Cycles: Hydrological cycle, carbon cycle, nitrogen cycle, sulfur cycle, and biogeochemical cycles.

Environmental Pollution: Types of pollution (air, water, soil, noise, and thermal pollution), sources, and impacts.

Chemistry of Pollutants: Properties and behavior of pollutants, including heavy metals, pesticides, and organic pollutants.

Environmental Monitoring and Regulations: Introduction to environmental monitoring, major environmental laws, and regulatory agencies.

Unit 2: Chemistry of Air Pollution

Composition of the Atmosphere: Major gases in the atmosphere, role of oxygen, nitrogen, CO₂, and trace gases.

Sources of Air Pollutants: Industrial processes, vehicles, natural sources (volcanic eruptions, wildfires).

Chemistry of Common Air Pollutants: Sulfur dioxide (SO₂), nitrogen oxides (NO_x), carbon monoxide (CO), ozone (O₃), particulate matter (PM), and volatile organic compounds (VOCs).

Photochemical Smog: Formation, effects on health and the environment, and control measures.

Indoor Air Pollution: Sources and effects of indoor pollutants, including radon, tobacco smoke, and household chemicals.

Unit 3: Chemistry of Water Pollution

Water Quality Parameters: pH, dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD), turbidity, and total dissolved solids (TDS).

Types of Water Pollutants: Organic, inorganic, and biological pollutants.

Sources of Water Pollution: Industrial waste, agricultural runoff, sewage, and oil spills.

Wastewater Treatment: Primary, secondary, and tertiary treatment processes, and their chemistry.

Eutrophication: Chemical processes in eutrophic water bodies, causes, and impacts of nutrient pollution.

Unit 4: Chemistry of Soil Pollution

Soil Composition: Organic and inorganic components of soil, soil pH, and soil texture.

Types of Soil Pollutants: Heavy metals (lead, cadmium, mercury), pesticides, fertilizers, and organic pollutants.

Sources of Soil Pollution: Agricultural practices, industrial waste disposal, and hazardous chemicals.

Soil Contamination and Toxicity: The chemistry of soil contaminants, bioavailability, and toxicity to plants, animals, and humans.

Soil Remediation: Chemical treatments for soil decontamination, bioremediation, and phytoremediation.

Unit 5: Green Chemistry and Environmental Sustainability

Principles of Green Chemistry: Introduction to green chemistry, prevention of waste, use of renewable resources, and energy efficiency.

Sustainable Chemical Processes: Green solvents, waste minimization, and eco-friendly chemical reactions.

Renewable Energy: Solar, wind, and bioenergy: chemical processes involved in sustainable energy production.

Climate Change and Chemistry: The role of chemistry in understanding and mitigating climate change, greenhouse gases, carbon footprint, and carbon sequestration.

Environmental Impact Assessment (EIA): Chemical methods for evaluating the environmental impact of industrial processes, development projects, and new chemical products.

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Lab Experiments and Equipment List

Lab Experiments

1. **Water Quality Testing:** Measurement of pH, turbidity, DO, BOD, and COD in water samples.
2. **Air Quality Monitoring:** Measurement of pollutants like sulfur dioxide (SO₂), nitrogen oxides (NO_x), and particulate matter (PM).
3. **Soil pH and Nutrient Content Testing:** Determining pH, moisture content, and nitrogen, phosphorus, and potassium levels in soil.
4. **Preparation of Standard Solutions:** Preparation of stock solutions and calibration curves for chemical analysis.
5. **Eutrophication Test:** Measuring the effects of nutrient-rich runoff on water bodies using simulated models.
6. **Titration of Water Samples:** To measure alkalinity, hardness, and dissolved oxygen content.
7. **Soil Remediation Techniques:** Experimental setup to test bioremediation and chemical treatments for contaminated soil.

Equipment List

- **pH Meters:** For measuring pH in soil and water.
- **Spectrophotometers:** For measuring light absorption in water samples, used to determine concentrations of pollutants.
- **BOD and COD Test Kits:** To determine biological oxygen demand and chemical oxygen demand.
- **Titration Setups:** For determining concentrations of different ions and pollutants in water and soil.
- **Water and Soil Sampling Kits:** For collection and transportation of samples to labs for analysis.
- **Air Pollution Monitoring Devices:** For monitoring particulate matter (PM), NO_x, SO₂, CO, and VOCs.
- **Fume Hoods:** For handling volatile or toxic substances safely in laboratory experiments.
- **Microscopes:** For observing biological pollutants in water samples (e.g., algae and bacteria).

Course Outcomes (COs)

CO1: Understand the basic principles of environmental chemistry and the role of chemical processes in environmental systems.

CO2: Analyze various types of pollutants and their sources in the air, water, and soil, and understand their environmental impact.

CO3: Develop practical skills in monitoring and analyzing environmental pollution using standard chemical techniques.

CO4: Understand the principles and applications of green chemistry in reducing environmental impact and promoting sustainability.

CO5: Evaluate the chemistry involved in climate change, renewable energy, and environmental sustainability.

Reference Books

1. "Environmental Chemistry" by Stanley E. Manahan
2. "Chemistry for Environmental Engineering and Science" by Mark J. Hammer and Mark A. Hammer
3. "Environmental Chemistry: A Global Perspective" by Gary W. VanLoon and Stephen J. Duffy
4. "Principles of Environmental Chemistry" by James M. Girard
5. "Green Chemistry: Theory and Practice" by Paul T. Anastas and John C. Warner
6. "Environmental Chemistry of Pollutants" by Venkat S. R. Anantharaman
7. "Environmental Chemistry: A Textbook" by Henk A. L. Kegley

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	2	2	2	1	1	1	2	2
CO2	3	3	3	3	2	3	2	2	2	2	3	3
CO3	3	3	3	3	2	2	3	2	2	3	2	3
CO4	3	3	3	2	3	3	1	1	2	3	3	3
CO5	3	3	3	3	2	2	2	2	1	3	3	3

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CE2536 - STRUCTURAL DESIGN

Objective:

The Structural Design course aims to provide students with a fundamental understanding of the principles and techniques involved in the design of structures. The course covers the application of various materials like steel, concrete, and timber for different structural components, including beams, columns, slabs, and foundations, with an emphasis on both theory and practical application.

Unit 1: Introduction to Structural Design

Basic Principles of Structural Design: Overview of structural engineering, types of structures (buildings, bridges, dams, towers), design philosophy (limit state design, working stress method).

Design Codes and Standards: Overview of IS codes, international design standards (e.g., AISC, Eurocode), and their importance in structural design.

Load Considerations: Types of loads acting on structures – dead loads, live loads, wind loads, seismic loads, temperature effects, and other environmental factors.

Structural Analysis: Methods of structural analysis including equilibrium equations, shear and bending moment diagrams, deflection, and compatibility.

Material Properties: Overview of material properties (steel, concrete, timber), their strengths, and behavior under different loading conditions.

Unit 2: Design of Structural Elements in Steel

Introduction to Steel Structures: Types of steel sections (I-beams, channels, angles), material properties, and design concepts for steel structures.

Design of Beams: Flexural design of beams under different loading conditions, use of IS 800 code for design.

Design of Columns: Axial load design of steel columns, combined axial load and bending, slenderness ratio, and the use of IS 800 for column design.

Design of Connections: Types of connections (welded, bolted), design of bolted connections and welded joints.

Design of Roof Trusses and Frames: Design and analysis of trusses for roofs, including load distribution, deflection checks, and stability considerations.

Unit 3: Design of Structural Elements in Concrete

Introduction to Concrete Structures: Material properties of concrete, types of concrete (normal, high-strength, reinforced, prestressed).

Reinforced Concrete Beams: Flexural design of beams for various load conditions using IS 456 code, calculation of moment of resistance, shear stress, and deflection.

Reinforced Concrete Slabs: Design of one-way and two-way slabs, cantilever, and simply supported slabs.

Design of Columns: Axial and bending stress in columns, design of short and long columns, lateral load effects.

Reinforced Concrete Foundations: Types of foundations, design of isolated footings, combined footings, and strap footings.

Unit 4: Design of Structural Elements in Timber

Introduction to Timber Structures: Timber as a building material, properties of timber, and types of timber sections.

Design of Timber Beams: Flexural design of timber beams under different loads using IS 883.

Design of Timber Columns: Axial load design of timber columns, effects of slenderness and lateral-torsional buckling.

Design of Joints and Connections in Timber: Various types of timber joints, design of nailed and bolted connections.

Design of Roof Frames in Timber: Design and analysis of timber roof structures, including trusses, rafters, and purlins.

Unit 5: Advanced Topics in Structural Design

Prestressed Concrete: Basic concepts of prestressed concrete, methods of prestressing, design of prestressed beams and slabs.

Seismic Design of Structures: Design considerations for seismic loads, response spectrum method, lateral load distribution, and dynamic analysis.

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Design of Composite Structures: Steel-concrete composite beams and columns, advantages, and design principles.

Sustainability in Structural Design: Use of green materials, energy-efficient design, and sustainability practices in structural engineering.

Computer-Aided Structural Design: Introduction to software tools (STAAD Pro, ETABS) for structural analysis and design.

Lab Experiments and Equipment List

Lab Experiments

1. **Testing of Steel Sections:** Determining the properties of steel sections under tension, compression, and bending.
2. **Concrete Mix Design:** Determining the proportions of different ingredients to make concrete of a specified strength.
3. **Testing of Concrete Beams:** Load testing of concrete beams to study bending behavior and determine ultimate load capacity.
4. **Testing of Steel Columns:** Buckling and load testing of steel columns under axial and lateral loads.
5. **Reinforced Concrete Slab Testing:** Load testing on reinforced concrete slabs to evaluate deflections, cracks, and failure modes.
6. **Timber Beam Testing:** Determining the strength and failure behavior of timber beams under bending loads.
7. **Design and Analysis Using Structural Software:** Hands-on practice using structural design software like STAAD Pro, ETABS, or SAP2000 for structural analysis and design.

Equipment List

- **Universal Testing Machine:** For tensile, compressive, and bending tests on materials.
- **Concrete Compression Testing Machine:** For testing the compressive strength of concrete.
- **Rebar Tension Testing Machine:** For testing the tensile strength of reinforcement bars.
- **Load Testing Frames:** For applying axial and bending loads on steel columns and beams.
- **Strain Gauges:** For measuring strain in structural components under loading conditions.
- **Deflection Measuring Instruments:** For measuring the deflection of beams and slabs under load.
- **Timber Testing Apparatus:** For testing bending strength and compressive strength of timber.
- **Structural Design Software:** STAAD Pro, ETABS, SAP2000 for analysis and design of structures.

Course Outcomes (COs)

CO1 Understand the principles of structural design and material properties in the context of building, bridges, and other civil structures.

CO2 Design structural elements like beams, columns, slabs, and foundations in steel, concrete, and timber, adhering to relevant design codes.

CO3 Develop the ability to analyze and design complex structural systems, including trusses, roof systems, and multi-storey buildings.

CO4 Gain practical knowledge in using software tools for structural design and analysis.

CO5 Understand the importance of sustainability and resilience in modern structural engineering practices, particularly in earthquake-resistant and energy-efficient design.

Reference Books

1. "Design of Reinforced Concrete Structures" by N. C. Sinha and S. S. R. Y. R.
2. "Structural Analysis" by R. C. Hibbeler
3. "Principles of Structural Design" by L. S. Negi
4. "Design of Steel Structures" by S. K. Duggal
5. "Timber Design" by S. R. Arora and P. K. Jain
6. "Design of Concrete Structures" by Arthur H. Nilson
7. "Structural Steel Design" by William T. Segui
8. "Introduction to Structural Engineering" by A. C. Ugural

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	2	2	1	1	1	2	1	2
CO2	3	3	3	3	2	3	2	2	2	2	3	3
CO3	3	3	3	3	3	2	3	3	2	3	3	3
CO4	2	2	2	2	1	1	1	1	1	2	2	2
CO5	3	3	3	3	3	2	3	2	2	3	3	3

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CE2522 - SOIL MECHANICS & FOUNDATION ENGINEERING

Objective:

The course aims to provide students with the fundamental knowledge of soil mechanics and the principles of foundation engineering. The course covers the behavior of soil under different loading conditions, the design of shallow and deep foundations, and the methods used to evaluate soil properties. Emphasis will be placed on practical applications and understanding the interaction between soil and structures.

Unit 1: Introduction to Soil Mechanics

Soil Formation and Classification: Origin of soil, mineral composition, soil structure, and soil types (sand, silt, clay, etc.).

Soil Properties: Physical properties of soil (density, void ratio, porosity, permeability, plasticity, shrinkage limit, etc.).

Soil Classification Systems: IS classification system, USCS (Unified Soil Classification System).

Phase Diagram of Soil: Understanding the relationships between soil mass, water content, and air voids.

Soil Compaction: Principles of compaction, compaction methods, and field compaction tests (Proctor test).

Unit 2: Permeability and Seepage

Permeability of Soils: Darcy's law, determination of permeability by laboratory methods (constant head and falling head tests), and field tests.

Factors Affecting Permeability: Particle size, soil structure, water content, and type of soil.

Seepage Analysis: Seepage through soil strata, Darcy's law for two-dimensional flow, flow nets, and applications in engineering.

Application in Dam and Embankment Design: Calculation of seepage in earth dams, effect of seepage on stability.

Unit 3: Shear Strength of Soils

Shear Strength Concepts: Mohr-Coulomb theory, cohesion, and internal friction angle of soils.

Laboratory Shear Tests: Direct shear test, triaxial compression test, unconfined compression test.

Factors Affecting Shear Strength: Type of soil, moisture content, density, and consolidation.

Effective Stress Principle: Concept of effective stress in soils and its role in soil shear strength.

Stability of Slopes: Methods of slope stability analysis (limit equilibrium, factor of safety, Taylor's stability charts).

Unit 4: Foundation Engineering

Shallow Foundations: Types of shallow foundations (isolated footings, combined footings, strip footings, raft foundations), design considerations, and settlement analysis.

Bearing Capacity of Soils: Definition of bearing capacity, factors affecting bearing capacity, Terzaghi's bearing capacity theory, plate load test, and empirical methods for determining bearing capacity.

Settlement of Foundations: Types of settlement (immediate, consolidation, and secondary settlement), factors affecting settlement, and methods for estimating settlements.

Deep Foundations: Types of deep foundations (piles, caissons, and piers), selection of foundation type, and pile capacity estimation (static and dynamic analysis).

Unit 5: Foundation Design and Applications

Design of Shallow Foundations: Design of isolated footings, combined footings, and raft foundations.

Pile Foundation Design: Types of piles, pile load analysis (static and dynamic), design of piles in cohesive and cohesionless soils.

Soil-Structure Interaction: Settlement analysis of foundations using elastic theory, consolidation settlements under various loading conditions, and structural analysis considering soil-structure interaction.

Special Foundations: Foundations for expansive soils, foundations on soft soils, and foundations in permafrost areas.

Geotechnical Site Investigation: Methods of site investigation (boring, trial pits, and geophysical methods), and interpretation of soil test results for foundation design.

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Course Outcomes (COs)

- CO1:** Understand the basic principles of soil mechanics, including soil classification, properties, and compaction.
- CO2:** Gain knowledge of soil permeability, seepage, and their role in design applications like dams and embankments.
- CO3:** Evaluate the shear strength of soils through laboratory tests and apply this knowledge in the design of foundations.
- CO4:** Apply the principles of soil mechanics in designing shallow and deep foundations, including their bearing capacity and settlement characteristics.
- CO5:** Understand the importance of soil-structure interaction, geotechnical site investigation, and the design of special foundations under varying conditions.

Reference Books

1. "Soil Mechanics and Foundations" by B. M. Das
2. "Principles of Geotechnical Engineering" by Braja M. Das
3. "Foundation Design: Principles and Practices" by Donald P. Coduto
4. "Geotechnical Engineering: Principles and Practices" by V.N.S. Murthy
5. "Soil Mechanics and Foundation Engineering" by K.R. Arora
6. "Foundation Analysis and Design" by Joseph E. Bowles
7. "Geotechnical Engineering: A Practical Problem-Solving Approach" by John N. C. D.
8. "Introduction to Soil Mechanics" by R. F. Craig

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1	2	2	1	1	2	1	2
CO2	2	3	2	2	2	2	1	2	1	2	3	2
CO3	3	2	3	3	2	2	2	2	3	3	2	3
CO4	3	3	3	3	3	3	2	2	2	3	3	3
CO5	3	3	3	3	2	3	3	2	3	3	3	3

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CE2537 -BUILDING SERVICES

Objective:

The course aims to familiarize students with the various essential services involved in the construction and operation of buildings. This includes electrical systems, HVAC (Heating, Ventilation, and Air Conditioning), plumbing, fire safety, and other key infrastructure that ensures the safety, comfort, and sustainability of built environments.

Unit 1: Introduction to Building Services

Introduction to Building Services: Definition, importance, and types of building services.

Types of Building Systems: Overview of electrical, HVAC, plumbing, firefighting, and other essential services in buildings.

Building Service Planning: Principles of service planning, system integration, and coordination in building designs.

Basic Terminology: Understanding key terminology such as MEP (Mechanical, Electrical, Plumbing), utilities, and other services.

Sustainability in Building Services: Overview of sustainable building practices, energy-efficient systems, and LEED certifications.

Unit 2: Electrical Services in Buildings

Electrical Systems: Overview of electrical distribution systems, wiring methods, and types of electrical circuits.

Power Supply and Distribution: Transformer installation, substations, electrical meter systems, and distribution boards.

Lighting Systems: Types of lighting, designing interior and exterior lighting, and energy efficiency in lighting systems (e.g., LED, CFL).

Electricity Distribution: Power supply to different zones of buildings, transformers, generators, and UPS systems.

Earthing and Grounding: Earthing requirements and techniques for electrical safety, grounding systems.

Electrical Safety: Protection devices, circuit breakers, and safety codes in buildings.

Unit 3: HVAC (Heating, Ventilation, and Air Conditioning) Systems

HVAC Overview: Basic concepts, types of HVAC systems (centralized and decentralized), components, and functions.

Heating Systems: Types of heating systems, such as central heating, floor heating, and radiators.

Ventilation: Principles of ventilation, mechanical ventilation, natural ventilation, and ventilation for different types of spaces.

Air Conditioning Systems: Types of air conditioning systems (window, split, package), components (compressors, evaporators, condensers), and their working principles.

Energy Efficiency in HVAC: Energy-saving strategies, efficient system design, and environmental impact of HVAC systems.

Air Quality Management: Managing indoor air quality (IAQ), air filtration, humidity control, and ventilation rates.

Unit 4: Plumbing and Sanitation Systems

Water Supply System: Design and installation of water supply systems (pumps, pipes, tanks, and distribution), sources of water, and water treatment.

Drainage Systems: Sanitary drainage systems, stormwater drainage, sewerage systems, and their design considerations.

Plumbing Fixtures: Types of plumbing fixtures (sinks, faucets, showers, toilets) and installation methods.

Rainwater Harvesting: Principles of rainwater harvesting, systems design, and applications in building construction.

Water Conservation: Techniques and technologies for reducing water consumption in buildings, including greywater recycling, low-flow devices, and water-efficient appliances.

Unit 5: Fire Protection and Safety Systems

Fire Safety Systems: Components of fire safety in buildings, including fire alarms, smoke detectors, and fire extinguishers.

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Fire Fighting Systems: Fire sprinklers, hose reels, hydrants, and pumps. Design considerations for different building types.

Escape Routes and Evacuation Plans: Designing safe and accessible escape routes, fire exits, staircases, and emergency plans.

Building Codes and Standards: National Building Code (NBC) and international standards for fire protection systems, fireproofing materials, and fire-resistance ratings.

Emergency Power Systems: Backup power systems (diesel generators, UPS), and their role in ensuring safety during power failures.

Fire Risk Assessment: Techniques for assessing and mitigating fire risks in buildings, including emergency drills.

Course Outcomes (COs)

CO1: Understand the principles and importance of building services such as electrical, HVAC, plumbing, and fire protection.

CO2: Design and plan electrical and lighting systems for buildings with attention to energy efficiency and safety.

CO3: Understand HVAC systems and their design, implementation, and maintenance for various building types.

CO4: Design and implement water supply, sanitation, and drainage systems, incorporating sustainable and efficient practices.

CO5: Implement and evaluate fire safety systems and ensure compliance with national and international building codes.

CO6: Analyze building systems holistically to ensure integration and optimize performance for safety, comfort, and sustainability.

Reference Books

1. **"Building Services Handbook"** by Fred Hall and Roger Greeno
2. **"Design of Building Services"** by R. J. Chudley and R. Greeno
3. **"Fundamentals of Building Services Engineering"** by F. Hall
4. **"Plumbing Engineering Design Handbook"** by American Society of Plumbing Engineers
5. **"HVAC Fundamentals"** by Samuel Sugarman
6. **"Fire Safety in Building Design"** by A. R. Smith
7. **"Electrical Systems in Buildings"** by E. W. Kimbrough
8. **"Building Services Engineering"** by Colin B. M. J. E.
9. **"Rainwater Harvesting Systems for Buildings"** by Subhash Bhatia

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	1	1	1	1	2	2	2	1	2
CO2	3	2	2	3	2	2	2	3	2	2	3	3
CO3	3	2	3	3	3	2	2	3	2	2	2	3
CO4	3	2	2	2	3	3	2	2	3	2	2	3
CO5	3	3	3	2	2	3	2	3	3	3	3	3
CO6	3	3	2	3	3	2	3	3	3	3	3	3

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MA2591-COMPUTER BASED NUMERICAL & STATISTICAL TECHNIQUES

Objective:

The course is designed to introduce students to the computational techniques for solving numerical problems and performing statistical analysis using computer-based tools. Students will learn algorithms and methods for solving engineering problems and analyzing data, which are essential for their future roles in industries and research.

Unit 1: Introduction to Numerical Methods

Introduction to Numerical Methods: Importance and application in engineering.

Error Analysis: Types of errors (round-off, truncation), propagation of errors, absolute and relative errors.

Solutions of Equations: Bisection method, Newton-Raphson method, Secant method, and their applications.

Convergence and Stability: Conditions for convergence of numerical methods and stability analysis.

Root-Finding Techniques: Graphical methods, fixed-point iteration method.

Unit 2: Numerical Differentiation and Integration

Numerical Differentiation: Derivation of numerical derivatives using forward, backward, and central difference methods.

Finite Difference Methods: Difference equations, boundary conditions, and the application in numerical solutions.

Numerical Integration: Trapezoidal rule, Simpson's rule, Gauss quadrature, and their implementation.

Error in Numerical Integration: Estimating error in numerical integration and choosing an optimal method.

Application: Solving engineering problems involving integration such as finding areas under curves.

Unit 3: Solving Linear Algebraic Equations

System of Linear Equations: Direct methods (Gaussian elimination, LU decomposition) and iterative methods (Jacobi, Gauss-Seidel, and Successive Over-Relaxation methods).

Matrix Operations: Matrix multiplication, inversion, and determinant calculation using numerical methods.

Eigenvalues and Eigenvectors: Power method, QR algorithm for finding eigenvalues and eigenvectors of matrices.

Applications: Solving structural analysis problems, electrical circuit analysis, and heat conduction problems.

Unit 4: Interpolation and Curve Fitting

Interpolation Methods: Lagrange's interpolation, Newton's interpolation, and Spline interpolation.

Curve Fitting: Least squares fitting of data to polynomials and other functions, regression analysis.

Numerical Differentiation and Integration using Interpolation: Application of interpolation methods in solving differential equations.

Barycentric Interpolation: Introduction to barycentric interpolation and its computational advantages.

Applications: Fitting experimental data to mathematical models and solving boundary value problems using interpolation techniques.

Unit 5: Statistical Methods and Data Analysis

Descriptive Statistics: Measures of central tendency (mean, median, mode), dispersion (variance, standard deviation).

Probability Distributions: Normal, binomial, Poisson distributions, and their applications.

Statistical Inference: Hypothesis testing, t-tests, chi-square tests, confidence intervals, regression analysis.

Data Fitting and Model Evaluation: Least squares method for fitting models to data, analysis of residuals, and goodness-of-fit tests.

Simulation and Monte Carlo Methods: Introduction to statistical simulation, random number generation, and Monte Carlo methods for solving complex engineering problems.

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Course Outcomes (COs)

- CO1:** Apply numerical methods to solve algebraic and transcendental equations.
- CO2:** Perform numerical differentiation and integration using appropriate techniques.
- CO3:** Solve systems of linear equations using direct and iterative methods.
- CO4:** Apply interpolation and curve fitting methods to analyze and predict data trends.
- CO5:** Perform statistical data analysis, hypothesis testing, and regression analysis using software tools.
- CO6:** Use Monte Carlo simulations to solve complex engineering problems.

Reference Books

1. **"Numerical Methods for Engineers"** by Steven C. Chapra and Raymond P. Canale
2. **"Introduction to Numerical Analysis"** by Josef Stoer and Roland Bulirsch
3. **"Numerical Methods in Engineering with Python"** by Jaan Kiusalaas
4. **"Applied Numerical Methods with MATLAB for Engineers and Scientists"** by Steven C. Chapra
5. **"Statistical Methods for Engineers"** by C. A. Bickel and K. A. D. Seitz
6. **"Applied Statistics and Probability for Engineers"** by Douglas C. Montgomery and George C. Runger
7. **"Monte Carlo Methods in Financial Engineering"** by Paul Glasserman

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2	1	2	1	2	3	2	2	3
CO2	3	3	2	2	2	3	1	3	3	3	2	2
CO3	3	2	3	3	2	3	2	3	3	2	3	3
CO4	3	3	2	3	3	2	2	3	3	3	2	2
CO5	3	3	3	2	2	3	2	3	2	3	3	2
CO6	2	2	3	2	3	3	2	2	3	3	2	3

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CM2502 - CLIMATE CHANGE SCIENCE

Objective:

This course introduces students to the science behind climate change, its impacts, and global efforts to mitigate it. The course explores the physical, chemical, and biological processes affecting Earth's climate system and human-driven influences that have accelerated climate change. It also covers the consequences of climate change on ecosystems, economies, and societies.

Unit 1: Introduction to Climate Change

Climate System and Components: Atmosphere, hydrosphere, lithosphere, and biosphere.

Earth's Energy Balance: Solar radiation, greenhouse effect, and energy exchange.

Climate vs. Weather: Understanding the distinction, timescales, and interactions.

Greenhouse Gases: Role of carbon dioxide, methane, nitrous oxide, and other gases in warming the Earth.

Natural vs. Anthropogenic Climate Change: Evidence for human-caused climate change vs. natural variations (volcanic activity, solar radiation).

Unit 2: Climate Change Mechanisms

Greenhouse Effect and Feedback Loops: Positive and negative feedback mechanisms, albedo effect, water vapor feedback.

Radiative Forcing: Direct and indirect effects of human activities (deforestation, industrial emissions).

Climate Models: Introduction to climate prediction models, uncertainties in forecasting.

Global Warming Potential (GWP): Comparison of various greenhouse gases and their effects.

Paleoclimate Data: Using ice cores, tree rings, and sediment layers to reconstruct past climates.

Unit 3: Impacts of Climate Change

Temperature and Precipitation Changes: Global warming trends, regional variations, and changes in precipitation patterns.

Sea-Level Rise: Causes (thermal expansion, glacial melt), implications for coastal areas.

Ocean Acidification: Impact on marine ecosystems and biota.

Extreme Weather Events: Increased frequency of hurricanes, droughts, floods, and heatwaves.

Ecosystem and Biodiversity Loss: Effects on terrestrial and marine ecosystems, extinction risk, and habitat loss.

Unit 4: Climate Change Mitigation and Adaptation

Mitigation Strategies: Reducing greenhouse gas emissions, renewable energy sources, carbon capture and storage (CCS).

International Agreements and Policies: The Kyoto Protocol, Paris Agreement, and their role in global climate policy.

Carbon Trading and Offsetting: Emissions trading schemes (ETS), carbon credits.

Adaptation Strategies: Building climate-resilient infrastructure, sustainable agriculture, water conservation.

Role of Technology: Geoengineering, bioengineering, and innovation in climate solutions.

Unit 5: Societal and Economic Impacts of Climate Change

Climate Change and Economic Development: Impact on industries, agriculture, and infrastructure.

Climate Justice: Ethical issues, equity in climate responsibility, and climate adaptation for vulnerable populations.

Public Perception and Media: How the public views climate change, the role of media in shaping opinions.

Climate Change and Health: Effects on public health (heat stress, vector-borne diseases, air pollution).

Global and Local Responses: Climate change action at local, national, and global levels.

Course Outcomes (COs)

CO1: Understand the science behind the climate system and the distinction between climate and weather.

CO2: Analyze the mechanisms of climate change, including greenhouse effects and radiative forcing.

CO3: Evaluate the impacts of climate change on ecosystems, biodiversity, and human societies.

CO4: Discuss global and local climate change mitigation and adaptation strategies.

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CO5: Assess the societal, economic, and health impacts of climate change and formulate strategies for addressing them.

CO6: Understand international policies and frameworks like the Paris Agreement and their implications for climate action.

Reference Books

1. **"Climate Change: A Very Short Introduction"** by Mark Maslin
2. **"The Discovery of Global Warming"** by Spencer Weart
3. **"Climate Change and Global Warming: The Science of Climate Change"** by William F. Ruddiman
4. **"Climate Change: Evidence and Causes"** by Richard A. Muller
5. **"The Economics of Climate Change: The Stern Review"** by Nicholas Stern
6. **"The Paris Agreement on Climate Change: A Commentary"** by Daniel Bodansky
7. **"Climate Change and the Global Harvest: Impacts of the Changing Climate on Agriculture and Food Security"** by Cynthia Rosenzweig and William D. Solecki

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2	1	2	2	2	2	1	2	1
CO2	3	2	3	2	1	3	2	2	3	2	2	2
CO3	3	3	3	2	3	2	2	3	3	2	3	3
CO4	3	3	3	3	2	3	3	3	3	3	3	2
CO5	3	3	3	2	3	2	2	3	3	2	3	2
CO6	3	2	3	3	1	3	2	3	3	3	2	2

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EN2502- ENVIRONMENTAL IMPACT ASSESSMENT

Objective:

This course introduces students to the principles, methodologies, and practices of Environmental Impact Assessment (EIA). It provides an understanding of the process of evaluating the environmental impacts of projects and activities, and the legal, social, and technical aspects involved in EIA. Students will also learn how to assess and mitigate the potential adverse impacts of proposed projects on the environment.

Unit 1: Introduction to Environmental Impact Assessment

Definition and Importance of EIA: Overview of the EIA process and its significance in sustainable development.

History and Evolution of EIA: Key milestones in the development of EIA regulations and practices worldwide.

EIA and Sustainable Development: The role of EIA in promoting sustainability, integrating environmental considerations into decision-making.

Key Terminologies and Concepts: EIA, Environmental Management Plan (EMP), Scoping, Screening, Baseline Data, and Mitigation Measures.

Legal and Institutional Framework: National and international EIA regulations, including the role of agencies like the Environmental Protection Agency (EPA).

Unit 2: EIA Process and Methodologies

Steps in the EIA Process: Screening, Scoping, Impact Prediction, Mitigation, and Monitoring.

Scoping and Screening: Defining the scope of EIA studies, determining which projects require EIA.

Baseline Data Collection: Methods for gathering environmental data on air, water, soil, biodiversity, etc.

Impact Prediction Methods: Use of models, matrices, and checklists to predict potential environmental impacts.

Public Participation: The role of stakeholders, including the community and NGOs, in the EIA process.

Unit 3: Environmental Impact Prediction and Evaluation

Types of Environmental Impacts: Direct, indirect, cumulative, and synergistic impacts on various environmental components (air, water, land, biodiversity).

Quantitative and Qualitative Impact Assessment: Methods for assessing impacts using both qualitative and quantitative approaches.

Impact Evaluation Techniques: Analytical methods for evaluating environmental impacts, including GIS, remote sensing, and environmental modeling.

Risk Assessment: Introduction to risk assessment and its application in predicting and managing potential environmental hazards.

Unit 4: Mitigation and Environmental Management Plans

Mitigation Strategies: Techniques for minimizing adverse environmental impacts (e.g., pollution control, habitat restoration, resource conservation).

Environmental Management Plans (EMP): Development of EMPs to manage and monitor environmental impacts during and after project implementation.

Sustainable Practices: Strategies for integrating sustainable practices in project planning and implementation.

Case Studies: Examples of EIA applications in large-scale infrastructure, industrial, and development projects.

Unit 5: Environmental Auditing, Monitoring, and Review

Environmental Auditing: The process of evaluating the effectiveness of environmental management practices.

Monitoring and Follow-up: Methods for monitoring the environmental performance of projects post-implementation.

EIA Review: The process of reviewing EIA reports, and ensuring compliance with legal and environmental standards.

Challenges and Future Trends in EIA: Emerging trends, such as strategic environmental assessments (SEA), and the challenges in conducting EIA in the face of climate change and environmental degradation.

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Ethical Issues in EIA: Ethical considerations in conducting EIA, including transparency, bias, and accountability.

Course Outcomes (COs)

CO1: Understand the principles and significance of Environmental Impact Assessment in sustainable development.

CO2: Demonstrate knowledge of the various steps involved in the EIA process.

CO3: Develop skills to collect and analyze baseline environmental data for EIA studies.

CO4: Use appropriate methodologies to predict, evaluate, and manage environmental impacts.

CO5: Design effective mitigation measures and develop an Environmental Management Plan (EMP) for projects.

Reference Books

1. **"Environmental Impact Assessment: A Practical Guide"** by Peter Wathern
2. **"Introduction to Environmental Impact Assessment"** by John Glasson, Riki Therivel, and Andrew Chadwick
3. **"Environmental Impact Assessment: Theory and Practice"** by Peter Morris and Riki Therivel
4. **"Environmental Impact Assessment Handbook: A Practical Guide for Practitioners"** by C. A. Brebbia and L. C. W. J. McCarter
5. **"Environmental Impact Assessment and Strategic Environmental Assessment: An Integrated Approach"** by J. M. Wood and P. A. L. W. McKeen
6. **"Environmental Impact Assessment: A Comprehensive Guide"** by R. K. Trivedi
7. **"Environmental Management: Science and Engineering for Industry"** by M. R. O'Callaghan and M. W. O'Callaghan
8. **"Environmental Law and Policy: Nature, Law and Society"** by Philippe Sands

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2	1	2	1	3	2	2	3	1
CO2	3	3	3	2	2	3	2	3	3	3	3	3
CO3	3	2	3	3	2	3	3	3	2	3	3	3
CO4	3	3	3	3	3	3	2	3	3	3	3	2
CO5	3	2	3	3	3	3	2	3	3	3	2	3

EN2503 - GLOBAL WARMING AND ECOSYSTEM IMPACTS

Objective:

This course explores the causes, consequences, and mitigation strategies for global warming, with a particular focus on its impact on ecosystems. Students will examine the science behind global warming, its effects on biodiversity, ecosystems, and the services they provide, as well as the social, economic, and policy implications. The course also covers adaptation and mitigation strategies aimed at reducing the impact of global warming on ecosystems.

Unit 1: Understanding Global Warming

Basics of Global Warming: Definition, causes, and natural vs. anthropogenic factors.

Greenhouse Effect and GHGs: Role of greenhouse gases (CO₂, CH₄, N₂O) in warming the Earth.

Sources and Sinks of Greenhouse Gases: Carbon cycle, deforestation, industrial emissions.

Global Temperature Trends: Current and projected global temperature increases.

Climate Models: Use of climate models to predict future warming scenarios.

Unit 2: Impacts of Global Warming on the Environment

Global Warming and Temperature Rise: Effects of temperature increases on global climates, ocean currents, and seasonal changes.

Sea-Level Rise: Impact of melting polar ice caps, glaciers, and thermal expansion of oceans.

Ocean Acidification: Impact of increased CO₂ on marine ecosystems, including coral reefs and fisheries.

Weather Extremes: Increased frequency of storms, droughts, floods, and heatwaves.

Changes in Precipitation Patterns: Alterations in regional rainfall, flooding risks, and desertification.

Unit 3: Ecosystem Impacts

Biodiversity and Ecosystem Services: Understanding the relationship between global warming and ecosystem functions such as carbon sequestration, water purification, and habitat provision.

Impact on Terrestrial Ecosystems: Effects of global warming on forests, grasslands, and deserts, including shifts in species distributions.

Impact on Aquatic Ecosystems: Impact of warming on freshwater and marine ecosystems, with particular focus on fisheries, coral reefs, and wetlands.

Disruption of Ecological Balance: Effects on species migration, life cycles, and extinction risks due to climate-induced changes.

Invasive Species and Global Warming: How climate change affects the spread of invasive species and disrupts ecosystems.

Unit 4: Social and Economic Impacts of Global Warming on Ecosystems

Human-Nature Interactions: Effects of ecosystem degradation on human societies (e.g., agriculture, water resources, health).

Economic Impact: The economic cost of ecosystem degradation, climate change-related disasters, and loss of biodiversity.

Climate Refugees: Migration due to environmental degradation (e.g., coastal flooding, agricultural failure).

Global vs. Local Impacts: Disparities in impacts on ecosystems and human societies in developed vs. developing countries.

Environmental Justice: Ethical issues and equity in the impacts of climate change and global warming on vulnerable populations.

Unit 5: Mitigation, Adaptation, and Policy Responses

Mitigation Strategies: Reduction of greenhouse gas emissions, renewable energy, carbon capture and storage (CCS).

Adaptation Strategies: Adapting to the changing climate, including ecosystem restoration, climate-resilient agriculture, and water management.

Global and National Policies: International treaties (e.g., Paris Agreement), national policies on climate change mitigation, and adaptation.

Role of Technology: Technological advancements to combat global warming and enhance ecosystem protection.

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Sustainable Development Goals (SDGs): Climate action and biodiversity conservation as part of the broader SDG framework.

Course Outcomes (COs)

CO1: Understand the scientific principles of global warming, its causes, and its effects on the environment.

CO2: Analyze the impacts of global warming on ecosystems, biodiversity, and ecosystem services.

CO3: Evaluate the social and economic consequences of ecosystem degradation due to global warming.

CO4: Understand and discuss strategies for mitigating and adapting to global warming, both at the local and global levels.

CO5: Assess the role of policy frameworks and international cooperation in combating global warming and protecting ecosystems.

Reference Books

1. "Global Warming: The Complete Briefing" by John Houghton
2. "Climate Change and Ecosystems" by T. S. P. B. Meira
3. "The Discovery of Global Warming" by Spencer Weart
4. "The Economics of Global Warming" by Richard S.J. Tol
5. "Ecosystems and Human Well-being: A Framework for Assessment" by Millennium Ecosystem Assessment
6. "Climate Change and Biodiversity" by Thomas E. Lovejoy and Lee Hannah
7. "Global Warming and the Built Environment" by Matthew K. Huber
8. "Sustainability: A History" by Jeremy L. Caradonna

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2	2	2	1	2	2	1	2	2
CO2	3	3	3	2	3	3	2	3	3	3	3	3
CO3	3	3	2	3	3	3	2	3	3	2	3	3
CO4	3	3	3	3	3	3	3	3	3	3	2	3
CO5	3	3	3	3	2	3	3	3	3	3	3	2

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BA2515 ELEMENTS OF ECONOMICS & PRINCIPLES OF MANAGEMENT SCIENCE

Course Objectives (CO)

1. To introduce the fundamental concepts of Economics and their relevance to managerial decision-making.
2. To provide knowledge of the microeconomic environment and its impact on business operations.
3. To develop an understanding of management principles and their application in organizations.
4. To familiarize students with organizational behavior and how decision-making impacts business processes.
5. To equip students with management science techniques for solving complex business problems.

Unit 1: Introduction to Economics

- Definition of Economics (Micro and Macro)
- Economic Systems (Market economy, Command economy, Mixed economy)
- Principles of Scarcity, Choice, and Opportunity Cost
- Demand and Supply Theory
- Law of Demand and Supply

Unit 2: Microeconomics and Managerial Economics

1. **Demand and Supply Analysis**
 - Law of Demand and Supply
 - Elasticity of Demand and Supply
 - Factors influencing demand and supply
2. **Theory of Production**
 - Production Function and Law of Diminishing Returns
 - Isoquants and Isocosts
 - Economies of Scale
3. **Market Structure**
 - Perfect Competition
 - Monopoly and Monopolistic Competition
 - Oligopoly

Unit 3: Introduction to Management and Its Functions

- Definition of Management
- Management vs. Leadership
- Roles and Functions of Managers
- Planning, Organizing, Leading, and Controlling (POLC)

Unit 4: Organizational Behavior and Decision Making

1. **Organizational Behavior**
 - Definition and importance of Organizational Behavior
 - Motivation Theories (Maslow, Herzberg, McGregor)
 - Leadership Theories (Trait Theory, Behavioral Theory, Contingency Theory)
2. **Decision-Making and Planning**
 - Decision-making process in organizations
 - SWOT Analysis
 - Strategic Planning

Unit 5: Principles of Management Science

- Concept of Management Science
- Linear Programming and Optimization Techniques
- Decision Theory and Simulation
- Applications in Business Decision Making

Course Outcomes (PO)

1. **PO1:** Understand and apply basic economic principles in the context of business.
2. **PO2:** Analyze business environments using microeconomic and managerial concepts.
3. **PO3:** Understand the key functions of management and their practical applications.
4. **PO4:** Evaluate the behavior of individuals and groups within organizations and their impact on management.
5. **PO5:** Use management science tools to improve decision-making and optimize business processes

Reference Books:

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1. **"Principles of Economics"** by N. Gregory Mankiw
2. **"Microeconomics"** by Paul A. Samuelson and William D. Nordhaus
3. **"Managerial Economics"** by William F. Samuelson and Stephen G. Marks
4. **"Microeconomics"** by Robert Pindyck and Daniel Rubinfeld
5. **"Management: Theory and Practice"** by G. R. Badger
6. **"Principles of Management"** by Charles W. L. Hill and Steven L. McShane
7. **"Organizational Behavior"** by Stephen P. Robbins and Timothy A. Judge
8. **"Managerial Decision Making"** by Edward Peter Stringham
9. **"Operations Research"** by Taha H.A.
10. **"Management Science: Principles and Applications"** by L.J. Krajewski and L.P. Ritzman

CO-PO Mapping Values

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	2	1	-	-	-	-	-	-	2
CO2	3	3	2	2	1	-	1	-	-	-	-	3
CO3	3	3	2	3	1	1	1	-	-	-	-	2
CO4	3	2	3	3	2	2	3	1	1	-	-	3
CO5	2	3	3	3	3	1	1	1	1	-	-	3

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CE2505 ENGINEERING GEOLOGY

Course Objectives

1. To introduce the principles of geology and their significance in civil engineering applications.
2. To familiarize students with geological structures and their impact on engineering projects.
3. To provide knowledge on geotechnical investigations and methods for assessing ground stability.
4. To explore the geological considerations for the safe and sustainable design of infrastructure.
5. To understand advanced tools like remote sensing, GIS, and geophysical methods in engineering geology.

Unit 1: Introduction to Engineering Geology

- Definition, scope, and importance of engineering geology in civil engineering.
- Physical properties and classification of minerals.
- Types of rocks: igneous, sedimentary, and metamorphic rocks; their characteristics and engineering applications.
- Geological processes and their impact: weathering, erosion, and deposition.
- Importance of geology in site selection and construction materials.

Unit 2: Structural Geology

- Concept of stress and strain in geological formations.
- Structural features: faults, folds, joints, and unconformities.
- Engineering significance of geological structures.
- Plate tectonics and its implications for civil engineering projects.
- Case studies: failures due to geological structures.

Unit 3: Geotechnical Investigations

- Methods of geological exploration: surface and subsurface methods.
- Core logging, seismic refraction, and resistivity methods.
- Engineering properties of rocks and soils.
- Role of groundwater in civil engineering: aquifers, water table, and permeability.
- Rock mechanics: classification and testing of rocks.

Unit 4: Geology for Infrastructure Development

- Geological considerations in the construction of dams, tunnels, and reservoirs.
- Landslides: causes, prevention, and mitigation.
- Earthquake geology and seismic zones in India.
- Case studies of infrastructure failure due to geological issues.
- Environmental geology and sustainable practices in civil engineering.

Unit 5: Applied Geology

- Remote sensing and GIS applications in geological studies.
- Use of geological maps and cross-sections in engineering projects.
- Engineering geology of soils: classification and properties.
- Quarrying and mining: impact on civil engineering.
- Introduction to geophysical methods in site investigations.

Course Outcomes

Upon completion of this course, students will be able to:

1. Identify and classify various types of rocks and minerals based on their engineering properties. (CO1)
2. Analyze geological structures and evaluate their influence on civil engineering projects. (CO2)
3. Conduct geotechnical investigations to determine site suitability for construction. (CO3)
4. Apply geological knowledge to address challenges in infrastructure development and environmental sustainability. (CO4)
5. Use advanced geological tools like GIS and remote sensing for problem-solving in engineering projects. (CO5)

Reference Books

1. **"Principles of Engineering Geology"** by K. M. Bangar.
2. **"Textbook of Engineering Geology"** by N. Chenna Kesavulu.
3. **"Engineering Geology"** by F. G. Bell.
4. **"Geology for Engineers"** by P. C. Varghese.
5. **"Engineering and General Geology"** by Parbin Singh.

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6. "Fundamentals of Engineering Geology" by S. K. Garg.
7. "Introduction to Geophysical Prospecting" by Milton B. Dobrin.

CO-PO Mapping Values

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	2	1	-	-	-	-	-	-	2
CO2	3	3	2	2	1	-	1	-	-	-	-	3
CO3	3	3	2	3	1	1	1	-	-	-	-	2
CO4	3	2	3	3	2	2	3	1	1	-	-	3
CO5	2	3	3	3	3	1	1	1	1	-	-	3

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ME2503 HEAT AND MASS TRANSFER

Course Objectives:

1. To provide an understanding of the fundamental principles of heat and mass transfer.
2. To equip students with the ability to analyze and solve heat transfer problems using conduction, convection, and radiation concepts.
3. To introduce students to applications of mass transfer processes in engineering systems.
4. To foster the application of heat and mass transfer principles in the design of heat exchangers and other industrial equipment.
5. To develop analytical, numerical, and experimental skills for solving real-world heat and mass transfer problems.

Unit 1: Introduction to Heat Transfer

- Modes of heat transfer: Conduction, Convection, and Radiation
- General heat conduction equation (Cartesian, Cylindrical, and Spherical coordinates)
- Fourier's law of heat conduction
- Thermal conductivity and thermal resistance concept
- One-dimensional steady-state heat conduction in slabs, cylinders, and spheres
- Heat transfer with and without internal heat generation
- Critical thickness of insulation

Unit 2: Convection Heat Transfer

- Newton's law of cooling
- Dimensional analysis and empirical correlations for convection
- Forced convection: Flow over flat plates, cylinders, and inside pipes
- Free convection: Heat transfer over vertical and horizontal plates
- Heat transfer in boiling and condensation

Unit 3: Radiation Heat Transfer

- Basic laws: Stefan-Boltzmann law, Planck's law, Wien's displacement law, and Kirchoff's law
- Blackbody and gray body radiation
- View factors and radiation heat exchange between surfaces
- Radiation shields
- Solar radiation and its applications

Unit 4: Heat Exchangers

- Classification of heat exchangers
- Overall heat transfer coefficient
- Log Mean Temperature Difference (LMTD) method
- Effectiveness-NTU method
- Performance analysis of heat exchangers (parallel flow, counterflow, and crossflow)

Unit 5: Mass Transfer

- Basic principles of mass transfer
- Fick's law of diffusion
- Diffusion through a stationary medium and equimolar counter-diffusion
- Convective mass transfer and mass transfer coefficients
- Analogies between heat, mass, and momentum transfer
- Applications: Drying, absorption, and distillation

List of Experiments:

1. **Heat Conduction in a Solid Rod**
2. **Natural Convection from a Vertical Plate**
3. **Forced Convection in a Pipe**
4. **Heat Transfer from a Pin-Fin**
5. **Radiation Heat Transfer**
6. **Boiling Heat Transfer**
7. **Mass Transfer by Diffusion**
8. **Performance of a Heat Exchanger**

List of Equipments:

1. **Thermal conductivity apparatus**

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2. **Natural convection test rig**
3. **Forced convection test rig**
4. **Pin-fin apparatus**
5. **Radiation setup**
6. **Boiling heat transfer apparatus**
7. **Mass transfer diffusion cell**
8. **Heat exchanger test rig**
9. **Calorimeter**

Course Outcomes (COs):

After successful completion of the course, students will be able to:

1. Understand the basic modes of heat transfer and their governing laws.
2. Solve problems involving steady-state and transient heat conduction.
3. Analyze convection heat transfer in various flow configurations.
4. Apply principles of radiation heat transfer in engineering applications.
5. Understand mass transfer mechanisms and solve related engineering problems.
6. Design and analyze heat exchangers for industrial applications.

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	-	-	-	-	-	-	2
CO2	3	3	3	2	2	-	-	-	-	-	-	2
CO3	3	2	2	3	2	-	-	-	-	-	-	2
CO4	3	3	3	3	3	-	-	-	-	-	-	3
CO5	3	3	2	3	2	-	-	-	-	-	-	2

(3 = High, 2 = Medium, 1 = Low, - = Not Mapped)

Reference Books

1. "Heat and Mass Transfer: Fundamentals and Applications" by Yunus A. Çengel and Afshin J. Ghajar
2. "Heat Transfer" by J.P. Holman
3. "Principles of Heat Transfer" by Frank Kreith and Raj M. Manglik
4. "Fundamentals of Engineering Heat and Mass Transfer" by R.C. Sachdeva
5. "Introduction to Heat Transfer" by Frank P. Incropera and David P. DeWitt

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CE2522 HEALTH, SAFETY, AND ENVIRONMENT

Course Objectives

1. To provide foundational knowledge of health, safety, and environmental concepts and their importance.
2. To identify and mitigate workplace hazards through systematic risk assessment.
3. To understand environmental challenges and sustainable practices in industries.
4. To implement effective safety and environmental management systems.
5. To adopt emerging technologies to enhance HSE performance.

Unit 1: Introduction to Health, Safety, and Environment

- Definition and importance of HSE in industries
- Evolution of health, safety, and environmental standards
- Legal and regulatory frameworks (OSHA, EPA, ISO standards)
- Principles of hazard identification and risk assessment (HIRA)
- Safety culture and human factors

Unit 2: Occupational Health and Safety

- Concepts of occupational health and hygiene
- Workplace hazards: physical, chemical, biological, and ergonomic
- Accident causation theories and prevention strategies
- Personal protective equipment (PPE) and its applications
- Case studies on workplace accidents and preventive measures

Unit 3: Environmental Management and Sustainability

- Overview of environmental pollution (air, water, soil, noise)
- Waste management techniques: recycling, reuse, and disposal
- Environmental Impact Assessment (EIA)
- Sustainable development and energy management
- Climate change and corporate environmental responsibility

Unit 4: Safety Management Systems

- Principles and components of safety management systems
- Emergency preparedness and response plans
- Fire safety: causes, prevention, and firefighting techniques
- Audits and inspections: types and methodologies
- Role of training in safety management

Unit 5: Integrated HSE Systems and Emerging Trends

- Integration of health, safety, and environment systems
- Industry 4.0 and its impact on HSE
- Advanced safety technologies: IoT, AI, and VR in safety management
- Behavioral-based safety (BBS)
- Future challenges in HSE

List of Experiments:

1. Air Quality Measurement and Analysis
2. Water Pollution Testing
3. Noise Pollution Measurement
4. Fire Safety Simulation
5. Ergonomics Evaluation
6. Toxicity Testing of Chemicals

List of Equipment:

1. Air quality monitors
2. Water testing kits (pH meters, turbidity meters)
3. Noise level meters (decibel meters)
4. Personal protective equipment (PPE)
5. Fire extinguishers and safety blankets
6. Emergency evacuation signage
7. Ergonomics equipment (chairs, desks, mouse pads)
8. First aid kits
9. Safety goggles, gloves, lab coats
10. Hazardous chemical handling kits (MSDS, storage containers)

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Course Outcomes

After completing this course, students will be able to:

1. Demonstrate knowledge of HSE principles and legal frameworks.
2. Analyze workplace hazards and propose effective control measures.
3. Apply environmental management practices to minimize industrial impacts.
4. Develop and implement safety management systems.
5. Integrate modern technologies to address HSE challenges.

CO-PO Mapping Values

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	3	3	2	2	1	1	3
CO2	3	3	3	3	2	3	3	2	2	1	1	3
CO3	3	3	3	3	3	3	3	2	2	2	2	3
CO4	3	3	2	2	2	3	3	3	2	1	1	3
CO5	3	2	3	3	2	3	3	3	3	3	2	3

(3 = High, 2 = Medium, 1 = Low, - = Not Mapped)

Reference Books:

1. **"Introduction to Health and Safety in Construction"** by Phil Hughes and Ed Ferrett
2. **"Environmental Health and Safety Management"** by Thomas P. Gormley
3. **"Fundamentals of Occupational Safety and Health"** by Mark A. Friend and James P. Kohn
4. **"Industrial Safety and Environmental Protection"** by K. M. Rani and B. S. Dhillon
5. **"Environmental Chemistry"** by Stanley E. Manahan
6. **"Safety at Work"** by John R. Ridley and James C. Channing

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SW2517 LAND AND WATER MANAGEMENT ENGINEERING

Course Objective:

The primary objective of this course is to provide students with the knowledge and skills required to manage land and water resources efficiently. The course will cover various techniques and strategies used in the sustainable management of land and water for agricultural, industrial, and urban applications. It will also provide insights into engineering practices that contribute to the optimal use of land and water resources.

Unit 1: Introduction to Land and Water Management

- Overview of land and water resources
- Role of land and water management in sustainable development
- Basic concepts of soil, water, and crop relations
- Importance of land and water management in agriculture and industry
- Environmental concerns in land and water management

Unit 2: Irrigation Engineering and Management

- Types of irrigation systems (surface, drip, sprinkler)
- Irrigation scheduling and water requirement estimation
- Design of irrigation canals and distribution systems
- Water use efficiency in irrigation
- Methods of water conservation in irrigation

Unit 3: Drainage and Flood Control Engineering

- Principles of drainage systems
- Design of surface and subsurface drainage
- Storm water management and flood control measures
- Watershed management techniques
- Erosion control and sediment management

Unit 4: Soil and Water Conservation Engineering

- Techniques of soil and water conservation
- Watershed management and planning
- Rainwater harvesting methods
- Control of soil erosion and sedimentation
- Role of vegetation in soil and water conservation

Unit 5: Environmental Impact and Sustainable Land and Water Management

- Environmental aspects of land and water use
- Sustainable practices in land and water management
- Climate change and its impact on water resources
- Wastewater treatment and reuse
- Role of technology in sustainable land and water management

Course Outcomes (CO):

By the end of this course, students will be able to:

CO1: Understand the principles of land and water management and their significance in sustainable development.

CO2: Analyze and design effective irrigation and drainage systems to optimize water use.

CO3: Evaluate the impact of land and water management on soil health, crop yield, and environmental sustainability.

CO4: Use modern engineering tools for the assessment and management of land and water resources.

CO5: Apply engineering techniques in flood control, watershed management, and water conservation.

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CO-PO Mapping Values Table

(CO)	(PO) 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	2	2	2	3	1	2	3
CO2	3	3	3	2	3	1	2	3
CO3	3	2	3	2	3	2	2	2
CO4:	3	2	3	3	3	2	2	3
CO5	3	3	2	3	3	2	2	2
CO6	3	2	3	3	2	3	1	3

(3 = High, 2 = Medium, 1 = Low, - = Not Mapped)

Reference Books

1. "Land and Water Management" by Dr. R. S. Bains
2. "Water Resources Engineering" by Larry W. Mays
3. "Irrigation and Water Power Engineering" by B.C. Punmia
4. "Principles of Irrigation Engineering" by P. N. Modi
5. "Drainage Engineering" by K. N. Duggal
6. "Flood Control Engineering" by Ven Te Chow
7. "Environmental Engineering: Water, Wastewater, Soil, and Waste Management" by M. L. Davis
8. "Sustainable Water Management" by I. A. M. Shiklomanov
9. "Soil and Water Conservation Engineering" by D. K. Sharma
10. "Principles of Soil Conservation" by K. N. Duggal

SW2507 GIS AND REMOTE SENSING

Course Objective (CO):

1. To introduce the fundamentals of GIS and Remote Sensing, their principles, techniques, and applications.
2. To develop proficiency in using GIS software and remote sensing tools for environmental and urban planning, agriculture, and other fields.
3. To equip students with the necessary skills to process and analyze spatial data for decision-making.

Unit 1: Introduction to GIS and Remote Sensing

- Definition, history, and evolution of GIS and Remote Sensing.
- Types of spatial data: raster and vector.
- Overview of Remote Sensing: sensors, platforms, and types of data.
- GIS components: hardware, software, and data.
- Basics of GPS and its applications.

Unit 2: GIS Data Models and Spatial Analysis

- Raster and vector data models.
- Spatial data structures and formats (Shape files, GeoTIFF, etc.).
- Geo referencing, map projections, and coordinate systems.
- Spatial analysis techniques: buffering, overlay, and proximity analysis.
- Introduction to GIS software (Arc GIS, QGIS).

Unit 3: Remote Sensing Data Processing and Analysis

- Types of remote sensing data: satellite and airborne.
- Image processing techniques: image enhancement, classification, and feature extraction.
- Use of spectral indices (NDVI, NDWI).
- Image interpretation and analysis.
- Data fusion and integration with GIS.

Unit 4: GIS Applications

- Land-use and land-cover mapping.
- Urban planning and zoning.
- Environmental monitoring and disaster management.
- Agriculture and forestry applications.
- Hydrological modeling and watershed management.

Unit 5: Advanced GIS Techniques and Emerging Trends

- Geospatial modeling and simulation.
- LiDAR and UAV (Unmanned Aerial Vehicle) applications.
- Web GIS and cloud-based GIS platforms.
- Geospatial Big Data and Machine Learning.
- Future trends in GIS and Remote Sensing.

Course Outcomes (COs):

1. **CO1:** Understand the basic concepts of GIS and Remote Sensing, including data collection and processing techniques.
2. **CO2:** Apply GIS software tools for spatial data analysis and visualization.
3. **CO3:** Use remote sensing data to interpret environmental features and create maps.
4. **CO4:** Analyze spatial data for practical applications like land-use planning, disaster management, and natural resource monitoring.
5. **CO5:** Evaluate and process remote sensing data using appropriate techniques to derive meaningful information.

List of Experiments:

1. **Experiment 1:** Introduction to GIS Software – Basic tools and functions in QGIS/ArcGIS.
2. **Experiment 2:** Spatial Data Creation and Editing – Working with vector and raster data.
3. **Experiment 3:** Georeferencing of Raster Images.
4. **Experiment 4:** Image Classification in Remote Sensing (Supervised and Unsupervised).
5. **Experiment 5:** NDVI Calculation using Remote Sensing Data.
6. **Experiment 6:** Buffer Analysis and Proximity Analysis in GIS.
7. **Experiment 7:** Overlay Analysis and Land Use Classification.

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8. **Experiment 8:** Remote Sensing Image Enhancement Techniques.
9. **Experiment 9:** Hydrological Modeling in GIS.
10. **Experiment 10:** UAV-based Data Collection and Analysis.

List of Equipment:

1. **GIS Software:**
 - ArcGIS, QGIS, Google Earth Pro.
2. **Remote Sensing Software:**
 - ENVI, ERDAS Imagine, SNAP.
3. **GPS Devices:**
 - Handheld GPS units, GPS receivers for field data collection.
4. **Satellites/Imagery Sources:**
 - Landsat, Sentinel, MODIS, World View imagery (for practical data).
5. **Drones (for UAV-based Data Collection).**
6. Computers with Spatial Data Processing Capabilities.
7. Spectro radiometers and Remote Sensing Sensors.

CO-PO Mapping:

COs \ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	3	2	2	2	2	2	1	2
CO2	3	3	2	2	3	2	2	1	3	3	2	3
CO3	2	3	2	1	3	2	2	2	1	2	2	3
CO4	3	3	3	3	3	2	3	2	2	2	3	2
CO5	2	3	3	2	3	3	3	1	1	3	3	2

(3 = High, 2 = Medium, 1 = Low, - = Not Mapped)

Reference Books:

1. **"Principles of Geographic Information Systems"** by Peter A. Burrough and Rachael A. McDonnell.
2. **"Remote Sensing and Image Interpretation"** by Thomas M. Lillesand and Ralph W. Kiefer.
3. **"Introduction to Remote Sensing"** by James B. Campbell.
4. **"Geographic Information Systems and Science"** by Paul A. Longley, Michael F. Goodchild, David J. Maguire, and David W. Rhind.
5. **"Remote Sensing for the Earth Sciences"** by Robert A. Schowengerdt.

CM2503 CLIMATE ADAPTATION TECHNOLOGIES

Course Objectives (CO):

1. To understand the concept of climate change and its impacts on various sectors.
2. To analyze the adaptation strategies required for climate resilience in sectors such as agriculture, water resources, and infrastructure.
3. To study the role of technologies in mitigating and adapting to climate change, focusing on innovation and sustainability.
4. To evaluate various tools and technologies for assessing climate vulnerability and risk management.
5. To develop critical thinking and practical skills for integrating climate adaptation strategies into policy and planning frameworks.

Unit 1: Introduction to Climate Change and Adaptation

- **Climate Science Basics:**
 - Greenhouse gases, climate change mechanisms, global warming.
 - Impacts of climate change on ecosystems and human societies.
- **Climate Change Vulnerability:**
 - Understanding regional and sectoral vulnerabilities.
 - Climate resilience and adaptive capacity.
- **Introduction to Climate Adaptation:**
 - Definition and principles.
 - Difference between mitigation and adaptation.

Unit 2: Climate Adaptation Strategies in Agriculture and Water Resources

- **Agriculture:**
 - Climate-smart agriculture.
 - Techniques like drought-resistant crops, soil management, and agro forestry.
 - Role of technology in precision farming.
- **Water Resources:**
 - Climate change impact on water availability and distribution.
 - Technologies for water conservation (rainwater harvesting, desalination).
 - Integrated water management and adaptive governance.

Unit 3: Climate Adaptation Technologies for Infrastructure and Urban Areas

- **Climate-Resilient Infrastructure:**
 - Design and construction of climate-resilient infrastructure (roads, buildings, bridges).
 - Technologies for early warning systems.
- **Urban Adaptation:**
 - Urban heat island mitigation.
 - Green infrastructure (rooftop gardens, green roofs, permeable pavements).
 - Adaptation to rising sea levels in coastal cities.

Unit 4: Tools and Technologies for Climate Risk Assessment

- **Vulnerability and Risk Assessment Tools:**
 - Climate vulnerability assessments.
 - GIS (Geographic Information Systems) and Remote Sensing for climate data.
 - Climate modeling techniques.
- **Climate Risk Management:**
 - Early warning systems, climate data integration.
 - Risk assessment and adaptation planning.

Unit 5: Policy, Governance, and Future Directions in Climate Adaptation

- **Climate Change Policy and Governance:**
 - National and international policies on climate change adaptation.
 - Role of governments and organizations (UNFCCC, IPCC).
 - Climate finance for adaptation technologies.
- **Future of Climate Adaptation Technologies:**
 - Emerging technologies in climate adaptation (AI, blockchain, etc.).
 - Innovations in climate resilience and adaptation in developing countries.

Course Outcomes (CO):

Upon completion of this course, students will be able to:

1. **CO1:** Describe the science of climate change and its global and local effects.
2. **CO2:** Identify and implement appropriate climate adaptation strategies in sectors such as agriculture and water resources.

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3. **CO3:** Evaluate and apply climate adaptation technologies in real-world scenarios.
4. **CO4:** Utilize tools and methods for climate vulnerability and risk assessments.
5. **CO5:** Analyze policies and governance structures for climate adaptation and integrate new technologies for future resilience.

CO-PO Mapping

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3					2	2	2			3
CO2	3	3			3		1		1	2		3
CO3	3	2	3	3	2		1	1		1	2	2
CO4	3	2	3	3	2	1	1			2	2	
CO5	3		3	2	3	1	1	1	2	2		3

(3 = High, 2 = Medium, 1 = Low, - = Not Mapped)

Reference Books:

1. **“Climate Change: A Very Short Introduction”** by Mark Maslin
2. **“Climate Change and Agriculture: An Economic Analysis of Global Impacts, Adaptation and Mitigation”** by David B. Lobell and Marshall Burke
3. **“Climate Adaptation: From Crisis to Opportunity”** by Dr. Saleemul Huq, et al.
4. **“The Technology of Climate Change”** by Michael J. P. Walker
5. **“Adaptation to Climate Change: From Resilience to Transformation”** by Mark Pelling
6. **“Climate Risk Management: Adaptation and Resilience”** by Joyashree Roy and William C. Turner II

CE2524-AIR POLLUTION CONTROL ENGINEERING

Course Objective:

The **Air Pollution Control Engineering** course is designed to equip students with the knowledge of air

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pollution sources, its effects on health and the environment, and various technologies available to control and mitigate air pollutants. Students will study the mechanisms of air pollution, regulatory frameworks, and design principles for controlling emissions in industrial, urban, and natural environments.

Unit 1: Introduction to Air Pollution

Definition and Types of Air Pollutants: Primary and secondary pollutants, particulate matter (PM), gaseous pollutants (SO₂, NO_x, CO, VOCs).

Sources of Air Pollution: Natural sources (volcanic eruptions, forest fires), anthropogenic sources (industries, transportation, agriculture).

Impact of Air Pollution: Human health effects, environmental impact (acid rain, global warming, smog formation).

Air Pollution Monitoring: Techniques for measuring pollutants (sampling methods, sensor technologies, remote sensing).

Air Quality Standards: Global and national air quality standards and limits (WHO, EPA, local regulations).

Unit 2: Atmospheric Chemistry and Behavior of Pollutants

Atmospheric Composition and Structure: Layers of the atmosphere, characteristics of troposphere and stratosphere.

Chemistry of Air Pollutants: Chemical reactions involved in the formation of ozone, smog, and acid rain.

Dispersion and Transport of Pollutants: Meteorological factors influencing pollutant dispersion (wind speed, temperature inversion, humidity).

Air Quality Modeling: Introduction to air quality models (Gaussian dispersion model, advanced photochemical models).

Unit 3: Air Pollution Control Technologies

Control of Particulate Matter (PM): Settling chambers, cyclone separators, electrostatic precipitators, fabric filters.

Control of Gaseous Pollutants: Scrubbers, catalytic converters, thermal oxidizers, biofilters.

Control of Odors: Adsorption techniques, biological treatment processes, chemical scrubbers.

Design Considerations: Selection of control technology based on pollutant type, concentration, and regulatory requirements.

Case Studies: Real-world applications of pollution control devices in industries (power plants, refineries, cement plants).

Unit 4: Air Pollution Prevention and Management Strategies

Source Reduction Techniques: Cleaner production methods, waste minimization, eco-design, and low-emission technologies.

Environmental Regulations and Policies: Local and international air quality regulations, emission inventories, permits, and control strategies.

Economic Aspects of Air Pollution Control: Cost-benefit analysis, market-based instruments (taxes, tradable permits), and incentives for pollution reduction.

Indoor Air Pollution: Sources of indoor air pollution, health effects, and control measures for residential and commercial buildings.

Unit 5: Emerging Technologies and Future Directions

Greenhouse Gas Emission Control: Technologies for CO₂ capture and storage (CCS), renewable energy sources, and sustainable urban planning.

Advanced Air Quality Control Technologies: Innovations in air filtration, hybrid systems, and nanotechnology for pollutant removal.

Air Pollution in Developing Countries: Challenges and solutions for air pollution control in developing nations, with a focus on transportation and industrial sectors.

Global Air Quality Management: International collaborations, strategies for global air pollution mitigation, and the role of the Paris Agreement.

Lab Experiments and Equipment List
Lab Experiments

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- Experiment 1: Measurement of Particulate Matter (PM) in Ambient Air**
- Experiment 2: Measurement of Gaseous Pollutants (SO₂, NO₂, CO, Ozone)**
- Experiment 3: Automatic Air Quality Monitoring Station**
- Experiment 4: Study of Cyclone Separator Efficiency**
- Experiment 5: Performance Testing of Electrostatic Precipitators (ESP)**
- Experiment 6: Fabric Filter Testing**
- Experiment 7: Efficiency of Wet Scrubber for SO₂ Removal**
- Experiment 8: Performance of a Catalytic Converter**
- Experiment 9: Efficiency of Biofilter for VOC Removal**
- Experiment 10: Air Pollution Dispersion Modeling**
- Experiment 11: Wind Tunnel Study for Air Pollution Dispersion**
- Experiment 12: Performance Evaluation of Nanofiltration Units for Air Purification**
- Experiment 13: Study of Green Infrastructure for Air Quality Improvement**

List of Required Equipment

Equipment List

1. **High-Volume Air Sampler**
2. **Gas Chromatograph (GC)**
3. **Chemiluminescence Analyzer**
4. **Ozone Analyzer**
5. **Sampling Pumps**
6. **Cyclone Separator**
7. **Particle Size Analyzer**
8. **Dust Collection Chamber**
9. **Electrostatic Precipitator (ESP)**
10. **Fabric Filter System**
11. **PM Monitor**
12. **Wet Scrubber Unit**
13. **SO₂ Gas Analyzer**
14. **Flow Meters**
15. **Catalytic Converter**
16. **Gas Sampling System**
17. **VOC Sensor**
18. **Biofilter System**
19. **Air Quality Modeling Software**
20. **Meteorological Data Collection System**
21. **Wind Tunnel**
22. **Pollutant Release System**
23. **Anemometer**
24. **Nanofiltration Unit**
25. **Environmental Sensors (for temperature, humidity, etc.)**
26. **Green Infrastructure Models (Green Walls, Green Roofs)**

Course Outcomes (COs)

Upon successful completion of the course, students will be able to:

- CO1:** Identify various types of air pollutants, their sources, and the impact on the environment and public health.
- CO2:** Understand the chemistry and behavior of air pollutants in the atmosphere.
- CO3:** Design and evaluate air pollution control technologies for particulate and gaseous emissions.
- CO4:** Apply regulatory frameworks and policies for air quality management and pollution control.
- CO5:** Explore emerging technologies and strategies for sustainable air pollution control and reduction.

Reference Books

1. **"Air Pollution Control Engineering"** by Noel de Nevers
2. **"Air Pollution: Its Origin and Control"** by Kenneth Wark and Cecil F. Warner
3. **"Fundamentals of Air Pollution Engineering"** by Richard C. Flagan and John H. Seinfeld
4. **"Introduction to Air Quality Engineering"** by William D. Grant
5. **"Environmental Engineering: Air Pollution and Control"** by Rao and Rao
6. **"Air Pollution and Its Impact on Health"** by M. C. D. G. J. J.
7. **"Environmental Pollution Control"** by S. S. K. Iyer

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	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C01	3	2	3	2	2	2	1	2	2	1	2	2
C02	3	3	3	2	3	3	2	3	3	3	3	3
C03	3	3	3	3	3	3	3	3	3	3	2	3
C04	3	3	2	3	3	3	2	3	3	2	3	3
C05	3	3	3	3	3	3	3	3	3	3	3	3

Course Objectives:

It is to provide students with a comprehensive understanding of waste types, collection, treatment, and disposal methods. It also aims to emphasize sustainable practices, regulatory frameworks, and innovative solutions for efficient waste management.

Unit 1: Introduction to Solid Waste Management

Definition and types of solid waste: Municipal solid waste, industrial waste, hazardous waste, biomedical waste, e-waste, etc.

Sources of solid waste: Residential, commercial, industrial, institutional, construction, and demolition.

Characteristics of solid waste: Physical, chemical, biological properties.

Solid Waste Management (SWM) hierarchy: Waste prevention, minimization, reuse, recycling, energy recovery, and disposal.

Role of solid waste management in environmental protection.

Unit 2: Collection and Segregation of Solid Waste

Collection systems: Curbside collection, containerized collection, collection routes.

Waste segregation techniques: Manual, mechanical, and automatic systems.

Collection containers and bins: Types, materials, and specifications.

Handling and transport of waste: Collection vehicles, transfer stations, and route optimization.

Health and safety considerations in collection and transport.

Unit 3: Waste Treatment and Disposal Technologies

Waste treatment methods:

Physical methods: Compaction, sorting, shredding.

Biological methods: Composting, anaerobic digestion.

Thermal methods: Incineration, pyrolysis, gasification.

Chemical methods: Stabilization, solidification.

Disposal techniques:

Landfilling: Design, operation, leachate management, landfill gas recovery.

Waste-to-Energy (WTE): Incineration, RDF (Refuse-Derived Fuel).

Recycling: Material recovery facilities (MRFs), recycling processes.

Environmental impacts of disposal methods.

Unit 4: Sustainable Waste Management and Policies

Sustainable waste management: Minimization, circular economy, zero waste approach.

Waste diversion strategies: Reduce, reuse, recycle, and recovery.

National and international policies on waste management: SWM rules, Extended Producer Responsibility (EPR).

Role of government and private sector: Regulatory frameworks, public-private partnerships.

Case studies: Successful waste management initiatives globally (e.g., Sweden, Japan).

Unit 5: Solid Waste Management in Urban and Rural Areas

Urban waste management challenges: Population density, informal settlements, and limited resources.

Rural waste management: Agricultural waste, open burning, collection difficulties.

Waste management in developing countries: Issues, challenges, and solutions.

Technological innovations in waste management: Smart waste bins, IoT in waste tracking, AI for waste sorting.

Public awareness and participation in waste management programs.

CO1: Understand the types, sources, and characteristics of solid waste, and the importance of waste management.

CO2: Explain the various waste collection, segregation, and transportation systems used in waste management.

CO3: Analyze and evaluate different waste treatment and disposal technologies.

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CO4: Demonstrate knowledge of sustainable waste management practices, including recycling and circular economy models.

CO5: Assess policies, regulations, and technological innovations related to waste management.

Reference Books:

1. **"Solid Waste Engineering"** by William A. Worrell and S. S. Gupta
2. **"Handbook of Solid Waste Management"** by George Tchobanoglous, Frank Kreith, and Ramani Narayan
3. **"Introduction to Waste Management"** by S. P. Ghosh
4. **"Municipal Solid Waste Management in Developing Countries"** by O. P. Gupta
5. **"Waste Management Practices: Municipal, Hazardous, and Industrial"** by John P. S. and Frank E. L.
6. **"Environmental Engineering: A Design Approach"** by Howard S. Peavy, David R. Rowe, and George Tchobanoglous.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2	2	2	1	2	2	1	2	2
CO2	3	3	3	2	3	3	2	3	3	3	3	3
CO3	3	3	2	3	3	3	2	3	3	2	3	3
CO4	3	3	3	3	3	3	3	3	3	3	2	3
CO5	3	3	3	3	2	3	3	3	3	3	3	2

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CE2526- WASTEWATER ENGINEERING AND MANAGEMENT

COURSE OBJECTIVES:

The objective of the Wastewater Engineering and Management course is to provide students with an understanding of wastewater treatment processes, from collection to advanced treatment technologies. It also aims to develop skills in designing and managing sustainable wastewater systems while addressing environmental regulations and impacts.

Unit 1: Introduction to Wastewater Engineering

Basic Concepts: Definition of wastewater, types of wastewater (domestic, industrial, stormwater), and the wastewater treatment process.

Sources of wastewater: Domestic, industrial, and commercial.

Characterization of wastewater: Physical, chemical, and biological properties.

Wastewater management: Needs and importance in urban and rural areas.

Wastewater quality parameters: pH, BOD (Biochemical Oxygen Demand), COD (Chemical Oxygen Demand), TSS (Total Suspended Solids), nitrogen, phosphorous, etc.

Overview of wastewater treatment plant (WWTP): Objectives, layout, and components.

Unit 2: Wastewater Collection Systems

Design of Sewer Systems: Types of sewer systems (separate, combined), design principles, and factors affecting design (population density, topography, etc.).

Materials used in sewer construction: Types of pipes, jointing methods, and their advantages.

Sewerage networks: Collection systems, layout, and maintenance.

Pump stations and force mains: Types of pumps, design, and selection.

Stormwater drainage systems: Importance, design, and management of stormwater.

Unit 3: Primary and Secondary Treatment of Wastewater

Primary Treatment: Screening, sedimentation, flotation, and grit removal.

Secondary Treatment: Biological treatment methods – activated sludge process (ASP), trickling filters, rotating biological contactors (RBC), and lagooning.

Suspended growth systems: Aeration tanks, secondary clarifiers, and their design principles.

Fixed-film processes: Trickling filters, RBC, and their operation and design.

Sludge management: Collection, treatment, and disposal of primary and secondary sludge.

Unit 4: Advanced Wastewater Treatment Technologies

Tertiary Treatment: Filtration, chemical precipitation, activated carbon adsorption, and advanced oxidation processes (AOPs).

Membrane Technologies: Microfiltration, ultrafiltration, nanofiltration, and reverse osmosis (RO) for wastewater treatment.

Nutrient Removal: Nitrogen and phosphorus removal methods, denitrification, and chemical precipitation.

Reuse of treated wastewater: Applications, standards, and challenges in wastewater reuse.

Disinfection: Methods of disinfection – chlorination, UV, ozonation, and their effectiveness.

Unit 5: Environmental Impact and Regulations

Environmental impact of wastewater discharges: Impact on water bodies, soil, and ecosystems.

Wastewater management regulations: Standards for wastewater treatment and effluent discharge (e.g., EPA standards, WHO guidelines).

Wastewater treatment and sustainability: Integrating sustainability into wastewater management practices.

Decentralized wastewater treatment systems: On-site treatment, decentralized systems for rural and remote areas.

Case studies: Successful wastewater management systems and technologies globally.

Course Objectives (COs):

CO1: To understand the principles of wastewater engineering and its management.

CO2: To learn the design and operation of wastewater collection systems.

CO3: To gain knowledge on primary and secondary treatment processes for wastewater.

CO4: To explore advanced wastewater treatment methods, including membrane technologies and nutrient removal.

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CO5: To understand the environmental impact of wastewater and the relevant regulations governing wastewater management.

Reference Books:

1. **"Wastewater Engineering: Treatment and Reuse"** by Metcalf & Eddy, George Tchobanoglous, and Franklin L. Burton.
2. **"Introduction to Environmental Engineering"** by Mackenzie L. Davis and David A. Cornwell.
3. **"Environmental Engineering: Principles and Practice"** by David A. Cornwell, Peter M. S. D. Johnson, and Michael S. Brown.
4. **"Wastewater Treatment and Reuse: Past, Present, and Future"** by Michael H. Gerardi.
5. **"Water and Wastewater Engineering"** by Mackenzie L. Davis.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	2	2	3	3	1	1	2	2	3
CO2	2	2	1	2	2	3	3	1	1	2	2	2
CO3	3	3	2	2	2	3	3	1	1	2	2	3
CO4	2	3	3	2	2	3	3	3	2	3	2	3
CO5	2	2	1	2	2	3	3	3	2	3	3	3

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CE2530- INDUSTRIAL WASTE WATER TREATMENT

COURSE OBJECTIVES:

The objective of the **Industrial Wastewater Treatment** course is to provide students with knowledge of industrial wastewater sources, pollutants, and treatment processes. It aims to develop skills in designing sustainable management strategies, advanced treatment technologies, and regulatory compliance for minimizing environmental impact.

Unit 1: Introduction to Industrial Wastewater Treatment

Introduction to Industrial Wastewater: Definition, sources, and characteristics of industrial wastewater.

Types of industries generating wastewater: Textiles, food and beverages, chemicals, pharmaceuticals, paper, and pulp industries.

Pollutants in industrial wastewater: Organic, inorganic, and hazardous pollutants.

Classification of industrial wastewater: Based on the type of industry, pollutants, and treatment methods.

Environmental regulations: Effluent discharge standards and guidelines for industrial wastewater treatment (EPA, WHO standards).

Unit 2: Physico-Chemical Treatment Methods

Preliminary Treatment: Screening, grit removal, and sedimentation.

Coagulation and Flocculation: Mechanisms, chemical agents, and applications in industrial wastewater.

Chemical precipitation: Removal of metals, phosphates, and other inorganic pollutants.

Adsorption: Use of activated carbon, ion exchange, and other adsorbents for removing pollutants.

Oxidation and Reduction: Use of chemical oxidants like chlorine, ozone, hydrogen peroxide for pollutant removal.

Unit 3: Biological Treatment Methods

Biological Treatment Principles: Aerobic vs. anaerobic processes, biodegradability, and microbial activity.

Activated Sludge Process (ASP): Design, operation, and modifications for industrial applications.

Fixed-film Processes: Trickling filters, rotating biological contactors (RBC).

Sequencing Batch Reactors (SBR): Design, operational parameters, and applications.

Bioreactors: Applications in treating specific industrial wastewater (e.g., dairy, distillery).

Unit 4: Advanced Treatment Methods

Membrane Filtration Technologies: Microfiltration, ultrafiltration, nanofiltration, and reverse osmosis (RO) in industrial wastewater treatment.

Electrochemical Treatment: Electrocoagulation, electroflotation, and electrooxidation for industrial wastewater.

Advanced Oxidation Processes (AOPs): Methods, mechanisms, and applications for removing organic pollutants.

Ion Exchange and Adsorption Technologies: Removal of heavy metals, dyes, and other contaminants.

Sludge Management: Treatment, stabilization, and disposal of industrial sludge.

Unit 5: Industrial Wastewater Management and Case Studies

Wastewater Management Strategies: Integrated wastewater management systems, water recycling, and reuse in industries.

Zero Liquid Discharge (ZLD): Concept, design, and implementation in industries.

Case Studies: Successful industrial wastewater treatment plant designs and operational strategies (e.g., textile, chemical, and paper industries).

Challenges and Innovations: Emerging technologies and challenges in treating industrial wastewater in developing countries.

Economic Considerations: Cost-benefit analysis, operational costs, and energy-efficient solutions in industrial wastewater treatment.

COURSE OUTCOMES:

CO1: Understand the sources, characteristics, and pollutants in industrial wastewater and their impact on the environment.

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CO2: Apply physico-chemical treatment methods such as coagulation, flocculation, sedimentation, and chemical precipitation to treat industrial wastewater.

CO3: Design and implement biological treatment processes, including activated sludge, sequencing batch reactors (SBR), and fixed-film systems for industrial applications.

CO4: Evaluate advanced treatment technologies such as membrane filtration, advanced oxidation processes (AOPs), and electrochemical methods for industrial wastewater treatment.

CO5: Develop and analyze wastewater management strategies, including water recycling, reuse, and zero liquid discharge (ZLD) systems.

Reference Books :

1. **"Industrial Wastewater Treatment: A Handbook for Engineers and Scientists"** by J. A. C. R. van der Meer
2. **"Wastewater Engineering: Treatment and Reuse"** by Metcalf & Eddy, George Tchobanoglous, and Franklin L. Burton
3. **"Industrial Wastewater Treatment: Problems and Solutions"** by L. R. F. Huggins and D. A. Drewes
4. **"Wastewater Treatment for Pollution Control"** by C. A. Burkhardt, G. T. P. V. Chasson
5. **"Introduction to Industrial Wastewater Treatment"** by D. R. S. Gupta

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	1	1	1	2	1	1	2
CO2	3	3	2	2	1	1	1	1	2	1	1	2
CO3	3	3	3	2	1	1	1	1	2	1	1	2
CO4	3	3	3	3	2	1	1	1	2	1	1	2
CO5	3	3	3	3	2	1	1	1	2	1	1	3

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BA2516 - CLIMATE POLICY AND ECONOMICS

Course Objectives:

The objective of the **Climate Policy and Economics** course is to equip students with an understanding of the interplay between climate policies and economic principles. It aims to develop skills to analyze, design, and evaluate sustainable solutions for climate mitigation and adaptation.

Unit 1: Introduction to Climate Change and Economics

Climate Change Basics: Causes, effects, and global challenges.

Economic Impacts of Climate Change: Costs of inaction, damage to infrastructure, health, agriculture, and ecosystems.

Economic Concepts in Climate Policy: Externalities, market failures, cost-benefit analysis, and valuation of environmental goods.

Unit 2: Climate Policy Frameworks

Global Climate Agreements: UNFCCC, Kyoto Protocol, and Paris Agreement.

National and Regional Policies: Renewable energy mandates, emissions trading systems (ETS), carbon taxes, and subsidies.

Policy Instruments: Command-and-control policies, market-based mechanisms, and voluntary actions.

Role of Stakeholders: Governments, businesses, NGOs, and international organizations.

Unit 3: Economics of Climate Mitigation

Mitigation Strategies: Renewable energy, energy efficiency, carbon capture, and reforestation.

Carbon Pricing: Cap-and-trade systems, carbon taxes, and their economic implications.

Cost of Mitigation vs. Cost of Inaction: Economic modeling and evaluation tools.

Innovation and Technology in Mitigation: Role of technological advancements in reducing emissions.

Unit 4: Economics of Climate Adaptation

Adaptation Strategies: Infrastructure resilience, disaster risk management, and sustainable agriculture.

Economic Evaluation of Adaptation Measures: Cost-effectiveness and prioritization.

Funding for Adaptation: Green Climate Fund (GCF) and other international funding mechanisms.

Social and Economic Equity in Adaptation: Vulnerable populations and the role of inclusivity in adaptation strategies.

Unit 5: Emerging Issues and Future Directions

Sustainable Development and Climate Policy: Linking climate action to sustainable development goals (SDGs).

Carbon Markets and Offsets: Emerging trends and controversies.

Climate Finance: Role of investments, green bonds, and public-private partnerships.

Geoengineering and Ethical Considerations: Economic and policy perspectives.

Case Studies: Success stories and lessons from global climate initiatives.

Course Outcomes (COs)

CO1: Understand the fundamentals of climate change, its economic impacts, and the concept of externalities.

CO2: Analyze global and regional climate policies, including international agreements and national strategies.

CO3: Evaluate economic instruments for climate mitigation, such as carbon taxes, emissions trading, and renewable energy policies.

CO4: Assess the economic aspects of climate adaptation measures and their effectiveness.

CO5: Develop strategies linking climate policy to sustainable development and equitable economic growth.

Reference Books :

1. **"Climate Economics: Economic Analysis of Climate, Climate Change, and Climate Policy"** Richard S. J. Tol
2. **"An Introduction to Climate Change Economics and Policy"** Felix R. FitzRoy and Elissaios Papyrakis

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3. **"Economics and Policy of Energy and Environmental Sustainability"** Narendra N. Dalei, Kakali Mukhopadhyay, and P. K. Joshi
4. **"The Climate Casino: Risk, Uncertainty, and Economics for a Warming World"** William Nordhaus
5. **"Global Climate Change: Economic and Policy Issues"** Ravi Kanbur and Henry Shue
6. **"Environmental Economics and Policy"** Tom Tietenberg and Lynne Lewis.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2	2	2	1	2	2	1	2	2
CO2	3	3	3	2	3	3	2	3	3	3	3	3
CO3	3	3	3	3	3	3	3	3	3	3	2	3
CO4	3	3	2	3	3	3	2	3	3	2	3	3
CO5	3	3	3	3	3	3	3	3	3	3	3	3

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CE2531- GREEN ENGINEERING PRINCIPLES

Course Objectives:

The objective of the **Green Engineering Principles** course is to impart knowledge of sustainable engineering practices that minimize environmental impact. It aims to develop skills in eco-friendly design, resource efficiency, and adherence to environmental standards for sustainable development.

Unit 1: Introduction to Green Engineering

Definition and Importance: Principles of green engineering and its role in sustainability.

Environmental Challenges: Climate change, pollution, resource depletion, and the role of engineering.

Life Cycle Thinking: Introduction to life cycle analysis (LCA) and systems thinking in engineering design.

Unit 2: Sustainable Design and Development

Design Principles: Incorporating environmental considerations in product and process design.

Energy Efficiency: Renewable energy technologies, energy conservation, and energy audits.

Green Building Concepts: Sustainable architecture and materials, LEED certification, and energy-efficient designs.

Unit 3: Materials and Processes in Green Engineering

Eco-Friendly Materials: Selection of renewable, biodegradable, and recyclable materials.

Cleaner Production Techniques: Waste minimization, pollution prevention, and green chemistry principles.

Advanced Manufacturing: Additive manufacturing, circular economy, and cradle-to-cradle design.

Unit 4: Environmental Impact Assessment and Risk Management

Environmental Assessment Tools: LCA, carbon footprint analysis, and eco-labeling.

Risk Management: Environmental risk assessment and mitigation techniques.

Regulatory Frameworks: Environmental laws, standards, and policies supporting green engineering.

Unit 5: Emerging Trends and Applications

Green Technologies: Applications in water treatment, air purification, and waste management.

IoT and Smart Technologies: Role in energy management and sustainable practices.

Case Studies: Success stories in green engineering practices across industries.

COURSE OUTCOMES (COS)

CO1: Understand the fundamental principles of green engineering and its importance in sustainability.

CO2: Apply sustainable design principles and energy-efficient practices in engineering systems.

CO3: Select eco-friendly materials and cleaner production techniques for minimizing environmental impacts.

CO4: Conduct environmental impact assessments and implement risk management strategies.

CO5: Explore emerging green technologies and innovative solutions for sustainable engineering practices.

Reference Books:

1. **"Green Engineering: Environmentally Conscious Design of Chemical Processes"** David T. Allen and David R. Shonnard
2. **"Sustainable Engineering: Principles and Practice"** David T. Allen and Michael R. Rimer
3. **"Introduction to Environmental Engineering and Science"** Gilbert M. Masters and Wendell P. Ela
4. **"Green Engineering: Innovative Technologies for Sustainable Development"** Riadh Habash
5. **"Cradle to Cradle: Remaking the Way We Make Things"** William McDonough and Michael Braungart
6. **"Sustainability: A Systems Approach"** Tony Clayton and Nicholas Radcliffe
7. **"Life Cycle Assessment Handbook: A Guide for Environmentally Sustainable Products"** Mary Ann Curran

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	3	2	3	2	2	1	2	2
CO2	3	3	3	3	3	3	3	3	2	2	2	3
CO3	3	2	3	3	3	2	3	2	3	2	2	3
CO4	3	3	3	3	3	3	3	3	3	3	2	3
CO5	3	3	3	3	3	3	3	3	3	3	3	3

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CE2580- Seminar –I

- 1. Topic Selection**
 - Identify a relevant topic related to the field of study (Engineering, Management, or interdisciplinary domains).
 - Topic should involve recent advancements, challenges, or trends in the chosen field.
 - Approval from the seminar guide or faculty advisor is mandatory.
- 2. Literature Review**
 - Conduct an extensive literature review of books, journals, research papers, and technical articles.
 - Compile and critically analyze findings to establish the context and importance of the topic.
- 3. Report Preparation**
 - Prepare a detailed seminar report, including:
 - Abstract
 - Introduction
 - Objectives
 - Literature survey
 - Methodology or analysis (if applicable)
 - Results and discussion (if applicable)
 - Conclusion and future scope
 - References (in standard format like IEEE, APA, or MLA).
- 4. Presentation Skills**
 - Develop effective presentation skills to convey ideas clearly and concisely.
 - Use visual aids (PowerPoint slides, posters, models, etc.) to enhance delivery.
- 5. Seminar Delivery**
 - Present the seminar topic to a panel of faculty and peers.
 - Answer questions and participate in discussions to validate the understanding of the topic.
- 6. Evaluation Criteria**
 - Quality of topic and relevance to the program.
 - Depth of literature review and analysis.
 - Clarity, structure, and format of the seminar report.
 - Presentation skills and effectiveness in delivering the seminar.
 - Ability to handle questions and discussions.

CE2508 ENVIRONMENTAL MODELING & SIMULATION

Course Objectives

1. To introduce students to the concepts and techniques used in environmental modeling and simulation.
2. To develop an understanding of the interaction between environmental systems.
3. To teach students how to apply models to solve real-world environmental problems.
4. To familiarize students with the tools and software used in environmental modeling.
5. To develop the skills for data analysis and simulation in environmental contexts.

Unit 1: Introduction to Environmental Modeling and Simulation

- Definition and importance of environmental modeling and simulation
- Types of models (deterministic, stochastic, dynamic, static)
- Simulation methods: Monte Carlo, System Dynamics, Cellular Automata, and Agent-Based Modeling
- Basics of Environmental Systems and their Interactions (Air, Water, Soil, etc.)
- Modeling tools and software (MATLAB, Simulink, Vensim, AnyLogic)

Unit 2: Environmental Systems and Modeling Techniques

- Ecological modeling: Population dynamics, biodiversity models
- Hydrological modeling: Watershed modeling, groundwater flow, and pollution transport
- Climate models: Global climate models, regional climate simulation
- Air quality models: Dispersion models, emissions, and pollution control strategies
- Soil models: Erosion models, nutrient cycling, soil pollution

Unit 3: Data Collection and Analysis for Environmental Modeling

- Remote sensing in environmental monitoring
- Geospatial data and spatial analysis
- Data sources: Satellite imagery, weather stations, sensor networks
- Statistical analysis methods in environmental data
- Handling and processing large-scale environmental data

Unit 4: Environmental Simulation Case Studies

- Case study of air pollution modeling (e.g., urban air quality models)
- Climate change modeling and forecasting
- Water resource management and simulation models
- Biodiversity conservation models
- Environmental risk assessment simulations (e.g., natural disaster modeling)

Unit 5: Advanced Topics in Environmental Modeling

- Advanced climate models and their applications
- Integrated environmental models (e.g., land use, water, air quality)
- Multi-criteria decision analysis (MCDA) in environmental modeling
- Coupled human-environmental systems models
- Use of artificial intelligence and machine learning in environmental simulations

List of Experiments and Equipment

Experiments:

1. **Air Quality Simulation:** Modeling air pollutant dispersion using Gaussian plume model.
2. **Water Quality Modeling:** Simulation of river or lake water quality under different pollutant loading scenarios.
3. **Climate Change Simulation:** Use of simple climate models to predict future temperature rise and impacts.
4. **Soil Erosion Modeling:** Simulating soil erosion using empirical models and GIS tools.
5. **Pollution Transport Simulation:** Simulating the movement of contaminants in groundwater.
6. **Hydrological Modeling:** Watershed simulation to predict runoff and flood scenarios.
7. **Ecological Modeling:** Population dynamics simulations for endangered species.
8. **Biodiversity Modeling:** Simulating habitat fragmentation and its effect on species diversity.

Equipment:

- MATLAB/Simulink software
- Vensim for system dynamics modeling
- AnyLogic for agent-based modeling
- Remote sensing tools (satellite data, drones)
- GIS software (ArcGIS, QGIS)

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- Environmental sensors (for temperature, humidity, air quality, water quality)
- Computers with high processing power for simulation

Course Outcomes (CO)

Upon completion of this course, students will be able to:

1. **CO1:** Understand the fundamentals of environmental modeling and simulation.
2. **CO2:** Apply different modeling techniques to environmental systems.
3. **CO3:** Collect, process, and analyze environmental data for modeling purposes.
4. **CO4:** Develop and validate environmental models using real-world case studies.
5. **CO5:** Apply advanced environmental simulation techniques to address global challenges like climate change and pollution.

CO-PO Mapping

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	2	2	1	3	2	2	1	2	1
CO2	3	3	2	3	2	2	2	2	3	3	3	2
CO3	2	2	3	2	1	2	3	2	2	1	3	2
CO4	3	3	2	2	2	3	2	2	2	2	3	2
CO5	2	3	3	2	2	2	3	3	3	2	2	2

(3 = High, 2 = Medium, 1 = Low, - = Not Mapped)

Reference Books

1. "Environmental Modeling: Fate and Transport of Pollutants in Water, Air, and Soil" by A. G. Johnson
2. "Introduction to Environmental Engineering and Science" by Gilbert M. Masters
3. "Environmental Systems and Processes: Principles, Modeling, and Design" by E. K. M. Mahaney
4. "Modeling the Environment" by Andrew Ford
5. "The Science and Engineering of Environmental Technology" by Rajeev Prakash

CE2509 Coastal Zone Management

Course Objectives (CO):

1. To understand the importance of coastal zones and the impact of human activities on them.
2. To explore the various management techniques and policies aimed at the sustainable use of coastal resources.
3. To familiarize students with coastal ecosystem dynamics, conservation strategies, and related environmental laws.
4. To develop skills for effective coastal zone management through real-world case studies and practical applications.
5. To study the effects of climate change and human interventions on coastal areas and the strategies for mitigating those effects.

Unit 1: Introduction to Coastal Zones

- Definition of Coastal Zones
- Importance of Coastal Zones in environmental, economic, and social contexts
- Coastal ecosystem services (fisheries, tourism, etc.)
- Threats to Coastal Zones (pollution, overexploitation, erosion, etc.)

Unit 2: Coastal Ecosystem and Biodiversity

- Coastal Ecosystems: Mangroves, Coral Reefs, Salt Marshes, Dunes, Estuaries
- Biodiversity of Coastal Areas
- Role of biodiversity in maintaining ecosystem services
- Case studies of successful conservation efforts

Unit 3: Integrated Coastal Zone Management (ICZM)

- Concept of Integrated Coastal Zone Management
- Planning, policies, and strategies in ICZM
- Stakeholder involvement: Government, NGOs, and local communities
- Tools and techniques for ICZM (GIS, remote sensing, etc.)

Unit 4: Impact of Climate Change on Coastal Areas

- Sea-Level Rise, Storm Surges, and Coastal Erosion
- Effects of global warming on coastal ecosystems (e.g., coral bleaching)
- Adaptive strategies for coastal zones
- Mitigation measures: Coastal protection structures, policy adaptation

Unit 5: Coastal Zone Management Policies and Legal Framework

- International and national laws on coastal protection (e.g., Coastal Zone Management Act)
- Environmental regulations and compliance
- Policies for sustainable development and disaster risk management
- Case studies of successful policies and legal frameworks in coastal zone management

Course Outcomes (CO):

CO1: Demonstrate an understanding of coastal zone characteristics, ecosystem functions, and threats to coastal resources.

CO2: Analyze and evaluate different coastal management approaches and their applications in real-world scenarios.

CO3: Apply principles of integrated coastal zone management (ICZM) in various case studies, addressing environmental, social, and economic concerns.

CO4: Develop skills in assessing the effects of climate change and human interventions on coastal zones, and propose mitigation strategies.

CO5: Critically review coastal management policies and suggest improvements based on scientific and socio-economic considerations.

CO-PO Mapping Values Table

(CO)	(PO) 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	2	2	2	3	1	2	3
CO2	3	3	3	2	3	1	2	3
CO3	3	2	3	2	3	2	2	2
CO4:	3	2	3	3	3	2	2	3
CO5	3	3	2	3	3	2	2	2
CO6	3	2	3	3	2	3	1	3

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(3 = High, 2 = Medium, 1 = Low, - = Not Mapped)

Reference Books:

1. “Coastal Management in the Caribbean” by K. R. Sharma
2. “Integrated Coastal Zone Management: A Global Perspective” by T. P. E. D. R. Johansson
3. “Coastal Ecosystems: Structure and Function” by P. D. J. Galloway
4. “Environmental Management in Coastal Zones” by P. K. Jha
5. “Coastal Zone Management: The Planning and Management of Marine and Coastal Environments” by R. C. Sudhakar
6. “Coastal Zone Management: Global Challenges and Opportunities” by D. M. S. K. Squires

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CE2510 NOISE POLLUTION

Course Objective:

1. To understand the causes and effects of noise pollution on the environment and human health.
2. To learn about various methods to control and mitigate noise pollution.
3. To understand the relationship between noise and regulations or standards governing noise levels.
4. To gain practical experience through experiments and projects focused on noise monitoring and control.

Unit 1: Introduction to Noise Pollution

- Definition of noise and noise pollution.
- Sources of noise pollution (industrial, transport, household, etc.).
- Characteristics of sound (frequency, intensity, etc.).
- Effects of noise pollution on human health, animals, and the environment.
- Noise levels and decibel scale.

Unit 2: Measurement and Monitoring of Noise

- Noise level measurement instruments (sound level meter, dosimeter, etc.).
- Calibration and operation of noise measurement devices.
- Types of noise measurements: Leq, Lmax, Lmin, L10, L90.
- Noise mapping and assessment techniques.
- Case studies of noise pollution monitoring in urban areas.

Unit 3: Noise Control and Mitigation

- Engineering and architectural controls for noise reduction (barriers, insulation, vibration damping, etc.).
- Legislative measures for noise pollution control.
- Noise abatement techniques for industries and transportation.
- Role of urban planning in noise control.
- Soundproofing and acoustic treatments for buildings.

Unit 4: Noise Pollution Standards and Regulations

- International and national noise standards (WHO, EPA, etc.).
- Noise regulations and limits for residential, industrial, and transportation areas.
- Noise pollution in different environments (urban, rural, industrial).
- Role of government and environmental agencies in noise control.

Unit 5: Environmental Impact of Noise Pollution

- Relationship between noise pollution and other environmental factors (air, water, soil pollution).
- Social and economic impacts of noise pollution.
- Sustainable noise management practices.
- Role of technology in reducing noise pollution.
- Case studies of successful noise reduction initiatives.

List of Experiments:

1. Measurement of Noise Levels in Different Locations:
2. Calibration of Sound Level Meter:
3. Noise Mapping using GIS:
4. Assessment of Noise Pollution in Traffic Areas:
5. Evaluation of Sound Insulation Materials:
6. Assessment of Noise Reduction in Buildings:

Equipment:

- **Sound Level Meter** (e.g., Brüel & Kjær, Larson Davis)
- **Dosimeter**
- **Data Logger**
- **Vibration Sensors**
- **Calibration Devices** (for sound meters)
- **Noise Barriers** (for experiments on noise mitigation)
- **GIS Software** (for noise mapping)
- **Noise Control Materials** (e.g., soundproof panels, acoustic foam)

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Course Outcome:

- CO1:** Students will be able to analyze and assess the impact of noise pollution on the environment.
CO2: Students will develop skills to measure and monitor noise pollution.
CO3: Students will understand noise regulations and control methods.
CO4: Students will apply their knowledge to reduce or mitigate noise pollution in various settings.
CO5: Students will apply their practical experience through experiments and projects focused on noise monitoring and control.

CO-PO Mapping:

COs \ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	3	2	2	2	2	2	1	2
CO2	3	3	2	2	3	2	2	1	3	3	2	3
CO3	2	3	2	1	3	2	2	2	1	2	2	3
CO4	3	3	3	3	3	2	3	2	2	2	3	2
CO5	2	3	3	2	3	3	3	1	1	3	3	2

(3 = High, 2 = Medium, 1 = Low, - = Not Mapped)

Reference Books:

1. **Environmental Noise Pollution: Noise Mapping, Public Health, and Policy** by Enda Murphy and James J. Gormley.
2. **Noise Pollution: A Modern Plague** by Richard H. Carmichael.
3. **Fundamentals of Acoustics** by Lawrence E. Kinsler and Austin R. Frey.
4. **Noise Control in Buildings** by David S. Kim and Richard H. Lyon.
5. **Acoustics and Noise Control** by P. R. K. Nambiar

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CE2511 ENVIRONMENTAL PROJECT MANAGEMENT

Course Objective:

- To understand the fundamentals of project management with an emphasis on environmental issues.
- To equip students with the knowledge and tools required to manage environmental projects effectively.
- To analyze environmental problems, devise sustainable solutions, and evaluate the risks and impacts associated with environmental projects.

Unit 1: Introduction to Project Management and Environmental Concepts

- Definition of a Project, Project Lifecycle, and Project Management Phases
- Key Principles of Project Management
- Sustainable Development Goals (SDGs) and Environmental Sustainability
- Environmental Impact Assessment (EIA)
- Environmental Policies and Regulations (National and International)

Unit 2: Environmental Project Planning and Scheduling

- Project Planning and Scope Definition for Environmental Projects
- Work Breakdown Structure (WBS) and Project Scheduling Tools (Gantt, PERT)
- Resource Allocation and Budgeting for Environmental Projects
- Risk Assessment and Environmental Risk Management

Unit 3: Environmental Project Implementation and Monitoring

- Project Execution and Deliverable Management
- Environmental Monitoring and Evaluation Techniques
- Stakeholder Management and Communication in Environmental Projects
- Legal and Ethical Considerations in Environmental Project Implementation

Unit 4: Environmental Risk Management and Sustainability

- Risk Identification and Analysis Techniques (Qualitative and Quantitative)
- Environmental Sustainability in Project Design and Implementation
- Life Cycle Assessment (LCA) and its Role in Project Decision-Making
- Sustainable Resource Management Practices

Unit 5: Project Closure and Evaluation

- Finalizing Project Deliverables and Reporting
- Post-Implementation Review and Environmental Impact Evaluation
- Project Documentation and Knowledge Management
- Closing a Project and Lessons Learned in Environmental Projects

Course Outcome:

Upon successful completion of the course, students will be able to:

1. Understand the principles of environmental project management and its relevance in sustainable development.
2. Plan, execute, and manage environmental projects from inception to completion.
3. Apply environmental assessment tools to identify and mitigate project impacts on the environment.
4. Implement risk management strategies specific to environmental projects.
5. Demonstrate leadership and decision-making skills in managing teams working on environmental projects.

CO-PO Mapping

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3					2	2	2			3
CO2	3	3			3		1		1	2		3
CO3	3	2	3	3	2		1	1		1	2	2
CO4	3	2	3	3	2	1	1			2	2	
CO5	3		3	2	3	1	1	1	2	2		3

(3 = High, 2 = Medium, 1 = Low, - = Not Mapped)

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Reference Books:

1. **"Environmental Project Management"** by Jean-Michel Gauthier
2. **"Sustainable Project Management"** by John M. W. Miller
3. **"Introduction to Project Management"** by Kathy Schwalbe
4. **"Project Management for Sustainability"** by J. Rodney Turner
5. **"Environmental Impact Assessment: A Practical Guide"** by J. P. McDonald

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CE2512 INDUSTRIAL PROCESS AND ENVIRONMENT POLLUTION

Course Objectives (CO):

1. To understand the industrial processes and their environmental implications.
2. To study the sources of pollution and its impact on the environment.
3. To explore pollution control methods in industrial processes.
4. To analyze the environmental regulations and policies for pollution control.
5. To evaluate sustainable practices in industrial operations for minimizing environmental harm.

Unit 1: Introduction to Industrial Processes and Pollution

- **Overview of Industrial Processes:** Types of industries and their processes (Chemical, Metallurgical, Textile, etc.)
- **Types of Pollution:** Air, water, soil, and noise pollution.
- **Sources of Pollution in Industries:** Emissions from raw materials, production processes, and waste disposal.

Unit 2: Air Pollution in Industrial Processes

- **Air Pollutants:** Types of pollutants (Gaseous, particulate matter).
- **Emission Sources:** Industrial chimneys, boilers, and furnaces.
- **Impact of Air Pollution:** Health impacts, environmental effects, acid rain, global warming.
- **Pollution Control Techniques:** Filters, electrostatic precipitators, scrubbers, and incinerators.

Unit 3: Water Pollution and Industrial Effluents

- **Water Pollutants:** Types (Organic, inorganic, and toxic substances).
- **Sources of Industrial Effluents:** Discharge from factories, mining operations, and agriculture.
- **Treatment of Industrial Effluents:** Primary, secondary, and tertiary treatment methods.
- **Wastewater Management:** Effluent treatment plants and water recycling techniques.

Unit 4: Solid Waste Management and Soil Pollution

- **Solid Waste from Industries:** Types and sources.
- **Environmental Impact:** Land degradation, loss of biodiversity, and health risks.
- **Waste Management Techniques:** Recycling, composting, incineration, landfilling.
- **Soil Contamination:** Heavy metals, chemicals, and hazardous wastes.

Unit 5: Environmental Regulations and Sustainable Industrial Practices

- **Environmental Laws and Policies:** National and international regulations (EPA, Kyoto Protocol, etc.).
- **Sustainable Practices in Industries:** Cleaner production, green chemistry, and energy-efficient technologies.
- **Case Studies:** Successful implementation of pollution control technologies and sustainable industrial practices.

Course Outcomes (PO):

Upon successful completion of this course, students will be able to:

CO1 (Understand and Analyze): Demonstrate an understanding of industrial processes and their relation to environmental pollution.

CO2 (Evaluate and Apply): Evaluate the sources and types of pollution in industries and apply methods for pollution control.

CO3 (Design and Implement): Design strategies for controlling pollution in industrial settings and apply regulatory guidelines for pollution control.

CO4 (Innovate and Adapt): Propose and adapt sustainable industrial processes that minimize environmental impact.

CO5 (Analyze and Interpret): Interpret data related to industrial emissions and pollution, and suggest remedial actions.

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CO-PO Mapping Values

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	2	1	-	-	-	-	-	-	2
CO2	3	3	2	2	1	-	1	-	-	-	-	3
CO3	3	3	2	3	1	1	1	-	-	-	-	2
CO4	3	2	3	3	2	2	3	1	1	-	-	3
CO5	2	3	3	3	3	1	1	1	1	-	-	3

(Note: 3 = Highly Relevant, 2 = Moderately Relevant, 1 = Slightly Relevant, "-" = Not Relevant)

Reference Books:

1. **"Industrial Pollution Control: A Practical Guide"** by L. M. Kumar and R. K. Sharma
2. **"Environmental Engineering: Water, Wastewater, and Air Pollution"** by Howard S. Peavy, Donald R. Rowe, and George Tchobanoglous
3. **"Introduction to Environmental Engineering and Science"** by Gilbert M. Masters
4. **"Environmental Pollution and Control"** by J. Jeffrey Peirce and Anuradha S. Sable
5. **"Handbook of Pollution Control and Waste Minimization"** by R. A. Massey

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**CE2599 a & CE2599 b CLIMATE SCIENCE AND ENGINEERING
[SPECIFIC PROJECT NAME]**

1. Introduction

- **Background of the Study:** Explanation of the importance of climate science and engineering in addressing global climate change.
- **Problem Statement:** Definition of the problem or challenge the project intends to solve.
- **Objectives of the Project:** What the project aims to achieve.

2. Course Objectives (CO)

The following are the general course objectives for a Climate Science and Engineering course:

1. To understand the principles of climate science and their role in engineering solutions.
2. To develop practical solutions to mitigate or adapt to climate change.
3. To apply scientific methods and engineering tools for climate modeling and analysis.
4. To explore the impact of human activities on climate and propose sustainable solutions.
5. To develop critical thinking and problem-solving skills in the context of climate science.

3. Project Outcomes (PO)

These outcomes are mapped to the course objectives and define the skills and knowledge the students will demonstrate upon completion of the project.

1. **PO1:** Apply knowledge of mathematics, science, and engineering to the design of climate-related solutions.
2. **PO2:** Design experiments, analyze data, and interpret results in the context of climate modeling.
3. **PO3:** Use modern tools for the analysis of climate data and simulations.
4. **PO4:** Understand the ethical, environmental, and societal impacts of engineering solutions related to climate change.
5. **PO5:** Communicate effectively through written reports and presentations.

4. Methodology

- **Research and Data Collection:** Gathering relevant data (temperature trends, emissions data, etc.).
- **Modeling and Simulations:** Using tools such as MATLAB, Python, or climate-specific software (e.g., EnergyPlus).
- **Analysis:** Statistical or machine learning techniques to analyze collected data and simulations.

5. Results and Discussion

- **Key Findings:** Presenting the outcomes of the analysis, models, and simulations.
- **Comparison with Literature:** How the project results align with existing research.
- **Impact Assessment:** Evaluating the environmental and social impacts of the findings and proposed solutions.

6. Conclusion

- **Summary of Key Points:** A recap of the project's objectives, methods, findings, and outcomes.
- **Recommendations:** Propose actionable solutions or further research based on the project's findings.
- **Future Work:** Suggestions for future studies or improvements to the current project.

7. References

- A list of all sources, research papers, articles, and textbooks used in the project.

8. Appendices

- Any supplementary information, such as raw data, charts, graphs, or additional models used in the project.

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Program Name: M.Tech – Sustainability in Building Maintenance

Program Duration: 2 years

Philosophy/Significance of the Course

Building maintenance consist of wide variety of tasks. It is important for a number of reasons such as safety, cost savings, sustainability and preventing various types of failures. So for design and construction of buildings there are lot of courses are available to learn but for maintenance there is some of the subjects only available.

Our course is designed to equip you with the technical know-how and management skills to be an effective leader in this specialist sector. Develop an understanding of the design, installation, operation and maintenance that make a building safe, comfortable, energy efficient and functional. This highly specialised skillset is always in demand. In fact, it's set to grow as buildings' designs evolve to become more sustainable.

Distinctiveness and uniqueness of the program

Building maintenance is often referred to as a "distinctive course" because it is a specialized field that requires a unique set of skills, knowledge, and expertise. The term emphasizes the specialized nature of the discipline within the broader construction, real estate, and facilities management industries. Building maintenance covers a wide range of tasks, from structural repairs to mechanical systems management. It requires knowledge in areas like plumbing, HVAC systems, electrical work, safety standards, and materials science, which makes it distinct from general construction or real estate management.

Building maintenance involves both preventive and corrective measures. Preventive maintenance aims to reduce the risk of failure and extend the life of a building and its systems, while corrective maintenance fixes issues when they arise. Both require specific technical knowledge and practical skills. Many building maintenance programs or courses are tailored to specific job functions, such as facilities management, building engineering, or specific technical fields (e.g., HVAC repair, plumbing). These courses are designed to develop competencies in particular areas, making it distinct from other construction-related education.

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REGULATION-2025**

(Choice Based Credit System)

M. Tech – Sustainability in Building Maintenance

PROGRAM OUTCOMES (POs):

Graduates of the Programme M. Tech. Sustainability in Building maintenance will be able to

- PO1 Research Aptitude:** An ability to independently carry out research / investigation and development work to solve practical problems.
- PO2 Technical Documentations:** An ability to write and present a substantial technical report/document.
- PO3 Technical Competence:** Students should be able to demonstrate a degree of mastery over the area as per the specialization of the programme. The mastery should be at a level higher than the requirements in the appropriate bachelor programme.
- PO4 Critical Analysis of Building Maintenance-related Problems:** Critically analyse complex Building Maintenance related problems and apply independent judgment for synthesizing information.
- PO5 Conceptualization and Evaluation of Innovative Engineering Solutions to Building Maintenance Issues:** Conceptualize and solve Building repair & Maintenance problems, evaluate potential solutions and arrive at technically feasible, economically viable and environmentally sound solutions with due consideration of health, safety and sociocultural factors.
- PO6 Life-long Learning:** Recognize the need for independent, life-long learning and adapt to emerging technologies in Building Maintenance +and solutions to novel problems.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs):

Graduates of the programme M. Tech. Sustainability in Building maintenance will

- PEO1** Perform exceptionally well in research and will be successful in the building maintenance field in both public and private sector enterprises.
- PEO2** Become consultants in Building maintenance and solve complex real-life issues related to analysis, design and maintenance of structures under various environmental conditions with modern equipment and software tools.
- PEO3** Become entrepreneurs and innovate processes and construction technologies by combining their skills in interdisciplinary management to fulfill societal demands and devise solutions that are technically sound, economically practical, and socially acceptable.

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- PEO4** Create cost-effective solutions for sustainability while understanding societal and ecological challenges through teamwork, ethics, and professionalism.
- PEO5** Possess the leadership skills, established ethical norms and guidelines, excellence in competency, and lifelong learning required for a prosperous professional career.

PROGRAM SPECIFIC OUTCOMES (PSOs):

M. Tech Building Maintenance program's specific goals are to create engineers who:

- PSO1** Possess knowledge of the latest advancements in composite materials, concrete technology, and construction technology, as well as their characteristics and real-world uses.
- PSO2** Achieve success in Building Maintenance by using ideas and solving diverse construction difficulties.
- PSO3** Professionally provide cost-effective solutions for a sustainable environment, using comprehensive understanding of societal and ecological challenges.

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M.Tech – SUSTAINABILITY IN BUILDING MAINTENANCE (Full Time)
CURRICULUM AND SYLLABI FOR SEMESTERS I to IV
SEMESTER I

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIODS
				L	T	P	C	
THEORY								
1	MA25101	Statistical Method for Engineers	FC	3	1	0	4	4
2	BM25101	Modern Construction Materials	PCC	3	1	0	4	4
3	BM25102	Building Services	PCC	3	0	0	3	3
4	BM25103	Maintenance Engineering & Management	PCC	3	1	0	4	4
5	BM25104	Advanced Concrete Technology	PCC	3	0	0	3	3
6	MC25101	Research Methodology and IPR	RMC	3	0	0	3	3
PRACTICALS								
7	BM2511	Advanced Construction Engineering & Computing Techniques laboratory	PCC	0	0	4	2	4
TOTAL				21	2	4	23	28

SEMESTER II

S.NO .	COURSE CODE	COURSE TITLE	CATE GOR Y	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
THEORY								
1	BM25201	Building Repair & Maintenance	PCC	3	1	0	4	4
2	BM25202	Non Destructive Testing	PCC	3	0	0	3	3
3	BM25203	Advanced Construction Techniques	PCC	3	0	0	3	3

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4	BM25204	Quality Control & Assurance in Construction	PCC	3	1	0	4	4
5		Professional Elective I	PEC	3	0	0	3	3
6		Professional Elective II	PEC	3	0	0	3	3
PRACTICALS								
7	BM2521	Advanced Material Testing Laboratory	PCC	0	0	4	2	4
8	BM2522	Seminar-I	EDC	0	0	2	1	2
9	BM2523	Practical Training-I	EDC	0	0	2	2	2
TOTAL				19	2	8	25	29

SEMESTER III

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
THEORY								
1		Professional Elective III	PEC	3	0	0	3	3
2		Professional Elective IV	PEC	3	0	0	3	3
3		Professional Elective V	PEC	3	0	0	3	3
PRACTICALS								
4	BM2531	Project Work I	EDC	0	0	12	6	12
5	BM2532	Seminar-II	EDC	0	0	2	1	2
6	BM2533	Practical Training-II	EDC	0	0	2	2	2
TOTAL				9	0	16	18	25

SEMESTER IV

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
PRACTICALS								

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1	BM2541	Project Work II	EDC	0	0	24	12	24
TOTAL				0	0	24	12	24

TOTAL CREDITS: 78

FOUNDATION COURSES (FC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	MA25101	Statistical Method for Engineers	FC	3	1	0	4	4

PROFESIONAL CORE COURSES (PCC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	BM25101	Modern Construction Materials	PCC	3	1	0	4	4
2	BM25102	Building Services	PCC	3	0	0	3	3
3	BM25103	Maintenance Engineering & Management	PCC	3	1	0	4	4
4	BM25104	Advanced Concrete Technology	PCC	3	0	0	3	3
5	BM25201	Building Repair & Maintenance	PCC	3	1	0	4	4
6	BM25202	Non Destructive Testing	PCC	3	0	0	3	3
7	BM25203	Advanced Construction Techniques	PCC	3	0	0	3	3
8	BM25204	Quality Control & Assurance in Construction	PCC	3	0	0	4	3
TOTAL CREDITS								28

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LIST OF PROFESSIONAL ELECTIVE COURSES (PEC)

SEMESTER II, ELECTIVE I

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	BM2501	Sustainable Construction	PEC	3	0	0	3	3
2	BM2502	Construction Equipment & Management	PEC	3	0	0	3	3
3	BM2503	Environmental Impact & Risk Assessment	PEC	3	0	0	3	3
4	BM2504	Value Engineering	PEC	3	0	0	3	3
TOTAL CREDITS							12	

SEMESTER II, ELECTIVE II

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	BM2505	Computer Application in Construction Management	PEC	3	0	0	3	3
2	BM2506	Design of Energy Efficient Buildings	PEC	3	0	0	3	3
3	BM2507	Shoring, Scaffolding & Formwork	PEC	3	0	0	3	3
4	BM2508	Project Safety Management	PEC	3	0	0	3	3
TOTAL CREDITS							12	

SEMESTER III, ELECTIVE III

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	BM2509	Green Building Techniques	PEC	3	0	0	3	3
2	BM2510	Management Information System	PEC	3	0	0	3	3
3	BM2512	Structural Health Monitoring	PEC	3	0	0	3	3

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4	BM2513	Material Procurement and Management	PEC	3	0	0	3	3
TOTAL CREDITS							12	

SEMESTER III, ELECTIVE IV

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	BM2514	Material Management	PEC	3	0	0	3	3
2	BM2515	Contract Laws & Regulations	PEC	3	0	0	3	3
3	BM2516	Lean Construction Concepts, Tools and Practice	PEC	3	0	0	3	3
4	BM2517	Maintainability	PEC	3	0	0	3	3
TOTAL CREDITS							12	

SEMESTER III, ELECTIVE V

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	BM2518	Repair Technology	PEC	3	0	0	3	3
2	BM2519	Human Resource Management in Construction	PEC	3	0	0	3	3
3	BM2520	Quality & Reliability Engineering	PEC	3	0	0	3	3
4	BM2521	Repair and Rehabilitation of structures	PEC	3	0	0	3	3
TOTAL CREDITS							12	

RESEARCH METHODOLOGY AND IPR COURSES (RMC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	

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1	MC25101	Research Methodology and IPR	MC	3	0	0	3	3
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EMPLOYMENT DEVELOPMENT COURSES (EDC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	BM2522	Seminar-I	EDC	0	0	2	1	2
2	BM2523	Practical Training-I	EDC	0	0	2	2	2
3	BM2531	Project Work I	EDC	0	0	12	6	12
4	BM2532	Seminar-II	EDC	0	0	2	1	2
5	BM2533	Practical Training-II	EDC	0	0	2	2	2
6	BM2541	Project Work II	EDC	0	0	24	12	24
TOTAL CREDITS								24

OBJECTIVES:

- To learn about statistical methods and their applications in engineering, including estimation theory, hypothesis testing, correlation, regression, randomized variables & probability distributions.

UNIT I INTRODUCTION TO STATISTICS

12

Introduction to Statistics – Measure of Central Tendency – Coefficient of Variation and its applications- Mean, Median & Mode, Standard Deviation – Moments, Skewness & Kurtosis-Efficiency & Sufficiency.

UNIT II HYPOTHESIS TESTING -I

12

Introduction to Testing Of Hypothesis- Sampling Distributions – Small & Large Scale Samples- Types Of Error – Large Scale Sample Tests –Z Test For Single Proportion – Difference Of Proportion- Single & Difference Mean.

UNIT III HYPOTHESIS TESTING-II

12

Introduction – Random Variables – Probability Mass Function, Distribution and Density Functions – Joint Probability Distribution and Joint Density Functions – Marginal, Conditional Distributions and Density Functions – Mathematical Expectation and its Properties – Covariance – Moment Generating Function – Characteristic Function.

UNIT IV DESIGN OF EXPERIMENTS & CORRELATION

12

Analysis of variance – One way and two way classifications – Completely randomized design – Randomized block design – Latin square design - Multiple and partial correlation – Method of least squares – Plane of regression – Properties of residuals – Coefficient of multiple correlation

UNIT V RANDOM VARIABLES & PROBABILITY DISTRIBUTIONS

12

Random Variables – Probability Mass Function – Distribution and Density Functions- Mathematical Expectation and Its Properties- Covariance-Binominal, Poisson, Normal, Gamma, Exponential & Weibull Distribution.

Total Periods: 60

OUTCOMES:

After completing this course, students should demonstrate competency in the following topics:

- CO1** Consistency, efficiency and unbiasedness of estimators, method of maximum likelihood estimation and Central Limit Theorem.
- CO2** Use statistical tests in testing hypotheses on data.
- CO3** Concept of linear regression, correlation, and its applications.

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- CO4** List the guidelines for designing experiments and recognize the key historical figures in Design of Experiments.
- CO5** Perform exploratory analysis of multivariate data, such as multivariate normal density, calculating descriptive statistics, testing for multivariate normality.

REFERENCES:

1. Gupta S.C., and Kapoor, V.K., “Fundamentals of Mathematical Statistics”, 12th Edition, Sultan Chand and Sons, 2020.
2. Jay L. Devore, “Probability and statistics for Engineering and the Sciences”, 8th Edition, Cengage Learning, 2014.
3. Johnson, R.A., Miller, I and Freund J., "Miller and Freund's Probability and Statistics for Engineers", 9th Edition, Pearson Education, Asia, 2016.
3. Johnson, R.A. and Wichern, D. W. “Applied Multivariate Statistical Analysis”, 6th Edition, Pearson Education, Asia, 2012.
5. Rice, J.A. "Mathematical Statistics and Data Analysis", 3rd Edition, Cengage Learning, 2015.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	2	2	3	3
CO2	3	3	3	3	2	2	1	2	2
CO3	3	3	3	3	2	2	1	2	2
CO4	3	3	3	3	2	2	1	3	3
CO5	3	3	3	3	2	2	1	3	3
Avg	3	3	3	3	2	2	1	2	2

*** 3- High, 2- Medium, 1- Low**

BM25101

MODERN CONSTRUCTION MATERIALS

L T P C

4 1 0 4

OBJECTIVES:

- To study and understand the properties of modern construction materials used in construction.

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- To understand special concretes, metals, composites, water proofing compounds, non weathering materials, and smart materials.

UNIT I METALS

12

Steels – New Alloy Steels – Aluminum and its Products –Coatings to reinforcement – anticorrosive polymer coating, anticorrosive treatment inhibited cement slurry coating, cement polymer composite coating and epoxy coated reinforcement – Applications.

UNIT II COMPOSITES

12

Composition – Constituents – mouldings methods – tooling materials and their selection - Plastics – Representative polymers - Reinforced Polymers – Fibre reinforced polymer (FRP) –Fibre process and moulding process – Application of carbon fibre reinforced polymers and glass fibre reinforced polymers.

UNIT III OTHER MATERIALS

12

Water Proofing Compounds – scope -, requirements, sampling, packing and marking - Non-weathering Materials – Flooring materials – soft coverings, wood flooring – resilient flooring – hard flooring – seamless chemical flooring – façade materials and their testing.

UNIT IV SPECIAL CONCRETES

12

Concretes, Behaviour of concretes - High Strength and High Performance Concrete – Fibre Reinforced Concrete- Light weight concrete - Polymer concrete - Hot and cold weather concrete - Self compacting concrete – Vacuum concrete - Alternate Materials to concrete.

UNIT V SMART AND INTELLIGENT MATERIALS

12

Smart and Intelligent Materials for intelligent buildings - Matrix tool for assessing the performance of intelligent buildings –Intelligent buildings technology – Thermal comfort , Visual comfort and Indoor air quality – climate , building-climate interaction - Energy Management systems – Smart BuildingsEvaluation Tool.

Total Periods: 60

OUTCOMES:

- CO1** Understand the roll of metals, composites and smart materials in construction projects
- CO2** Evaluate the different types of composites with its properties and behavior in construction
- CO3** Analyze the different types of smart and modern materials with its usages
- CO4** Apply the materials for maintaining the indoor thermal comfort and indoor air quality

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CO5 Apply the appropriate smart and intelligent materials in construction Industry

REFERENCE BOOKS:

1. Ashby, M.F. and Jones.D.R.H.H. “Engineering Materials 1: An introduction to Properties, applications and designs”, Elsevier Publications, 2005.
2. Deucher, K.N, Korfiatis, G.P and Ezeldin, A.S, “Materials for civil and Highway Engineers”, Prentice Hall Inc., 1998.
3. Mamlouk, M.S. and Zaniewski, J.P., “Materials for Civil and Construction Engineers”, Prentice Hall Inc., 1999.
4. Santhakumar.A.R., “Concrete Technology”, Oxford University press, New Delhi
5. Shan Somayaji, “Civil Engineering Materials”, Prentice Hall Inc., 2001
6. Shetty M.S, “Concrete Technology: Theory and Practice”, S.Chand & Company Ltd., 2005.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	1	3	3	3	2	2	2	3
CO2	1	1	2	2	2	2	2	2	3
CO3	2	2	2	3	3	2	1	1	3
CO4	2	2	2	3	3	2	2	1	1
CO5	2	2	3	3	3	2	1	1	1
Avg	2	2	2	3	3	2	2	1	2

*** 3- High, 2- Medium, 1- Low**

BM25102

BUILDING SERVICES

L T P C

3 0 0 3

OBJECTIVES:

- Building services are the essential services provided in the buildings for improving functioning of the buildings in efficient manner for the desired use of the building.

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- The electrical services, mechanical services such as air conditioning, lighting, ventilation, fire protection, acoustics and sound insulations, elevators, escalators, as well as civil engineering services such as water supply, sanitary services, etc. have become most essential services for residential, industrial, high rise, hotels, motels, monumental buildings.

UNIT I INTRODUCTION TO BUILDING SERVICES

9

Definitions, Objective and uses of services, Applications of services for different types building considering, Classification of building services, Types of services and selection of services, Natural and artificial lighting- principles and factors, Necessity of Ventilation, Types of ventilation – Natural and Mechanical, Factors to be considered in the design of Ventilation.

UNIT II ELECTRICAL SERVICES AND LAYOUT IN DIFFERENT TYPES OF BUILDING

9

Technical terms and symbols for electrical installations and accessories of wiring, Types of insulation, electrical layout for residence, small work shop, show room, school building, etc.

UNIT III MECHANICAL SERVICES IN BUILDINGS

9

Introduction of mechanical services Lift: Definition, Types of Lifts, Design Considerations, Location, Sizes, Component parts Elevators & Escalators-Different types of elevators and Escalators, Freight elevators, Passenger elevators, Hospital elevators, Uses of different types of elevators Escalators. Air Conditioning -Definition, Purpose, Principles, Temperature Control, Air Velocity Control, Humidity Control, Air Distribution system, Cleaners, Filters, Spray washers, Electric preceptors, Types of Air Conditioners, (Central type, Window Type, Split Unit.

UNIT IV FIRE PROTECTION

9

Introduction, Causes of fire and Effects of fire, General Requirements of Fire Resisting Building as per IS: 1642:1989 and NBC 2005, Characteristics of Fire Resisting Materials , Maximum Travel Distance, Fire Fighting Installations for Horizontal Exit, Roof Exit / Fire Lifts, External Stairs.

UNIT V MISCELLANEOUS SERVICES AND GREEN BUILDINGS PROVISIONS

9

Plan for Rain Water Harvesting in the New Buildings, Concept of GREEN Buildings, Components of Green Building, Components of Grey Water System, Management of Grey Water System and Distribution Pattern, Solar Power System

Total Periods: 45

OUTCOMES:

On the successful completion of this course, the students will be able:

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- CO1** To acquire different learning outcomes in managing building services provisions in big construction sites.
- CO2** Synchronize the construction activities with installation of building services engineering.
- CO3** Select the suitable electrical as well mechanical services for particular requirements of buildings.
- CO4** To acquire knowledge about fire protection.
- CO5** Ensure green building applications to the new constructions.

TEXT BOOKS:

1. R. Udaykumar ; A text book on Building Services;Eswar Press, Chennai
2. S. M. Patil ; Building Services ; Seema Publication, Mumbai Revised edition

REFERENCE BOOKS:

1. Bureau of Indian Standards; National Building Code of India – 2005;BIS, New Delhi
2. Dr. B. C. Punmia ; Building Construction ; Laxmi Publications (P) Ltd., New Delhi
3. P. C. Varghese ; Building Construction ; PHI Learning (P) Ltd., New Delhi
4. P. S. Gahlot ; Building repair and Maintenance Management ; CBS Publishers & Distribution(P) Ltd.
5. IS: 1642:1989, Code of Practice for Fire Safety of Building
6. NBC 2016, National Building Code of India

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	2	2	2	2
CO2	2	2	3	2	3	2	2	3	2
CO3	2	3	1	2	3	2	2	3	2
CO4	3	1	1	2	3	2	2	3	2
CO5	3	2	3	2	3	2	2	3	2
Avg	3	2	3	2	3	2	2	3	2

*** 3- High, 2- Medium, 1- Low**

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BM25103

MAINTENANCE ENGINEERING & MANAGEMENT

L T P C

4 1 0 4

OBJECTIVES:

- To learn about the maintenance objectives, functions, systems & overhaul & repair. To acquire knowledge in inspection decisions and NUCREC method.

UNIT I INTRODUCTION

9

Objectives and Functions of maintenance. Factors influencing plant availability, Maintenance control, Maintenance Strategies, Organization for Maintenance. Failure Statistics: Breakdown time distributions, Running-in failures, Time independent failures, Wear-out failures, Failure Probability, Survival Probability and age specific failure rates.

UNIT II OVERHAUL AND REPAIR

9

Meaning and difference, optimal overhaul / Repair / Replace maintenance policy for equipment subject to breakdown. Replacement Decisions: Deterministic and stochastic replacement situations, failure and preventive replacement, Optimal Interval between preventive replacement of equipment subject to breakdown, group replacement.

UNIT III MAINTENANCE SYSTEMS

9

Fixed time maintenance, Condition based Maintenance, Operate to failure, Opportunity Maintenance, Design out maintenance, Total Productive Maintenance. Maintenance Planning: Establishing maintenance plan and schedule, illustrative examples, Preventive Maintenance: Designing a Technically sound preventive maintenance program, failure data, FMECA, Maintenance to prevent failures, lubrication program development.

UNIT IV INSPECTION DECISIONS

9

Optimal Inspection frequency (for maximization of profit and minimization of downtime). Shut down planning using CPM & PERT.

UNIT V NUCREC METHOD OF PRIORITIZING MAINTENANCE WORK. SPARE PARTS MANAGEMENT

9

Classification of spares, traditional approach to spares inventory, MUSIC-3D Approach to spares inventory, optimum number of spares to satisfy given service level, simulation technique.

Total Periods: 45

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OUTCOMES:

At the end of the course the students shall have the abilities to:

- CO1** Explain maintenance objectives and functions, factors influencing Plant Availability, Need for maintenance plan and organization, Functions of maintenance control and determine Failure probability, Survival probability and Age specific failure rates of equipments and components.
- CO2** Determine the optimal overhaul/repair/replacement maintenance policy for an equipment subject to breakdown and optimal interval between preventive replacements for individual and group replacement of equipments.
- CO3** Explain different maintenance systems and the steps involved in establishing a maintenance plan and designing a technically sound preventive maintenance and lubrication program.
- CO4** Determine the optimal inspection frequency for maximization of profit and minimization of down time and the critical path using CPM and PERT.
- CO5** Explain the NUCREC method of prioritizing maintenance work, classification of spares and the costs associated with spares inventory, perform EOQ computations, explain MUSIC - 3D approach to spares management, determine the optimal number of spares to satisfy given service level and apply simulation technique for spares inventory.

TEXT BOOKS:

1. A KELLY AND M J HARRIS, “Management of Industrial Maintenance”, Butterworth’s Co, Ltd.

REFERENCE BOOKS:

1. AKS JARDINE “Maintenance, Replacement and Reliability” Pitman publishing Co.
2. A KELLY, “Maintenance planning and control”, Butterworth Co, Ltd.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	3	3	2	2	2	3	2	2
CO2	2	3	3	2	3	3	2	2	1
CO3	3	3	3	2	3	3	2	3	1

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CO4	3	3	3	2	2	2	3	2	2
CO5	3	3	3	2	3	2	2	2	1
Avg	3	3	3	2	3	2	2	2	1

*** 3- High, 2- Medium, 1- Low**

BM25104

ADVANCED CONCRETE TECHNOLOGY

L T P C

3 0 0 3

OBJECTIVE:

- To study the properties of concrete making materials, tests, mix design, special concretes, and various methods for making concrete.

UNIT I CONCRETE MAKING MATERIALS

9

Aggregates: Classification, IS specifications, Properties, Grading, Methods of combining, Testing - Cement: Grade, Chemical composition, tests - Hydration of cement - Structure of hydrated cement - Special cements - Water - Chemical admixture - Mineral admixture.

UNIT II MIX DESIGN

9

Principles of concrete mix design - Methods of concrete mix design: IS method, ACI method, DOE method - Mix design for special concretes - Statistical quality control - Sampling and acceptance criteria as per IS 456 - 2000.

UNIT III CONCRETING METHODS

9

Extreme weather concreting - Vacuum dewatering - Underwater Concreting - 3D printing - Curing methods - Maturity of concrete.

UNIT IV SPECIAL CONCRETES

9

Light weight concrete - Fiber reinforced concrete - Polymer concrete - High performance concrete - Self-compacting concrete - Geopolymer concrete - Waste material-based concrete - Ready mixed concrete - Roller compacted concrete.

UNIT V TESTS ON CONCRETE

9

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Hardened concrete: Strength, Elastic properties, Creep and shrinkage - Durability of concrete - Permeability - Chemical attack - Acid attack - Frost damage - Alkali silica reaction - Corrosion tests - Non-destructive testing techniques - Microstructure of concrete.

Total Periods: 45

OUTCOMES:

On completion of the course, the student is expected to be able to:

- CO1** Explain the properties of the constituent materials of concrete
- CO2** Understand the factors influencing concrete mix and apply the guidelines to do mix designs for concrete by various methods
- CO3** Explore the various methods of concreting and curing
- CO4** Define special concretes and their applications for practical purpose
- CO5** Study the behavior of concrete at its hardened state, describe and carry out tests relevant to the use of concrete on site

REFERENCES:

1. Gambhir M. L. "Concrete Technology", Fifth Edition, McGraw Hill Education, 2017.
2. Gupta B. L. and Amit Gupta, "Concrete Technology", Jain Book Agency, 2010.
3. Neville A. M., "Properties of Concrete", Prentice Hall, London, 2019.
4. Shetty M. S., "Concrete Technology", Revised Edition, S. Chand and Company Ltd., Delhi, 2019.
5. Job Thomas, "Concrete Technology", Cengage Learning India Private Ltd., New Delhi, 2015.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	3	3	2	2	2	2	1	1
CO2	2	3	3	2	3	3	2	2	2
CO3	3	3	3	2	3	3	3	1	2
CO4	3	3	3	2	2	2	2	2	1
CO5	3	3	3	2	3	2	2	2	2
Avg	3	3	3	2	3	2	2	2	2

*** 3- High, 2- Medium, 1- Low**

A) ADVANCED CONSTRUCTION ENGINEERING LABORATORY

OBJECTIVES:

This course provides a thorough knowledge of material selection through the material testing based on specification

LIST OF EXPERIMENTS:

1. Mix design and testing of concrete.
2. Effect of mineral and chemical admixtures in concrete at fresh and hardened state with relevance to workability, strength and durability.
3. Flow characteristics of self compacting concrete.
4. 4. Permeability tests on hardened concrete and RCPT
5. NDT on hardened concrete - UPV, rebound hammer and core test.
6. NDT on Welded steel connections (Demonstration)

Total Periods: 30

OUTCOMES:

On completion of this laboratory course

- Students will be able to test the concrete mixes designed as per IS, ACI and BS methods.
- Students will also be able to know various tests on hardened concrete.

(B) ADVANCED COMPUTING TECHNIQUES LABORATORY

OBJECTIVES:

- This course gives an exposure to students in utilizing the sophisticated spread sheets programs, estimation software and other package programs.

LIST OF EXPERIMENTS:

1. Quantity takeoff, Preparation and delivery of the bid or proposal of an engineering construction project.
2. Design of a simple equipment information system for a construction project.
2. Scheduling of a small construction project using Primavera scheduling systems including reports and tracking.

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3. Scheduling of a small construction project using tools like MS project scheduling systems including reports and tracking.
4. Simulation models for project risk analysis.

Total Periods: 30

OUTCOME:

- On completion of this laboratory course the students will be able to do the scheduling of constructions projects using tools primavera and MS projects.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	3	3	3	2	2	1	2
CO2	3	1	3	3	3	2	3	1	2
CO3	2	1	3	3	3	2	3	1	2
CO4	2	2	3	3	3	2	3	2	2
CO5	2	1	3	3	3	2	2	2	2
Avg	2	2	3	3	3	2	3	1	2

*** 3- High, 2- Medium, 1- Low**

BM25201

BUILDING REPAIR AND MAINTENANCE

L T P C

4 1 0 4

OBJECTIVES:

- To know about principles of construction and detection of defects, and advice on the course of action to be taken: planning.
- General knowledge of the life serviceability and strength of building materials in common use for the purpose of maintenance.
- The preparation of schedules of dilapidation and bill of quantity with specifications for repairs. Methods of measurement, recording and pricing of work.
- Building survey for assessment of damage due to fire, explosion, earthquake or any other peril for insurance purpose and preparation of estimate for insurance claim.
- Maintenance of plants providing services and refurbishment cost in use and prediction of performance in building.

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UNIT I INTRODUCTION

12

General - Quality of concrete - Damages in RC structures - Deterioration of structures- Damage assessment of structures - Repairs and Rehabilitation - Repair management - Maintenance of structures

UNIT II CRACKS & CORROSION DETERIORATION OF CONCRETE STRUCTURES

12

Wall cracks - Causes of concrete cracking- Types of cracks-Aggressive Deteriorating Chemical Agents .Corrosion of reinforcing bar -Carbonation – Chlorides- Sulphate attack - Alkali Silica Reaction- Causes and stages of disintegration. Effects of corrosion on reinforced concrete- Modeling of corrosion deterioration - Chemical Process of Corrosion- Causes of corrosion - Experimental investigation - Corrosion control.

UNIT III DAMAGE ASSESSMENT OF STRUCTURES

12

Process of concrete manufacture - Preparation of concrete and their control- Factors adversely affecting the durability of concrete -Testing of concrete- Damage assessment procedure- Non-destructive testing - methods of survey- Methods and inspection techniques.

UNIT IV REPAIR METHODS

12

Repair , Rehabilitation, Retrofitting- Repair materials- Factors affecting the selection of a repair material- Essential parameters for repair materials -Classification of repair materials- Patch repairing- Polyester resins -Acrylic concrete and mortar - Quick setting compounds -Bituminous materials-Ferro cement - SIFLON & SIMCON -Grouts –Shotcrete- Bonding agents -Polymer Latex Emulsion - Epoxy Latex - Epoxy Bonding Agents - Surface Coatings & Sealants.

UNIT V STRENGTHENING OF RC MEMBERS

12

Introduction -Crack injection repair to concrete structures- Jacketing - Plate bonding - Strengthening of foundation -Techniques to restore original strength - Stitching - Repair procedure for corrosion damaged elements - Treatment of distressed floor in Toilets / kitchen - Strengthening solution using FRP Plates.

Total Periods: 60

OUTCOMES:

At the end of this course, the student will able to

CO1 To understand the terms used in repair and rehabilitation of structures.

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- CO2** To learn about the causes, types and control of cracks in concrete structures and also to know about causes, effect and control of corrosion of reinforcement in concrete.
- CO3** To understand the composition, properties and tests on concrete and the various damage assessment procedure to assess the quality of concrete structure.
- CO4** To know about the different repair methods used in structures.
- CO5** To learn about the strengthening techniques for RC elements using different methods.

REFERENCE BOOKS:

1. B.G. Blake, Building Repairs, B.T. Batsford Press (1999) U.K.
2. 2) Lan A. Melvice, Repairs and Maintenance of Houses, Estate Gazette (1999)
3. 3) R.N. Raikar, Learning from Failures, Dhanpatrai & Sons (2008), New Delhi
4. 4) Malcolm Hollis, Surveying for Dilapidation, Estate Gazette (1999).
5. 5) Jagadish, K.S, Reddy, B.V. Venkatarama & Rao, K.S. Nanjunda, Alternative Building Materials and Technologies, New Age Publisher (2007), New Delhi.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	2
CO2	1	1	3	3	1	2	2	2	1
CO3	1	1	3	3	1	2	2	3	1
CO4	1	1	3	3	1	2	3	2	2
CO5	1	1	3	3	1	2	2	2	1
Avg	1	1	3	3	1	2	2	2	1

*** 3- High, 2- Medium, 1- Low**

OBJECTIVES:

- Principles of various NDT Techniques. The equipment required for the NDT and the procedure followed in NDT techniques. Applications of NDT and recent trends in NDT.

UNIT 1 VISUAL INSPECTION & EDDY CURRENT TESTING**9**

Scope and advantages of NDT, comparison of NDT with DT, classifications of NDT Visual Inspection Equipment – Magnifying Glass Magnifying Mirror, Microscope Bore scope, endoscopes or endoprobes flexible fiber optic bore scope, Video Image scope, Eddy Current Testing – Principle, Advantages , Disadvantages factors affecting Eddy Current Response- Material Conductivity Permeability – Frequency – Geometry – Proximity (Lift off) – Typical Applications, limitations, Types of Probes.

UNIT 2 LIQUID PENETRANT TESTING**9**

Liquid penetration testing – Introduction, Principle, Equipment, Procedures, and Characteristics of penetrants – developers – Evaluation – hazards Precautions, advantages, limitations and applications.

UNIT 3 MAGNETIC PARTICLE TESTING**9**

Principle of Magnetic Particle Testing – different methods to generate magnetic fields – Magnetic Particle Testing Equipment – Magnetic Particle Testing Procedures method of De- Magnetization – Magnetic Particle Medium – Evaluation of indications and acceptance standards – magnetic particle test-applications, advantages and limitations.

UNIT 4 RADIOGRAPHIC TESTING**9**

X-ray radiography principle, equipment & methodology – Type of Industrial radiation sources and application – Radiographic exposure factors and technique – GAMA Ray and x-Ray Equipment – Radiographic Procedure – Radiograph Interpretation, Radiography Image Quality Indicators – Radiographic Techniques – Film Processing – Methods of viewing Radiographs- Radiographic Testing Procedures for welds. Precautions against radiation hazards.

UNIT 5 ULTRASONIC TESTING**9**

Introduction, Principle of operation – Type of ultrasonic propagation – Ultrasonic Probes. Types of Transducers – Ultrasonic Testing Techniques. Method for Evaluating Discontinuities – Ultrasonic Testing Procedures for different component – applications, advantages and limitations, Documentation.

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Application in inspection of castings, Forgings, Extruded steel parts, bars, pipes, rails and dimensions measurements.

Total Periods: 45

OUTCOMES:

At the end of the course the student will be able to understand

- CO1** To know about principles of various NDT Techniques
- CO2** To learn about the equipment required for the NDT
- CO3** To acquire knowledge in the procedure followed in NDT techniques.
- CO4** To learn about applications of NDT
- CO5** To know about recent techniques in NDT testing.

TEXTBOOKS:

1. J Prasad, C G K Nair, “Non-Destructive Testing and Evaluation of Materials”, Tata McGraw Hill Education Private Limited.

REFERENCES:

1. American Metals Society, “Non Destructive Examination and Quality Control”, Metals Handbook, Vol.17, 9th Ed, Metals Park, OH, 1989.
2. Bray, Don.E and Stanley, Roderic.K, “Non Destructive Evaluation: A Tool in Design, Manufacturing, and Service. Revised”, CRC Press New York, Edition 1997.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	2	3	3	3	2	3	3
CO2	2	2	2	2	2	2	3	3	3
CO3	2	2	3	3	3	3	2	3	3
CO4	2	2	2	3	3	3	2	3	3
CO5	2	2	2	3	3	3	3	3	3
Avg	2	2	2	3	3	3	2	3	3

*** 3- High, 2- Medium, 1- Low**

OBJECTIVES:

- To study and understand the latest construction techniques applied to engineering construction for sub structure, super structure, and special structures.
- To gain the knowledge about the rehabilitation and strengthening techniques.
- To learn about the various demolition techniques.

UNIT I SUB STRUCTURE CONSTRUCTION	9
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Box jacking - Pipe jacking - Under water construction of diaphragm walls and basement - Tunneling techniques - Piling techniques - Driving well and caisson - sinking cofferdam - cable anchoring and grouting - Driving diaphragm walls, Sheet piles - Laying operations for built up offshore system - Shoring for deep cutting - Large reservoir construction - well points - Dewatering for underground open excavation.

UNIT II SUPER STRUCTURE CONSTRUCTION FOR BUILDINGS	9
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Vacuum dewatering of concrete flooring – Concrete paving technology – Techniques of construction for continuous concreting operation in tall buildings of various shapes and varying sections – Erection techniques of tall structures, Large span structures – launching techniques for heavy decks – in-situ prestressing in high rise structures, Post tensioning of slab- Aerial transporting – Handling and erecting lightweight components on tall structures.

UNIT III CONSTRUCTION OF SPECIAL STRUCTURES	9
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Erection of lattice towers - Rigging of transmission line structures – Construction sequence in cooling towers, Silos, chimney, and sky scrapers - Bow string bridges, Cable stayed bridges – Launching and pushing of box decks – Construction of jetties and break water structures – Construction sequence and methods in domes – Support structure for heavy equipment and machinery in heavy industries – Erection of articulated structures and space decks.

UNIT IV REHABILITATION AND STRENGTHENING TECHNIQUES	9
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Seismic retrofitting - Strengthening of beams, columns, slab and masonry wall - Protection methods of structures, Mud jacking and grouting for foundation – Micro piling and underpinning for strengthening floor and shallow profile - Sub grade water proofing, Soil Stabilization techniques.

UNIT V DEMOLITION

9

Demolition Techniques, Demolition by Machines, Demolition by Explosives, Advanced techniques using Robotic Machines, Demolition Sequence, Dismantling Techniques, Safety precaution in Demolition and Dismantling.

Total Periods: 45

OUTCOMES:

- On completion of the course, the student is expected to be able to

- CO1** Classify the modern construction techniques used in the sub structure construction.
- CO2** Demonstrate knowledge and understanding of the principles and concepts relevant to super structure construction for buildings
- CO3** Summarize the concepts used in the construction of special structures
- CO4** Distinguish Various strengthening and repair methods for different cases.
- CO5** Identify the suitable demolition technique for demolishing a building.

REFERENCES:

1. Robertwade Brown, Practical foundation engineering hand book, McGraw Hill Publications, 2000.
2. Concrete Structures: Repair, Rehabilitation and Strengthening, Dr. Mohamed A. El-Reedy, 2020
3. Patrick Powers. J., Construction Dewatering: New Methods and Applications, John Wiley & Sons, 1992.
4. Peter H.Emmons, “Concrete repair and maintenance illustrated”, Galgotia Publications Pvt. Ltd., 2001.Press, 2008.
5. Sankar, S.K. and Saraswati, S., Construction Technology, Oxford University, New Delhi, 2008.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	3
CO2	1	1	3	3	1	2	2	2	2
CO3	1	1	3	3	1	2	3	2	3
CO4	1	1	3	3	1	2	3	2	3

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CO5	1	1	3	3	1	2	2	2	2
Avg	1	1	3	3	1	2	3	2	3

*** 3- High, 2- Medium, 1- Low**

BM25204 QUALITY CONTROL & ASSURANCE IN CONSTRUCTION L T P C
4 1 0 4

OBJECTIVES:

- To study the concepts of and control techniques in construction.
- To study the of design philosophy, design of special elements, flat slabs and yield line based design, and ductile detailing quality assurance.

UNIT I QUALITY MANAGEMENT 12

Introduction – Definitions and objectives – Factor influencing construction quality - Responsibilities and authority - Quality plan - Quality Management Guidelines – Quality circles.

UNIT II QUALITY SYSTEMS 12

Project Introduction - Quality system standard – ISO 9000 family of standards – Requirements – Preparing Quality System Documents – Quality related training – Implementing a Quality system – Third party Certification.

UNIT III QUALITY PLANNING 12

Techniques of Quality Policy, Objectives and methods in Construction industry - Consumers satisfaction, Ergonomics - Time of Completion - Statistical tolerance – Taguchi's concept of quality – Codes and Standards – Documents – Contract and construction programming – Inspection procedures - Processes and products – Total QA / QC programme and cost implication.

UNIT IV QUALITY ASSURANCE AND CONTROL 12

Objectives - Regularity agent, owner, design, contract and construction oriented objectives, methods - Techniques and needs of QA/QC - Different aspects of quality - Appraisals, Factors influencing construction quality - Critical, major failure aspects and failure mode analysis, -Stability methods and tools, optimum design - Reliability testing, reliability coefficient and reliability prediction.

UNIT V QUALITY IMPROVEMENT TECHNIQUES 12

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Selection of new materials - Influence of drawings, detailing, specification, standardization - Bid preparation - Construction activity, environmental safety, social and environmental factors - Natural causes and speed of construction - Life cycle costing - Value engineering and value analysis.

Total Periods: 60

OUTCOMES:

- CO1** Acquire knowledge on quality systems and quality documents
- CO2** Understand the elements of quality planning and the implication of quality planning
- CO3** Analyze the various factors influencing the quality of construction
- CO4** Dissect the various failure mode analysis
- CO5** Design the safety programmes and contractual obligations.

REFERENCE BOOKS:

1. James, J.O' Brian, "Construction Inspection Handbook – Quality Assurance and Quality Control", Van Nostrand, New York, 2009.
2. Kwaku, A., Tena, Jose, M. Guevara, "Fundamentals of Construction Management and Organisation", Reston Publishing Co., Inc., Virginia, 2005.
3. Juran Frank, J.M. and Gryna, F.M., "Quality Planning and Analysis", Tata McGraw Hill, 1993
4. Hutchins.G, ISO 9000, Viva Books, New Delhi, 2002
5. Clarkson H. Oglesby, "Productivity Improvement in Construction", McGraw-Hill, 2009.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	2	2	2	1	3	2	2
CO2	3	3	3	2	2	1	2	1	1
CO3	3	2	3	3	2	1	1	3	2
CO4	3	2	3	3	2	1	3	2	2
CO5	3	2	2	2	3	2	1	2	2
Avg	3	2	3	2	2	1	2	2	2

*** 3- High, 2- Medium, 1- Low**

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BM2521

ADVANCED MATERIAL TESTING LABORATORY

L T P C

0 0 4 2

OBJECTIVES:

- Familiarize the students with physical, chemical and mechanical properties of cement concrete constituents and understand the mix design of high grade concrete.
- Analyze the stress-strain curve of high strength concrete and develop correlation between cube and cylinder of high strength concrete.
- Determine the mechanical properties of high strength concrete and knowledge on cyclic loading on steel.
- To study the behavior of self-compacting concrete and existing RC structures reinforcement details and corrosion levels.

LIST OF EXPERIMENTS:

1. Determination on Test on Cements.
2. Determination on Test on Fine & Coarse Aggregates.
3. Mix Design of High Strength Concrete.
4. Correlation between Cube & Cylinder of High Strength Concrete.
5. Split Tensile Strength of High Strength Concrete.
6. Modulus of rupture of Concrete Beam.
7. Stress Strain Curve of High Strength Concrete.
8. Flow Test on Self-Compacting Concrete.
 - a) Slump flow test
 - b) L-Box Test
 - c) V-Funnel Test
 - d) U-Box Test
9. Determine the rheological properties of cements and mortars using Marsh Cone Test
10. Determine the location of Reinforcement details in existing RC Structures.
11. Determine the corrosion levels in existing RC Structures using Profometer.

Total Periods: 60

OUTCOMES:

At the end of the course, students will be able to

- CO1** Design high grade concrete and identify, carry out laboratory tests related to the use of concrete on site
- CO2** Develop correlation between cube and cylinder of high strength concrete and analyze the stress-strain curve.

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- CO3** Interpret the mechanical properties of high strength concrete and examine the effect of cyclic loading on steel
- CO4** Assess the quality of existing concrete members by Non-Destructive testing methods and study the behavior of beams under flexure.
- CO5** Analyze the behavior of Self-Compacting Concrete and understanding reinforcement details and corrosion levels in existing RC structures.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	3	3	3	2	2	1	2
CO2	3	1	3	3	3	2	3	1	2
CO3	2	1	3	3	3	2	3	1	2
CO4	2	2	3	3	3	2	3	2	2
CO5	2	1	3	3	3	2	2	2	2
Avg	2	2	3	3	3	2	3	1	2

*** 3- High, 2- Medium, 1- Low**

BM2522

SEMINAR-I

L T P C

0 0 2 1

GUIDELINES:

The students will work under the guidance of a group of staff members. The students will be asked to talk on any topic of their choice related to Sustainability in Building Maintenance and to engage in dialogue with the audience. A brief copy of their talk also should be submitted. Similarly, the students will have to present a seminar of not less than fifteen minutes and not more than thirty minutes on the technical topic. They will also answer the queries on the topic. The students as audience also should interact. Evaluation will be based on the technical presentation and the report and also on the interaction during the seminar.

Total Periods: 30

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OUTCOMES:

- On completion of the course, the student is expected to be able to

- CO1** Identify latest developments in the field of Building maintenance
CO2 Identify latest developments in the field of Building Repair materials
CO3 Presentation Skills and ability to answer the queries during Interaction
CO4 Acquire technical writing abilities for seminars, conferences and journal publications
CO5 Use modern tools to present the technical details

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	3	3	2	3
CO2	2	3	3	3	2	3	3	2	3
CO3	2	3	3	3	2	3	3	2	3
CO4	3	3	3	3	2	3	3	2	3
CO5	3	3	3	3	2	3	3	2	3
Avg	3	3	3	3	2	3	3	2	3

*** 3- High, 2- Medium, 1- Low**

BM2523

PRACTICAL TRAINING –I

L T P C

0 0 2 2

GUIDELINES:

The students individually undertake training in reputed engineering companies doing construction during the summer vacation for a specified duration of four weeks. At the end of training, a detailed report on the work done should be submitted within ten days from the commencement of the semester. The students will be evaluated through a viva-voce examination by a team of internal staff.

OUTCOMES:

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On completion of the course, the student is expected to be able to

- CO1** Describe the real-time challenges and procedures in the construction industry
- CO2** Realize the various functions of construction activities
- CO3** Develop skills in facing and solving the problems experiencing in the Building Maintenance field
- CO4** Presentation of work carried out in Practical Training
- CO5** Report Preparation

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	2	3	2	2	3	2	2
CO2	2	2	2	3	2	2	3	2	2
CO3	3	2	2	3	2	2	3	2	2
CO4	3	3	3	3	3	3	3	3	3
CO5	3	3	3	3	3	3	3	3	3
Avg	3	2	2	3	2	2	3	2	2

*** 3- High, 2- Medium, 1- Low**

BM2531

PROJECT WORK – I

L T P C

0 0 12 6

GUIDELINES:

The student individually works on a specific topic approved by the head of the division under the guidance of a faculty member who is familiar in this area of interest. The student can select any topic which is relevant to the area of building maintenance. The topic may be theoretical or case studies. At the end of the semester, a detailed report on the work done should be submitted which contains clear definition of the identified problem, detailed literature review related to the area of work and methodology for carrying out the work. The students will be evaluated through a viva-voce examination by a panel of examiners including one external examiner.

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Total Periods: 180

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	3	3	3	3	3	3	3
CO2	3	3	3	3	3	3	3	3	3
CO3	3	3	3	3	3	3	3	3	3
CO4	3	3	3	3	3	3	3	3	3
CO5	3	3	3	3	3	3	3	3	3
Avg	3	3	3	3	3	3	3	3	3

*** 3- High, 2- Medium, 1- Low**

OUTCOMES:

On completion of the course, the student is expected to be able to

- CO1** Apply the knowledge gained from theoretical and practical courses by providing solutions
- CO2** Summarize the importance of literature review
- CO3** Identify the problem
- CO4** Solve the identified problem based on the formulated methodology
- CO5** Interpret and present the findings of the work conducted

BM2532

SEMINAR-II

L T P C

0 0 2 1

OBJECTIVE:

- To work on a specific technical topic in Sustainability in Building Maintenance in order to acquire the skills of oral presentation and to acquire technical writing abilities for seminars and conferences.

SYLLABUS:

The students will work for two hours per week guided by a group of staff members. They will be asked to talk on any topic of their choice related to Sustainability in Building Maintenance and to engage

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in dialogue with the audience. A brief copy of their talk also should be submitted. Similarly, the students will have to present a seminar of not less than fifteen minutes and not more than thirty minutes on the technical topic. They will also answer the queries on the topic. The students as audience also should interact. Evaluation will be based on the technical presentation and the report and also on the interaction during the seminar.

Total Periods: 30

OUTCOMES:

- CO1** Identify latest developments in the field of Building Maintenance
- CO2** Identify latest developments in the field of Building repair materials and techniques
- CO3** Presentation Skills and ability to answer the queries during Interaction
- CO4** Acquire technical writing abilities for seminars, conferences and journal publications
- CO5** Use modern tools to present the technical details

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	3	3	2	3
CO2	2	3	3	3	2	3	3	2	3
CO3	2	3	3	3	2	3	3	2	3
CO4	3	3	3	3	2	3	3	2	3
CO5	3	3	3	3	2	3	3	2	3
Avg	3	3	3	3	2	3	3	2	3

*** 3- High, 2- Medium, 1- Low**

BM2533

PRACTICAL TRAINING- II

L T P C

0 0 2 2

SYLLABUS:

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The students individually undertake training in reputed engineering companies doing construction during the summer vacation for a specified duration of four weeks. At the end of training, a detailed report on the work done should be submitted within ten days from the commencement of the semester. The students will be evaluated through a viva-voce examination by a team of internal staff.

OUTCOMES:

On completion of the course, the student is expected to be able to

- CO1** Describe the real-time challenges and procedures in the construction industry
- CO2** Realize the various functions of construction activities
- CO3** Develop skills in facing and solving the problems experiencing in the Building Maintenance field
- CO4** Presentation of work carried out in Practical Training
- CO5** Report Preparation

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	2	3	2	2	3	2	2
CO2	2	2	2	3	2	2	3	2	2
CO3	3	2	2	3	2	2	3	2	2
CO4	3	3	3	3	3	3	3	3	3
CO5	3	3	3	3	3	3	3	3	3
Avg	3	2	2	3	2	2	3	2	2

*** 3- High, 2- Medium, 1- Low**

BM2541

PROJECT WORK –II

L T P C

0 0 24 12

Objectives:

- To solve the identified problem based on the formulated methodology.

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- To develop skills to analyze the research problem.
- To develop skills to discuss the test results, and make conclusions.

SYLLABUS:

The student should continue the phase I work on the selected topic as per the formulated methodology/ Undergo internship. At the end of the semester, after completing the work to the satisfaction of the supervisor and review committee, a detailed report should be prepared and submitted to the head of the department. The students will be evaluated based on the report and the viva-voce examination by a panel of examiners including one external examiner.

Total Periods: 360

OUTCOMES:

On completion of the course, the student will be able to

- CO1** Discover the potential research areas
- CO2** Apply the knowledge gained from theoretical and practical courses to be creative, well planned, organized and coordinated
- CO3** Identify the problem.
- CO4** solve the identified problem based on the formulated methodology
- CO5** Interpret and present the findings of the work conducted

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	3	3	3	3	3	3	3
CO2	3	2	2	3	3	3	3	3	3
CO3	2	2	2	3	3	3	3	3	3
CO4	3	2	3	3	3	3	3	3	3
CO5	3	2	3	3	3	3	3	3	3
Avg	3	2	3	3	3	3	3	3	3

*** 3- High, 2- Medium, 1- Low**

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TIRUVANNAMALAI**

BM2501

SUSTAINABLE CONSTRUCTION

L T P C

3 0 0 3

OBJECTIVE:

- To impart knowledge about sustainable construction and to understand the concepts of sustainable materials, energy calculations, green buildings and environmental effects.

UNIT I INTRODUCTION

9

Introduction and definition of Sustainability - Carbon cycle - role of construction material: concrete and steel, etc. - CO₂ contribution from cement and other construction materials.

UNIT II MATERIALS USED IN SUSTAINABLE CONSTRUCTION

9

Construction materials and indoor air quality - No/Low cement concrete - Recycled and manufactured aggregate - Role of QC and durability - Life cycle and sustainability.

UNIT III ENERGY CALCULATIONS

9

Components of embodied energy - calculation of embodied energy for construction materials - Energy concept and primary energy - Embodied energy via-a-vis operational energy in conditioned building - Life Cycle energy use

UNIT IV GREEN BUILDINGS

9

Control of energy use in building - ECBC code, codes in neighboring tropical countries - OTTV concepts and calculations – Features of LEED and TERI – Griha ratings - Role of insulation and thermal properties of construction materials - influence of moisture content and modeling Performance ratings of green buildings - Zero energy building.

UNIT V ENVIRONMENTAL EFFECTS

9

Non-renewable sources of energy and Environmental aspects – energy norm, coal, oil, natural gas - Nuclear energy - Global temperature, Green house effects, global warming - Acid rain: Causes, effects and control methods - Regional impacts of temperature change.

Total Periods: 45

OUTCOMES:

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On completion of the course, the student is expected to be able to

- CO1** Summarize the various sustainable materials used in construction.
- CO2** Explain the method of estimating the amount of energy required for building
- CO3** Interpret the features of LEED, TERI and GRIHA ratings of buildings
- CO4** Relate the concept and performance of zero energy buildings
- CO5** Select less carbon emission materials for construction.

REFERENCES:

1. Charles J Kibert, Sustainable Construction : Green Building Design & Delivery, 4th Edition , Wiley Publishers 2016.
2. Steve Goodhew, Sustainable Construction Process, Wiley Blackwell,UK, 2016
3. Craig A. Langston & Grace K.C. Ding, Sustainable Practices in the Built Environment, Butterworth Heinemann Publishers, 2011
4. William P Spence, Construction Materials, Methods & Techniques (3e), Yesdee Publication Pvt. Ltd, 2012.
5. New Building Materials and Construction World magazine.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	3	3	3	2	1	2	1
CO2	2	2	2	3	3	2	2	2	2
CO3	2	1	2	3	3	2	1	3	1
CO4	2	1	2	3	3	2	2	3	2
CO5	1	1	2	2	3	2	2	3	2
Avg	2	1	2	3	3	2	2	3	2

*** 3- High, 2- Medium, 1- Low**

BM2502

CONSTRUCTION EQUIPMENT AND MANAGEMENT

L T P C

3 0 0 3

OBJECTIVE:

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- To study and understand the various types of equipment's used for earthwork, tunneling, drilling, blasting, dewatering, material handling conveyors and its applications in construction projects.

UNIT I CONSTRUCTION EQUIPMENT SELECTION

9

Identification – Planning of equipment – Selection of Equipment - Equipment Management in Projects - Maintenance Management – Equipment cost – Operating cost – Cost Control of Equipment - Depreciation Analysis – Replacement of Equipment- Replacement Analysis - Safety Management.

UNIT II EQUIPMENT FOR EARTHWORK

9

Fundamentals of Earth Work Operations - Earth Moving Operations - Types of Earth Work Equipment - Tractors, Motor Graders, Scrapers, Front end Waders – Dozer, Excavators, Rippers, Loaders, trucks and hauling equipment, Compacting Equipment, Finishing equipment.

UNIT III OTHER CONSTRUCTION EQUIPMENT

9

Equipment for Dredging, Trenching, Drag line and clamshells, Tunneling – Equipment for Drilling and Blasting - Pile driving Equipment - Erection Equipment - Crane, Mobile crane - Types of pumps used in Construction - Equipment for Dewatering and Grouting – Equipment for Demolition.

UNIT IV ASPHALT AND CONCRETING EQUIPMENT

9

Aggregate production- Different Crushers – Feeders - Screening Equipment - Handling Equipment - Batching and Mixing Equipment - Pumping Equipment – Ready mix concrete equipment, Concrete pouring equipment. Asphalt Plant, Asphalt Pavers, Asphalt compacting Equipment.

UNIT V MATERIALS HANDLING EQUIPMENT

9

Forklifts and related equipment - Portable Material Bins – Material Handling Conveyors – Material Handling Cranes- Industrial Trucks.

Total Periods: 45

OUTCOME:

On completion of the course, the student is expected to be able to

- CO1** Develop knowledge on the planning of equipment and selection of equipment
- CO2** Explain the knowledge on fundamentals of earth work operations, earth moving operations and types of earth work equipment

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- CO3** Develop the knowledge on special construction equipment's
- CO4** Apply the knowledge on asphalt and concrete plants
- CO5** Apply the knowledge and select the proper materials handling equipment

REFERENCES:

1. Peurifoy, R.L., Schexnayder, C. and AviadShapira., Construction Planning, Equipment and Methods, McGraw Hill, Singapore, 2010.
2. Granberg G.,Popescu M Construction Equipment and Management for Engineers Estimators and Owners, Taylor and Francis Publishers, New York, 2006
3. Deodhar, S.V. Construction Equipment and Job Planning, Khanna Publishers, New Delhi, 2001.
4. Arora S.P. and Bindra S.P., Building Construction, Planning Techniques and Method of Construction, DhanpatRai and Sons, 2010.
5. Sharma S.C. Construction Equipment and Management, Khanna Publishers, New Delhi, 2019.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	3	2	3	2	2	2	2
CO2	2	3	2	3	2	2	3	2	3
CO3	3	2	3	2	3	1	3	2	3
CO4	3	3	1	3	2	2	3	2	3
CO5	3	3	3	2	1	2	3	2	3
Avg	3	3	3	2	2	2	3	2	3

*** 3- High, 2- Medium, 1- Low**

BM2503

ENVIRONMENTAL IMPACT & RISK ASSESSMENT

L T P C

3 0 0 3

OBJECTIVES:

- To impart the knowledge and skills required for understanding the various impacts of infrastructure projects on the environment and expose the students to the various methodologies available to

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assess and predict the impacts, and to develop the skill to prepare Environmental Impact Assessment report.

UNIT I OVER VIEW

9

Sustainable Development challenges and need- Key approaches for Impact Assessment; Emerging Dimensions and future Directions-Strategic Environmental Assessment, Technological Assessment and Risk Assessment.

UNIT II ASSESSING ENVIRONMENTAL IMPACTS

9

Historical development, Legal and Regulatory aspects in India, Types and Objectives, Components, Process of E I A, Prediction and Assessment: tools, impact on air, water, soil & Noise, Role of Biodiversity impact Assessment, Identification, Prediction & Evaluation of Impacts on Biodiversity, Techniques of Biodiversity impact assessment, E I A Report Preparation.

UNIT III SOCIO- ECONOMIC IMPACT ASSESSMENT

9

Health Assessment: Impact of Environment on Health - Developing framework for Health impact analysis, tools, and techniques - Socio-Economic Impact Assessment.: Overview and Scope of Social Impact Assessment, S I A model and the planning process, Land acquisition –Legal aspects, Resettlement & Rehabilitation and Development.

UNIT IV ENVIRONMENTAL MANAGEMENT PLAN

9

Integrated Analysis of Environmental, Social, and Health Impacts - Challenges for Integrated Approach - Scope for Integrated approach in economic analysis - CBA, Social CBA, and Cost effectiveness Analysis - Analytic Hierarchy process-based Approach - Emerging Dimensions and Future Directions.

UNIT V CASE STUDIES

9

EIA for Mining, extraction of natural resources and power generation, Primary Processing and Material production, Material Processing, Manufacturing/Fabrication, Service Sectors, Physical Infrastructure including Environmental Services, Building/Construction Projects/Area Development Projects & townships.

OUTCOMES:

On the successful completion of the course, students will be able to:

- CO1** Understand the necessity to study the impacts that will be caused by projects or industries and the methods to assess these impacts Understand

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- CO2** Describe the legal requirements of environmental impact assessment for projects
- CO3** Prepare terms of reference for environmental impact and socio-economic impact for any developmental project
- CO4** Prepare environmental management plan and risk mitigation plan by considering environmental aspects, impacts and potential hazards respectively for any project
- CO5** Analyze the environmental impacts specified in the EIA & Report and suggest suitable mitigation measures for any developmental projects

REFERENCES:

1. Canter, L.W., "Environmental Impact Assessment", McGraw Hill, New York, 1996.
2. Anjaneyulu, Yerramilli, and Valli Manickam, "Environmental impact assessment methodologies", Hyderabad: BS Publications, 2007.
3. Lawrence, D.P., "Environmental Impact Assessment – Practical Solutions to recurrent problems", Wiley-Interscience, New Jersey, 2003.
4. Petts, J., "Handbook of Environmental Impact Assessment", Vol., I and II, Blackwell science, London, 1999.
5. World Bank – Source Book on Environmental Impact Assessment, 2010

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	2	2	2	1	3	2	3
CO2	3	3	3	2	2	1	3	3	2
CO3	3	2	3	2	3	1	3	2	3
CO4	3	2	3	2	3	1	3	2	2
CO5	3	2	2	2	3	1	3	3	3
Avg	3	2	3	2	3	1	3	2	3

*** 3- High, 2- Medium, 1- Low**

BA2404

VALUE ENGINEERING

L T P C

3 0 0 3

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OBJECTIVES:

- Estimation of project budget using capitalized income approach.
- Analyze a building using LCC methodology

UNIT – I INTRODUCTION

9

Introduction to Value engineering (VE), definition, objectives of value engineering, reasons for unnecessary costs, VE techniques and methodology, interface with the other programs.

UNIT – II PROJECT BUDGETING

9

Elements of the project budget, need for cost control, meaning of capitalization, capitalization process, and capitalized income approach to construction project budgeting.

UNIT – III LIFE CYCLE COST

9

Life cycle cost (LCC) and building costs, LCC technology and examples, LCC methodology, LCC formats and analysis and weighted evaluation – application of LCC to buildings.

UNIT – IV VALUE ENGINEERING

9

Value engineering and total project management, level of effort, team selection, value engineering job plan, and work plan phases.

UNIT – V PROJECT LEVEL FUNCTION

9

Classifying function, defining function, project level function system technique (fast) diagram, creativity and fixation, interpersonal skills, generation of ideas, brainstorming, rules for brainstorming, Delphi technique, application of Delphi technique to civil engineering projects.

Total Periods: 45

OUTCOMES:

At the end of the course, student will be able to

- CO1** Establish value engineering techniques and methodology
- CO2** Outline the capitalized income approach to project budgeting
- CO3** Apply Life Cycle Cost (LCC) analysis to buildings
- CO4** Draw value engineering job plan and work plan phases
- CO5** Examine the concept of Delphi techniques and rules for brainstorming

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REFERENCE BOOKS:

1. Tenah, K.A. (1985). "The Construction Management Process", Reston Publishing Company, Inc. Virginia
2. Dell’Isola, Alphonse (1997). “Value Engineering: Practical Applications.” R.S. Means Company, Inc: Kingston, MA.
3. Oberiender, G. D. (1993). “Project Management for Engineering and Construction”. McGraw-Hill, Inc.: New York.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	3	2	2	2	2	1	1
CO2	1	2	2	3	3	3	2	2	2
CO3	1	1	1	2	2	2	2	2	2
CO4	1	2	2	3	3	3	1	1	1
CO5	1	1	1	3	3	2	1	1	1
Avg	1	2	2	3	3	2	2	1	1

*** 3- High, 2- Medium, 1- Low**

BM2505 COMPUTER APPLICATION IN CONSTRUCTION MANAGEMENT L T P C
3 0 0 3

OBJECTIVE:

- To study and understand the optimization techniques, inventory models and scheduling techniques applied to construction engineering.

UNIT I INTRODUCTION

6

Overview of IT Applications in Construction – Construction process – Computerization in Construction – Developing application with database software.

UNIT II OPTIMIZATION TECHNIQUES

6

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Linear and Integer Programming - Branch and Bound Techniques – Application to Equipment Replacement, Material Transportation and Work Assignment Problems – Software applications.

UNIT III INVENTORY MODELS

6

Deterministic and Probabilistic Inventory Models - Case Study- Software applications.

UNIT IV SCHEDULING APPLICATIONS

6

PERT and CPM - Advanced planning and scheduling concepts –BIM based planning, scheduling, visualization and control of construction projects - BIM and AR/VR integration - Case Studies.

UNIT V INDUSTRIAL ENGINEERING PROBLEMS

6

Sequencing problems – Johnson’s procedure, ‘n’ jobs and ‘m’ machines, two jobs ‘m’ machine, and Graphical procedure - Simulation – Enterprises – Introduction to ERP and EIP systems.

Theory Periods: 30

LIST OF EXPERIMENTS:

1. Formulate a relation Database Management system for a simple construction project using MS Access.
2. Formulate a LPP, and solve using MS Excel Solver.
3. Prepare a sample inventory control technique for a simple construction project using MS Excel.
4. Schedule a simple construction project using CPM and PERT and perform schedule performance analysis over Microsoft project.
5. Using graphical method, perform sequencing operations for two jobs and ‘m’ machines.
6. Prepare a simple simulation experiment and solve using MS Excel.
7. Using BIM Tools, perform an automated Quantity Takeoff for a simple construction project.

Practical Periods: 30

Total Periods: 60

OUTCOMES:

On completion of the course, the student is expected to be able to

CO1 Use of software's in construction Industry.

CO2 Apply various optimization techniques.

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CO3 Apply Deterministic and Probabilistic Inventory Models.

CO4 Analyze the scheduling concepts.

CO5 Solve problems using simulation and ERP systems.

REFERENCES:

1. Billy E. Gillet., Introduction to Operations Research – A Computer Oriented Algorithmic Approach, McGraw Hill, 2008.
2. Feigenbaum, L., Construction Scheduling with Primavera Project Planner Prentice Hall Inc., 2002.
3. Ming Sun and Rob Howard, “Understanding I.T. in Construction, Spon Press, Taylor and Francis Group, 2004.
4. Paulson, B.R., Computer Applications in Construction, McGraw Hill, 1995.
5. Tarek Hegazy, Computer-Based Construction Project Management, Pearson New International Edition, 2013.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	3	1	2	1	2	1	1
CO2	3	1	1	2	2	3	3	2	2
CO3	2	1	1	3	2	3	3	2	2
CO4	2	1	1	3	3	3	3	2	2
CO5	3	2	1	3	2	2	3	1	2
Avg	3	1	1	2	2	2	3	2	2

*** 3- High, 2- Medium, 1- Low**

BM2506

DESIGN OF ENERGY EFFICIENT BUILDINGS

L T P C

3 0 0 3

OBJECTIVE:

- To understanding the concept of energy consumption in buildings and design a energy efficient building.

UNIT I INTRODUCTION

9

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Climate adapted and climate rejecting buildings – Heat Transfer – Measuring Conduction – Thermal Storage – Measurement of Radiation – The Greenhouse Effect – Convection – Measuring latent and sensible heat – Psychrometry Chart – Thermal Comfort – Microclimate, Site Planning and Development – Temperature – Humidity – Wind – Optimum Site Locations – Sun Path Diagrams – Sun Protection – Types of Shading Devices – Design responses to energy conservation strategies.

UNIT II PASSIVE SOLAR HEATING AND COOLING

9

General Principles of passive Solar Heating – Key Design Elements – Sunspace – Direct gain – Trombe Walls, Water Walls – Convective Air loops – Concepts – Case Studies – General Principles of Passive Cooling – Ventilation – Principles – Case studies – Courtyards – Roof Ponds – Cool Pools – Predicting ventilation in buildings – Window Ventilation Calculations – Room Organization Strategies for Cross and Stack Ventilation – Radiation – Evaporation and dehumidification – Wind Catchers – Mass Effect – Zoning – Load Control – Air Filtration and odor removal.

UNIT III DAYLIGHTING AND ELECTRICAL LIGHTING

9

Materials, components and details – Insulation – Optical materials – Radiant Barriers – Glazing materials – Glazing Spectral Response – Day lighting – Sources and concepts – Building Design Strategies – Case Studies – Daylight apertures – Light Shelves – Codal requirements – Day lighting design – Electric Lighting – Light Distribution – Electric Lighting control for day lighted buildings – Switching controls – Coefficient of utilization – Electric Task Lighting – Electric Light Zones – Power Adjustment Factors.

UNIT IV HEAT CONTROL AND VENTILATION

9

Hourly Solar radiation – Heat insulation – Terminology – Requirements – Heat transmission through building sections – Thermal performance of Building sections – Orientation of buildings – Building characteristics for various climates – Thermal Design of buildings – Influence of Design Parameters – Mechanical controls – Examples. Ventilation – Requirements – Minimum standards for ventilation – Ventilation Design – Energy Conservation in Ventilating systems – Design for Natural Ventilation – Calculation of probable indoor wind speed.

UNIT V DESIGN FOR CLIMATIC ZONES

9

Energy efficiency – An Overview of Design Concepts and Architectural Interventions – Embodied Energy – Low Embodied Energy Materials – Passive Downdraft Evaporative Cooling – Design of Energy Efficient Buildings for Various Zones – Cold and cloudy – Cold and sunny – Composite – Hot and dry – Moderate – Warm and humid – Case studies of residences, office buildings and other buildings in each zones – Commonly used software packages in energy efficient building analysis and design - Energy Audit – Certification.

OUTCOMES:

On completion of this course, the student is expected to be able to

- CO1** Explain environmental energy supplies on buildings
- CO2** Explain the passive solar heating, cooling system
- CO3** Discuss the various aspects of day-lighting and electrical lighting in a building
- CO4** Predict and design building ventilation and heat control for indoor comfort
- CO5** Design a building for climatic zone and apply simulation programs of buildings to perform energy calculations

REFERENCES:

1. Energy Conservation Building Code, Bureau of Energy Efficiency, New Delhi, 2018.
2. Brown, G.Z. and DeKay, M., Sun, Wind and Light - Architectural Design Strategies, John Wiley and Sons Inc, 3rd Edition, 2014.
3. Handbook on Functional Requirements of Buildings Part 1 to 4 SP : 41 (S and T) 1995
4. Residential Energy: Cost Savings and Comfort for Existing Buildings by John Krigger and Chris Dorsi, Published by Saturn Resource Management, 2013.
5. Majumdar, M (Ed), Energy - Efficient Buildings in India, Tata Energy Research Institute, Ministry of Non-Conventional Energy Sources, 2009.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	3	2	1	1	3	2	1
CO2	3	2	3	3	1	1	3	2	1
CO3	3	2	3	3	1	1	3	2	2
CO4	2	2	3	3	1	1	3	3	3
CO5	3	2	3	3	1	2	2	3	3
Avg	3	2	3	3	1	1	3	2	2

*** 3- High, 2- Medium, 1- Low**

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BM2507

SHORING, SCAFFOLDING & FORMWORK

L T P C

3 0 0 3

OBJECTIVE:

- To disseminate knowledge about detailed planning.
- To impart knowledge about materials used in formwork.
- To learn design of formwork and shores.
- To disseminate knowledge about erection of form work.
- To impart knowledge about design of formwork for domes, shells, and tunnels.

UNIT I PLANNING, SITE EQUIPMENT & PLANT FOR FORM WORK

9

Introduction - Forms for foundations, columns, beams walls etc., General objectives of formwork building - Planning for safety - Development of a Basic System - Key Areas of cost reduction - Planning examples. Overall Planning - Detailed planning - Standard units - Corner units - Pass units - Calculation of labour constants - Formwork hours - Labour Requirement - Overall programme - Detailed programme - Costing - Planning crane arrangements - Site layout plan - Transporting plant - Formwork beams - Scaffold frames - Framed panel formwork - Formwork accessories.

UNIT II MATERIALS ACCESSORIES PROPRIETARY PRODUCTS & PRESSURES

9

Lumber - Types - Finish - Sheathing boards working stresses - Repetitive member stress - Plywood - Types and grades - Jointing Boarding - Textured surfaces and strength - Reconstituted wood - Steel - Aluminum - Hardware and fasteners - Nails in Plywood - Allowable withdrawal load and lateral load. Pressures on formwork - Examples - Vertical loads for design of slab forms - Uplift on shores - Laterals loads on slabs and walls.

UNIT III DESIGN OF FORMS AND SHORES

9

Basic simplification - Beam formulae - Allowable stresses - Deflection, Bending - Lateral stability - Shear, Bearing - Design of Wall forms - Slab forms - Beam forms - Column forms - Examples in each. Simple wood stresses - Slenderness ratio - Allowable load vs length behaviour of wood shores - Form lining Design Tables for Wall formwork - Slab Formwork - Column Formwork - Slab props - Stacking Towers - Free standing and restrained - Rosett Shoring - Shoring Tower - Heavy Duty props.

UNIT IV BUILDING AND ERECTING THE FORM WORK

9

Carpentry Shop and job mill - Forms for Footings - Wall footings - Column footings - Sloped footing forms - Strap footing - Stepped footing - Slab form systems - Sky deck and Multiflex - Customized slab table - Standard Table module forms - Swivel head and uniportal head - Assembly sequence - Cycling

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with lifting fork - Moving with table trolley and table prop. Various causes of failures - ACI - Design deficiencies - Permitted and gradual irregularities.

UNIT V FORMS FOR DOMES AND TUNNELS, SLIP FORMS AND SCAFFOLDS 9

Hemispherical, Parabolic, Translational shells - Typical barrel vaults Folded plate roof details - Forms for Thin Shell roof slabs design considerations - Building the forms - Placing concrete - Form removed - Strength requirements - Tunnel forming components - Curb forms invert forms - Arch forms - Concrete placement methods - Cut and cover construction - Bulk head method - Pressures on tunnels - Continuous Advancing Slope method - Form construction - Shafts. Slip Forms - Principles -Types - advantages - Functions of various components - Planning -Desirable characteristics of concrete - Common problems faced - Safety in slip forms special structures built with slip form Technique - Types of scaffolds - Putlog and independent scaffold -Single pole scaffolds - Truss suspended - Gantry and system scaffolds.

Total Periods: 45

OUTCOMES:

On completion of this course, the student is expected to be able to

- CO1** Explain detailed planning of formwork, plant, and site equipment.
- CO2** Select material accessories for formwork connection and analyze pressures on formworks.
- CO3** Design the forms and shores.
- CO4** Apply the knowledge of erecting forms for beams, slabs, columns, walls, and causes of failures.
- CO5** Apply the knowledge of forms and their erection for domes and tunnels, types of slip forms, and scaffolds.

REFERENCES:

1. Austin, C.K., Formwork for Concrete, Cleaver -Hume Press Ltd., London, 1996.
2. Hurd, M.K., Formwork for Concrete, Seventh Edition, American Concrete Institute, Detroit, 2016
3. Michael P. Hurst, Construction Press, London and New York, 2003.
4. Robert L. Peurifoy and Garold D. Oberlender, Formwork for Concrete Structures, McGraw - Hill, 2010.
5. Kumar Neeraj Jha, Formwork for Concrete Structures, 2017

COs-POs & PSOs MAPPING:

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	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	1	3	2	1	2	3	2
CO2	3	3	2	1	2	1	3	2	3
CO3	2	3	3	1	1	1	3	3	3
CO4	3	3	3	1	1	2	3	2	3
CO5	3	2	3	1	1	2	3	2	3
Avg	3	3	2	1	1	1	3	2	3

*** 3- High, 2- Medium, 1- Low**

BM2508

PROJECT SAFETY MANAGEMENT

L T P C

3 0 0 3

OBJECTIVES:

- To study and understand the various safety concepts and requirements applied to construction projects.

UNIT I CONSTRUCTION ACCIDENTS

9

Accidents and their Causes – Human Factors in Construction Safety – Costs of Construction Injuries – Occupational and Safety Hazard Assessment – Legal Implications.

UNIT II SAFETY PROGRAMMES

9

Problem Areas in Construction Safety – Elements of an Effective Safety Programme – Job-Site Safety Assessment – Safety Meetings – Safety Incentives.

UNIT III CONTRACTUAL OBLIGATIONS

9

Safety in Construction Contracts – Substance Abuse – Safety Record Keeping - Occupational Safety and Health Administration Manuals, Laws and Act - Indian and International Practices.

UNIT IV DESIGNING FOR SAFETY

9

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Safety Culture – Safe Workers – Safety and First Line Supervisors – Safety and Middle Managers – Top Management Practices, Company Activities and Safety – Safety Personnel – Sub contractual Obligation – Project Coordination and Safety Procedures – Workers Compensation.

UNIT V OWNERS' AND DESIGNERS' OUTLOOK

9

Owners and Designers – Roles and responsibility in ensuring safety – Preparedness – Role of the designer in ensuring safety – Safety clause in the design document.

Total Periods: 45

OUTCOME:

On completion of the course, the student is expected to be able to

- CO1** Develop knowledge on accidents and their causes.
- CO2** Develop knowledge about safety programs and job-site safety assessments.
- CO3** Apply the knowledge of contractual obligations.
- CO4** Explain about designing for safety and safety procedures.
- CO5** Develop the knowledge of owners' and designers' responsibilities.

REFERENCES:

1. Jimmy W. Hinze, Construction Safety, Prentice Hall Inc., 1997.
2. Richard J. Coble, Jimmie Hinze and Theo C. Haupt, Construction Safety and Health Management, Prentice Hall Inc., 2001.
3. Sathyanarayanan Rajendran and Mandi Kime, Construction Project Safety-Management Best Practices Handbook, 2013.
4. Safety, Health and Environmental Handbook, CPWD, 2019.
5. Bhattacharjee S.K. Safety Management in Construction (Principles and Practice), Khanna Publishers, New Delhi 2011

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	1	1	-	1	2	2	2
CO2	2	1	2	2	-	1	2	2	2

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CO3	1	1	1	-	-	1	2	1	2
CO4	1	1	2	1	1	1	2	2	2
CO5	1	1	2	1	1	1	2	2	2
Avg	2	1	2	2	1	1	2	2	2

*** 3- High, 2- Medium, 1- Low**

BM2509

GREEN BUILDING TECHNIQUES

L T P C

3 0 0 3

OBJECTIVES:

- Exposure to the green building technologies and their significance.
- Educate about the Sun-earth relationship and its effect on climate.
- Enhance awareness of end-use energy requirements in the society.

UNIT I ENERGY USAGE

9

Overview of the significance of energy use and energy processes in building - Indoor activities and environmental control - Internal and external factors on energy use and the attributes of the factors - Characteristics of energy use and its management - Macro aspect of energy use in dwellings and its implications

UNIT II INDOOR MANAGEMENT

9

Indoor environmental requirement and management - Thermal comfort - Ventilation and air quality – Airconditioning requirement – Visual perception - Illumination requirement - Auditory requirement.

UNIT III ENERGY IMPACT

9

Climate, solar radiation and their influences - Sun-earth relationship and the energy balance on the earth's surface - Climate, wind, solar radiation, and temperature - Sun shading and solar radiation on surfaces - Energy impact on the shape and orientation of buildings.

UNIT IV ENERGY UTILIZATION

9

End-use, energy utilization and requirements - Lighting and day lighting - End-use energy requirements - Status of energy use in buildings Estimation of energy use in a building. Heat gain and thermal performance of building envelope - Steady and non steady heat transfer through the glazed window and the wall - Standards for thermal performance of building envelope - Evaluation of the overall thermal transfer.

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UNIT V ENERGY MANAGEMENT

9

Energy management options - Energy audit and energy targeting - Technological options for energy management

Total Periods: 45

OUTCOMES:

At the end of the course the students will be able to:

- CO1** Infer the fundamentals of energy use and energy processes in building.
- CO2** Identify the energy requirement and its management.
- CO3** Interpret the Sun-earth relationship vis-a-vis its effect on climate.
- CO4** Estimate the overall energy use of a building.
- CO5** Interpret the audit procedures of energy and set the energy targets

REFERENCE BOOKS:

1. Bryant Edwards (2005): Natural Hazards, Cambridge University Press, U.K.
2. Carter, W. Nick, 1991: Disaster Management, Asian Development Bank, Manila.
3. Sahni, Pardeep et.al. (eds.) 2002, Disaster Mitigation Experiences and Reflections, Prentice Hall of India, New Delhi.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	3	3	3	2	3	3	3
CO2	2	2	2	3	3	2	1	2	2
CO3	2	1	2	3	3	2	1	2	2
CO4	2	1	2	3	3	2	1	2	2
CO5	1	1	2	3	3	2	2	1	2
Avg	2	1	2	3	3	2	2	2	2

*** 3- High, 2- Medium, 1- Low**

BM2510

MANAGEMENT INFORMATION SYSTEMS

L T P C

3 0 0 3

OBJECTIVE:

- To study the concepts of information systems and their applications, system development and information systems, implementation and control and system audit.

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UNIT I INTRODUCTION

9

Information Systems – Establishing the Framework – Business Models – Information System Architecture – Evolution of Information Systems – Database Management Systems

UNIT II SYSTEM DEVELOPMENT

9

Modern Information System – System Development Life Cycle – Structured Methodologies- Designing Computer Based Methods, Procedures, Control – Designing Structured Programs.

UNIT III INFORMATION SYSTEMS

9

Integrated Construction Management Information System – Project Management Information System – Functional Areas, Finance, Marketing, Production, Personnel – Levels, DSS, EIS, and ES – Comparison, Concepts and Knowledge Representation – Managing International Information System.

UNIT IV IMPLEMENTATION AND CONTROL

9

Control – Testing Security – Coding Techniques – Defection of Error – Validating – Cost Benefit analysis – Assessing the value and risk of Information System.

UNIT V SYSTEM AUDIT

9

Software Engineering qualities – Design, Production, Service, Software specification, Software Metrics, Software quality assurance – Systems Methodology – Objectives – Time and Logic, Knowledge and Human Dimension – Software life cycle models – Verification and Validation.

Total Periods: 45

OUTCOMES:

On completion of the course, the student is expected to be able to

- CO1** Apply the knowledge of engineering fundamentals in various information systems in management.
- CO2** Design the structured programs by computer-based methods.
- CO3** Explain management information system for projects.
- CO4** Perform the validation of the computer model and assess the risk of information system.
- CO5** Perform software quality assurance, verification and validation.

REFERENCES:

1. Card and Glass, Measuring Software Design Quality, Prentice Hall, 1990.
2. Gordon B. Davis, Management Information System: Conceptual Foundations, Structure and Development, McGraw Hill, 1985.

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3. Joyce J Elam, Case series for Management Information Systems, Simon and Schuster, Custom Publishing, 1996.
4. Kenneth C Laudon and Jane Price Laudon, Management Information Systems - Organisation and Technology, Prentice Hall, 2015.
4. Michael W. Evans and John J Marciniak, Software Quality assurance and Management, John Wiley and Sons, 1987.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	2	2	1	1	3	3	3
CO2	3	3	2	2	1	3	1	2	2
CO3	2	2	1	3	3	3	1	2	2
CO4	2	1	1	2	3	2	1	2	2
CO5	3	2	2	3	3	3	2	3	3
Avg	2	2	2	2	3	3	3	1	1

*** 3- High, 2- Medium, 1- Low**

BM2512

STRUCTURAL HEALTH MONITORING

L T P C

3 0 0 3

OBJECTIVES:

At the end of the course, the students will be able to.

- Decide the appropriate repair, strengthening, rehabilitation and retrofitting technique required for a case study building.
- Use an appropriate health monitoring technique and demolition technique.

UNIT I INTRODUCTION

6

Introduction to SHM: An Overview of Structural Health Monitoring and Smart Materials

UNIT II FINITE ELEMENT FORMULATION

12

Vibration Control for SHM: Vibration Control using SHM – Introduction to FE formulation, Constitutive Relationship, Element Stiffness Matrix for High Precision Finite Element, Element Mass Matrix for High Precision Finite Element, Developing Actuator and Sensor Influence Matrix, Estimating Sensor Voltage,

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Active Control of Damping, A Case study of Performance Estimation for Different Patches, SHM of Ribbon Reinforced Composite Laminate

UNIT III LAMINATES

9

SHM using Piezo and Magnetostrictive Layers: Delamination Sensing using Piezo Sensory Layer, Voltage Response from Piezopatch, Electrical Impedance Method basic theory, A Case Study: Results and Discussions, SHM using Magnetostrictive Sensory Layer, Basics of Magnetization and Hysteresis, Delamination Sensing using Magnetostrictive Sensory Layer, Constitutive relationship with composite relationship, MS Layer in symmetric Laminate, MS Layer Away from the Mid plane in Asymmetric Laminate, Case Studies related to MS Layer based SHM

UNIT IV ANALYSIS OF LDV

9

SHM using LDV: Experimental Modal Analysis using LDV – Introduction, What is LDV?, Velocity and Displacement Measurement using LDV, Case Study for Symmetric Laminate, Case Study for Cross-ply

UNIT V SMART AND INTELLIGENT MATERIALS

9

Smart and Intelligent Materials for intelligent buildings - Matrix tool for assessing the performance of intelligent buildings –Intelligent buildings technology – Thermal comfort , Visual comfort and Indoor air quality – climate , building-climate interaction - Energy Management systems – Smart Buildings Evaluation Tool.

Total Periods: 45

OUTCOMES:

- CO1** Examine the various vibration-based techniques for structural health monitoring.
- CO2** Apply the fiber-optic and piezoelectric sensors for structural health monitoring.
- CO3** Analyze the influence of MS layer in structural health monitoring
- CO4** Determine the effects of LDV in structural health monitoring
- CO5** Apply the appropriate smart and intelligent materials in construction Industry

REFERENCE BOOKS:

1. Daniel Balageas, Claus-Peter Fritzen, Alfredo Güemes, Structural Health Monitoring, John Wiley and Sons, 2006.

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2. Douglas E Adams, Health Monitoring of Structural Materials and Components-Methods with Applications, John Wiley and Sons, 2007.
3. J.P. Ou, H.Li and Z.D. Duan, Structural Health Monitoring and Intelligent Infrastructure, Vol-1, Taylor and Francis Group, London, U.K, 2006.
4. Victor Giurgutiu, Structural Health Monitoring with Wafer Active Sensors, Academic Press Inc, 2007.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	2
CO2	1	1	3	3	1	2	2	2	1
CO3	1	1	3	3	1	2	2	3	1
CO4	1	1	3	3	1	2	3	2	2
CO5	1	1	3	3	1	2	2	2	1
Avg	1	1	3	3	1	2	2	2	1

*** 3- High, 2- Medium, 1- Low**

BM2513

MATERIAL PROCUREMENT & MANAGEMENT

L T P C

3 0 0 3

OBJECTIVES:

- This course focuses on the core principles of project procurement management, material planning and evaluation methods of materials consumed in various infrastructure domains. Students are exposed to effective techniques for successfully allocating risks and delivering projects which help in acquiring future projects.

UNIT I INTRODUCTION

9

Importance of material management and its role in construction industry, scope, objectives and functions, integrated approach to materials management, Role of materials manager.

UNIT II CLASSIFICATION AND CODIFICATION OF MATERIALS OF CONSTRUCTION 9

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ABC, FSN, VED, SOS analysis - Procedure and its use, Standardization in materials and their management, Procurement - Identification of sources of procurement, vendor analysis. Materials Requirement Planning (MRP), Purchase procedure, legal aspects.

UNIT III INVENTORY MANAGEMENT

9

Store Purchase Manual, Contractors Obligation, Inventory Control techniques. EOQ, Advantages and limitation of use of EOQ, Periodic ordering, order point control, safety stock, stock outs, application of ABC analysis in inventory control, Just In Time (JIT) Management, Indices used for assessment of effectiveness of inventory management.

UNIT IV STORES MANAGEMENT

9

Receipt and inspection, care and safety in handling, loss on storage, wastage, Bulk purchasing, site layout and site organization, scheduling of men, materials and equipment. Quality Control – Conventional methods of quality control of Construction materials. Statistical method of quality control, sampling techniques in quality control process. Quality management and its economics.

UNIT V PROJECT PROCUREMENT PROCESSES

9

Materials Management Systems (MMS) and its scope in materials planning, procurement, inventory control, cost control etc. Project evaluation: Discounted Cash Flow, Real Options Theory. Project delivery methods, Competitive bidding. Risk allocation and management. Integrated project delivery. Contract negotiation.

Total Periods: 45

OUTCOMES:

On the successful completion of the course, students will:

- CO1** Identify the need and role of material management
- CO2** Classify materials, identify sources of procurement, conduct vendor analysis
- CO3** Exercise control for effective management of inventory
- CO4** Manage stores and plan site layout
- CO5** Exercise quality control on materials and MMS in planning, procurement, inventory and cost control, evaluate projects and manage risks

REFERENCES:

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1. Chitale A.K. and R.C. Gupta, “Material Management – Text and Cases”, Prentice Hall of India Pvt. Ltd., 2007
2. Denise Bower, “Management of Procurement”, Construction Management Series, Thomas Telford Publishing, 2003
3. Jhamb L.C., “Inventory Management”, Everest Publishing house, 2005
4. Peter Holm Andreasen, “Dynamics of Procurement Management – A Complexity Approach”, Copenhagen Business School, 2012
5. Peter Baily, David Farmer, Barry Crocker, David Jessop & David Jones, “Procurement Principles and Management”, FT Prentice Hall, 2010

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	2	3	3
CO2	2	2	2	-	1	-	2	3	3
CO3	2	1	2	3	-	1	1	2	2
CO4	2	1	2	2	-	1	1	1	1
CO5	2	1	2	1	-	1	1	1	1
Avg	2	1	2	2	1	1	1	2	2

*** 3- High, 2- Medium, 1- Low**

BM2514

MATERIAL MANAGEMENT

L T P C

3 0 0 3

OBJECTIVES:

- To understand reducing the price of items, operating cost but in enhancing the profits.
- To ensure the efficiency with which the capital of the company is utilized resulting in higher return on investment.
- To gain knowledge in procuring of quality materials.
- To understand handling of materials effectively and efficiently.

UNIT I MATERIAL CLASSIFICATION

9

Material Classification- Organizing for materials management – basis for, forming organizations – conventional and modern approaches to organizing, materials management. Materials identification –

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classifying of materials – codification of materials – standardization – simplification and variety reduction of materials

UNIT II MATERIAL PURCHASING

9

Material Purchasing– Planning Purchasing Materials – Norms of Vendor, Rating – Cei Methodology – Material Selection And Development – Purchasing Procedures And Methods – Legal Aspects – Insurance of Materials.

UNIT III PROCUREMENT MANAGEMENT

9

Supply Management – Sources Of Supply – Out Sourcing Material, Management- Procurement Organization - Procurement Planning – Functions of Material Management - Inventory Control.

UNIT IV STORE MANAGEMENT

9

Storing of Materials-Management of stores – location – different types of stores – methods of storing – safety and security of materials – stores equipment – materials handling equipment – factors affecting materials handling

UNIT V WASTE MANAGEMENT

9

Scrap & Obsolete Materials-Management of surplus obsolete and scrap materials – reasons for accumulation of surplus obsolete and scrap materials– methods of disposal – regulations and procedures

Total Periods: 45

OUTCOMES:

- CO1** Outline the meaning of materials management and are able to plan and manage material flows
- CO2** Identify the need for coordination of planning, sourcing, purchasing, moving, storing and controlling of materials.
- CO3** Minimize procurement costs and improves opportunities for reducing the overall project costs.
- CO4** Illustrate the various waste management techniques in construction.
- CO5** Develop a suitable method of disposal of construction wastes adhering to the rules and regulation

REFERENCE BOOKS:

1. A.K. Datta, “Materials Management: Procedures, Text and Cases”, PHI Learning Pvt. Ltd., 2004.
2. Arnold, “Introduction To Materials Management”, Pearson Education India,2009
3. Richard J. Tersine, “Principles Of Inventory And Materials ,Management”, Prentice Hall,2004

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4. Richard J. Tersine, “Modern Materials Management”, John Hardin Campbell - 2007
5. P. Gopalakrishnan, “Handbook of Materials Management”, PHI Learning Pvt. Ltd.2004

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	2	2	3	3	2	3	3
CO2	1	2	2	2	2	2	3	3	3
CO3	2	2	3	3	3	3	2	2	2
CO4	3	3	3	3	3	3	3	3	3
CO5	2	2	2	3	3	3	2	2	2
Avg	2	2	2	3	3	3	2	3	3

*** 3- High, 2- Medium, 1- Low**

BM2515

CONTRACT LAWS & REGULATIONS

L T P C

3 0 0 3

OBJECTIVES:

- To study the various types of construction contract and their legal aspects and provisions.
- To learn concepts in Tenders.
- To learn concepts in Arbitration and legal requirements.
- To study the concepts in labour regulations.

UNIT I CONSTRUCTION CONTRACTS

9

Indian Contracts Act – Elements of Contracts – Types of Contracts – Features – Suitability – Design of Contract Documents – International Contract Document – Standard Contract Document – Law of Torts.

UNIT II TENDERS

9

Prequalification – Bidding – Accepting – Evaluation of Tender from Technical, Contractual and Commercial Points of View – Contract Formation and Interpretation – Potential Contractual Problems – World Bank Procedures and Guidelines – Tamilnadu Transparency in Tenders Act.

UNIT III ARBITRATION

9

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Comparison of Actions and Laws – Agreements – Subject Matter – Violations – Appointment of Arbitrators – Conditions of Arbitration – Powers and Duties of Arbitrator – Rules of Evidence – Enforcement of Award – Costs.

UNIT IV LEGAL REQUIREMENTS

9

Insurance and Bonding – Laws Governing Sale, Purchase and Use of Urban and Rural Land – Land Revenue Codes – Tax Laws – Income Tax, Sales Tax, Excise and Custom Duties and their Influence on Construction Costs – Legal Requirements for Planning – Property Law – Agency Law – Local Government Laws for Approval – Statutory Regulations.

UNIT V LABOUR REGULATIONS

9

Social Security – Welfare Legislation – Laws relating to Wages, Bonus and Industrial Disputes, Labour Administration – Insurance and Safety Regulations – Workmen’s Compensation Act – Indian Factory Act – Tamilnadu Factory Act – Child Labour Act - Other Labour Laws.

TOTAL PERIODS: 45

OUTCOMES:

On completion of the course, the student is expected to be able to

- CO1** Design the construction contracts
- CO2** Develop a skill for the tendering process.
- CO3** Explain the duties of the arbitrator.
- CO4** Develop an idea on the various legal requirements to be met in relation to land and construction.
- CO5** Identify and apply the provisions provided in the labour welfare schemes.

REFERENCES:

1. Gajaria G.T., Laws Relating to Building and Engineering Contracts in India, 2000.
2. Jimmie Hinze, Construction Contracts, McGraw Hill, 3rd Edition, 2013.
3. Ali D. Haidar, Handbook of Contract Management in Construction, Springer Cham, 1st Edition, 2021
4. Patil. B.S, Civil Engineering Contracts and Estimates, Universities Press (India) Private Limited, 4th Edition 2015.
5. Dharmendra Rautray, Principles of Law of Arbitration in India, Wolters Kluwer, 2018.

COs-POs & PSOs MAPPING:

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	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	3	3	2	2	2	3	2	2
CO2	3	3	3	3	2	1	3	3	3
CO3	3	3	3	3	2	1	3	2	2
CO4	2	2	2	2	2	1	3	3	2
CO5	3	2	2	3	2	2	3	3	2
Avg	3	3	3	3	2	1	3	3	2

*** 3- High, 2- Medium, 1- Low**

BM2516 LEAN CONSTRUCTION CONCEPTS, TOOLS AND PRACTICE L T P C
3 0 0 3

OBJECTIVES:

- To impart knowledge about the basics of lean construction.
- To impart knowledge about the lean principles.
- To impart knowledge about the core concepts of lean construction.
- To impart knowledge about the lean tools and techniques.
- To impart knowledge about the basics of lean implementation in the construction industry.

UNIT I INTRODUCTION

9

Introduction and overview of the construction project management -Review of Project Management& Productivity Measurement Systems – Productivity in Construction– Daily Progress Report-The state of the industry for its management practices –construction project phases - Essential features of contemporary construction management techniques - The problems with current construction management techniques– Current production planning.

UNIT II LEAN MANAGEMENT

9

Introduction to lean management – Toyota's management principle-Evolution of lean in the construction industry - Production theories in construction –Lean construction value - Value in construction - Target value design – Lean project delivery system- Forms of waste in the construction industry – Waste Elimination.

UNIT III CORE CONCEPTS IN LEAN

9

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Concepts in lean thinking – Principles of lean construction – Variability and its impact – Traditional construction and lean construction – Traditional project delivery - Lean construction and workflow reliability – Work structuring – Production control.

UNIT IV LEAN CONSTRUCTION TOOLS AND TECHNIQUES

9

Value Stream Mapping – Work sampling – Last planner system – Flow and pull-based production – Last Planner System – Look ahead schedule – constraint analysis – weekly planning meeting- Daily Huddles – Root cause analysis – Continuous improvement – Just in time.

UNIT V LEAN CONSTRUCTION IMPLEMENTATION

9

Lean construction implementation- Enabling lean through information technology – Lean in design - Design Structure Matrix Location Based Management System-BIM (Building Information Modelling) - IPD (Integrated Project Delivery) – Sustainability through lean construction approach.

Total Periods: 45

OUTCOMES:

On completion of this course, the student is expected to be able to

- CO1** Explains the contemporary management techniques and the issues in the present scenario.
- CO2** Apply the basics of lean management principles and their evolution from the manufacturing industry to the construction industry.
- CO3** Develops a better understanding of core concepts of lean construction tools and techniques and their importance in achieving better productivity.
- CO4** Apply lean techniques to achieve sustainability in construction projects.
- CO5** Apply lean construction techniques in design and modeling

REFERENCES:

1. Corfe, C. and Clip, B., Implementing lean in construction: Lean and the sustainability agenda, CIRIA, 2013.
2. Shang Gao and Sui Pheng Low, Lean Construction Management: The Toyota Way, Springer, 2014.
3. Dave, B., Koskela, L., Kiviniemi, A., Owen, R., and Tzortzopoulos, P., Implementing lean in construction: Lean construction and BIM, CIRIA, 2013.

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4. Ballard, G., Tommelein, I., Koskela, L. and Howell, G., Lean construction tools and techniques, 2002.
5. 5. Salem, O., Solomon, J., Genaidy, A. and Luegring, M., Site Implementation and Assessment of Lean Construction Techniques, Lean Construction Journal, 2005. 6. Lincoln H. Forbes , Syed M. Ahmed, Lean Project Delivery and Integrated Practices in Modern Construction, Routledge Publishers, 2nd Edition, 2020.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	1	2	3	2	2	3	2	2
CO2	2	1	1	3	3	2	3	3	2
CO3	2	1	1	3	1	2	3	1	2
CO4	3	1	1	3	2	3	3	2	1
CO5	2	1	1	3	2	2	3	2	1
Avg	2	1	1	3	2	2	3	2	2

*** 3- High, 2- Medium, 1- Low**

BM2517

MAINTAINABILITY

L T P C

3 0 0 3

OBJECTIVES:

- To enable the students know about the basic concept of maintainability engineering.
- To impart knowledge on various maintenance models, maintenance policies and replacement model of various equipment.
- To provide knowledge on logistics for the effective utilization of existing resources and facilities availability of spares parts.

UNIT I MAINTENANCE CONCEPT

9

Maintenance definition – Need for maintenance – Maintenance objectives and challenges – Tero technology – Maintenance costs – Scope of maintenance department.

UNIT II MAINTENANCE MODELS

9

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Proactive/Reactive maintenance – Imperfect maintenance – Maintenance policies – PM versus b/d maintenance – PM schedule and product characteristics – Inspection models-Optimizing profit/downtime – Replacement decisions.

UNIT III MAINTENANCE LOGISTICS

9

Human factors – Maintenance staffing: Learning curves – Simulation – Maintenance resource requirements: Optimal size of service facility – Optimal repair effort – Maintenance planning and scheduling – Spare parts planning.

UNIT IV MAINTENANCE QUALITY

9

Maintenance excellence – Five Zero concept –FMECA –Root cause analysis – System effectiveness – Design for maintainability – Reliability Centered Maintenance.

UNIT V TOTAL PRODUCTIVE MAINTENANCE

9

TPM features – Chronic and sporadic losses – Equipment defects – Six major losses – Overall Equipment Effectiveness – TPM pillars – Autonomous maintenance – TPM implementation

Total Periods: 45

OUTCOMES:

The students will be able to

- CO1** Understand the various terms and terminologies about the maintenance concept.
- CO2** Understand the various maintenance modes
- CO3** Understand the various maintenance logistics meant for the execution of various services.
- CO4** To learn about the maintenance quality.
- CO5** Apply their knowledge in areas where the down time, over replacement are existing and could lead to improve the productivity and quality

REFERENCES

1. Andrew K.S.Jardine & Albert H.C.Tsang, “Maintenance, Replacement and Reliability”, Taylor and Francis,2006.
2. Bikas Badhury & S.K.Basu, “Tero Technology: Reliability Engineering and Maintenance Management”, Asian Books,2003
3. Seichi Nakajima, “Total Productive Maintenance”, Productivity Press, 1993.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	1	1	2	2	2	2	2
CO2	1	1	2	2	2	2	2	2	2

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CO3	1	1	2	2	1	1	2	2	2
CO4	1	1	2	2	2	2	2	2	2
CO5	1	1	1	1	1	2	2	3	3
Avg	1	1	2	2	2	2	2	2	2

*** 3- High, 2- Medium, 1- Low**

BM2518

REPAIR TECHNOLOGY

L T P C

3 0 0 3

OBJECTIVES:

- To understand the need for repair and maintenance, types, selection of suitable tools and equipments for the systems.
- To understand types, objectives, benefits and safety of the material handling.
- To develop the procedures and mechanisms used for repair and maintenance of machines and transmission systems.
- To learn the concept and components of hydraulic and pneumatic systems, identification of problems associated with them and their rectification.
- To understand the selection, operation and maintenance procedures for repair and maintenance of heavy duty machines used in industries.

UNIT I REPAIR AND MAINTENANCE

9

Introduction to repair and maintenance, tools and equipments used for repairs; extractors, jacks, presses, ropes, rope slings and fittings, stages in repair procedure, Maintenance objectives, Scope of maintenance, Types of maintenance, suitability of adoption of maintenance system, Minimizing downtime, Replacement decisions.

UNIT II MATERIAL HANDLING

9

Objectives and benefits of Material handling, Relationship between layout and Material handling, Principles of material handling, Unit load concept, Classification of material handling equipments, Equipment selection. Safety rules for material handling jobs, Factories Act, Safety regulations & Product safety – case studies.

UNIT III REPAIR OF MACHINES AND TRANSMISSION

9

Repair of bed housing type parts, equipment repair and acceptance procedures, repair of parts and mechanisms of production equipments, shafts, bearings, couplings, clutches, gears, pulleys, pistons, cylinders, and connecting rods, recovery and strengthening of machine elements, repair of filled joints.

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UNIT IV MAINTENANCE OF HYDRAULIC AND PNEUMATIC SYSTEM 9

Concepts of hydraulics, classification of the accessories used in hydraulic system, problems in various valves and auxiliaries, Rectifying the problems, constructional details of pumps and motors, Identifying the problems, Pump Maintenance & Trouble Shooting, Classification the hydraulic circuits and its development, identify various components of pneumatic system, repair and maintenance of air Compressors: Single Acting and Double Acting, Components of Pneumatic System, air receiver and pressure control.

UNIT V REPAIR AND MAINTENANCE OF HEAVY DUTY MACHINES 9

Selection, operation and maintenance of Industrial Trucks, Mobile Cranes, Tower crane, Checklist, Competent persons, planning the movement of storage and retrieval of common goods of various shapes and sizes in a general store of a big industry, Problems identification in power press, shear and furnace and their repair.

Total Periods: 45

OUTCOMES:

At the end of the course the students shall have the abilities to

- CO1** Explain the need for repair and maintenance, types, selection of suitable tools and equipments for the systems.
- CO2** Explain types, objectives, benefits and safety of the material handling.
- CO3** Explain the procedures and mechanisms used for repair and maintenance of machines and transmission systems
- CO4** Explain the concept and components of hydraulic and pneumatic systems, identification of problems associated with them and their rectification.
- CO5** Explain the selection, operation and maintenance procedures for repair and maintenance of heavy duty machines used in industries.

TEXT BOOKS:

1. B.Celberg, G.Pekeu, "Repair of Industrial equipment", Mir Publishers, Moscow current lieterature.

REFERENCE BOOKS:

1. H.P.Garg, "Industrial Maintenance", S. Chand Limited, 1987

COs-POs & PSOs MAPPING:

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	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	3	3
CO2	1	2	2	2	3	3	2	2	3
CO3	3	3	3	3	3	3	2	2	2
CO4	2	2	2	3	3	3	2	3	3
CO5	2	2	2	3	3	3	3	3	3
Avg	2	2	2	3	3	3	2	3	3

*** 3- High, 2- Medium, 1- Low**

BM2519

HUMAN RESOURCE MANAGEMENT IN CONSTRUCTION

LTPC

3 0 0 3

OBJECTIVES:

- Implement practices and techniques for evaluating performance, structuring teams, coaching and mentoring people,
- Understand the role of the leader and leadership principles and attitudes
- Demonstrate an understanding of professional and ethical responsibilities; and
- Demonstrate commitment to quality, timeliness, and continuous improvement.
- Clearly understand their future managerial role, with emphasis on the management of the human resources and with a multi-cultural perspective

UNIT I MANPOWER PLANNING

9

Manpower planning and forecasting – Recruitment, selection process-Sources- Induction- Orientation and Training -Manpower Planning process - Organising, Staffing, directing, and controlling — Factors influencing supply and demand of human resources – Role of HR manager – Personnel Principles.

UNIT II ORGANISATION

9

Elements of an organisation- Management process in organisations- Planning- Organising-Staffing- Directing- Controlling – Delegation of authority – responsibility – accountability – lines and staff organization, Workforce diversity- international dimensions of Organisation- Organisational structure- determinants of organisational design

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UNIT III HUMAN RELATIONS AND ORGANISATIONAL BEHAVIOUR 9

Basic individual psychology – Approaches to job design and job redesign – Self managing work teams – Intergroup – Conflict in organizations – Leadership-Engineer as Manager –aspects of decision making – Significance of human relation and organizational – Individual in organization – Motivation – Personality and creativity – Group dynamics, Team working – Communication and negotiation skills.

UNIT IV WELFARE MEASURES 9

Establishing Pay plans - Basics of compensation - factors determining pay rate - Current trends in compensation - Job evaluation – Incentives- Practices in Indian organisations - Statutory benefits - non-statutory (voluntary) benefits - Insurance benefits - retirement benefits and other welfare measures to build employee commitment – Laws related to welfare measures.

UNIT V MANAGEMENT AND DEVELOPMENT METHODS 9

Management Development - On-the-job and off-the-job- Management Developments - Performance appraisal in practice. Managing careers: Career planning and development - Managing promotions and transfers. of operations – Developing policies, practices and establishing process pattern – Competency upgradation and their assessment – New methods of training and development – Performance Management.

Total Periods: 45

OUTCOMES:

On completion of the course, the student is expected to be able to

- CO1** Implement practices and techniques for evaluating performance, structuring teams, coaching and mentoring people
- CO2** Understand the role of the leader and leadership principles and attitudes
- CO3** Demonstrate an understanding of professional and ethical responsibilities
- CO4** Demonstrate commitment to quality, timeliness, and continuous improvement.
- CO5** Clearly understand their future managerial role, with emphasis on the management of the human resources and with a multi-cultural perspective

REFERENCES

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1. Dwivedi R.S, Human Relations and Organisational Behaviour, A Global perspective, 5th Edition, Macmillian India Ltd., 2008.
2. Joseph J. Famiularo, “Handbook of Human Resources Administration”, McGraw-Hill International Edition, 2nd Edition 1986.
3. D. Longford, M.R. Hancock, R. Fellows & A. W. Gale, Human Resource Management In Construction CRC Press, 2014.
4. Carleton Counter II and Jill Justice Coulter, “The Complete Standard HandBook of Construction Personnel Management ", Prentice Hall, Inc., New Jersey, 1989.
5. Michael Armstrong, Stephen Taylor, Armstrong’s Handbook of Human Resource Management Practice, 15th Edition, CRC Press, 2020.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	1	2	2	1	2	2	2
CO2	3	3	1	1	1	2	3	3	3
CO3	2	2	2	1	2	2	2	3	3
CO4	1	1	1	1	1	3	2	2	2
CO5	3	3	1	1	3	3	2	2	2
Avg	2	2	1	1	2	2	2	2	2

*** 3- High, 2- Medium, 1- Low**

BM2520

QUALITY & RELIABILITY ENGINEERING

L T P C

3 0 0 3

OBJECTIVES:

- To explain quality, reliability, quality control and statistical quality control.
- To explain and apply statistical tools in quality control and solve related numerical problems.
- To analyze failure data, hazard models, system reliability and solve related numerical problems.
- To apply reliability improvement and allocation methods to engineering systems.
- To explain maintainability and availability concepts to improve the system effectiveness.

UNIT I INTRODUCTION

9

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Basic Concepts: Definitions of Quality and Reliability, Parameters and Characteristics, Quality control, Statistical Quality Control, Reliability concepts. Concepts in Probability and Statistics Events, Sample Space, Probability rules, Conditional probability, Dependent and Independent Events, Application of Probability concepts in Quality Control, Problems.

UNIT II STATISTICAL ASPECTS AND PROBABILITY DISTRIBUTIONS **9**

Statistical Tools in Quality Control, the Concept of Variation, Graphical Tools for data representation and analysis, Discrete and Continuous Distributions, Normal, Poisson, Binomial, Weibull Distribution, Problems. Failure Data Analysis: Introduction, Failure Data, Quantitative measures, MTTF, MTBF, Bath tub Curve, Mean Life, Life Testing, Problems, Introduction to Failure Mode and Effect Analysis.

UNIT III HAZARD MODELS **9**

Introduction, Constant Hazard, Linearly increasing hazard, the Weibull model, (Derivation not required, emphasis to be on applications). System reliability Series, Parallel and Mixed Configurations, Block Diagram Concept, r-out-of-n structure, solving problems using mathematical models.

UNIT IV RELIABILITY IMPROVEMENT AND ALLOCATION **9**

Difficulty in achieving reliability, Methods for improving reliability during design, Different techniques available to improve reliability, Optimization, Reliability-Cost trade off, Elements of a typical reliability program, setting overall reliability goals, Reliability Apportionment, Prediction and Analysis, Problems.

UNIT V MAINTAINABILITY AND AVAILABILITY **9**

Introduction, Formulas, Techniques available to improve maintainability and availability, trade-off among reliability, maintainability and availability, Simple problems.

Total Periods: 45

OUTCOMES:

At the end of the course the students shall have the abilities to

- CO1** Explain quality, reliability, quality control and statistical quality control.
- CO2** Explain and apply statistical tools in quality control and solve related numerical problems.
- CO3** Analyze failure data, hazard models and system reliability and solve related numerical problems
- CO4** Apply reliability improvement and allocation methods to engineering systems

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CO5 Explain maintainability and availability concepts to improve the system effectiveness.

TEXT BOOKS:

1. Halpern, Seigmund,"The Assurance Sciences", Prentice Hall International, New Jersey, U.S.A. (1978).
2. Srinath, L.S,"Concepts in Reliability Engineering", Affiliated East-West Press Private Limited, New Delhi, India, (1985).

REFERENCE BOOKS:

1. Juran, J.M. and Gryna, F.M,"Quality Planning and Analysis", Tata McGraw Hill publishing Company Ltd., New Delhi, India. (1982).
2. Blanchard, Benjamin S,"Logistics Engineering and Management", Prentice Hall International, New Jersey, U.S.A. (1986).
3. Kraus, John W, "Maintainability and Reliability", Handbook of Reliability I Engineering and Management, Editors -Ireson. W .a. and Coombs, C.F. McGraw Hill, Book Company Inc., U .S.A. (1988).

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	2	2	2	2	1	2	2
CO2	1	2	3	3	3	3	2	3	3
CO3	2	2	3	3	3	3	2	3	3
CO4	1	2	2	3	3	3	1	2	2
CO5	1	2	2	2	3	3	2	3	3
Avg	1	2	2	3	3	3	2	3	3

*** 3- High, 2- Medium, 1- Low**

BM2521

REPAIR AND REHABILITATION OF STRUCTURES

L T P C

3 0 0 3

OBJECTIVES:

- To make the students to gain the knowledge on quality of concrete, durability aspects, causes of deterioration, assessment of distressed structures, repairing of structures and demolition procedures.

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- To understand the rehabilitation of structures.

UNIT I MAINTENANCE AND REPAIR STRATEGIES 9

Maintenance, repair and rehabilitation, Facets of Maintenance, importance of Maintenance various aspects of Inspection, Assessment procedure for evaluating a damaged structure, causes of deterioration

UNIT II SERVICEABILITY AND DURABILITY OF CONCRETE 9

Quality assurance for concrete construction concrete properties- strength, permeability, thermal properties and cracking. - Effects due to climate, temperature, chemicals, corrosion - design and construction errors - Effects of cover thickness and cracking

UNIT III MATERIALS AND TECHNIQUES FOR REPAIR 9

Special concretes and mortar, concrete chemicals, special elements for accelerated strength gain, Expansive cement, polymer concrete, sulphur infiltrated concrete, Ferro cement and polymers coating for rebars loadings from concrete, mortar and dry pack, vacuum concrete, Guniting and Shotcrete, Epoxy injection, Mortar repair for cracks, shoring and underpinning. Methods of corrosion protection, corrosion inhibitors, corrosion resistant steels and cathodic protection.

UNIT IV REPAIRS TO STRUCTURES 9

Repair of structures distressed due to earthquake – Strengthening using FRP- Strengthening and stabilization techniques for repair. Repairs to overcome low member strength – Deflection – Cracking – Chemical disruption - weathering corrosion – wear – fire - leakage and marine exposure

UNIT V DEMOLITION OF STRUCTURES 9

Engineered demolition techniques for structures – application – sequence – damage assessment – controlled blasting for demolition – Techniques for residual stress measurements. Case studies

Total Periods: 45

OUTCOMES:

At the end of the course student will be able to

- CO1** Summarize the assessment procedure for evaluation of a damaged structure
- CO2** Evaluate the durability properties of concrete
- CO3** Outline the effects of various factors on concrete

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CO4 Identify the various materials and techniques for corrosion protection

CO5 Examine the effect of strengthening and stabilization techniques on repair

REFERENCE BOOKS:

1. Denison Campbell, Allen and Harold Roper, "Concrete Structures, Materials, Maintenance and Repair", Longman Scientific and Technical UK, 1991.
2. Allen R.T and Edwards S.C, "Repair of Concrete Structures", Blakie and Sons, UK, 1987.
3. Raikar, R.N., "Learning from failures - Deficiencies in Design, Construction and Service" - R&D Centre (SDCPL), RaikarBhavan, Bombay, 1987.
4. Santhakumar A.R., "Concrete Technology" Oxford University Press, Printed in India by Radha Press, New Delhi, 2007.
5. Peter H.Emmons, "Concrete Repair and Maintenance Illustrated", Galgotia Publications pvt. Ltd., 2001.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	2
CO2	1	1	3	3	1	2	2	2	1
CO3	1	1	3	3	1	2	2	3	1
CO4	1	1	3	3	1	2	3	2	2
CO5	1	1	3	3	1	2	2	2	1
Avg	1	1	3	3	1	2	2	2	1

*** 3- High, 2- Medium, 1- Low**

MC25101

RESEARCH METHODOLOGY AND IPR

L T P C

3 0 0 3

OBJECTIVES:

- To impart scientific, statistical and analytical knowledge for carrying out research work effectively.

UNIT I RESEARCH PROBLEM FORMULATION

9

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Objectives of research, types of research, research process, approaches to research; conducting literature review- information sources, information retrieval, tools for identifying literature, Indexing and abstracting services, Citation indexes, summarizing the review, critical review, identifying research gap, conceptualizing and hypothesizing the research gap

UNIT II RESEARCH DESIGN AND DATA COLLECTION

9

Statistical design of experiments- types and principles; data types & classification; data collection - methods and tools

UNIT III DATA ANALYSIS, INTERPRETATION AND REPORTING

9

Sampling, sampling error, measures of central tendency and variation,; test of hypothesis- concepts; data presentation- types of tables and illustrations; guidelines for writing the abstract, introduction, methodology, results and discussion, conclusion sections of a manuscript; guidelines for writing thesis, research proposal; References – Styles and methods, Citation and listing system of documents; plagiarism, ethical considerations in research

UNIT IV INTELLECTUAL PROPERTY RIGHTS

9

Concept of IPR, types of IPR – Patent, Designs, Trademarks and Trade secrets, Geographical indications, Copy rights, applicability of these IPR; , IPR & biodiversity; IPR development process, role of WIPO and WTO in IPR establishments, common rules of IPR practices, types and features of IPR agreement, functions of UNESCO in IPR maintenance.

UNIT V PATENTS

9

Patents – objectives and benefits of patent, concept, features of patent, inventive steps, specifications, types of patent application; patenting process - patent filling, examination of patent, grant of patent, revocation; equitable assignments; Licenses, licensing of patents; patent agents, registration of patent agents.

Total Periods: 45

OUTCOMES:

Upon completion of the course, the student can

- CO1** Describe different types of research and identify, review and define the research problem
- CO2** Select suitable design of experiments and describe types of data and the tools for collection of data
- CO3** Explain the process of data analysis; interpret and present the result in suitable form

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CO4 Explain about Intellectual property rights, types and procedures

CO5 Execute patent filing and licensing

REFERENCES:

1. Cooper Donald R, Schindler Pamela S and Sharma JK, “Business Research Methods”, Tata McGraw Hill Education, 11e (2012).
2. Soumitro Banerjee, “Research methodology for natural sciences”, IISc Press, Kolkata, 2022,
3. Catherine J. Holland, “Intellectual property: Patents, Trademarks, Copyrights, Trade Secrets”, Entrepreneur Press, 2007.
4. David Hunt, Long Nguyen, Matthew Rodgers, “Patent searching: tools & techniques”, Wiley, 2007.
5. The Institute of Company Secretaries of India, Statutory body under an Act of parliament, “Professional Programme Intellectual Property Rights, Law and practice”, September 2013.

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PROGRAM NAME: M.TECH – NATURAL LANGUAGE PROCESSING

PROGRAM DURATION: 2 YEARS

PHILOSOPHY OF THE PROGRAM:

The philosophy of the program focuses on equipping students with advanced skills in computational linguistics, machine learning, and AI for processing and understanding human language. It covers topics like text analytics, speech recognition, machine translation, and conversational AI. This program prepares graduates for careers in research, academia, and industries developing language-based AI applications such as chatbots, voice assistants, and translation systems.

SIGNIFICANCE OF THE PROGRAM:

NLP bridges the gap between human language and machines, enabling computers to understand, interpret, and generate human text. From chatbots and virtual assistants to sentiment analysis and machine translation, NLP fuels advancements across various industries. It facilitates more effective communication by enabling personalized experiences, improving accessibility, and breaking down language barriers. This program plays a crucial role in fields like linguistics, psychology, and social sciences, providing valuable insights into human language and cognition.

DISTINCTIVENESS OF THE NLP:

This program stands out for its unique integration of linguistics, machine learning, and artificial intelligence, offering a deep focus on language-related AI technologies. Unlike general AI programs, it specializes in cutting-edge topics like sentiment analysis, conversational AI, and language modeling. The program bridges the gap between human communication and AI systems, offering innovative solutions for real-world applications. Its interdisciplinary approach ensures graduates are well-equipped to lead advancements in NLP research and industry.

NEED OF THE PROGRAM:

The program addresses the growing demand for experts who can develop intelligent systems to process and analyze human language. With the rapid expansion of AI-driven technologies like chatbots, virtual assistants, and automated translation tools, the need for specialized knowledge in NLP has never been greater. It is essential for driving innovation in AI applications and solving real-world linguistic challenges. It empowers businesses to analyze vast amounts of text data, such as customer reviews, social media posts, and news articles, to gain valuable insights and improve decision-making. The program is ideal for those aiming to advance in cutting-edge NLP research or build innovative AI-driven language applications.

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M.TECH – NATURAL LANGUAGE PROCESSING

REGULATIONS – 2025

CHOICE BASED CREDIT SYSTEM

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1: Apply domain knowledge and skills of AI and ML to succeed in professional careers and/or higher studies

PEO2: Design and implement hardware and/or software-related engineering solutions applying the principles and practices of computing, grounded in mathematics and science, to meet industrial & societal demands and/or productively engage in research

PEO3: Function as an individual or in multi-disciplinary teams, by creatively solving problems and communicating effectively

PEO4: Contribute to society, by functioning ethically and responsibly, and involving in professional and social activities for sustainable developments

PROGRAM OUTCOMES (POs)

PO1. An ability to independently carry out research /investigation and development work to solve practical problems.

PO2. An ability to write and present a substantial technical report/document.

PO3. Students should be able to demonstrate a degree of mastery over the area as per specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.

PO4. An ability to apply modern NLP tools and techniques to solve engineering problems, and provide sustainable development in societal and environmental contexts, while following ethical engineering practices.

PO5. An ability to demonstrate an understanding of engineering and management principles along with an ability to adapt to technological changes towards execution and management of projects, either as an individual or as a team member.

PROGRAM SPECIFIC OUTCOMES (PSO)

PSO: Design, develop and deploy artificial intelligence and machine learning based software-intensive solutions, by applying acquired knowledge of NLP and allied thrust areas, and by conducting research-based investigations.

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CURRICULUM AND SYLLABI FOR SEMESTERS I to IV

SEMESTER I

S.NO	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIODS
				L	T	P	C	
THEORY								
1	MA24101	Mathematical Foundation for Data Science.	FC	3	1	0	4	4
2	NP24102	Advanced Data Structures & Algorithms	PCC	3	0	0	3	3
3	NP24103	Foundations of Natural Language Processing	PCC	3	0	0	3	3
4	ML24105	Machine Learning for Multimodal Data	PCC	3	0	0	3	3
5	AL24106	Advanced Artificial Intelligence	PCC	3	0	0	3	3
6	RM24107	Research Methodology and IPR	RMC	3	0	0	3	3
PRACTICALS								
7	NL2451	Advanced Data Structures Laboratory	PCC	0	0	4	2	4
8	ML2452	Machine Learning Laboratory	PCC	0	0	4	2	4
TOTAL				18	1	8	23	27

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SEMESTER II

S.NO	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIODS
				L	T	P	C	
THEORY								
1	NP24108	Advanced Data Visualization Technique	PCC	3	0	0	3	3
2	NL24109	Advanced Natural Language Processing	PCC	3	0	0	3	3
3	NT24110	Text Analytics in NLP	PCC	3	0	0	3	3
4	IR24111	Information Retrieval & Text Mining	PCC	3	0	0	3	3
5	ND24112	Machine Translation & NLP System Design	PCC	3	0	2	4	5
6		Professional Elective I	PEC	3	0	0	3	3
PRACTICALS								
7		Professional Elective II	PEC	3	0	0	3	3
8	NP2453	Advanced Natural Language Processing Laboratory	PCC	0	0	4	2	4
9	NS2454	Internship & Seminar	EEC	0	0	4	2	4
TOTAL				21	0	10	26	31

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SEMESTER III

S.NO	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL CONTACT PERIODS
				L	T	P	C	
THEORY								
1	DL24113	Advanced Deep Learning	PCC	3	0	2	4	5
2		Professional Elective III	PEC	3	0	0	3	3
3		Professional Elective IV	PEC	3	0	0	3	3
4		Open Elective	OEC	3	0	0	3	3
PRACTICALS								
5	NL2466	Project Work-Phase I	EEC	0	0	12	6	12
TOTAL				12	0	14	19	26

SEMESTER IV

S.NO	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL CONTACT PERIODS
				L	T	P	C	
PRACTICALS								
1	NL2477	Project Work-Phase II	ECC	0	0	24	12	24

TOTAL CREDITS: 80

FOUNDATION COURSES (FC)

S.NO	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK				TOTAL CONTACT PERIODS
				L	T	P	C	
1	MA24101	Mathematical Foundation of Data Science	FC	3	1	0	4	4

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PROFESIONAL CORE COURSES (PCC)

S.NO	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK				TOTAL CONTACT PERIODS
				L	T	P	C	
1	NP24102	Advanced Data Structures & Algorithms	PCC	3	0	0	3	3
2	NP24103	Foundations of Natural Language Processing	PCC	3	0	0	3	3
3	ML24105	Machine Learning for Multimodal Data	PCC	3	0	0	3	3
4	AL24106	Advanced Artificial Intelligence	PCC	3	0	0	3	3
5	NP24108	Advanced Data Visualization Technique	PCC	3	0	0	3	3
6	NL24109	Advanced Natural Language Processing	PCC	3	0	0	3	3
7	NT24110	Applications in Multilingual NLP	PCC	3	0	0	3	3
8	IR24111	Information Retrieval & Text Mining	PCC	3	0	0	3	3
9	ND24112	Machine Translation & NLP System Design	PCC	3	0	2	4	5
TOTAL CREDITS							28	

LIST OF PROFESSIONAL ELECTIVE COURSES (PEC)
SEMESTER II, ELECTIVE I

S.NO	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK				TOTAL CONTACT PERIODS
				L	T	P	C	
1	NP2481	Cognitive Computing	PEC	3	0	0	3	4

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2	BT2482	Blockchain Technology	PEC	2	0	2	3	4
3	IP2483	IoT Programming	PEC	2	0	2	3	3
4	NP2484	Principles Of Programming Languages	PEC	3	0	0	3	3
TOTAL CREDITS							12	

SEMESTER II, ELECTIVE II

S.NO	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK				TOTAL CONTACT PERIODS
				L	T	P	C	
1	NP2491	Speech And Audio Processing	PEC	3	0	0	3	3
2	ML2492	Cloud & Edge Computing for ML Applications	PEC	3	0	0	3	3
3	NP2493	Image Processing	PEC	3	0	0	3	3
4	NP2494	Multimedia Communication Networks	PEC	3	0	0	3	3
TOTAL CREDITS							12	

SEMESTER III, ELECTIVE III

S.NO	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK				TOTAL CONTACT PERIODS
				L	T	P	C	
1	NS2455	Social Networks & Open-Source Intelligence	PEC	3	0	0	3	3
2	NA2466	Mobile Application Development	PEC	2	0	2	3	4
3	NR2477	Robotic Programming	PEC	2	0	2	3	4

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4	NT2488	IOT projects	PEC	2	0	2	3	4
TOTAL CREDITS							12	

SEMESTER III, ELECTIVE IV

S.NO	COURSE CODE	COURSE TITLE	CATEG ORY	PERIODS PER WEEK				TOTAL CONTACT PERIODS
				L	T	P	C	
1	ND2432	Data Privacy	PEC	3	0	0	3	3
2	NC2433	Deep Learning & Computer Vision	PEC	3	0	0	3	3
3	NA2434	Agile Software Development	PEC	2	0	2	3	4
4	NR2435	Reinforcement Learning	PEC	3	0	0	3	3
TOTAL CREDITS							12	

LIST OF OPEN ELECTIVES

S.NO	COURSE CODE	COURSE TITLE	CATEGOR Y	PERIODS PER WEEK				TOTAL CONTACT PERIODS
				L	T	P	C	
1	OEC2491	Intellectual Property Rights	OEC	3	0	0	3	3
2	OEC2492	Health, Sports & Financial Analytics	OEC	3	0	0	3	3
3	OEC2493	Business Data Analytics	OEC	3	0	0	3	3
4	OEC2494	Ethical Management	OEC	3	0	0	3	3
5	OEC2495	Digital forensics	OEC	3	0	0	3	3
6	OEC2496	Operations Research	OEC	3	0	0	3	3

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7	OEC2497	IOT for Smart Systems	OEC	3	0	0	3	3
8	OEC2498	Embedded Automation	OEC	3	0	0	3	3
9	OEC2499	Software Architecture & Design	OEC	3	0	0	3	3
10	OEC24910	Sustainable Management	OEC	3	0	0	3	3
TOTAL CREDITS							30	

RESEARCH METHODOLOGY AND IPR COURSES (RMC)

S.NO .	COURSE CODE	COURSE TITLE	CATEG ORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	RM24107	Research Methodology and IPR	RMC	3	0	0	3	3

EMPLOYABILITY ENHANCEMENT COURSES (EEC)

S.NO .	COURSE CODE	COURSE TITLE	CATEG ORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	NS2454	Internship And Seminar	EEC	3	0	0	3	3
2	NL2466	Project work-I	EEC	0	0	12	6	12
3	NL2477	Project Work -II	EEC	0	0	24	12	24
TOTAL CREDITS							21	

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SEMESTER 1

MA24101 MATHEMATICAL FOUNDATION FOR DATA SCIENCE L T P C
4 1 0 4

COURSE OBJECTIVES:

- To understand and apply matrix techniques for engineering applications.
- To understand and apply, vector spaces, basis and dimensions, linear transformations
- To familiarize the student with basic concepts of Graph Theory: Isomorphism.
- To solve the problems in Formulation of optimization problems, convex set, convex functions, convex optimization.
- To acquaint the student with mathematical tools needed in evaluating search methods for Multivariable unconstrained problems.

UNIT – I: MATRICES

12

Eigen values, Eigen vectors, and Diagonalization, singular value decomposition. Positive definite Matrices-Minima, Maxima and saddle points, Test of Positive definiteness, semi-definite and indefinite Matrices.

UNIT - II: LINEAR ALGEBRA

12

Geometry of linear equations, real vector spaces (R^n , matrices etc.) and subspaces, linear independence, basis and dimensions, linear transformations, orthogonality, Orthogonal basis, Gram Schmidt Process projections and least square applications

UNIT - III: GRAPH THEORY

12

Graph Theory: Isomorphism, Planar graphs, graph colouring, Hamilton circuits and Euler cycles. Permutations and Combinations with and without repetition. Specialized techniques to solve combinatorial enumeration problems.

UNIT - IV: OPTIMIZATION TECHNIQUES

12

Introduction- Formulation of optimization problems, classification of optimization problems, overview of analytical solution for unconstrained optimization problems, constrained optimization, convex set, convex functions, convex optimization problem, Kuhn- Tucker conditions, Search methods – Overview of single variable search methods.

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UNIT - V: MULTI – VARIABLE OPTIMIZATION

12

Search methods for Multivariable unconstrained problems - Optimality criteria, unidirectional search – direct search methods- evolutionary search method, Hook-Jeeves pattern search method, gradient based methods –Cauchy’s steepest descent method, Newton’s method.

Total Periods: 60

COURSE OUTCOMES:

At the end of the course, the student should be able to:

CO1. Apply the idea of reducing complex problems into simple form using matrix technique.

CO2. Basic application of vector spaces, basis and dimensions, linear transformations.

CO3. Equips the students to have basic knowledge and understanding of Graph Theory, Isomorphism of Graphs.

CO4. Apply the concept of Formulation of optimization problems, convex set, convex functions, convex optimization.

CO5. Basic application of mathematical tools needed in evaluating search methods for Multivariable unconstrained problems.

REFERENCE BOOKS:

1. Gilbert Strang, Linear Algebra and its Applications, Fourth Edition, Cambridge University Press. 2009.

2. Howard Anton and Chris Rorrers,|| Elementary Linear Algebra||, Tenth Edition, 2010 John Wiley & Sons, Inc.

3. Kalyanmoy Deb, —Optimization for Engineering Design Algorithms and Examples||, Prentice Hall of India, New Delhi, 2004.

4. S.S. Rao, —Optimization Theory and Applications||, Second Edition, New Age International (P) Limited Publishers, 1995 5. M. AsgharBhatti, —Practical Optimization Methods: with Mathematics Applications||, Springer Verlag Publishers, 2000.

CO-PO Mapping:

COs	POs					PSO
	1	2	3	4	5	
1	2	3		2		2

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2	2	3		2		2
3	1	2	3	2		2
4	1	3	2		1	2
5	1		3	2	3	2

NP24102 ADVANCED STRUCTURES AND ALGORITHMS

L T P C 3 0 0 3

COURSE OBJECTIVES

- To understand the usage of algorithms in computing.
- To learn and use hierarchical data structures and its operations
- To learn the usage of graphs and its applications.
- To select and design data structures and algorithms that are appropriate for problems.
- To study about NP hard and NP Completeness of problems.

UNIT I ALGORITHM ANALYSIS AND DESIGN TECHNIQUES

10

Algorithms in computing, growth of functions, probabilistic analysis, and randomized algorithms - Randomized Divide and Conquer: Medium-Finding and Quicksort - Finding closest Pairs of Points-A Randomized Approach: Load Balancing. Greedy Algorithms - Huffman Trees and Codes, Dynamic Programming, Amortized Analysis, Approximation Algorithms - Greedy Algorithms and Bounds on the Optimum: A Load Balancing Algorithm, The Center Selection Problems, Arbitrarily Good Approximations: The Knapsack Problem.

UNIT II DATA STRUCTURES AND CONCURRENCY

9

Review of elementary data structures – List, Stack, Queue–Concurrent linked lists– coarse-grained synchronization – fine-grained synchronization – lazy synchronization – non-blocking synchronization – concurrent queues – bounded partial queues – Unbounded total Queue, Unbounded Lock Free queues – concurrent stacks: Unbounded Lock -Free Stack with Backoff.

UNIT III ADVANCED CONCURRENT STRUCTURES

9

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Hashing – closed-address and open-addressed hash sets –LockFree Hash set- Lock based closedaddress Concurrent hash set, Cuckoo Hashing. Lock-based concurrent skip lists –An Unbounded Heap-Based Concurrent Priority Queue – skip list based unbounded priority queues.

UNIT IV NON-LINEAR DATA STRUCTURES

9

Binary search trees, Balanced Search Trees - Height balanced and Weight balanced search trees, Red Black Tree, B-Tree, K-D Trees, Treaps and Randomized Search Trees (RSTs), Multiway Tries (MWTs) ,Ternary Search Trees (TSTs), Coding Tree.

UNIT V NP AND COMPUTATIONAL INTRACTABILITY

8

Polynomial time reduction – Satisfiability Problem - NP-Completeness - Sequencing Problem - Partitioning Problem-Graph - Graph Coloring - Strongly connected components- Network Flow - Maximum Flow and Minimum Cut – Bipartite Matching-Reductions and – Graph Coloring.

TOTAL PERIODS: 45

REFERENCES

1. The Art of Multiprocessor Programming by Maurice Herlihy and Nir Shavit, revised first edition, Morgan Kaufman, 2012. Note: be sure not to buy the unrevised edition
2. Thomas H Cormen, Charles E Leiserson, Ronald L Rivest and Clifford Stein, “Introduction to Algorithms”, Prentice Hall of India, New Delhi, Second Edition, 2009.
3. Advanced Data Structures by Peter Brass

COURSE OUTCOMES:

Upon the completion of the course the students should be able to

CO 1: Apply knowledge of computing and mathematics to analyze the problem, requirements and algorithms

CO 2: Solve the complex engineering problems using hierarchical data structures

CO 3: Apply suitable design strategy in a way that demonstrates comprehension of the tradeoffs involved in design choices for problem-solving

CO 4: Classify the type of the problem as P, NP Complete or NP Hard.

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CO-PO Mapping:

COs	POs					PSO
	1	2	3	4	5	
1	2		2			1
2	2		2			2
3	2		2			1
4			2			2
5			2	1	1	2

ML24105 MACHINE LEARNING FOR MULTIMODAL DATA L T P C 3 0 0 3

COURSE OBJECTIVES

- To learn the fundamentals of image, video and audio analysis
- To understand the working principles of machine Learning algorithms
- To enable design and implementation of machine learning solutions to classification and clustering problems.
- To gain experience on applying machine learning techniques for image, video, speech and text analysis

UNIT I IMAGE AND VIDEO ACQUISITION, REPRESENTATION AND STORAGE

9

Human eye physiology, Image acquisition devices, Image formats, Images in the Spatial Domain, Images in the Frequency Domain, Image Segmentation, Video principles, state-of-the-art features in image and video; 3D motion and 2D optical flow

UNIT II AUDIO AND TEXT ACQUISITION, REPRESENTATION AND STORAGE

9

Speech production and perception and acquisition-Articulatory phonetics - Acoustic phonetics and signals - Time domain and frequency domain processing; state-of-the-art

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speech features. Text preparation - Extracting terms from tokens - Vector space representation - Similarity Computation

UNIT III MACHINE LEARNING – SUPERVISED LEARNING

9

Learning from Examples, VC Dimension, Bias-Variance trade-off, Cross validation, Logistic regression, Bayesian Classifier, Neural Networks – Single and multilayer perceptron, Support Vector Machines - linear and non-linear kernel functions, Ensemble Methods-bagging, boosting & stacking.

UNIT IV MACHINE LEARNING – UNSUPERVISED LEARNING

9

Clustering: K-Means algorithm, Hierarchical clustering, Gaussian Mixture Models, Expectation Maximization Algorithm; Feature Extraction Methods: Curse of Dimensionality, Principal Component Analysis, Independent Component Analysis

UNIT V CASE STUDIES

9

Automatic face recognition - Object detection, recognition and tracking, Speech, speaker, and language identification/recognition, Text summarization

TOTAL PERIODS: 45

TEXT BOOK

1. Francesco Camastra, Alessandro Vinciarelli, Machine Learning for Audio, Image and Video Analysis -Theory and Applications, Springer, 2015
2. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006
3. Charu C. Aggarwal, Machine Learning for Text. Springer, 2018

REFERENCES:

1. Milan Sonka, Vaclav Hlavac, Roger Boyle, Image processing, analysis, and machine vision, Cengage Learning, Fourth edition, 2015
2. Reinhard Klette Concise Computer Vision: An Introduction into Theory and Algorithms, Springer, 2014
3. Lawrence R. Rabiner, Biing-Hwang Juang, and B. Yegnanarayana, Fundamentals of Speech Recognition, Pearson, 2008
4. Tom Mitchell, Machine Learning, McGraw Hill, 3rd Edition, 1997.
4. Charu C. Aggarwal, Data Classification Algorithms and Applications, CRC Press, 2014.
5. Ethem Alpaydin, Introduction to Machine Learning, MIT Press, Prentice Hall of India, Third Edition 2014.

COURSE OUTCOMES

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At the end of the course, the students should be able to

CO 1: Demonstrate the image, video and speech and text data acquisition, representation, and storage approaches

CO 2: Analyse Machine Learning algorithms and their limitations

CO3: Apply Machine Learning algorithms on image, video, audio, and text data

CO4: Relate current research papers on Machine Learning and understand the issues raised by current research.

CO-PO Mapping:

COs	POs					PSO
	1	2	3	4	5	
1	3		1			
2	3		1	2	1	1
3	3		1	2	1	1
4	3	2	2	1	2	3
5			2	1	1	2

NP23103 FOUNDATIONS OF NATURAL LANGUAGE PROCESSING L T P C
3 0 0 3

COURSE OBJECTIVES:

1. To teach the fundamentals of NLP, and also to make them for understanding CFG, PCFG in NLP.
2. To know the role of semantics of sentences and pragmatic.
3. To teach the basic concepts of speech processing along with analysis and modeling.

UNIT-I Introduction

9

Origins and challenges of nlp – language modeling: grammar-based lm, statistical lm –regular expressions, finite-state automata – english morphology, transducers for lexicon and rules, tokenization, detecting and correcting spelling errors, minimum edit distance.

UNIT-II Word Level Analysis

9

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Unsmoothed n-grams, evaluating n-grams, smoothing, interpolation and back off – word classes, part-of-speech tagging, rule-based, stochastic and transformation-based tagging, issues in pos tagging – hidden markov and maximum entropy models.

UNIT-III Syntactic Analysis

9

Context free grammars, grammar rules for english, treebanks, normal forms for grammar – dependency grammar – syntactic parsing, ambiguity, dynamic programming parsing – shallow parsing – probabilistic cfg, probabilistic cyk, probabilistic lexicalized cfgs – feature structures, unification of feature structures.

UNIT-IV Semantics and Pragmatics

9

Requirements for representation, first-order logic, description logics – syntax-driven semantic analysis, semantic attachments – word senses, relations between senses ,thematic roles, selectional restrictions – word sense disambiguation, wsd using supervised, dictionary & thesaurus, bootstrapping methods – word similarity using thesaurus and distributional methods.

UNIT-V Basic Concepts of Speech Processing

9

Speech fundamentals: articulatory phonetics – production and classification of speech sounds; acoustic phonetics – acoustics of speech production; review of digital signal processing concepts; short-time fourier transform, filter- bank and lpc methods.

TOTAL PERIODS: 45

TEXT BOOKS:

1. Daniel Jurafsky, James H. Martin—Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech, Pearson Publication, 2014.
2. Steven Bird, Ewan Klein and Edward Loper, —Natural Language Processing with Python, First Edition, O'Reilly Media, 2009.
3. Lawrence Rabiner And Biing-Hwang Juang, “Fundamentals Of Speech Recognition”, Pearson Education, 2003.

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4. Daniel Jurafsky And James H Martin, “Speech And Language Processing – An Introduction To Natural Language Processing, Computational Linguistics, And Speech Recognition”, Pearson Education, 2002

REFERENCE BOOKS:

1. Frederick Jelinek, “Statistical Methods Of Speech Recognition”, MIT Press, 1997.
2. Breck Baldwin, —Language Processing with Java and Ling Pipe Cookbook, Atlantic Publisher, 2015
3. Richard M Reese, —Natural Language Processing with Java, OReilly Media, 2015.
4. Nitin Indurkha and Fred J. Damerau, —Handbook of Natural Language Processing, Second Edition, Chapman and Hall/CRC Press, 2010.
5. Tanveer Siddiqui, U.S. Tiwary, —Natural Language Processing and Information Retrieval, Oxford University Press, 2008.

COURSE OUTCOMES:

- CO1. Learn the fundamentals of natural language processing
- CO2. Understand the use of CFG and PCFG in NLP
- CO3. Understand the role of semantics of sentences and pragmatic
- CO4. Introduce Speech Production And Related Parameters Of Speech.
- CO5. Show The Computation And Use Of Techniques Such As Short Time Fourier Transform, Linear Predictive Coefficients and Other Coefficients in The Analysis Of Speech.

CO-PO Mapping:

COs	POs					PSO
	1	2	3	4	5	
1	1	1	1	1		1
2	1	3	1	1		2

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3	2	1	2	1		2
4	1	1	1	1	2	2
5	1	1	1	1		1

RM24107

RESEARCH METHODOLOGY IN IPR

L P T C 3 0 0 3

COURSE OBJECTIVES

To enable the students to understand the formulation of research problem.
To be familiar with data collection and literature survey process
To know the statistical concepts in experimentation
To acquire knowledge in writing research proposal
To understand about patent rights and its importance

UNIT I RESEARCH PROBLEM FORMULATION

9

Meaning of research, Objectives of Research, Types of research, Significance of Research, Search process, Selecting the problem, Necessity of defining the problem, Meaning of Research design, Need for research design, features of a good design, Different research designs.

UNIT II LITERATURE SURVEY

9

Quantitative and Qualitative data, Scaling, Scaling Techniques, Experiments and Surveys, Collection of primary and secondary data, Data preparation process. Research problems, Effective literature studies approaches, Survey for existing literature, Procedure for reviewing the literature, analysis and assessment.

UNIT III DESIGN OF EXPERIMENTS

9

Strategy of Experimentation Typical applications of experimental design, Guidelines for designing experiments; Basic statistical concepts - Statistical concepts in experimentation, Regression approach to analysis of variance.

UNIT IV RESEARCH PROPOSAL AND WRITING

9

Contents of a research proposal, Writing a research report Research writing in general, Referencing. Writing a bibliography, Presentation and assessment by a review committee, Plagiarism, Research ethics.

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UNIT V INTELLECTUAL PROPERTY RIGHTS

9

Intellectual Property- Definition, WTO, Fundamentals of Patent, Copyright, The rights of the owner, Term of copyright, Register of trademark, Procedure for trade mark, Term of trademark; New Developments in IPR- Administration of patent system, IPR of Biological Systems, Computer Software.

TOTAL PERIODS: 45

COURSE OUTCOMES

1. Upon the completion of the course, students will be able to
2. Identify research problems.
3. Collect and prepare suitable data for research.
4. Design experiments for different statistical concepts.
5. Research proposals reports and apply their research work for patent through IPR

REFERENCES

1. C.R Kothari and Gaurav Garg. "Research Methodology Methods and Techniques", 4th Edition, New Age International Publishers, 2019.
2. Ranjit Kumar, "Research Methodology" A step by Step Guide for beginners, 2nd Edition, Pearson Education, 2010.
3. Douglas C. Montgomery, "Design and Analysis of Experiments", 9th edition, Wiley Publishers, 2017.
4. Neeraj Pandey and Khushdeep Dharni, "Intellectual Property rights", PHI Learning, 2014.
5. Dr.R. Radhakrishnan and Dr.S.Balasubramanian, "Intellectual Property Rights, text and cases", Excel Books, New Delhi.

CO-PO Mapping:

COs	POs					PSO
	1	2	3	4	5	
1	3	3	3			3

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2	3	3	1			3
3	3	2	3	2		3
4	3	3	2		1	3
5	3	3	3	2	3	3

AL24106 ADVANCED ARTIFICIAL INTELLIGENCE L T P C 3 0 0 3

COURSE OBJECTIVES

- To understand the role of logic in artificial intelligence
- To understand knowledge and reasoning under certainty and uncertainty
- To enable design and implement AI principles for problem solving, inference and perception
- To implement algorithms on simple and complex decision making

UNIT I KNOWLEDGE AND REASONING

9

Propositional Logic: Simple logic – Propositional theorem proving – Effective propositional model checking – Agents based on propositional logic – First order logic – Representation – syntax and semantics – using first order logic – knowledge engineering in first order logic; Propositional vs first order inference - Unification and lifting – forward and backward chaining – resolution.

UNIT II PLANNING AND REPRESENTATION

9

Classical planning: definition – algorithms for planning as state space search – planning graphs – classical planning approaches – analysis; Knowledge representation: Ontological engineering - categories and objects - events – mental events and mental objects – reasoning systems for categories
– reasoning with default information – Case study: The internet shopping world.

UNIT III UNCERTAIN KNOWLEDGE

9

Uncertainty - Quantifying uncertainty - Acting under uncertainty – basic probability notation – inference using full joint distributions – independence - Bayes’ rule and its use

UNIT IV PROBABILISTIC REASONING

9

Representing knowledge in an uncertain domain – semantics of Bayesian networks –

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efficient representation of conditional distributions – exact and approximate inference in Bayesian networks - relational and first order probability models – Time and uncertainty – inference in temporal models- Bayesian Learning: Learning with complete and hidden data – Expectation Maximization Algorithm; Hidden Markov Models – Kalman filters – dynamic Bayesian networks – multiple object tracking.

UNIT V DECISION MAKING AND LEARNING

9

Reinforcement Learning : Basics of Reinforcement learning – Active and Passive reinforcement learning – Generalization – Applications ;Making simple decisions: combining beliefs and desires – The basis of utility theory – Utility functions – multi attribute utility functions – decision networks -The value of information – Decision theoretic expert systems; Sequential Decision problems – value iteration – policy iteration – Partially Observable MDPs ; Decisions with Multiple Agents : Game Theory – Mechanism Design.

TOTAL PERIODS :45

TEXT BOOK

1. Artificial Intelligence: A modern approach, by Stuart Russell and Peter Norvig, Third Edition, Pearson Series in Artificial Intelligence. 2020.

REFERENCES

1. Artificial Intelligence Engines: A Tutorial Introduction to the Mathematics of DeepLearning by James V Stone, Sebtel Press, 2019.
2. Artificial Intelligence by Example: Acquire advanced AI, machine learning, and deeplearning design skills by Denis Rothman, 2nd Edition, 2020.
3. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006
4. Ethem Alpaydin, Introduction to Machine Learning, MIT Press, Prentice Hall of India, Third Edition 2014.

COURSE OUTCOMES

At the end of the course, the students should be able to

- CO 1: Identify the role of propositional and first order logic in Artificial Intelligence
- CO 2: Demonstrate an understanding of knowledge and reasoning under certainty and uncertainty.
- CO 3: Apply principles of AI in solutions that require problem solving, inference, perception
- CO 4: Implement algorithms on simple and complex decision making.

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CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	1		1			
2	2	2	1	1	1	1
3	1		1	1	1	1
4			1			1
5			1	1	1	1

ML2452 MACHINE LEARNING LABORATORY L T P C 0 0 4 2

COURSE OBJECTIVES

- To learn the fundamentals and practice Python codes for Data Science.
- To understand and implement machine learning algorithms using Python
- To apply machine learning algorithms for image, video and speech and text recognition

LIST OF EXERCISES

1. Python packages for Machine Learning
2. Image processing using OpenCV
3. Text processing using NLTK
4. Speech processing - time and frequency domain features
5. Implement classification using Logistic regression
6. Implement classification using Multilayer perceptron
7. Implement classification using SVM
8. Implement Adaboost and Bagging ensemble techniques
9. Implement K-means and hierarchical Clustering
10. Mini project - build image/video/speech/text application, that maps to the sustainable development goals 7, 11, 13 etc. (7: Affordable and clean energy, 11: Sustainable cities and communities, 13: Climate action)

TOTAL PERIODS: 45

COURSE OUTCOMES

At the end of the course, the students should be able to

CO 1: Use Python packages for Machine Learning applications

CO 2: Implement image/video/speech /text processing techniques using tools such as

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openCV and NTLK

CO3: Build and validate machine learning models using supervised and unsupervised learning methods

CO4: Design and develop Machine Learning based solutions for image, video and speech and text analysis through research/investigations.

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	3		1			
2	3		1	2	1	1
3	3		1	2	1	1
4	3	2	2	1	2	3

NL2451 ADVANCED DATA STRUCTURES LABORATORY
0 0 4 2

L T P C

COURSE OBJECTIVES

- To analyze the existing algorithms and design efficient algorithms
- To learn the implementation of linear data structures for concurrency
- To learn the implementation of advanced data structures such as search trees and
- To explore the advanced concurrent data structures such as hash table, Priority Queue
- To learn to apply principles of efficient algorithm design and learn various advanced algorithms

LIST OF EXERCISES

1. Insertion sort and merge sort with complexity analysis
2. Implementation and applications of classic linear data structures, namely, linked lists, queues, and stacks
3. Implementation of various locking and synchronization mechanisms for concurrent linked lists

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4. Implementation of various locking and synchronization mechanisms for concurrent queues and stacks.
5. Application of binary search trees/2-D Tree
6. Application of Heaps
7. Dynamic programming (e.g. Longest common subsequence, 0/1 Knap-sack)
8. Recursive backtracking (e.g. N-Queen's problem)
9. Greedy technique (e.g. Job/event scheduling, Minimum spanning tree)
10. Implementation and demonstration of hashing and concurrent hashing
11. Implementation of approximation algorithms / randomized algorithms.
12. Implementation of graph algorithm
13. Mini project: For a given application, identify and solve using suitable data structures & design techniques and database

TOTAL PERIODS: 45

COURSE OUTCOMES

After the completion of this course, students will be able to

CO1: Analyze the existing algorithms.

CO2: Implement and apply concurrency in linked lists, stacks, queues, heap and hashing.

CO3: Implement advanced search trees and graph algorithms

CO4: Implement approximation, randomized, parallel algorithm, Greedy and Dynamic Algorithms

CO5: Solve real time applications using advanced algorithm concepts

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	

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1	2		1			1
2	2		1			1
3	2		1			1
4	2		1			1
5	2	3	3			3

SEMESTER II

NP24108 ADVANCED DATA VISUALIZATION TECHNIQUES L T P C
4 0 0 4

COURSE OBJECTIVES:

- To understand how to accurately represent voluminous complex data set in web and from other data sources.
- To understand the methodologies used to visualize large data sets.
 - To understand the various process involved in data visualization.
 - To apply data visualization techniques.
 - To understand the different security aspects involved in data visualization.

UNIT – I: INTRODUCTION

9

Information visualization – effective data analysis – traits of meaningful data – visual perception –making abstract data visible – building blocks of information visualization – analytical interaction – analytical navigation – optimal quantitative scales – reference lines and regions – trellises and crosstabs – multiple concurrent views – focus and context – details on demand – over-plotting reduction – analytical patterns – pattern examples.

UNIT - II: DATA VISUALIZATION TECHNIQUES

9

Mapping - Time series - Connections and correlations - Scatterplot maps - Trees, Hierarchies and Recursion - Networks and Graphs, Info graphics.Univariate data visualization: Bar chart,

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histograms, frequency polygram, box plots, dot plots - Bivariate data visualization - Multivariate data visualization: Histogram matrix, scatter plot matrix, multiple box plot and trellis plot - Visualizing groups - Dynamic techniques

UNIT - III: VISUALIZING DATA PROCESS

9

Acquiring data, - Where to Find Data, Tools for Acquiring Data from the Internet, Locating Files for Use with Processing, Loading Text Data, Dealing with Files and Folders, Listing Files in a Folder, Asynchronous Image Downloads, Advanced Web Techniques, Using a Database, Dealing with a Large Number of Files. Parsing data - Levels of Effort, Tools for Gathering Clues, Text Is Best, Text Markup Languages, Regular Expressions (regexps), Grammars and BNF Notation, Compressed Data, Vectors and Geometry, Binary Data Formats, Advanced Detective Work.

UNIT - IV: INTERACTIVE DATA VISUALIZATION

9

Drawing with data – Scales – Axes – Updates, Transition and Motion – Interactivity - Layouts – Geomapping – Exporting, Framework – T3, .js, tablo.

UNIT - V: SECURITY DATA VISUALIZATION

9

Port scan visualization - Vulnerability assessment and exploitation - Firewall log visualization - Intrusion detection log visualization -Attacking and defending visualization systems – Creating Security visualization system.

TOTAL: 45 PERIODS

COURSE OUTCOMES:

At the end of the course, the student should be able to:

CO1: Understand the representation of complex and voluminous data.

CO2: Design and use various methodologies present in data visualization.

CO3: Understand the various process and tools used for data visualization.

CO4: Use interactive data visualization to make inferences.

CO5: Discuss the process involved and security issues present in data visualization

TEXT BOOKS:

1. Scott Murray, —Interactive data visualization for the webl, O“Reilly Media, Inc., 2013.
2. Stephen Few, "Now you see it: Simple Visualization techniques for quantitative analysis", Analytics Press, 2009.

REFERENCE BOOKS:

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1. Ben Fry, —Visualizing Data, O'Reilly Media, Inc., 2007.
2. Greg Conti, —Security Data Visualization: Graphical Techniques for Network Analysis, NoStarch Press Inc, 2007.
3. Nathan Yau, "Data Points: Visualization that means something", Wiley, 2013.

CO-PO MAPPING:

COs	POs					PSO
	1	2	3	4	5	
1	2	3	1			2
2	2	3	1			2
3	2	3	2			2
4	2	3	2			2
5	2	3	3	1	2	2

NL24109 ADVANCED NATURAL LANGUAGE PROCESSING L T P C

COURSE OBJECTIVES: 3 0 0 3

- Understand and implement advanced NLP models like Word2Vec, GloVe, and Transformers (BERT, GPT).
- Learn sequence-to-sequence models and attention mechanisms for tasks like summarization and translation.
- Apply transfer learning and fine-tuning techniques on pretrained models for real-world NLP applications.
- Explore practical applications such as chatbots, question-answering systems, and sentiment analysis.

UNIT I WORD REPRESENTATIONS 10

Word2Vec: Skip - gram and CBOW-GloVe: Global Vectors for Word Representation - FastText: Subword Information in Word Embeddings - Embedding evaluation techniques - Word

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Analogies with Word2Vec - Character-level models and subword embeddings - Dimensionality Reduction in Word Embeddings - Word Embeddings for Language Modeling - Continuous Bag of Words (CBOW) Model - Training Word Embeddings using Neural Networks - Subword Embeddings for Rare Words - Comparison of Word2Vec, GloVe, and FastText - Fine-tuning Embeddings for Domain-specific Tasks - Evaluating Embedding Models with Semantic Tasks - Embedding Visualization Techniques (e.g., t-SNE)

UNIT II TRANSFORMERS

9

Transformer Architecture (Vaswani et al.) - Attention Mechanism: Self-Attention and Multi-head Attention - Positional Encoding in Transformers - BERT (Bidirectional Encoder Representations from Transformers) - GPT (Generative Pretrained Transformer) - RoBERTa: Robustly Optimized BERT Pretraining Approach - Pretrained models and their applications in NLP - Transformers for Text Classification – Fine tuning Pretrained Models - Tokenization Techniques in Transformers - Performance Comparison of BERT, GPT, and RoBERTa - Transfer Learning in NLP

UNIT III SEQUENCE MODELING

9

Recurrent Neural Networks (RNNs) - Long Short-Term Memory (LSTM) Networks - Gated Recurrent Units (GRUs) - Encoder-Decoder Architecture - Sequence-to-sequence models with attention mechanisms - Challenges in training RNNs for NLP tasks - Bidirectional RNNs for Sequence Processing - Sequence Padding and Masking in Deep Learning - Attention Mechanisms in Sequence Models - Variational Autoencoders for Sequence Generation - Sequence Labeling Tasks - Applications of Seq2Seq Models in NLP (e.g., Translation)

UNIT IV ADVANCED TOPICS IN NLP

9

Fine-tuning BERT for text classification - Fine-tuning GPT for text generation - Adapting Pretrained Models to Specific Tasks - Domain-Specific Fine-tuning - Knowledge Distillation in NLP - Pretraining Strategies for NLP Models - Hyperparameter Tuning in Fine-tuning Pretrained Models - BERT vs GPT in Fine-tuning for Different Tasks - Fine-tuning for Low-Resource Languages - Transfer Learning with Transformer Models

UNIT V APPLICATIONS OF ADVANCED NLP

8

Text Summarization (Extractive and Abstractive) - Question Answering Systems - Dialog Systems and Chatbots - Information Extraction Techniques - Named Entity Recognition (NER) -

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Sentiment Analysis - Machine Reading Comprehension - Conversational AI Models - Multi-turn Dialogue Systems

TOTAL PERIODS: 45

Textbooks:

1. **"Speech and Language Processing"** by Daniel Jurafsky and James H. Martin
2. **"Deep Learning for Natural Language Processing"** by Palash Goyal, Sumit Pandey, and Karan Jain

References:

1. **"Neural Network Methods in Natural Language Processing"** by Yoav Goldberg
2. **"Natural Language Processing with Transformers"** by Lewis Tunstall, Leandro von Werra, and Thomas Wolf
3. **"Pattern Recognition and Machine Learning"** by Christopher Bishop

COURSE OUTCOMES:

- CO 1: Ability to use word embeddings like Word2Vec, GloVe, and FastText.
CO 2: Expertise in using transformer models (BERT, GPT, RoBERTa).
CO 3: Proficiency in sequence-to-sequence models for translation and summarization.
CO 4: Mastery of transfer learning, fine-tuning, and zero-shot learning.
CO 5: Ability to implement advanced applications like text summarization and chatbots.

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	2		2			1
2	2		2			2
3	2		2			1
4			2			2

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NT24110

TEXT ANALYTICS IN NLP

L T P C 3 0 0 3

COURSE OBJECTIVES:

- To make students learn about different text analytics techniques, such as text classification, sentiment analysis, and information extraction.
- To make students understand the principles of natural language processing, such as part-of-speech tagging, syntactic parsing, and semantic analysis.
- To understand the applications of text analytics and natural language processing in various fields, such as customer service, social media analysis, and healthcare.
- To develop critical thinking skills and the ability to evaluate and analyze different text analytics and natural language processing techniques.

UNIT - I

9

Language in Cognitive Science: Definitions of language, Language as a rule-governed dynamic system, Knowledge of language, Modes of language: spoken and written, Language system as expression and content Language Analysis and Computational Linguistics: What is Language Analysis?, Form, Function and Meaning in Language Analysis, Levels of Linguistic Analysis: Phonetics, Phonology, Morphology, Syntax, Semantics, Discourse, Pragmatics, Lexicology

UNIT II

9

Shallow Parsing and Tools for NLP: Morphological Analysis, Tokenization & PoS Tagging, Chunking & Multi word expression (MWE), Named-Entity Recognition, Lemmatizer & Stemming, Morphological Synthesis Deep Parsing and Tools for NLP: Syntactic Parsing Techniques and algorithms, Semantic Parsing, Information Extraction, Automatic Summarization, Anaphora Resolution, Pragmatics and Discourse analysis

UNIT - III

9

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Text Classification: Bag of words representation. Vector space model and cosine similarity. Relevance feedback and Rocchio algorithm. Versions of nearest neighbor and Naive Bayes for text, Text Classification Using Support Vector Machine (SVM), Statistical Parsing Language Learning: Classification problems in language: word-sense disambiguation, sequence labelling. Hidden Markov models (HMM's). Viterbi algorithm for determining most-probable state sequences, Training the parameters of HMM's. Use of HMM's for speech recognition.

UNIT - IV

9

NLP with ANN: Issues in using ANN with text, understanding word and sentence embedding, Introduction to NLTK, Binary encoding, TF, TF-IDF encoding, Latent Semantic analysis encoding, Latent Dirichlet Allocation, Word2Vec models (Skipgram, CBOW, Glove, one hot Encoding), Sequence-to-sequence models (Seq2Seq) - GloVe: Global Vectors for Word Representation

UNIT - V

9

Speech Processing: Articulatory Phonetics, Speech Sounds and Phonetic Transcription, Acoustic Phonetics, Phonology, Computational Phonology, Automatic Speech Recognition (ASR), Speech Recognition Approaches, Text to Speech (TTS) system, Speech Synthesis Approaches
NLP Applications: Lexicon, Dictionaries, thesaurus, Transliteration, Spell Checker, Grammar Checker, Domain identification, Language identification, Auto suggest/ Auto complete, Machine Translation, Question answering & dialogue agents, OCR, Hand Writing Recognition, Sentiment analysis

TOTAL PERIODS :45

Textbooks:

1. Bird S, Klein E, Loper E. Natural language processing with Python: analyzing text with the natural language toolkit. " O'Reilly Media, Inc."; 2009.
2. Thanaki J. Python natural language processing. Packt Publishing Ltd; 2017.

Reference books:

1. Hardeniya N, Perkins J, Chopra D, Joshi N, Mathur I. Natural language processing: python and NLTK. Packt Publishing Ltd; 2016.
2. Srinivasa-Desikan B. Natural Language Processing and Computational Linguistics: A practical guide to text analysis with Python, Gensim, spaCy, and Keras. Packt Publishing Ltd; 2018

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COURSE OUTCOMES:

CO1. Ability to understand and apply different text analytics techniques to analyze and classify text data.

CO2. Ability to apply different natural language processing techniques to process and understand natural language text.

CO3. Ability to evaluate and analyze different text analytics and natural language processing techniques for different applications.

CO4. Students will be prepared for advanced courses and research in natural Language processing and text analysis.

CO-PO Mapping:

COs	POs					PSO
	1	2	3	4	5	
1	2	2	2	1	1	1
2	2	2	2	2	1	2
3	2	2	2	2	1	1
4	3	2	2	3	2	2
5	3	2	3	3	2	2

IR24111 INFORMATION RETRIEVAL AND TEXT MINING

L T P C

COURSE OBJECTIVES:

3 0 0 3

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- Understand the fundamentals of information retrieval and text mining.
- Explore various retrieval models and their applications.
- Implement text mining techniques for document classification and clustering.
- Analyze sentiment and opinion mining techniques.

UNIT I FUNDAMENTALS OF INFORMATION RETRIEVAL

10

Vector Space Models - TF-IDF and BM25 - Boolean and Probabilistic Models - Term Frequency and Inverse Document Frequency - Relevance Feedback and Query Expansion - Inverted Indexing and its Applications - Precision and Recall in Information Retrieval - Evaluation Measures: MAP, NDCG - Ranking Algorithms in Search Engines - Text Preprocessing Techniques - Web Crawling and Indexing - Retrieval Models in Search Engines

UNIT II WEB SEARCH AND OPTIMIZATION

9

Crawling and Indexing Web Data - Link Analysis: PageRank and HITS - Query Optimization Techniques - Information Retrieval with NoSQL Databases - Query Expansion and Relevance Feedback - Personalized Search Engines - Cross-lingual Search and Retrieval - Real-Time Search Engines - Performance Evaluation in Web Search - Search Engine Algorithms

UNIT III TEXT MINING TECHNIQUES

8

Text Clustering and Classification - Supervised vs. Unsupervised Learning in Text Mining - Latent Dirichlet Allocation (LDA) - Non-Negative Matrix Factorization (NMF) - Text Preprocessing: Tokenization, Lemmatization - Sentiment Analysis and Opinion Mining - Text Classification Algorithms (SVM, Naive Bayes) - Document Similarity Techniques - Named Entity Recognition (NER) - Text Summarization Methods

UNIT IV SENTIMENT ANALYSIS AND OPINION MINING

8

Lexicon-based vs. Machine Learning Approaches - Sentiment Polarity Classification - Aspect-Based Sentiment Analysis - Fine-Grained Sentiment Analysis - Sentiment Analysis for Social Media - Opinion Mining from Reviews - Emotion Detection in Text - Sentiment Analysis in Multilingual Texts

UNIT V EMERGING TRENDS IN TEXT MINING

9

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Neural IR Models - Cross-Lingual Information Retrieval - Multimodal IR (Text + Image + Audio) - Knowledge Graphs for Text Mining - Deep Learning for Text Mining - Text Mining in Healthcare and Finance - Legal and Ethical Issues in Text Mining - Real-Time Text Mining

TOTAL PERIODS: 45

Textbooks:

1. **"Introduction to Information Retrieval"** by Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze
2. **"Search Engines: Information Retrieval in Practice"** by Bruce Croft, Donald Metzler, and Trevor Strohman

References:

1. **"Text Mining: Classification, Clustering, and Applications"** by Michael W. Berry and Malu Castellanos
2. **"Foundations of Statistical Natural Language Processing"** by Christopher D. Manning and Hinrich Schütze
3. **"Mining the Web: Discovering Knowledge from Hypertext Data"** by Soumen Chakrabarti

COURSE OUTCOME

- CO1:** Apply retrieval models like TF-IDF and BM25 in search engines.
CO2: Implement text mining techniques for clustering and topic modeling.
CO3: Use sentiment analysis techniques for social media and reviews.
CO4: Understand emerging trends in information retrieval and text mining.
CO5: Build real-world applications using text mining and IR techniques.

CO-PO Mapping:

COs	POs					PSO
	1	2	3	4	5	

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1	2	1				1
2	2	1				2
3	2	1	2			1
4	2		2	1		2
5	2	1	1	1	2	2

ND24112 MACHINE TRANSLATION AND NLP SYSTEMS DESIGN L T P C

COURSE OBJECTIVES: 3 0 2 5

- To understand the fundamental principles of Machine Translation and its various types.
- To gain hands-on experience in implementing machine translation models using traditional and neural-based approaches.
- To learn how to evaluate the effectiveness of machine translation systems using standard metrics.
- To explore real-world challenges in multilingual NLP systems and develop solutions for low-resource languages.

UNIT I FUNDAMENTALS OF MACHINE TRANSLATION 10

History and Development of Machine Translation - Overview of Rule-Based Machine Translation (RBMT) - Statistical Machine Translation (SMT) Techniques - Neural Machine Translation (NMT) and Sequence-to-Sequence Models - Phrase-Based Translation vs. Word-Based Translation - Data Collection and Preprocessing for MT - Challenges in MT for Low-Resource Languages - Machine Translation Paradigms: SMT, NMT, Hybrid Systems - Role of Alignment Models in Translation - Word Alignment and Sentence Alignment Techniques - Bilingual Corpora and Parallel Datasets in MT - Evaluation Metrics for Machine Translation Systems - Post-Editing in MT Systems - Language Modeling in MT - Building Basic Translation Models Using Open-Source Tools (e.g., Moses)

UNIT II NEURAL MACHINE TRANSLATION 8

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Introduction to NMT and its Benefits Over SMT - Sequence-to-Sequence Model in NMT - Attention Mechanism and Its Role in NMT - Encoder-Decoder Architecture - Bidirectional RNNs and Their Application in NMT - Transformer Architecture in NMT (Vaswani et al.) - Pretrained Models in NMT: mBERT, MarianMT - Data Preprocessing and Tokenization for NMT - Fine-Tuning Pretrained Models for Translation - Transfer Learning in NMT for Low-Resource Languages - Beam Search Decoding in NMT - Evaluation of NMT Models using BLEU and Other Metrics

UNIT III MULTILINGUAL AND LOW-RESOURCE MACHINE TRANSLATION 10

Multilingual NMT Models and Their Advantages - Transfer Learning in Multilingual MT Systems - Handling Code-Switching in MT - Challenges in Low-Resource Languages and Solutions - Zero-Shot Translation in NMT - Cross-Lingual Transfer Learning - Use of Unsupervised Translation Methods - Data Augmentation Techniques for Low-Resource Languages - Evaluation of Multilingual MT Systems - Case Studies: NMT for African Languages, Indian Languages, etc. - Language-agnostic Representations for MT - Fine-Tuning mBERT for Multilingual Tasks

UNIT IV MT SYSTEM DESIGN AND EVALUATION

7

System Architecture of an MT Pipeline - Data Collection for MT: Challenges and Techniques - Text Preprocessing: Tokenization, Normalization, and Lemmatization - Building Custom Translation Datasets - Model Training for NMT Systems - Evaluation Techniques for MT: BLEU, METEOR, TER - Error Analysis in MT Outputs - Post-Editing and Improving MT Outputs - Real-Time Translation System Design - Incorporating User Feedback into MT Systems - Optimization of MT Systems for Scalability - Testing and Deploying an MT System

UNIT V ADVANCED TOPICS IN MT AND NLP SYSTEMS

10

Neural MT with Reinforcement Learning - Multimodal Machine Translation - Integrating Speech Recognition with MT for Real-Time Translation - MT for Domain-Specific Applications (e.g., Legal, Medical) - Ethical Issues in Machine Translation - Privacy Concerns and Data Security in MT - Bias in MT Models and Mitigation Strategies - Speech-to-Text and Text-to-Speech Integration in MT - Adapting MT Systems for Specific User Groups (e.g., children, elderly) - Building Chatbots with Translation Capabilities - Future Trends in Machine Translation: Multimodal and Cross-Lingual Models - Open Source Tools and Frameworks for MT (e.g., OpenNMT, Fairseq)

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TOTAL PERIODS: 45

TEXTBOOK

1. "Neural Machine Translation" by Philipp Koehn
2. "Statistical Machine Translation" by Philipp Koehn

REFERENCE

1. "Speech and Language Processing" by Daniel Jurafsky & James H. Martin
2. "Neural Networks for Pattern Recognition" by Christopher Bishop
3. "Statistical Methods for Machine Translation" by R. K. Gupta, S. S. Saini, & S. A. Khan

Course Outcomes

CO1: Develop an understanding of the theoretical foundations of Machine Translation.

CO2: Implement machine translation systems using rule-based, statistical, and neural machine translation approaches.

CO3: Apply preprocessing techniques for multilingual data.

CO4: Evaluate machine translation outputs using standard evaluation metrics.

CO5: Build and optimize a real-time multilingual translation system.

PRACTICAL EXPERIMENTS

1. Explore the history and development of machine translation (MT) systems. Implement a basic rule-based or statistical machine translation (SMT) system for a given pair of languages.
2. Build a neural machine translation system using sequence-to-sequence models with attention mechanism. Train the model on a parallel corpus and evaluate its translation performance.
3. Collect a multilingual dataset and perform preprocessing tasks such as tokenization, sentence segmentation, and alignment. Prepare the data for training a machine translation model.
4. Implement common MT evaluation metrics like BLEU, METEOR, and TER. Use these metrics to evaluate the performance of your translation models and compare their results.
5. Apply transfer learning techniques to build an MT system for a low-resource language. Investigate the challenges and propose solutions for effective translation.
6. Implement a NMT system with attention and evaluate its performance on a real-world dataset (e.g., English-French, English-Spanish). Compare it with traditional rule-based and SMT models.
7. Implement post-editing techniques to improve the translation output. Perform error analysis to identify common translation mistakes and propose methods to handle them.

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8. Design and implement a real-time translation system using a pre-trained translation model. Evaluate its speed and accuracy for live translation tasks.

TOTAL PERIODS: 30

TOTAL PERIODS: 45+30=75

SOFTWARE REQUIREMENTS

OpenNMT,Fairseq,MarianNMT,TensorFlow

COURSE OUTCOMES

CO1: Understand the history and evolution of machine translation systems.

CO2: Develop and implement neural machine translation models with advanced techniques like attention mechanisms.

CO3:Evaluate the performance of machine translation systems using standard metrics such as BLEU and METEOR.

CO4: Apply transfer learning to build effective translation systems for low-resource languages.

CO5: Design and deploy real-time machine translation systems for practical use cases.

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	2	2	2	1	1	1
2	2	2	2	2	1	2
3	2	2	2	2	1	1
4	3	2	2	3	2	2
5	3	2	3	3	2	2

NP2453

ADVANCED NLP LABORATORY

L T P C

0 0 4 2

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COURSE OBJECTIVES:

- To provide hands-on experience with advanced Natural Language Processing (NLP) techniques and tools.
- To implement state-of-the-art NLP models using libraries like Hugging Face, SpaCy, and NLTK.
- To build applications using transformer-based models and fine-tuned pre-trained models.
- To explore various advanced NLP tasks, including text classification, named entity recognition, and question answering.

EXPERIMENTS

1. Implement the Word2Vec model using a given corpus of text and explore word similarities, analogies, and clustering. Compare its performance with the GloVe model.
2. Train a custom Named Entity Recognition (NER) model using SpaCy on a labeled dataset. Identify and classify entities such as persons, organizations, and locations in text.
3. Fine-tune a pre-trained BERT or RoBERTa model for sentiment analysis. Evaluate the model's performance on a sentiment-labeled dataset (positive, negative, neutral).
4. Implement extractive and abstractive text summarization using pre-trained models like BERTSUM or GPT-3. Evaluate the quality of generated summaries using ROUGE scores.
5. Build a question answering system using a pre-trained BERT model. Fine-tune the model on a question-answering dataset (such as SQuAD) and evaluate its performance.
6. Implement a text generation model using GPT-2 or GPT-3. Fine-tune the model for a specific domain or task and evaluate its ability to generate coherent and relevant text.
7. Perform topic modeling using Latent Dirichlet Allocation (LDA) and Non-Negative Matrix Factorization (NMF). Extract topics from a collection of documents and evaluate their coherence.
8. Fine-tune a multilingual BERT (mBERT) model for a text classification task. Evaluate its ability to handle multiple languages and analyze its performance on a multilingual dataset.
9. Develop a multi-language text classification system using a pre-trained multilingual transformer model like mBERT. Evaluate its performance across different languages and categories.

TOTAL PERIODS: 60

SOFTWARE REQUIREMENTS

Anaconda (with Jupyter Notebooks), Kaggle Kernels

Course Outcome:

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CO1: Gain hands-on experience in working with multilingual models.

CO2: Develop an end-to-end text classification pipeline.

CO3: Apply state-of-the-art models to real-world data and evaluate their performance.

CO4: Evaluate the effectiveness of text classification models using performance metrics.

CO5: Implement data preprocessing, model fine-tuning, and evaluation techniques for NLP tasks.

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	2	2	2	1	1	1
2	2	2	2	2	1	2
3	2	2	2	2	1	1
4	3	2	2	3	2	2
5	3	2	3	3	2	2

NS2454

INTERNSHIP AND SEMINAR

L T P C 0 0 4 2

COURSE OBJECTIVES:

The objective of this course is to enable the students to

- Ability to identify and solve practical and real-world IT applications
- Ability to survey literature, identify gaps, and write well-documented report
- Display effective technical presentation skills

COURSE OUTCOMES:

On successful completion of this course, the students should be able to

CO1: Ability to learn and apply new and existing concepts for solving engineering problems

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CO2: function effectively on multidisciplinary teams or as an individual

CO3: clearly communicate ideas through preparation of a well-documented internship report, and oral demonstration of technical knowledge

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	1		3	2		3
2		1			2	1
3	1	2	1		2	1

SEMESTER III

DL24113

ADVANCED DEEP LEARNING

L T P C 3 0 2 5

COURSE OBJECTIVES:

- Develop and Train Deep Neural Networks.
- Develop a CNN, R-CNN, Fast R-CNN, Faster-R-CNN, Mask-RCNN for detection and recognition
- Build and train RNNs, work with NLP and Word Embeddings
- The internal structure of LSTM and GRU and the differences between them
- The Auto Encoders for Image Processing

UNIT 1 DEEP LEARNING CONCEPTS

6

Fundamentals about Deep Learning. Perception Learning Algorithms. Probabilistic modelling. Early Neural Networks. How Deep Learning different from Machine Learning. Scalars. Vectors. Matrixes, Higher Dimensional Tensors. Manipulating Tensors. Vector Data. Time Series Data. Image Data. Video Data

UNIT 2 NEURAL NETWORKS

9

About Neural Network. Building Blocks of Neural Network. Optimizers. Activation Functions. Loss Functions. Data Pre-processing for neural networks, Feature Engineering. Overfitting and Underfitting. Hyper parameters.

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UNIT 3 CONVOLUTIONAL NEURAL NETWORK

10

About CNN. Linear Time Invariant. Image Processing Filtering. Building a convolutional neural network. Input Layers, Convolution Layers. Pooling Layers. Dense Layers. Backpropagation Through the Convolutional Layer. Filters and Feature Maps. Backpropagation Through the Pooling Layers. Dropout Layers and Regularization. Batch Normalization. Various Activation Functions. Various Optimizers. LeNet, AlexNet, VGG16, ResNet. Transfer Learning with Image Data. Transfer Learning using Inception Oxford VGG Model, Google Inception Model, Microsoft ResNet Model. R-CNN, Fast R-CNN, Faster R-CNN, Mask-RCNN, YOLO

UNIT 4 NATURAL LANGUAGE PROCESSING USING RNN

10

About NLP & its Toolkits. Language Modeling . Vector Space Model (VSM). Continuous Bag of Words (CBOW). Skip-Gram Model for Word Embedding. Part of Speech (PoS) Global Co-occurrence Statistics-based Word Vectors. Transfer Learning. Word2Vec. Global Vectors for Word Representation GloVe. Backpropagation Through Time. Bidirectional RNNs (BRNN) . Long Short Term Memory (LSTM). Bi-directional LSTM. Sequence-to-Sequence Models (Seq2Seq). Gated recurrent unit GRU.

UNIT 5 DEEP REINFORCEMENT & UNSUPERVISED LEARNING

10

About Deep Reinforcement Learning. Q-Learning. Deep Q-Network (DQN). Policy Gradient Methods. Actor-Critic Algorithm. About Autoencoding. Convolutional Auto Encoding. Variational Auto Encoding. Generative Adversarial Networks. Autoencoders for Feature Extraction. Auto Encoders for Classification. Denoising Autoencoders. Sparse Autoencoders.

LIST OF EXPERIMENTS:

- 1: Feature Selection from Video and Image Data
- 2: Image and video recognition
- 3: Image Colorization
- 4: Aspect Oriented Topic Detection & Sentiment Analysis
- 5: Object Detection using Autoencoder

COURSE OUTCOMES:

CO1: Feature Extraction from Image and Video Data

CO2: Implement Image Segmentation and Instance Segmentation in Images

CO3: Implement image recognition and image classification using a pretrained network

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(Transfer Learning)

CO4: Traffic Information analysis using Twitter Data

CO5: Autoencoder for Classification & Feature Extraction

TOTAL : 45+30=75

PERIODS

REFERENCES

1. Deep Learning A Practitioner's Approach Josh Patterson and Adam Gibson O'Reilly Media, Inc.2017
2. Learn Keras for Deep Neural Networks, Jojo Moolayil, Apress,2018
3. Deep Learning Projects Using TensorFlow 2, Vinita Silaparasetty, Apress, 2020
4. Deep Learning with Python, FRANÇOIS CHOLLET, MANNING SHELTER ISLAND,2017
5. Pro Deep Learning with TensorFlow, Santanu Pattanayak, Apress,2017

SEMESTER II, ELECTIVE I

NP2481

COGNITIVE COMPUTING

L T P C
3 0 0 3

COURSE OBJECTIVES:

- To familiarize Use the Innovation Canvas to justify potentially successful products.
- To learn various ways in which to develop a product idea.
- To understand about how Big Data can play vital role in Cognitive Computing
- To know about the business applications of Cognitive Computing
- To get into all applications of Cognitive Computing

UNIT I FOUNDATION OF COGNITIVE COMPUTING

9

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Foundation of Cognitive Computing: cognitive computing as a new generation, the uses of cognitive systems, system cognitive, gaining insights from data, Artificial Intelligence as the foundation of cognitive computing, understanding cognition Design Principles for Cognitive Systems: Components of a cognitive system, building the corpus, bringing data into cognitive system, machine learning, hypotheses generation and scoring, presentation, and visualization services

UNIT II NATURAL LANGUAGE PROCESSING IN COGNITIVE SYSTEMS 9

Natural Language Processing in support of a Cognitive System: Role of NLP in a cognitive system, semantic web, Applying Natural language technologies to Business problems Representing knowledge in Taxonomies and Ontologies: Representing knowledge, Defining Taxonomies and Ontologies, knowledge representation, models for knowledge representation, implementation considerations

UNIT III BIG DATA AND COGNITIVE COMPUTING 9

Relationship between Big Data and Cognitive Computing: Dealing with human-generated data, defining big data, architectural foundation, analytical data warehouses, Hadoop, data in motion and streaming data, integration of big data with traditional data Applying Advanced Analytics to cognitive computing: Advanced analytics is on a path to cognitive computing, Key capabilities in advanced analytics, using advanced analytics to create value, Impact of open source tools on advanced analytics

UNIT IV BUSINESS IMPLICATIONS OF COGNITIVE COMPUTING 9

Preparing for change ,advantages of new disruptive models , knowledge meaning to business, difference with a cognitive systems approach , meshing data together differently, using business knowledge to plan for the future , answering business questions in new ways , building business specific solutions , making cognitive computing a reality , cognitive application changing the market

The process of building a cognitive application: Emerging cognitive platform, defining the objective, defining the domain, understanding the intended users and their attributes, questions and exploring insights, training and testing

UNIT V	APPLICATION OF COGNITIVE COMPUTING	9
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Building a cognitive health care application: Foundations of cognitive computing for healthcare, constituents in healthcare ecosystem, learning from patterns in healthcare Data, Building on a foundation of big data analytics, cognitive applications across the health care eco system, starting with a cognitive application for healthcare, using cognitive applications to improve health and wellness, using a cognitive application to enhance the electronic medical record Using cognitive application to improve clinical teaching

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COURSE OUTCOMES:

CO1: Explain applications in Cognitive Computing.

CO2: Describe Natural language processor role in Cognitive computing.

CO3: Explain future directions of Cognitive Computing

CO4: Evaluate the process of taking a product to market

CO5: Comprehend the applications involved in this domain.

TOTAL:45 PERIODS

REFERENCES

1. Judith H Hurwitz, Marcia Kaufman, Adrian Bowles, “Cognitive computing and Big DataAnalytics”, Wiley, 2015
2. Robert A. Wilson, Frank C. Keil, “The MIT Encyclopedia of the Cognitive Sciences”, TheMIT Press, 1999.
3. Noah D. Goodman, Joshua B. Tenenbaum, The ProbMods Contributors, “Probabilistic Models of Cognition”, Second Edition, 2016, <https://probmods.org/>.

CO-PO Mapping

CO	POs					
	PO1	PO2	PO3	PO4	PO5	PSO
1	1	1	1	1	1	1
2	1	3	3	1	1	1
3	1	3	2	2	1	1
4	1	3	3	1	1	1
5	2	2	3	3	1	1

BT2482

BLOCKCHAIN TECHNOLOGY

L T P C 2 0 2 3

COURSE OBJECTIVES

- To learn the fundamentals of Blockchain.
- To obtain knowledge about technologies of Blockchain.
- To understand the different models used in Blockchain.

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- To learn solidity programming to understand Ethereum based blockchain.
- To learn to code Hyperledger Fabric to establish enterprise based blockchain.

UNIT I INTRODUCTION

6

Basic Cryptographic primitives used in Blockchain –Secure- Collision Resistant hash functions - Digital signature – Public-key cryptosystems - Zero-knowledge proof systems - Need for Distributed Record-Keeping - Modelling faults and adversaries- Byzantine Generals problem - Consensus and their scalability problems

UNIT II TECHNOLOGIES BORROWED IN BLOCKCHAIN

6

Technologies Borrowed in Blockchain –hash pointers- Consensus- Byzantine Models of fault tolerance- Digital cash etc.- Bitcoin blockchain - Wallet - Blocks - Merkle Tree - hardness of mining
- Transaction verifiability - Anonymity - forks - Double spending.

UNIT III MODELS FOR BLOCKCHAIN

6

Models f-GARAY model -RLA Model -Proof of Work (PoW) as a random oracle - Formal treatment of consistency- Liveness and Fairness - Proof of Stake (PoS) based Chains -Hybrid models (PoW +PoS)

UNIT IV ETHEREUM

6

Ethereum -Ethereum Virtual Machine (EVM) -Wallets for Ethereum -Solidity - Smart Contracts - The Turing Completeness of Smart Contract Languages and verification challenges

UNIT V HYPERLEDGER FABRIC

6

Hyperledger fabric- the plug and play platform and mechanisms in permissioned blockchain
- applications of blockchain in cyber security- integrity of information- E-Governance and other contract enforcement mechanisms - Limitations of blockchain as technology and myths vs reality of blockchain technology
Case studies that map sustainable development goals such as 9, 11 (9: Industry, innovation and infrastructure, 11: Sustainable cities and communities)

TOTAL PERIODS: 30

LAB COMPONENT

**TOTAL LAB
PERIODS: 30**

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LIST OF EXPERIMENTS

List of Experiments using Solidity

Environment
setupBasic
syntax
First and sample
applicationComments
Types, Variables and Variable
scopeOperators
Loops and Decision
makingStrings,
Arrays and Enum
Structs, Mappings and
ConversionsEther units and
special variables Mini project

List of Experiments using Hyperledger Fabric

Introduction
Hyperledger Fabric Model
How Fabric networks are
structuredIdentity
Membership Service Provider
(MSP)Policies
The Ordering Service
Smart Contracts and Chaincode
Fabric chaincode
lifecyclePrivate data
Channel
capabilities
Security
Model
Use Cases

TEXTBOOKS

1. S.Shukla, M.Dhawan, S.Sharma, S. Venkatesan “Blockchain Technology: Cryptocurrency andApplications, Oxford University Press 2019.
2. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder,

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” Bitcoin and cryptocurrency technologies: a comprehensive introduction”, Princeton University Press, 2016.

REFERENCES

1. Joseph Bonneau et al, SoK: “Research perspectives and challenges for Bitcoin and cryptocurrency”, IEEE Symposium on Security and Privacy, 2015
2. J.A.Garay et al, “The bitcoin backbone protocol - analysis and applications”, EUROCRYPT 2015, Volume 2.
3. R.Pass et al, “Analysis of Blockchain protocol in Asynchronous networks”, EUROCRYPT 2017.
4. Pass et al, ” Fruitchain- a fair blockchain”, PODC 2017

COURSE OUTCOMES

At the end of the course, the student should be able to

CO1: Explain the fundamentals of Blockchain

CO2: Illustrate the technologies used to build

Blockchain

CO3: Describe the models of

Blockchain

CO4: Apply the solidity language to develop smart contracts

CO5: Demonstrate Hyperledger fabric for the enterprise applications

CO-PO Mapping:

COs	POs					PSO
	1	2	3	4	5	1
1	1		1	2		3
2	1		1	2		3
3	1		1	2		3
4	1	3	3	3	3	3
5	1	3	3	3	3	3

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COURSE OBJECTIVES:

The objective of this course is to enable the students

- To use micro python for IoT
- To understand the concepts of Raspberry pi
- To develop IoT devices using Node MCU
- To develop IoT devices using ESP32

UNIT I ADVANCED C AND INTRODUCTION TO MICRO-PYTHON 6

RTOS, Hard Real Time System, Soft Real Time System – Introduction to Hardware Programming - Introduction to Micro-Python – Micro Python Coding Fundamentals - Beyond desktop computing – Physical computing: Raspberry pi – Raspbian OS – Installation and Configuration – Remote Access – VNC, Remote Desktop Configuration

UNIT II WORKING WITH MICRO COMPUTER RASPBERRY PI 6

Architecture of raspberry Pi – Versions – Pin Diagrams – Networking with raspberry Pi- Micro Python over Raspberry Pi – GPIO pins – Digital Sensors

UNIT III MICROCONTROLLER PROGRAMMING 6

Introduction to Microcontroller – Architecture – Pin Diagram - Advantages and limitations of Micro Controller – Sketch – Programming in Arduino – Handling analog and digital sensors – PWM pins

UNIT IV ADVANCED MICRO CONTROLLER PROGRAMMING 6

Node MCU (ESP8266) – Architecture – Pin Diagram – Configuring ESP8266 – Flashing the controller - Programming in ESP8266 – ESP32 – Architecture – Pin diagram – Configuring ESP32 – Flashing – Programming with ESP32

UNIT V CLOUD INTERFACE 6

Google Cloud API: Drive API, Sheet API, OAuth – AWS: DynamoDB Interfacing – MongoDB DBInterfacing – FireBase Interfacing.

TOTAL PERIODS: 30

LAB COMPONENT

TOTAL PERIODS: 30

LIST OF EXPERIMENTS:

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1. Practicing structures, pointers, and far-pointers in C
2. Programming in Micro-Python
3. Circuit design for sensors
4. Programming in Sketch with Arduino Uno
5. Programming with Raspberry Pi (Digital Sensors)
6. Arduino - Raspberry Pi interface programming - master-slave architecture
7. Programming with Analog and digital sensors in ESP8266
8. Programming with ESP32
9. Program for cloud Interfacing and analytics

TEXTBOOKS:

1. Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, UniversitiesPress, 2015, ISBN: 9788173719547
2. Simon Monk, Programming the Raspberry Pi, Second Edition: Getting Started with Python, Second Edition, McGraw-Hill,
3. Coleman Cottrill, ESP8266 Programming Language: Nodemcu Programming, ESP8266 For Beginners, Independently Published, April 2021
4. Agus Kurniawan, Internet of Things Projects with ESP32, Packt Publishing, 2019
5. Reema Thareja, —Programming in C, Oxford University Press, Second Edition, 2016.
6. Kernighan, B.W and Ritchie, D.M, —The C Programming language, Second Edition, Pearson Education, 2006

REFERENCES:

1. Peter Waher, 'Learning Internet of Things', Packt Publishing, 2015
2. Peter Friess, 'Internet of Things – From Research and Innovation to Market Deployment', River Publishers, 2014
3. Juneja, B. L and Anita Seth, —Programming in C++, CENGAGE Learning India Pvt. Ltd., 2011
4. Peter Friess, 'Internet of Things – From Research and Innovation to Market Deployment', River Publishers, 2014
5. N. Ida, Sensors, Actuators and Their Interfaces, SciTech Publishers, 2014.
6. Getting Started with Raspberry Pi, Matt Richardson & Shawn Wallace, O'Reilly (SPD), 2014, ISBN: 9789350239759
7. Raspberry Pi Cookbook, Software and Hardware Problems and solutions, Simon Monk, O'Reilly (SPD), 2016, ISBN 9789352133895

OUTCOMES:

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On successful completion of this course, the students should be able to
CO1: Design and implement applications using arrays, strings and functions and exposure to micro python

CO2: Understanding and working with raspberry pi

CO3: Configure and programming with

Node MCUCO4: Configure and

programming with ESP32

CO5: Interface the IoT devices with cloud technologies

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	1
1	2		3	3	2	3
2	1		2	1	2	3
3	1	1	3	3	2	3
4	1	1	3	3	2	3
5	3	1	3	3	2	3

NP2484

PRINCIPLES OF PROGRAMMING LANGUAGES

L T P C

3 0 0 3

COURSE OBJECTIVES:

- To understand and describe syntax and semantics of programming languages
- To understand data, data types, and basic statements
- To understand call-return architecture and ways of implementing them
- To understand object-orientation, concurrency, and event handling in programming languages
- To develop programs in non-procedural programming paradigms

UNIT I SYNTAX AND SEMANTICS

9

Evolution of programming languages – describing syntax – context – free grammars –attribute grammars – describing semantics – lexical analysis – parsing – recursive-descent – bottom- up parsing

UNIT II DATA, DATA TYPES, AND BASIC STATEMENTS

9

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Names – variables – binding – type checking – scope – scope rules – lifetime and garbage collection – primitive data types–strings–array types– associative arrays–record types– union types – pointers and references – Arithmetic expressions – overloaded operators – type conversions – relational and boolean expressions – assignment statements – mixed- mode assignments – control structures – selection – iterations – branching – guarded statements

UNIT III SUBPROGRAMS AND IMPLEMENTATIONS 9

Subprograms – design issues – local referencing – parameter passing – overloaded methods – generic methods – design issues for functions – semantics of call and return – implementing simple subprograms – stack and dynamic local variables – nested subprograms – blocks – dynamic scoping

UNIT IV OBJECT-ORIENTATION, CONCURRENCY, AND 9
EVENTHANDLING

Object-orientation – design issues for OOP languages – implementation of object-oriented constructs – concurrency – semaphores – monitors – message passing – threads – statement level concurrency – exception handling – event handling

UNIT V FUNCTIONAL AND LOGIC PROGRAMMING LANGUAGES 9

Introduction to lambda calculus – fundamentals of functional programming languages –Programming with Scheme – Programming with ML – Introduction to logic and logic programming
– Programming with Prolog – multi-paradigm languages

COURSE OUTCOMES:

CO1: Describe syntax and semantics of programming languages

CO2: Explain data, data types, and basic statements of programming languages

CO3: Design and implement subprogram constructs

CO4: Apply object-oriented, concurrency, and event handling programming constructs

CO5: Develop programs in Scheme, ML, and Prolog and Understand and adopt new programming language

TOTAL : 45 PERIODS

REFERENCES:

1. Robert W. Sebesta, “Concepts of Programming Languages”, Eleventh Edition, Addison Wesley, 2012
2. W. F. Clocksin and C. S. Mellish, “Programming in Prolog: Using the ISO Standard”, Fifth Edition, Springer, 2003

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3. Michael L.Scott, “Programming Language Pragmatics”, Fourth Edition, Morgan Kaufmann,2009.
4. R.KentDybvig,“TheSchemeprogramminglanguage”,FourthEdition,MITPress,2009
5. Richard A. O'Keefe, “The craft of Prolog”, MIT Press,2009
6. W.F.ClocksinandC.S.Mellish,“ProgramminginProlog:UsingtheISOStandard”,Fifth Edition, Springer,2003

CO-PO Mapping

CO	POs					
	PO1	PO2	PO3	PO4	PO5	PSO
1	2	1	3	1	3	2
2	2	1	3	1		
3	3	3	2	2	3	1
4	2	3	2	3	3	3
5	3	2		3	3	3

SEMESTER II,ELECTIVE II

NLP2491

SPEECH AND AUDIO PROCESSING

L T P C 3 0 0 3

COURSE OBJECTIVES:

To provide an in-depth understanding of speech signal processing, feature extraction, and enhancement techniques.

To explore and implement automatic speech recognition (ASR) and text-to-speech (TTS) systems.

To apply machine learning and deep learning models in speech processing tasks.

To evaluate and apply speech technology in real-world applications such as voice assistants and emotion recognition.

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UNIT I BASICS OF SPEECH PROCESSING

10

Speech Production Mechanism - Speech Signal Analysis: MFCC, Spectrograms - Fundamentals of Acoustics in Speech - Analysis of Speech Signals and Features - Speech Signal Representation - Prosodic Features in Speech - Articulatory and Acoustic Phonetics - Temporal and Frequency Domain Speech Features - Noise Robustness in Speech Processing - Speech Segmentation Techniques - Mel-frequency Cepstral Coefficients (MFCC) and Feature Extraction - Voice Activity Detection (VAD) - Fundamental Frequency (F0) Estimation - Prosody Modeling in Speech Processing - Speech Signal Transformation and Enhancement

UNIT II AUTOMATIC SPEECH RECOGNITION

10

Acoustic Models for Speech Recognition - Hidden Markov Models (HMM) in ASR - Neural Network-Based ASR - Deep Neural Networks (DNN) for ASR - End-to-End ASR Models - Language Modeling in ASR - Decoding in Speech Recognition: Viterbi Algorithm - Feature Normalization in ASR - Speech-to-Text Systems - Evaluation Metrics for ASR: Word Error Rate (WER) - Speaker Adaptation in ASR Systems - Multi-Channel Speech Recognition - Data Augmentation in ASR - ASR for Low-Resource Languages - Acoustic and Language Model Integration

UNIT III TEXT-TO-SPEECH

7

Concatenative TTS - Parametric TTS Systems - Neural Network-Based TTS (Tacotron, WaveNet) - Speech Synthesis Models - Prosody in TTS - Voice Quality and Emotion in TTS - Speech Signal Synthesis Techniques - TTS Feature Representation - TTS Training Using Deep Learning - Waveform Generation in TTS Systems - Synthesis of Natural Sounding Speech - End-to-End Text-to-Speech Systems - Voice Cloning and Personalized TTS - Multilingual TTS Systems - Evaluation of TTS Quality

UNIT IV SPEAKER IDENTIFICATION AND VERIFICATION

10

Speaker Recognition: Identification vs. Verification - Voice Biometrics and Its Applications - Feature Extraction for Speaker Identification - Speaker Model Creation and Adaptation - Techniques in Speaker Verification - Gaussian Mixture Models (GMM) in Speaker Recognition - i-vectors for Speaker Identification - Deep Learning for Speaker Verification - Speaker Adaptation in Verification Systems - Evaluation of Speaker Recognition Systems - Speaker Diarization - Application of Speaker Identification in Forensic Analysis - Multimodal Speaker Recognition - Robustness of Speaker Recognition to Noise - Real-time Speaker Verification

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UNIT V EMERGING APPLICATIONS

8

Voice Assistants and Virtual Agents (e.g., Alexa, Siri) - Speech-based Emotion Recognition - Speech Translation Systems - Real-time Speech Recognition for Live Applications - Voice-Activated User Interfaces - Speech-based Sentiment Analysis - Audio Content Analysis and Processing - Applications in Healthcare: Medical Speech Processing - Forensic Applications of Speech Technology - Cross-Lingual Speech Recognition - Privacy and Security Issues in Voice Assistants - Multimodal Speech Processing (Audio + Visual) - Speech Enhancement for Hearing Aids - Audio-Visual Speech Recognition Systems - Speech Processing in Autonomous Vehicles

TOTAL PERIODS: 45

Textbooks:

1. **"Speech and Language Processing"** by Daniel Jurafsky and James H. Martin
2. **"Fundamentals of Speech Recognition"** by Lawrence Rabiner and Biing-Hwang Juang

References:

1. **"Deep Learning for Speech and Language Processing"** by S. M. Dhamdhere and M. R. N. Prasad
2. **"Voice Biometrics for Multimedia Systems"** by Mohammed G. S. H.
3. **"Text to Speech Synthesis"** by Paul Taylor

Course Outcomes:

CO1: Understand the core concepts and techniques in speech signal processing.

CO2: Gain proficiency in implementing speech recognition (ASR) and synthesis (TTS) systems.

CO3: Develop the ability to apply machine learning models for improving speech processing tasks.

CO4: Critically analyze the performance of speech processing systems using standard metrics.

CO5: Explore advanced applications of speech technology in fields like emotion recognition, voice assistants, and multilingual speech processing.

CO-PO Mapping

	POs	PSO
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COs	1	2	3	4	5	
1	2		2			1
2	2		2			2
3	2		2			1
4			2			2

ML249 CLOUD & EDGE COMPUTING FOR ML APPLICATIONS L T P C 3 0 0 3

COURSE OBJECTIVES

- To learn the fundamentals of cloud and edge computing
- To understand the principles of deploying machine learning algorithms in cloud and edgecomputing
- To enable design and implementation of machine learning solutions using cloud and edgecomputing
- To gain experience on recent trends in integrating cloud and edge computing techniques formachine learning applications

Unit I - Cloud and Edge computing fundamentals 9

Introduction to Cloud computing, Edge computing, Edge intelligence, Cloud Edge computing, Usecases

Unit II - Machine Learning and Deep Learning Models in the Cloud 9

Defining Structure and Training Machine Learning Models in cloud, Serverless Machine Learning, using containers with Machine Learning Models, Benefits of Serverless Computing for Machine Learning

Unit III - Functions-as-a-Service and Event-Driven Programming in Cloud 9

Software-as-a-Service, Machine Learning as a service (MLaaS), MLaaS functionality, Benefits of MLaaS, MLaaS service providers, Machine Learning Operations (MLOps), Key phases in MLOps implementation

Unit IV - Edge Machine Learning 9

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Machine learning and edge bringing AI to IoT, Challenges in machine learning and edge computing integration, Edge devices and ML frameworks, Hardware Inference Model in EdgeMachine Learning – VPU, GPU, TPU

Unit V – Recent trends in Cloud and Edge computing for Machine learning **9**

Cloud APIs for computer vision, Scalable inference serving on cloud, Embedded ML at Edge –Shallow RNN, ProtoNN, Resource-efficient Machine learning

Case studies that map sustainable development goals such as 9, 11 (9: Industry, innovation and infrastructure, 11: Sustainable cities and communities)

TOTAL PERIODS: 45

TEXT BOOK

1. John Biggs, Vicente Herrera García, Building Intelligent Cloud Applications, O'Reilly publisher, 2019
2. Daniel Situnayake, Jenny Plunkett, AI at the Edge, O'Reilly publisher, 2022

REFERENCES:

1. Anirudh Koul, Siddha Ganju and Meher Kasam, Practical Deep Learning for Cloud, Mobile, and Edge, O'Reilly Publication, 2020
2. Mark Treveil, Nicolas Omont, Clément Stenac, Kenji Lefevre, Du Phan, Joachim Zentici, Adrien Lavoillotte, Makoto Miyazaki, Lynn Heidmann, Introducing MLOps, O'Reilly, 2020
3. Kai Hwang, Cloud Computing for Machine Learning and Cognitive Applications, MIT press, 2017
4. Web resources: https://github.com/bisonai/awesome-edge-machine-learning/tree/master/Papers/ML_Algorithms_For_Edge

COURSE OUTCOMES

At the end of the course, the students should be able to

CO1: Demonstrate the application of cloud and edge computing in Machine learning applications
CO2: Analyse the various services and architecture of cloud and edge computing

CO3: Apply Machine Learning algorithms using cloud and edge computing

CO4: Relate current research papers on cloud and edge computing to understand the issues raised by current research

CO-PO Mapping:

COs	POs					PSO
	1	2	3	4	5	1

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1	2		1	1		
2	2		1	2	1	1
3	2		1	2	1	1
4	3	2	2	1	2	3

NP2493

IMAGE PROCESSING

L T P C 3 0 0 3

COURSE OBJECTIVES:

- To study fundamental concepts of digital image processing.
- To get exposed to simple image enhancement techniques in Spatial and Frequency domain..
- To become familiar with image compression
- To study the image segmentation and Morphological Processing.
- To expose student's in recognition methods.

UNIT I INTRODUCTION

9

Examples of fields that use digital image processing, fundamental steps in digital image processing, components of image processing system. Digital Image Fundamentals: A simple image formation model, image sampling and quantization, basic relationships between pixels. Color Image Processing: Color fundamentals, color models, pseudo color image processing, basics of full-color image processing, color transforms, smoothing and sharpening, color segmentation

UNIT II IMAGE ENHANCEMENT

9

Image enhancement in the spatial domain: Basic gray-level transformation, histogram processing, enhancement using arithmetic and logic operators, basic spatial filtering, smoothing, and sharpening spatial filters, combining the spatial enhancement methods. Filtering in the Frequency Domain: Preliminary Concepts, Extension to functions of two variables, Image Smoothing, Image Sharpening, Homomorphic filtering. A model of the image degradation/restoration process, noise models, restoration in the presence of noise-only spatial filtering.

UNIT III WAVELETS AND IMAGE COMPRESSION

9

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Wavelets and Multiresolution Processing. Fundamentals, image compression models, error-free compression, lossy predictive coding, image compression standards

UNIT IV IMAGE SEGMENTATION

9

Detection of Discontinuities, Edge Linking and Boundary Detection, Thresholding, Region-Based Segmentation, Segmentation by Morphological Watersheds, The Use of Motion in Segmentation Morphological Image Processing: Preliminaries, dilation, erosion, open and closing, hit or miss transformation, basic morphologic algorithms.

9

UNIT V REPRESENTATION AND OBJECT RECOGNITION

Representation, Boundary Descriptors, Regional Descriptors, Use of Principal Components for Description. Object Recognition: Patterns and patterns classes, recognition based on decision– theoretic methods, matching, optimum statistical classifiers, neural networks, structural methods – matching shape numbers, string matching.

COURSE OUTCOMES:

CO1: Apply knowledge of mathematics for image understanding and analysis.

CO2: Design and analysis of techniques / processes for image Enhancement.

CO3: Design and analysis of techniques / processes for image compression.

CO4: Able to expose to current trends in field of image segmentation.

CO5: Design, realize and troubleshoot various algorithms for image processing case studies.

TOTAL: 45 PERIODS

REFERENCES

1. Digital Image Processing, Rafeal C.Gonzalez, Richard E.Woods, fourth Edition, Pearson Education/PHI, 2018
2. Image Processing, Analysis, and Machine Vision, Milan Sonka, Vaclav Hlavac and Roger Boyle, fourth Edition, Thomson Learning, 2015
3. Introduction to Digital Image Processing with Matlab, Alasdair McAndrew, Thomson Course Technology, 2021
4. Computer Vision and Image Processing, Adrian Low, Second Edition, B.S.Publications,2022.
5. 5. Digital Image Processing using Matlab, Rafeal C.Gonzalez, Richard E.Woods, Steven L. E ddins, Pearson Education,2006

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CO-PO Mapping:

CO	POs					
	PO1	PO2	PO3	PO4	PO5	PSO
1	1	2	2	2	2	3
2	3	3	3	2	3	3
3	1	1	2	2	3	3
4	2	2	2	1	1	1
5	1	1	2	2	3	3

NP2494 MULTIMEDIA COMMUNICATION NETWORKS L T P C 3 0 0 3

COURSE OBJECTIVES:

- To recapitulate the fundamentals of networking and understand the requirements for multimedia communication.
- To learn guaranteed service model.
- To learn communication protocols that are frequently used in IoT ecosystems.
- To explore the support provided for multimedia communication in 3G and 4G networks.
- To study about VoIP and real time multimedia network applications.

UNIT I INTRODUCTION

9

Switched Networks and Shared media Networks – Circuit Switching, Packet Switching and Virtual Circuits – Flow Control and Congestion Control – TCP/IP reference model – Network Externalities
– Service Integration – Elastic and Inelastic Traffic – Playback Applications – Additional Requirements For Inelastic Traffic – Core Networks And Access/Edge Networks.

Suggested Activities:

- Flipped classroom on network externalities and Economies of scale.
- External learning – Inter-continental backbone network and Autonomous Systems model of the Internet.
- Assignments on computing the playout time of packets.

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Suggested Evaluation Methods:

- Quiz and discussion on network externalities and economies of scale.
- Assignments on proprietary protocols used in IoT and M2M.
- Assignments on problems related to playout time of multimedia applications.

UNIT II GUARANTEED SERVICE MODEL

9

Best Effort Service Model and Its Limitations – Qos Metrics – Diffserv and Intserv Networks – Queuing Techniques – WFQ and Its Variants – RED – Qos Aware Routing – Call Admission Control – RSVP – Policing and Traffic Shaping Algorithms – Multicast Routing – IGMP, Protocol Independent Multicast – PIM SM and PIM DM Variants.

Suggested Activities:

- Flipped classroom on IntServ and DiffServ networks.
- External learning – Exploring the ways of using DSCP in IP header.
- Assignments on finish time problems related to WFQ and its variants.

Suggested Evaluation Methods:

- Quiz and discussion on IntServ and DiffServ networks.
- Assignments on configuring a router in such a way that DSCP fielder is exploited to provideQoS.
- Assignments on problems related to the virtual finish and actual finish of packets in WFQand its variants.

UNIT III MULTIMEDIA TRANSPORT

9

End To End Solutions – Laissez Faire Approach – Multimedia over TCP – Significance of UDP – Multimedia Streaming – Audio and Video Streaming – Accessing Audio And Video from a Web Server And Media Server – Removing Jitter at the Receiver – Recovering from Packet Loss – Forward Error Correction and Interleaving – Interactive And Non-Interactive Multimedia – Transcoding – RTSP – RTP/RTCP.

Suggested Activities:

- External learning – Exploring various media players available and the ways to customizethem.
- Exploring the ways to configure RTP.

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- Flipped classroom on RTP and RTCP.

Suggested Evaluation Methods:

- Assignments on media players available and configuring them.
- Configuring RTP and RTSP.
- Quiz and discussion on RTP and RTCP.

UNIT IV MULTIMEDIA OVER WIRELESS NETWORKS

9

Architecture of IP Multimedia Subsystem in 3G Networks – Application, Control and Data Planes in IMS Networks – Session Control, AAA, Real Time Data Transfer and Policy Control Protocols of IMS Networks – Relay Node and Multiple Radio Access Technologies in LTE – Voice Over IP Basics – IMS Volte Architecture – IP Multimedia Service Identity Module, Private Identity, Public Identity (ISIM, IMPI And IMPU) – SIP User Agent (SIP UAC And SIP UAE) – Real Time Polling Service and Extended Real Time Polling Service in IEEE 802.16/Wimax Networks.

Suggested Activities:

- Flipped classroom on IMSVoLTE architecture.
- External learning – Multimedia support in 5G networks.
- Analyzing the protocols of IP media subsystem

Suggested Evaluation Methods:

- Quiz and discussion on IMSVoLTE architecture.
- Assignments on multimedia support in 5G networks.
- Assignments on analyzing the headers of IP multimedia subsystem.

UNIT V MULTIMEDIA NETWORKED APPLICATIONS

9

H.322 Standard – Protocol Stack And Call Setup – Session Initiation Protocol – Components, Messages And Operation – Supporting Protocols For SIP – Media Gateway Access Protocol, Resource Reservation Protocol, Session Description Protocol – Case Study – Video Conferencing– Military Surveillance – Interactive TV – Video On Demand – Smart Phone.

Suggested Activities:

- ☐ Flipped classroom on SCIBus and S.100.
- ☐ External learning – Multimedia access networks and edge networks.

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- ☐ Exploring the ways to configure SIP.

Suggested Evaluation Methods:

- ☐ Quiz and discussion on SCIBus and S.100.
- ☐ Assignments on multimedia access networks and edge networks.
- ☐ Configuring SIP using suitable commands.

TOTAL PERIODS: 45

COURSE OUTCOMES:

On completion of the course, the students will be able to: **CO1:**Deploy the right multimedia communication models.

CO2:Apply QoS to multimedia network applications at the network level with efficient scheduling and routing techniques.

CO3:Apply QoS to multimedia network applications at the end system level with efficient scheduling and routing techniques.

CO4:Understand IP multimedia subsystem and IP initiatives in cellular networks to support multimedia traffic.

CO5:Design and implement VoIP based solutions for multimedia transport.

CO6:Develop the real-time multimedia network applications.

REFERENCES:

1. Mario Marques da Silva, “Multimedia Communications and Networking”, CRC Press, 2012
2. K. R. Rao, Zoran S. Bojkovic, Bojan M. Bakmaz, “Wireless Multimedia Communication Systems: Design, Analysis and Implementation”, CRC Press, 2017
3. Jim Kurose, Keith Ross, “Computer Networking: A Top Down Approach”, Pearson Education, 2017
4. K. R. Rao, Zoran S. Bojkovic, Dragorad A. Milovanovic, “Introduction to Multimedia Communications Applications, Middleware, Networking”, John Wiley and Sons, 2009

CO-PO Mapping

CO	POs					
	PO1	PO2	PO3	PO4	PO5	PSO
1	1	1	1	2	3	2

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2	2	2	2	2	1	1
3	2	2	2	2	1	1
4	1	2	2	3	1	1
5	1	2	2	2	3	3

SEMESTER III, ELECTIVE III

NA2466

MOBILE APPLICATION DEVELOPMENT

L T P C 2 0 2 3

COURSE OBJECTIVES

- To learn how to use Flutter to rapidly develop a mobile app that runs on both iOS and Android devices.
- To understand the use of Flutter, including building a UI, using animations, and creating a database app.
- To explore state management and asynchronous programming in Dart & Flutter.
- To gain knowledge on how to use Flutter with Firebase.

UNIT I: GETTING STARTED WITH FLUTTER

3

Introducing Flutter – Widget Lifecycle Events – Widget Tree and Element Tree – Dart Basics

Lab Component: Installing the Flutter SDK – Configuring the Android Studio Editor – Creating a Hello World App – Creating a Starter Project Template.

UNIT II: FLUTTER USER INTERACTION, STYLES, AND ANIMATIONS

3

Flutter UI – User Interaction and gestures – Flutter forms – Flutter Animations

Lab Component: Adding Animation to an App - Creating an App's Navigation – Creating Scrolling Lists and Effects – Building Layouts – Applying Interactivity – Writing Platform-Native Code.

UNIT III: STATE MANAGEMENT

3

Routing in Flutter – Declarative routing and named routes – Routing animations – Flutter State Management.

Lab Component: Building a 2D game with Flutter state management

UNIT IV: ASYNCHRONOUS DART

3

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Business Logic Components - Async Dart – Sinks and streams – Using streams in blocs – AsyncFlutter.

Lab Component: Build interactive live streams using Async Dart with Flutter.

UNIT V: BEYOND FOUNDATIONS

3

HTTP and Flutter – JSON Serialization – Working with Firebase in Flutter – Dependency Injection in the App - Kotlin Foundations – Simple APP Design.

Lab Component: Mini-project using Firebase Cloud Firestore and Flutter

Projects that map sustainable development goals such as 9, 11 (9: Industry, innovation and infrastructure, 11: Sustainable cities and communities)

TOTAL PERIODS: 60

TEXT BOOKS

1. Eric Windmill, “Flutter in Action”, Manning Publications; 1st Edition, 2019.
2. Marco L. Napoli, “Beginning Flutter: A Hands On Guide to App Development”, Wrox, 1st Edition, 2019.

REFERENCES:

1. Alberto Miola, Felix Angelov, Matej Rešetár, Rémi Rousselet, “Flutter Complete Reference: Create beautiful, fast and native apps for any device”, Kindle Edition, 2020.
2. Simone Alessandria, Brian Kayfitz, “Flutter Cookbook: Over 100 proven techniques and solutions for app development with Flutter 2.2 and Dart”, 1st Edition, Kindle Edition, 2021.
3. Barry Burd, “Flutter For Dummies”, 1st Edition, Kindle Edition, 2020.
4. Simone Alessandria, Flutter Projects: A practical project-based guide to building real-world cross-platform mobile applications and games, publisher: packt publishing, www.packt.com.
5. Carmine Zaccagnino, Programming Flutter: Native, Cross-Platform Apps the Easy Way (The Pragmatic Programmers), publisher: packt publishing, www.packt.com.
6. John Horton, Android Programming with Kotlin for Beginners: Build Android apps starting from zero programming experience with the new Kotlin programming language Packt Publishing Limited, 2019.

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WEB REFERENCES:

- <https://aws.amazon.com/mobile/mobile-application-development/>
- [https://flutter.dev/\(Applications\)](https://flutter.dev/(Applications))
- <http://ai2.appinventor.mit.edu/> (Applications)
- <https://developer.android.com/guide>
- https://en.wikipedia.org/wiki/Android_10

COURSE OUTCOMES

At the end of the course, the students should be able to

CO1: Demonstrate how to use Flutter for creating cross-platform mobile apps, including how to setup the SDK and simulators; add widgets, components, and navigation.

CO2: Customize the app using sophisticated user interfaces, styles and animations.

CO3: Apply state management techniques and asynchronous programming in Dart & Flutter.

CO4: Build Firebase's real-time database in a Flutter App.

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	1
1	2		1	3		
2	2		3	2		
3	2		3	2		
4	3	3	3	3	3	3

NR2477

ROBOTIC PROGRAMMING

L T P C 2 0 2 3

COURSE OBJECTIVES

- To introduce the fundamental aspects of Robotics.
- To analyse trade-offs among position control, velocity control, and force control in simulation problem.
- To study the importance of Robotic Operating System
- To develop the programming skills related to kinematics, control, optimization.
- To have hands on interfacing the different sensors and actuators through ROS.

Unit I Fundamentals of Robotics

3

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Robot - Definition - Robot Anatomy - Coordinate Systems, Robot Parts and their Functions- Need for Robots-Drives-Effectors-Sensors-Forward and Inverse Kinematics- Different Applications.

Unit II Simulation of programmable robots **3**

Programming functions for perception vs reality of robots-Control functions for different desired behaviour-Differential drive functions for mobility in robot-API interface functions for sensors and actuators-Functions for regulating the velocity in robot

Unit III Robot Operating System **3**

Introduction to open-source meta operating system for robot-Working with ROS System- Programs for publishing and subscribing messages-Log messages. Graph resource names- Launch files- Parameters and Services. Installing ROS on an ARM based Board.

Unit IV ROS Advanced Concepts **3**

ROS file system-Understanding the Parameter server and Dynamic parameters-actionlib- Pluginlib- PCL Library-OpenGL Library-Selenium Library for Robot Framework-- toolchains-nodelets- Gazebo Framework and Plugin-ROS transform frames-ROS visualization and Debugging Tools.

Unit V Interfacing sensors and actuators through ROS **3**

Arduino-ROS Interface-Interfacing 9DoF Razor IMU with Arduino-ROS-Interfacing a GPS system- Interfacing Servomotors-Interfacing with Laser Rangefinder-Kinect sensor to view objects in 3D- Collision Avoidance Algorithms for Navigation-Gazebo architecture and interface with ROS.

LAB COMPONENT

TOTAL PERIODS: 60

LIST OF EXPERIMENTS:

1. Study of different components used in robot.
2. Study of different functions used to perform different types of movement in robot.
3. Study of Robot Operating System.
4. Study of visualization in ROS.
5. Installing ROS in Arduino based board.

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6. Program to access the different sensors and actuators through ROS.
7. Study of Navigation stack in ROS.
8. Implementation of localization using Navigation stack in ROS.

9. Program to perform visualization.
10. Program to control the robots in simulation.
11. Study of Mobile Autonomous robot.
12. Program to implement collision avoidance

Projects that map to sustainable development goals such as 9, 11, 14, 15 (9: Industry, innovation and infrastructure, 11: Sustainable cities and communities, 14: Life below water, 15: Life on land)

TEXTBOOKS:

1. Programming Robots with Robotic Operating System-A practical introduction to the Robot Operating System, Morgan Quigley, Brian Gerkey & William D. Smart, O'REILLY.
2. Robot Operating Systems for Absolute Beginners- Robotic Programming made easy, Lentin Joseph, Apress,
3. ROS Robot Operating System From the basic concepts to Practical Programming and Robot Applications, YoonSeokPyo, HanCheol Cho, RyuWoon Jung & TaeHoon Lim
4. Robot Operating System - The Complete Reference (Volume 4) , Janusz Kacprzyk, Polish Academy of Sciences, Warsaw, Poland, Springer

REFERENCES:

1. Robot Operating System(ROS) The complete Reference(Volume 1) , Anis Kouba, Springer, Artificial Intelligence Studies.
2. Robot Operating System(ROS) The complete Reference(Volume 2) , Anis Kouba, Springer, Artificial Intelligence Studies.
3. Robot Operating System(ROS) The complete Reference(Volume 3) , Anis Kouba, Springer, Artificial Intelligence Studies

OUTCOMES:

On successful completion of this course, the student will be able to:

CO1: Perform hands-on introduction to the many engineering principles involved for Robotics. CO2: Design systems with various flexures, actuators and sensor systems for

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Robots

CO3: Implement various control function for mobility of Robot.

CO4: Analyze the manipulation and navigation problems using knowledge of coordinateframes, kinematics, optimization, control, and uncertainty.

COs	POs					PSO
	1	2	3	4	5	
1	3	1	2	1	3	3
2	3	1	2	1	3	3
3	3	1	2	1	3	3
4	3	1	2	1	3	3

NS2455 SOCIAL NETWORKS & OPEN SOURCE INTELLIGENCE L T P C 2 0 2 3

COURSE OBJECTIVES

- To learn and understand the structure of social networks using graph theory
- To understand the fundamental network model related to links and communities
- To learn and understand the intelligence produced from publicly available information
- To understand the application of social network models in game theory and epidemic networks

UNIT I GRAPH THEORY AND SOCIAL NETWORKS

6

Introduction to graphs, Strong and Weak Ties, Networks in real-time context, Positive and negative relationships

UNIT II INFORMATION NETWORKS AND WORLD WIDE WEB

6

Structure of web: web as directed graph, Bow-Tie structure of the web, Link analysis and web search: Searching the web, link analysis using Hub and Authority model, Page rank

UNIT III NETWORK DYNAMICS: POPULATION MODELS

6

Information cascades: Herding experiment, Bayes' rule for Herding, General cascades, sequential decision making, Power laws and Rich-Get-Richer Phenomena, Long tail

UNIT IV NETWORK DYNAMICS: STRUCTURAL MODELS

6

Cascading behaviour in networks: Modeling Diffusion through a Network, Cascades and Clusters, Diffusion, Thresholds, and the Role of Weak Ties, The Small-World Phenomenon:

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Six degrees of separation, Decentralized Search and Modelling the process

UNIT V APPLICATIONS OF GAME THEORY FOR SOCIAL NETWORKS

6

Game theory: Introduction to game strategies, Evolutionary game theory, Modelling network traffic using game theory, Network dynamics in epidemics: SIR, SIS models

TOTAL LECTURE PERIODS: 30

LAB COMPONENT

TOTAL LAB PERIODS: 30

LIST OF EXPERIMENTS:

1. OSINT preparation
2. Google dorks
3. Reverse searches
4. OSINT on a person
5. OSINT on social media
6. OSINT Documentation and reporting

Tools: OSINT framework, Maltego

TEXT BOOK

1. Networks, Crowds and Markets by David Easley and Jon Kleinberg, Cambridge University Press, 2010
2. Michael Bazzell, Open-Source Intelligence Techniques: Resources for Searching and Analyzing Online Information, 7th Edition, 2019, IntelTechniques.com

REFERENCES

1. Stanley Wasserman, Katherine Faust Social Network Analysis: Methods and Applications Volume 8 of Structural Analysis in the Social Sciences, ISSN 0954- 366X, Cambridge University Press, 1994
2. John Scott, Social Network Analysis, 3rd Edition, SAGE, 2012
3. Borko Furht, Handbook of Social Network Technologies and Applications, 1st Edition, Springer, 2010.
4. Peter Mika, Social Networks and the Semantic Web, First Edition, Springer 2007.
5. Guandong Xu, Yanchun Zhang and Lin Li, Web Mining and Social Networking – Techniques and applications, First Edition Springer, 2011.

COURSE OUTCOMES

At the end of the course, the students should be able to

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CO1: Explain and analyze the structure and evolution of social networks
CO2: Explain the framework of network analysis
CO3: Analyze the network dynamics using population and structural models
CO4: Apply models for real-time applications like network traffic and epidemic network
CO5: Demonstrate various OSINT techniques using appropriate tools

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	1		1			1
2			1			1
3	1		1			1
4	1		1			1
5	1	3		2	3	1

NT2488

IOT PROJECTS

L T P C 1 0 4 3

COURSE OBJECTIVES:

- To understand how to Independently Design, Code and Build IOT products.
- To learn to work with Microprocessors and Micro controllers (Arduino Uno, Raspberry PI, NodeMCU), Sensors, Relays, Displays, and more
- To develop code using Arduino IDE
- To establish communication using Ethernet and Wifi shields
- To build connectivity to cloud IOT Platforms, Persist Data, Program Triggers

UNIT 1 - Working with Electronics

3

Electronic boards and its working – Arduino, Raspberry, ESP32, Node MCU, Sensors – Temperature, photo resistor, motion detector, sound sensors, Working with relays – Switches, buzzer, LED display.

UNIT 2 - Working with Arduino IDE, Raspberry, and Python

3

Communication – Ethernet, Bluetooth, WiFi, NFC and Zigbee, Arduino IDE – Variables, keywords, Control structures, Functions Raspberry and Python – Python libraries for board interfacing.

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UNIT 3 - Industry approach to Edge computing and Data analytics in IoT **3**

Blending data analytics and IoT, Python for Data analytics, Machine learning, Data visualization, Industrial challenges with respect to IoT.

UNIT 4 - Working with Cloud DB, Platforms and services **3**

Cloud platform for IoT, Carriots/Spark/Thinkspeak/AWS and services. IoT communication models

– Request/Response, Publish/Subscribe, Push/Pull, Exclusive Pair

UNIT 5 - IoT Computing model implementation **3**

Construction of IoT application based on computing models: Cloud computing, Fog Computing, Edge computing, MIST computing with mobile interfacing through Android application.

TOTAL LECTURE PERIODS: 15

TOTAL LAB PERIODS: 60

LIST OF EXPERIMENTS:

Each student will be given an IoT project as a context of the following:

Smart Cities- Smart Waste Management, Smart Street Lights, Smart Street Parking, Security without Surveillance, Connected Vehicles. Healthcare- Baby Monitoring, Elderly Monitoring, Mood Enhancing, Disease Treatment and Progression Monitoring, Enhance Adherence, Challenges. Agriculture- Precision Agriculture, Connected Livestock, Food Safety. Manufacturing and Logistics- Smart Manufacturing- Smart Packaging, Smart Label. Smart Electricity Grid- Managing Supply and Demand. Home Automation

Projects that map to sustainable development goals 2, 3, 6, 7, 9, 11, 13 (2: Zero hunger, 3: Good health and well being, 6: Clean water and sanitation, 7: Affordable and clean energy, 9: Industry, innovation and infrastructure, 11: Sustainable cities and communities, 13: Climate action)

1. Processes: PSP, Scrum, DevOps: 15%
2. Management: Estimation, WBS, Planning, Tracking: 10%
3. Risk Management: 15%
4. Coding Standards and Configuration Management: 15%
5. Automation of routine tasks: 15%

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- 6. Meetings: “Customer” meetings, review meetings, brainstorming: 10%
- 7. Presentations: 10%
- 8. Documentations: 10%

COURSE OUTCOMES:

On successful completion of this course, the student will be able to:

CO1: Develop an individual IoT project associated with Industry 4.0

CO2: Understand the connectivity to cloud IOT Platforms, Persist Data, Program Triggers through IoT devices.

CO3: Develop mobile application to interface with IoT devices.

CO4: Communicate efficiently in presentations and prepare project report.

CO-PO Mapping:

COs	POs					PSO
	1	2	3	4	5	
1	3	1	3	3		3
2	3	1	3	3		3
3	3	1	3	3		3
4		3		1	3	2

SEMESTER III,ELECTIVE IV

NA2434

AGILE SOFTWARE DEVELOPMENT

L T P C 2 0 2 3

OBJECTIVES:

- To provide students with understanding of agile software development to create highquality software
- To provide a good understanding of Agile scrum framework.
- To do a detailed examination of Agile testing techniques.
- To provide a good understanding Agile design practices
- To understand the benefits of Agile and its current trend in the industry.

UNIT I: FUNDAMENTALS OF AGILE

12

The Genesis of Agile, Introduction and background, Agile Manifesto and Principles, Overview of Agile methodologies: Scrum,Kanban, Extreme Programming, Pair

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Programming, Feature Driven development, Lean Software Development, Agile project management, Design and development practices in Agile projects, Test Driven Development, Pair Programming.

Use Case: Any business Scenario.

UNIT II: AGILE SCRUM FRAMEWORK **12**

Introduction to Scrum, Project phases, Agile Estimation, Product backlog, Sprint backlog, Iteration planning, User story definition, Characteristics and content of user stories, Acceptance tests and Verifying stories, Project velocity, Burn down chart, Sprint planning and retrospective, Daily scrum, Scrum roles – Product Owner, Scrum Master, Scrum Team.

Use Case: Identify Product backlog, user stories, Sprint tasks. Usage of Agile tool such as AgileFantor Jira.

UNIT III: AGILE TESTING **12**

The Agile lifecycle and its impact on testing, Test-Driven Development (TDD), xUnit framework and tools for TDD, Testing user stories - acceptance tests and scenarios, Planning and managing testing cycle, Exploratory testing, Risk based testing, Regression tests, Test Automation, Tools to support the Agile tester.

Use Case: Design Test Cases aligned to xUnit framework for TDD

UNIT IV: AGILE SOFTWARE DESIGN AND DEVELOPMENT **12**

Agile design practices, Role of design Principles including Single Responsibility Principle, Open Closed Principle, Liskov Substitution Principle, Dependency Inversion Principle in Agile Design, Need and significance of Refactoring, Refactoring Techniques, Continuous Integration, Automated build tools, Version control.

Use Case: Refactor a given design for next sprint requirements.

UNIT V: INDUSTRY TRENDS **12**

Market scenario and adoption of Agile, Continuous Integration, Roles in an Agile project, Agile in Distributed teams, Business benefits, Challenges in Agile, Risks and Mitigation, Agile projects on Cloud, Balancing Agility with Discipline.

Use Case: Execute Continuous integration using a tool such as Jenkins.

TOTAL PERIODS: 60

TEXT BOOKS:

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1. Agile Software Development, Principles, Patterns and Practices, Robert C. Martin, Publisher:Prentice Hall
2. Agile Testing: A Practical Guide for Testers and Agile Teams, Lisa Crispin, Janet Gregory,Publisher: Addison Wesley

REFERENCES:

1. Agile Software Development with Scrum, Ken Schawber, Mike Beedle, Publisher: Pearson.
2. Agile Software Development: The Cooperative Game, By Alistair Cockburn
Publisher:Addison Wesley
3. Craig Larman, —Agile and Iterative Development: A Manager’s Guide, Addison-Wesley,2004.

Other Online Courses

1. www.it-ebooks.info/tag/agile
2. <http://martinfowler.com/agile.html>

COURSE OUTCOMES:

At the end of the course, the students should be able to

CO1: Point out the impact of social aspects on software development success.

CO2: Realize the importance of iterative software development processes: how to plan them, howto execute them.

CO3: Realize the impact of Agile design and its testing

CO4: Show how agile approaches can be scaled up to the enterprise level.

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	1		1	1	1	1
2	1	1	1	1	1	1
3	1	2	1	1	1	1
4	1	1	1	2	1	1

ND2432

DATA PRIVACY

L T P C 3 0 0 3

COURSE OBJECTIVES

- To introduce the basics of information privacy

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- To make the learner aware of the globally used privacy principles and guidelines
- To make the learner understand various privacy risk, threats, violation and adopt privacy by design for building IT solutions
- To understand and apply privacy enhancing technologies.

UNIT I Privacy in Digital Age **7**

What is privacy?: Nissenbaum's Contextual Integrity, Alan Westin's privacy states, Ryan Calo's Harm Dimensions, Sloove's taxonomy-Privacy, security and data governance – The data life cycle: harms in each stage, solution in each stage- Online issues causing privacy breach- Identity and Anonymity – Tracking and Surveillance

UNIT II Recent Technologies, Privacy Principles, Standards and Regulations **7**

Technologies With Privacy Considerations: Cloud Computing, Wireless Technology and Devices, Location-Based Services (LBS), Surveillance of Individuals, Video/Audio Surveillance, Data Surveillance, Biometric Recognition, AI and VR, Standards-ISO/IEC 27701, PCIDSS- Regulations: PIPEDA, EU DPD, GDPR, Hongkong's Personal Data Ordinance- India's Privacy Data protection bill- Based on domain: HIPPA, CALOPPA- COPPA- e Privacy Directive

UNIT III Privacy Risk Quantification **11**

Risk analysis: FAIR method, organizational risk, domain redux-Quantifying risk: Risk of Exposure, Opportunity Frequency, Probability of Action, Attempt Frequency, Vulnerability, Action Frequency, Population Magnitude, Risk, Adverse Consequence Risk, Case Study : Risk of Secondary Use of the population data

UNIT IV Adopting Privacy By Design **11**

Foundational principles – Challenges of principles - The Privacy by Design Process: requirements engineering for privacy, understanding quality attributes, identify information needs, High level design, Low Level Design and Implementation, imposing controls, Testing and validation - Integrating Privacy by Design into the Business

UNIT V Privacy Enhancing Technologies **9**

Data Oriented Strategies: Separate, minimize, abstract, hide – Techniques: Aggregation, De-identification, Encryption, Identity and access management, Authentication -Process Oriented Strategies: Informing the Individual, User Control, Policy and Process, Demonstrate Compliance.

TOTAL PERIODS: 45

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REFERENCES

1. An Introduction to Privacy for Technology Professionals
,ExecutiveEditorTravis D.BreauxYashavant P. Kanetkar, K, Let Us C, BPB Publications, 2011.
2. Strategic Privacy by Design R. Jason Cronk, CIPP/US, CIPM, CIPT, FIP
3. Privacy in Technology Standards and Practices for Engineers and Security and ITProfessionals by JC Cannon
4. Alessandro Acquisti, Leslie John, and George Loewenstein, “What Is Privacy Worth?” Carnegie Mellon University- Heinz College, Heinz CMU, 2010, accessed July 5,2018, www.heinz.cmu.edu/~acquisti/papers/acquisti-ISRworth.pdf

COURSE OUTCOMES

At the end of the course, the student should be able to:

CO1: Explain the basics of privacy and possible breaches across the life cycle

stagesCO2: Appreciate the role of Privacy Principles, Standards and Regulations

CO3: Explain the role of available Privacy Principles, Standards &

RegulationsCO4: Perform risk analysis

CO5: Apply privacy by design

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	1		1	1		
2	2		1	2		1
3	2			2		1
4	3		2	2		2
5	2		1	1		2

NC2433 DEEP LEARNING & COMPUTER VISION L T P C 3 0 0 3

COURSE OBJECTIVES

- To study the fundamentals of computer vision and deep learning
- To introduce the concepts of Convolutional neural networks
- To learn various deep learning models for real time applications.

UNIT I INTRODUCTION TO VISUAL FEATURES AND REPRESENTATIONS

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9

Image Formation, Image Representation: Linear Filtering, Correlation, Convolution; Visual Features: Edge, Blobs, Scale Space and Scale Selection, SIFT, SURF, HoG, LBP, Optical Flow.

UNIT II CONVOLUTIONAL NEURAL NETWORKS

9

Convolutions and Pooling: Color Channel Ordering Formats, Convolutional Layers, Filter Size, Padding, Stride, Pooling layers; A complete DNN architecture; Transfer learning: LeNet5, AlexNet, VGG, Inception, GoogLeNet, ResNet.

UNIT III VISUALIZATION AND UNDERSTANDING DNNs

9

Visualization of Kernels; Backprop-to-image/Deconvolution Methods; Deep Dream, Hallucination, Neural Style Transfer; CAM, Grad-CAM, Grad-CAM++; Recent Methods (IG, Segment-IG, SmoothGrad).

UNIT IV DNNs FOR RECOGNITION, VERIFICATION, SEGMENTATION

9

CNNs for Recognition and Verification: Siamese Networks, Triplet Loss, Contrastive Loss, Ranking Loss; CNNs for Detection: Object Detection, R-CNN, Fast R-CNN, Faster R-CNN, YOLO, SSD; CNNs for Segmentation: FCN, SegNet, U-Net, Mask-RCNN.

UNIT V RECURRENT NEURAL NETWORK, ATTENTION AND RECENT MODELS

9

Review of RNNs; CNN + RNN Models for Video Understanding: Spatio-temporal Models, Action/Activity Recognition; Attention Models in Vision; Vision and Language: Image Captioning, Visual QA, Visual Dialog; Spatial Transformers: Transformer Networks; Deep Generative Models: GANs, VAEs; Recent Trends: Zero-shot, One-shot, Few-shot Learning.

TOTAL PERIODS: 45

TEXT BOOK

1. Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, 2016.
2. Simon Prince, Computer Vision: Models, Learning, and Inference, 2012.

REFERENCES:

1. Michael Nielsen, Neural Networks and Deep Learning, 2016
2. Yoshua Bengio, Learning Deep Architectures for AI, 2009
3. Richard Szeliski, Computer Vision: Algorithms and Applications, 2010.
4. David Forsyth, Jean Ponce, Computer Vision: A Modern Approach, 2002.

COURSE OUTCOMES

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At the end of the course, the students should be able to

CO 1: Explain the fundamentals of computer vision

techniques.CO 2: Describe deep learning architecture models.

CO 3: Apply the deep learning algorithms and solve real-world problems.

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	1		1		1	1
2	1		2	1	1	2
3	3	3	2	1	2	3

NR2435

REINFORCEMENT LEARNING

L T P C 3 0 0 3

COURSE OBJECTIVES

- Introduce the different basic elements of Reinforcement Learning (RL).
- Study about Tabular methods and Q-networks.
- Study about policy optimization.
- Learn current advancements and applications in RL.

UNIT I Basics of Reinforcement Learning (RL)

9

Elements of RL, RL framework, Markov property, Partially Observable Markov Decision Process, Policies, value functions and Bellman equations.

UNIT II Tabular methods

9

Planning with dynamic Programming, Monte Carlo control, Temporal-Difference learning methods

- TD(0), SARSA, Q-Learning.

UNIT III Q-networks

8

Deep Q-networks - DQN, DDQN, Dueling DQN, Prioritised Experience Replay.

UNIT IV Policy optimizations

9

Optimal policies and optimal value functions, Bellman optimality equations, Vanilla Policy Gradient, REINFORCE algorithm and stochastic policy search, Actor-critic methods - A2C and A3C, Advanced policy gradient - PPO, TRPO, DDPG.

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UNIT V Recent Advancements and Applications

10

Meta-learning, Multi-Agent RL, Model-based RL approach, Code Standards and Python Libraries used in RL: SuperSuit, Stable Baselines3, Pistonball and MAgent

RL for real-world problems: Implementation of value iteration, discount grid, action selection and Q-Learning.

TOTAL PERIODS: 45

TEXTBOOK

1. Richard S. Sutton and Andrew G. Barto, "Reinforcement Learning: An Introduction", MIT Press, 2nd edition, 2018.

REFERENCES:

2. Russell, Stuart J., and Peter Norvig. "Artificial intelligence: a modern approach." Pearson Education Limited, 2016.
3. Kevin P. Murphy, "Machine Learning: A Probabilistic Perspective".
4. Csaba Szepesvari, "Algorithms for Reinforcement learning".
5. Wiering, Marco, and Martijn Van Otterlo. "Reinforcement learning." Adaptation, learning, and optimization 12 (2012): 3
6. Li, Yuxi. "Deep reinforcement learning." arXiv preprint arXiv:1810.06339 (2018).
7. Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. "Deep learning." MIT press, 2016.

COURSE OUTCOMES

At the end of the course, the students should be able to

CO 1: Explain the elements of reinforcement learning.

CO 2: Apply tabular methods and Q-networks to solve classical

problems. CO 3: Interpret policy gradient methods from vanilla to

more complex cases. CO 4: Implement real-world problems applying code standards.

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	1		1			
2	2		1			
3	2		1			

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4	1		1	1		1
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OPEN ELECTIVES

OEC2499

SOFTWARE ARCHITECTURE & DESIGN

L T P C 3 0 0 3

COURSE OBJECTIVES

- To understand the role of architecture and the architect in the software development cycle.
- To learn the design, development, and assessment activities.
- To introduce architectural patterns and tactics, architecture assessment techniques, architecture driven design, and techniques for documenting architectures.
- To provide a basis for understanding cutting-edge techniques and concepts, using open-source projects as case studies.

UNIT I SOFTWARE ARCHITECTURE & DESIGN

9

The role of the software architect- Evolving role and responsibilities- Business implications of architecture solution: Licensing, deployment, and billing - Defining architecture -Different types of architecture – application architecture, enterprise architecture - Evolutionary Architecture - Architecture design, design process and the pattern decision relationship - Architecture principles - Internet Architecture and its impact- Quality Assurance - Quality Attribute Workshops (QAWs).

UNIT II SOFTWARE ARCHITECTURE PATTERNS

9

Purpose of architecture patterns - Using patterns practical considerations – Benefits & Limitation of patterns - Types of architecture patterns: Layered architecture, Event Driven Architecture, Microkernel, Microservices, Blackboard, Nano Services, Space-based architecture, Model View Controller- Pattern analysis: agility, deployment, testability, performance, scalability, ease of deployment - Considerations when choosing an architectural pattern.

UNIT III SOFTWARE DESIGN PATTERNS – I

9

Purpose - Iterators –Comprehensions –generators –Coroutines –case study. Decorator pattern – observer pattern –strategy pattern –state pattern –singleton pattern –template pattern

UNIT IV SOFTWARE DESIGN PATTERNS – II

9

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Adaptor pattern –facade pattern –flyweight pattern –command pattern –abstract factory pattern – composite pattern - Patterns for Agile Development- Restful Service Pattern - Concurrency –threads
–multi-processing – async i/o.

UNIT V ARCHITECTURE AS A SERVICE

9

Serverless architectures: Serverless benefits and drawbacks - Future of Serverless - Examples of serverless architecture systems - Reference architecture: mobile backend, real time processing, web applications, IoT backend, real time stream processing - Functions as a service (FaaS) and Platform as a Service (PaaS) – comparison and opportunities - FaaS: state, execution duration, startup latency, gateway- FaaS providers: IBM OpenWhisk, AWS Lambda, Google Cloud Functions, Microsoft Azure Functions, Iron.io and Web task-Frameworks: Serverless, Nano services – Case Studies.

TOTAL PERIODS: 45

TEXT BOOKS

1. R. N. Taylor, N. Medvidovic and E. M. Dashofy 2009, Software Architecture: Foundations, Theory, and Practice, Wiley [ISBN: 9780470167748]
2. Len Bass 2012, Software Architecture in Practice, 3rd Ed., Addison-Wesley Professional [ISBN: 9780321815736]
3. Dusty Phillips, “Python 3 Object-Oriented Programming: Build robust and maintainable software with object-oriented design patterns in Python 3.8”, Third Edition, Packt Publishing, 2018.
4. Mark Richards 2015, Software Architecture Patterns, O' Reilly [ISBN: 9781491924242]
5. Gregor Hohpe 2016, 37 Things One Architect Knows About IT Transformation: A Chief Architect's Journey, CreateSpace Independent Publishing Platform [ISBN: 9781537082981]

REFERENCES

1. Stephen F. Lott, “Mastering Object-oriented Python”, Second Edition, Packt Publishing, 2019.
2. Mark Lutz, “Programming Python: Powerful Object-Oriented Programming”, Fourth Edition, O'Reilly Media, 2011.

WEB REFERENCES

1. <http://www.oodeesign.com/>
2. <http://ui-patterns.com/patterns/>
3. http://sourcemaking.com/design_petterns
4. http://www.dmoz.org/computers/programming/methodologies/patterns_and_anti-patterns/
5. Serverless: <https://serverless.com/>

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6. Serverless Google Cloud Platform: <https://cloud.google.com/functions/>
7. Serverless Reference Architectures with AWS Lambda:
<http://www.allthingsdistributed.com/2016/06/aws-lambda-serverless-reference-architectures.html>
8. The Open Group Architecture Framework:
<http://www.opengroup.org/subjectareas/enterprise/togaf>

COURSE OUTCOMES

At the end of the course, the students should be able to

- CO 1: Discuss the evolving role of a software architect in the digital age.
CO 2: Appraise the different approaches to software design and patterns that can be applied to solve a problem in a variety of contexts.
CO 3: Examine how architecture patterns capture structural and behavioural information of a system.
CO 4: Design a software architecture solution from a presented case study.
CO 5: Evaluate how Architecture as a Service (AaaS), serverless architecture systems and other emerging trends are impacting the field of software design and architecture.

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	1					
2	1	1	1	1		
3	1	1	1	1		
4	3	3	3	3	3	3
5	3	3	2	3	3	3

OEC2492 HEALTH, SPORTS & FINANCIAL ANALYTICS L T P C 3 0 0 3

COURSE OBJECTIVES:

- To explore the various forms of electronic health care information.
- To learn the techniques adopted to analyze healthcare data.
- To understand the predictive models for clinical and sports data
- To understand the importance of sports & financial analytics.
- To use analytics, improve financial situations.

UNIT I HEALTHCARE ANALYTICS I

9

Introduction to Healthcare Data Analytics- Electronic Health Records– Components of EHR- Coding Systems- Benefits of EHR- Barrier to Adopting HER Challenges- Phenotyping

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Algorithms-Health Informatics on FHIR.

UNIT II HEALTHCARE ANALYTICS II **9**

Natural Language Processing and Data Mining for Clinical Text- Mining the Biomedical- Visual Analytics for Healthcare - Social Media Analytics for Healthcare - Case Study :COVID19 PandemicData.

UNIT III SPORTS ANALYTICS & APPLICATIONS **9**

Understanding sports data - Creating performance metrics for players and teams - Forecasting and prediction - Machine learning in sports- Sports data visualization-Sports Analytics algorithms for performance prediction.

Data Analytics Tools: Rating Sports Teams - The Ratings Percentage Index (RPI) - From Point Ratings to Probabilities - Ranking Great Sports Collapses – Case studies: Comparing Players from Different Eras

UNIT IV FINANCIAL DATA ANALYTICS & RISK MODELING **9**

Definition, relevance and scope financial Analytics, Components of Financial Analytics, Features of Financial Analytics recent trends in financial analytics, building models using accounting and financial data. Measuring and modeling risk, Modeling Credit Risk.

UNIT V BUSINESS VALUATION ANALYTICS **9**

Cash Flow statement – prepare and analyze, model and forecast financial statement & business valuation. Capital budgeting – application and issues

Case studies that map sustainable development goals such as 3, 6, 7, 8 (3: Good health and wellbeing, 6: Clean water and sanitation, 7: Affordable and clean energy, 8: Decent work and economic growth)

TOTAL PERIODS: 45

REFERENCES

1. Chandan K. Reddy and Charu C Aggarwal, “Healthcare data analytics”, Taylor & Francis, 2015.
2. FHIR - <https://dl.acm.org/doi/book/10.5555/3281321>
3. Massaad, Elie, and Patrick Cherfan. “Social Media Data Analytics on Telehealth During theCOVID-19 Pandemic.” Cureus vol. 12,4 e7838. 26 Apr. 2020, doi:10.7759/cureus.7838
4. Wayne L. Winston, “Mathletics: How Gamblers, Managers, and Sports Enthusiasts Use Mathematics in Baseball, Basketball, and Football”, Pearson, 2012. (Part IV).
- 5.<https://www.mentionlytics.com/blog/social-media-monitoring-analytics-healthcare-systems>
7. K. Apostolou and C. Tjortjis, "Sports Analytics algorithms for performance prediction,"

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2019 10th International Conference on Information, Intelligence, Systems and Applications (IISA), 2019, pp. 1-4, doi: 10.1109/IISA.2019.8900754.

8. <https://www.hindawi.com/journals/sp/2022/3547703/>

9. Singh, Nitin. "Sport analytics: a review." learning 9 (2020): 11.

10. Dixon, Mathew F., Halperin, Igor, Bilokon, Paul, "Machine Learning in Finance", O'Reilly, 2019

Course Outcomes:

On successful completion of the course the learner will be able to,

CO1: Apply analytics for decision making in healthcare services.

CO2: Apply data mining to develop efficient clinical decision support systems

.CO3: Recognize, formulate, and analyze decision-making in sports

CO4: Explain various concepts and applications of analytics in

Finance CO5: Use business value analytics in improving financial outcomes.

CO-PO Mapping:

COs	POs					PSO
	1	2	3	4	5	
1	2	2	2	1		2
2	2	2	2	1		2
3	2	2	2	1		2
4	2	2	2	1		2
5	2	2	2	1		2

OEC2493

BUSINESS DATA ANALYTICS

L T P C

3 0 0 3

COURSE OBJECTIVES

- To understand the basics of business analytics and its life cycle.
- To gain knowledge about fundamental business analytics.
- To learn modeling for uncertainty and statistical inference.
- To understand analytics using Hadoop and Map Reduce frameworks.
- To acquire insight on other analytical frameworks.

UNIT I OVERVIEW OF BUSINESS ANALYTICS

9

Introduction – Drivers for Business Analytics – Applications of Business Analytics: Marketing

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and Sales, Human Resource, Healthcare, Product Design, Service Design, Customer Service and Support

– Skills Required for a Business Analyst – Framework for Business Analytics Life Cycle for Business Analytics Process – Converting real time decision making problems into hypothesis.

UNIT II ESSENTIALS OF BUSINESS ANALYTICS 9

Descriptive Statistics – Using Data – Types of Data – Data Distribution Metrics: Frequency, Mean, Median, Mode, Range, Variance, Standard Deviation, Percentile, Quartile, z-Score, Covariance, Correlation – Data Visualization: Tables, Charts, Line Charts, Bar and Column Chart, Bubble Chart, Heat Map – Data Dashboards – Solve numerical problems on basic statistics.

UNIT III MODELING UNCERTAINTY AND STATISTICAL INFERENCE 9

Modeling Uncertainty: Events and Probabilities – Conditional Probability – Random Variables – Discrete Probability Distributions – Continuous Probability Distribution – Statistical Inference: Data Sampling – Selecting a Sample – Point Estimation – Sampling Distributions – Interval Estimation – Hypothesis Testing – Real time applications.

UNIT IV ANALYTICS USING HADOOP AND MAPREDUCE FRAMEWORK 9

Introducing Hadoop – RDBMS versus Hadoop – Hadoop Overview – HDFS (Hadoop Distributed File System) – Processing Data with Hadoop – Introduction to MapReduce – Features of MapReduce
– Algorithms Using Map-Reduce: Matrix-Vector Multiplication, Relational Algebra Operations, Grouping and Aggregation – Extensions to MapReduce.

UNIT V OTHER DATA ANALYTICAL FRAMEWORKS 9

Overview of Application development Languages for Hadoop – PigLatin – Hive – Hive Query Language (HQL) – Introduction to Pentaho, JAQL – Introduction to Apache: Sqoop, Drill and Spark, Cloudera Impala – Introduction to NoSQL Databases – Hbase and MongoDB.

TOTAL PERIODS: 45

COURSE OUTCOMES

On successful completion of this course, the student will be able to

CO1: Identify the real world business problems and model with analytical solutions.

CO2: Solve analytical problem with relevant mathematics background knowledge.

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CO3: Convert any real world decision making problem to hypothesis and apply suitable statistical testing.

CO4: Solve simple applications involving analytics using Hadoop and MapReduce. CO: Summarize open source frameworks for modeling and storing data.

REFERENCE BOOKS

1. Vignesh Prajapati, “Big Data Analytics with R and Hadoop”, Packt Publishing, 2013.
2. Umesh R Hodeghatta, Umesha Nayak, “Business Analytics Using R – A Practical Approach”, Apress, 2017.
3. Anand Rajaraman, Jeffrey David Ullman, “Mining of Massive Datasets”, Cambridge University Press, 2012.
4. Jeffrey D. Camm, James J. Cochran, Michael J. Fry, Jeffrey W. Ohlmann, David R. Anderson, “Essentials of Business Analytics”, Cengage Learning, second Edition, 2016.
5. U. Dinesh Kumar, “Business Analytics: The Science of Data-Driven Decision Making”, Wiley, 2017.
6. A. Ohri, “R for Business Analytics”, Springer, 2012
7. Rui Miguel Forte, “Mastering Predictive Analytics with R”, Packt Publication, 2015.

OEC 2495

DIGITAL FORENSICS

L T P C 3 0 0 3

COURSE OBJECTIVES:

- To learn the security issues network layer and transport layer.
- To get exposed to security issues of the application layer.
- To learn computer forensics.
- To be familiar with forensics tools.
- To learn to model and interpret forensics data.

UNIT I NETWORK AND TRANSPORT LAYER SECURITY

9

Introduction, Network layer security: IPSec protocol – Authentication header – Key management protocol, Transport layer security: SSL and TLS, Introduction to E-mail security, Introduction to firewalls: Terminology – Types of firewalls.

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UNIT II UNDERSTANDING DIGITAL FORENSICS AND INVESTIGATION 9

Overview of digital forensics, Preparation for digital investigation, Professional conduct, preparing digital forensics investigation, conducting an investigation, Procedures for private sector investigations.

UNIT III DATA ACQUISITION AND PROCESSING 9

Understanding storage formats, determining acquisition methods, Contingency planning, using acquisition tools and validating, Identifying and collecting digital evidence, preparing for a search, Storing digital evidence.

UNIT IV DIGITAL FORENSICS ANALYSIS AND VALIDATION 9

Determining the data to collect and analyze, validating forensics data, addressing data hiding techniques, performing live acquisition.

UNIT V E-MAIL AND SOCIAL MEDIA INVESTIGATIONS 9

Introduction, Role of client and server in E-Mail, Investigating E-mail crimes: Understanding forensics linguistics – Examining E-mail headers and messages – Tracing E-mail files, Social mediaforensics on mobile devices: Forensics tools for social media investigations– OSINT.

TOTAL: 45 PERIODS

TEXT BOOKS:

1. Man Young Rhee, Internet Security: Cryptographic Principles, Algorithms and Protocols, Wiley Publications, 2003.
2. Warren G. Kruse II and Jay G. Heiser, “Computer Forensics: Incident Response Essentials”, Addison Wesley, 2002.

REFERENCES:

1. Nelson, Phillips, Enfinger, Steuart, Computer Forensics and Investigations”, Cengage Learning, India Edition, 2008.
2. John R. Vacca, Computer Forensics, Cengage Learning, 2005.
3. Richard E. Smith, Internet Cryptography, 3rd Edition Pearson Education, 2008.
4. Marjie T. Britz, Computer Forensics and Cyber Crime: An Introduction, 3rd Edition, Prentice Hall, 2013.

OUTCOMES:

Upon completion of this course, the students will be able to

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CO1: Explain the security issues network layer and transport layer.
CO2: Explain computer forensics.
CO3: Make use of forensics tools.
CO4: Analyze and validate forensics data.

COs	POs					PSO
	1	2	3	4	5	
1	1					
2	1			1		
3	2		1	2		2
4	3		2	1		2

OEC2498 EMBEDDED AUTOMATION L T P C 3 0 0 3

COURSE OBJECTIVES:

- ☐ To learn about the process involved in the design and development of real-time embedded system
- ☐ To develop the embedded C programming skills on 8-bit microcontroller
- ☐ To study about the interfacing mechanism of peripheral devices with 8-bit microcontrollers
- ☐ To learn about the tools, firmware related to microcontroller programming
- ☐ To build a home automation system

UNIT - I INTRODUCTION TO EMBEDDED C PROGRAMMING 9

C Overview and Program Structure - C Types, Operators and Expressions - C Control Flow - C Functions and Program Structures - C Pointers And Arrays - FIFO and LIFO - C Structures - Development Tools

UNIT - II AVR MICROCONTROLLER 9

ATMEGA 16 Architecture - Nonvolatile and Data Memories - Port System - Peripheral Features : Time Base, Timing Subsystem, Pulse Width Modulation, USART, SPI, Two Wire Serial Interface, ADC, Interrupts - Physical and Operating Parameters

UNIT – III HARDWARE AND SOFTWARE INTERFACING WITH 8-BIT SERIES CONTROLLERS 9

Lights and Switches - Stack Operation - Implementing Combinational Logic - Expanding I/O - Interfacing Analog To Digital Convertors - Interfacing Digital To Analog Convertors - LED Displays : Seven Segment Displays, Dot Matrix Displays - LCD Displays - Driving Relays - Stepper Motor Interface - Serial EEPROM - Real Time

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Clock - Accessing Constants Table - Arbitrary Waveform Generation - Communication Links - System Development Tools

UNIT – IV VISION SYSTEM

9

Fundamentals of Image Processing - Filtering - Morphological Operations - Feature Detection and Matching - Blurring and Sharpening - Segmentation - Thresholding - Contours - Advanced Contour Properties - Gradient - Canny Edge Detector - Object Detection - Background Subtraction

UNIT – V HOME AUTOMATION

9

Home Automation - Requirements - Water Level Notifier - Electric Guard Dog - Tweeting Bird Feeder - Package Delivery Detector - Web Enabled Light Switch - Curtain Automation - Android Door Lock - Voice Controlled Home Automation - Smart Lighting - Smart Mailbox - Electricity Usage Monitor - Proximity Garage Door Opener - Vision Based Authentic Entry System

TOTAL: 45 PERIODS

COURSE OUTCOMES:

On successful completion of this course, students will be able to

CO1: analyze the 8-bit series microcontroller architecture, features and pin details

CO2: write embedded C programs for embedded system application

CO3: design and develop real time systems using AVR microcontrollers

CO4: design and develop the systems based on vision mechanism

CO5: design and develop a real time home automation system

REFERENCES:

1. Dhananjay V. Gadre, "Programming and Customizing the AVR Microcontroller", McGraw-Hill, 2001.
2. Joe Pardue, "C Programming for Microcontrollers ", Smiley Micros, 2005.
3. Steven F. Barrett, Daniel J. Pack, "ATMEL AVR Microcontroller Primer : Programming and Interfacing", Morgan & Claypool Publishers, 2012
4. Mike Riley, "Programming Your Home - Automate With Arduino, Android and Your Computer", the Pragmatic Programmers, Llc, 2012.
5. Richard Szeliski, "Computer Vision: Algorithms and Applications", Springer, 2011.

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6. Kevin P. Murphy, "Machine Learning - a Probabilistic Perspective", the MIT PressCambridge, Massachusetts, London, 2012.

OEC24910 SUSTAINABLE MANAGEMENT L T P C3 0 0 3

COURSE OBJECTIVES:

- To provide students with fundamental knowledge of the notion of corporate sustainability.
- To determine how organizations impacts on the environment and socio-technical systems, the relationship between social and environmental performance and competitiveness, the approaches and methods.

UNIT I MANAGEMENT OF SUSTAINABILITY

9

Management of sustainability -rationale and political trends: An introduction to sustainability management, International and European policies on sustainable development, theoretical pillarsin sustainability management studies.

UNIT II CORPORATE SUSTAINABILITY AND RESPONSIBILITY

9

Corporate sustainability parameter, corporate sustainability institutional framework, integration of sustainability into strategic planning and regular business practices, fundamentals of stakeholder engagement.

UNIT III SUSTAINABILITY MANAGEMENT: STRATEGIES AND APPROACHES 9

Corporate sustainability management and competitiveness: Sustainability-oriented corporate strategies, markets and competitiveness, Green Management between theory and practice, Sustainable Consumption and Green Marketing strategies, Environmental regulation and strategic postures; Green Management approaches and tools; Green engineering: clean technologies and innovation processes; Sustainable Supply Chain Management and Procurement.

UNIT IV SUSTAINABILITY AND INNOVATION

9

Socio-technical transitions and sustainability, Sustainable entrepreneurship, Sustainable pioneers in green market niches, Smart communities and smart specializations.

UNIT V SUSTAINABLE MANAGEMENT OF RESOURCES, COMMODITIES AND COMMONS

9

Energy management, Water management, Waste management, Wild Life Conservation, Emerging trends in sustainable management, Case Studies.

TOTAL: 45 PERIODS

COURSE OUTCOMES:

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- CO1: An understanding of sustainability management as an approach to aid in evaluating and minimizing environmental impacts while achieving the expected social impact.
- CO2: An understanding of corporate sustainability and responsible Business Practices
- CO3: Knowledge and skills to understand, to measure and interpret sustainability performances.
- CO4: Knowledge of innovative practices in sustainable business and community management
- CO5: Deep understanding of sustainable management of resources and commodities

REFERENCES:

1. Daddi, T., Iraldo, F., Testa, Environmental Certification for Organizations and Products: Management, 2015
2. Christian N. Madu, Handbook of Sustainability Management 2012
3. Petra Molthan-Hill, The Business Student's Guide to Sustainable Management: Principles and Practice, 2014
4. Margaret Robertson, Sustainability Principles and Practice, 2014
5. Peter Rogers, An Introduction to Sustainable Development, 2006

OEC2497 IoT FOR SMART SYSTEMS L T P C 3 0 0 3

COURSE OBJECTIVES:

- To study about **Internet of Things** technologies and its role in real time applications.
- To introduce the infrastructure required for IoT
- To familiarize the accessories and communication techniques for IoT.
- To provide insight about the embedded processor and sensors required for IoT
- To familiarize the different platforms and Attributes for IoT

UNIT I INTRODUCTION TO INTERNET OF THINGS 9

Overview, Hardware and software requirements for IOT, Sensor and actuators, Technology drivers, Business drivers, Typical IoT applications, Trends and implications.

UNIT II IOT ARCHITECTURE 9

IoT reference model and architecture -Node Structure - Sensing, Processing, Communication, Powering, Networking - Topologies, Layer/Stack architecture, IoT standards, Cloud computing for IoT, Bluetooth, Bluetooth Low Energy beacons.

UNIT III PROTOCOLS AND WIRELESS TECHNOLOGIES FOR IOT 9

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PROTOCOLS:

NFC, SCADA and RFID, Zigbee MIPI, M-PHY, UniPro, SPMI, SPI, M-PCIe GSM, CDMA, LTE, GPRS, small cell.

Wireless technologies for IoT: WiFi (IEEE 802.11), Bluetooth/Bluetooth Smart, ZigBee/ZigBee Smart, UWB (IEEE 802.15.4), 6LoWPAN, Proprietary systems-Recent trends.

UNIT IV IOT PROCESSORS

9

Services/Attributes: Big-Data Analytics for IOT, Dependability, Interoperability, Security, Maintainability.

Embedded processors for IOT : Introduction to Python programming -Building IOT with RASPBERRY PI and Arduino.

UNIT V CASE STUDIES

9

Industrial IoT, Home Automation, smart cities, Smart Grid, connected vehicles, electric vehicle charging, Environment, Agriculture, Productivity Applications, IOT Defense

TOTAL: 45 PERIODS

COURSE OUTCOMES:

At the end of this course, the students will have the ability to

CO1: Analyze the concepts of IoT and its present developments.

CO2: Compare and contrast different platforms and infrastructures available for IoT

CO3: Explain different protocols and communication technologies used in IoT

CO4: Analyze the big data analytic and programming of IoT

CO5: Implement IoT solutions for smart applications

REFERENCES:

1. Arshdeep Bahga and Vijai Madisetti : A Hands-on Approach “Internet of Things”, Universities Press 2015.
2. Oliver Hersent, David Boswarthick and Omar Elloumi “ The Internet of Things”, Wiley, 2016.
3. Samuel Greengard, “ The Internet of Things”, The MIT press, 2015.
4. Adrian McEwen and Hakim Cassimally “Designing the Internet of Things “Wiley, 2014.
5. Jean- Philippe Vasseur, Adam Dunkels, “Interconnecting Smart Objects with IP: The Next Internet” Morgan Kaufmann Publishers, 2010.
6. Adrian McEwen and Hakim Cassimally, “Designing the Internet of Things”, John

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Wiley and sons, 2014.

7. Lingyang Song/DusitNiyato/ Zhu Han/ Ekram Hossain,” Wireless Device-to-Device Communications and Networks, CAMBRIDGE UNIVERSITY PRESS,2015.
8. OvidiuVermesan and Peter Friess (Editors), “Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems”, River Publishers Series in Communication, 2013.
9. Vijay Madiseti , ArshdeepBahga, “Internet of Things (A Hands on-Approach)”, 2014.
- 10.Zach Shelby, Carsten Bormann, “6LoWPAN: The Wireless Embedded Internet”, John Wileyand sons, 2009.
- 11.Lars T.Berger and Krzysztof Iniewski, “Smart Grid applications, communications andsecurity”, Wiley, 2015.
- 12.JanakaEkanayake, KithsiriLiyanage, Jianzhong Wu, Akihiko Yokoyama and Nick Jenkins, “Smart Grid Technology and Applications”, Wiley, 2015.
- 13.UpenaDalal,”Wireless Communications & Networks,Oxford,2015.

OEC2491 INTELLECTUAL PROPERTY RIGHTS L T P C3 0 0 3

COURSE OBJECTIVE

- To understand intellectual property rights and its valuation.

UNIT I INTRODUCTION

9

Intellectual property rights - Introduction, Basic concepts, Patents, Copyrights, Trademarks, Trade Secrets, Geographic Indicators; Nature of Intellectual Property, Technological Research, Inventions and Innovations, History - the way from WTO to WIPO, TRIPS.

UNIT II PROCESS

9

New Developments in IPR, Procedure for grant of Patents, TM, GIs, Patenting under Patent Cooperation Treaty, Administration of Patent system in India, Patenting in foreign countries.

UNIT III STATUTES

9

International Treaties and conventions on IPRs, The TRIPs Agreement, PCT Agreement, The Patent Act of India, Patent Amendment Act (2005), Design Act, Trademark Act, Geographical Indication Act, Bayh- Dole Act and Issues of Academic Entrepreneurship.

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UNIT IV STRATEGIES IN INTELLECTUAL PROPERTY 9

Strategies for investing in R&D, Patent Information and databases, IPR strength in India, Traditional Knowledge, Case studies.

UNIT V MODELS 9

The technologies Know-how, concept of ownership, Significance of IP in Value Creation, IP Valuation and IP Valuation Models, Application of Real Option Model in Strategic Decision Making, Transfer and Licensing.

TOTAL: 45 PERIODS

COURSE OUTCOMES

CO1: Understanding of intellectual property and appreciation of the need to protect it

CO2: Awareness about the process of patenting

CO3: Understanding of the statutes related to IPR

CO4: Ability to apply strategies to protect intellectual property

CO5: Ability to apply models for making strategic decisions related to IPR

REFERENCES

1. Sople Vinod, Managing Intellectual Property by (Prentice hall of India Pvt.Ltd), 2006.
2. Intellectual Property rights and copyrights, EssEss Publications.
3. Primer, R. Anita Rao and Bhanoji Rao, Intellectual Property Rights, Lastain Book company.
4. Edited by Derek Bosworth and Elizabeth Webster, The Management of IntellectualProperty, Edward Elgar Publishing Ltd., 2006.
5. WIPO Intellectual Property Hand book.

OE2494 ETHICAL MANAGEMENT L T P C3 0 0 3

COURSE OBJECTIVE

- To help students develop knowledge and competence in ethical management and decisionmaking in organizational contexts.

UNIT I ETHICS AND SOCIETY 9

Ethical Management- Definition, Motivation, Advantages-Practical implications of ethical management. Managerial ethics, professional ethics, and social Responsibility- Role of culture and society's expectations- Individual and organizational responsibility to society and the community.

UNIT II ETHICAL DECISION MAKING AND MANAGEMENT IN A CRISIS 9

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Managing in an ethical crisis, the nature of a crisis, ethics in crisis management, discuss case studies, analyze real-world scenarios, develop ethical management skills, knowledge, and competencies. Proactive crisis management.

UNIT III STAKEHOLDERS IN ETHICAL MANAGEMENT

9

Stakeholders in ethical management, identifying internal and external stakeholders, nature of stakeholders, ethical management of various kinds of stakeholders: customers (product and service issues), employees (leadership, fairness, justice, diversity) suppliers, collaborators, business, community, the natural environment (the sustainability imperative, green management, Contemporary issues).

UNIT IV INDIVIDUAL VARIABLES IN ETHICAL MANAGEMENT

9

Understanding individual variables in ethics, managerial ethics, concepts in ethical psychology- ethical awareness, ethical courage, ethical judgment, ethical foundations, ethical emotions/intuitions/intensity. Utilization of these concepts and competencies for ethical decision- making and management.

UNIT V PRACTICAL FIELD-GUIDE, TECHNIQUES AND SKILLS

9

Ethical management in practice, development of techniques and skills, navigating challenges and dilemmas, resolving issues and preventing unethical management proactively. Role modelling and creating a culture of ethical management and human flourishing.

TOTAL: 45 PERIODS

COURSE OUTCOMES

CO1: Role modelling and influencing the ethical and cultural context.

CO2: Respond to ethical crises and proactively address potential crises

situations.CO3: Understand and implement stakeholder management decisions.

CO4: Develop the ability, knowledge, and skills for ethical management.

CO5: Develop practical skills to navigate, resolve and thrive in management situations

REFERENCES

1. Brad Agle, Aaron Miller, Bill O' Rourke, The Business Ethics Field Guide: the essential companion to leading your career and your company, 2016.
2. Steiner & Steiner, Business, Government & Society: A managerial Perspective, 2011.
3. Lawrence & Weber, Business and Society: Stakeholders, Ethics, Public Policy, 2020.

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COURSE OBJECTIVES

- Solve linear programming problem and solve using graphical method.
- Solve LPP using simplex method
- Solve transportation, assignment problems
- Solve project management problems
- Solve scheduling problems

UNIT I LINEAR PROGRAMMING

9

Introduction to Operations Research – assumptions of linear programming problems -
Formulation of linear programming problem – Graphical method

UNIT II ADVANCES IN LINEAR PROGRAMMING

9

Solutions to LPP using simplex algorithm- Revised simplex method - primal dual relationships
–Dual simplex algorithm - Sensitivity analysis.

UNIT III NETWORK ANALYSIS – I

9

Transportation problems -Northwest corner rule, least cost method, Voges's
approximation method - Assignment problem -Hungarian algorithm

UNIT IV NETWORK ANALYSIS – II

9

Shortest path problem: Dijkstra's algorithms, Floyd's algorithm, systematic method -CPM/PERT

UNIT V NETWORK ANALYSIS – III

9

Scheduling and sequencing - single server and multiple server models - deterministic
inventory models - Probabilistic inventory control models

TOTAL PERIODS: 45

OUTCOMES

On successful completion of this course, the student will be able to

CO1: To formulate linear programming problem and solve using graphical
method.CO2: To solve LPP using simplex method
CO3: To formulate and solve transportation, assignment
problemsCO4: To solve project management problems
CO5: To solve scheduling problems

REFERENCES

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1. Harvey M Wagner, Principles of Operations Research: Prentice Hall of India 2010
2. Hitler Libermann, Operations Research: McGraw Hill Pub. 2009
3. Pant J C, Introduction to Optimisation: Operations Research, Jain Brothers, Delhi, 2008
4. Pannerselvam, Operations Research: Prentice Hall of India 2010
5. Taha H A, Operations Research, An Introduction, PHI, 2008

NL2466

PROJECT PHASE 1

L T P C 0 0 12 6

COURSE OBJECTIVES:

The objective of this course is to enable the students to

- ☐ Identify and formulate complex engineering problem, survey literature, identify suitable technologies and computing resources required and identify research gaps
- ☐ Demonstrate technical presentation skills, both through written and oral communication
- ☐

COURSE OUTCOMES:

On successful completion of this course, the students should be able to

CO1: Demonstrate an ability to define a precise problem statement with objectives and scope, and define a complex/ open-ended problem in engineering terms

CO2: demonstrate the skills for performing literature survey, identify gaps, analyze the technical content, and prepare a well-documented dissertation report

CO3: Communicate orally, showing knowledge on the subject and sensitivity towards social impact of the dissertation where applicable

COs	POs					PSO
	1	2	3	4	5	
1	2	1	1		2	2
2	2	3	1			1
3	2	3	2	3	3	2

NL2477

PROJECT PHASE II

L T P C 0 0 24 12

COURSE OBJECTIVES:

The objective of this course is to enable the students to

- ☐ Design an innovative product or an effective solution by applying current knowledge and adopt to emerging applications of engineering and technology

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- ☐ Effective technical presentation and knowledge skills
- ☐ Write technical paper in scientific journal style & format

COURSE OUTCOMES:

On successful completion of this course, the students should be able to

CO1: Apply knowledge to design suitable software/hardware, conduct experiments and utilize modern tools for developing working models / process / systems for solving real world engineering problems

CO2: Demonstrate communication skills through effective oral presentation showing knowledge on the subject, and where applicable, sensitivity towards social impact of the dissertation

CO3: Write a dissertation paper in scientific journal style and publish in refereed journal, or patent.

CO-PO Mapping

COs	POs					PSO
	1	2	3	4	5	
1	3		3	3		3
2	3	2	3	3	3	3
3	2	3	1	1	2	1

PROGRAM NAME: M.TECH - AI IN VLSI TECHNOLOGY

PROGRAM DURATION: 2 YEARS

PROGRAM DESCRIPTION/PHILOSOPHY:

The integration of AI and VLSI design to create smarter, faster, and more energy-efficient semiconductor chips. It emphasizes automation, customization, and interdisciplinary learning, enabling students to master AI-driven tools for chip design, layout, verification, and testing. This program addresses the industry's growing need for custom AI chips used in smartphones, autonomous vehicles, wearable, and IoT devices.

DISTINCTIVENESS OF THE AI IN VLSI TECHNOLOGY

This program lies in its ability to produce next-generation engineers who are skilled in both AI and VLSI design. By focusing on AI-driven design automation, verification, optimization, and hardware-software co-design, the program prepares students for roles at the forefront of semiconductor innovation. Unlike traditional VLSI engineers, graduates of this program are equipped to design AI-specific hardware accelerators, support edge AI devices, and develop the next generation of neuromorphic chips. This unique blend of skills positions them to be industry leaders in fields like autonomous vehicles, IoT, robotics, and smart devices.

NEED OF THE PROGRAM

The future of chip design is AI-driven, and this program prepares students to be at the forefront of this revolution. This program uniquely positions graduates to:

- Lead the development of custom AI hardware for edge devices.
- Innovate in AI-driven VLSI automation and optimization.
- Work on hardware-software co-design for AI-based applications.

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M.TECH - AI IN VLSI TECHNOLOGY
REGULATIONS – 2025
CHOICE BASED CREDIT SYSTEM

PROGRAM OUTCOMES (POs)

- PO1** An ability to independently carry out research / investigation and development work to solve practical problems.
- PO2** An ability to write and present a substantial technical report/document.
- PO3** Students should be able to demonstrate a degree of mastery over the area as per the specialization of the programme. The mastery should be at a level higher than the requirements in the appropriate bachelor programme.
- PO4** Understand the fundamentals involved in the Designing and Testing of electronic circuits in the VLSI domain.
- PO5** Provide solutions through research to socially relevant issues for modern Electronic Design Automation (EDA) tools with knowledge, techniques, skills and for the benefit of the society
- PO6** Interact effectively with the technical experts, applying **AI** and **VLSI technology** to solve complex engineering challenges in emerging technology fields in industry and society.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs):

PEO 1: Strong Foundation in VLSI Design and AI Integration

They will be able to integrate AI algorithms into the VLSI design flow for automation, optimization, and enhancement of system performance in digital circuit design.

PEO 2: Proficiency in AI-Driven Design Automation Tools

They will be equipped to apply **machine learning algorithms**, **reinforcement learning**, and other AI techniques to solve practical problems in VLSI design, such as timing optimization, placement, routing, power optimization, and yield prediction.

PEO 3: Research and Innovation in VLSI Design Automation

Graduates will be prepared to work in research and development teams, contributing to the advancement of AI applications in the semiconductor industry and academic research.

PEO 4: Professional Development and Lifelong Learning

Graduates will be committed to **professional development** and **lifelong learning** to stay updated with the latest trends in both VLSI technology and AI.

PEO 5: Ethical and Responsible Practices in AI and VLSI Design

Graduates will demonstrate an understanding of the ethical, social, and environmental implications of using AI in VLSI design and will practice **ethical decision-making** in their professional careers.

PROGRAMME SPECIFIC OBJECTIVES (PSOs):

M.Tech. AI IN VLSI TECHNOLOGY specific goals are to create engineers who will be able:

- PSO1** To analyze, design and develop solutions by applying foundational concepts of AI and VLSI Design.
- PSO2** To apply design principles and best practices for developing quality products for scientific and business applications.
- PSO3** To adapt to emerging information and communication technologies (ICT) to innovate ideas and solutions to existing/novel problems

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CURRICULUM AND SYLLABI FOR SEMESTERS I to IV
SEMESTER I

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS WEEK				PER C	TOTAL PERIODS
				L	T	P	C		
THEORY									
1	MA25101	Statistics And Graph Theory	FC	3	1	0	4	4	
2	AV25101	Introduction to VLSI Design	PCC	3	0	0	3	3	
3	AV25102	Scripting language in VLSI	PCC	3	1	0	4	4	
4	AV25103	Advanced Digital Logic Design	PCC	3	1	0	4	4	
5	AV25104	Introduction to Artificial Intelligence	PCC	3	0	0	3	3	
6		Professional Elective I	PEC	3	0	0	3	3	
PRACTICALS									
7	AV2511	Advanced VLSI laboratory	PCC	0	0	4	2	4	
8	AV2512	AI laboratory	PCC	0	0	4	2	4	
TOTAL				18	3	8	25	29	

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SEMESTER II

S.NO .	COURSE CODE	COURSE TITLE	CATE GOR Y	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
THEORY								
1	AV25201	High-Performance Computing in VLSI	PCC	3	1	0	4	4
2	AV25202	VLSI Processor Design	PCC	3	0	0	3	3
3	AV25203	ASIC and FPGA Testing	PCC	3	0	0	3	3
4	AV25204	AI Algorithms in VLSI	PCC	3	1	0	4	4
5		Professional Elective I	PEC	3	0	0	3	3
6		Professional Elective II	PEC	3	0	0	3	3
PRACTICALS								
7	AV2521	AI in VLSI Laboratory	PCC	0	0	4	2	4
8	AV2522	Seminar	EDC	0	0	2	1	2
TOTAL				18	2	6	23	26

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SEMESTER III

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
THEORY								
1	AV25301	System-on-Chip (SoC) Design Verification	PCC	3	0	0	3	3
2		Professional Elective II	PEC	3	0	0	3	3
3		Professional Elective III	PEC	3	0	0	3	3
4	MC25101	Research Methodology and IPR	MC	3	0	0	3	3
PRACTICALS								
5	AV2531	VLSI Design Automation using AI Laboratory	PCC	0	0	4	2	4
6	AV2532	Project Work I	EDC	0	0	12	6	12
TOTAL				12	0	16	20	28

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SEMESTER IV

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
PRACTICALS								
1	AV2541	Project Work II	EDC	0	0	24	12	24
TOTAL				0	0	24	12	24

TOTAL CREDITS: 80

FOUNDATION COURSES (FC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	MA25101	Statistics And Graph Theory	FC	3	1	0	4	4

MANDATORY COURSE (FC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	MC25101	Research Methodology and IPR	MC	3	0	0	3	3

ARUNAI UNIVERSITY**TIRUVANNAMALAI****PROFESIONAL CORE COURSES (PCC)**

S.NO	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	AV25101	Introduction to VLSI Design	PCC	3	0	0	3	3
2	AV25102	Scripting language in VLSI	PCC	3	1	0	4	4
3	AV25103	Advanced Digital Logic Design	PCC	3	1	0	4	4
4	AV25104	Introduction to Artificial Intelligence	PCC	3	0	0	3	3
5	AV25201	High-Performance Computing (HPC) in VLSI	PCC	3	1	0	4	3
6	AV25202	VLSI Processor Design	PCC	3	0	0	3	3
7	AV25203	ASIC and FPGA Testing	PCC	3	0	0	3	3
8	AV25204	AI Algorithms in VLSI	PCC	3	1	0	4	4
9	AV25301	System-on-Chip (SoC) Design Verification	PCC	3	0	0	3	3
TOTAL CREDITS							31	

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LIST OF PROFESSIONAL ELECTIVE COURSES (PEC)

SEMESTER I, ELECTIVE I

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	AV2501	Hybrid Quantum-Classical VLSI Design	PEC	3	0	0	3	3
2	AV2502	Neural Networks in VLSI Design Automation	PEC	3	0	0	3	3
3	AV2503	Programming for AI and VLSI	PEC	3	0	0	3	3
4	AV2504	Programmable Architecture	PEC	3	0	0	3	3
TOTAL CREDITS								12

SEMESTER II, ELECTIVE II

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	AV2505	Design for Testability (DFT) and Yield Optimization	PEC	3	0	0	3	3
2	AV2506	Advanced CMOS Technology and Process Integration	PEC	3	0	0	3	3
3	AV2507	VLSI Design Automation	PEC	3	0	0	3	3
TOTAL CREDITS								9

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SEMESTER III, ELECTIVE III

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	AV2508	FPGA Embedded Design	PEC	3	0	0	3	3
2	AV2509	Network on Chip	PEC	3	0	0	3	3
3	AV2510	Signal Integrity for High Speed Design	PEC	3	0	0	3	3
TOTAL CREDITS								9

EMPLOYMENT DEVELOPMENT COURSES (EDC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	AV2522	Seminar	EDC	0	0	2	1	2
2	AV2532	Project Work I	EDC	0	0	1 2	6	12
3	AV2541	Project Work II	EDC	0	0	2 4	12	24
TOTAL CREDITS								19

MA25101

STATISTICS AND GRAPH THEORY

L T P C

3 1 0 4

OBJECTIVES

- To learn about statistical methods and their applications in engineering.
- To introduce graph as mathematical model to solve connectivity related problems.
- To introduce fundamental graph algorithms.
- To familiarize the students with the formulation and construction of a mathematical model for a linear programming problem in a real life situation.
- To provide knowledge and training using non-linear programming under limited resources for engineering and business problems.

UNIT I LINEAR ALGEBRA

12

Vectors and Spaces: Vector operations, linear independence, basis, and dimension. Matrices and Transformations, Matrix operations, eigenvalues, eigenvectors, singular value decomposition (SVD).

UNIT II PROBABILITY AND STATISTICS

12

Probability Basics: Random variables, probability distributions, Bayes' theorem. Statistical Analysis Mean, variance, covariance, and correlation. Markov Chains and Stochastic Processes.

UNIT III GRAPHS

12

Graphs and graph models, Graph terminology and special types of graphs, Matrix representation of graphs and graph isomorphism, Connectivity, Euler and Hamilton paths.

UNIT IV GRAPH ALGORITHM

12

Graph Algorithms, Directed graphs, Some basic algorithms, Shortest path algorithms, Depth, First search on a graph, Theoretic algorithms, Performance of graph theoretic algorithms – Graph theoretic computer languages.

UNIT V LINEAR AND NON LINEAR PROGRAMMING

12

Linear Programming: Formulation, Graphical solution, Simplex method, Two-phase method, Transportation and Assignment Models. NonLinear Programming: Constrained Problems, Equality

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constraints, Lagrangean Method , Inequality constraints ,Karush , Kuhn-Tucker (KKT) conditions – Quadratic Programming

TOTAL PERIODS: 60

OUTCOMES

At the end of the course, students will be able to

CO1: apply the concept of statistics methods and their applications in engineering

CO2: apply graph ideas is solving connectivity related problems.

CO3: apply fundamental graph algorithms to solve certain optimization problems.

CO4: formulate and construct mathematical models for linear programming problems and solve the transportation and assignment problems.

CO5: model various real life situations as optimization problems and effect their solution through Non-linear programming.

TEXTBOOK

- 1."Linear Algebra and Its Applications" by Gilbert Strang
- 2."Probability and Statistics for Engineers and Scientists" by Ronald E. Walpole et al.
- 3.Taha H.A, “Operation Research: An Introduction”, Ninth Edition, Pearson Education, New Delhi, 2010.
- 4.Gupta P. K, and Hira D.S., “Operation Research”, Revise Edition, S. Chand and Company Ltd., 2012.
- 5.Sharma J.K., “Operation Research”, 3rd Edition, Macmillan Publishers India Ltd., 2009.
- 6.Douglas B. West, “Introduction to Graph Theory”, Pearson Education, New Delhi, 2015.
- 7.Balakrishna R., Ranganathan. K., “ A text book of Graph Theory”, Springer Science and Business Media, New Delhi, 2012.
- 8.Narasimha Deo, “Graph Theory with Applications to Engineering and Computer Science”, Prentice Hall India,1997.

COs-POs & PSOs MAPPING:

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	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	2	2	3	3
CO2	3	3	3	3	2	2	2	2	2
CO3	3	3	3	3	2	2	2	2	2
CO4	3	3	3	3	2	2	2	3	3
CO5	3	3	3	3	2	2	2	3	3
Avg	3	3	3	3	2	2	2	2.6	2.6

*** 3- High, 2- Medium, 1- Low**

AV25101

INTRODUCTION TO VLSI DESIGN

L T P C

3 0 0 3

OBJECTIVES

- Understanding the fundamentals of CMOS technology and its role in VLSI design.
- Gaining proficiency in digital logic design, including combinational and sequential circuits.
- Familiarizing with VLSI design methodologies, design flow, and automation tools.
- Developing skills in layout design, physical design, and fabrication processes.
- Exploring power, performance, and area (PPA) optimization techniques in VLSI circuits.
- Introducing design verification, testing, and fault tolerance techniques for reliable systems.

UNIT 1 OVERVIEW OF VLSI TECHNOLOGY

9

History, evolution, and importance of VLSI in modern electronics, Design Process: VLSI design flow from specification to fabrication, VLSI Design Levels: System-level design, logic-level design, and physical design, Moore's Law: Impact on scaling and technology trends.

UNIT II CMOS TECHNOLOGY

9

Basics of MOSFETs: Working principle, types (NMOS, PMOS), and characteristics, CMOS Inverter: Operation, voltage transfer characteristics, and sizing, Technology Scaling: Effects of scaling on device characteristics and performance, Process Flow: Basic fabrication steps of CMOS ICs (oxidation, lithography, etching, etc.).

UNIT III LAYOUT DESIGN AND FABRICATION

9

IC Layout Design: Introduction to the physical design of VLSI circuits, layout of standard cells, and layout design rules. Design Rule Checking (DRC): Checking the validity of layout in terms of fabrication constraints. Parasitic Effects: Impact of parasitic capacitances, resistances, and inductances on the performance of circuits. Fabrication Steps: Lithography, etching, doping, and packaging.

UNIT IV CMOS CIRCUIT DESIGN AND OPTIMIZATION

9

Static CMOS Design: Full-static CMOS logic circuits, sizing, and optimization. Dynamic CMOS Design: Dynamic logic circuits and their trade-offs. Design Optimization: Speed, area, and power trade-offs in CMOS circuit design. High-Speed and Low-Power Design Techniques: Optimization methods for performance and power efficiency in CMOS circuits.

UNIT V TESTING AND VERIFICATION OF VLSI CIRCUITS

9

Testing Techniques: Fault models, scan chains, built-in self-test (BIST), and design for testability (DFT). Verification Methods: Functional verification, timing verification, and simulation-based verification.

TOTAL PERIODS: 45

OUTCOMES

At the end of the course, students will be able to

CO1: Understand the Fundamentals of VLSI Design

CO2: Understand CMOS Technology and its Applications

CO3: Understand Layout Design and Fabrication Processes

CO4: Implement CMOS Circuit Design and Optimization

CO5: Understand Testing and Verification Techniques for VLSI Circuits

TEXT BOOKS:

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TIRUVANNAMALAI

1. "CMOS VLSI Design: A Circuits and Systems Perspective" by Neil H.E. Weste and David Harris
2. "Principles of CMOS VLSI Design: A Systems Perspective" by Neil Weste and Kamran Eshraghian
3. "Digital Integrated Circuits: A Design Perspective" by Jan M. Rabaey, Anantha Chandrakasan, and Borivoje Nikolic
4. "VLSI Design" by Douglas Pucknell and Kamran Eshraghian
5. "Analog Integrated Circuit Design" by Tony Chan Carusone, David A. Johns, and Kenneth W. Martin
6. "VLSI Physical Design Automation" by Sadiq M. Sait and Habib Youssef
7. "SPICE for Circuits and Electronics Using PSPICE" by Muhammad H. Rashid

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	2	2	2	3
CO2	3	2	2	2	2	2	2	2	3
CO3	3	2	1	1	2	2	1	1	3
CO4	3	2	2	1	2	2	2	1	1
CO5	3	2	1	1	2	2	1	1	1
Avg	3	2	1.6	1.4	2	2	1.6	1.4	2.2

*** 3- High, 2- Medium, 1- Low**

OBJECTIVES

- Provide a strong foundation in key scripting languages such in VLSI design automation.
- Teach how to automate repetitive tasks in VLSI design processes, such as simulation setup, synthesis, verification, and timing analysis, using scripting languages.
- Explore how to use scripting languages to interact with Electronic Design Automation (EDA) tools.
- Enable students to develop scripts for automating complex workflows, improving efficiency and reducing human error in VLSI design projects.
- Train students to write custom test benches and verification environments using scripting languages, facilitating better test coverage and faster design verification.
- Emphasize the integration of scripting languages with the complete VLSI design flow, from high-level design to verification, and from synthesis to implementation.
- Encourage students to use scripting to solve practical problems in VLSI design, such as automating parameter tuning, handling large datasets, and managing simulation outputs.

UNIT I INTRODUCTION TO VLSI DESIGN AUTOMATION 12

Introduction to VLSI design and its components, Importance of scripting in the VLSI design flow, Overview of simulation, synthesis, and layout design flows.

UNIT II INTRODUCTION TO TCL AND SCRIPTING WITH EDA TOOLS 12

Basic TCL commands and syntax, File I/O operations in TCL, Automation of synthesis and simulation tasks.

UNIT III PERL FOR TEXT PARSING AND AUTOMATION 12

Introduction to Perl syntax, Using regular expressions for log file parsing, Automating post-processing of simulation results.

UNIT IV PYTHON SCRIPTING FOR VLSI DESIGN 12

Python basics and file handling, Working with libraries for data analysis, Generating and running testbenches with Python.

UNIT V SHELL SCRIPTING FOR VLSI DESIGN AUTOMATION 12

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Writing Bash scripts to automate design tasks, Creating Makefiles for managing simulation and synthesis tasks, Testbench Automation and Regression Testing.

TOTAL PERIODS: 60

OUTCOMES

On the successful completion of this course, the students will be able:

- CO1** Proficiently write and debug scripts and Automate tasks in VLSI design flow
- CO2** Integrate scripting with EDA tools and Write custom testbenches for verification..
- CO3** Handle simulation and synthesis results and Debug and troubleshoot design issues using scripts.
- CO4** Optimize design flows using scripts.
- CO5** Communicate design automation strategies.

TEXTBOOK

1. "TCL Scripting for VLSI Design" by William B. Smith
2. "Python for Hardware Design and Automation" by N.A. Patil and A.V. Bhuvaneshwari
3. "VLSI Design Automation: Tools and Techniques" by J. Bhasker

REFERENCE

1. "VHDL for Engineers" by Kenneth L. Short
2. "FPGA Prototyping by VHDL Examples" by Pong P. Chu
3. "SystemVerilog for Verification: A Guide to Learning the Testbench Language Features" by Chris Spear and Greg Tumbush

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	2	2	3	2	2	2	2
CO2	3	2	2	2	3	2	2	3	2

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CO3	3	2	2	2	3	2	2	3	2
CO4	3	2	2	2	3	2	2	3	2
CO5	3	2	1	2	3	2	2	3	2
Avg	3	2	1.8	2	3	2	2	2.6	2

*** 3- High, 2- Medium, 1- Low**

AV25103

ADVANCED DIGITAL LOGIC DESIGN

L T P C

3 1 0 4

OBJECTIVES

- Understand and Apply Advanced Digital Logic Concepts
- Design and Optimize Sequential Logic Systems
- Master Hardware Description Languages (HDL)
- Timing and Performance Analysis
- Design for Low-Power and High-Performance Digital Systems
- Utilize FPGA and ASIC Design Flow
- Understand System-Level Digital Design
- Collaborate and Communicate Design Concepts Effectively

UNIT I REVIEW OF BASIC DIGITAL LOGIC DESIGN

12

Boolean algebra and simplification techniques, Combinational circuits: gates, multiplexers, adders, subtractors, Karnaugh maps (K-map) and Quine–McCluskey method, Designing and optimizing combinational circuits, Sequential circuits: flip-flops, registers, counters

UNIT II ADVANCED COMBINATIONAL LOGIC DESIGN

12

Arithmetic Logic Units (ALU): Design of multi-bit ALUs, signed arithmetic, fast adders (carry-lookahead, carry-save), Multiplexers and Demultiplexers: Advanced multiplexing techniques, designing multi-level multiplexers, Decoders and Encoders: High-performance decoding, priority encoders, Code

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Converters: Binary to Gray, BCD to 7-segment, Memory Structures: ROM, RAM, and PLDs (Programmable Logic Devices)

UNIT III SEQUENTIAL LOGIC DESIGN

12

Sequential Circuits: Synchronous vs. asynchronous circuits, Designing Counters: Ripple counters, Johnson counters, and up/down counters, Finite State Machines (FSM): Moore and Mealy state machines, state diagram design, state minimization, and optimization, Pipelining and Register Transfer Level (RTL) Design: Using pipelining for high-performance design, Timing and Clocking: Setup time, hold time, propagation delay, and clock skew

UNIT IV DIGITAL SYSTEM DESIGN AND OPTIMIZATION

12

Timing Analysis: Critical path, delay analysis, and clock period calculations, Synchronous Design: Clocking schemes, clock distribution networks, skew and jitter management, Optimization Techniques: Power consumption optimization, area optimization, and speed optimization in digital circuits, Algorithmic State Machines (ASM): Design and optimization of state machines using ASM charts, Low-Power Design: Techniques for minimizing power in combinational and sequential circuits

UNIT V ADVANCED TOPICS IN LOGIC DESIGN

12

FPGA-Based Design: Overview of FPGA architecture, design flow, and optimization techniques, HDL Design (VHDL/Verilog): Writing and testing advanced RTL designs using hardware description languages, Synthesis and Simulation: Behavioral, structural, and gate-level modeling; timing simulation, High-Level Synthesis (HLS): From algorithm to hardware design using high-level synthesis tools.

TOTAL PERIODS: 60

OUTCOMES

CO1: Design and analyze advanced combinational logic circuits.

CO2: Master sequential logic circuit design.

CO3: Apply timing and performance analysis techniques.

CO4: Implement digital designs using HDL.

CO5: Design digital systems for low power and high performance.

TEXTBOOKS

1. "Digital Design: Principles and Practices" by John F. Wakerly

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2. "Advanced Digital Logic Design" by William G. O'Dell
3. "Digital Logic and Microprocessor Design with VHDL" by Enoch O. Hwang
4. "Digital System Design Using VHDL" by Charles H. Roth Jr.
5. "Digital Logic Design" by M. Morris Mano and Michael D. Ciletti

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	3	2	2	2	3	2	2
CO2	3	3	3	2	2	2	2	2	1
CO3	3	3	3	2	2	2	2	3	1
CO4	3	3	3	2	2	2	3	2	2
CO5	3	3	3	2	2	2	2	2	1
Avg	3	3	3	2	2	2	2.4	2.2	1.4

*** 3- High, 2- Medium, 1- Low**

AV25104 INTRODUCTION TO ARTIFICIAL INTELLIGENCE

L T P C

3 0 0 3

OBJECTIVES

- Explore advanced AI techniques and algorithms.
- Understand the role of deep learning, optimization, and generative models.
- Learn to apply AI to real-world challenges like vision, language, and decision-making.
- Gain exposure to cutting-edge research and emerging AI trends.

UNIT I ADVANCED SEARCH AND PLANNING

9

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Optimization-Based Search:Local search, hill climbing, simulated annealing.Genetic algorithms and evolutionary computation.Automated Planning:STRIPS and PDDL.Heuristic planning and partial-order planning.Real-time planning for robotics and dynamic systems.

UNIT II PROBABILISTIC REASONING AND GRAPHICAL MODELS

9

Bayesian Networks:Structure learning and inference.Applications in decision-making.Markov Models:Hidden Markov Models (HMMs), conditional random fields (CRFs).Monte Carlo Methods:Sampling techniques, particle filters.

UNIT III DEEP LEARNING FOR AI

9

Neural Networks:Advanced architectures: ResNet, Transformer, LSTM.Convolutional Neural Networks (CNNs):Applications in computer vision.Techniques like transfer learning and data augmentation.Generative Models:Variational autoencoders (VAEs).Generative adversarial networks (GANs).Diffusion models.Reinforcement Learning and Deep RL:Policy gradient methods, DDPG, PPO.

UNIT IV NATURAL LANGUAGE PROCESSING (NLP)

9

Advanced NLP Techniques:Transformer architecture (BERT, GPT).Attention mechanisms and self-attention.Applications:Language translation, summarization, chatbots.Semantic search and conversational AI.

UNIT V MULTI-AGENT SYSTEMS AND GAME THEORY

9

Introduction to Multi-Agent Systems:Collaborative vs. competitive agents.Game Theory:Nash equilibrium, mixed strategies.Applications:Autonomous vehicles, market simulations.

TOTAL PERIODS: 45

OUTCOMES

Upon completing this course, students will be able to:

CO 1: Apply search algorithms and optimization techniques to solve complex problems.

CO 2: Demonstrate knowledge of knowledge representation and reasoning methods for building intelligent systems.

CO 3: Analyze the fundamentals of machine learning algorithms and their applications in predictive modeling.

CO 4: Develop simple AI models for specific tasks, such as classification, clustering, and language processing.

CO 5: Evaluate the ethical implications of AI technologies and propose responsible AI solutions.

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1. "Artificial Intelligence: A Modern Approach" by Stuart Russell and Peter Norvig
2. "Introduction to Artificial Intelligence" by Wolfgang Ertel
3. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
4. "Artificial Intelligence: Foundations of Computational Agents" by David L. Poole and Alan K. Mackworth
5. Pattern Recognition and Machine Learning, by Christopher M. Bishop

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	3	2	2	2	2	1	1
CO2	3	3	3	2	2	2	2	2	2
CO3	3	3	3	2	2	2	3	1	2
CO4	3	3	3	2	2	2	2	2	1
CO5	3	3	3	2	2	2	2	2	2
Avg	3	3	3	2	2	2	2.2	1.6	1.6

*** 3- High, 2- Medium, 1- Low**

AV2511 ADVANCED VLSI LABORATORY

L T P C

0 0 4 2

OBJECTIVES

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Students will be equipped to design, verify, and optimize complex VLSI systems, preparing them for careers or research in the semiconductor industry.

EXPERIMENTS

1. a. Design, simulation, and characterization of CMOS inverters.
b. Analyze parameters like noise margin, propagation delay, and power dissipation.
2. Design and simulation of NAND, NOR, XOR, and multiplexer circuits.
3. Design and analyze flip-flops and latches for performance parameters.
4. Implementation and simulation of shift registers and counters.
5. Layout creation for combinational and sequential circuits using DRC and LVS.
6. Perform floorplanning and placement of standard cells using CAD tools.
7. Experiment with global and detailed routing techniques.
8. Analyze timing paths and identify critical path delays.
9. Implement power optimization techniques such as clock gating and multi-Vt.
10. Design and analyze SRAM cells for stability, read/write operations, and power.
11. Study of DRAM and flash memory structures and basic operations.

TOTAL PERIODS: 60

TOOLS REQUIRED

1. Design Tools (EDA Software): Cadence Tools/Synopsys Tools/Mentor Graphics (Siemens), Xilinx Tools
2. SPICE Simulators
3. Physical Design and Layout Tools: Magic VLSI/K Layout
4. Hardware Platforms: FPGA Development Boards

OUTCOMES

By completing this laboratory course, students will:

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CO1: Design, simulate, and optimize advanced CMOS circuits.

CO2: Develop skills in layout, physical design, and verification.

CO3: Gain hands-on experience with industry-standard tools for VLSI design.

CO4: Understand low-power design, timing analysis, and testing techniques.

CO5: Build a foundation for research or careers in ASIC, FPGA, and VLSI design.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	3	3	3	2	2	1	2
CO2	3	1	3	3	3	2	3	1	2
CO3	2	1	3	3	3	2	3	1	2
CO4	2	2	3	3	3	2	3	2	2
CO5	2	1	3	3	3	2	2	2	2
Avg	2.2	1.4	3	3	3	2	2.6	1.4	2

*** 3- High, 2- Medium, 1- Low**

AV2511 AI LABORATORY

L T P C

0 0 4 2

OBJECTIVES

- To provide students with a strong foundation in Artificial Intelligence and Machine Learning principles, including supervised, unsupervised, and reinforcement learning paradigms, as well as advanced techniques like neural networks and deep learning.

- To enable students to apply AI algorithms and tools to solve practical engineering problems, particularly in areas like VLSI design automation, optimization, and verification.

EXPERIMENTS

1. Introduction to AI and Python Programming: Basics of Python programming for AI, libraries like NumPy, Pandas, and Matplotlib.
2. Supervised Learning Algorithms: Implement linear regression, logistic regression, and decision trees for classification tasks and Evaluate models using accuracy, precision, recall, and confusion matrices
3. Unsupervised Learning Algorithms: Implement clustering algorithms (e.g., K-means, hierarchical clustering).
4. Implement a simple neural network using libraries such as TensorFlow or Keras.
5. Implement GANs for generating synthetic data.
6. Implement basic reinforcement learning algorithms (e.g., Q-learning, Deep Q Networks).
7. Apply machine learning techniques for optimizing VLSI design tasks like placement, routing, and timing analysis.
8. Implement AI-based solutions for predictive maintenance, anomaly detection, and sensor data analysis in IoT systems.
9. Study the techniques to detect and mitigate bias in AI models.
10. Implement AI techniques, such as SHAP and LIME, to interpret model predictions and decision-making processes

TOTAL PERIODS: 45

Tools Required

Programming Languages and Libraries

1. **Python:** The primary programming language for AI, with a wide range of libraries and frameworks for machine learning and deep learning.
2. **NumPy/Pandas:** For data manipulation and analysis.
3. **Matplotlib/Seaborn:** For data visualization and plotting.
4. **TensorFlow/Keras:** For deep learning model development and training.

5. **PyTorch:** A flexible deep learning framework, widely used in research for building custom neural networks.
6. **Scikit-learn:** For classical machine learning algorithms, model evaluation, and preprocessing.

Hardware Platforms

1. **GPUs** (e.g., NVIDIA Tesla, Google TPUs): For accelerating deep learning model training and inference.
2. **Raspberry Pi/Jetson Nano:** For deploying AI models on edge devices and IoT platforms.
3. **FPGAs:** For hardware-accelerated deep learning inference and VLSI design optimization.

OUTCOMES

By completing the Advanced AI Laboratory course, students will:

CO1: Master advanced AI and deep learning techniques for real-world applications.

CO2: Be proficient in building and training complex models for NLP, computer vision, reinforcement learning, and generative modeling.

CO3: Understand the ethical, biased, and fairness issues in AI, and apply techniques to mitigate them.

CO4: Be able to deploy AI models in production environments, including cloud platforms and edge devices.

CO5: Develop expertise in using modern AI tools, libraries, and frameworks for cutting-edge research and development.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	2	2	1	2	2	2	1
CO2	3	2	2	2	1	2	2	2	1
CO3	3	2	2	2	1	2	2	2	1
CO4	3	2	2	2	1	2	2	2	1

CO5	3	2	2	2	1	2	2	2	1
Avg	3	2	2	2	1	2	2	2	1

* 3- High, 2- Medium, 1- Low

AV25201 HIGH-PERFORMANCE COMPUTING (HPC) IN VLSI

L T P C

3 1 0 4

OBJECTIVES

- Understand the role of HPC in VLSI system design and optimization.
- Learn advanced techniques for high-speed circuit and system design.
- Explore parallel computing approaches and hardware accelerators.
- Examine the integration of HPC concepts in VLSI for AI, big data, and cloud applications.

UNIT I INTRODUCTION TO HIGH-PERFORMANCE COMPUTING AND VLSI 12

Overview of HPC: Definition, importance, and applications. Challenges in high-performance computing. VLSI and HPC: Role of VLSI in enabling HPC. Evolution of computing architectures: Single-core to multi-core processors. Performance Metrics: Speed, latency, throughput, and power consumption.

UNIT II PARALLEL COMPUTING ARCHITECTURES 12

Multi-core and Many-core Architectures: Design challenges and trade-offs. GPUs, TPUs, and FPGAs. Instruction-Level Parallelism (ILP): Pipelining, superscalar processing, and out-of-order execution. Memory Hierarchies: Cache coherence, memory bandwidth, and latency optimization.

UNIT III HIGH-SPEED VLSI CIRCUIT DESIGN 12

Advanced CMOS Design: High-speed logic design techniques. Power-delay trade-offs in VLSI. Clock Distribution: High-performance clocking schemes. Clock skew, jitter, and synchronization challenges. Interconnect Design: Signal integrity, crosstalk, and delay optimization.

UNIT IV HARDWARE ACCELERATORS FOR HPC 12

ASICs and FPGAs: FPGA programming for HPC applications. Case study: Cryptography and AI acceleration. Application-Specific Processors: TPUs and their role in AI workloads. Designing AI Accelerators: Challenges in deep learning hardware: Latency, throughput, and power.

UNIT V HPC ALGORITHMS AND IMPLEMENTATION IN VLSI**12**

Parallel Algorithms: Sorting, matrix multiplication, and FFT. Mapping algorithms to hardware. HPC in Signal and Image Processing: VLSI architectures for DSP, convolution, and filtering. Big Data Processing: Hadoop and Spark hardware optimization using VLSI.

TOTAL PERIODS: 60**OUTCOMES**

Upon successful completion of this course, students will be able to:

CO 1: Explain the fundamentals of High-Performance Computing and its application in solving large-scale VLSI design problems.

CO 2: Develop parallel algorithms for tasks such as logic synthesis, placement, and routing in VLSI design automation.

CO 3: Utilize HPC tools and frameworks to optimize the PPA metrics of VLSI circuits effectively.

CO 4: Apply machine learning and data-driven techniques in conjunction with HPC for VLSI design challenges.

CO 5: Evaluate and analyze the computational efficiency of HPC-based solutions for VLSI design and verification.

REFERENCE BOOKS:

1. “High-Performance Computing” by Charles Severance and Kevin Dowd.
2. “Digital Integrated Circuits” by Jan M. Rabaey.
3. “Computer Architecture: A Quantitative Approach” by John L. Hennessy and David A. Patterson.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	1	1	2	2	3	2	2
CO2	2	2	1	1	1	2	2	2	1

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CO3	2	2	1	1	1	2	2	3	1
CO4	2	2	1	1	1	2	3	2	2
CO5	2	2	1	1	1	2	2	2	1
Avg	2	2	1	1	1.2	2	2.4	2.2	1.6

*** 3- High, 2- Medium, 1- Low**

AV25202

VLSI PROCESSOR DESIGN

L T P C

3 0 0 3

OBJECTIVES

- Understand the fundamentals of processor architecture and its VLSI implementation.
- Learn techniques for designing high-performance and energy-efficient processors.
- Explore modern processor design trends, including multi-core architectures and AI accelerators.
- Gain hands-on experience with hardware description languages and EDA tools for processor design.

UNIT I FUNDAMENTALS OF PROCESSOR DESIGN

9

Introduction to Processor Architecture: Processor types: CISC, RISC, DSP, and VLIW architectures. Pipeline vs. non-pipeline processors. Basic Components: ALU, control unit, registers, and memory interface. Instruction set architecture (ISA) and instruction encoding. VLSI and Processor Design: Role of VLSI in processor design. Design challenges: Power, performance, and area.

UNIT II PIPELINED PROCESSOR DESIGN

9

Pipelining Concepts: Instruction-level parallelism and pipeline stages. Hazards: Data, control, and structural hazards. Pipeline Optimization: Forwarding, branch prediction, and speculative execution. MIPS pipeline architecture.

UNIT III MULTI-CORE AND PARALLEL PROCESSING

9

Multi-Core Processors: Design challenges in multi-core architectures. Shared vs. distributed memory architectures. Cache Design: Cache coherence protocols (MESI).Memory hierarchy optimization for performance. Interconnect Networks: Network-on-Chip (NoC) design principles. Latency and throughput trade-offs.

UNIT IV LOW-POWER PROCESSOR DESIGN

9

Power Optimization Techniques:Dynamic voltage and frequency scaling (DVFS).Clock gating and power gating.Sub-threshold and near-threshold design.Energy-Efficient Architectures:Techniques for battery-powered devices and IoT.

UNIT V SPECIALIZED PROCESSORS AND AI INTEGRATION

9

Application-Specific Processors:DSP processors and GPUs. Tensor Processing Units (TPUs).AI Accelerator:Neural network processing units (NNPUs).Hardware implementation of AI algorithms.Edge AI Processors:Processors for mobile and IoT applications.

TOTAL PERIODS: 45

OUTCOMES

CO1: Demonstrate an understanding of processor architecture fundamentals and their implementation in VLSI technology.

CO2: Design and optimize pipelined processors while resolving hazards to achieve high performance.

CO3: Analyze and implement multi-core processor architectures, including memory hierarchies and interconnect networks.

CO4: Develop low-power processor designs using advanced techniques such as clock gating and dynamic voltage scaling.

CO5: Explore specialized processors, including AI accelerators and edge computing hardware, and integrate them into VLSI systems.

TEXTBOOKS

1. “Computer Architecture: A Quantitative Approach” by John L. Hennessy and David A. Patterson.
2. “Digital Design and Computer Architecture” by David Money Harris and Sarah L. Harris.
3. “CMOS VLSI Design: A Circuits and Systems Perspective” by Weste and Harris.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	2	3	3	3	2	3	3
CO2	2	2	2	2	2	2	3	3	3
CO3	2	2	2	3	3	3	2	3	3
CO4	2	2	2	3	3	3	2	3	3
CO5	2	2	2	3	3	3	3	3	3
Avg	2	2	2	2.8	2.8	2.8	2.4	3	3

* 3- High, 2- Medium, 1- Low

AV25203

ASIC AND FPGA TESTING

L T P C

3 0 0 3

OBJECTIVES:

- To understand the importance of testing in the VLSI design lifecycle and its impact on product reliability, yield, and cost.
- To explore various testing techniques, including functional testing, structural testing, and defect-oriented testing.
- To learn fault modeling methods such as stuck-at faults, bridging faults, and delay faults, and their applications in fault diagnosis.
- To gain an in-depth understanding of DFT techniques like scan design, built-in self-test (BIST), and boundary scan, and their integration into ASIC and VLSI designs.

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- To familiarize students with industry-standard tools and automated test equipment (ATE) for efficient and accurate testing.
- To analyze timing issues, power constraints, and their testing implications in high-speed and low-power circuits.
- To understand yield analysis and learn strategies to enhance yield during fabrication and testing.
- To introduce students to advanced topics like testing for mixed-signal circuits, 3D ICs, and AI-assisted testing techniques.
- To apply theoretical concepts through practical assignments, projects, and case studies on ASIC and VLSI testing.

UNIT I INTRODUCTION TO ASIC AND FPGA

9

Basics of ASIC and FPGA: Definition, architecture, and design flow, Comparison: ASIC vs. FPGA, Testing in VLSI Design: Importance of testing in ASIC and FPGA development, Testing lifecycle and challenges

UNIT II: TESTING BASICS AND FAULT MODELS

9

Introduction to Testing Concepts: Goals of testing: defect detection, fault tolerance, and yield improvement, Terminologies: faults, errors, defects, testability. Fault Models

UNIT III TEST GENERATION TECHNIQUES

9

Combinational Circuits Testing: ATPG (Automatic Test Pattern Generation), Fault simulation techniques. Sequential Circuits Testing: Scan-based testing, BIST (Built-In Self-Test), Advanced Testing Techniques: Delay testing, Power-aware testing

UNIT IV DESIGN FOR TESTABILITY (DFT)

9

Concept of DFT, Importance of DFT in ASIC and FPGA, Techniques to improve testability. Scan Design: Scan insertion and testing, Boundary scan (IEEE 1149.1). BIST in Depth: Memory BIST (MBIST), Logic BIST (LBIST)

UNIT V: FPGA-SPECIFIC TESTING

9

FPGA Testing Challenges: Differences between ASIC and FPGA testing, Testing during prototyping and production. Configuration Testing: Bit stream verification, Configuration memory testing. Dynamic Reconfiguration Testing: Partial reconfiguration validation, Post-reconfiguration consistency checks.

TOTAL PERIODS: 45

OUTCOMES

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By completing this syllabus, students will:

CO1: Explain the significance of testing in the design and production of ASICs and FPGAs, including its impact on reliability, yield, and manufacturing cost.

CO2: Develop and apply fault models (e.g., stuck-at faults, bridging faults, and delay faults) to simulate and diagnose circuit defects in ASICs and FPGAs.

CO3: Analyze timing and power-related issues in digital and analog circuits and develop testing strategies to address them.

CO4: Integrate DFT features such as scan chains, boundary scan, and built-in self-test (BIST) into ASIC and FPGA designs to enhance testability.

CO5: Perform post-configuration testing of FPGAs to ensure functionality and performance under real-world operating conditions.

TEXTBOOKS

1. "Digital Systems Testing and Testable Design" by Miron Abramovici, Melvin A. Breuer, and Arthur D. Friedman
2. "Essentials of Electronic Testing for Digital, Memory, and Mixed-Signal VLSI Circuits" by Michael L. Bushnell and Vishwani D. Agrawal

REFERENCES

1. "Introduction to VLSI Design" by Eugene D. Fabricius.
2. "ASIC Design and Testing" by Ming-Bo Lin.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	1	1	3	1	2	3	2	3
CO2	2	1	1	3	1	2	2	2	2
CO3	2	1	1	3	1	2	3	2	3

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CO4	2	1	1	3	1	2	3	2	3
CO5	2	1	1	3	1	2	2	2	2
Avg	2	1	1	3	1	2	2.6	2	2.6

*** 3- High, 2- Medium, 1- Low**

AV25204

AI ALGORITHMS IN VLSI

L T P C

3 1 0 4

OBJECTIVES

- Learn the basics of AI techniques and how they are integrated into VLSI design processes for optimization, synthesis, testing, and manufacturing.
- Explore various AI algorithms (e.g., machine learning, deep learning, genetic algorithms) and their applications to optimize key design parameters such as power, performance, and area (PPA).
- Learn how AI algorithms can be used to automate fault detection, diagnosis, and yield prediction during VLSI testing and verification.

UNIT I MACHINE LEARNING ALGORITHMS FOR VLSI DESIGN

12

Supervised Learning Unsupervised Learning Reinforcement Learning (RL)Q-Learning,Deep Q-Networks (DQN): Applied for complex routing problems where deep neural networks predict and optimize routing decisions at each step.

UNIT II EVOLUTIONARY ALGORITHMS FOR VLSI OPTIMIZATION

12

Genetic Algorithms (GA) Optimization of circuit placement, routing, floorplanning, and power optimization.Particle Swarm Optimization (PSO)Application: Floorplanning, power optimization, and circuit placement.

UNIT III DEEP LEARNING ALGORITHMS IN VLSI

12

Convolutional Neural Networks (CNN), Application: Layout analysis, defect detection in manufacturing, and image-based circuit inspection. Recurrent Neural Networks (RNN) and

LSTM,Application: Time-series prediction for performance metrics like power consumption and timing analysis.

UNIT IV OPTIMIZATION TECHNIQUES USING AI**12**

Neural Network-Based Optimization,Power and area optimization in logic synthesis and placement. Bayesian OptimizationApplication: Tuning hyperparameters and optimizing complex multi-objective VLSI designs.

UNIT V FAULT DETECTION AND DIAGNOSIS USING AI**12**

Anomaly DetectionAlgorithms: Autoencoders Neural Networks for Fault Diagnosis

TOTAL PERIODS: 60**OUTCOMES**

Upon successful completion of the course, students will be able to:

CO1: Identify and explain the role of various AI techniques (e.g., machine learning, genetic algorithms, reinforcement learning) in the VLSI design and optimization process.

CO2: Apply AI algorithms for optimizing power, delay, area, and other design metrics in VLSI systems through hands-on projects and simulations.

CO3: Utilize machine learning and evolutionary algorithms (e.g., genetic algorithms) to optimize circuit layouts and placements, addressing congestion and timing issues.

CO4: Use AI techniques like neural networks and anomaly detection to diagnose faults and predict yield issues in VLSI systems during the testing and validation phases.

CO5:Gain the ability to design and implement AI-specific hardware using VLSI techniques, focusing on low-power, high-performance designs.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	1	3	2	2
CO2	3	3	2	2	2	1	2	1	1

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CO3	3	2	2	2	2	1	1	3	2
CO4	3	2	2	2	2	1	3	2	2
CO5	3	2	2	2	2	2	1	2	2
Avg	3	2	2	2	2	1.2	2	2	1.8

*** 3- High, 2- Medium, 1- Low**

AV2521**AI IN VLSI LABORATORY****L T P C****0 0 4 2****OBJECTIVES**

- To provide a comprehensive understanding of how AI methodologies can enhance VLSI design automation, optimization, and testing workflows.
- To familiarize students with AI frameworks (e.g., TensorFlow, PyTorch) and their application in VLSI tasks such as placement, routing, power optimization, and fault detection.
- To teach students how to leverage AI techniques for optimizing performance metrics such as area, power, timing, and yield in VLSI circuits.
- To explore AI-driven approaches for efficient fault detection, test pattern generation, and fault classification in digital circuits.
- To enable students to apply AI for predictive modeling, defect detection, and process optimization in semiconductor fabrication.
- To prepare students for advanced research and development by exploring emerging applications of AI in analog/mixed-signal design, thermal management, and hardware accelerators.
- To equip students with the technical expertise required to solve real-world challenges in the integration of AI and VLSI design, preparing them for roles in industry and academia.

EXPERIMENTS

1. Implement optimization algorithms (e.g., reinforcement learning, genetic algorithms) for design space exploration.

2. Develop reinforcement learning models for optimizing placement in a VLSI design and evaluate wirelength, timing, and congestion.
3. Apply AI algorithms to automate routing processes-Use supervised learning to improve routing efficiency.
4. Implement AI models to generate optimized test patterns for ATPG and analyze fault coverage and detection efficiency.
5. Apply AI techniques to optimize power gating and clock gating strategies.
6. Use AI techniques to automate layout generation and optimization for analog circuits.

TOTAL PERIODS: 45

Tools Required

1. **AI Frameworks:** TensorFlow, PyTorch, Scikit-learn, Keras.
2. **VLSI CAD Tools:** Cadence Innovus, Synopsys Design Compiler, Mentor Graphics, Xilinx.
3. **Hardware Platforms:** GPUs for AI model training, FPGA boards for prototyping.
4. **Data Processing:** Pandas, NumPy, Matplotlib for data preprocessing and visualization.
5. **Simulation Tools:** ModelSim, QuestaSim for circuit validation.

OUTCOMES

By the end of this lab, students will be able to:

CO1: Integrate AI techniques into VLSI design and testing workflows.

CO2: Develop and apply AI models for tasks like fault detection, optimization, and verification.

CO3: Use AI-assisted CAD tools to solve real-world VLSI challenges.

CO4: Automate critical processes in VLSI workflows using AI methodologies.

CO5: Bridge the gap between traditional VLSI design and emerging AI-driven techniques.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
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CO1	2	2	3	3	3	2	3	2	2
CO2	2	2	3	3	3	2	3	2	2
CO3	2	2	3	3	3	2	3	2	2
CO4	2	2	3	3	3	2	3	2	2
CO5	2	2	3	3	3	2	3	2	2
Avg	2	2	3	3	3	2	3	2	2

*** 3- High, 2- Medium, 1- Low**

AV2522**SEMINAR****L T P C****0 0 2 1****GUIDELINES:**

The students will work under the guidance of a group of staff members. The students will be asked to talk on any topic of their choice related to AI in VLSI and to engage in dialogue with the audience. A brief copy of their talk also should be submitted. Similarly, the students will have to present a seminar of not less than fifteen minutes and not more than thirty minutes on the technical topic. They will also answer the queries on the topic. The students as audience also should interact. Evaluation will be based on the technical presentation and the report and also on the interaction during the seminar.

TOTAL PERIODS: 30**OUTCOMES:**

On completion of the course, the student is expected to be able to

CO1 Identify latest developments in the field of AI in VLSI

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CO2 Identify latest developments in the field of AI in VLSI

CO3 Presentation Skills and ability to answer the queries during Interaction

CO4 Acquire technical writing abilities for seminars, conferences and journal publications

CO5 Use modern tools to present the technical details

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	3	3	2	3
CO2	2	2	2	3	2	3	3	2	3
CO3	2	2	2	3	2	3	3	2	3
CO4	3	3	3	3	2	3	3	2	3
CO5	3	3	3	3	2	3	3	2	3
Avg	2.6	2.6	2.6	3	2	3	3	2	3

*** 3- High, 2- Medium, 1- Low**

AV25301

SYSTEM-ON-CHIP (SOC) DESIGN VERIFICATION

L T P C

3 0 0 3

OBJECTIVE:

- Familiarize students with the methodologies, tools, and challenges involved in verifying complex System-on-Chip (SoC) designs.
- Equip students with techniques for functional correctness, performance analysis, and compliance testing in SoC designs.

- Teach students about assertion-based verification, coverage-driven verification, and formal verification for SoCs.
- Provide hands-on experience with EDA tools for SoC verification.
- Introduce students to modern approaches such as hardware-software co-verification, low-power verification, and AI/ML-assisted verification.
- Train students to analyze and debug verification failures and optimize test coverage for SoC designs.

UNIT I INTRODUCTION TO SOC DESIGN VERIFICATION

9

Overview of SoC Architecture: Components of an SoC: CPU, GPU, memory, peripherals, interconnects, Challenges in SoC verification, Introduction to verification methodologies (UVM, OVM, etc.). SoC Verification Basics: Types of verification: functional, performance, power, and formal verification. Stages of verification: unit-level, IP-level, subsystem-level, and SoC-level.

UNIT II VERIFICATION METHODOLOGIES AND PLANNING

9

Verification Planning and Management: Test planning, verification coverage, and metrics. Assertion-based verification (ABV) and Coverage-Driven Verification (CDV). Verification Environment Development: Components of a verification environment: testbenches, drivers, monitors, and checkers. Introduction to Universal Verification Methodology (UVM) and Open Verification Methodology (OVM).

UNIT III SIMULATION-BASED VERIFICATION

9

Simulation Techniques: Directed and constrained random testing, Debugging and failure analysis in simulation. Protocol and Interface Verification: Verification of standard interfaces (e.g., AXI, PCIe, USB). Verification IPs (VIPs) for interface testing.

UNIT IV FORMAL VERIFICATION

9

Introduction to Formal Verification: Basic concepts: property checking, equivalence checking. Use of assertions in formal verification (SystemVerilog Assertions - SVA). Tools and Techniques: Symbolic model checking and SAT-based methods. Tools for formal verification.

UNIT V ADVANCED VERIFICATION TECHNIQUES

9

Low-Power Verification: Techniques for power-aware verification. Use of UPF/CPF for low-power design testing. Performance and Timing Verification: Analysis of performance bottlenecks in SoC designs, Tools for timing analysis and optimization. Verification of AI/ML Accelerators in SoCs: Verification challenges for AI and ML cores, Specialized tools and methodologies.

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TOTAL PERIODS: 45

OUTCOMES:

On completion of the course, the student is expected to be able to

CO1: Gain expertise in SoC verification methodologies like simulation, formal, and low-power verification.

CO2: Develop hands-on skills with industry-standard EDA tools for SoC verification.

CO3: Understand how to verify complex interfaces and embedded software-driven designs.

CO4: Learn best practices for test planning, coverage analysis, and debugging in SoC environments.

CO5: Be equipped to address real-world challenges in SoC verification for high-performance and low-power designs.

TEXTBOOKS

1. System-on-Chip Verification: Methodology and Techniques by Prakash Rashinkar, Peter Paterson, Leena Singh.
2. Writing Testbenches Using SystemVerilog by Janick Bergeron.
3. Principles of Verifiable RTL Design: A Functional Coding Style Supporting Verification Processes in Verilog by Lionel Bening and Harry Foster.

REFERENCES

1. Digital Design and Verification with SystemVerilog by Ben Cohen.
2. Formal Verification: An Essential Toolkit for Modern VLSI Design by Erik Seligman and Tom Schubert.
3. UVM Cookbook: A Step-by-Step Guide to Building Your First UVM Testbench by Ray Salemi.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	2	3	3	2	1	2	1

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CO2	2	2	2	3	3	2	2	2	2
CO3	2	2	2	3	3	2	1	3	1
CO4	2	2	2	3	3	2	2	3	2
CO5	2	1	1	2	3	2	2	3	2
Avg	2	1.8	1.8	3	3	2	1.6	2.6	1.6

*** 3- High, 2- Medium, 1- Low**

MC25101 RESEARCH METHODOLOGY AND IPR

L T P C

3 0 0 3

OBJECTIVES:

- To arrange the conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose
- To gather information in a measured and systematic manner to ensure accuracy and facilitate data analysis
- To transform and model the collected data to discover useful information for decision- making
- To create public awareness about the benefits of Intellectual property among students
- To Provide legal certainty to inventors/ Patent applicants

UNIT I RESEARCH DESIGN

9

Overview of research process and design, Use of Secondary and exploratory data to answer the research question, Qualitative research, Observation studies, Experiments and Surveys.

UNIT II DATA COLLECTION AND SOURCES

9

Measurements, Measurement Scales, Questionnaires and Instruments, Sampling and methods. Data - Preparing, Exploring, examining and displaying.

UNIT III DATA ANALYSIS AND REPORTING

9

Overview of Multivariate analysis, Hypotheses testing and Measures of Association. Presenting Insights and findings using written reports.

UNIT IV INTELLECTUAL PROPERTY RIGHTS

9

Intellectual Property – The concept of IPR, Evolution and development of concept of IPR, IPR development process, Trade secrets, utility Models, IPR & Bio diversity, Role of WIPO and WTO. IPR establishments, Right of Property, Common rules of IPR practices, Types and Features of IPR Agreement, Trademark, Functions of UNESCO in IPR maintenance.

UNIT V PATENTS

9

Patents – objectives and benefits of patent, Concept, features of patent, Inventive step, Specification, Types of patent application, process E-filing, Examination of patent, Grant of patent, Revocation, Equitable Assignments, Licences, Licensing of related patents, patent agents, Registration of patent agents.

TOTAL PERIODS: 45

OUTCOMES:

CO1: Ability to arrange the conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose

CO2: Ability to gather information in a measured and systematic manner to ensure accuracy and facilitate data analysis

CO3: Ability to transform and model the collected data to discover useful information for decision- making

CO4: Ability to awareness about the benefits of Intellectual property

CO5: Ability to take up legal certainty while applying for Patent

REFERENCES:

1. Cooper Donald R, Schindler Pamela S and Sharma JK, “Business Research Methods”, Tata McGraw Hill Education, 11e (2012).
2. Catherine J. Holland, “Intellectual property: Patents, Trademarks, Copyrights, Trade Secrets”, Entrepreneur Press, 2007.
3. David Hunt, Long Nguyen, Matthew Rodgers, “Patent searching: tools & techniques”, Wiley, 2007.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	2	3	3	3	3	3	3
CO2	3	2	2	3	3	3	3	3	3
CO3	3	2	2	3	3	3	3	3	3
CO4	3	3	3	3	3	3	3	3	3
CO5	3	3	3	3	3	3	3	3	3
Avg	3	2.4	2.4	3	3	3	3	3	3

* 3- High, 2- Medium, 1- Low

AV2531

VLSI DESIGN AUTOMATION USING AI LABORATORY L T P C

0 0 4 2

OBJECTIVES

- To provide a comprehensive understanding of how AI methodologies can enhance VLSI design automation, optimization, and testing workflows.
- To equip students with the technical expertise required to solve real-world challenges in the integration of AI and VLSI design, preparing them for roles in industry and academia.

EXPERIMENTS

1. Implement a scan chain for a simple DUT (like a counter or multiplier).
2. Design a Built-in Self-Test (BIST) controller for a memory or ALU.
3. Use AI/ML models to automate floorplanning.
4. Use Machine Learning models (like regression models) to predict power consumption.

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5. Design and automate a complete 32-bit RISC processor.
6. Design and Automation of Memory (SRAM, DRAM)

TOTAL PERIODS: 45

Tools Required

1. **EDA Tools:** Synopsys Design Compiler, IC Compiler, Cadence Innovus, Mentor Questa
2. **Verification Tools:** UVM, Synopsys VCS, Cadence Xcelium
3. **Scripting Tools:** Python, Perl, TCL, Shell Script
4. **AI/ML Tools:** TensorFlow, Scikit-learn.

OUTCOMES

By the end of this lab, students will be able to:

CO1: Integrate AI techniques into VLSI design and testing workflows.

CO2: Develop and apply AI models for tasks like fault detection, optimization, and verification.

CO3: Use AI-assisted CAD tools to solve real-world VLSI challenges.

CO4: Automate critical processes in VLSI workflows using AI methodologies.

CO5: Bridge the gap between traditional VLSI design and emerging AI-driven techniques.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	3	3	3	2	2	1	2
CO2	2	2	3	3	3	2	3	1	2
CO3	2	2	3	3	3	2	3	1	2
CO4	2	2	3	3	3	2	3	2	2
CO5	2	2	3	3	3	2	2	2	2

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Avg	2	2	3	3	3	2	2.8	1.4	2
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* 3- High, 2- Medium, 1- Low

AV2532

PROJECT WORK –I

L T P C

0 0 12 6

OBJECTIVES:

- Introduce students to the initial research and feasibility study of a VLSI design automation project.
- Guide students in defining the scope of the project, including setting specific objectives, identifying the design problem, and proposing potential solutions.
- Familiarize students with design tools and methodologies required for VLSI design automation
- Encourage students to explore and document the state-of-the-art in VLSI design automation techniques, including AI/ML integration, hardware/software co-design, and verification.
- Enhance communication skills by requiring students to present their project proposal in a professional manner, including key design metrics, methodologies, and expected outcomes.

GUIDELINE

- The student should do the phase I work on the selected topic as per the formulated methodology. Students present their **project proposal** in a professional manner, including key design metrics, methodologies, and expected outcomes.

OUTCOMES:

Upon successful completion of **Project Phase 1**, students will be able to:

CO1:Develop a comprehensive project proposal that includes a clear design objective, the proposed methodology, tools, and expected results.

CO2:Understand and explore VLSI design automation techniques, including synthesis, placement, routing, and verification methodologies.

CO3:Conduct a literature review on state-of-the-art VLSI design automation methods and propose novel improvements.

CO4:Use appropriate design tools for initial design and analysis (e.g., Cadence, Synopsys).

CO5: Prepare and present a professional project proposal, demonstrating an ability to communicate complex technical concepts.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	3	3	3	3	3	3	3
CO2	3	2	2	3	3	3	3	3	3
CO3	2	2	2	3	3	3	3	3	3
CO4	3	2	3	3	3	3	3	3	3
CO5	3	2	3	3	3	3	3	3	3
Avg	2.8	2	2.6	3	3	3	3	3	3

* 3- High, 2- Medium, 1- Low

AV2541

PROJECT WORK –II

L T P C

0 024 12

OBJECTIVES:

- Allow students to implement and **test their designs** based on the proposals from Phase 1.
- Guide students through the **synthesis, placement, and routing** of the design, ensuring that the design meets timing, power, and area constraints.
- Emphasize **verification** methodologies (e.g., UVM, coverage-driven verification) to validate the correctness of the design.
- Introduce students to the concepts of **performance optimization**, including techniques to reduce power consumption, improve speed, and optimize area.
- Prepare students for **real-world application** by guiding them through a complete VLSI design flow from RTL to GDSII or verification results.

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GUIDELINE

The student should continue the phase I work on the selected topic as per the formulated methodology/ Undergo internship. At the end of the semester, after completing the work to the satisfaction of the supervisor and review committee, a detailed report should be prepared and submitted to the head of the department. The students will be evaluated based on the report and the viva-voce examination by a panel of examiners including one external examiner.

OUTCOMES:

Upon successful completion of **Project Phase 2**, students will be able to:

CO1:Implement and test VLSI designs, including synthesis, placement, routing, and verification.

CO2:Use VLSI design tools to perform the full design flow, from RTL to GDSII or to achieve functional verification.

CO3:Validate the design using advanced verification techniques, including UVM-based testbenches, functional coverage, and random testing.

CO4:Optimize VLSI designs for power, area, and timing using various optimization techniques (e.g., clock gating, power gating, layout optimization).

CO5:Prepare a detailed report and Present a comprehensive final presentation

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	2	3	3	3	3	3	3
CO2	3	2	2	3	3	3	3	3	3
CO3	3	2	2	3	3	3	3	3	3
CO4	3	2	2	3	3	3	3	3	3
CO5	3	2	2	3	3	3	3	3	3
Avg	3	2	2	3	3	3	3	3	3

* 3- High, 2- Medium, 1- Low

AV2501

HYBRID QUANTUM-CLASSICAL VLSI DESIGN

L T P C

3 0 0 3

OBJECTIVES

- Provide students with a solid foundation in both quantum computing and classical VLSI design concepts
- Enable students to design hybrid systems that combine quantum computing and classical VLSI techniques for applications in optimization, AI, and cryptography.
- Teach students how to use quantum algorithms to address common VLSI design challenges such as placement, routing, and testing.
- Provide hands-on experience with simulation tools for hybrid quantum-classical systems, including software tools for designing and testing quantum circuits and integrating them with classical VLSI designs.
- Expose students to the latest research, trends, and challenges in integrating quantum computing and classical VLSI design, preparing them to contribute to future advancements in quantum VLSI systems.

UNIT I INTRODUCTION TO HYBRID QUANTUM-CLASSICAL SYSTEMS

9

Overview of Quantum Computing Basics of quantum mechanics: qubits, superposition, entanglement. Quantum gates and circuits. Quantum algorithms (e.g., Grover's search, Shor's factoring algorithm). Classical VLSI Design Fundamentals of VLSI design: logic gates, digital circuits, CMOS technology. Classical VLSI design automation tools (e.g., synthesis, place and route).

UNIT II Quantum Computing and VLSI Design

9

Quantum Circuits for VLSI: Quantum logic gates and their implementation. Quantum error correction and fault tolerance in VLSI. Quantum-to-classical interface in VLSI design. Quantum Hardware Architectures Types of quantum processors Quantum hardware design challenges in VLSI Quantum communication and networking components.

UNIT III HYBRID QUANTUM-CLASSICAL ARCHITECTURES

9

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Integrating Quantum and Classical CircuitsQuantum-classical interface in hybrid systems.Co-design of classical VLSI and quantum processors for hybrid applications.Hybrid architecture for optimization problems.Design Considerations for Hybrid SystemsPower, area, and performance trade-offs in hybrid systems.Optimization techniques for hybrid quantum-classical circuits.Clocking, synchronization, and interconnects between classical and quantum systems.

UNIT IV: HYBRID QUANTUM-CLASSICAL ALGORITHMS

9

Quantum-Classical Co-Design for VLSI OptimizationApplication of quantum computing to classical VLSI design problems (e.g., placement, routing, optimization).Quantum computing for simulation and verification in VLSI design.Quantum Algorithms for VLSI Tasks:Quantum machine learning for design automation.Hybrid quantum-classical algorithms for performance analysis and testing.

UNIT V: APPLICATIONS

9

Future Trends and Challenges,Quantum computing and VLSI convergence: Opportunities and challenges.Potential applications of hybrid systems in AI, cryptography, and communication.

TOTAL PERIODS: 45

OUTCOMES

By the end of the course, students will be able to:

CO1: Understand the fundamentals of both quantum computing and classical VLSI design.

CO2: Apply quantum computing concepts to solve VLSI design problems.

CO3: Design and implement hybrid quantum-classical circuits for VLSI systems.

CO4: Explore new frontiers of computing by integrating quantum and classical systems in real-world applications.

CO5: Use industry-standard tools for simulating and designing hybrid quantum-classical VLSI systems and Demonstrate the impact of hybrid systems in the fields of optimization, AI, and cryptography.

TEXTBOOKS

1. "Quantum Computation and Quantum Information" by Michael A. Nielsen and Isaac L. Chuang
2. "Digital Integrated Circuits: A Design Perspective" by Jan M. Rabaey
3. "Principles of Quantum Computation and Information" by Benoît O. Boucher

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4. "Quantum Computer Science: An Introduction" by N. David Mermin

REFERENCES

1. "CMOS VLSI Design: A Circuits and Systems Perspective" by Neil H. E. Weste, David Harris
2. "Quantum Algorithms via Linear Algebra" by Richard J. Lipton and Kenneth W. Regan
3. "Quantum Information and Quantum Computation" by J. Watrous

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	3	2	3	2	2	2	2
CO2	2	3	2	3	2	2	3	2	3
CO3	3	2	3	2	3	1	3	2	3
CO4	3	3	1	3	2	2	3	2	3
CO5	3	3	3	2	1	2	3	2	3
Avg	2.8	2.8	2.4	2.4	2.2	1.8	3	2	3

*** 3- High, 2- Medium, 1- Low**

AV2502

NEURAL NETWORKS IN VLSI DESIGN AUTOMATION

L T P C

3 0 0 3

OBJECTIVES

- To introduce the fundamentals of neural networks and their role in automating VLSI design processes.
- To explore how neural network architectures can enhance physical design tasks such as placement, routing, and optimization in VLSI circuits.
- To understand the implementation challenges and solutions for integrating neural networks into hardware, emphasizing digital, analog, and mixed-signal systems.

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- To equip students with the ability to apply machine learning and neural network models to solve real-world VLSI design automation problems.
- To provide insights into recent research trends and the future scope of artificial intelligence in electronic design automation (EDA).

UNIT I INTRODUCTION TO NEURAL NETWORKS 9

Basics of neural networks and neurocomputing. Different architectures of neural networks used in VLSI.

UNIT II VLSI DESIGN CONCEPTS 9

Overview of VLSI design principles and the role of design automation. Use of graph algorithms and clustering in VLSI physical design.

UNIT III NEURAL NETWORK IMPLEMENTATION IN VLSI 9

Design of neural networks using VLSI circuits. Hardware-oriented neural network models Digital, analog, and mixed-signal implementations of neural networks.

UNIT IV OPTIMIZATION TECHNIQUES 9

Application of machine learning and neural network models to improve VLSI design processes like floorplanning, placement, and routing. Methods for timing optimization, buffer insertion, and low-power design.

UNIT V NN-VLSI CO-DESIGN 9

Weight Compression and Pruning, Quantization, fixed-point arithmetic On-Chip Learning, Advances in VLSI architectures for neural networks.

TOTAL PERIODS: 45

OUTCOMES

Upon successful completion of the course, students will be able to:

CO1: Explain the fundamental concepts of neural networks and their relevance to VLSI design.

CO2: Apply neural network models to optimize various phases of VLSI physical design, including floorplanning, placement, and routing.

CO3: Design and simulate neural network models for hardware implementation using advanced tools and techniques.

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CO4:Evaluate the impact of neural network-based design automation on the performance, power efficiency, and scalability of VLSI circuits.

CO5:Conduct independent research to develop innovative solutions for VLSI design automation using AI and ML techniques

TEXTBOOKS

1. "Algorithms for VLSI Design Automation" by N. A. Sherwani:
2. "VLSI Design of Neural Networks" by Springer.
3. "VLSI Architectures for Neural Networks" from IEEE Xplore.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	1	2	3	2	3	2	2
CO2	3	2	2	3	2	2	3	2	3
CO3	3	2	3	2	3	1	3	2	3
CO4	3	2	1	3	2	2	3	2	3
CO5	3	2	3	2	1	2	3	2	3
Avg	3	2	2	2	2.4	2	3	2	2.8

*** 3- High, 2- Medium, 1- Low**

AV2503

PROGRAMMING FOR AI AND VLSI

L T P C

3 0 0 3

OBJECTIVES

- Equip students with programming skills to build AI systems.
- Learn essential frameworks, tools, and libraries for AI development.
- Apply AI algorithms to solve real-world problems.
- Learn the basics of VLSI design and its applications
- Explore digital design techniques and hardware description languages (HDLs). Develop an understanding of fabrication processes and optimization for AI hardware.

UNIT I PROGRAMMING FUNDAMENTALS FOR AI

9

Python for AI: Data types, control structures, functions, and modules. Libraries: NumPy, Pandas (data manipulation). Matplotlib, Seaborn (data visualization). Data Handling and Preprocessing, Data cleaning, normalization, and transformation. Feature extraction and selection. Working with datasets: CSV, JSON, and large datasets. Tools: Scikit-learn, OpenCV (image data handling).

UNIT II MACHINE LEARNING PROGRAMMING

9

Implementing supervised learning: Linear regression, SVM, decision trees. Implementing unsupervised learning: K-means, DBSCAN. Introduction to TensorFlow and PyTorch.

UNIT III HDL FOR VLSI

9

Introduction to Hardware Description Languages (HDLs). Programming in Verilog/VHDL: Basic syntax, combinational and sequential circuits. Writing testbenches for simulation. Physical Design Automation: Partitioning, floorplanning, and placement. Routing and clock distribution. Design Rule Checking (DRC).

UNIT IV HARDWARE OPTIMIZATION AND VERIFICATION METHODOLOGIES

9

Hardware Optimization: Minimization techniques for area, power, and speed. Logic synthesis and optimization tools. Verification Methodologies: Static and dynamic verification. Introduction to Universal Verification Methodology (UVM)

UNIT V AI AND VLSI INTEGRATION

9

Designing hardware accelerators for AI: Basics of AI chips (TPUs, GPUs). Implementing neural networks in hardware. Low-power design techniques for AI. Edge AI with VLSI technology: Smart sensors, IoT devices.

TOTAL PERIODS: 45

OUTCOMES

CO1: Understand and apply foundational programming concepts, libraries, and tools for artificial intelligence.

CO2: Implement machine learning algorithms and evaluate their performance on real-world datasets.

CO3: Apply hardware description languages (HDLs) such as Verilog/VHDL for modeling and simulation of digital circuits.

CO4: Understand and implement physical design techniques, including partitioning, floorplanning, placement, and routing.

CO5: Develop efficient VLSI-based hardware accelerators for AI and machine learning applications.

REFERENCES

1. "Python Machine Learning" by Sebastian Raschka and Vahid Mirjalili
2. "Deep Learning with Python" by François Chollet
3. "Artificial Intelligence with Python" by Prateek Joshi
4. "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron
5. "Verilog HDL: A Guide to Digital Design and Synthesis" by Samir Palnitkar
6. "Digital Design Using VHDL: A Systems Approach" by William J. Dally and R. Curtis Harting
7. "HDL Programming: VHDL and Verilog" by Nazeih M. Botros

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	2	2	2	1	3	2	3
CO2	3	3	3	2	2	1	3	3	2
CO3	3	2	3	2	3	1	3	2	3

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CO4	3	2	3	2	3	1	3	2	2
CO5	3	2	2	2	3	1	3	3	3
Avg	3	2	3	2	3	1	3	2	3

*** 3- High, 2- Medium, 1- Low**

AV2504**PROGRAMMABLE ARCHITECTURE****L T P C****3 0 0 3****OBJECTIVES**

The student shall develop

- an overview and deeper insight into the research and development that is underway to meet future needs of flexible processors
- to learn the concepts of implementation, synthesis and placement of modules in reconfigurable architectures
- to understand the communication techniques and System on Programmable Chip for reconfigurable architectures
- to learn the process of reconfiguration management
- to familiarize the applications of reconfigurable architectures

UNIT I INTRODUCTION**9**

General purpose computing – domain specific processors – Application Specific Processors – reconfigurable computing – fields of application – evolution of reconfigurable systems – simple Programmable Logic Devices – Complex Programmable Logic Devices – Field Programmable Gate Arrays – coarse grained reconfigurable devices.

UNIT II IMPLEMENTATION, SYNTHESIS AND PLACEMENT**9**

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UNIT III COMMUNICATION AND SOPC 9

UNIT IV RECONFIGURATION MANAGEMENT 9

UNIT V APPLICATIONS 9

TOTAL PERIODS: 45

At the end of this course, the students should will be able to:

CO2: compare the tradeoffs that are necessary to meet the area, power and timing criteria of reconfigurable systems

CO3: analyze the algorithms related to placement and partitioning

CO4:analyze the communication techniques and system on programmable chip for reconfigurable architectures

CO5: analyze the principles of Network and System on a Programmable Chip

1. Christophe Bobda, “Introduction to Reconfigurable Computing: Architectures, Algorithms and Applications”, Springer 2007.

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2. Scott Hauck and Andre Dehon, “Reconfigurable Computing: The Theory and Practice of FPGA Based Computation”, Elsevier 2008
3. M. Gokhale and P. Graham, “Reconfigurable Computing: Accelerating Computation with Field-Programmable Gate Arrays”, Springer, 2005.
4. Nikoloas Voros Et Al. “Applied Reconfigurable Computing: Architectures, Tools and Applications” Springer, 2018.
5. Koen Bertels, João M.P. Cardoso, Stamatis Vassiliadis, “Reconfigurable Computing: Architectures and Applications”, Springer 2006.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	1	1	2	2	2	2	1	1
CO2	2	2	1	3	3	3	2	2	2
CO3	2	1	1	2	2	2	2	2	2
CO4	2	2	2	3	3	3	1	1	1
CO5	2	1	1	3	3	2	1	1	1
Avg	2	1.4	1.2	2.6	2.6	2.4	1.6	1.4	1.4

*** 3- High, 2- Medium, 1- Low**

AV2505 DESIGN FOR TESTABILITY (DFT) AND YIELD OPTIMIZATION L T P C

3 0 0 3

OBJECTIVES

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- To provide students with a thorough understanding of Design for Testability (DFT) principles and their importance in ensuring high-quality VLSI design.
- To familiarize students with various DFT techniques, including scan design, built-in self-test (BIST), and boundary scan.
- To equip students with the knowledge to develop test strategies for fault detection, isolation, and repair in ICs.
- To introduce students to Yield Optimization techniques, focusing on improving manufacturing yield through redundancy, defect tolerance, and error correction.
- To provide hands-on experience with EDA tools for test generation, fault simulation, and yield analysis.
- To expose students to emerging trends and research challenges in DFT and yield optimization for next-generation ICs.

UNIT I INTRODUCTION TO DFT

9

Basics of Design for Testability (DFT), Structural Testing Concepts, Introduction to Fault Models (Stuck-at, Transition, Delay Faults)

UNIT II DFT TECHNIQUES

9

Full-Scan and Partial-Scan Designs Scan Design Techniques (Clocked Full-Scan, LSSD, Random-Access Scan) RTL Design for Testability, Special-Purpose Scan Designs (Enhanced Scan, Snapshot Scan, Error-Resilient Scan)

UNIT III TEST PATTERN GENERATION

9

Automatic Test Pattern Generation (ATPG), Fault Coverage Analysis, Techniques for Test Pattern Optimization **Built-In Self-Test (BIST)**-BIST Architecture and Implementation, Linear Feedback Shift Register (LFSR) for Test Pattern Generation, Response Compaction and Signature Analysis

UNIT IV DESIGN FOR YIELD (DFY)

9

Yield Concepts and Calculation, Sources of Yield Loss in ICs, Techniques for Yield Enhancement (Redundancy, Error Correction), Statistical Process Control for Yield Improvement

UNIT V TESTING TECHNIQUES

9

Logic and Fault Simulation, Functional Testing vs. Structural Testing, Test Cost, Test Time, and Test Coverage Optimization.

TOTAL PERIODS: 45

OUTCOMES

At the end of this course, the students should will be able to:

CO1: Apply DFT methods such as scan design, BIST, and boundary scan to make circuits more testable.

CO2: Identify, analyze, and categorize faults and defects in VLSI circuits using models like stuck-at faults, transition faults, and delay faults.

CO3: Design and implement testable circuits at the RTL and gate levels, integrating scan chains and BIST hardware into designs.

CO4: Use Automatic Test Pattern Generation (ATPG) tools for generating optimized test patterns that maximize fault coverage while minimizing test time and cost.

CO5: Apply methods to improve IC yield through redundancy, reconfiguration, and statistical process control.

TEXTBOOK

1. "Design for Testability for Digital ICs Using Multi-Cycle Test and Scan Design" by Alfred Crouch
2. "Introduction to Yield Analysis and Defect Tolerance in VLSI Circuits" by V. Agrawal and M. Bushnell
3. CMOS VLSI Design: A Circuits and Systems Perspective" by Neil H.E. Weste, David Harris

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	3	1	2	1	2	1	1
CO2	3	1	1	2	2	3	3	2	2
CO3	2	1	1	3	2	3	3	2	2
CO4	2	1	1	3	3	3	3	2	2
CO5	3	2	1	3	2	2	3	1	2

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Avg	3	1	1	2	2	2	3	2	2
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*** 3- High, 2- Medium, 1- Low**

AV2506 ADVANCED CMOS TECHNOLOGY AND INTEGRATION**L T P C****3 0 0 3****OBJECTIVES**

- This includes device physics, fabrication processes, and the role of scaling in device performance and integration.
- Students will explore the challenges of short-channel effects, power consumption, reliability, and the physics of advanced MOSFETs such as FinFETs and nano-scale CMOS devices.
- Students will learn about the latest tools and methodologies for simulating CMOS device behavior, interconnect performance, and CMOS circuit design.

UNIT I INTRODUCTION TO CMOS TECHNOLOGY**9**

Basics of CMOS devices , Scaling trends and challenges ,Process technologies from 65 nm to 7 nm.

UNIT II CMOS FABRICATION PROCESS**9**

Overview of semiconductor materials, Steps in the CMOS fabrication process: oxidation, ion implantation, lithography, etching, deposition, Advanced techniques: FinFET, SOI, and 3D integration.

UNIT III ADVANCED CMOS DEVICES**9**

Short-channel effects ,High-k dielectrics and metal gates, FinFETs and Tri-Gate transistors, Nano-scale MOSFETs.

UNIT IV CMOS INTEGRATION AND CHALLENGES**9**

Power consumption and energy-efficient designs, Interconnect scaling and resistance-capacitance delays, Parasitic effects and noise management, Advanced packaging technologies.

UNIT V APPLICATIONS OF ADVANCED CMOS**9**

Application in consumer electronics, IoT, mobile devices, and high-performance computing, CMOS integration for RF circuits, mixed-signal designs, and memory systems.

TOTAL PERIODS: 45

OUTCOMES

Upon successful completion of the course, students should be able to:

CO1: This includes the ability to analyze device characteristics, fabrication processes, and performance metrics of CMOS technologies at different scales.

CO2: Students will gain expertise in understanding the physical limitations of CMOS and the innovations that have been developed to overcome these challenges, such as FinFETs and high-k materials.

CO3: Students will be proficient in understanding process flows, integration methodologies, and fabrication issues specific to modern CMOS technology.

CO4: They will be able to analyze CMOS device and circuit behavior.

CO5: Students will be familiar with current research in beyond-CMOS technologies and how these innovations may impact future integrated circuit designs.

TEXTBOOK

1. CMOS VLSI Design: A Circuits and Systems Perspective" by Neil Weste and David Harris
2. "Advanced CMOS Process Technology" by Nader J. S. R. L.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	1	2	2	1	3	3	1
CO2	3	2	1	2	2	3	3	2	2
CO3	3	2	1	3	2	3	3	2	2
CO4	3	2	1	3	3	3	3	2	2
CO5	3	2	1	3	2	2	3	1	2
Avg	3	2	1	2.6	2.2	2.4	3	2	1.8

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*** 3- High, 2- Medium, 1- Low**

AV2507	VLSI DESIGN AUTOMATION	L T P C
		3 0 0 3

OBJECTIVES

- to introduce the VLSI design methodologies and design methods.
- to introduce data structures and algorithms required for VLSI design.
- to study algorithms for partitioning and placement.
- to study algorithms for floor planning and routing.
- to study algorithms for modelling, simulation and synthesis.

UNIT I	INTRODUCTION	9
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Introduction to VLSI Design Methodologies – VLSI Design Cycle – New Trends in VLSI Design Cycle – Physical Design Cycle – New Trends in Physical Design Cycle – Design Styles – Review of VLSI Design Automation Tools

UNIT II	DATA STRUCTURES AND BASIC ALGORITHMS	9
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Introduction to Data Structures and Algorithms – Algorithmic Graph Theory and Computational Complexity – Tractable and Intractable Problems – General Purpose Methods for Combinatorial Optimization.

UNIT III	ALGORITHMS FOR PARTITIONING AND PLACEMENT	9
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Layout Compaction – Problem Formulation – Algorithms for Constraint Graph Compaction – Partitioning – Placement – Placement Algorithms.

UNIT IV	ALGORITHMS FOR FLOORPLANNING AND ROUTING	9
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Floorplanning – Problem Formulation – Floorplanning Algorithms – Routing – Area Routing – Global Routing – Detailed Routing.

UNIT V MODELLING, SIMULATION AND SYNTHESIS 9

Simulation – Gate Level Modeling and Simulation – Logic Synthesis and Verification – Binary Decision Diagrams – High Level Synthesis.

TOTAL PERIODS: 45

OUTCOMES:

At the end of this course, the students should be able to:

CO1: use various VLSI design methodologies

CO2: understand different data structures and algorithms required for VLSI design.

CO3: develop algorithms for partitioning and placement.

CO4: develop algorithms for floorplanning and routing.

CO5: design algorithms for modelling, simulation and synthesis.

REFERENCES

1. Sabih H. Gerez, “Algorithms for VLSI Design Automation”, Second Edition, Wiley-India, 2017.
2. Naveed a. Sherwani, “Algorithms for VLSI Physical Design Automation”, 3rd Edition, Springer, 2017.
3. Charles J. Alpert, Dinesh P. Mehta and Sachin S Sapatnekar, “Handbook of Algorithms for Physical Design Automation, CRC Press, 1st Edition, 2.
4. N.a. Sherwani, "Algorithms for VLSI Physical Design Automation", Kluwer Academic Publishers, 2002.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	1	2			2	3	3
CO2	2	2	1	2			1	2	2
CO3	2	2	1	2	2		1	2	2
CO4	2	2	1	2	2	1	1	3	3
CO5	2	2	1	2	2	1	1	3	3
Avg	2	2	1	2	2	1	1.2	2.6	2.6

*** 3- High, 2- Medium, 1- Low**

AV2508

FPGA EMBEDDED DESIGN

L T P C

3 0 0 3

COURSE OBJECTIVES:

- To acquire the knowledge about system specification and modelling
- To learn the formulation of partitioning
- To study the different technical aspects about prototyping and emulation

UNIT I

SYSTEM SPECIFICATION AND MODELLING

9

Embedded Systems, Hardware/Software Co-Design, Co-Design for System Specification and Modeling, Co-Design for Heterogeneous Implementation - Processor Synthesis, Single-Processor Architectures with One ASIC, Single-Processor Architectures with Many ASICs, Multi-Processor Architectures, Comparison of Co-Design Approaches, Models of Computation,

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Requirements for Embedded System Specification

UNIT II HARDWARE/SOFTWARE PARTITIONING 9

The Hardware/Software Partitioning Problem, Hardware-Software Cost Estimation, Generation of The Partitioning Graph, Formulation of The HW/SW Partitioning Problem, Optimization, HW/SW Partitioning Based On Heuristic Scheduling, HW/SW Partitioning Based On Genetic Algorithms.

UNIT III HARDWARE/SOFTWARE COSYNTHESIS 9

The Co-Synthesis Problem, State-Transition Graph, Refinement and Controller Generation, Distributed System Co-Synthesis

UNIT IV PROTOTYPING AND EMULATION 9

Introduction, Prototyping and Emulation Techniques, Prototyping and Emulation Environments, Future Developments in Emulation and Prototyping, Target Architecture, Architecture Specialization Techniques, System Communication Infrastructure, Target Architectures and Application System Classes, Architectures for Control-Dominated Systems, Architectures for Data- Dominated Systems, Mixed Systems and Less Specialized Systems.

UNIT V DESIGN SPECIFICATION AND VERIFICATION 9

Concurrency, Coordinating Concurrent Computations, Interfacing Components, Verification, Languages for System-Level Specification and Design System-Level Specification, Design Representation for System Level Synthesis, System Level Specification Languages,

Heterogeneous Specification and Multi-Language Co-Simulation

TOTAL PERIODS: 45

OUTCOMES:

At the end of this course, the students will be able to

CO1: Describe The Broad Range of System Architectures and Design Methodologies that

currently exist and define their fundamental attributes.

CO2: Discuss the Dataflow Models as a State-of-the-Art Methodology to Solve Co-Design Problems and to Optimize the balance between Software and Hardware.

CO3: Understand in Translating between Software and Hardware Descriptions through Co-Design Methodologies.

CO4: Understand the State-of-The-Art practices in developing Co-Design Solutions to problems using modern Hardware/Software Tools for building prototypes.

CO5: Understand the Concurrent Specification from an Algorithm, Analyze its behavior and partition the Specification into Software (C Code) and Hardware (HDL) Components

TEXTBOOKS

1. Patrick Schaumont, “A Practical Introduction to Hardware/Software Co-design”, Springer, 2010.
2. Ralf Niemann, “Hardware/Software Co-Design for Data Flow Dominated Embedded Systems”, Kluwer Academic Publisher, 1998.
3. Jorgen Staunstrup, Wayne Wolf, “Hardware/Software Co-Design: Principles and Practice”, Kluwer Academic Publisher, 1997.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	1	2			2	3	3
CO2	3	2	1	2			1	2	2
CO3	3	2	1	2	2		1	2	2
CO4	3	2	1	2	2	1	1	3	3

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CO5	3	2	1	2	2	1	1	3	3
Avg	3	2	1	2	2	1	1	2.6	2.6

*** 3- High, 2- Medium, 1- Low**

AV2509 NETWORK ON CHIP

L T P C

3

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3

OBJECTIVES:

The students should be made to:

- Understand the concept of Network - on - Chip
- Learn router architecture designs
- Study fault tolerance Network - on – Chip

UNIT I INTRODUCTION TO NOC

9

Introduction to NOC – OSI Layer Rules in NOC - Interconnection Networks in Network-On-Chip Network Topologies - Switching Techniques - Routing Strategies - Flow Control Protocol Quality- of-Service Support

UNIT II ARCHITECTURE DESIGN

9

Switching Techniques and Packet Format - Asynchronous FIFO Design - GALS Style of Communication - Wormhole Router Architecture Design - VC Router Architecture Design - Adaptive Router Architecture Design

UNIT III ROUTING ALGORITHM

9

Packet Routing-QOS, Congestion Control and Flow Control – Router Design – Network Link Design – Efficient and Deadlock-Free Tree-Based Multicast Routing Methods - Path-Based Multicast Routing For 2D and 3D Mesh Networks- Fault-Tolerant Routing Algorithms - Reliable and Adaptive Routing Algorithms

UNIT IV TEST AND FAULT TOLERANCE OF NOC

9

Design-Security in Networks-On-Chips-Formal Verification of Communications in Networks-On Chips-Test and Fault Tolerance For Networks-On-Chip Infrastructures-Monitoring Services For Networks-On-Chips

UNIT V THREE-DIMENSIONAL INTEGRATION OF NETWORK-ON-CHIP

9

Three-Dimensional Networks-On-Chips Architectures – A Novel Dimensionally-Decomposed Router for On-Chip Communication in 3D Architectures - Resource Allocation For QOS On-Chip Communication – Networks-On-Chip Protocols-On-Chip Processor Traffic Modeling For Networks-On-Chip

TOTAL PERIODS: 45

OUTCOMES:

At the end of this course, the students will be able to:

CO1: Compare different architecture design

CO2: Discuss different routing algorithms

CO3: Explain three dimensional Networks on Chip architectures

CO4: Test and design fault tolerant NOC

CO5: Design three dimensional architectures of NOC

REFERENCES

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3

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CO1	2	2	1	2			2	3	3
CO2	2	2	1	2			1	2	2
CO3	2	2	1	2	2		1	2	2
CO4	2	2	1	2	2	1	1	3	3
CO5	2	2	1	2	2	1	1	3	3
Avg	2	2	1	2	2	1	1	2.6	2.6

*** 3- High, 2- Medium, 1- Low**

AV2510 SIGNAL INTEGRITY FOR HIGH SPEED DESIGN**L T P C****3 0 0 3****OBJECTIVES:**

- To identify sources affecting the speed of digital circuits.
- To introduce methods to improve the signal transmission characteristics

UNIT I SIGNAL PROPAGATION ON TRANSMISSION LINES**9**

Transmission line equations, wave solution, wave vs. circuits, initial wave, delay time, Characteristic impedance , wave propagation, reflection, and bounce diagrams Reactive terminations – L, C , static field maps of micro strip and strip line cross-sections, per unit length parameters, PCB layer stackups and layer/Cu thicknesses, cross-sectional analysis tools, Zo and Td equations for microstrip and stripline Reflection and terminations for logic gates, fan-out, logic switching , input impedance into a transmission-line section, reflection coefficient, skin-effect, dispersion.

UNIT II MULTI-CONDUCTOR TRANSMISSION LINES AND CROSS-TALK**9**

Multi-conductor transmission-lines, coupling physics, per unit length parameters ,Near and far-end cross-talk, minimizing cross-talk (stripline and microstrip) Differential signalling,

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termination, balanced circuits ,S-parameters, Lossy and Lossless models.

UNIT III NON-IDEAL EFFECTS

9

Non-ideal signal return paths – gaps, BGA fields, via transitions , Parasitic inductance and capacitance , Transmission line losses –routing parasitic, Common-mode current, differential-mode current , Connectors.

UNIT IV POWER CONSIDERATIONS AND SYSTEM DESIGN

9

SSN/SSO , DC power bus design , layer stack up, SMT decoupling ,, Logic families, power consumption, and system power delivery , Logic families and speed Package types and parasitic,SPICE, IBIS models ,Bit streams, PRBS and filtering functions of link-path componentsEye diagrams , jitter , inter-symbol interference Bit-error rate ,Timing analysis.

UNIT V CLOCK DISTRIBUTION AND CLOCK OSCILLATORS

9

Timing margin, Clock slew, low impedance drivers, terminations, Delay Adjustments, canceling parasitic capacitance, Clock jitter.

TOTAL PERIODS : 45

OUTCOMES:

At the end of the course the student will be able to

CO1: identify sources affecting the speed of digital circuits.

CO2: identify methods to improve the signal transmission characteristics

CO3:characterise and model multiconductor transmission line

CO4: analyze clock distribution system and understand its design parameters

CO5: analyze nonideal effects of transmission line

REFERENCES

1. H. W. Johnson and M. Graham, High-Speed Digital Design: A Handbook of Black Magic, Prentice Hall, 1993.
2. Douglas Brooks, Signal Integrity Issues and Printed Circuit Board Design, Prentice Hall PTR , 2003.

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3. S. Hall, G. Hall, and J. McCall, High-Speed Digital System Design: A Handbook of Interconnect Theory and Design Practices, Wiley-Interscience, 2000.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	2	2	1	1	2	3	3
CO2	3	3	2	2	1	1	1	2	2
CO3	3	3	2	2	2	1	1	2	2
CO4	3	3	2	2	2	1	1	3	3
CO5	3	3	2	2	2	1	1	3	3
Avg	3	3	2	2	1.8	1	1	2.6	2.6

*** 3- High, 2- Medium, 1- Low**

**ARUNAI UNIVERSITY
TIRUVANNAMALAI
REGULATION – 2025
CHOICE BASED CREDIT SYSTEM
M.TECH. AGRO-ELECTRONICS**

PROGRAM DURATION: 2 YEARS

PROGRAM DESCRIPTION/PHILOSOPHY:

Agro-Electronics is an exciting and impactful field, merging agriculture with advanced electrical and electronic solutions to promote sustainable farming practices. It combines engineering principles with agricultural sciences to develop smart, automated, and technologically-driven solutions tailored for the agricultural industry.

This program provides an opportunity for the graduates to support agricultural efficiency, environmental sustainability, and food security.

DISTINCTIVENESS OF THE PROGRAM:

The M.Tech Agro-Electronics is a distinct and interdisciplinary program that blends agriculture, electronics, and technology to address modern agricultural challenges through technological innovation.

It integrates electronics, sensors, AI, and automation with agricultural sciences. It equips students to revolutionize traditional agriculture practices through technology-driven, sustainable, and precision-based solutions, making them future-ready for both agriculture and technology industries.

Graduates of this program possess skills that span agriculture, electronics, and IT industries, making them versatile for careers in Agri-tech start-ups, Government agricultural departments, Agro-electronics product development and Research & development in smart agriculture.

PROGRAM OUTCOMES (POs):

After going through the two years of study, our Agro-Electronics Graduates will exhibit their ability to:

- | | |
|------------|--|
| PO1 | Research Aptitude: An ability to independently carry out research / investigation and development work to solve practical problems. |
| PO2 | Technical Documentations: An ability to write and present a substantial technical report/document. |
| PO3 | Technical Competence: Students should be able to demonstrate a degree of mastery over the area as per the specialization of the programme. The mastery should be at a level higher than the requirements in the appropriate bachelor programme. |
| PO4 | Critical Analysis of Building Maintenance-related Problems: Critically analyse complex Building Maintenance related problems and apply independent judgment for synthesizing information. |
| PO5 | Conceptualization and Evaluation of Innovative Engineering Solutions to Building Maintenance Issues: Conceptualize and solve Building repair & Maintenance problems, evaluate potential solutions and arrive at technically feasible, economically viable and |

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environmentally sound solutions with due consideration of health, safety and sociocultural factors.

PO6 Life-long Learning: Recognize the need for independent, life-long learning and adapt to emerging technologies in Building Maintenance +and solutions to novel problems.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs):

1. Equip students with knowledge of advanced electronics, IoT, and renewable energy applications in agriculture.
2. Train students to design, develop, and implement electronic solutions that address challenges in sustainable agriculture and food security.
3. Promote environmentally friendly practices in agricultural technology by leveraging renewable energy and minimizing waste.
4. Enable graduates to work collaboratively with agricultural stakeholders to address rural energy needs and improve resource efficiency.
5. Prepare graduates for R&D in Agri-tech innovations and start-ups.

PROGRAM SPECIFIC OUTCOMES (PSOs):

On completion of Agro-Electronics program, the student will have the following Program Specific Outcomes.

1. **Application of Electronics in Agriculture:** Ability to Develop and implement electronic systems, sensors, and IoT-based technologies for precision agriculture, resource optimization, and environmental monitoring.
2. **Integration of Automation and Mechanization:** Ability to Design and operate automated agricultural machinery and robotics to enhance productivity, reduce labor dependency, and ensure sustainable farming practices.
3. **Adaptation of Advanced Technologies:** Ability to apply modern advancements like drones, wireless communication, and cloud computing to develop real-time agricultural monitoring and decision-making systems.

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M.Tech –AGRO-ELECTRONICS (Full Time)

CURRICULUM AND SYLLABI FOR SEMESTERS I to IV

SEMESTER I

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIODS
				L	T	P	C	
THEORY								
1	MA25101	Statistical Methods for Engineers	FC	3	1	0	4	4
2	AG25101	System Analysis and Soft Computing in Agriculture	PCC	3	0	0	3	3
3	AG25102	AI and Data Analytics in Agriculture	PCC	3	0	0	3	3
4	AG25103	Post-Harvest Technology and Storage Systems	PCC	3	0	0	3	3
5	AG25104	Agricultural Drones and UAVs	PCC	3	0	0	3	3
6	MC25101	Research Methodology and IPR	RMC	3	0	0	3	3
PRACTICALS								
7	AG25105	Agro Technology Laboratory	PCC	0	0	4	2	4
TOTAL				18	1	4	21	23

SEMESTER II

S.NO .	COURSE CODE	COURSE TITLE	CATE GOR Y	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
THEORY								
1	AG25201	Embedded System Design	PCC	2	0	2	4	4
2	AG25202	Irrigation Engineering and Automation	PCC	3	0	0	3	3
3	AG25203	Organic Farming and Sustainable Practices	PCC	3	0	0	3	3
4	AG25204	Food Supply Chain Management	PCC	3	0	0	3	3

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5		Professional Elective I	PEC	3	0	0	3	3
6		Professional Elective II	PEC	3	0	0	3	3
PRACTICALS								
7	AG25205	Sustainable Agriculture Laboratory	PCC	0	0	4	2	4
8	AG25206	Seminar-I	EDC	0	0	2	1	2
TOTAL				17	0	8	22	25

SEMESTER III

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
THEORY								
1		Open Elective I	OEC	3	0	0	3	3
2		Professional Elective III	PEC	3	0	0	3	3
3		Professional Elective IV	PEC	3	0	0	3	3
PRACTICALS								
4	AG25301	Project Work Phase I	EDC	0	0	12	6	12
TOTAL				9	0	12	15	21

SEMESTER IV

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
PRACTICALS								
1	AG25401	Project Work Phase II	EDC	0	0	24	12	24
TOTAL				0	0	24	12	24

TOTAL CREDITS: 70

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FOUNDATION COURSES (FC)**

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	MA25101	Statistical Methods for Engineers	FC	3	1	0	4	4

PROFESIONAL CORE COURSES (PCC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	AG25101	System Analysis and Soft Computing in Agriculture	PCC	3	0	0	3	3
2	AG25102	AI and Data Analytics in Agriculture	PCC	3	0	0	3	3
3	AG25103	Post-Harvest Technology and Storage Systems	PCC	3	0	0	3	3
4	AG25104	Agricultural Drones and UAVs	PCC	3	0	0	3	3
5	AG25201	Embedded System Design	PCC	2	0	2	4	3
6	AG25202	Irrigation Engineering and Automation	PCC	3	0	0	3	3
7	AG25203	Organic Farming and Sustainable Practices	PCC	3	0	0	3	3
8	AG25204	Food Supply Chain Management	PCC	3	0	0	3	3
TOTAL CREDITS							25	

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LIST OF PROFESSIONAL ELECTIVE COURSES (PEC)

SEMESTER II, ELECTIVE I

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	AG25001	Climate-Resilient Agriculture	PEC	3	0	0	3	3
2	AG25002	Advanced Landscape Architecture	PEC	3	0	0	3	3
3	AG25003	Environmental Policy and Law	PEC	3	0	0	3	3
TOTAL CREDITS								9

SEMESTER II, ELECTIVE II

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	AG25004	Advanced Farm Equipment Design	PEC	3	0	0	3	3
2	AG25005	3D Printing in Agri-Tech	PEC	3	0	0	3	3
3	AG25006	Digital Twin Technology in Farming	PEC	3	0	0	3	3
TOTAL CREDITS								9

SEMESTER III, ELECTIVE III

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	AG25007	Autonomous Vehicles in Agriculture	PEC	3	0	0	3	3
2	AG25008	Geospatial Data Analysis for Agriculture	PEC	3	0	0	3	3
3	AG25009	Hydroponics and Vertical Farming	PEC	3	0	0	3	3
TOTAL CREDITS								9

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SEMESTER III, ELECTIVE IV**

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	AG25010	Climate Change and Adaptation	PEC	3	0	0	3	3
2	AG25011	Block chain for Agri Supply Chain	PEC	3	0	0	3	3
3	AG25012	Bio-Electronics and Bio Sensors for Agriculture	PEC	3	0	0	3	3
TOTAL CREDITS								9

RESEARCH METHODOLOGY AND IPR COURSES (RMC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	MC25101	Research Methodology and IPR	MC	3	0	0	3	3

EMPLOYMENT DEVELOPMENT COURSES (EDC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	AG25206	Seminar-I	EDC	0	0	2	1	2
2	AG25301	Project Work I	EDC	0	0	12	6	12
3	AG25401	Project Work II	EDC	0	0	24	12	24
TOTAL CREDITS								19

OBJECTIVES:

- To learn about statistical methods and their applications in engineering, including estimation theory, hypothesis testing, correlation, regression, randomized variables & probability distributions.

UNIT I INTRODUCTION TO STATISTICS

12

Introduction to Statistics – Measure of Central Tendency – Coefficient of Variation and its applications- Mean, Median & Mode, Standard Deviation – Moments, Skewness & Kurtosis-Efficiency & Sufficiency.

UNIT II HYPOTHESIS TESTING -I

12

Introduction to Testing Of Hypothesis- Sampling Distributions – Small & Large Scale Samples- Types Of Error – Large Scale Sample Tests –Z Test For Single Proportion – Difference Of Proportion- Single & Difference Mean.

UNIT III HYPOTHESIS TESTING-II

12

Introduction – Random Variables – Probability Mass Function, Distribution and Density Functions – Joint Probability Distribution and Joint Density Functions – Marginal, Conditional Distributions and Density Functions – Mathematical Expectation and its Properties – Covariance – Moment Generating Function – Characteristic Function.

UNIT IV DESIGN OF EXPERIMENTS & CORRELATION

12

Analysis of variance – One way and two way classifications – Completely randomized design – Randomized block design – Latin square design - Multiple and partial correlation – Method of least squares – Plane of regression – Properties of residuals – Coefficient of multiple correlation

UNIT V RANDOM VARIABLES & PROBABILITY DISTRIBUTIONS

12

Random Variables – Probability Mass Function – Distribution and Density Functions- Mathematical Expectation and Its Properties- Covariance-Binominal, Poisson, Normal, Gamma, Exponential & Weibull Distribution.

Total Periods: 60

OUTCOMES:

After completing this course, students should demonstrate competency in the following topics:

- CO1** Consistency, efficiency and unbiasedness of estimators, method of maximum likelihood estimation and Central Limit Theorem.
- CO2** Use statistical tests in testing hypotheses on data.
- CO3** Concept of linear regression, correlation, and its applications.

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- CO4** List the guidelines for designing experiments and recognize the key historical figures in Design of Experiments.
- CO5** Perform exploratory analysis of multivariate data, such as multivariate normal density, calculating descriptive statistics, testing for multivariate normality.

REFERENCES:

1. Gupta S.C., and Kapoor, V.K., “Fundamentals of Mathematical Statistics”, 12th Edition, Sultan Chand and Sons, 2020.
2. Jay L. Devore, “Probability and statistics for Engineering and the Sciences”, 8th Edition, Cengage Learning, 2014.
3. Johnson, R.A., Miller, I and Freund J., "Miller and Freund's Probability and Statistics for Engineers", 9th Edition, Pearson Education, Asia, 2016.
3. Johnson, R.A. and Wichern, D. W. “Applied Multivariate Statistical Analysis”, 6th Edition, Pearson Education, Asia, 2012.
5. Rice, J.A. "Mathematical Statistics and Data Analysis", 3rd Edition, Cengage Learning, 2015.

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	2	2	3	3
CO2	3	3	3	3	2	2	1	2	2
CO3	3	3	3	3	2	2	1	2	2
CO4	3	3	3	3	2	2	1	3	3
CO5	3	3	3	3	2	2	1	3	3
Avg	3	3	3	3	2	2	1	2	2

*** 3- High, 2- Medium, 1- Low**

AG25101 SYSTEM ANALYSIS AND SOFT COMPUTING IN AGRICULTURE L T P C
3 0 0 3

COURSE OBJECTIVES:

- To introduce the students to the application of systems concept to agricultural engineering problems, planning and management.
- Soft computing techniques for modeling different problems in the field agricultural engineering

UNIT I	SYSTEM CONCEPTS	9
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UNIT II LINEAR PROGRAMMING & DYNAMIC PROGRAMMING 9

UNIT III SIMULATION 9

UNIT IV NEURAL NETWORKS 9

UNIT V FUZZY LOGIC AND GENETIC ALGORITHM 9

Genetic Algorithm (GA) - Basic concepts, working principle, procedures, flow chart, Genetic representations, encoding, Initialization and selection, Genetic operators, Mutation - applications

TOTAL: 45 PERIODS

1. Vedula, S., and Majumdar, P.P. Water Resources Systems – Modeling Techniques and Analysis Tata McGraw Hill, New Delhi, Fifth reprint, 2010.
2. Robert M Peart and W David Shoup, Agricultural Systems Management – Optimizing efficiency and performance, CRC Press, 2013.
3. Gupta, P.K., and Man Mohan, “Problems in Operations Research”, (Methods and Solutions), Sultan Chand and Sons, New Delhi, 1995.

1. Chaturvedi, M.C., “Water Resources Systems Planning and Management”, Tata McGraw Hill, New Delhi, 1997.
2. Taha, H.A., “Operations Research”, McMillan Publication Co., New York, 1995.
3. Hiller, F.S., and Liebermann, G.J., “Operations Research”, CBS Publications and Distributions, New Delhi, 1992.
4. Timothy J. Ross, “Fuzzy Logic with Engineering Applications” Wiley India.

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5. S. Rajsekaran & G.A. Vijayalakshmi Pai, “Neural Networks, Fuzzy Logic and Genetic Algorithm: Synthesis and Applications” Prentice Hall of India.

COURSE OUTCOMES:

After completion of this course, students will be able to:

- CO1 Gain major knowledge on system concepts.
- CO2 Apply the concepts of linear and dynamic programming for agricultural applications.
- CO3 Apply simulation techniques for irrigation design.
- CO4 Apply the concepts of neural networks for agricultural applications.
- CO5 Have knowledge about the fuzzy logic and genetic algorithms technique.

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	2	2	-	-	-	-	-
CO2	2	3	3	2	-	-	-	-	-
CO3	2	3	3	2	-	-	-	-	-
CO4	2	3	3	2	-	-	-	-	-
CO5	2	3	3	2	-	-	-	-	-
AVERAGE	2	3	3	2	-	-	-	-	-

- 1 – Low, 2 – Medium, 3 – High, ‘-’ – No correlation

AG25102

AI AND DATA ANALYTICS IN AGRICULTURE

L T P C

3 0 0 3

COURSE OBJECTIVES:

- To equip students with the knowledge and skills to harness artificial intelligence (AI) and data-driven approaches for modern agriculture

UNIT I

INTRODUCTION

9

Overview of Agriculture 4.0 - Role of AI and data analytics in modern agriculture - Benefits of AI-driven solutions: efficiency, productivity, and sustainability - Basics of AI and Machine Learning - Key concepts: supervised, unsupervised, and reinforcement learning - AI tools and technologies in agriculture - Introduction to Data Analytics - Data collection and types of agricultural data (soil, weather, crop, market) - Data pre-processing and cleaning techniques for agricultural datasets - Overview of big data in agriculture.

UNIT II

PREDICTIVE ANALYTICS IN AGRICULTURE

9

Crop Yield Prediction - Techniques for yield forecasting (regression models, neural networks) - Factors influencing crop yield (soil, water, weather, fertilizers) - Weather Forecasting and its Applications - Machine learning for accurate weather predictions - Impact of weather on crop planning and disease

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control - Pest and Disease Detection - AI-based image recognition for pest and disease identification - Predictive models for early pest outbreak warnings.

UNIT III PRECISION AGRICULTURE AND SMART FARMING 9

AI in Precision Farming - Role of AI in optimizing inputs (fertilizers, water, and pesticides) - IoT integration with AI for smart farming systems - Smart Irrigation Systems - AI for water management and irrigation scheduling - Predictive models for water usage optimization.

UNIT IV AGRICULTURAL SUPPLY CHAIN AND MARKET ANALYTICS 9

Supply Chain Optimization - AI for inventory management and logistics - Demand forecasting and market trend analysis - Blockchain and Traceability in Agriculture - Role of blockchain in ensuring supply chain transparency - Use of AI in managing and analyzing blockchain data - Agri-Market Analytics - Price prediction models for agricultural commodities - AI tools for market analysis and decision-making.

UNIT V ROBOTICS IN AGRICULTURE 9

Fundamental of Robotics - types – application. Agricultural robots - types- function - application. Future trends in automation in agriculture.

TOTAL: 45 PERIODS

TEXTBOOKS:

1. J. Satendra Kumar and A. Mohan, 'Artificial Intelligence in Agriculture', Apple Academic Press.
2. B. Dutta, P. Choudhary, and A. Pathak, 'Data Analytics for Smart Agriculture', Springer.
3. Rohitash Chandra, R. Phadtare, 'Machine Learning for Agriculture: Foundations and Applications', CRC Press.

REFERENCES:

1. David Rose and Jason Chilvers, 'The Fourth Industrial Revolution in Agriculture', Routledge.
2. X. Li and S. P. Simon, 'Big Data Analytics in Agriculture', Elsevier.

COURSE OUTCOMES:

After completion of this course, students will be able to:

- CO1 Gain major knowledge on AI and data analytics.
- CO2 Analyze agricultural data to make strategic decisions.
- CO3 Understand the role of AI in precision farming.
- CO4 Apply the concepts of blockchain for agricultural applications.
- CO5 Have knowledge about robotics in agriculture.

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	2	2	3	-	-	3	3
CO2	3	3	3	2	3	-	-	3	3
CO3	3	3	3	2	3	-	-	3	3

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CO4	3	3	3	2	3	-	-	3	3
CO5	3	3	3	2	3	-	-	3	3
AVERAGE	3	3	3	2	3	-	-	3	3

- 1 – Low, 2 – Medium, 3 – High, ‘-‘ – No correlation

AG25103 POST-HARVEST TECHNOLOGY AND STORAGE SYSTEMS L T P C
3 0 0 3

COURSE OBJECTIVES:

- The students would be exposed to fundamental knowledge in engineering properties of agricultural materials, different Post Harvest operations and processing methods of harvested crops and storage of crops.

UNIT I FUNDAMENTALS OF POST HARVESTING 9

Post-harvest technology – introduction –objectives –post harvest losses of cereals, pulses and oilseeds – importance - optimum stage of harvest. Threshing – traditional methods mechanical threshers – types- principles and operation-moisture content –measurement –direct and indirect methods – moisture meters – equilibrium moisture content.

UNIT II PSYCHROMETRY AND DRYING 9

Psychrometry – importance – Psychrometric charts and its uses – Drying – principles and theory of drying – thin layer and deep bed drying – Hot air drying – methods of producing hot air – Types of grain dryers – selection – construction, operation and maintenance of dryers – Design of dryers.

UNIT III CLEANING AND GRADING 9

Principles - air screen cleaners – adjustments - cylinder separator - spiral separator – magnetic separator - colour sorter - inclined belt separator – length separators - effectiveness of separation and performance index.

UNIT IV SHELLING AND HANDLING 9

Principles and operation – maize sheller, husker sheller for maize – groundnut decorticator – castor sheller – material handling – belt conveyor –screw conveyor – chain conveyor – bucket elevators – pneumatic conveying.

UNIT V CROP PROCESSING 9

Paddy processing – parboiling of paddy – methods – merits and demerits – dehusking of paddy – methods – merits and demerits – rice polishers –types – constructional details – polishing –layout of modern rice mill - wheat milling – pulse milling methods – oil seed processing – millets processing - Traditional storage structures (bins, silos, warehouses) - Modern storage systems (cold storage, modified atmosphere storage, and hermetic storage).

TOTAL: 45 PERIODS

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TEXTBOOKS:

1. Chakraverty, A. Post harvest technology for Cereals, Pulses and oilseeds. Oxford & IBH publication Pvt Ltd, New Delhi, Third Edition, 2000.
2. Sahay, K.M., and Singh, K.K. Unit operations of Agricultural Processing. Vikas publishing house Pvt. Ltd., New Delhi, 1994.

REFERENCES:

1. Pande, P.H. Principles of Agriculture Processing. Kalyani Publishers, Ludhiana, 1994.
2. Henderson, S.M. and R.L. Perry. Agricultural Process Engineering. John Wiley and Sons, New York. 1955.

COURSE OUTCOMES:

After completion of this course, students will be able to:

- CO1 Understand the importance of post-harvest processing and determine moisture content of products
- CO2 Perform drying of agricultural products and analyze performance of dryers
- CO3 Recognize the working principles of grain cleaning and grading devices and able to select suitable equipment for cereal grains, oilseeds, and pulses
- CO4 Understand the operation of post-harvest equipment like shellers, conveyors
- CO5 Describe the different Post Harvest operations and processing methods of harvested crops and their storage methods.

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	2	1	1	-	-	1	-
CO2	2	2	3	1	1	-	-	1	-
CO3	2	2	3	1	1	-	-	1	-
CO4	2	2	3	1	1	-	-	1	-
CO5	2	2	3	1	1	-	-	1	-
AVERAGE	3	2	3	1	1	-	-	1	-

- 1 – Low, 2 – Medium, 3 – High, ‘-’ – No correlation

AG25104

AGRICULTURAL DRONES AND UAVS

L T P C
3 0 0 3

COURSE OBJECTIVES:

- To provide an understanding of drones and Unmanned Aerial Vehicles (UAVs) and their applications in agriculture.
- To equip students with knowledge of drone technologies, operations, and data analysis for agricultural purposes.

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- To develop skills in designing and implementing drone-based solutions for precision agriculture, crop monitoring, and resource management.

UNIT I INTRODUCTION TO AGRICULTURAL DRONES AND UAVs 9

History and evolution of drones - Types of drones: fixed-wing, multirotor, and hybrid UAVs - Basic components of drones - Applications of Drones in Precision farming, Crop health monitoring, Soil analysis, irrigation management, and pest control - Benefits and Challenges - Advantages of drone-based solutions in agriculture - Regulatory, technical, and environmental challenges.

UNIT II HARDWARE AND SOFTWARE OF AN AGRICULTURAL DRONE 9

Sensors used in agricultural drones: Multispectral and hyperspectral sensors, RGB and thermal cameras, LiDAR sensors - Payloads for agricultural applications - Flight Control Systems - GPS, IMU, and autopilots - Drone communication and navigation - Flight planning and mission execution software - Data acquisition and analysis tools (e.g., Pix4D, Drone Deploy) - Integration with Geographic Information Systems (GIS).

UNIT III DRONE APPLICATIONS IN PRECISION AGRICULTURE 9

Crop Health Monitoring - Normalized Difference Vegetation Index (NDVI) analysis - Disease and pest detection using drone imagery - Soil and Field Analysis - Soil moisture mapping - Terrain mapping for irrigation and drainage planning - Drone-based pesticide and fertilizer spraying - Seeding using drones for inaccessible areas - Livestock Monitoring.

UNIT IV DRONE OPERATIONS, SAFETY AND REGULATIONS 9

Flight planning - Drone control techniques and piloting basics - Safety Protocols - Handling emergencies and ensuring safe operations - Risk management in agricultural drone usage - Drone Regulations: Global and regional drone regulations - Licensing and permissions for agricultural drone operations - Ethical considerations in drone use.

UNIT V DATA ANALYSIS AND FUTURE TRENDS IN AGRICULTURAL DRONES 9

Data Processing and Analysis - Image stitching and orthomosaic creation - Extracting actionable insights from drone data (e.g., crop health, yield predictions) - Cloud-based platforms for data storage and sharing - IoT-enabled drone systems for real-time monitoring - AI-based analysis for predictive farming - Future Trends in Agricultural Drones.

TOTAL: 45 PERIODS

TEXTBOOKS:

1. K.R.Krishna, Agricultural Drones – A Peaceful Pursuit, Apple Academic Press.
2. Sundaravadivazhagan.B, Gnanasankaran.N, PethuruRaj.C, Intelligent Robots and Drones for Precision Agriculture, Springer edition, 2024

REFERENCES:

1. Simon Rose, ‘Agricultural drones’, May 2019, Capstone press.

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2. Donald Swett, Agricultural Drones : Cultivating Efficiency and Profits in Modern Farming (The Intelligent Future Series) Kindle Edition

COURSE OUTCOMES:

After completion of this course, students will be able to:

- CO1 Understand the basics of agricultural drones and UAVs.
CO2 Understand the hardware components and software of agricultural drones.
CO3 Acquire knowledge about usage of drones in precision agriculture.
CO4 Demonstrate compliance with drone safety precautions.
CO5 Study about the future technologies in drones.

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	3	2	3	2	3	3	3
CO2	3	3	3	2	3	2	3	3	3
CO3	3	3	3	2	3	2	3	3	3
CO4	3	3	3	2	3	2	3	3	3
CO5	3	3	3	2	3	2	3	3	3
AVERAGE	3	3	3	2	3	2	3	3	3

- 1 – Low, 2 – Medium, 3 – High, ‘-’ – No correlation

MC25101 RESEARCH METHODOLOGY AND IPR L T P C
3 0 0 3

UNIT I RESEARCH DESIGN 9

Overview of research process and design, Use of Secondary and exploratory data to answer the research question, Qualitative research, Observation studies, Experiments and Surveys.

UNIT II DATA COLLECTION AND SOURCES 9

Measurements, Measurement Scales, Questionnaires and Instruments, Sampling and methods. Data - Preparing, Exploring, examining and displaying.

UNIT III DATA ANALYSIS AND REPORTING 9

Overview of Multivariate analysis, Hypotheses testing and Measures of Association. Presenting Insights and findings using written reports and oral presentation.

UNIT IV INTELLECTUAL PROPERTY RIGHTS 9

Intellectual Property – The concept of IPR, Evolution and development of concept of IPR, IPR development process, Trade secrets, utility Models, IPR & Bio diversity, Role of WIPO and WTO in

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IPR establishments, Right of Property, Common rules of IPR practices, Types and Features of IPR Agreement, Trademark, Functions of UNESCO in IPR maintenance.

UNIT V PATENTS

9

Patents – objectives and benefits of patent, Concept, features of patent, Inventive step, Specification, Types of patent application, process E-filing, Examination of patent, Grant of patent, Revocation, Equitable Assignments, Licences, Licensing of related patents, patent agents, Registration of patent agents.

TOTAL : 45 PERIODS

REFERENCES:

1. Cooper Donald R, Schindler Pamela S and Sharma JK, “Business Research Methods”, Tata McGraw Hill Education, 11e (2012).
2. Catherine J. Holland, “Intellectual property: Patents, Trademarks, Copyrights, Trade Secrets”, Entrepreneur Press, 2007.
3. David Hunt, Long Nguyen, Matthew Rodgers, “Patent searching: tools & techniques”, Wiley, 2007.
4. The Institute of Company Secretaries of India, Statutory body under an Act of parliament, “Professional Programme Intellectual Property Rights, Law and practice”, September 2013.

AG25151

AGRO TECHNOLOGY LABORATORY

**L T P C
0 0 4 2**

LIST OF EXPERIMENTS:

1. IoT sensor integration to monitor temperature, humidity and soil pressure.
2. IoT enabled irrigation system.
3. Crop monitoring with IoT devices.
4. Yield prediction using machine learning.
5. GIS based land use mapping.
6. Remote sensing for crop health analysis.
7. Design and simulation of a Wireless Sensor Network.
8. LoRaWAN-Based Farm Monitoring System.
9. Crop Disease Detection Using AI.

TOTAL : 60 PERIODS

COURSE OBJECTIVES:

- To introduce the Building Blocks of an embedded System and Software Tools
- To emphasize the role of Input/output interfacing with Bus Communication protocol.
- To illustrate the ISR and scheduling for the multitasking process.
- To explain the basics of a Real-time operating system
- To analyze the applications based on embedded design approaches

UNIT I INTRODUCTION TO EMBEDDED SYSTEMS 6

Introduction to Embedded Systems –Structural units in Embedded processor, selection of processor & memory devices- DMA — Memory management methods- Timer and Counting devices, Real Time Clock, In-circuit emulator, Target Hardware Debugging.

UNIT II EMBEDDED NETWORKING 6

Embedded Networking: Introduction, I/O Device Ports & Buses– Serial Bus communication protocols RS232 standard – RS485 – CAN Bus- Serial Peripheral Interface (SPI) – Inter- Integrated Circuits (I²C).

UNIT III INTERRUPTS THE SERVICE MECHANISM AND DEVICE DRIVER 6

Programmed-I/O busy-wait approach without interrupt service mechanism-ISR concept-interrupt sources – multiple interrupts – context and periods for context switching, interrupt latency and deadline – Introduction to Device Drivers.

UNIT IV RTOS-BASED EMBEDDED SYSTEM DESIGN 6

Introduction to basic concepts of RTOS- Task, process & threads, interrupt routines in RTOS, Multiprocessing and Multitasking, Preemptive and non-preemptive scheduling, Task communication-shared memory, message passing- Interprocess Communication- Introduction to process synchronization using semaphores.

UNIT V EMBEDDED SYSTEM APPLICATION DEVELOPMENT 6

Embedded Product Development Life Cycle - Case Study: Precision Agriculture -Autonomous car.

30 PERIODS

LAB COMPONENT: 30 PERIODS

1. Laboratory exercise: Use any Embedded processor/IDE/open source platform to give hands-on training on basic concepts of embedded system design:
 - a) Introduction to IDE and Programming Environment.
 - b) Configure timer block for signal generation (with given frequency).
 - c) Interrupts programming example using GPIO.
 - d) I²C communication with peripherals
 - e) Master-slave communication between processors using SPI.
 - f) Networking of processor using Wi-Fi.
 - g) Basic RTOS concept and programming

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2. Assignment: Introduction to VxWorks, µC/OS-II, RT Linux
3. Embedded systems-based Mini project.

TOTAL: 30+30 = 60 PERIODS

COURSE OUTCOMES:

After completion of the above subject, students will be able to understand

CO1: The hardware functional and software strategies required to develop various Embedded systems

CO2: The basic differences between various Bus communication standards

CO3: The incorporation of the interface as Interrupt services

CO4: The various scheduling algorithms through a Real-time operating system.

CO5: The various embedded concepts for developing automation applications.

TEXTBOOKS:

1. Rajkamal, 'Embedded system-Architecture, Programming, Design, McGraw-Hill Edu, 3rd edition 2017.
2. Peckol, "Embedded system Design", John Wiley & Sons, 2010.

REFERENCES:

1. Shibu. K.V, "Introduction to Embedded Systems", TataMcgraw Hill, 2nd edition 2017.
2. Lya B.Das," Embedded Systems", Pearson Education, 1st edition 2012.
3. Parag H.Dave,Himanshu B.Dave," Embedded Systems-Concepts ,Design and Programming, Pearson Education,2015, 1st edition.
4. Elicia White, "Making Embedded systems", O'Reilly Series ,SPD,2011, 1st edition.
5. Jonathan W. Valvano, 'Embedded Microcomputer Systems Real-time Interfacing', Cengage Learning , 3rd edition 2010.
6. Tammy Noergaard, "Embedded Systems Architecture", Newnes, 2nd edition, 2013.

List of Open Source Software/ Learning websites:

1. <https://nptel.ac.in/courses/108102045>
2. https://ece.uwaterloo.ca/~dwharder/icsrts/Lecture_materials/A_practical_introduction_to_real-time_systems_for_undergraduate_engineering.pdf
3. <https://www.circuitbasics.com/basics-of-the-i2c-communication-protocol/>
4. https://www.tutorialspoint.com/embedded_systems/es_interrupts.htm
5. <https://www.theengineeringprojects.com/2016/11/examples-of-embeddedsystems.html#:~:text=Embedded%20Product%3A%20Automatic%20Washing%20Machine,done%20by%20your%20machine%20itself.>

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	2	2	1	-	-	-	-
CO2	3	2	3	2	1	-	-	-	-

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CO3	3	3	2	3	1	-	-	-	-
CO4	3	2	2	2	1	-	-	-	-
CO5	3	2	1	2	1	-	3	3	3
AVERAGE	3	2.2	2	2.2	1	-	3	3	3

- 1 – Low, 2 – Medium, 3 – High, ‘-’ – No correlation

AG25202 IRRIGATION ENGINEERING AND AUTOMATION

**L T P C
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COURSE OBJECTIVES:

- To inculcate water resources development and various parameters required for irrigation scheduling and its requirement.
- To understand different kinds of irrigation system and choose appropriate system for a given environment.
- To introduce different types of water control and diversion structures for planning the irrigation system.
- To equip students with knowledge of automation technologies in modern irrigation systems.
- To develop skills in designing efficient and sustainable irrigation systems integrated with smart technologies.

UNIT I WATER RESOURCES AND IRRIGATION REQUIREMENT 9

Water Resources- River basins-Development and Utilization in India and Tamil Nadu-Irrigation – duty and delta - Rooting characteristics - Moisture use of crop, Evapotranspiration - ET plot - Crop water requirement - Effective rainfall - Scheduling - Irrigation requirement - Irrigation frequency, Irrigation efficiencies.

UNIT II METHODS OF IRRIGATION 9

Methods of Irrigation – Surface and Subsurface methods – Drip and Sprinkler - Hydraulics and design - Erodible and non-erodible, Kennedy’s and Lacey’s theories, Materials for lining water courses and field channel, Water control and diversion structure - Underground pipeline irrigation system

UNIT III DIVERSION AND IMPOUNDING STRUCTURES 9

Head works –Weirs and Barrage –Types of impounding structures - Factors affecting, location of dams - Forces on a dam -Design of Gravity dams- Earth dams, Arch dams – Spillways -Energy dissipaters.

UNIT IV AUTOMATION IN IRRIGATION SYSTEMS 9

Benefits of automation in irrigation management - IoT integration for remote monitoring and control - Types of Automated Systems - Time-based systems, Volume-based systems, Real-time and adaptive control systems - Integration of IoT platforms for data analytics - Precision Irrigation - Concept and importance in modern agriculture - Variable Rate Irrigation (VRI) - GIS and GPS applications in

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irrigation - Greenhouse Irrigation Systems - Automated micro-irrigation systems - Environmental control in greenhouses.

UNIT V SUSTAINABLE AND EFFICIENT IRRIGATION PRACTICES 9

Water Conservation in Irrigation - Minimizing water losses: seepage, evaporation, runoff - Use of mulching and cover crops for water retention - Irrigation Scheduling and Management - Scientific scheduling methods: crop water stress index, soil moisture depletion - Automation-based scheduling using real-time data - Environmental and Economic Considerations - Cost-benefit analysis of automated and traditional irrigation systems - Energy-Efficient Irrigation Systems - Solar-powered irrigation systems - Optimizing pump performance and energy use.

TOTAL: 45 PERIODS

TEXTBOOKS:

1. Dilip Kumar Majumdar., “Irrigation Water Management”, Prentice-Hall of India, New Delhi, 2008.
2. Michael, A.M., “Irrigation Engineering”, Vikas Publishers, New Delhi, 2008.
3. Garg, S.K., “Irrigation Engineering,” Laxmi Publications, New Delhi, 2008.

REFERENCES:

1. **IoT Based Automated Irrigation System for Agricultural Activities”, IEEE,**
<https://ieeexplore.ieee.org/xpl/conhome/10249210/proceeding>
2. “Automation in Irrigation - A Review”, <https://www.researchgate.net/journal/International-Journal-of-Current-Microbiology-and-Applied-Sciences/23197706?tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6InB1Ym90aWw9uIn9>

COURSE OUTCOMES:

After completion of this course, students will be able to:

- CO1: Gain knowledge and skills on Planning, design, operation and management of Water Resources and Irrigation Requirement.
- CO2: Acquire knowledge on different methods of irrigation
- CO3: Understand the different types of Diversion and Impounding Structures
- CO4: Understand the different automation technologies in modern irrigation systems.
- CO5: Develop efficient and sustainable irrigation systems integrated with smart technologies.

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	2	2	-	-	-	-	-
CO2	3	2	2	2	-	-	-	-	-
CO3	3	2	2	2	-	-	-	-	-
CO4	3	2	2	2	-	-	3	3	3

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CO5	3	2	2	2	-	3	3	3	3
AVERAGE	3	2	2	2	-	3	3	3	3

- 1 – Low, 2 – Medium, 3 – High, ‘-’ – No correlation

AG25203 ORGANIC FARMING AND SUSTAINABLE PRACTICES

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COURSE OBJECTIVES:

1. To understand the principles and benefits of organic farming.
2. To explore sustainable practices for soil, crop, and pest management.
3. To analyze certification and marketing strategies for organic produce.
4. To promote environmentally friendly and economically viable farming methods.

UNIT I PRINCIPLES OF ORGANIC FARMING

9

Organic farming basics: Definition, importance, and global trends - Agroecology: Ecological principles in organic farming for soil and biodiversity conservation - Recycling of organic materials - Use of renewable resources and ecological harmony - Transition barriers, yield gaps, and market acceptance.

UNIT II SOIL AND CROP MANAGEMENT

9

Nutrient management in organic farming – Composting Techniques, types, and importance - Vermicomposting Process, advantages, and challenges - Biofertilizers: Rhizobium, Azotobacter, and phosphate-solubilizing bacteria - Crop management techniques: Mixed cropping and intercropping - Crop rotation for soil fertility enhancement - Soil health indicators: Microbial activity and organic matter content.

UNIT III PEST AND DISEASE MANAGEMENT

9

Biological pest control - Role of parasitoids and predators in pest suppression - Commonly used biopesticides (e.g., neem-based products, Trichoderma) - Natural pest repellents - Botanical extracts and essential oils - Case studies: Successful IPM approaches in organic farms.

UNIT IV CERTIFICATION AND MARKETING

9

Organic certification standards - India's NPOP standards and APEDA guidelines - International standards - USDA, EU organic regulations - Challenges in certification - Cost, complexity, and farmer adoption - Marketing organic produce - Supply chain considerations for perishable products - Role of cooperatives and e-commerce platforms in expanding market reach.

UNIT V TECHNOLOGICAL ADVANCEMENTS

9

Role of technology in organic farming - Smart sensors for pest and nutrient monitoring - Automated irrigation systems for organic farms - Enhancing traceability and transparency in organic supply chains - AI in predicting pest outbreaks, smart packaging for organic produce.

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COURSE OUTCOMES:

At the end of the course, students will be able to

CO1: Understand the fundamentals of organic farming and its benefits.

CO2: Manage soil and crops sustainably using organic practices.

CO3: Implement IPM for pest and disease control.

CO4: Navigate certification processes and market organic produce effectively.

CO5: Explore advanced technologies for enhancing organic farming.

TEXT BOOKS:

1. Lampkin, N. *Organic Farming*. Blackwell Science, 2002.
2. Pimentel, D. *Agroecology and Sustainable Agriculture*. CRC Press, 2014.

REFERENCE BOOKS:

1. Gliessman, S. R. *Agroecology: The Ecology of Sustainable Food Systems*. CRC Press, 2007.
- Scialabba, N., & Müller-Lindenlauf, M. *Organic Agriculture and Climate Change*. FAO, 2010.

AG25204 FOOD SUPPLY CHAIN MANAGEMENT

**L T P C
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COURSE OBJECTIVES:

- To understand the components and dynamics of food supply chains.
- To explore strategies for efficient food production, distribution, and storage.
- To identify challenges in food safety, sustainability, and traceability.
- To integrate modern technologies like IoT, blockchain, and AI in managing food supply chains.
- To enable students to design sustainable and efficient food supply systems.

UNIT I INTRODUCTION

9

Overview of Food Supply Chains: Definition, scope, and importance - Suppliers, producers, processors, distributors, retailers, and consumers - Unique Characteristics of Food Supply Chains - Types of Food Supply Chains - Fresh produce, processed foods, frozen products, and organic foods - Current Trends and Challenges - Globalization, urbanization, and climate change impacts.

UNIT II OPERATIONS AND LOGISTICS IN FOOD SUPPLY CHAINS

9

Procurement and Sourcing - Supplier selection, contracts, and raw material management - Planning, scheduling, and optimization of Production and Processing - Cold Chain Management: Technologies and challenges in temperature-sensitive logistics - Distribution and Transportation - Cost optimization, route planning, and multi-modal transportation - Inventory Management - Stock control techniques, demand forecasting, and warehouse design.

UNIT III FOOD SAFETY, QUALITY AND REGULATIONS

9

Food Safety Standards and Regulations: HACCP, ISO 22000, FSSAI, and Codex Alimentarius - Quality Assurance in Food Supply Chains - Inspection, testing, and certification - Traceability and Recall Systems: Technologies and processes for tracking food products - Risk Management - Identifying and

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mitigating risks in food safety and quality - Case Studies: Major food safety incidents and lessons learned.

UNIT IV SUSTAINABILITY AND ETHICAL ISSUES

9 Sustainable

Practices in Food Supply Chains: Reducing food waste, sustainable sourcing, and renewable energy use - Ethics in Food Supply Chains - Fair trade, labor rights, and animal welfare - Recycling and reusing food waste - Adaptation strategies for food production and distribution - Corporate Social Responsibility (CSR)

UNIT V TECHNOLOGY AND INNOVATION IN FOOD SUPPLY CHAINS 9

Role of IoT, AI, and data analytics in optimizing food supply chains - Ensuring transparency and building consumer trust - Robotics and advanced logistics systems - Extending shelf life and improving safety - Vertical farming, lab-grown food, and alternative proteins.

TOTAL: 45 PERIODS

COURSE OUTCOMES:

By the end of this course, students will:

CO1: Understand the structure and dynamics of food supply chains.

CO2: Be able to address challenges related to food safety, sustainability, and traceability.

CO3: Use modern technologies to optimize food production and distribution.

CO4: Design sustainable and efficient food supply chain systems.

CO5: Be prepared for careers in food supply chain management, logistics, or sustainability roles.

TEXTBOOKS:

1. John Mangan, Chandra Lalwani, Tim Butcher, and Roya Javadpour, 'Global Logistics and Supply Chain Management', Wiley press.
2. Joel D. Wisner, Keah-Choon Tan, and G. Keong Leong, 'Principles of Supply Chain Management: A Balanced Approach', Cengage Learning

REFERENCES:

1. Samir Dani, 'Food Supply Chain Management and Logistics: From Farm to Fork', Kogan Page.
2. Madeleine Pullman and Zhaohui Wu, 'Food Supply Chain Management: Economic, Social, and Environmental Perspectives', Routledge.

AG25205

SUSTAINABLE AGRICULTURE LABORATORY

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LIST OF EXPERIMENTS:

1. Advanced Soil Fertility Mapping Using GIS
2. Biochar Application in Soil for Carbon Sequestration
3. Soil Microbial Diversity and Organic Farming
4. Drone-Assisted Precision Agriculture for Crop Monitoring
5. Variable Rate Technology (VRT) for Seed and Fertilizer Application

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6. Integrating IoT Sensors for Climate-Resilient Crop Management
7. Water Quality Management in Aquaponics
8. Wind Energy for Irrigation and Farm Operations
9. Biogas Production from Agricultural Waste
10. Integrated Pest Management (IPM) with Biological Controls

TOTAL: 60 PERIODS

AG25001	CLIMATE-RESILIENT AGRICULTURE	L T P C
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COURSE OBJECTIVES:

- To understand the impact of climate change on agriculture and food security.
- To explore strategies for enhancing the resilience of agricultural systems to climate variability.
- To study climate-smart agricultural practices and technologies.
- To analyze policy frameworks and programs supporting climate-resilient agriculture.

UNIT I INTRODUCTION TO CLIMATE CHANGE AND AGRICULTURE 9

Climate change fundamentals - Definition, natural vs. anthropogenic causes, greenhouse gases, and global warming potential - Agriculture and climate vulnerability - Impact on cropping patterns, pest outbreaks, soil degradation, and water resources - Food security and climate change - Global and regional impacts on food production and supply chains.

UNIT II CLIMATE-SMART AGRICULTURE (CSA) 9

Core principles of CSA - Productivity, adaptation, and mitigation - Techniques for CSA - Agroforestry systems and their role in carbon sequestration - Crop diversification and selection of climate-resilient crop varieties - Livelihood diversification for smallholder farmers - Technological integration - IoT for real-time monitoring, data analytics for decision-making.

UNIT III WATER AND SOIL MANAGEMENT 9

Efficient water use - Precision irrigation methods (drip, sprinkler, smart sensors), and water budgeting - Soil health management - Mulching, conservation tillage, and biochar applications - Techniques for improving soil organic carbon - Case studies: Successful soil and water conservation projects in arid and semi-arid regions.

UNIT IV POLICY AND FRAMEWORKS 9

Global policies: Paris Agreement, Kyoto Protocol, and IPCC recommendations - United Nations' Sustainable Development Goals (SDGs) related to agriculture - India's National Mission for Sustainable Agriculture (NMSA) - Regional programs addressing droughts and floods - Case studies: Policy-driven success stories in climate-resilient farming.

UNIT V TECHNOLOGICAL INNOVATIONS 9

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Early warning systems - Satellite-based weather prediction and disaster preparedness - Advances in precision farming - Role of drones in crop health monitoring and mapping - Use of sensors and IoT for adaptive irrigation and nutrient management - Crop modeling for climate scenario simulation and AI-based decision support systems.

COURSE OUTCOMES:

At the end of the course, students will be able to

CO1: Understand the effects of climate change on agricultural systems.

CO2: Implement climate-smart technologies to enhance resilience.

CO3: Manage soil and water resources effectively under changing climates.

CO4: Analyze policy frameworks supporting sustainable agriculture.

CO5: Explore technological innovations for mitigating climate impacts.

TEXT BOOKS:

1. Wollenberg, E., et al. *Climate-Smart Agriculture*. Springer, 2016.
2. Lal, R., *Climate Resilient Agriculture for Ensuring Food Security*. Springer, 2020.

REFERENCE BOOKS:

1. Campbell, B. M., et al. *Sustainable Agriculture and Climate Change*. Springer, 2011.
2. Pathak, H., et al. *Climate Change and Agriculture in India*. Springer, 2015.

AG25002

ADVANCED LANDSCAPE ARCHITECTURE

**L T P C
3 0 0 3**

COURSE OBJECTIVES:

- To understand the paradigms in landscape architecture in the post-industrial revolution era and to understand the multifaceted dimensions of landscape architecture such as ecology, environment and sustainability.
- To study the contemporary landscape and the manifestation in the western and Indian contexts.

UNIT I

ADVANCED LANDSCAPE DESIGN PRINCIPLES

9

Historical and contemporary design approaches - Integrating sustainability in landscape design - Site-specific design strategies - Adaptive reuse of landscapes - Form, function, and materiality in landscapes - Concepts of visual harmony and spatial organization - Analysis of iconic and contemporary landscape projects.

UNIT II

SUSTAINABLE LANDSCAPE PLANNING

9

Biodiversity conservation in urban and rural landscapes - Restoration ecology principles - Rainwater harvesting and stormwater management systems - Designing for flood resilience and drought adaptation - Role of green roofs, walls, and urban forests - Climate-responsive planning and urban heat mitigation - Materials and construction techniques for sustainable landscapes.

UNIT III

ADVANCED TECHNOLOGIES

9

GIS: Applications of GIS in landscape analysis and planning - Remote sensing for site and environmental assessment - BIM tools for landscape architecture projects - Use of IoT and sensors in managing landscape systems - Advanced software tools for 3D modeling and rendering.

Designing inclusive and accessible public landscapes - Participatory design methods - Strategies for preserving historical landscapes - Integrating traditional knowledge into modern design - Impact of landscapes on mental well-being - Designing therapeutic and healing gardens - Enhancing identity and sense of place through landscape design.

Climate adaptation strategies - Regenerative landscapes - Scenario-based planning and modeling - Legal and regulatory aspects of landscape projects - Collaboration with multidisciplinary teams - On-site observations and experiential learning.

At the end of the course, students will be able to

- CO1** Equip the students to do landscape working drawings and preparation of bill of quantities and estimation.
- CO2** Understand the design solutions for larger sites and express the same using models.
- CO3** Understand the different concepts of landscape formulations.

1. Simonds, J. O. 1961. Landscape Architecture: The Shaping of Man's Natural Environment. F.W. Dodge Cooperation, London Harris.C.W. and Din, N.T. 1997. Time Saver Standards For Landscape Architecture. Mcgraw – Hill International Edition, Arch. Series
2. Starke .B. and Simonds, J. O. 2013. Landscape Architecture: A Manual of Site Planning and Design. 5th edition. McGraw-Hill Professional.

1. Shaheer, M., Dua, G.W. and Pal, A.2012. Landscape Architecture in India: A Reader. Indian Journal of Landscape Architecture.
2. Reid, G. W. 1993. From Concept to Form: In Landscape Design. John Wiley & Sons.

1. To understand the fundamentals of environmental policies and legal frameworks.
2. To analyze the role of laws in protecting natural resources and biodiversity.
3. To evaluate global and national environmental treaties, agreements, and regulations.
4. To explore the relationship between environmental laws and sustainable development.

UNIT I	INTRODUCTION TO ENVIRONMENTAL POLICIES	9
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UNIT II ENVIRONMENTAL LEGISLATION IN INDIA 9

UNIT III GLOBAL ENVIRONMENTAL AGREEMENTS 9

UNIT IV JUDICIAL INTERVENTIONS AND CASE STUDIES 9

UNIT V ENVIRONMENTAL LAW AND SUSTAINABILITY 9

COURSE OUTCOMES:

CO1: Understand the principles and evolution of environmental policies and laws.

CO2: Analyze key environmental legislation in India.

CO3: Evaluate global treaties and agreements on environmental conservation.

CO4: Assess judicial interventions and case studies in environmental protection.

CO5: Explore the role of laws in promoting sustainability and addressing future challenges.

1. Divan, S., & Rosencranz, A. *Environmental Law and Policy in India*. Oxford University Press, 2021.
2. Shastri, S. C. *Environmental Law*. Eastern Book Company, 2022.

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REFERENCE BOOKS:

1. Birnie, P., & Boyle, A. *International Law and the Environment*. Oxford University Press, 2009.
2. Agrawal, A., & Narain, S. *State of India's Environment: A Citizen's Report*. CSE, 2021.

AG25004 ADVANCED FARM EQUIPMENT DESIGN

**L T P C
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COURSE OBJECTIVES:

1. To understand the principles of designing advanced agricultural machinery.
2. To analyze the role of automation and robotics in modern farming.
3. To apply mechanical and electronic systems for efficient equipment design.
4. To explore energy-efficient and sustainable equipment for agriculture.

UNIT I BASICS OF FARM EQUIPMENT DESIGN 9

Design principles: Functional, ergonomic, and economic considerations - Classification of farm equipment: Tillage, planting, harvesting, and post-harvest equipment - Material selection: Strength, corrosion resistance, and cost-effectiveness.

UNIT II MECHANICAL DESIGN ASPECTS 9

Power transmission systems: Gears, belts, and chains - Hydraulic systems: Principles, components, and applications in farm machinery - Finite Element Analysis (FEA): Stress analysis of critical components.

UNIT III ELECTRONICS AND AUTOMATION 9

Sensor integration: Types (proximity, load, position) and applications in equipment - Embedded systems for automation: Microcontrollers and programmable logic controllers (PLCs) - Robotics in agriculture: Autonomous tractors, robotic harvesters, and UAVs.

UNIT IV ENERGY-EFFICIENT EQUIPMENT 9

Solar-powered machinery - Design considerations and challenges - Biofuel engines: Principles and applications in agricultural tractors - Hybrid power systems: Combining renewable and conventional energy sources.

UNIT V CASE STUDIES AND FUTURE TRENDS 9

Innovative designs: Self-cleaning seed drills, precision planters, and smart irrigation systems - Sustainable practices: Equipment designed for minimal soil compaction and energy use - Future trends: AI-driven farm equipment and smart integration with IoT.

COURSE OUTCOMES:

At the end of the Semester, students will be able to

CO1: Understand the fundamentals of advanced farm equipment design.

CO2: Apply mechanical principles to optimize performance.

CO3: Integrate electronics and automation for precision farming.

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CO4: Design energy-efficient machinery for sustainable farming.

CO5: Explore future trends and innovative solutions in equipment design.

TEXT BOOKS:

1. Kepner, R. A., Bainer, R., & Berger, E. L. *Principles of Farm Machinery*. CBS Publishers, 2005.
2. Srivastava, A. K., et al. *Engineering Principles of Agricultural Machines*. ASABE, 2006.

REFERENCE BOOKS:

1. Hunt, D. R. *Farm Power and Machinery Management*. Iowa State University Press, 2001.
2. Singh, G. *Farm Machinery: Principles and Applications*. Jain Brothers, 2019.

AG25005

3D PRINTING IN AGRI-TECH

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COURSE OBJECTIVES:

- To introduce students to the principles of 3D printing and its potential applications in agriculture.
- To develop an understanding of the materials suitable for 3D printing in agricultural settings, including sustainable and biodegradable options.
- To enable students to design, prototype, and implement 3D-printed solutions for farm-specific challenges.
- To explore the integration of 3D printing with other technologies, such as IoT and precision farming.
- To promote innovation in agri-tech with a focus on cost-effectiveness, environmental sustainability, and scalability.

UNIT I INTRODUCTION TO 3D PRINTING AND AGRI-TECH

9

Fundamentals of 3D Printing - Principles of additive manufacturing - Overview of 3D printing technologies (FDM, SLA, SLS, etc.) - Evolution of 3D printing in various industries - Intersection of 3D Printing and Agriculture - Role of 3D printing in agri-tech innovation - Overview of digital fabrication in modern farming - Case studies of successful applications in agriculture. 3D Printing Ecosystem - Software, hardware, and materials involved - Understanding slicing and G-code generation for agri-tech use.

UNIT II MATERIALS FOR 3D PRINTING IN AGRICULTURE

9

Material Science Basics - Properties of 3D printing materials: polymers, metals, composites, and bioplastics - Comparison of material performance for agricultural applications - Sustainable and Biodegradable Materials - PLA and other biodegradable polymers for agri-tech - Recycling agricultural waste for 3D printing filaments - Advanced Materials - Conductive and smart materials for sensors and IoT devices - Soil-compatible and weather-resistant materials.

UNIT III APPLICATIONS OF 3D PRINTING IN AGRI-TECH

9

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Farm Equipment and Tools - Customization and prototyping of tools for small and large-scale farms - Repair and replacement of equipment parts using 3D printing - Precision Agriculture - Creation of drone components for precision farming - Development of IoT-enabled devices and sensors - Soil and Water Management - Design and printing of drip irrigation systems and nozzles - Fabrication of soil-testing kits and water-quality sensors - Crop Production and Protection - Biodegradable plant pots and seed trays - Pest traps and precision pesticide delivery systems.

UNIT IV DESIGN AND PROTOTYPING FOR AGRI-TECH

9

CAD for 3D Printing - Basics of CAD software for designing agricultural components - Optimization techniques for agri-tech applications - Prototyping Process - Iterative design and testing for farm equipment - Case study: Prototyping a 3D-printed irrigation part - Farm-Specific Customizations - Creating site-specific designs for farming challenges - Cost and time analysis of custom 3D-printed solutions - Rapid Prototyping Techniques - Integration of 3D printing with CNC machining and laser cutting.

UNIT V FUTURE TRENDS, CHALLENGES, AND SUSTAINABILITY

9

Advanced Trends in 3D Printing - Integration of AI and machine learning in design and printing - Development of 4D printing for dynamic and responsive agricultural tools - Economic and Environmental Impact - Cost analysis of implementing 3D printing in agriculture - Energy efficiency and carbon footprint of 3D printing in agri-tech - Challenges and Solutions - Scaling 3D printing for industrial agricultural applications - Addressing limitations in material strength and durability - Research and Innovation - Ongoing research in bio-based 3D printing for agriculture - Role of 3D printing in achieving sustainable development goals (SDGs).

TOTAL: 45 PERIODS

COURSE OUTCOMES:

- CO1 Demonstrate a comprehensive understanding of 3D printing and agri-tech
- CO2 Identify and select appropriate materials for 3D printing in agriculture.
- CO3 Understand the applications of 3D printing in agriculture.
- CO4 Utilize CAD software and 3D printers to design and fabricate prototypes tailored to agricultural needs.
- CO5 Explore advanced trends in 3D printing.

TEXTBOOKS:

1. Paul.H.Jones, “Sustainable Agriculture and Agribusiness in 3D Printing Technology”, Academic Press.
2. Ian Gibson, David Rosen, Brent Stucker, “Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing”, Springer.
3. Chee Kai Chua, Kah Fai Leong, “3D Printing and Additive Manufacturing: Principles and Applications”, World Scientific.

COURSE OBJECTIVES:

- To introduce students to the concept, components, and architecture of digital twin technology, especially its relevance in agriculture.
- To develop the ability to design and implement digital twin models for farming operations using advanced tools and technologies.
- To explore various applications of digital twins in precision farming, livestock management, irrigation, and supply chain optimization.
- To equip students with knowledge of data collection, integration, and analytics for real-time decision-making in agriculture.
- To address challenges in implementing digital twin solutions in agriculture and study emerging trends and innovations in this field.

UNIT I INTRODUCTION TO DIGITAL TWIN TECHNOLOGY

9

Definition and principles of digital twin technology - Evolution of digital twins in industry and agriculture - Key components: physical asset, virtual model, and data flow - Internet of Things (IoT) in digital twins - Role of sensors, actuators, and data acquisition systems - Integration of AI, machine learning, and big data analytics - Potential applications in precision farming, irrigation, and crop management - Benefits: real-time monitoring, predictive analytics, and decision-making.

UNIT II DESIGNING DIGITAL TWINS FOR FARMING

9

Virtual modeling tools and platforms (e.g., CAD, GIS) - Data sources: drones, satellites, and ground sensors - Communication protocols and standards - Digital Representation of Agricultural Assets - Modeling soil, crops, livestock, and equipment - Creating a dynamic virtual environment for farming operations - Simulating farm processes using digital twins - Calibration of digital models with real-world data.

UNIT III APPLICATIONS OF DIGITAL TWIN TECHNOLOGY IN AGRICULTURE

9

Precision Farming - Real-time monitoring of soil health, crop growth, and pest activity - Optimizing irrigation and fertilization schedules using predictive models - Livestock Management - Digital twins for tracking animal health and behaviour - Enhancing productivity through predictive analytics - Farm Equipment and Machinery - Maintenance and performance optimization of tractors and harvesters - Predictive maintenance using digital twin models - Supply Chain and Post-Harvest Management - Tracking produce from farm to market using digital twins - Improving storage and logistics with real-time insights.

UNIT IV TECHNOLOGIES ENABLING DIGITAL TWINS IN FARMING

9

IoT and Sensor Networks - Role of IoT devices in collecting farm data - Sensor types: soil moisture, temperature, pH, and livestock wearables - Data Management and Analytics - Big data tools for processing large datasets - Cloud computing and edge computing in agriculture - Visualization and

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Interaction - Using AR/VR for immersive interactions with digital twins - Dashboards and visualization tools for decision-making - Integration with Smart Farming Ecosystems - Role of 5G and advanced communication technologies - Interoperability with existing agricultural systems.

UNIT V FUTURE TRENDS, CHALLENGES, AND CASE STUDIES

9

Emerging Trends in Digital Twins for Farming - Integration of blockchain for traceability - Autonomous farming systems powered by digital twins - Development of cognitive digital twins using AI - Challenges and Solutions - Data privacy, security, and ethical concerns - Scalability and cost-effectiveness for small-scale farmers - Overcoming technology adoption barriers in rural areas - Case Studies and Applications - Digital twin applications in large-scale farming projects - Success stories from smart farms globally - Comparative analysis of traditional and digital twin-enabled farming.

TOTAL: 45 PERIODS

COURSE OUTCOMES:

- CO1 Understand the basics of Digital Twin technology.
- CO2 Design and develop digital twin models for farming operations using simulation tools, IoT, and data analytics platforms.
- CO3 Apply digital twin technology in real world applications of agriculture.
- CO4 Analyze the technologies enabling digital twins in farming.
- CO5 Explore advanced trends in digital twin technology.

TEXTBOOKS:

1. Shyam Nath, 2014, Digital Twin Technologies and Smart Farming, Springer.
2. Rajesh Singh, Anita Gehlot, Bhupendra Singh, "Internet of Things for Agriculture 4.0", CRC Press.
3. Latief Ahmad, S. A. Wani, "Smart Agriculture: IoT-Based Smart Sensors and Devices", CRC Press.

AG25007

AUTONOMOUS VEHICLES IN AGRICULTURE

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COURSE OBJECTIVES:

- To provide an overview of autonomous vehicles (AVs) and their role in modern agriculture, including applications, technologies, and benefits.
- To explore the sensor systems, navigation technologies, and perception models critical for autonomous vehicles in agriculture.
- To examine the practical applications of autonomous vehicles in various agricultural tasks.
- To explore into the control systems, AI algorithms, and machine learning techniques used to operate autonomous vehicles.
- To analyze the challenges, regulations, and emerging trends shaping the future of autonomous vehicles in agriculture.

UNIT I	INTRODUCTION	9
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UNIT I	INTRODUCTION	9
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UNIT II SENSORS AND NAVIGATION SYSTEMS 9

UNIT II	SENSORS AND NAVIGATION SYSTEMS	9
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UNIT III AUTONOMOUS VEHICLE APPLICATIONS 9

UNIT III AUTONOMOUS VEHICLE APPLICATIONS 9

UNIT IV	AUTONOMOUS VEHICLE CONTROL SYSTEMS	9
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UNIT IV	AUTONOMOUS VEHICLE CONTROL SYSTEMS	9
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UNIT V	CHALLENGES AND REGULATIONS	9
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UNIT V	CHALLENGES AND REGULATIONS	9
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TOTAL: 45 PERIODS

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COURSE OUTCOMES:

By the end of this course, students will:

- Understand the key technologies and components that enable autonomous vehicles in agriculture.
- Gain practical knowledge of how autonomous vehicles are applied to various agricultural tasks such as planting, irrigation, harvesting, and pest management.
- Be able to assess the challenges and opportunities of implementing autonomous vehicles in farming operations.
- Be familiar with the regulatory landscape and the ethical considerations surrounding the use of autonomous vehicles in agriculture.
- Be prepared to contribute to future innovations in the field of autonomous agricultural systems.

TEXTBOOKS:

1. Guangyu Li, Wenbo Zhu, and Han Ding, “Autonomous Vehicles in Agricultural Robotics”, Springer.
2. Harold D. Maynard, John W. Rausch, “Autonomous Vehicles and Advanced Driver Assistance Systems”
3. Rajesh Singh, Anita Gehlot, Bhupendra Singh, “Internet of Things for Agriculture 4.0”, CRC Press.
4. Latief Ahmad, S. A. Wani, “Smart Agriculture: IoT-Based Smart Sensors and Devices”, CRC Press.

AG25008	GEOSPATIAL DATA ANALYSIS FOR AGRICULTURE	L T P C
		3 0 0 3

COURSE OBJECTIVES:

- To introduce the fundamentals of geospatial technologies and their applications in agriculture.
- To analyze geospatial data using Geographic Information Systems (GIS) and remote sensing.
- To explore the role of geospatial tools in precision agriculture and resource management.
- To develop decision-making strategies using geospatial data integration.

UNIT I	BASICS OF GEOSPATIAL TECHNOLOGIES	9
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Introduction to geospatial technologies: GIS, GPS, and remote sensing. Geospatial data types: Raster and vector data models. Coordinate systems and map projections: UTM, WGS84, and transformations. Basics of GPS: Working principles, accuracy factors, and agricultural applications.

UNIT II	REMOTE SENSING IN AGRICULTURE	9
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Principles of remote sensing: Electromagnetic spectrum and image acquisition. Satellite platforms and sensors: Landsat, Sentinel, MODIS, and hyperspectral sensors. Remote sensing indices: NDVI, SAVI, and EVI for crop health monitoring. Applications: Crop area estimation, drought assessment, and soil mapping.

UNIT III	GIS FOR AGRICULTURAL APPLICATIONS	9
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GIS basics: Spatial data handling, analysis, and visualization. Mapping agricultural resources: Soil properties, water availability, and land use patterns. Spatial analysis techniques: Overlay, buffering, and interpolation. GIS software tools: QGIS, ArcGIS, and Google Earth Engine.

UNIT IV INTEGRATING GEOSPATIAL DATA IN PRECISION AGRICULTURE 9

Geospatial data integration: Combining GPS, GIS, and remote sensing for decision-making. Precision farming applications: Variable rate technology (VRT), site-specific nutrient management. Monitoring and forecasting systems: Real-time data integration and analytics. Case studies: Smart irrigation and pest management using geospatial tools.

UNIT V: FUTURE TRENDS AND ADVANCEMENTS

9

Cloud-based geospatial analysis: Google Earth Engine and AWS for agriculture. Machine learning integration: AI models for geospatial data analysis in farming. IoT and geospatial systems: Real-time monitoring and data collection. Emerging technologies: Drones and UAVs in geospatial mapping for agriculture.

TOTAL: 45 hours

COURSE OUTCOMES:

- CO1: Understand geospatial technologies and their applications in agriculture.
- CO2: Analyze agricultural data using remote sensing and GIS tools.
- CO3: Integrate geospatial data for precision agriculture applications.
- CO4: Develop geospatial solutions for resource management in agriculture.
- CO5: Explore future trends and emerging technologies in geospatial analytics.

TEXT BOOKS

1. Lillesand, T. M., & Kiefer, R. W. *Remote Sensing and Image Interpretation*. John Wiley & Sons, 2015.
2. Burrough, P. A., & McDonnell, R. A. *Principles of Geographical Information Systems*. Oxford University Press, 2015.

REFERENCE BOOKS

1. Bastiaanssen, W. G., & Molden, D. *Remote Sensing for Agriculture and Hydrology*. Springer, 2019.
2. Kumar, L. *Geospatial Technologies for Agriculture*. Springer, 2021.

COURSE OBJECTIVES

- To introduce advanced soilless farming techniques.
- To explore the design and implementation of hydroponic systems.
- To understand the role of automation in vertical farming for resource optimization.

UNIT I INTRODUCTION TO HYDROPONICS AND VERTICAL FARMING 9

Soilless farming: Definition, principles, and advantages. Types of hydroponics: Deep water culture, nutrient film technique, aeroponics. Evolution and importance of vertical farming in urban agriculture.

UNIT II SYSTEM DESIGN AND IMPLEMENTATION 9

Components of hydroponic systems: Grow beds, reservoirs, and pumps. Nutrient solutions: Composition, preparation, and recycling techniques. Vertical farming setups: Layered systems, lighting requirements, and space management.

UNIT III AUTOMATION IN CONTROL SYSTEMS 9

Role of IoT and sensors in monitoring pH, temperature, and humidity. Automated irrigation and nutrient delivery systems. Integration of AI for predictive crop management.

UNIT IV CHALLENGES AND SUSTAINABILITY 9

Resource efficiency: Water, energy, and nutrient use. Economic feasibility: Cost-benefit analysis for hydroponic setups. Challenges: Pest management, scalability, and environmental impact.

UNIT V CASE STUDIES AND FUTURE DIRECTIONS 9

Successful hydroponic and vertical farming projects globally. Innovations in LED lighting and energy-efficient farming. Future trends: Urban farming hubs and sustainable food systems.

COURSE OUTCOMES (COs)

CO1: Understand the principles of hydroponics and vertical farming.

CO2: Design and implement efficient soilless farming systems.

CO3: Integrate automation for optimizing resource use.

CO4: Address sustainability challenges in modern farming systems.

CO5: Analyze global case studies and explore emerging trends.

TEXT BOOKS

1. Resh, H. M. *Hydroponic Food Production*. CRC Press, 2019.
2. Kozai, T., et al. *Plant Factory: An Indoor Vertical Farming System for Efficient Quality Food Production*. Academic Press, 2016.

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AG25010

CLIMATE CHANGE AND ADAPTATION

L T P C

3 0 0 3

COURSE OBJECTIVES:

- To know the basics, importance of global warming
- To know the concept of mitigation measures against global warming
- To learn about the global warming and climate change

UNIT I EARTH'S CLIMATE SYSTEM

9

Role of ozone in environment ozone layer ozone depleting gases Green House Effect, Radiative effects of Greenhouses Gases Hydrological Cycle Green House Gases and Global Warming Carbon Cycle

UNIT II ATMOSPHERE AND ITS COMPONENTS

9

Importance of Atmosphere - Physical Chemical Characteristics of Atmosphere - Vertical structure of the atmosphere- Composition of the atmosphere Atmospheric stability- Temperature profile of the atmosphere - Lapse rates - Temperature inversion - effects of inversion on pollution dispersion.

UNIT III IMPACTS OF CLIMATE CHANGE

9

Causes of Climate change: Change of Temperature in the environment Melting of ice Pole-sea level rise-Impacts of Climate Change on various sectors Agriculture, Forestry and Ecosystem Water Resources Human Health Industry, Settlement and Society Methods and Scenarios Projected Impacts for Different Regions Uncertainties in the Projected Impacts of Climate Change Risk of Irreversible Changes.

UNIT IV CLIMATE CHANGES AND ITS CAUSES

9

Climate change and Carbon credits - CDM - Initiatives in India-Kyoto Protocol Intergovernmental Panel on Climate change - Climate Sensitivity and Feedbacks - The Montreal Protocol - UNFCCCIPCC - Evidences of Changes in Climate and Environment - on a Global Scale and in India.

UNIT V CLIMATE CHANGE AND MITIGATION MEASURES

9

Clean Development Mechanism -Carbon Trading -examples of future Clean Technology - Biodiesel - Natural Compost - Eco-Friendly Plastic - Alternate Energy -Hydrogen - Bio-fuels - Solar Energy - Wind - Hydroelectric Power -Mitigation Efforts in India and Adaptation funding Key Mitigation Technologies and Practices-Energy Supply - Transport - Buildings- Industry-Agriculture - Forestry - Carbon sequestration- Carbon capture and storage (CCS) - Municipal solid Waste (MSW) & Bio waste, Biomedical, Industrial waste International and Regional cooperation.

TOTAL: 45 PERIODS

COURSE OUTCOMES:

- CO1 Demonstrate an understanding of how the threats and opportunities of predicted climate changes will influence specific sectors at global and regional scale
- CO2 Identify the relationship between atmosphere and its components
- CO3 Analyze the impacts of climate change on environment parameters

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- CO4 Evaluate the scientific insights underlying the assessment reports of the IPCC, with a focus on impacts, adaptation and mitigation
- CO5 Critically evaluate the relative opportunities and needs for mitigation and adaptation (including vulnerability assessments) in a variety of sectoral contexts

TEXTBOOKS:

1. Sangam Shrestha, Mukand S. Babel and Vishnu Prasad Pandey, 2014, Climate Change and Water Resources, CRC Press an imprint of the Taylor & Francis Group.
2. Intergovernmental Panel on Climate Change: <https://www.ipcc.ch/>

REFERENCES:

1. Adaptation and mitigation of climate Scientific Technical Analysis, Cambridge University Press, Cambridge, 2006
2. Atmospheric Science, J.M. Wallace and P.V. Hobbs, Elsevier / Academic Press 2006
3. Jan C. van Dam, Impacts of Climate Change and Climate Variability on Hydrological Regimes, Cambridge University Press, 2003

AG25011	BLOCK CHAIN FOR AGRI SUPPLY CHAIN	L T P C
		3 0 0 3

COURSE OBJECTIVES:

- To understand blockchain technology and its applications in agriculture.
- To enhance traceability and transparency in the agri-supply chain.
- To explore smart contracts for secure agricultural trade.

UNIT I	INTRODUCTION TO BLOCKCHAIN TECHNOLOGY	9
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Basics of blockchain: Distributed ledgers, blocks, and consensus mechanisms. Types of blockchain: Public, private, and hybrid. Key features: Immutability, transparency, and security.

UNIT II	AGRI-SUPPLY CHAIN BASICS	9
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Overview of supply chain management in agriculture. Challenges: Fragmentation, inefficiencies, and lack of transparency. Role of blockchain in addressing these challenges.

UNIT III	SMART CONTRACTS IN AGRICULTURE	9
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Basics of smart contracts: Features and applications. Automating agricultural payments and trade using smart contracts. Case studies: Blockchain-enabled contract farming systems.

UNIT IV	TRACEABILITY AND SUSTAINABILITY	9
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Traceability: Ensuring product origin, quality, and safety. Blockchain for carbon footprint tracking in agriculture. Integration with IoT for real-time data capture.

UNIT V	CASE STUDIES AND FUTURE DIRECTIONS	9
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Successful blockchain implementations - Agri-commodity trading and traceability platforms. Challenges - Adoption barriers, scalability, and regulations. Future trends -Tokenized assets and decentralized finance (DeFi) in agriculture.

TOTAL: 45 PERIODS

COURSE OUTCOMES (COs)

- CO1: Understand blockchain technology and its core principles.
CO2: Apply blockchain solutions to agri-supply chain challenges.
CO3: Develop smart contracts for secure and efficient transactions.
CO4: Enhance traceability and sustainability in agricultural systems.
CO5: Analyze global use cases and explore future blockchain trends.

TEXT BOOKS

1. Mougayar, W. *The Business Blockchain: Promise, Practice, and Application of the Next Internet Technology*. Wiley, 2016.
2. Ge, L., et al. *Blockchain Applications in Food Supply Chain Management*. Springer, 2021.

AG25012	BIOELECTRONICS AND BIOSENSORS FOR AGRICULTURE	L T P C
		3 0 0 3

COURSE OBJECTIVES:

- To introduce the principles of bioelectronics and biosensors for agricultural applications.
- To design and develop biosensors for real-time monitoring of soil and crop health.
- To integrate biosensors with IoT for precision farming.

UNIT I	BASICS OF BIOELECTRONICS	9
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Fundamentals of bioelectronics - Definitions, components, and significance. Principles of bioelectric signals - Generation, detection, and amplification. Bioelectronic interfaces - Transducers, amplifiers, and signal processing.

UNIT II	BIOSENSOR TECHNOLOGIES	9
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Types of biosensors - Electrochemical, optical, piezoelectric, and thermal. Design principles - Selectivity, sensitivity, and stability. Applications - Soil nutrient detection, pesticide residue analysis, and water quality monitoring.

UNIT III	BIOSENSORS IN AGRICULTURE	9
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Soil and crop health monitoring using biosensors. Development of lab-on-a-chip devices for rapid testing. Role of biosensors in precision irrigation and fertilizer application.

UNIT IV	INTEGRATION WITH IoT	9
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Data acquisition systems for biosensor outputs. Integration with IoT for real-time decision-making.
Wireless communication protocols for biosensor networks.

UNIT V CASE STUDIES AND FUTURE DIRECTIONS

9

Advanced biosensors for pest and disease monitoring. Biosensors for greenhouse gas emission tracking in agriculture. Emerging trends: Nano-biosensors and bio-microelectromechanical systems (BioMEMS).

TOTAL: 45 PERIODS

COURSE OUTCOMES (COs)

CO1: Understand the principles of bioelectronics and biosensors.

CO2: Develop biosensors for monitoring soil and crop health.

CO3: Integrate biosensors with IoT for precision farming.

CO4: Analyze case studies of successful biosensor applications.

CO5: Explore future trends in agricultural bioelectronics.

TEXT BOOKS

1. Turner, A. P. F. *Biosensors: Fundamentals and Applications*. Oxford University Press, 2008.
- Rouf, M. A., et al. *Biosensors for Sustainable Food*. Springer, 2020.

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Program Name: B.Sc ORGANIC FARMING

Program Duration: 3years

Philosophy/Significance of the Course

1. Environmental Benefits: To reduce pollution to save nature.
2. Soil Health management: Improves soil fertility, Structure of soil and Biodiversity.
3. Climate change : To help Agriculture sector adapt climate changes..
4. Use certified organic seeds: Use organic seeds but not GMOs.
5. Use Natural and recycled materials: Use natural and recycled products..

Distinctiveness and uniqueness of the program

This course enables the students to understanding and application of organic farming in environment, human health, animal welfare. It helps to maintain environment health by reducing the level of pollution, green house gas emissions and soil erosion. The organic farming theory and practicals designing and evaluating policy solutions to a variety of environmental issue, soil health, climate change and to produce safe foods.

1. Climate Change Mitigation and Adaptation: Understanding economic implications and solutions.
2. Biodiversity Conservation: Analysing economic benefits and costs of preserving ecosystems.
3. Human Well-being and Environmental Quality: Exploring links between economic development and environmental health.
4. Sustainability and food safety: Sustain natural resource for future generation and reduce the risk of food contamination.

Course Description: This course explores the organic farming principles and policies related to environmental issues, Human and animal Health Welfare, Food safety and environmental sustainability

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REGULATION-2025

(Choice Based Credit System)

B.Sc –ORGANIC FARMING

PROGRAMME OUTCOMES (POs):

10+2 any group of the Programme B.Sc Organic farming will be able to

- PO1 :** To Understand organic farming theories and practicals.
- PO2 :** To Analyze Soil and water management.
- PO3 :** To Evaluate Biodynamic farming and Farm machinery.
- PO4 :** To Develop urban farming and terrace farming
- PO5 :** To Communicate food safety ideas effectively
- PO6 :** To Understand organic farming theories and practicals.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs):

10 +2 any group of the Programme B.Sc Organic farming will be able to

- PEO1** Perform exceptionally well in research and will be successful in the biofarming
- PEO2** Become consultants in crop cultivation environmental conditions with transgenic plants.
- PEO3** Become Entrepreneurial in Bio-business
- PEO4** Sustainability and food safety: Sustain natural resource for future generation and reduce the risk of food contamination.
- PEO5** Possess the leadership skills, established ethical norms and guidelines, excellence in competency, and lifelong learning required for a prosperous professional career.

PROGRAMME SPECIFIC OBJECTIVES (PSOs):

B.Sc Organic Farming program's specific goals are to create Graduates

- PSO1** Climate Change Mitigation and Adaptation: Understanding economic implications and solutions.
- PSO2** Biodiversity Conservation: Analysing economic benefits and costs of preserving ecosystems.
- PSO3** Human Well-being and Environmental Quality: Exploring links between economic development and environmental health.

Subjects / Papers to be included:

- Paper 1: Introductory to Organic farming
- Paper 2: Organic Agromicals
- Paper 3: Nursery Technology.
- Paper 4: Soil and water management
- Paper 5: Farm machinery
- Paper 6: Biopesticides.
- Paper 7: Biodynamic Farming.
- Paper 8: Residue management and Manure production.
- Paper 9: Bioinoculant technology
- Paper 10: Environmental studies.
- Paper 11: Plant Genetics and plant breeding.
- Paper 12: Urban farming and terrace gardening
- Paper 13: Mushroom cultivation.
- Paper 14: Value education.
- Paper 15: Vermicompost
- Paper 16: Genetic engineering and transgenic plants.
- Paper 17: Soil and Agricultural microbiology
- Paper 18: Sericulture
- Paper 19: Apiculture
- Paper 20: Bioremediation.
- Paper 21: Herbal technology
- Paper 22: Agroforestry
- Paper 24: Entrepreneurship and Bio-business
- Paper 25: Core Practical
- Paper 26: Internship
- Paper 27: Project work

Pedagogy of the course selected – Assessment methods

1. Lectures and discussions
2. Emphasize repetition
3. Field work – To Agricultural Land
4. Guest lectures from experts in Organic farming
5. Field trips to Natural Farming
6. Hand on Training in mushroom cultivation, Biopesticides and Biofertilizers production.
7. Online resources and multimedia materials
8. Collaborative learning and group projects
9. Reflective journaling and self-assessment
10. Flipped classroom approach

Assessment:

1. Assessments and exams (40%)
2. Group projects and presentations (30%)
3. Research paper (20%)
4. Class participation and engagement (10%)

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BACHELOR OF SCIENCE**

**B.Sc. ORGANIC
FARMINGDEGREECOURSE**

The Course of Study and the Scheme of Examinations

S.NO	PART	SUBJECT CODE	STUDYCOMPONENTS		INS.H RS/ WEEK	CREDI T	TITLEOFTHEPAPER	MAXIMUMMARKS		
			CourseTitle							
			SEMESTERI					CIA	Uni. Exam	Total
1.	I		Language	Paper-1	6	4	Tamil/OtherLanguages	25	75	100
2.	II		English	Paper-1	6	4	EnglishI	25	75	100
3.	III	BOF2501	CoreTheory	Paper-1	6	4	Introductory organic farming	25	75	100
4.	III	BOFP2501	CorePractical	Practical-1	3	2	Practical- I	25	75	100
5.	III	BOFA2501	Allied-1	Paper-1	4	3	Organic Agronomicals	25	75	100
	III	BOFAP2501	Allied-1	Practical-1	3	2	Allied practicals-I	25	75	100
7.	IV	BOFS2501	Skill	Paper-1	2	2	Nursery technology	25	75	100
			Sem.Total		30	21		175	525	700
			SEMESTERII					CIA	Uni. Exam	Total
8.	I		Language	Paper-2	6	4	Tamil/OtherLanguages	25	75	100
9.	II		English	Paper-2	6	4	EnglishII	25	75	100
10.	III	BOF2502	CoreTheory	Paper-2	6	4	Soil and water management	25	75	100
11.	III	BOFP2502	CorePractical	Practical-2	3	2	Practical- II	25	75	100
12.	III	BOFA2502	Allied-2	Paper-2	4	3	Farm Mechinary	25	75	100
13.	III	BOFAP2502	Allied Practical -2	Practical-2	3	2	Allied practicals -II	25	75	100
14.	IV	BOFS2502	Skill	Paper-2	2	2	Biopesticides	25	75	100
			Sem.Total		30	21		175	525	700
			SEMESTERIII					CIA	Uni. Exam	Total
15.	I		Language	Paper-3	6	4	Tamil/OtherLanguages	25	75	100
16.	II		English	Paper-3	6	4	English III	25	75	100
17.	III	BOF2503	CoreTheory	Paper-3	4	4	Biodynamic Farming	25	75	100
18.	III	BOFP2503	CorePractical	Practical-3	3	2	Practical-III	25	75	100

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19.	III	BOFA2503	Allied-3	Paper-3	4	3	Residue Management and Manure Production	25	75	100
20.	III	BOFAP2503	Allied Practical-3	Practical-3	3	2	Allied practicals - III	25	75	100
21.	IV	BOFS2503	SkillBased Subject	Paper-3	2	2	Bioinoculant technology	25	75	100
22.	IV	BES2503	EVS	Paper-1	2	2	EVS	25	75	100
			Sem.Total		30	23		200	600	800
			SEMESTER IV							
23.	I		Language	Paper-4	6	4	Tamil/Other Languages	25	75	100
24.	II		English	Paper-4	6	4	English –IV	25	75	100
25.	III	BOF2504	CoreTheory	Paper-4	4	4	Plant Genetics and Plant Breeding	25	75	100
26.	III	BOFP2504	CorePractical	Practical-4	3	2	Practical- IV	25	75	100
27.	III	BOFA2504	Allied-4	Paper-4	4	3	Urban Farming and Terrace Gardening	25	75	100
28.	III	BOFAP2504	Allied Practical-4	Practical-2	3	2	Allied Practicals- IV	25	75	100
29.	IV	BOFS2504	SkillBased Subject	Paper-4	2	2	Mushroom Cultivation	25	75	100
30.	IV	BVE04	Value Education	Paper-2	2	2	Value Education	25	75	100
			Sem.Total		30	23		200	600	800
			SEMESTER V							
31.	III	BOF2505	CoreTheory	Paper-5	6	4	Vermicompost	25	75	100
32.	III	BOF2506	CoreTheory	Paper-6	6	4	Genetic engineering and Transgenic plants	25	75	100
33.	III	BOF2507	CoreTheory	Paper-7	5	4	Soil and Agricultural Microbiology	25	75	100
34.	III	BOFP2505	CorePractical	Practical-5	4	2	Practical -5	25	75	100
35.	III	BOFP2506	CorePractical	Practical-6	4	2	Internship	25	75	100
36.	III	BOFE2501	InternalElective	Paper-1	3	3	Sericulture	25	75	100
37.	IV	BOFS2505	SkillBasedSubject	Paper-5	2	2	Apiculture	25	75	100
			Sem.Total		30	21		175	525	700
			SEMESTER VI							
38.	III	BOF2508	CoreTheory	Paper-8	7	4	Bioremediation	25	75	100
39.	III	BOF2509	CoreTheory	Paper-9	7	4	Herbal technology	25	75	100
40.	III	BOFP2507	CorePractical	Practical-7	4	2	Practical-6	25	75	100
41.	III	BOFP2508	CompulsoryProject	Paper-10	5	5	Group/IndividualProject	25	75	100
42.	III	BOFE2502	InternalElective	Paper-2	4	3	Agroforestry	25	75	100
43.		BOFS2506	SkillBasedSubject	Paper -6		2	Entrepreneurship and Bio-business	25	75	100

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			Sem.Total		30	20		150	600	900
										4300

Part	Subject	Papers	Credits	Total credits	Marks	Total Marks
Part I	Languages	4	4	16	100	400
Part II	English	4	4	16	100	400
Part III	Allied (odd semester)	2	3	6	100	200
	Allied (even semester)	2	3	6	100	200
	Allied practicals	4	2	8	100	400
	Elective	2	3	6	100	200
	Core Paper	9	4	36	100	900
	Core Practical	7	2	14	100	700
	Compulsory Project Group/ Individual Project	1	5	5	100	100
Part IV	EVS	1	2	2	100	100
	Skill Based	6	2	12	100	600
	Value Education	1	2	2	100	100
	TOTAL	43	36	129	1200	4300

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B. SC. DEGREE IN ORGANIC FARMING

SEMESTER I

CORE PAPER 1: INTRODUCTORY ORGANIC FARMING

Course Objectives:

To learn about basic concept and principles of Organic farming.

Course Out Comes:

CO1: To provide knowledge about basic concepts and principles related to organic farming.

CO2: To study the scope, applications and needs of organic farming.

CO3: To develop concepts of conventional farming and organic farming.

CO4: To help students understand various organic farming practices in the national and international level.

CO5: To know about Certification and Accreditation process

UNIT - I

History and development, IFOAM Organic farming – definition – need – scope – principles characteristics - relevance to modern agriculture Principle- health, fairness, ecology and care Methods, advantages and limitations. Different eco friendly farming systems- biological farming, natural farming, regenerative agriculture – permaculture - biodynamic farming. Relevance of organic farming to A.P, India, and global agriculture. Need of Organic farming in present context and future prospects- advantages - barriers.

UNIT - II

Initiatives taken by the central and state governments, NGOs and other organizations for promotion of organic agriculture in India. Organic nutrient sources and their fortification organic manures- methods of composting Green manures- bio fertilisers – types, methods of application – benefits and limitations.

UNIT - III

Nutrient use in organic farming-scope and limitations. Nutrient management in organic farming. Organic ecosystem and their concepts. Choice of crops and varieties in organic farming – crop rotations – need and benefits – multiple cropping.

UNIT - IV

Fundamentals of insect, disease and weed management under organic mode of production cultural-biological methods-non chemical pest & disease management. Botanicals- pyrethrum,

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neem seed kernel extract, neem seed powder, soluble neem formulations, neem oil.
Operational structure of NPOP – other agencies for organic production.

UNIT - V

Inspection – certification - labelling and accreditation procedures for organic products.
Processing, - economic consideration and viability. Marketing and export potential of organic products – national economy.

Text Books

1. Arun K. Sharma. 2002. A Hand book of organic farming. Agrobios, India. 627p.
2. Palaniappan, S.P and Annadurai, K.1999. Organic farming-Theory and Practice. Scientific publishers, Jodhpur,India. 257p.
3. Mukund Joshi and Prabhakarasetty, T.K. 2006. Sustainability through organic farming. Kalyani publishers, New Delhi. 349p.
4. Balasubramanian, R., Balakishnan, K and Siva Subramanian, K. 2013. Principles and practices of organic farming. Satish Serial Publishing House. 453p

References Books

- 1.Tarafdar, J.C., Tripathi, K.P and Mahesh Kumar, 2009. Organic agriculture. Scientific Publishers, India. 369p.
2. Tiwari, V.N., Gupta, D.K., Maloo, S.R and Somani, L.L. 2010. Natural, organic, biological, ecological and biodynamic farming. Agrotech Publishing Academy, Udaipur. 420p.
3. Dushyent Gehlot. 2005. Organic farming- standards, accreditation, certification and inspection. Agrobios, India. 357p

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	M	S	M	S	S	M	S	M	S
CO2	M	S	M	S	S	S	M	S	S	M
CO3	S	M	S	M	S	S	S	S	M	S
CO4	S	S	S	S	M	S	M	M	S	M
C05	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

CORE PRACTICAL -I

INTRODUCTORY ORGANIC FARMING

Course Objectives:

To develop experimental knowledge about organic farming.

Course Out Comes:

CO1: The course will enrich the basic concept and principles related to organic farming.

CO2: The course makes the student to understand different concepts of conventional and organic farming.

CO3: Development of theoretical concept through field experiments and field visit.

CO4: The student will learn about real life issue related to farming through mini visit of organic farms and organic farm industry visit.

CO5: Details about post harvest management systems.

LIST OF EXPERIMENTS

1. Visit to organic farm to study the various components, identification and utilisation of organic products.
2. Compost making- aerobic and anaerobic methods
3. Vermicompost preparation
4. Preparation of enriched farm yard manure
5. Visit to organic clusters and bio control lab to study the maintenance of bio- fertilizers/bio-inoculant cultures
6. Biological nitrogen fixers.

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7. Methods of application of Bio-pesticides (Trichocards, BT, NPV)
8. Preparation of neem products and other botanicals for pest and disease control
9. Preparation of green pesticides (panchagavya, beezamrutam, jeevamrutam, ghanajeevamrutam, dravajeevamrutam).
10. Different methods of biofertiliser applications.
11. Quality analysis of biofertilisers/bioinoculants and compost
12. Case studies of Indigenous Technical knowledge e (ITK) for nutrient , insect, pest, disease and weed management
13. Economic analysis of organic production system
14. Study of post harvest management in organic farming
15. Study of quality parameters of organic produce
16. Visit to organic farms to study the various components and their utilization

Text Books:

1. Veeresh G. K. (2011). Organic Farming. Publisher: Foundation Books. ISBN: 9788175968813
<https://doi.org/10.1017/UPO9788175968813>
2. Reddy S. R. (2017). Principles of Organic Farming. Publisher: Kalayani, ISBN :9327274474.

References Books

1. Rateaver B. (1993). Organic method primer update: A practical explanation: the how and why for the beginner and the experience (Conservation gardening and farming). Publisher: The Rateavers; Special edition, ISBN: 0915966018.
2. Gershuny G. and Martin D. L. (2018). The Rodale Book of Composting, Newly Revised and Updated: Simple Methods to Improve Your Soil, Recycle Waste, Grow Healthier Plants, and Create an Earth-Friendly Garden (Rodale Classics). Publisher: Rodale Books; Updated edition, ISBN: 1635651026.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	S	S	S	M	S	S	M	S	M
CO2	M	S	S	M	S	S	M	S	S	S
CO3	S	M	S	S	S	S	M	S	S	S

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CO4	S	S	M	S	S	M	S	S	M	S
CO5	S	M	S	S	M	S	S	M	S	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

ALLIED PAPER-01

ORGANIC AGRONOMICALS

Course Objectives:

To help students to understand the concept of Agronomicals and produce crops National and International level.

Course Out Comes

CO1: To provide knowledge about basic concepts and principles related to agriculture and crop production.

CO2: To develop concepts of agronomic classification and classification of crop plants.To provide the knowledge of crop and crop production method.

CO3: To help students to understand various crop production practices in the national and international level.

CO4: Learning Outcomes :On successful completion of this course, the students will be able to :

- Develop critical understanding on various aspects of agronomy.
- Explain the nutrition and application of nutrients to plants.

CO5: Explain the cropping methods and crop rotation.Realize various weed management practices. Explain the different aspects of crop harvesting.

UNIT – I

Basics of Agronomy:Agronomy- definition, scope; role of Agronomist andrelationship of Agronomy with other sciences. Tillage - definition, objects of tillage, types of tillage, tillage implements and factors affecting tillage, Effect of tillage on soil andcrop growth. Tilth- definition, characteristics and ideal tilth; modern concepts of tillage, minimum, zero and stubble mulch tillage, importance ofpuddling. Seed - definition, characteristics of quality seed, seed treatment andits objectives; methods of sowing seed and sowing implements.

UNIT – II

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Crop communities: Effect of plant population on growth and yield, planting geometry viz., solid, paired and skipped rowplanting. Role of plant nutrients in crop production, Importance of manures and fertilizers and its classification. Methods and time of application of manures, fertilizers and green manuring. Nutrient use efficiency, meaning and factors affecting nutrient use efficiency.

UNIT – III

Growth and development :Growth and development- definition, growth curve and factors affecting growth and development. Plant ideotypes - definition and types of ideotypes. Crop rotation, its definition, principles and advantages of crop rotation. Study of crop adaptation and its distribution.

UNIT-IV

Weed management: Weeds, its definition, characteristics of weeds, merits and demerits of weeds. Classification of weeds, meaning of crop weed competition and its period in different crops. Principles and methods of weed management viz., cultural, mechanical, chemical, biological weed control methods and integrated weed management. Classification of herbicides, its selectivity and resistance, Allelopathic effect of weed.

UNIT-V

Crop harvesting: Crop harvesting, signs of maturity in different field crops; physiological and crop maturity. Methods of threshing crops, cleaning, drying and storage of field crops. Rainfed agriculture: Introduction, types, History of rainfed agriculture and watershed in India. Problems and prospects of rainfed agriculture in India; Soil and climatic conditions prevalent in rainfed areas.

Text Books :

1. Gopal Chandra De. 1980., Fundamentals of Agronomy. Oxford and IBH Publishing Co. Ltd., Bangalore.
2. Panda, S.C., 2006. Agronomy, Agribios Publication, New Delhi.
3. Reddy, S.R. 2011. Principles of Agronomy Kalyani Publishers, Ludhiana, India.

References Books:

1. Sankaran, S. and V.T. Subbiah Mudliyar, 1991. Principles of Agronomy. The Bangalore Printing and Publishing Co. Ltd., Bangalore.
2. Rao V.S., 2006. Principles of Weed Science. Oxford and IBH Publishing Co., New Delhi, India.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
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CO1	M	S	S	S	M	S	M	S	M	S
CO2	S	S	M	S	S	M	S	S	S	S
CO3	S	M	S	S	S	S	M	S	M	S
CO4	S	S	S	M	S	M	S	S	S	M
CO5	S	M	S	S	S	S	M	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

ALLIED PRACTICALS-I

ORGANIC AGRONOMY

Course Objectives:

The course will enrich knowledge about basic concepts and principles related to agriculture and crop production.

Course Out Comes

CO1: The course will enrich knowledge about basic concepts and principles related to agriculture and crop production.

CO2: The course makes the student to understand different concepts of agronomic classification and classification of crop plants.

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CO3: Development of practical experience through field experiments and field visit.

CO4: The student will learn about Practical approaches of agricultural practices.

CO5: To know about application methods of Herbicides.

LIST OF EXPERIMENTS

1. Identification of seeds and crop plants at different growth stages.
2. Identification of different tillage implements.
3. Identification of fertilizers and pesticides.
4. Identification of weed flora in different field crops.
5. Study of agro climatic zones of Andhra Pradesh and India.
6. Operational tillage viz., primary, secondary, inter-tillage, sowing, harvesting, harvesting implements, Working with them.
7. Calculation of plant population, seed rate, fertilizer and herbicide dose for different field crops.
8. Determination of purity and germination percentage of seed, Methods of seed germination.
9. Study of viability test and practice of seed treatments in different field crops.
10. Study of yield contributing characters and yield estimation in different field crops.
11. Methods of application of herbicides in different field crops.
12. Measurement of air and soil temperatures, tabulation and variation extended.

References Books:

1. Reddy S.R. (2014). Principles of Crop Production. Kalyani Publishers, ISBN-10: 9327218582.
2. An Introduction to Agriculture and Agronomy. 2015.
<http://www.newagepublishers.com/samplechapter/001757>
3. Onwueme I. C. and Sinha, T. D. (1999). Field Crop Production in Tropical Africa. Netherlands: CTA, Wageningen, Pp. 1-14.
4. Bassey E. (2019). Fundamental Principles of Crop Production. ISBN; 9781706242604, 1706242603. Publisher: Independently Published
5. Jena, J. and Jena, T. (2020). Glimpses of Crop Production. Publisher: Jain Brothers ISBN: 9788194484646

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Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	M	S	S	S	M	S	S	S	M
CO2	S	S	S	M	S	S	M	S	S	S
CO3	M	S	M	S	M	S	S	M	S	S
CO4	S	M	S	S	M	S	M	S	S	M
CO5	S	M	S	S	S	S	M	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

SKILL BASED SUBJECT 1 - NURSERY TECHNOLOGY

Course Objectives:

The course makes the student to understand the method of preparation and management of a Nursery.

Course Outcome:

CO1: The course will provide knowledge about basic concepts and principles about Nursery technology.

CO2: The course makes the student to understand the method of preparation and management of a Nursery.

CO3: Development of practical experience of disease and pest control through field experiments and nursery visit.

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CO4: The student will learn about real life issue related to nursery management and marketing.

CO5: Details of Micropropagation.

UNIT-I

Nursery: definition, objectives and scope and building up of infrastructure for nursery. Seed: structure and types- seed dormancy; causes and methods of breaking dormancy-seed storage: seed banks, factors affecting seed viability, sowing/ raising of seeds and seedling-transplanting of seedlings.

UNIT-II

Vegetative propagation: air-layering/soil layering, cutting, selection of cutting, selection of stock plant, collecting season, treatment of cutting, rooting medium and planting of cuttings. Storage and marketing procedures. Gardening: definition, objectives and scope-different types of gardening- landscape and home gardening-parks and its components-plant material and designs. Manuring, watering, management of pests and diseases and harvesting.

UNIT-III

Nursery care management: Planning and layout of nurseries - Media for propagating and growing nursery plants - Propagation structures and equipments - Sanitation Care and handling of nursery plants - nutrition management for nursery plants - Pest and disease management of nursery plants.

UNIT-IV

Sexual propagation and Asexual propagation: Introduction - Advantages and limitations of Sexual propagation and Asexual propagation. Raising nursery for seedlings-techniques, portray nursery. Propagation by specialized vegetative structures: Bulbs - tubers - Tuberous roots and stems - Corms - Rhizomes - Runners - Offsets - Suckers.

UNIT-V

Propagation on its own root systemsCuttings : Types of cuttings - use of plant growth regulators in rooting Layering. Types of layering.Propagation on the root system of other plants:Grafting : Methods of grafting. Budding Methods of budding union and formation.Bud certification programme. Factors affecting graft. Methods of culturing plant tissues and organsMicro propagation: Requirements Limitations. Advantages and Limitations.

Text Books:

1. Bose T.K.&Mukherjee,D.1972.GardeningInIndia,Oxford&IBHPublishingCo.,NewDelhi.
- Sandhu,M.K.1989.PlantPropagation,WileEasternLtd.,Bangalore,Madras.
2. EdmondMusser&andres,FundamentalsofHorticulture,McGrawHillBookCo.,NewDelhi.
3. Agrwal,P.K.1993.HandBookOfseedTechnology,Dept,OfAgricultureand

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Cooperation, National Seed Corporation Ltd., New Delhi.

4. Janick Jules. 1979. Horticultural Science (3rd Ed), W.H. Freeman and Co. San Francisco, USA

References Books:

1. Adriance, W. and R. Brison. 1979. Propagation of Horticultural plants, Tata McGraw Hill Publishing Com. Pvt. Ltd., New Delhi.
2. Hartmann, H.T. and D.E. Kester. 1975. Plant Propagation, Englewood cliffs, New Jersey, Prentice Hall.
3. Jitendra Singh 2004, Basic Horticulture, Kalyan Publications, New Delhi.
4. N. Kumar. 1993. Introduction to Horticulture, Rajalakshmi Publications, Nagercoil. 5. Sadhu, M. 1991. Plant Propagation.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	M	S	M	S	S	M	S	M	S
CO2	S	S	S	S	M	S	S	S	S	S
CO3	S	M	M	S	S	M	S	M	S	S
CO4	M	S	S	S	M	S	M	S	S	M
CO5	S	M	S	S	S	S	M	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

SEMESTER - II

CORE PAPER-2

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SOIL AND WATER MANAGEMENT

Course objective

- To discuss the distribution and association of microorganism in various ecosystems and to know about the role of microorganism in water pollution and water quality.
- To acquire knowledge about the role of microorganism in water pollution and water quality
- To get knowledge about soil fertility and nutrition level of soil
- The objective of this course is to impart the knowledge of hydrology that deals with the occurrence, distribution, movement and properties of water on the earth.

Course out Comes

CO1: Describe about the structure and function of soil and understand the role of microbes in various environments.

CO2: Details about the soil nutrition.

CO3: Assessment of water quality management

CO4: Details about soil fertility.

CO5: Water quality assessment.

UNIT-I

Soil: Pedological and Edaphological concept and components of soil. Important soil forming minerals and rocks, weathering of rocks and minerals. Soil forming factors and processes.

Development of soil profile. Physical properties of soil and their significance. Chemical properties of soil, cation and anion exchange phenomenon and their importance in agriculture. Soil biota, Soil microbial ecology, types of organisms in different soils; Soil microbial biomass; Microbial interactions.

UNIT-II

Soil fertility: Definition concept, and factors influencing soil fertility. Essential plant nutrients: Criteria of essentiality, forms of nutrients in soil, functions, deficiency Symptoms, Critical levels of deficiency and toxicity. Soil fertility evaluation, soil and plant analysis, tissue tests. Soil testing: Soil test in crop response and correlation, fertilizer recommendation based on soil test.

UNIT-III

Mechanism of nutrient: transport to plants and factors influencing nutrient availability to plants. Mineralization and immobilization of Nitrogen, fixation and availability of N, P and K. **Fertilizers** - definition, classification, characteristics, reactions of fertilizer in soil and fertilizers use efficiency. Organic manures: Preparation, properties and use in crop production, nutrient enriched compost, vermi-compost and green manure.

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UNIT-IV

Properties of water :Origin of water on earth, Unique properties of water (Polarity, Cohesion, Density, Surface Tension, Viscosity, Heat capacity, Boiling and freezing points, Temperature, Taste, Odour, Colour), Importance of water.

UNIT V

water quality parameters: Water resources distribution Water as a resource, Concept of valuing water, Types of water resources, Inland water distribution and importance, Ground water distribution and importance, Major cations (Ca, Mg, Na, K), Major Anions (bicarbonates, sulphates, chlorides), Dissolved Gases in water (DO, CO₂), Biochemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD), Microbial water quality-coliform bacteria, Indicator organisms.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	S	M	S	S	M	S	S	M	S
CO2	S	M	S	M	S	S	S	M	S	S
CO3	S	S	M	S	M	S	M	S	S	M
CO4	S	S	S	M	S	S	S	M	S	S
CO5	S	M	S	S	S	S	M	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

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CORE PRACTICAL-II

Course Objectives:

To learn about experimental Knowledge about soil and analyses soil and water quality.

Course Out Comes:

CO1: Determination of soil components.

CO2: Identification of green manure.

CO3: Analyses water quality.

CO4: known about surrounding eco systems.

CO5: visit to water plant.

1. Determination of soil pH
2. Determination of organic carbon, nitrate, phosphate and potassium by the use of rapid soil test kit .
3. Identification of green manure crops
4. Calculation of soil porosity
5. Chemical analysis of soil for total and available nutrients
6. Determination of soil moisture by gravimetric method
7. Determination of water quality on basis of odor and color
8. Perception of stakeholders regarding drinking water quality available in the institution/College
9. Estimation of water temperature of different ecosystems
10. Visit to a any nearby drinking water supply scheme/source

Text Books

1. Murthy, J.V.S.(1994). Watershed management in India. Wiley eastern Publishers, New Delhi.
2. Gupta, U.S. (1975). Physiological Aspect of Dry land Farming. Oxford & IBH Publishing Co., New Delhi.
3. Sankara Reddi, G.H. and Yellamanda Reddy, T. (1996). Efficient use of Irrigation water. Kalyani Publishers, Ludhiana.
4. Misra, R.D. and Ahmed, M. (1987). Manual on Irrigation Agronomy. Oxford & IBH Publishing Co., New Delhi.
5. Brady, N.C. (1995). The nature and Properties of soil. MacMillan Publishing Company, New York. Sahai, V.N. (1990). Fundamentals of Soil. Kalyani Publishers, Ludhiana.
6. Tisdale, S.L., Nelson, W.L. and Beaton, J.D. (1993). Soil Fertility and Fertilizers. MacMillan Publishing Company, New York.

References ooks

1. .Sivanappan Rk Padma Kumari O and Kumar, V 1987. Drip Irrigation— .Keerthi Publishing House Pvt., Ltd., Coimbatore. Bansil, P.C. 2004. Water Management in India. Concept Publishing Company, India

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2. Brebbia, C.A. 2013. Water Resources Management VII. WIT Press.
3. Mays, L.W. 2006. Water Resources Sustainability. The McGraw-Hill Publications.
4. 4..Schward and Zhang, 2003. Fundamentals of Groundwater. John Willey and Sons.
5. Vickers, A. 2001. Handbook of Water Use and Conservation. Water Plow Press.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	M	S	M	S	S	M	S	S	S
CO2	M	S	M	S	S	S	M	S	M	S
CO3	S	M	S	M	S	M	S	S	S	M
CO4	S	S	S	S	M	S	S	M	S	S
CO5	S	M	S	S	S	S	M	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

ALLIED PAPER-2

FARM MACHINERY

Course Objectives

To understand how principles of management are applied to farm machinery systems
To make them more effective and profitable.

Course Out Comes

CO1:The student will be able to understand how farm machinery is selected and operated to make them economically viable.

CO2:The student will be able to understand how farm machinery is selected and operate

CO3: To learn about selection of machines and powers.

CO4: Details of Network analysis.

CO5: to know about the cost of machine

UNIT- I

Importance and objectives of farm mechanization in Indian agriculture, its impact, strategies, myths and future needs. Estimation of operating cost of tractors and farm machinery. Management and performance of power, operator, labour. Economic performance of machinery, field capacity, field efficiency and factors affecting field efficiency.

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UNIT-II

Tractor power performance in terms of PTO, drawbar and fuel consumption. Power requirement problems to PTO, DBHP.

UNIT-III

Selection of farm machinery, size selection, and timeliness of operation, optimum width and problem related to its power selection. Reliability of agricultural machinery. Replacement of farm machinery and inventory control of spare parts.

UNIT- IV

Systems approach to farm machinery management and application of programming techniques to farm machinery selection and scheduling. Network Analysis: Transportation, CPM and PERT, dynamic programming, Markov chain.

UNIT-V

Equipment for land development and soil conservation– Cost of operation of farm machinery – Tractor and implement selection.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	M	S	S	M	S	S	M	S	S
CO2	M	S	M	S	S	S	M	S	M	S
CO3	S	M	S	M	S	M	S	S	S	M
CO4	S	S	S	S	M	S	S	M	S	S
C05	S	M	S	S	S	S	M	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

ALLIED PRACTICAL –II

Course Objectives:

To learn experimental Knowledge about farm machinery and tools used in Farms.

Course Out Comes:

CO1: To study different farm operations.

CO2: Details of mechanical weed control.

CO3: Analyses subsoiler.

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CO4: known about seeding.

CO5: calibrate drills and planters.

1. Study of Different Farm Operations and Familiarization with Farm Machines and Equipment
2. Study on Animal and Tractor Drawn Mould Board Plough
3. Study on Disk Plough
4. Study on Animal and Tractor Drawn Disk Harrows
5. Study on Rotary Tillage Tools
6. Study on Sub-soiler and Chisel Plough
7. Study on Mechanical Weed Control Machines and Equipments
8. Study on Seeding and Planting Machines
9. Calibration of Seed Drills and Planters
10. Study on Paddy Trans-planters
11. Study on Mowers and Reapers
12. Study on Mechanical Threshers
13. Study on Combined Harvesters
14. Study on Plant Protection Machines and Equipments
15. Measurement of Power Requirement of Farm Machines and Equipments

References Books

1. Car Culpin C. 1996. Profitable Farm Mechanization. Lock Wood and Sons, London.veille LA. 1980. Selecting Farm Machinery. Louisiana Cooperative Extn. Services Publication.
2. FAO. 1990. Agricultural Engineering in Development: Selection of Mechanization Inputs.
3. FAO, Agri service Bulletin
4. Hunt D. 1979. Farm Power and Machinery Management. Iowa State University Press, USA
5. Kapoor VK. 2012. Operation Research: Concepts, Problems and Solutions. Sultan Chand and Sons, India.
6. Singh S and Verma SR. Farm Machinery Maintenance and Management. DIPA, IC KAB-I, New Delhi.
7. Senthilkumar, T., R. Kavitha and V.M.Duraisamy 2015. A TEXT BOOK OF FARM MACHINERY, ThannambikkaiPublicaons, Coimbatore. ISBN: 978-9381102305
8. Jagadishwar Sahay, 2010. ELEMENTS OF AGRICULTURAL ENGINEERING. Standard Publishers Distributors, New Delhi. ISBN: 978 – 818040440.
9. Ojha, T.P and A.M.Michael 2005. PRINCIPLES OF AGRICUTURAL ENGINEERING VOL I. Jain Brothers, New Delhi. ISBN: 978-8186321638

Mapping with Programme Outcomes

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COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	M	S	M	S	M	S	S	S	M
CO2	S	S	S	S	M	S	S	M	S	S
CO3	S	M	S	S	S	S	M	S	M	S
CO4	M	S	M	S	M	S	S	M	S	S
CO5	S	S	S	M	S	M	S	S	S	M

- PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

SILL BASED SUBJECT-2

BIOPESTICIDES

Course Objective

1. To learn about the importance of Biopesticides.
2. To provide knowledge of Mass production technology of bio-pesticides.
- 3.To gain knowledge of pest management.

Course Outcomes:

- CO1: Quality control of biopesticides..
- CO2: Management of pesticides in environment.
- CO3: Safety and protective measurement of biopesticides.
- CO4: To study applications of pesticides.
- CO5: Care and precautions handling the pesticides.

UNIT-1

Pesticides and their history, Classification of Pesticides. (insecticides,herbicides,rodenticides,bactericides,fungicides and larvicides)Biodegeradable and non-biodegeradablepesticides,effects and benefits of pesticides.

UNIT-2

Pests,their types,harmful and beneficial pests, Identification of pests,natural pests killers,household remedy for pests removal.Impact of pesticides on physiological and pharmacological system of human body.effect on liver and kidney,immunotoxic effects of pesticides.effect on endocrine system.

UNIT-3

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Integrated pest management, biological pest control, natural methods, pest predators, importance of pest management. Advantages over chemical pesticides, availability of biopesticides in the market, bioavailability and environmental safety.

UNIT-4

Mixing, loading and application of pesticides in house, veterinary practices and agriculture crops, preparation of diluted solution and their quantity, spray methods.

UNIT-5

Storage of pesticides, care and precautions during handling of pesticides, safety and protective measures during applications

Text Books

1. R.P. Soundarajan pesticides. Advance in chemical and botanical pesticides. Published by in Tech Janeza Trdine
2. Insecticides and pesticides Techniques for crop protection .Drew Adams, 2017.
3. Pesticides and insecticides Development and use ,Bobby Jones, 2018
4. Principles of Crop production, by Reddy SR, Kalyani publications.
5. Modern techniques of raising field crops Chhidda Singh, Prem Singh and Rajbir Singh
6. Crops of India N.R. Das
7. Principles Of Crop Production, by S.R REDDY, C NAGAMANI, Kalyani Publications.

Reference books:

1. A Manual on Crop Production Technology (Kharif and Kharif), Lokesh Kumar Jain
2. Agronomy of field crops S.R. Reddy.
3. Crop Production Technology I & II – Kharif and Rabi Crops – As per 5th Deans Committee Recommendations, B. S. Lalitha, N. Mavarkar, B. R. Premalatha, 2020

Mapping with Programme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	M	S	M	S	M	S	S	S	M
CO2	S	S	S	S	M	S	S	M	S	S
CO3	S	M	S	S	S	S	M	S	M	S

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CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

**SEMESTER –III
CORE PAPER- 3**

BIODYNAMIC FARMING

Course Objectives:

To study about conversion of farm into Biodynamic Farm and identify the insect pests.

Course Out comes:

CO1:To study crop rotation

CO2:Identify insect pests.

CO3: Converting a farm to Biodynamic.

CO4: To know about modern concept and approaches.

CO5:Different Management Strategies

UNIT I

Introduction , history, principles and advantages, biodynamic preparation: crop rotation , physical chemical and biological factors .Soil and climate influence on crop growth.Concept of soil plant realations.Use of plant growth hormones and growth regulators for better adaptation in stressed condition.

UNIT II

Weeds pest and diseases. Identify the diseases and the insect pests of important field crops. Identify the diseases of important horticultural crops. Select the insect pests of important horticultural crops. Recognize the different symptoms of important fungal diseases. Identify the different symptoms of important bacterial diseases. Identify the different symptoms of important viral diseases. Plan, select and execute different management strategies against seed borne diseases.

UNIT III

Converting a farm to biodynamic , cow horn manure and cow horn silica :preparation storage and application ,preparation storage and application of jivaamrit , bijaamrit, plant based preparation , panchgayapreparation and application, Dasakavya preparation , storage and application .

UNIT IV

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Modern concepts and approaches. Crops models – definitions, concepts and types (empirical mechanic) model application. Precision agriculture - definition , base concepts , scope and approach. Technologies for precision agriculture-GIS GPS , sensor and remote sensing . Modern crop production and approaches.

UNIT V

MANAGEMENT STRATEGIES. Plan, Select And Execute Different Management Strategies Against Soil Borne Diseases. Describe Different Management Strategies Against Air Borne Diseases. Different Management Strategies Against Sucking Insect-Pests. Different Management Strategies Against Lepidopteron Insect-Pests. Recognize The Probable Reasons Of Atypical Symptoms (Wilting / Leaf Curling /Yellowing Of Leaves Etc.) Observed In Plants. Demonstrate The Difference Between Fungal And Bacterial Wilt. Differentiate Between Viral And Nutrient Deficiency Symptoms. 4 Explain Various Fertilizer Application Techniques For Nutrient Deficiencies After Soil Testing And Analysis

Text Books

1. Selvaraj, N, AnitaB, Anusha B and sarawathi G. M , Organic horticulture creating a more sustainable farming, Horticultural Research station, Udthagamandalam.
2. Rudolf (2004) what is Biodynamic? Publisher steinerbooks, 2004 ISBN:0880109890
3. Waldin M (2015) . Biodynamic gardening ,Publisher:Dorling Kindersley Ltd,

Masson V., MassonP, and Blais (2014) A biodynamic Manual Practical Instruction for farmers and gardeners;

Mapping with Programme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	M	S	M	S	M	S	S	S	M
CO2	S	S	S	S	M	S	S	M	S	S
CO3	S	M	S	S	S	S	M	S	M	S
CO4	M	S	M	S	M	S	S	M	S	S
CO5	S	S	S	M	S	M	S	S	S	M

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

CORE PRACTICALS-3

BIODYNAMIC FARMING

Course Objectives:

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To learn about experimental Knowledge to create Biodynamic farming.

Course Out comes

CO1: To prepare manure waste

CO2: Categorization of compost.

CO3: To produce compost.

CO4: Evaluation of compost

CO5: Storage and application.

1. Preparation of cow horn manure.
2. Preparation of cow horn silica.
3. Compost preparation through plant materials
4. Poison preparation,
5. storage and application of compost
6. Dasagavya preparation storage and application.
7. Bijiaamrit and jivaamrit preparation storage and application methods .

References Books

1. Organic Farming: Theory and Practices by Palanippan, S.P. and Anaadurai,
2. Recent Developments in Organic farming by Gulati and Barik.
3. Hand book of Agriculture, ICAR New Delhi
4. Balasubramaniyan P & Palaniappan S.P. 2001. Principles and Practices of Agronomy. Agrobios. Fageria N.K. 1992. Maximizing Crop Yields. Marcel Dekker.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	S	M	S	S	M	S	S
CO2	S	M	S	M	S	M	S	S	S	M
CO3	S	S	S	S	M	S	S	M	S	S
CO4	S	M	S	S	S	S	M	S	M	S
CO5	M	S	M	S	M	S	S	M	S	S

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PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

ALLIED PAPER-3

RESIDUE MANAGEMENT PRACTICES& MANURE PRODUCTION

Course Objectives:

To study about Residue management and manure production.

Course Out comes:

CO1: To study crop rotation

CO2: Identify insect pests.

CO3: Converting a farm to Biodynamic.

CO4: To know about modern concept and approaches.

CO5: Different Management Strategies

UNIT I

Introduction sources, compostion of solid waste , Economic consideration ;wastes as a wealth and source of income , planning system , valorization of organic solid waste , recycling of organic waste .

UNIT II

Animal feed , composting , anaerobic digestion , rendering, rapid thermophilic digestion, immobilized enzyme reaction, process, sanitary land filling. Energy recovery

UNIT IV

Commercial fertilizers;composition,relative fertilizervalueandcost;cropresponseto different nutrients, residual effects and fertilizer use efficiency; agronomic, chemical and physiological, fertilizer mixtures and grades; nano-fertilizer materials and application; methods of increasing fertilizer use efficiency; nutrient interactions; precision nutrient management; Forms of fertilizers (Conventional and Water soluble fertilizers), Nano fertilizers, Customized slow fertilizers.

UNIT IV

Origin, geographical distribution, economic importance, soil and climatic requirements,

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varieties, cultural practices Intercropping, pest and disease management and yield of Kharif crops. Cereals – rice, maize, sorghum, pearl millet and finger millet, Small millets* pulses- pigeon pea, mung bean, urdbean, Cowpea, kidney bean and horse gram*; oilseeds- groundnut, soybean,

UNIT V

General Vermiculture. Introduction to vermiculture. definition, meaning, history, economic important, their value in maintenance of soil structure, role as four's of recycling reduce, reuse, recycle, restore. His role in bio transformation of the residues generated by human activity and production of organic fertilizers. How does nature works. The matter and humus cycle (product, qualities). Ground population, transformation process in organic matter. Choosing the right worm. Useful species of earthworms. Local species of earthworms. Exotic species of earthworms. Complementary activities of auto evaluation

Text books:

1. Bhatt J. V. & S. R. Khambata (1959) "Role of Earthworms in Agriculture" Indian Council of Agricultural Research, New Delhi.
2. Dash, M.C., B.K. Senapati, P.C. Mishra (1980) "Vermis and Vermicomposting" Proceedings of the National Seminar on Organic Waste Utilization and Vermicomposting Dec. 5-8, 1984, (Part School of Life Sciences, Sambalpur University, Jyoti Vihar, Orissa.
3. Edwards, C.A. and J.R. Lofty (1977) "Biology of Earthworms" Chapman and Hall Ltd. London. Balasubramanian P & Palaniappan SP. 2001. Principles and Practices of Agronomy.

Fageria NK. 1992. Maximizing Crop Yields. Marcel Dekker.

4. Havlin JL, Beaton JD, Tisdale SL & Nelson WL. 2006. Soil Fertility and Fertilizers. 7th Ed. Prentice Hall.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	M	S	M	S	M	S	S	S	M
CO2	S	S	S	S	M	S	S	M	S	S
CO3	S	M	S	S	S	S	M	S	M	S
CO4	M	S	M	S	M	S	S	M	S	S
CO5	S	S	S	M	S	M	S	S	S	M

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PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

5.

ALLIED PRACTICAL -3

Course Objectives:

To learn experimental knowledge about Residue management and manure production.

RESIDUE MANAGEMENT PRACTICES & MANURE PRODUCTION

Course Out Comes

CO1: To evaluate source of waste

CO2: Categorization of Waste.

CO3: To produce vermicompost.

CO4: Evaluation of Waste

CO5: Recycling of waste.

1. Evaluation of the source of waste
2. Categorization of waste in different categories
3. Vermicompost production
4. Litter decomposition of different plant specie
5. Economic evaluation of waste management practices.
6. Recycling of waste in organic manure or any other useful material

References Books

1. Waste management ,k kseth 2012
2. Agriculture Waste management bydevinderreddy, mandapatirojachdeepthi
3. Kumar, A. (2005) Verms and Vermitechnology, APH Publishing. 11. Lekshmy, M. S., Santhi R. (2012) Vermitechnology, Sara Publications, New Delhi, India,

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	M	S	M	S	M	S	S	S	M

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CO2	S	S	S	S	M	S	S	M	S	S
CO3	S	M	S	S	S	S	M	S	M	S
CO4	M	S	M	S	M	S	S	M	S	S
CO5	S	M	S	M	S	S	M	S	S	M

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

SKILL BASED SUBJECT-3

BIOINOCULANTS TECHNOLOGY

Course Objectives:

To study about plant growth promoting bacteria and their field applications.

Course Out Comes:

CO1: To Understand plant growth promoting bacteria.

CO2: Categorization of Waste.

CO3: production and field application of Rhizobacteria

CO4: Cyanobacteria as nitrogen fixer.

CO5: Role of Mycorrhizae in nitrogen fixing.

Unit I

Concept and significance of bioinoculants. Types of bioinoculants. General account on symbiotic, symbiotic nitrogen fixing bacteria, phosphate solubilizing bacteria and mycorrhizae.

Unit II:

Mass cultivation and application of cyanobacteria, *Azospirillum*, *Azotobacter*, *Rhizobium*, *Azolla* and *Mycorrhizal* bioinoculants.

Unit III:

Concept and significance of biopesticides. Types and their application. Bioherbicides and bioinsecticides. Integrated pest management.

Unit IV:

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Mass production and application of herbal, bacterial, fungal and viral biopesticides. Methods of making bio-compost, vermicompost and its application.

Unit V:

National and International scenario of bioinoculants. Quality control, storage and marketing, small and large-scale strategies of bioinoculants.

TextBooks:

1. Dubey RC, A Text book of Biotechnology, 5th Edition, Chand & Company Pvt Ltd, New Delhi, India, 2014.
2. Kumaresan V, Biotechnology, 6th Edition, Saras Publications Pvt Ltd, Nagercoil, Tamil Nadu, India, 2013.
3. Subba Rao NS, Biofertilizers in Agriculture and Forestry, 3rd Edition. CBS Publishers and Distributors Pvt Ltd, New Delhi, India, 2019.

References Books:

1. Bhattacharya DP and Purohit P, Organic Farming: Biocontrol and Biopesticide Technology, 1st Edition, Agrobios, Jodhpur, Rajasthan, India, 2012.
2. Sadasivam K, Kumar K and Govindarajan K, Biofertilizers Technology, 1st Edition, Scientific Publishers Pvt Ltd, Chennai, India, 2004

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	S	M	S	S	M	S	M	S	S
CO2	S	S	M	S	M	S	S	S	S	S
CO3	M	S	S	M	S	M	S	M	S	S
CO4	S	M	S	S	S	S	M	S	S	M
CO5	S	S	M	S	M	S	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

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ENVIRONMENTAL STUDIES

Course Objectives:

To study about our environment , Natural resources and Biodiversity.

Course Out Comes:

CO1: To Understand Natural resources.

CO2: To learn ecosystem, biodiversity and its conservation

CO3: To analyse environmental pollution and management

CO4: To discuss social ethics.

CO5: Human population and the environment

UNIT-I:

INTRODUCTION TO ENVIRONMENTAL SCIENCES: NATURAL RESOURCES:

Environmental Sciences – scope and importance - Public awareness - Forest resources - Water resources - Mineral resources - Food resources - conflict over resource sharing - Exploitation - Land use pattern - Environmental impact - fertilizer - Pesticide Problems - case studies.

UNIT-II:

ECOSYSTEM, BIODIVERSITY AND ITS CONSERVATION:

Ecosystem - concept - structure and function of the ecosystem - producers, consumers and decomposers - Food chain - Food web - Ecological pyramids - Energy flow - Forest, Grassland, desert and aquatic ecosystem.

Biodiversity - Definition - genetic, species and ecosystem diversity - Values and uses of biodiversity - biodiversity at global, national (India) and local levels - Hotspots, threats to biodiversity - conservation of biodiversity - In situ & Ex situ.

UNIT-III

ENVIRONMENTAL POLLUTION AND MANAGEMENT

Environmental Pollution - Causes -

Effects and control measures of Air, Water, Marine, soil, solid waste, Thermal, Nuclear pollution and Disaster Management - Floods, Earth quake, Cyclone and Land slides. Role of individuals in prevention of pollution - pollution case studies.

UNIT-IV

SOCIAL ISSUES – AND THE ENVIRONMENTS

Urban issues - Energy - water conservation – Rain water harvesting, Climate change, Green

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house effect , Acid rain , Ozone layer depletion , Waste land reclamation ,
Environmental legislation and law, Environmental protection Act. 1986 - Air, Water, Wildlife
and forest conservation Act Enforcement

UNIT-V

HUMAN POPULATION AND THE ENVIRONMENT

Population growth and Explosion- Family welfare programme Human rights health and
environment , human rights - value education, human rights education , child welfare
environmental information system, variation of population among nation, urbanization .

1. Kumarasamy, K. A. Alagappa moose and M. Vasanthi, 2004. Environmental
studies Bharathidasan university PUB, 1, Trichy
2. Rajamannar, 2004, Environmental studies, EVR college PUB, Trichy
3. Kalavathy, s. (ed.) 2004, Environmental studies, Bishop Heber college. PUB., Trichy

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	M	S	M	S	S
CO5	S	S	M	S	S	S	M	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

SEMESTER IV

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CORE PAPER -4

PLANT GENETICS AND PLANT BREEDING

Course Objectives:

To provide the necessary skills in Plant Genetics and Plant Breeding.

Course outcomes:

At end of the course, the students will be able to

CO1: Outline the History, Gene Structure of Plant Genetics

CO2: Discuss the important Transgenic Plants for crop improvement

CO3: Elaborate the knowledge about Plant Breeding.

CO4: Appreciate the Hybrid breeding and methods.

CO5: Describe the Concept of plant ideotype and its role in crop improvement.

UNIT I

Introduction and History of plant genetics, Structure of plant genetic materials (DNA and RNA) Vectors used for the gene cloning in plants and methods. Mechanism of genetically modified crops and their applications- Current status and future prospects of plant genetics.

UNIT II

Transgenic plants for crop improvement (dicots and monocots), Insect resistance, insecticide, Resistance to virus, Resistance to other diseases, Recombinant DNA techniques for the production of transgenic plants, Procedure and protocols of producing transgenic plants. Transgenics for quality and storage, Transgenic plants to study regulated gene expression, Commercial transgenic crops.

UNIT III

History of Plant Breeding (Pre and post-Mendelian era); Objectives of plant breeding, characteristics improved by plant breeding; Patterns of Evolution in Crop Plants- Centres of Origin-biodiversity and its significance. Heritability and genetic advance, genotype- environment interaction; General and specific combining ability; Types of gene actions and implications in plant breeding.

UNIT-IV

Plant introduction and role of plant genetic resource in plant breeding. Breeding methods in cross pollinated crops; Hybrid breeding - genetical and physiological basis of

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heterosis and inbreeding, production of inbreds, breeding approaches for improvement of inbreds, predicting hybrid performance; seed production of hybrid and their parent varieties/inbreds.

UNIT-V

Concept of plant ideotype and its role in crop improvement; Transgressive breeding. Special breeding techniques- Mutation breeding; Breeding for abiotic and biotic stresses. Cultivar development- testing, release and notification, maintenance breeding, Participatory Plant Breeding, Plant breeders' rights and regulations for plant variety protection and farmers rights.

Text Books:

1. Allard, R.W. 1960. Principles of Plant Breeding. John Wiley & Sons. Inc. New York.
2. Backcock, E.B. 2001 Genetics and Plant breeding. Agrobios (India), Jodhpur
3. Briggs, F.N & Knowles, P.F 1967. Introduction to Plant Breeding. Reinhold Publ. Co., New York/ Amsterdam/ London.

References Books

1. Chopra, V.L. 2000. Plant Breeding. Theory and Practicals IBH Publ. Co. Pvt. Ltd., New Delhi.
2. Jain H.K. & Kharkwal, M.C. (Eds.) 2004. Plant Breeding: Mendelian to Molecular Approaches. -. Narosa Publishing. House, New Delhi, Chennai, Mumbai, Calcutta.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

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ALLIED PAPER-4

URBAN FARMING AND TERRACE GARDENING

Course Objectives:

To provide the necessary skills in Urban farming and terrace gardening.

Course Outcomes:

CO1: To gain knowledge about the introduction and benefits of urban farming.

CO2: To learn the process and methods of growing plants without soil.

CO3: To know how to create terrace gardening and its advantages.

CO4: To obtain knowledge about organic garden soil, composting and safe, pesticide-free, terrace gardening.

CO5: Describe the potting soil, seedling and Transplantation of terrace gardening.

UNIT I

Introduction and concept of urban farming and its types. Benefits of urban farming (economic, environmental, social, cultural, educational, skill building, job training benefits, health, nutrition and food accessibility benefits). Challenges of urban farming. Government initiatives in urban farming.

UNIT II

Hydroponics, aeroponics, aquaponics: merits and limitations, costs and challenges. Backyard gardens – tactical gardens – street landscaping – forest gardening, greenhouse, urban beekeeping.

UNIT III

Introduction and Plan of Terrace Garden, Tools and equipment's required for the preparation of Terrace garden, Methods of Terrace gardening. Advantages of Terrace gardening.

UNIT IV

Organic garden soil, Composting and its types.
Application of fertilizers in Terrace garden. Plant disease management, Pest management,
Precautions during Terrace gardening.

UNIT V

Preparing the potting soil, Preparation of seedling and Watering,
Transplantation. Application of biopesticide and weed control. Common vegetables and fruits
for terrace garden.

Text books:

1. Michelle Nowak (May 2004). "Urban Agriculture on the Rooftop". City

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Farmer, Canada's Office of Urban Agriculture. Retrieved March 12, 2014.

2. Louise Lundberg Scandinavian Green Roof Institute (2009).

"The benefits of Rooftop Gardens" (PDF). Retrieved March 12, 2014.

3. Ong, Boon Lay (May 2003). "Green plot ratio: an ecological measure for architecture and urban planning". *Landscape and Urban Planning*. **63** (4): 197–211. doi:10.1016/S0169-2046(02)00191-3

4. Akrong M O. 2012. Urban Agriculture. LAP Academic Publishing.

Reference books

1. Trent University (October 30, 2009). "Rooftop Gardens" (PDF). Retrieved March 12, 2014.

2. Martellozzo F and J S Landry. 2020. Urban Agriculture. Scitus Academics LIC.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	M	S	M	S	S
CO5	S	S	M	S	S	S	M	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

SKILL BASED SUBJECT -4

MUSHROOM CULTIVATION

Course Objectives: To provide the necessary skills in Mushroom cultivation.

Course outcomes:

At end of the course, the students will be able to

CO1: Outline the History, Structure and Cultivation of Mushroom

CO2: Explain about the Spawn preparation.

CO3: Elaborate the Cultivation of important mushroom varieties.

CO4: Appreciate Nutritional value of mushrooms.

CO5: Describe the economics aspects of mushroom cultivation.

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UNIT I:

Introduction, History of Mushroom Cultivation - Morphology and life cycle of Mushroom - Edible and Non-Edible Mushroom (Most commonly cultivated Mushrooms in the World, Distribution and Production in various Countries).

UNIT II:

Spawn; Types Spawn, Preparation of Spawn, Mushroom Bed Preparation and factors affecting Mushroom bed preparation, Compost: Materials used for Compost preparation, Compost Technology in Mushroom production - Casing; Raw material used for casing, preparation of Casing Material. Important Sanitation during various stages of Mushroom cultivation.

UNIT III:

Cultivation of important Mushroom: General process for the cultivation of *Agaricus bisporus* (White button Mushroom), *Pleurotus flabellatus* (Oyster Mushroom), *Volvarellae* (Paddy Straw Mushroom).

UNIT IV:

Mushroom nutritional value; (Proteins, Amino acids, Vitamins, Minerals, Carbohydrates) - Pests and diseases of Edible Mushrooms (Environmental, Fungal, Bacterial, Viral, Insect Pests and Nematode diseases and competitor Moulds).

UNIT V:

Economics of mushroom cultivation (fixed assets, recurring expenditure, labour, economics of cultivation throughout the year and seasonal growing formulation of project report for getting finance from funding agencies). Precautions in mushroom cultivation (precaution to be taken while selecting the area, spawn preparation, spawn run, during cropping harvest etc.). Mushroom recipes (Western and Indian recipes, pickles, powders, jam etc.)

Text Books

1. Mushroom production and processing Technology, Pathak Yadav Gour (2010) Published by Agrobios (India).
2. Mushroom - the art of cultivation, Harander Sing (1991). Sterling Publishers.

Reference Books

1. Biology and conservation of mushroom, Kaul TN (2001). Oxford and IBH Publishing Company, New Delhi.
2. Changs. T. W. A. Hanyanes 1978. "Biology and cultivation of Mushrooms" Acad press. N. Y.
3. Kannaiyan. 2001. Handbook of Edible Mushrooms "TNAU Publication 4. Weinheim: Verlag Chemie, Berlin.
4. Zadrazil. F & K. Grabbe 1983 "Edible Mushroom, Biotechnology" Vol. 3,

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Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	M	S	M	S	S
CO5	S	S	M	S	S	S	M	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

VALUE EDUCATION

Course Objectives:

To provide the necessary skills in Value Education.

Course out comes:

At end of the course, the students will be able to

CO1: Outline of the Value Education.

CO2: Provide knowledge of the Family values.

CO3: Elaborate the Ethical values.

CO4: Discuss the important Social values.

CO5: Describe the Effect of international affairs on values of life.

UNIT-I

Value Education - Definition - relevance to present day - Concept of Human Values - selfintrospection- Selfesteem.

UNIT-II

Familyvalues-Components,structureandresponsibilitiesoffamily-Neutralization of anger - Adjustability - Threats of family life - Status of women infamily and society - Caring for needy and elderly - Time allotment for sharing ideasandconcerns.

UNIT-III

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Ethical values-Professional ethics-Mass media ethics-Advertising ethics-Influence of ethics on family life - psychology of children and youth - Leadership qualities- Personality development.

UNIT-IV

Social values-Faith, service and secularism-Social sense and commitment-Students and Politics - Social awareness, Consumer awareness, Consumer rights and responsibilities - Redressal mechanisms.

UNIT-V

Effect of international affairs on values of life/ Issue of Globalization - Modern warfare - Terrorism. Environmental issues - mutual respect of different cultures, religions and their beliefs.

Reference Books

1. T. Anchukandam and J. Kuttainimathathil (Ed) Grow Free Live Free, Krisitu Jyoti Publications, Bangalore (1995)
2. DBNI, NCERT, SCERT, Dharma Bharti National Institute of Peace and Value Education, Secunderabad, 2002.
3. Daniel and Selvamony - Value Education Today, (Madras Christian College, Tambaram and ALACHE, New Delhi, 1990)
4. Mani Jacob (Ed) Resource Book for Value Education, Institute for Value Education, New Delhi 2002.
5. S. Ignacimuthu - Values for Life - Better Yourself Books, Mumbai, 1991.
6. M. M. M. Mascaron has Centre for Research Education Science and Training for Family Life Promotion - Family Life Education, Bangalore, 1993.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	M	S	M	S	S
CO5	S	S	M	S	S	S	M	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

CORE PRACTICAL -4

EXPERIMENTS IN PLANT GENETICS AND PLANT BREEDING

Course Objectives:

To learn about experimental Knowledge to create Biodynamic farming.

Course Out comes

CO1: General principles of Electrophoresis

CO2: To isolate RNA and Protein.

CO3: To Isolation of plant pigments.

CO4: Evaluation of Floral structure of crops.

CO5: Procedure for germination test.

LIST OF EXPERIMENTS

General principles of Electrophoresis

1. Isolation of RNA from plants
2. SDS – PAGE gel Electrophoresis
3. Isolation of plant pigments by column chromatography
4. Paper chromatography
5. Study of floral structure of self pollinated crops.
6. Study of floral structure of cross pollinated crops.
7. Germplasm collection
 - a. Cereals - Paddy
 - b. Vegetables - *Capsicum*
 - c. Pulses - Green gram, Blackgram, Redgram
8. Hybridization and Emasculation – different types Solitary flower
 - a. 'V' cut method
 - b. Slit method
 - c. Round cut method
 - d. Bagging
9. To study the procedure of germination test of seeds any crops plant species.

References:

1. Adriance, W. and R. Brison. 1979. Propagation of Horticultural plants, Tata McGraw Hill Publishing Co. Pvt. Ltd., New Delhi.
2. Jitendra Singh 2004, Basic Horticulture, Kalyan Publications, New Delhi.
3. Chopra, V. L. 2000. Plant Breeding. Theory and Practicals (2nd edition), Oxford & IBH Publ. Co. Pvt. Ltd., New Delhi.
4. Frankel, R. & Galun, E. 1977. Pollination Mechanisms, Reproduction and Plant Breeding. Springer-Verlag, Berlin/ Heidelberg/ New York.
5. Sharma, J. R. 1994 Principles and Practice of Plant Breeding, Tata-McGraw-Hill Publ. Co. Ltd, New Delhi.

Mapping with Programme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M

CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

ALLIED PRACTICAL-4

URBAN FARMING AND TERRACE GARDENING

Objectives: To provide the necessary skills in Urban farming and terrace gardening.
course Out comes:

1. To gain knowledge about the introduction and benefits of urban farming.
2. To learn the process of methods of growing plant without soil.
3. To know how to create the terrace gardening and advantages.
4. To obtain knowledge about the organic garden soil, composting and safe, pesticide-free, of terrace gardening.
5. Describe the potting soil, seedling and Transplantation of terrace gardening.

UNIT I

Introduction and concept of urban farming and its types. Benefits of urban farming (economic, environmental, social, cultural, educational, skill building, job training benefits, health, nutrition and food accessibility benefits). Challenges of urban farming. Government initiatives in urban farming.

UNIT II

Hydroponics, aeroponics, aquaponics: merits and limitations, costs and challenges. Backyard gardens – tactical gardens – street landscaping – forest gardening, greenhouse, urban beekeeping.

UNIT III

Introduction and Plan of Terrace Garden, Tools and equipment's required for the preparation of Terrace garden, Methods of Terrace gardening. Advantages of Terrace gardening.

UNIT IV

Organic garden soil,

Composting and its types.

Application of fertilizers in Terrace garden. Plant disease management,
Precautions during Terrace gardening.

Pest management,

UNIT V

Preparing the potting soil, Preparation of seedling and Watering,
Transplantation. Application of biopesticide and weed control. Common vegetables and fruits
for terrace garden.

Reference books:

1. Michelle Nowak (May 2004). "Urban Agriculture on the Rooftop". City Farmer, Canada's Office of Urban Agriculture. Retrieved March 12, 2014.
2. Louise Lundberg Scandinavian Green Roof Institute (2009). "The benefit of Rooftop Gardens" (PDF). Retrieved March 12, 2014.
3. Ong, Boon Lay (May 2003). "Green plot ratio: an ecological measure for architecture and urban planning". *Landscape and Urban Planning*. **63** (4): 197–211. doi:10.1016/S0169-2046(02)00191-3
4. Akrong M O. 2012. Urban Agriculture. LAP Academic Publishing.

Reference books:

1. Trent University (October 30, 2009). "Rooftop Gardens" (PDF). Retrieved March 12, 2014.
2. Martellozzo F and J S Landry. 2020. Urban Agriculture. Scitus Academics LIC.

Mapping with Programme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

ALLIED PRACTICAL-4

EXPERIMENTS IN URBAN FARMING AND TERRACE GARDENING

Course Objectives:

To learn about experimental Knowledge to create Urban farming and Terrace Gardening.

Course Out comes

CO1:Preparingthepottingsoil mixtures

CO2: seedlingandWatering

CO3: To practices methods of layering.

CO4: Evaluation of differentmethodsoflayering

CO5: Propagationtechniqueoffruit crops

- 1.Preparingthepottingsoil mixtures
2. PreparationofseedlingandWatering
3. Transplantation
4. CommonvegetablesforUrbanGardening
5. Practisingindifferentmethodsoflayering
- 6.Practisingindifferentmethodsofgrafting
- 7.Practisingindifferentmethodsofbudding
- 8.Practisinginnutritionmanagementfornurseryplants
9. Practisingpropagationtechniqueoffruit crops
- 10.Applicationofbiopesticideandweedcontrol

Reference:

1. Michelle Nowak (May 2004). "Urban Agriculture on the Rooftop". City Farmer,Canada'sOfficeof Urban Agriculture.RetrievedMarch 12,2014.
2. LouiseLundbergScandinavianGreenRoofInstitute(2009). "ThebenefitsofRooftopGardens"(PDF). RetrievedMarch12, 2014.
3. Ong,BoonLay(May2003)."Greenplotratio:anecologicalmeasureforarchitectureandurbanplanning". *LandscapeandUrbanPlanning*. **63** (4):197–211.doi:10.1016/S0169-2046(02)00191-3
4. "What to Consider When Designing and Building Your Own Rooftop or BalconyGarden – Projex Group".Projex Group. 2017-11-17. Retrieved 2018-05-23.
5. Trent University (October 30, 2009). "Rooftop Gardens"(PDF). Retrieved March12,2014.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
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CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

SEMESTER -V

CORE PAPER-5

VERMICOMPOSTING

Course objective

- Students will be able to compost in a limited space and describe the decomposing process.
- The interested students will get the knowledge of composting
- They will also turn towards organic farming,
- They will help to maintain the environment pollution free.
- They will get the knowledge of biodiversity of local earthworms.

Course Outcomes

CO1: On completion of this course, students will;

CO2: Compare and contrast the uses of vermicompost to the soil.

CO3: Recommend different species of earthworms after acquiring knowledge on its biology.

CO4: Design the vermicomposting process

CO5: Assess the Best Practices of Vermicomposting

UNIT-I

General Vermiculture . Introduction to vermiculture. definition, meaning, history, His role in bio transformation of the residues generated by economic important, their value in maintenance of soil structure, role as fourr's of recycling reduce, reuse, recycle, restore.human activity and production of organic fertilizers. How does nature works. The matter and humus cycle (product, qualities). Ground population, transformation process in organic matter. Choosing the right worm. Useful species of earthworms. Local species of earthworms. Exotic species of earthworms. Complementary activities of auto evaluation.

UNIT-II

Earthworm Biology and Rearing. Key to identify the species of earthworms. Biology of Eisenia fetida. a) Taxonomy Anatomy, physiology and reproduction of Lumbricidae. b) Vital cycle of Eisenia fetida: alimentation, fecundity, annual reproducer potential and limit factors (gases, diet, humidity, temperature, PH, light, and climatic factors). Complementary activities of auto evaluation. Biology of Eudrilus eugeniae. a) Taxonomy Anatomy, physiology and reproduction of Eudrilidae. b) Vital cycle of Eudrilus eugeniae: alimentation, fecundity, annual reproducer potential and limit factors (gases, diet, humidity, temperature, PH, light, and climatic factors). Complementary activities of auto evaluation.

UNIT-III

Vermicomposting Technology. Small Scale Earthworm farming for home gardens - Earthworm compost for home gardens Conventional commercial composting- Earthworm composting large scale. Earthworm Farming (Vermiculture), Extraction (harvest), vermin composting harvest and processing. Nutritional Composition of Vermicompost for plants, comparison with other fertilizers. Vermi wash collection, composition & use. Enemies of Earthworms, Sickness and worm's enemies. Frequent problems. How to prevent and fix them. Complementary activities of auto evaluation.

UNIT-IV

Applications of Vermiculture - Vermiculture Bio-technology, use of vermi castings in organic farming/horticulture, as feed/bait for capture/culture fisheries; forest regeneration. Application quantity of vermicompost in Agricultural fields- crops, fruits, vegetables & flowers. By-products and value.

UNIT-V

Agriculture Product Marketing. Co-operative Credit Societies-Function Structure Importance and Problems, District Central Co-operative Banks- Functions, Structure , Importance and Problems, State Co-operative Banks- Role and urban, Urban Co-operative Banks, Nationalized Banks, Regional Rural Bank, Need and Importance, Structure, development and present position of Co-operative Marketing, Problems and Remedies of co-operative marketing, Role of NAFED, E-Marketing, Role of Co-operative Processing Societies in Development, Sugar Co-operatives – Present Position and Problems, Dairy Co-operatives- Present Position and Problems, Poultry Farming, Cotton Industry, Horticultural, Floricultural

and Medical Plants

Process Industry, Fisheries, Meaning & definition of Agriculture Transport, Role of Transportation in agriculture, Types of Transportation, Importance of Transportation in Agriculture.

Text Books

1. Ismail S. A. (2005). The Earthworm Book, Second Revised Edition. Other India Press, Goa, India.
2. Rathoure A. K., Bharati P. K. and Ray J. (2020). Vermitechnology, Farm and Fertilizer. Vermitechnology, Farm and Fertilizer Discovery Publishing House Pvt Ltd.
- Christy M. V. 2008. Vermitechnology, (1st Edition), MJP Publishers.
3. The complete technology book on Vermiculture and Vermicompost with manufacturing Process, machinery equipment details and Plant Layout. AB Press.
4. Keshav Singh (2014). A Textbook of vermicompost: Vermiwash and Biopesticide

References:

1. Bhatt J. V. & S. R. Khambata (1959) "Role of Earthworms in Agriculture" Indian Council of Agricultural Research, New Delhi.
 2. Dash, M. C., B. K. Senapati, P. C. Mishra (1980) "Vermisand Vermicomposting" Proceedings of the National Seminar on Organic Waste Utilization and Vermicomposting Dec. 5-8, 1984, (Part B), School of Life Sciences, Sambalpur University, Jyoti Vihar, Orissa.
 3. Edwards, C. A. and J. R. Lofty (1977) "Biology of Earthworms" Chapman and Hall Ltd., London.
 4. Lee, K. E. (1985) "Earthworms: Their ecology and Relationship with Soils and Land Use" Academic Press, Sydney.
 5. Kevin, A and K. E. Lee (1989) "Earthworm for Gardeners and Fisherman" (CSIRO, Australia, Division of Soils)
 6. Rahudakar V. B. (2004). Gandulkhatashivay Naisargeek Paryay, Atul Book Agency, Pune.
 7. Satchel, J. E. (1983) "Earthworm Ecology" Chapman Hall, London.
 8. Wallwork, J. A. (1983) "Earthworm Biology" Edward Arnold (Publishers) Ltd. London.
 8. Christy, M. V. (2008) Vermitechnology, 1st edition, MJP Publishers.
- Dash, M. C. (2012) Charles Darwin's

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S

CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

CORE PAPER -6

GENETIC ENGINEERING AND TRANSGENIC PLANT

Course Objectives

Understanding the basic concepts of genetics.

Helping students to develop their analytical.

Quantitative and problem solving skills from classical to molecular genetics.

Course Outcomes

CO1:Techniques used in DNA Technology

CO2:DNA cloning methods, using vectors.

CO3:Yeast and algal biomass .

CO4:Gene transfer methods in plants

CO5:Production of virus free plants- shoot meristem culture.

UNIT I

Techniques used in DNA Technology: Gel Electrophoresis, PFGE, Southern and Western blotting, Dot blots, Chemical synthesis of genes, DNA chip technology. Isolation of genes, Sequencing of genes: Maxam& Gilbert's method, Sanger's method and next- generation sequencing technologies, Brief account of proteomics and genomics.

UNIT II

DNA cloning methods, using vectors (Plasmids, phages, cosmids, phagemids, transposons, artificial chromosomes, BAC, YAC, MAC), cloning in bacteria and eukaryotes, genomic and C-DNALibraries.Gene amplification by PCR: different types, DNA finger printing, molecular probes: General features and applications.

UNIT III

Gene transfer methods in plants: plasmid mediated, electroporation, cation precipitation, liposomes, microinjection and particles gun technology, expression of transgenes. Transgenic plants: production of transgenic plants with respect to insect resistance, herbicide resistance, resistance against biotic and abiotic factors, transgenics for male sterility and edible vaccines

UNIT IV

Yeast and algal biomass as source of single cell protein, oils and vitamins, microbial fermentation technology in food industry. Plant and microbial biopesticides, bioremediation and phytoremediation.

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UNIT V

Production of virus free plants- shoot meristem culture. Thermotherapy, cryotherapy and chemotherapy. Virus indexing. Maintenance of virus free stocks. Applications and limitations. Transgenic plants - insect resistant and herbicide resistant plant for crop improvement. Molecular farming

References:

1. Shain- Dow Kung & Ray Wu (Ed) : Transgenic Plants Hamish
- A. Collin and Sue Edwards : Plant Cell Culture Peter M. Gresshoff : Plant Genome. Analysis Razdan M. K : An Introduction to Plant Tissue Culture
- Jogdand S. N : Advances in Biotechnology

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
C05	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

SOIL AND ENVIRONMENTAL MICROBIOLOGY

Course Objectives

- Explain the role of microorganisms in soil fertility.
- Discuss the benefits of interactions among soil microbes and acquire awareness about microbes as biofertilizers and biocontrol agents.
- Create awareness. about components of environment, environmental pollution, and detection methods.
- Acquire in depth knowledge about solid and liquid waste treatments.
- Develop knowledge about organic matter degradation, bioremediation, and the environment risk assessment.

Course Outcomes

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CO1: On completion of this course, students will

CO2: Depict diversity and significance of soil microbes and predict the role of microbes in biological nitrogen fixation.

CO3: Utilize the knowledge of microbial interactions, with beneficial application of biofertilizers for sustainable agriculture and benefits of biopesticides.

CO4: Explain the different types of microorganisms in water. Identify the causes of water pollution and the methods for quality assessment of water and control of water borne diseases.

CO5: Apply knowledge about waste treatments and microbial decomposition and bio-remediation process in environmental cleanup.

- Plan a clear approach on environmental issues. Control pollution and explain protection laws to public.

UNIT I

Soil Microbiology– Soil as Microbial Habitat, Soil profile and properties, Soil formation, Diversity, and distribution of major group of microorganisms in soil. Quantification of soil microflora, role of microorganism in soil fertility. Mineralization of Organic & Inorganic Matter in Soil. Biological Nitrogen fixation- Chemistry and Genetics of BNF. Phytopathology and Disease cycle of Plant pathogens - Tikka and Citrus canker, Types of disease symptoms, Structural and Inducible biochemical defenses - Systemic Acquired Resistance (SAR), pathogenesis related (PR) proteins, Plantibodies, Phenolics, Phytoalexins

UNIT II

Microbial Interactions - Mutualism, Commensalism, Amensalism, Synergism, Competition, Rhizosphere- Rhizosphere effect, Mycorrhizae – Types, Endophytes, PGPR- Plant growth promoting bacteria– symbiotic (*Bradyrhizobium*, *Rhizobium*, *Frankia*), Non-Symbiotic (*Azospirillum*, *Azotobacter*, Mycorrhizae, MHBs, Phosphate solubilizers, algae), Novel combination of microbes as biofertilizers, PGPRs. Biofertilizers and Biocontrol agents – Types, benefits and application. Advantages, social and environmental aspects - Bt crops, golden rice.

UNIT III

Components of Environment: Hydrosphere, lithosphere, atmosphere, and biosphere – definitions with examples; Energy flow in the ecosystem- Carbon, Nitrogen, Sulfur and Phosphorous cycles. Physical factors affecting distribution of microorganisms in various environments. Predisposing factors for Environmental diseases – infectious (water and air borne) and pollution related, spread and control of these diseases. Treatment and safety of drinking (potable) water, methods to detect potability of water samples. Space microbiology - Microbiological research in space environment.

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UNIT IV

Waste management – Solid waste - Types - management - Factors affecting solid waste generation rates. Industrial effluent treatment, primary, secondary, tertiary, and advanced treatment process. Quality assessment of decontaminated matters and other biological effluents. Biological reference standards. Utilization of Solid Waste as Food, Feed and Fuel- Composting, Vermicomposting, Bio manure and Biogas production. E waste management.

UNIT V

Degradation of organic matter - lignin, cellulose, hemicellulose, pectin, common pesticides- herbicides (2,4-D) and pesticides (DDT), heavy metals. Biodegradation of Xenobiotics - Recalcitrant Halocarbons, Recalcitrant TNTs, PCBs and Synthetic polymers. Biodegradation of Hydrocarbons. Biodeterioration of Textiles and Leather. Pollution Control Bodies and Environmental laws in India. Environmental impact assessment, EIA guidelines, US Environment protection Agency norms.

Text Books

Subba Rao. N. S. (2017). Soil Microbiology. (5th Edition). MedTech Publishers
Daniel. C. J. (2006). Environmental Aspects of Microbiology. (2nd Edition). Bright Sun Publications.
Rangaswami. G. and Mahadevan. A. (2006). Diseases of Crop Plants in India. (4th Edition). Prentice–Hall of India Pvt. Ltd.
Sharma P. D. (2010). Microbiology and Plant pathology. (2nd Edition). Rastogi Publications.
Subba Rao. N.S. (2005). Soil microorganisms and Plant Growth. (4th Edition). Oxford and IBH Publishing Pvt. Ltd.

References Books

Pepper I. L., Gerba C. P. and Gentry T. J. (2014). Environmental Microbiology (1st Edition). Academic Press, Elsevier.
Bitton, G. (2011). Wastewater Microbiology. (4th Edition). Wiley-Blackwell.
Bridgewater L. (2012). Standard Methods for the Examination of Water and Wastewater. American Public Health Association.
Shrivastava A.K. (2003). Environment Auditing. A. P. H. Publishing Corporation.
Tinsley, S. and Pillai, I. (2012). Environmental Management Systems – Understanding Organizational Drivers and Barriers.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S

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CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

➤ PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

**ELECTIVE –I
SERICULTURE**

Course Objectives

To study about cultivation of silkworm.

Course Out Comes

CO1 Acquire knowledge on the concepts of origin, growth and study of Sericulture as science

and scientific approach of mulberry plant.

CO2 Describe the morphology and physiology of silkworm.

CO3 Discuss effective management of silkworm diseases.

CO4 Demonstrate field skills in mulberry cultivation and silkworm rearing with an emphasis on technological aspects.

CO5 Demonstrate entrepreneurship abilities, innovative thinking, planning, and setting up small-scale enterprises.

UNIT-I

General introduction to Sericulture, its distribution in India. Botanical distribution and taxonomical characters of mulberry varieties and species. Biology of Mulberry plant and Mulberry crop cultivation and protection.

UNIT-II

Silkworm- biology-morphology of silkworm. Life cycle of silkworm- egg, larva, pupa, and moth.

UNIT-III

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Silkworm pathology: Introduction to Parasitism, Commensalism, Symbiosis and Parasite relationship - Mulberry Silkworm Diseases: Introduction, types, Pebrine, Grasserie, Muscardine,

Flacherie, Symptoms and Pathogens, Mode of Infection, Prevention and Control - Non – mulberry silkworm diseases: Pebrine, Bacterial and viral diseases. Brief Account of Pests and Predators of Silkworms, Nature of damage and control measures.

UNIT-IV

Rearing of silkworm. Cocoon assessment and processing technologies. Value added products of mulberry and silkworms.

UNIT-V

Entrepreneurship and rural development in sericulture: Planning for EDP, Project formulation, Marketing, Insectary facilities and equipments: Location, building specification, air conditioning and environmental control, furnishings and equipment, sanitation and equipment, subsidiary facilities.

Text Books

- 1 Ganga, G. and Sulochana Chetty (2010). Introduction to Sericulture, J., Oxford and IBH Pub. Co. Pvt. Ltd., New Delhi.
- 2 Dr. R. K. Rajan & Dr. M. T. Himantharaj (2005). Silkworm Rearing Technology, Central Silk Board, Bangalore.
- 3 Dandin S B, Jayant Jayaswal and Giridhar K (2010). Handbook of Sericulture technologies, Central Silk Board, Bangalore.
- 4 M. C. Devaiah, K. C. Narayanaswamy and V. G. Maribashetty (2010). Advances in Mulberry Sericulture, CVG Publications, Bangalore
- 5 T.V. Sathe and Jadhav. A.D. (2021). Sericulture and Pest Management, Daya Publishing House.

References Books

- 1 S. Morohoshi (2001). Development Physiology of Silkworms 2nd Edition, Oxford & IBH Publishing Co. Pvt. Ltd. New Delhi
- 2 Hamamura, Y (2001). Silkworm rearing on Artificial Diet. Oxford & IBH publishing 50 Co., Pvt. Ltd. New Delhi.
- 3 M. Johnson, M. Kesary (2019). Sericulture, 5th. Edition. Saras Publications.
- 4 Manisha Bhattacharyya (2019). Economics of Sericulture, Rajesh Publications.
- 5 Muzafar Ahmad Bhat, Suraksha Chanotra, Zafar Iqbal Buhroo, Abdul Aziz and Mohd. Azam (2020). A Textbook on Entrepreneurship Development Programme in

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Sericulture, IP Innovative Publication.

Web Resources

- 1 [https://egyankosh.ac.in > bitstream](https://egyankosh.ac.in/bitstream)
- 2 [https://archive.org > details > SericultureHandbook](https://archive.org/details/SericultureHandbook)
- 3 <https://www.academic.oup.com>
- 4 <https://www.sericulture.karnataka.gov.in>
- 5 <https://www.silks.csb.gov.in>

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

SKILL BASED SUBJECT – 3

APICULTURE

Course Objectives

To study about cultivation of Honey bee.

Course Out Comes

CO1 To understand the biology of honey bees.

CO2 To study on honey bee colony establishment.

CO3 To develop knowledge on honey extraction.

CO4 To understand the diseases of honey bees and their control.

CO5 To gain information on financial assistance and funding agencies for bee keeping industry.

UNIT-I

Biology of Bees: Honeybee – Systematic position – Species of Honey bees – Life history of Honey bee – behaviour – swarming –Pheromone.

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UNIT-II

Social life in Bees: Bee colony – Castes – natural colonies and their yield – Types of bee hives – Structure – location, care and management.

UNIT-III

Bee Rearing: Apiary – Care and Management – Artificial bee hives – types – construction of space frames – Selection of sites – Handling – Maintenance – Instruments employed in Apiary –
Extraction instruments.

UNIT-IV

Bee Economy: Honey – Composition – uses – Bee wax and its uses – yield in national and international market – Diseases of honeybees and their control methods. Economics of bee culture.

UNIT-V

Entrepreneurship: venture – Preparing proposals for financial assistance and funding agencies – Bee Keeping Industry – Recent Efforts, Modern Methods in employing artificial Beehives for cross pollination in horticultural gardens.

Text Books

1. Dewey M. Caron. (2013). Honey Bee Biology and Beekeeping. Revised Edition. Wicwas Press, Kalamazoo. ISBN 10: 1878075292
2. R. A. Morse. (1993). Rearing queen honey bees. Wicwas press, NY. ISBN-10 : 1878075055
3. Ted Hooper. (2010). Guide to Bees & Honey: The World's Best Selling Guide to Beekeeping. Northern Bee Books. Oxford. ISBN 10: 1904846513
4. Jayashree K. V., Tharadevi C.S. and Arumugam N. (2014) Apiculture. Saras Publication
5. Raj H. (2020). Vinesh Text Book of Apiculture. S. Vinesh and Co.

References Books

- 1 Dewey M. Caron. (2020). The Complete Bee Handbook: History, Recipes, Beekeeping Basics, and More, Rockridge Press. ISBN-10 : **1646119878**
- 2 Joachim Petterson. (2016). Beekeeping: A Handbook on Honey, Hives & Helping the Bees, Weldon Owen.
- 3 Eva Crane. (1999). The World History of Beekeeping and Honey Hunting. Routledge. India. ISBN-10 : 0415924677
- 4 Pagar B. S. (2016). Textbook Of Apiculture. Sahitya Sagar.
- 5 Sehgal P.K. (2018). Text Book of Sericulture, Apiculture and Entomology. Kalayani.

Web Resources

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1. BeeKeepingBasics. Retrieved from: <https://denton.agrilife.org/files/2013/08/beekeepingbasics.pdf>
2. Beekeeping as an Entrepreneurship, Retrieved from: <https://lupinepublishers.com/agriculture-journal/pdf/CIACR.MS.ID.000270.pdf>
3. Raising Bumble Bees at Home: A Guide to Getting Started. Retrieved from: <https://www.ars.usda.gov/ARSUserFiles/20800500/BumbleBeeRearingGuide.pdf>
4. Apiculture – Biology for Everybody (homeomagnet.com)
5. Apiculture: Introduction to Apiculture (iasri.res.in)

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

CORE PRACTICAL V

Course Objectives

- To study about the chemical and biological principles of living systems.

Course Outcomes

- Students will be able to understand and integrate knowledge of chemical and biological principles of living systems.
- Students will develop quantitative skills in order to collect, analyze and interpret experimental data.
- Understand the importance of recombinant DNA technology. Learn isolation of DNA and its separation on an agarose gel.
- Understand restriction and ligase enzymes and their application in gene cloning.
- Identify plant pathogens

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List of Experiments

1. Isolation and identification of *Helicobacter pylori*.
2. Detection and isolation of anti infectives from plants
3. Extraction of bioactive principles from plant.
4. Estimation of its antimicrobial activity using standard guidelines
5. Plasmid DNA Isolation from *E.coli*
6. Plasmid DNA Isolation from plant sources.
7. Restriction Digestion and analysis (web cutter).
8. Competent Cell preparation.
9. DNA Ligation.
10. Bacterial transformation. 8. Agarose Gel Electrophoresis.
11. Quantifying DNA by zymogram (computer approach).
12. SDS – PAGE.

References:

1. DNA Cloning: A Practical Approach by D.M. Glover and B.D. Hames, IRL Press, Oxford. 1995. 2. Molecular cloning: Sambrook et al.,

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

➤ PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

➤

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SEMESTER –VI

**CORE PAPER – 8
BIOREMEDIATION**

Course Objectives

To learn about environmental cleaning process.

Course Out Comes

CO1 Describe the nature and importance of bioremediation and use in real world applications.

CO2 Describe the typical composition of waste water and application of efficient technologies for water treatment.

CO3 Explain the fundamentals of treatment technologies and the considerations for its design and implementation in treatment plants.

CO4 Explain the potential of microbes in ore extraction and acquaint students with methods of reducing health risks caused by xenobiotics.

CO5 Familiarize the role of plants and their associated microbes in remediation and management of environ.

UNIT -I

Bioremediation - process and organisms involved. Bioaugmentation - Ex-situ and in-situ processes; Intrinsic and engineered bioremediation. Major pollutants and associated risks; organic pollutant degradation. Microbial aspects and metabolic aspects. Factors affecting the process.

Recent developments and significance.

UNIT –II

Microbes involved in aerobic and anaerobic processes in nature. Water treatment, BOD, COD, dissolved gases, removal of heavy metals, total organic carbon removal. Secondary waste water treatments- use of membrane bioreactor.

UNIT –III

Composting of solid wastes, anaerobic digestion – methane production and important factors involved, Pros and cons of anaerobic process, sulphur, iron and nitrate reduction, hydrocarbon degradation, degradation of nitroaromatic compounds. Bioremediation of dyes, bioremediation in paper and pulp industries.

UNIT –IV

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Microbial leaching of ores - process, microorganisms involved and metal recovery with special reference to copper and iron. Biotransformation of heavy metals and xenobiotics. Petroleum biodegradation - reductive and oxidative. Dechlorination. Biodegradable of plastics and super bug.

UNIT- V

Phytoremediation of heavy metals in soil - Basic principles of phytoremediation - Uptake and transport, Accumulation and sequestration. Phytoextraction. Phytodegradation. Phytovolatilization. Rhizodegradation. Phytostabilization –Organic and synthetic amendments in multi metal contaminated mine sites. Role of Arbuscular mycorrhizal fungi and plant growth promoting rhizobacteria in phytoremediation.

Text Books

1. Bhatia H.S. (2018). A Text book on Environmental Pollution and Control. (2nd Edition). Galgotia Publications.
2. Chatterjee A. K. (2011). Introduction to Environmental Biotechnology. (3rd Edition). Printice-Hall, India.
3. Pichtel, J. (2014). Waste Management Practices: Municipal, Hazardous, and Industrial, 2nd edition, CRC Press.
4. Liu, D.H.F and Liptak, B.G (2005). Hazardous Wastes and Solid Wastes, Lewis Publishers.
- .
5. Rajendran, P. & Gunasekaran, P. (2006). Microbial Bioremediation. 1st edition. MJ Publishers

References Books

1. Sangeetha J., Thangadurai D., David M. and Abdullah M.A. (2016). Environmental Biotechnology: Biodegradation, Bioremediation, and Bioconversion of Xenobiotics for Sustainable Development. (1st Edition). Apple Academic Press.
2. Singh A. and Ward O. P. (2004). Biodegradation and Bioremediation. Soil Biology. 51 Springer.
3. Singh A., Kuhad R. C., and Ward O. P. (2009). Advances in Applied Bioremediation (1st Edition). Springer-Verlag Berlin Heidelberg, Germany.
4. Atlas, R.M & Bartha, R. (2000). Microbial Ecology. Addison Wesley Longman Inc.
5. Rathoure, A.K. (Ed.). (2017). Bioremediation: Current Research and Applications. 1st edition. I.K. International Publishing House Pvt. Ltd.

Web Resources

1. Bioremediation- Objective, Principle, Categories, Types, Methods, Applications (microbenotes.com)
2. <https://agris.fao.org/agris-search>
3. <https://www.sciencedirect.com/topics/earth-and-planetary-sciences/bioremediation>
4. <https://www.intechopen.com/chapters/70661>

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5. <https://microbiologysociety.org/blog/bioremediation-the-pollution-solution.html>

Mapping with Programme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

**CORE PAPER -9
HERBAL TECHNOLOGY
Course Objectives**

To identify and exploit the potential benefits of plants for the betterment of human life

Course Out Comes

1. After studying unit-1, the student will be able to get acquainted with the basics of Pharmacognosy
2. After studying unit-2, the student will be able to Gain knowledge of medicinal plants
3. After studying unit-3, the student will be able to Understand the use of various medicinal plants
4. After studying unit-4, the student will be able to Appreciate the Herbal medicines used to treat human ailments
5. After studying unit-5, the student will be able to Understand the Propagation methods of medicinal plants.

UNIT - I

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Pharmacognosy: Pharmacology - Definition and history; Indian systems of medicine - Siddha, ayurvedha, and Unani systems; Classification of Crude drugs; Chemistry of Drugs; Future of pharmacognosy.

UNIT - II

Classification of medicinal plants; locally available medicinal plants - Vernacular name and family - Geographical Distribution; Crude drugs - chemical composition - pharmaceutical uses - processing and marketing.

UNIT - III

Medicinal plant parts as herbal medicines: Leaves - Adathoda, Eucalyptus; Flower - Clove; fruits/seeds - Nux vomica Nutmegs, Gooseberry - unorganized drugs; Gum - Acacia - Resin - Turpentine, fixed oil - castor oil. Underground stem - ginger, Alpinia; Roots - Rauwolfia, Belladonna; Aerial parts - Bark - Cinchona.

UNIT - IV

Herbal medicines for Human ailments - Drugs acting on Blood pressure, cardiac diseases, cerebral diseases, Respiratory diseases - Drugs acting on Nervous system - Depressants. - stimulants - Urogenital system - Psychoactive plants; Drug adulteration - methods of Drug evaluation.

UNIT - V

Propagation of medicinal plants - cultivation of medicinal and aromatic plants - Micro and macropropagation - conservation of rare medicinal plants - seed banks - Role of biotechnology in medicinal plants - Herbal food - Food processing - packaging - Herbal sale and Export of medicinal plants - marketing - Intellectual property rights - Export laws.

Text Books

1. Herbal Drug Technology, 2nd Edition. Agarwal, S.S and Paridhavi, M (2012). Universities Press (India) Private Ltd.
2. Herbal Plants and their Applications in Cosmeceuticals. Kuntal Das (2014). CBS publishers and distributors Pvt. Ltd., Chennai.

Reference books

1. Pharmacognosy 12th edition. George Edward Trease and W.C. Evans. English Language

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Books Society, Baelliere Tindall.

2. Pharmacognosy by 2nd Edition. Handa, S.S. and Kapoor, V.K. Vallabh Prakashan Publishers,
New Delhi.

3. Indian Medicinal plants. Jain, S.K (1980).

4. Pharmacognosy by 12th edition. Kokate, C.K., Durohit, A.P. and Gokhale, S.R. Nirali Prakasham Publishers, Pune.

5. Introduction to Medical Botany and Pharmacognosy. Kumar N.C. (1993).

Mapping with Programme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

CORE PRACTICAL -6

1. Degradation of para nitro phenol using *Pseudomonas putida*
2. Demonstration of DNA finger-printing technique
3. Biodiesel production using micro-algae
4. Isolation and Identification of Bioremediation Microbes from the polluted environmental sites.
5. Growth profiling and biochemical of the microbes isolated from the polluted sites.
6. Degradation of oil by *Bacillus* species or the microorganisms isolated from the oil contaminated sites.
7. Compare the aerobic and anaerobic degradation of pollutants or xenobiotic compounds.
8. Biostimulation demonstration by adding nutrients to the selected microbial cultures and examining its effect on bioremediation.
9. To perform preliminary phytochemical screening of crude drugs.
10. Determination of the alcohol content of Asava and Arista .
16. Evaluation of excipients of natural origin
17. Incorporation of prepared and standardized extract in cosmetic formulations like creams, lotions and shampoos and their evaluation.

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18. Incorporation of prepared and standardized extract in formulations like syrups, mixtures and tablets and their evaluation as per Pharmacopoeial requirements.
19. Monograph analysis of herbal drugs from recent Pharmacopoeias
20. Determination of Aldehyde content
21. Determination of Phenol content
22. Determination of total alkaloids

References Books

1. "Bioremediation Principles and Applications" by Ronald L. Crawford and DonL. Crawford.
2. "Environmental Biotechnology: Principles and Applications" by Bruce E. Rittmann and Perry L. McCarty.
3. "Microbial Ecology: Fundamentals and Applications" by Ronald M. Atlas and Richard Bartha.
4. Textbook of Pharmacognosy by Trease & Evans.
5. Textbook of Pharmacognosy by Tyler, Brady & Robber.
6. Pharmacognosy by Kokate, Purohit and Gokhale
7. Essential of Pharmacognosy by Dr.S.H.Ansari
8. Pharmacognosy & Phytochemistry by V.D.Rangari
9. Pharmacopoeal standards for Ayurvedic Formulation (Council of Research in Indian Medicine & Homeopathy)
10. Mukherjee, P.W. Quality Control of Herbal Drugs: An Approach to Evaluation of Botanicals. Business Horizons Publishers, New Delhi, India, 2002.

Mapping with Programme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

**CORE PAPER - 10
COMPULSORY PROJECT
GROUP PROJECT**

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Group projects enable students to get hands-on training in microbiological techniques needed for research. Thus the students can share diverse perspectives resulting in pooling of knowledge and skills. Group work may approach tasks and solve problems in novel, interesting ways, thereby converting established theoretical concepts to practical skills. If structured properly, it will promote team work and collaboration. Group projects also will help students to choose a research design, solve real life problems and benefit the society at large. Thus group project facilitates the students to convert ideas to practice thereby creating a research culture among students.

Guidelines for group project:

A research problem need to be selected based on creative ability and scientific thought.

A brief description of the problem needs to be given.

Hypothesis statement should be framed.

Objectives by which the project work is to be carried out should be clearly stated.

Methodology has to be designed to test the hypothesis.

Results obtained need to be replicable.

Documented report has to be submitted on completion of the project.

Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

INTERNAL ELECTIVE -2

AGRO-FORESTRY

Objective To impart knowledge on the concept of agroforestry land use including diagnosis & design methodologies.

UNIT I

Agroforestry objectives, importance, potential and impediments in implementation. Land

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capability classification and land evaluation.

UNIT II

Overview of global agro-forestry systems, shifting cultivation, taungya system, multiple and mixed cropping, alley cropping, shelter-belts and windbreaks, energy plantations and homestead gardens. Productin potential of different silvi-pasture system.

UNIT III

Concepts of community forestry and social forestry, linear strip plantations.

UNIT IV

Diagnosis and Design – Trends in Agroforestry systems research and development.

UNIT V

Practical Survey and analysis of land use systems in the adjoining areas. Design and plan of suitable models for improvement.

Suggested Readings

Dwivedi AP. 1992. Agroforestry: Principles and Practices. Oxford & IBH. Nair PKR, Rai

MR & Buck LE. 2004. New Vistas in Agroforestry. Kluwer.

Nair PKR. 1993. An Introduction to Agroforestry. Kluwer.

Ong CK & Huxley PK. 1996. Tree Crop Interactions – A Physiological Approach. ICRAF.

Thampan PK. 1993. Trees and Tree Farming. Peekay Tree Crops Development Foundation.

Young A. 1997. Agroforestry for Soil Management. CABI.

Mapping with Programme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M
CO5	S	S	M	S	S	M	S	S	M	S

PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

SKILL BASED SUBJECT -4

ENTREPRENEURSHIP AND BIO-BUSINESS

Course Objectives

To know about entrepreneurship and Bio business

Course out comes

CO1 Understanding basic concepts in the area of entrepreneurship, the role and importance

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of entrepreneurship for economic development

CO2 Developing personal creativity and entrepreneurial initiative, adopting the key steps in the elaboration of business idea.

CO3 Understanding the stages of the entrepreneurial process and the resources needed for the successful development of entrepreneurial ventures.

CO4 Explain the central components of successful business strategies in biotechnology, and create a business plan.

CO5 Understand the various funding resources and develop as Entrepreneur

UNIT-I

Bio Entrepreneurship: Introduction to bio-business, SWOT analysis of bio-business. Ownership, Development of Entrepreneurship; Stages in entrepreneurial process; Government schemes and funding. Small scale industries: Definition; Characteristics; Need and rationale.

UNIT-II

Entrepreneurship Opportunity in Agricultural Biotechnology: Business opportunity, Essential requirement, marketing strategies, schemes, challenges and scope-with case study on Plant cell and tissue culture technique, polyhouse culture. Herbal bulk drug production, Nutraceuticals, value added herbal products. Bioethanol production using Agricultural waste, Algal Source. Integration of system biology for agricultural applications. Biosensor development in Agriculture management.

UNIT-III

Entrepreneurship Opportunity in Industrial Biotechnology: Business opportunity, Essential requirement, marketing strategies, schemes, challenges, and scope- Pollution monitoring and Bioremediation for Industrial pollutants. Integrated compost production- microbe enriched compost. Bio pesticide/ insecticide production. Biofertilizer. Single cell protein.

UNIT-IV

Therapeutic and Fermented products: Stem cell production, stem cell bank, production of monoclonal/polyclonal antibodies, secondary metabolite production – antibiotics, probiotic and prebiotics.

UNIT-V

Project Management, Technology Management and Startup Schemes: Building Biotech business challenges in Indian context-biotech partners (BIRAC, DBT, Incubation

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centers.etc.), operational biotech parks in India. Indian Company act for Bio business-schemes and subsidies. Project proposal preparation, Successful start-ups-case study.

Text Books

1. Craig Shimasaki. (2014). Biotechnology Entrepreneurship: Starting, Managing, and Leading Biotech Companies. Academic Press.
2. Ashton Acton, O. (2012). Biological Pigments– Advances in Research and Application Scholarly Editions: Atlanta, Georgia.
- 126
3. Jennifer Merritt, Jason Feifer (2018). Start Your Own Business, 7th edition, Entrepreneur Press publisher.
4. Peter F. Drucker (2006). Innovation and Entrepreneurship. Harper Business publisher.
5. Leah Cannon (2017). How to Start a Life Science Company: A Comprehensive Guide for First-Time Entrepreneurs. International Kindle paperwhite.

References Books

- 1 Crueger, W, and Crueger. A.(2000). Biotechnology: A Text Book of Industrial microbiology, 2nd Edition, Sinauer Associates: Sunderland.Mass.
- 2 Paul S Teng. (2008). Bioscience Entrepreneurship in AsiaWorld Scientific Publishing Company.
- 3 Charles E. Bamford, Garry D. Bruton (2015). ENTREPRENEURSHIP: The Art, Science, and Process for Success, 2nd Edition, McGraw Hill publisher.
- 4 Yali Friedman (2014). Building Biotechnology: Biotechnology Business, Regulations, Patents, Law, Policy and Science 4th Edition, Logos press publication.
- 5 Stephanie A. Wisner (2022). Building Backwards to Biotech: The Power of Entrepreneurship to Drive Cutting-Edge Science to Market, International Kindle paperwhite.

Web Resources

- 1 <https://www.bio-rad.com/webroot/web/pdf/lse/literature/Biobusiness.pdf>
- 2 <https://www.crg.eu/biobusiness-entrepreneurship>
- 3 <https://www.entrepreneur.com>
- 4 <https://www.birac.nic.in>
- 5 <https://www.springer.co>

Mapping with Programme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	M	S	M	S	M	S	S
CO2	S	M	S	S	S	S	M	S	S	M
CO3	S	S	M	S	M	S	S	S	M	S
CO4	M	S	S	M	S	S	M	S	S	M

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CO5	S	S	M	S	S	M	S	S	M	S
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PO Programme outcome, Co-Course outcome, S- Strong, M- Medium, L-Low

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Program Name: M.Sc. Clinical Research & Health Informatics (Integrated Programme)

Program Duration: 5 years

Philosophy/Significance of the Course

A clinical research requires a systematic approach with diligent planning, execution and sampling in order to obtain reliable and validated results, as well as an understanding of each research methodology is essential for researchers. Indeed, selecting an inappropriate study type, an error that cannot be corrected after the beginning of a study, results in flawed methodology. Health informatics is a specialized field within the healthcare industry that combines communications, information and healthcare with the goal of improving patient care. It involves using resources such as health information systems in collecting, storing, retrieving, analyzing and utilizing healthcare information for a variety of purposes

Our course is designed is to produce high-quality graduates with significant communication, leadership, intellectual skills backed by ethics and social responsibility. Also this course is to introduce the principles of clinical research, including skills and knowledge on why and how to do research, how to build a good research environment, and where to find additional resources, tools, and reference packages to support a research project.

Distinctiveness and uniqueness of the program

M.Sc Clinical research & health informatics is a distinctive course, since it includes co-curricular and extracurricular activities to make learning also it offers soft skills training programs to help students develop corporate manners and other skills needed for success in the global marketplace. Clinical research plays a pivotal role in advancing medical knowledge and improving patient outcomes. By investigating the safety and efficacy of new interventions, clinical trials provide valuable insights that guide treatment decisions and enhance the quality of care. Furthermore, research findings inform evidence-based guidelines, empowering healthcare providers to deliver personalized and effective therapies tailored to individual patient needs. As a result, clinical research serves as a catalyst for transformative breakthroughs that have the potential to revolutionize healthcare delivery and disease management. The integration of artificial intelligence in clinical research holds promise for streamlining data analysis and accelerating the discovery of novel therapeutic interventions. Health informatics plays a crucial role in healthcare by combining information technology, computer science, clinical knowledge, and management science to improve patient care, enhance efficiency, and reduce overall costs.

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M.Sc. Clinical Research & Health Informatics (Integrated Programme)

Programme Educational Objectives

PEO 1 Graduates will be accomplished professionals, innovators, or entrepreneurs actively involved in bio-medical research, academics, and technology development.

PEO 2 Graduates will function in their profession with ethics, social awareness, and responsibility.

PEO 3 Graduates will engage with professionals in allied health and environmental sciences for collaborations and knowledge exchange.

PEO 4 Graduates will be successful in pursuing higher studies in allied health science.

PEO 5 Graduates will pursue career paths in academia or research.

Program Outcomes (POs):

At the end of programme student should be able to:

PO1 Gain Knowledge of basic and applied aspects of clinical research and allied sciences.

PO2 Demonstrate basic laboratory skills in clinical research and understanding of advanced analytical tools to gather scientific evidence.

PO3 Ability to comprehend basic analytical problems, apply critical thinking skills and offer scientific solutions.

PO4 Demonstrate and execute time management during completion of academic exercises, assignments, and research projects.

PO5 Demonstrate awareness of roles, responsibilities, and ethical standards as per academia and industry alignments.

PO6 Demonstrate leadership abilities, problem solving capability and decision making with high ethical standards

PO7 Demonstrate good interpersonal communication skills.

PO8 Demonstrate eagerness for tackling critical societal issues concerning human and planetary health.

PO9 Recognize the need for and prepare to engage in life-long learning to adapt with technological changes.

Program Specific Outcomes (PSOs) for Clinical Research

PSO1. Demonstrate knowledge on good clinical practice, record-keeping practices within all stages of clinical trials and post marketing processes to ensure compliance with research approvals and professional and ethical standards of practice.

PSO2 Apply elements of research methodology to design and execute basic experiments and work towards developing solutions for emerging health, social and environmental problems.

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PSO3 Apply knowledge of clinical research to pursue a wide range of careers in higher academia and research, public health, environmental organizations, societal organizations, relevant industries, and entrepreneurial activities.

Abbreviations

Cat	Category
L	Lecture
T	Tutorial
P	Practical
Cr	Credits
UC	University Core
PC	Program Core
PRJ	Project Work (including Seminars, Dissertation, and Internship)
PE	Professional Elective
TC	Total Credits

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Overview of Credit Allocation/ Credit

Break up Category-wise Credit distribution

Category	Number of Credits	Credit Percentage (%)
University Core (UC)	6	3
Programme Core (PC)	177	81
Professional Elective (PE)	5	2
Projects (PRJ)	20	9
Mandatory Non-Credit Courses	10 (7 Courses)	5
Total	218	100

- ❖ University core subjects are those subjects that are mandatory for all similar programmes.
- ❖ Program Core courses in a curriculum are program specific. To be eligible for the degree, students must successfully finish each of the PC-listed courses.
- ❖ Program Elective courses provide the students the opportunity to study courses that are more complex and specialized, in their field of specialization.
- ❖ University electives are courses that a student can opt for from outside of his/ her programme, from across the university. This allows students to pursue their interests in other subjects as well. The number of credits that a student may take under University Elective is regulated.

Programme Structure

The term "Program Structure" refers to a list of courses (Core, Elective, and Open Elective) that make up an academic program, describing the syllabus, credits, hours of instruction, assessment, and examination systems, minimum amount of credits necessary for program graduation, etc.

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SEMESTER I

Cat	Course Code	Course Title	L	T	P	C
PC	25CI101	Human Anatomy and Physiology	3	1	2	6
PC	25CI102	Fundamentals of Clinical Research	3	1	2	6
PC	25CI103	Community Health and Disease	3	1	2	6
MNC	25CI104	Ability Enhancement/Co-curricular I	1	1	-	2
Total Credits						20

SEMESTER II

Cat	Course Code	Course Title	L	T	P	C
PC	25CI201	Principles of Biochemistry	2	1	1.5	4.5
PC	25CI202	Computer Application and Bioinformatics	2	1	1.5	4.5
PC	25CI203	Fundamentals of Biostatistics	2	1	1.5	4.5
UC	25CI204	Ability Enhancement/Co-curricular II	1	1	0	2
PC	25CI205	Biosafety Guidelines	2	1	1.5	4.5
Total Credits						20

SEMESTER III

Cat	Course Code	Course Title	L	T	P	C
PC	25CI301	Pharmacology and Toxicology	2	1	1.5	4.5
PC	25CI302	Design and Management of Clinical Trials	2	1	1.5	4.5
PC	25CI303	Clinical Sampling and Analytical Techniques	2	1	2	5
UC	25CI304	Environmental Science	3	1	-	4
MNC	25CI305	Ability Enhancement/Co-curricular III	0	-	-	2
Total Credits						20

SEMESTER IV

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Cat	Course Code	Course Title	L	T	P	C
PC	25CI401	Pharmacovigilance	2	1	1.5	4.5
PC	25CI402	Clinical Pharmacokinetics	2	1	1.5	4.5
PC	25CI403	Bioethics and Drug Safety	2	1	1.5	4.5
PC	25CI404	Regulatory Aspects in Clinical Research	2	1	1.5	4.5
MNC	25CI405	Ability Enhancement/Co-curricular IV	0	-	-	2
Total Credits						20

SEMESTER V

Cat	Course Code	Course Title	L	T	P	C
PC	25CI501	Global Regulations in Clinical Trials	2	1	1.5	4.5
PC	25CI502	Epidemiology and Global Health (with FND)	2	1	1.5	4.5
PC	25CI503	Immunology	2	1	1.5	4.5
PC	25CI504	Bioinstrumentation	2	1	1.5	4.5
MNC	25CI505	Ability Enhancement/Co-curricular V	0	-	-	2
MNC	25CI506	Industrial Training/Survey/Project			4	Qualifying
Total Credits						20

SEMESTER VI

Cat	Course Code	Course Title	L	T	P	C
PC	25CI601	Pharmacotherapeutics	4	1	1	6
PC	25CI602	Biopharmaceutics	4	1	1	6
PC	25CI603	Data Management Technologies	4	1	1	6
MNC	25CI604	Ability Enhancement/Co-curricular VI	0	-	-	2
MNC	25CI605	Industrial Training/Survey/Project			4	Qualifying

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Total Credits		20
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SEMESTER VII

Cat	Course Code	Course Title	L	T	P	C
PC	25CI701	Health Economics and Outcome Research	4	1	-	5
PC	25CI702	Emerging Technologies in Clinical Trials	3	1	1	5
PC	25CI703	Research Methodology	3	1	1	5
PC	25CI704	Good laboratory and Manufacturing Practices	3	1	1	5
Total Credits						20

SEMESTER VIII

Cat	Course Code	Course Title	L	T	P	C
PC	25CI801	Fundamentals of Computer Applications & Data Base Management	7	-	2	9
PC	25CI802	Health & Bioinformatics	6	2	-	8
PC	25CI803	Disease classification system	3	2	-	5
PC	25CI804	Bio Statistics	6	2	-	8
Total Credits						30

SEMESTER IX

Cat	Course Code	Course Title	L	T	P	C
PC	25CI901	Health Care Financing	8	1	-	9
PC	25CI902	Clinical Workflow & Process Redesigning	3	2	-	5

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PC	25CI903	Hospital Administration	7	2	-	9
Total Credits						23

SEMESTER X

Cat	Course Code	Course Title	L	T	P	C
PRJ	25CI100	Industrial Training / Research Project	-	-	15	20
PE	-	Professional Elective	3	2	0	5
Total Credits						25

*Q – Qualifying

List of Electives as per NEP Minor electives

ELECTIVES

Course code	Course Title	L	T	P	TC
25a606a	Ayush and Lifestyle	3	2	0	5
25a606b	Site Management Operations	3	2	0	5

List of Electives as per NEP

S.No	Electives	Credits
1	Ability Enhancement/Co-curricular-I	2
2	Ability Enhancement/Co-curricular-II	2
3	Minor Elective (Exploratory Elective) either in Ist or IIInd Semester	4.5
4	Skill Enhancement/Vocational-III: Environmental Science	4
5	Skill Enhancement/Vocational-IV: Leadership and Teamwork	2
6	Ability Enhancement/Co-curricular-III	2
7	Ability Enhancement/Co-curricular-IV	2

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8	Industrial Training/Survey/Project	Qualifying
9	Ability Enhancement/Co-curricular-V	2
10	Ability Enhancement/Co-curricular-VI	Qualifying
11	Industrial Training/Survey/Project	Qualifying
12	Industrial Internship/Startup - BSc Honours students Research Project-BSc Honours with Research students	15
13	Minor elective	5

Modes of Evaluation: Quiz/Assignment/Presentation/ Written Exam/ Examination

Scheme: Continuous Assessment- 50%, mid semester examination- 20% and End term examination- 30%

Theory Assessment:

	Continuous Assessment/Internal Assessment (50)				Mid Term Exam	End Term Exam	Total
Components	Surprise Test/ Quiz	Assignments	Group Discussion /Presentations	Project Based Learning/ Tutorials based learning			
Weightage (%)	10	10	10	20	20	30	100

Laboratory Assessment:

	Continuous Assessment/Internal Assessment			End Term Examination		
Components	Experimental Performance	Vivavoce	Lab record	Major Experiments (Practical)	Vivavoce	Total
Weightage (%)	30	20	20	20	10	100

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SEMESTER I

25CI101 HUMAN ANATOMY AND PHYSIOLOGY L-T-P-C: 3-1-2-6

COURSE OBJECTIVES

- To cover human anatomy and lay out the physiology of various systems including cardiovascular, respiratory, reproductive, skeletal, nervous systems, special senses, among others.
- To help students to explore the amazing inner workings of the body, which is a complex collection of interacting systems that carry out the important functions that let you move, think, feel, and live.

UNIT – I: Introduction to Human Anatomy and Physiology 12 hours

Introduction to anatomy and physiology, Level of Organization (atomic, molecular), Cellular Level of Organization, Tissue Level of Organization, Cell Structure and functions, cell-cell communication, intracellular cell signalling including Adenyl cyclase, ATP, AMP, GMP, Tyrosine Kinase, Phosphorylase, cell-cell signalling, cell division, homeostasis, negative and positive feedback. Integumentary System: Anatomy and functions of skin- Skeletal System: Axial skeleton system, appendicular skeleton system, neuromuscular junction, joins, muscle contraction.

UNIT- II: Circulatory System and Special Senses 12 hours

Body fluids and blood: Body fluids, composition, and functions of blood, hemopoiesis, formation of hemoglobin, anemia, mechanisms of coagulation, blood grouping, Rh factors, transfusion, its significance, and disorders of blood. Lymphatic system: Lymphatic organs and tissues, lymphatic vessels, lymph circulation and functions of lymphatic system, Immune System, and its classification. Special Senses: Structure and functions of eye, ear, nose and tongue and their disorders.

UNIT III: Nervous and Cardiovascular System 12 hours

Nervous system: Organization of nervous system, neuron, neuroglia, classification and properties of nerve fibre, electrophysiology, action potential, nerve impulse, receptors, synapse, neurotransmitters. Central nervous system: Meninges, ventricles of brain and cerebrospinal fluid. Structure and functions of brain, spinal cord. Peripheral Nervous System: Classification of peripheral nervous system: structure and functions of sympathetic and parasympathetic nervous system. Origin and functions of spinal and cranial nerves. Cardiovascular System Heart: Anatomy of heart, blood circulation, blood vessels, structure and functions of artery, vein and capillaries, elements of conduction system of heart and heartbeat, its regulation by autonomic nervous system, cardiac output, cardiac cycle. Regulation of blood pressure, electrocardiogram, and disorders of heart.

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UNIT IV: Digestion, Respiration and Hormone Regulation

12 Hours

Digestive system: Anatomy of GI Tract with special reference to anatomy and functions of stomach, (Acid production in the stomach, regulation of acid production through parasympathetic nervous system, pepsin role in protein digestion) small intestine and large intestine, Energetics Formation and role of ATP, Creatinine Phosphate and BMR. Respiratory system: Anatomy of respiratory system with special reference to anatomy of lungs, mechanism of respiration, regulation of respiration Lung Volumes and capacities transport of respiratory gases, artificial respiration, and resuscitation methods. Endocrine system: Classification of hormones, mechanism of hormone action, structure and functions of pituitary gland, thyroid gland, parathyroid gland, adrenal gland, pancreas, pineal gland, thymus, and their disorders

UNIT V: Genitourinary System and Introduction to genetics

12 Hours

Urinary system: Anatomy of urinary tract with special reference to anatomy of kidney and nephrons, functions of kidney and urinary tract, physiology of urine formation, role of RAS in kidney and disorders of kidney. Reproductive system: Anatomy of male and female reproductive system, Functions of male and female reproductive system. Introduction to genetics Chromosomes, genes and DNA, protein synthesis, genetic pattern of inheritance

Total Periods: 60

Practical Course Content

Practical physiology is complimentary to the theoretical discussions in physiology. Practical allows the verification of physiological processes discussed in theory classes through experiments on living tissue, intact animals, or normal human beings. This is helpful for developing an insight on the subject.

1. Study of compound microscope.
2. Microscopic study of epithelial and connective tissue
3. Microscopic study of muscular and nervous tissue
4. Identification of axial bones
5. Identification of appendicular bones
6. Introduction to hemocytometry.
7. Enumeration of white blood cell (WBC) count
8. Enumeration of total red blood corpuscles (RBC) count
9. Determination of bleeding time, clotting time.
10. Estimation of hemoglobin content
11. Determination of blood group and erythrocyte sedimentation rate (ESR).
12. Determination of heart rate and pulse rate.
13. Recording of blood pressure.
14. Study of various organ system with the help of models

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15. Determination of respiratory volumes with the help of spirometer

Total Periods: 60

Reference and Textbooks

1. Principles of Anatomy and Physiology by Tortora Grabowski. Palmetto, GA, U.S.A. Wiley Publisher 16th edition, 2020.
2. Ross and Wilson: Anatomy and physiology in Health and Illness, 11th Edition, ChurchHill Livingstone, 2011.
3. West, J.B.: Best and Taylor's Physiological Basis of Medical Practice, 11th Edition, 2007
4. Physiological basis of Medical Practice-Best and Tailor. Williams & Wilkins Co, Riverview, MI USA Wolters Kluwer India Pvt. Ltd.; Thirteenth edition (1 January 2011); Wolters Kluwer India Pvt. Ltd.
5. Textbook of Medical Physiology- Arthur C, Guyton, and John. E. Hall. Miamisburg, OH, U.S.A. 11th Edition, Elsevier Saunders, 2006.
6. Keel and Neil: Samson and Wright's Applied Physiology (12th edition), Oxford University Press. London. 2004
7. Essentials of Medical Physiology by K. Sembulingam and P. Sembulingam. Jaypee brothers' medical publishers, New Delhi.

COURSE OUTCOMES

On completion of this course, the students will be able to

- CO1.** Explain the gross morphology, structural organization, and body systems.
- CO2.** Discuss and correlate body systems with various homeostatic mechanisms and their imbalances.
- CO3.** Identify and distinguish various types of cells, tissues, and organs.
- CO4.** Perform various experiments related to various organ systems.
- CO5.** Analyze the coordinated working pattern of different organs of each system.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3- Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
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Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	3	-	2
CO 2	3	-	-	-	-	-	-	-	1	3	-	2
CO 3	3	3	-	-	1	-	1	-	1	3	-	2
CO4	-	3	-	2	-	1	1	-	1	-	2	1
CO5	-	3	2	2	-	1	1	-	1	-	2	1
Average	3	3	2	2	-	1	1	-	1	-	2	1

25CI102 FUNDAMENTALS OF CLINICAL RESEARCH L-T-P-C: 3-1-2-6

COURSE OBJECTIVES: This module covers the basics of clinical research, clinical trial definition, the drug development process and ethical regulations governing clinical research. Learners would also be exposed to scope of clinical research, career prospects, and overview of documentations.

UNIT I: Introduction to Clinical Research

14 hours

Glossary, Overview of clinical research and its importance Historical perspectives and ethical considerations; Types of clinical research studies(observational, experimental, etc.) Drug Discovery and Development Process: Drug discovery and target identification; Preclinical testing and animal models; Investigational new drug (IND) application process

UNIT II: Clinical Trial Fundamentals

10 hours

Good Clinical Practices: Phases of clinical trials (Micro dosing, Phase I to Phase IV), Randomization and blinding, Outcome measures and endpoints, Tertiary endpoints.

UNIT III: Clinical Trials Designs

12 hours

ICH-E6 Regulations, Clinical trial design (observational and interventional) protocol consent in clinical trials, placebo, bias, and methods to prevent bias, monitoring, problems, and solutions of controlled clinical trials.

UNIT IV: Research Design and Methodology

10 hours

Study designs (observational, experimental, cross-sectional, cohort, case-control, etc.); Sampling techniques and sample size determination; Data collection methods

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(questionnaires, interviews, medical records, etc.)

UNIT V: Regulation in Clinical Research

14 hours

Introduction of Clinical Trial Regulation; European Medicine Agency; Food and Drug Administration (US FDA); NDA and ANDA filing, Drug and cosmetic act; Schedule Y; ICMR Guidelines; Good Clinical Practice (GCP) guidelines; Protection of human subjects and privacy

Total Periods: 60

Practical

Introduction to Clinical Research: and Ethical considerations and regulations, Research Study Designs

1. Observational studies (cross-sectional, case-control, cohort) Developing a Research Question and Hypothesis
2. Experimental studies (randomized controlled trials), Prospective vs. retrospective studies, Developing a Research Question and Hypothesis
3. Formulating a research question, Developing testable hypotheses, Importance of feasibility and relevance
4. Study Protocol Development, Components of a study protocol, Informed consent process and documentation
5. Data collection methods and tools,
6. Basic statistical concepts: Commonly used statistical tests and software,
7. Ethical Considerations and Human Subjects Protection

Total Periods: 60

Reference and Textbooks

1. Friedman, L.M., Furberg, C.D., DeMets, D.L., Reboussin, D.M. and Granger, C.B., 2015. Fundamentals of clinical trials. Springer.
2. Bacchieri, A. and Della Cioppa, G., 2007. Fundamentals of clinical research: bridging medicine, statistics, and operations. Milan: Springer.
3. Rees, J., 2004. The fundamentals of clinical discovery. Perspectives in Biology and Medicine, 47(4), pp.597-607.
4. Spriet, A., Dupin-Spriet, T. and Simon, P., 1994. Methodology of clinical drug trials. Karger Publishers.
5. Chow, S.C. and Liu, J.P., 2008. Design and analysis of clinical trials: concepts and methodologies (Vol. 507). John Wiley & Sons.
6. EDITION, T., 1988. Principles and Practice of Pharmaceutical Medicine.
7. Karlberg, J.P.E. and Speers, M.A., 2010. Reviewing clinical trials: a guide for the ethics committee. Hong Kong.

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8. Kayser, O. and Warzecha, H. eds., 2012. Pharmaceutical biotechnology: drug discovery and clinical applications. John Wiley & Sons.

COURSE OUTCOMES:

On completion of this course, the students will be able to

CO1. Explain the process of drug discovery and drug development and extrapolate data from animals to humans.

CO2. Define the terms in clinical trials and understand the phases of clinical trials, their importance in clinical research and trial related issues.

CO3. Describe the study design and methodologies in clinical trials, requirements, and regulations.

CO4. Understand the roles of regulatory agencies EMA, US FDA, industry sponsors, in clinical research.

CO5. Apply practical knowledge base for various approaches to clinical research.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	3	-	2
CO 2	3	-	-	-	-	-	-	-	1	3	-	2
CO 3	3	3	-	-	1	-	1	-	1	3	-	2
CO4	3	3	2	2	1	1	1	-	1	3	2	2
CO5	3	3	2	2	1	1	1	-	1	3	2	2
Average	3	3	2	2	1	1	1	-	1	3	2	2

25CI103

COMMUNITY HEALTH AND DISEASE

L-T-P-C: 3-1-2-6

COURSE OBJECTIVES:

This course focuses on the principles and practices of community health and disease prevention. It examines the social, environmental, and behavioral factors that influence

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health at the community level.

UNIT I Introduction to Community Health

12 Hours

Definition and scope of community health, social determinants of health, Health disparities in communities, Epidemiology of Common Diseases, Measures of disease frequency and association

UNIT II Community Health Assessment and Study designs

12 Hours

Community Health Assessment, Methods of assessing community health needs, Collection, and analysis of health data, Study designs in epidemiology risk factors and causation of diseases, Disease surveillance and outbreak investigation.

UNIT III Health Centres and Disease Prevention

12 Hours

Identifying health priorities in communities, Health Promotion and Disease Prevention, Health Promotion and Disease Prevention, Health behavior theories and models Health Centres: Community Health Centres, Primary Health Centres, Hospitals, and their organization. Role of Health Centres in Disease Surveillance and Disease Prevention

UNIT IV Disease Surveillance Programs

14 Hours

Role of Health Centres in Disease Surveillance and Disease Prevention. Strategies for health promotion and behavior change, Primary, secondary, and tertiary prevention approaches, Community-Based Interventions, Community organizing and mobilization, Designing and implementing community health programs, Evaluation of community interventions.

UNIT V Environment Health Safety

10 Hours

Introduction to Environmental Health and Safety, Environmental risk factors and health effects, Water and air quality management, Occupational health, and safety

Total Periods: 60

Practical Course Content

Practical activities and experiences enhance student's understanding of community health and disease:

1. Field Visits: Visit local community health centers, public health departments, or non-governmental organizations (NGOs)
2. Community Health Assessments, identify health needs, collect, and analyze

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data, and develop recommendations for improving community health.

3. **Health Promotion Campaigns:** Plan and implement a health promotion campaign targeting a specific health issue in the community.
4. **Participatory Research:** Engage in participatory research projects that involve community members as active partners in the research process.
5. **Volunteer Work:** Volunteer with community organizations or public health agencies that focus on community health.
6. **Reflection and Discussion:** Reflect and describe various challenges, successes, and ethical considerations encountered during practical activities related to community health.

Total Periods: 60

References

1. Harkness, G.A. and DeMarco, R.F., 2016. Community and public health nursing: Evidence for practice. Wolters Kluwer.
2. Dicker, R.C., Coronado, F., Koo, D. and Parrish, R.G., 2006. Principles of epidemiology in public health practice; an introduction to applied epidemiology and biostatistics. Third edition
3. Hunt, R., 2009. Introduction to community-based nursing. Lippincott Williams & Wilkins.
4. Grim, M.L. and Fertman, C.I. eds., 2010. Health promotion programs: from theory to practice. John Wiley & Sons.
5. Brown, A.E.C., Hobart, T.R. and Morrow, C.B. eds., 2019. Bioethics, Public Health, and the Social Sciences for the Medical Professions: An Integrated, Case-Based Approach. Springer.
6. Szklo, M. and Nieto, F.J., 2014. Epidemiology: beyond the basics. Jones & Bartlett Publishers.

COURSE OUTCOMES: On completion of this course, the students will be able to

CO1: Explain and understand the concept of community health, its significance, and epidemiological studies.

CO2: Identify and distinguish the social, environmental, and behavioral determinants of health in community.

CO3: Describe the principles and strategies for preventing diseases at community level.

CO4: Describe the ethical considerations in community health and disease prevention.

CO5: Analyze and evaluate the public health intervention programs like health promotion

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campaigns and disease surveillance in community health and disease prevention.

Co-Relationship Matrix

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	-	-	-
CO 2	3	-	-	-	-	-	-	-	1	-	-	-
CO 3	-	-	-	-	3	-	-	-	1	-	-	-
CO4	-	3	-	2	-	1	1	-	1	-	-	-
CO5	-	3	2	2	3	1	1	3	1	-	2	-
Average	3	3	2	2	1	1	1	3	1	-	2	-

25CI104

Ability Enhancement/Co-curricular I

L-T-P-C:2-1-1.5-4.5

SEMESTER II

25CI201

PRINCIPLES OF BIOCHEMISTRY L-T-P-C:2-1-1.5-4.5

Course Objectives

The main objective of the course will be to expose students to thermodynamics basis, bioenergetics of reaction, the basic role of biomolecules and their chemical interactions inside the cell. It provides deeper insight into structures, properties and functions of major biomolecules and metabolic pathways in the living systems.

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Unit-I Introduction to biophysics

12 Hours

Introduction and history of biophysics, main features of quantum theory, elementary particles, and their interactions. Bioenergetics, endergonic and exergonic reactions, Laws of thermodynamics, entropy, enthalpy, Gibb's free energy, standard Gibb's free energy, ATP, and different high energy compounds. Properties and role of water. Buffers - action, capacity, relationship between pH & pKa (Henderson -Hasselbalch equation) and its importance.

Unit-II Proteins & Enzymes

11 Hours

Amino acids and peptides- classification, chemical and physical properties, Introduction to protein structure and function, secondary, tertiary, and quaternary structure of proteins, fibrous and globular proteins, protein folding and Anfinsen's experiment. Amino acid metabolism-Amino acid deamination and transamination, urea cycle. Introduction to enzymes, classification of enzymes, mechanism of action, Michaelis-Menten equation and significance of K_m , V_{max} and K_{cat} .

Unit-III Carbohydrates

12 Hours

Monosaccharides-structure of aldoses and ketoses, open and ring structure of sugars, conformations of sugars, mutarotation, anomers, epimers and enantiomers. Disaccharides- maltose, lactose, and sucrose. Polysaccharides-homo and heteropolysaccharides, structural and storage polysaccharides. Anabolism and catabolism, glycolysis citric acid cycle and gluconeogenesis.

Unit-IV Lipids

10 Hours

Definition, biological functions, general formulae, nomenclature and properties of fatty acids, essential and non-essential fatty acids, classification of lipids, building blocks of lipids - fatty acids, glycerol, ceramide, saponification number and iodine number, suitability of triglycerides as storage lipids, saponification number and iodine number. Introduction to lipid micelles, monolayer and bilayer, transport of fatty acids.

Total Periods: 45

Practical

1. To prepare different solutions-based molarity, normality, and percentage.
2. To prepare buffer solution and pH measurement
3. Qualitative test for carbohydrates[4]. Qualitative test for amino acids.
4. Titration of Amino acid (Neutral) with a strong base and acid.
5. Quantitative estimation of protein by Bradford/Bicinchoninic acid method
6. Assay of salivary amylase.

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7. Qualitative test for lipids.
8. Quantitative test for lipids- Salkowski/Lieberman-Burchard test
9. Colorimetric estimation of urea/blood urea nitrogen (BUN).

Total Periods: 45

Reference Books

1. Lehninger: Principles of Biochemistry (2017) 7th ed., Nelson, D.L. and Cox, M.M., W.H.Freeman, and Company (New York), ISBN 13: 978-1464126116.
2. Textbook of Biochemistry with Clinical Correlations an Indian Adaptation (2022) 7th ed., Devlin, T.M., John Wiley & Sons, Inc., ISBN: 978-9354641558.
3. Biochemistry (2019) 9th ed., Berg, J.M., Tymoczko, J.L. and Stryer, L., W.H Freeman and Company (New York), ISBN 13: 978-1319114671.
4. Principles and Techniques of Biochemistry and Molecular Biology (2018) 8th ed., Wilson, K. and Walker, J. Cambridge University Press, ISBN 13: 978-1316614761.
5. Introduction to Practical Biochemistry, Sawhney, S.K. and Singh R. so Narosa Publishing House (New Delhi), ISBN-13: 978-8173193026.

Course Outcomes

Course outcomes: After the completion of course, the students will be able to:

CO1: Understand thermodynamics basis of life, bioenergetics of a reaction and pathway and different intermolecular interactions in structural organization of proteins.

CO2: Describe the structure and functions of different chemical building blocks (carbohydrates, proteins, and lipids) of life.

CO3: Identify and draw structures of various types of biomolecules (carbohydrate, lipids, and proteins).

CO4: Classify enzymes in different categories and explain what enzyme does, how enzyme works and primary biochemical pathways leading to synthesis and catabolism of major biomolecules.

CO5: Demonstrate skills to prepare the solutions, buffers and identify and analyze any biological molecules in the given sample.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3- Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
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Course Outcomes												
CO 1	2	-	-	-	-	-	-	-	1	-	-	-
CO 2	2	-	-	-	-	-	-	-	1	-	-	-
CO 3	2	2	-	-	-	-	-	-	1	-	-	-
CO4	2	2	2	2	1	-	1	-	1	-	2	-
CO5	2	2	2	2	1	-	1	-	1	-	2	-
Average	2	2	2	2	1	-	1	-	1	-	2	-

25CI202 COMPUTER APPLICATIONS AND BIOINFORMATICS

L-T-P-C:2-1-1.5-4.5

Course Objectives

Theory:

- The basic objective is to give students an introduction to the basic practical techniques of Computer Application and Bioinformatics.
- Emphasis will be given to the application of Computer Application and Bioinformatics and biological databases to problem solving in real research problems.
- The students will become familiar with the use of a wide variety of internet applications, computer application, biological database, sequence alignment, biological database management, molecular docking and drug designing and will be able to apply these methods to research problems.

Practical:

- The aim is to provide practical training in bioinformatics methods including accessing the major public sequence databases, structure database, drug database.
- Use of the different computational tools to find sequences, protein structure and drug molecules.
- Analysis of protein and nucleic acid sequences by various software packages.
- It also provides a step by step, theoretical and practical introduction to the development of useful tools for automation of complex computer jobs and making these tools accessible on the network from a web browser

Unit I: Introduction to Computer Application

5 hours

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Basics of computer hardware, software, and networking. Operating Systems software and Application Systems software. Windows features, Microsoft office, data format for biological samples.

Unit II: Biological Databases

10 hours

Definition and History and Applications of Computational Biology and Bioinformatics, Internet resources, various databases and bioinformatics tools, organization of databases. Sequencing database, 3D Structure Database, Chemical Structure database, Gene Expression database, Derived Databases, Structure classification database, Protein-Protein interaction database and Pathway database.

Unit III: Sequence Alignment and Phylogenetic Tree

10 hours

File formats, Basic concepts of sequence analysis, Scoring matrices, Pair wise sequence alignments, Multiple sequence alignment, Database Searches: Keyword-based searches and Sequence-based searches. Phylogenetic Trees: phylogenetic tree representation, building phylogenetic trees,

Unit IV: Structure Prediction

10 hours

Overview and Introduction to Protein Structure, Sequence-Sequence Alignment Methods, Sequence Based Secondary Structure Prediction. Visualization of structures using Rasmol or ADT. Fundamentals of the methods for 3D structure prediction, Homology/comparative Modeling. AI based protein structure prediction.

Unit V: Molecular Docking and Drug Designing

10 hours

General approach to discovery of new drugs, lead discovery, lead modification physiochemical, principles of drug action, 3D database search, computer aided drug design, AI based drug screening, docking, molecular modelling in drug design, structure-based drug design.

Total Periods: 45

Practicals

1. Referencing in Scientific literature and their practical usage, PubMed
2. Sequence retrieval
3. Biological Databases: Study of different biological databases (esp. the ones given below), Format.
4. Pair wise sequence alignment, Local and Global alignment – Algorithms
5. DOT matrix analysis
6. Databases search for homologous sequence using (BLAST) and (FASTA)

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7. MSA: (Clustal W, Clustal X), Algorithms-MSA, Progressive alignment etc, Problems with MSA method, Statistics behind MSA
8. MUSCLE, T-COFFEE
9. Protein structure prediction tools (2D and 3D structure prediction)
10. Molecular Docking using Autodock
11. Drug Designing using Chems sketch
12. AI application in computational biology and bioinformatics

Total Periods: 45

Recommended Books/ Resources:

1. Bioinformatics and Computational Biology-A Primer for Biologists by Basant K. Tiwary. 2022, ISBN: 978-981-16-4240-1
2. Lopes H, editor. Computational Biology and Applied Bioinformatics. InTech; 2011. Available from: <http://dx.doi.org/10.5772/772>
3. Encyclopedia of Bioinformatics and Computational Biology-ABC of Bioinformatics. Shoba Ranganathan, Kenta Nakai, Christian Schonbach. August 21, 2018, ISBN:9780128114148
4. Introduction to Bioinformatics, Teresa Attwood, David Parry-Smith, Pearson Education. ISBN: 978-8178085074
5. Bioinformatics: A Practical Guide to the Analysis of Genes and Proteins, Second Edition, Andreas D. Baxevanis, B. F. Francis Ouellette. A John Wiley & Sons, Inc., Publication. ISBN: 978-0471478782
6. Jianyuan Deng and others, Artificial intelligence in drug discovery: applications and techniques, Briefings in Bioinformatics, Volume 23, Issue 1, January 2022, bbab430, <https://doi.org/10.1093/bib/bbab430>

Course Outcomes

After completion of the course, students will be able to

CO1: Understand the basics of computer application and bioinformatics.

CO2: Identify and define the basic concepts of bioinformatics and its significance in biological data analysis.

CO3: Demonstrate the ability to choose the methods to symbolize and manage the different types of biological database, sequence alignment and phylogenetic tree analysis.

CO4: Overview about biological macromolecular structures and structure prediction methods, molecular docking, and drug designing.

CO5: Develop competency in bioinformatics for solving different biological problems, data handling process and data retrieval process from different, biological databases, usage of different software for analyzing biological data, sequence alignment, molecular

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docking, and drug designing.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	2	-	-	-	-	-	-	-	1	-	-	-
CO 2	2	2	-	-	-	-	1	-	1	-	2	-
CO 3	2	2	-	-	-	-	1	-	1	-	2	-
CO4	2	2	2	2	1	-	1	-	1	-	2	-
CO5	2	2	2	2	1	-	1	-	1	-	2	-
Average	2	2	2	2	1	-	1	-	1	-	2	-

25CI203 FUNDAMENTALS OF BIOSTATISTICS L-T-P-C:2-1-1.5-4.5

Course Objectives

This course introduces the fundamental principles and techniques of Biostatistics. It emphasizes the application of statistical methods in Health Sciences & Technology. Students will learn to analyze and interpret data, conduct hypothesis testing, and apply statistical techniques to make evidence-based decisions in various healthcare and research settings. The course also provides hands-on experience with statistical software and practical exercises to reinforce the concepts learned.

Unit I: Introduction to Biostatistics and Data Types

5 Hours

Understand the importance of biostatistics in healthcare and clinical research. Define different types of data. Variables: continuous, nominal, ordinal. Scales of measurement.

Unit II: Descriptive Statistics and Probability Distributions

15 Hours

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Measures of central tendency: mean, median and mode. Measures of variability: range, variance, and standard deviation. Frequency distributions and graphical representations of data: histograms, box plots, and scatter plots. Basic principles of Probability. Discuss probability distributions: discrete and continuous, normal distribution.

Unit III: Sampling Techniques and Sampling Distributions

5 Hours

Understand different sampling techniques and their applications. Discuss sampling distributions and the Central Limit Theorem. Calculate confidence intervals and understand their interpretation.

Unit IV: Hypothesis Testing – Parametric and Non-parametric methods 10 Hours

One-Sample Tests: formulate null and alternative hypotheses, conduct hypothesis tests for one-sample mean and proportion, Interpret test results and inferencing. Two-Sample Tests: hypothesis testing for two independent samples, Compare means and proportions between groups. Analysis of Variance (ANOVA), Understand the principles of analysis of variance, one-way ANOVA, and result interpretation. Apply post hoc tests for multiple comparisons. Chi square tests, Kruskal-Wallis, Fischer's Exact Test. Compare parametric and nonparametric tests.

Unit V: Correlation and Regression Analysis

10 Hours

Understand the concepts of correlation and regression. Correlation coefficients. Perform simple linear regression and assess the model's goodness of fit. Multiple regression for multiple variables. Understand the principles and applications of survival analysis. Estimate survival probabilities using Kaplan-Meier curves. Apply Cox proportional hazards regression.

Total Periods: 45

Practical

1. Perform basic data manipulation tasks such as importing, exporting, and cleaning data using suitable statistical software (e.g., R, SPSS, or SAS).
2. Calculate measures of central tendency (mean, median, mode) and variability (range, variance, standard deviation), construct frequency distributions and generate box plots, scatter plots, histograms using the given data.
3. Formulate null and alternative hypotheses. Perform one-sample and two-sample hypothesis tests. Interpret and communicate results using appropriate statistical language.
4. Conduct one-way ANOVA to compare means across multiple groups and perform posthoc tests for multiple comparisons and interpret the results.
5. Apply nonparametric tests and compare the results with their parametric counterparts.

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6. Perform chi-square tests to analyze categorical data. Test for independence and homogeneity in contingency tables and interpret results to assess associations between variables.
7. Calculate correlation coefficients between variables, interpret and evaluate the strength of the relationship.
8. Perform simple linear regression analysis. Make predictions using regression models.
9. Design a basic study and analyze the experimental data using appropriate statistical tests and draw conclusions.

Total Periods: 45

Reference Books

1. Dawson, B. and Trapp, R.G. 2019. Basic and Clinical Biostatistics. 5th ed. Lange medical books/McGraw-Hill Inc.
2. Zar, Jerrold H. 2014. Biostatistical Analysis. 5th ed. Pearson Education.
3. Selvin, S. 2011. Biostatistics: Statistical Tools for Epidemiological Research. Oxford University Press.
4. Lepš, J. and Šmilauer, P. 2020. Biostatistics with R: An introductory guide for Field Biologists. 1st ed. Cambridge University Press.

Course Outcomes

Upon completion of the course, students will be able to

- CO1.** Understand the basic concepts and principles of biostatistics and its relevance to healthcare and clinical research.
- CO2.** Apply appropriate statistical techniques to analyze and investigate scientific questions in healthcare and research settings.
- CO3.** Design and conduct basic experiments or studies to investigate scientific questions in the relevant disciplines.
- CO4.** Apply statistical software tools to perform data analysis, interpret the output, and effectively communicate the findings of statistical analyses in healthcare and clinical research.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3- Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
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Course Outcomes												
CO 1	2	-	-	-	-	-	-	-	1	-	-	-
CO 2	2	2	-	-	-	-	1	-	1	-	2	-
CO 3	2	2	-	-	-	-	1	-	1	-	2	-
CO4	2	2	2	2	1	-	1	-	1	-	2	-
Average	2	2	2	2	1	-	1	-	1	-	2	-

25CI204	Ability Enhancement/Co-curricular-I	Credits 2
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25CI205	Elective Course	L-T-P-C:2-1-1.5-4.5
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25CI205

BIOSAFETY GUIDELINES

L-T-P-C: 2-1-1.5-4.5

Course Objective: This course provides details of biosafety principles and guidelines for working with biological agents in laboratory and healthcare settings. This course will cover the important concepts of containment measures, risk assessment, personal protective equipment(PPE), regulatory frameworks and waste management. Students will gain a comprehensive understanding of biosafety practices ensuring safe storage, handling, and disposal of biological materials.

UNIT I. Introduction to Biosafety

13 Hours

Definition and importance of biosafety, Historical overview and major biosafety incidents, Regulatory frameworks and guidelines (e.g., WHO, CDC, OSHA) Risk Assessment and Management, Principles of risk assessment Identification and characterization of biological hazards, Biosafety levels and facility design considerations, Containment and Engineering Controls, Primary, secondary, and tertiary containment measures, Laboratory design and engineering controls, biological safety cabinets and other containment equipment, Digital monitoring systems and smart technologies for real-time biosafety monitoring

UNIT II Personal Protective Equipment (PPE) and Transport of Biological Materials

12 Hours

Personal Protective Equipment (PPE), Selection and use of PPE based on risk assessment Proper donning and doffing of PPE, Maintenance and disposal of PPE, Innovative PPE technologies (e.g., smart textiles, antimicrobial coatings) and their application in biosafety. Safe Handling and Transport of Biological Materials, Packaging, labeling, and documentation requirements, Shipping regulations and transport considerations, Security measures for handling select agents and toxins, Implementation of blockchain and traceability systems for secure tracking and monitoring of biological materials.

UNIT III. Waste Management and Decontamination

10 Hours

Introduction to waste management and decontamination, proper disposal of biohazardous waste, Decontamination techniques and disinfectants, Spill response and emergency procedures, Biosafety in Specific Settings, Healthcare-associated infections (HAIs), Biosafety in clinical laboratories and diagnostic facilities, Biosafety in animal research facilities, Novel decontamination methods (e.g., advanced oxidation processes, plasma sterilization).

UNIT IV Emerging Trends and Technologies in Biosafety

10 Hours

Advanced containment systems and technologies, Genome editing and biosafety implications, Biosafety challenges in the era of globalization and bioterrorism threats,

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Biosafety in Specific Settings, including the application of automation, robotics, and artificial intelligence in biosafety practices, Gene drive technology, viral vectors, and their biosafety considerations.

Total Periods: 45

Course Outcomes: Upon completion of this course, the student will be able to

CO1: Explain the importance of biosafety and its role in preventing occupational hazards and the spread of infectious diseases.

CO2: Familiarize students with national and international biosafety guidelines, regulations, and standards.

CO3: Develop knowledge and skills to assess risks associated with biological agents and implement appropriate containment measures.

CO4: Identify different types of personal protective equipment (PPE) and their proper use in biosafety practices.

CO5: Explain the techniques for safe handling, storage, and transport of biological materials.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3- Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 1	2	-	-	-	-	-	-	-	1	-	-	-
CO 2	2	-	-	-	-	-	-	-	1	-	-	-
CO 3	2	2	2	-	1	-	1	-	1	-	2	-
CO4	2	2	2	1	1	-	1	1	1	1	2	1
CO5	2	2	2	1	1	-	1	1	1	1	2	1
Average	2	2	2	1	1	-	1	1	1	1	2	1

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SEMESTER III**

25CI301

PHARMACOLOGY AND TOXICOLOGY

L-T-P-C

2-1-1.5-4.5

Course Objectives

The main purpose of the subject is to understand what drugs do to the living organisms and how their effects can be applied to therapeutics. The subject covers the information about the drugs like, mechanism of action, physiological and biochemical effects (pharmacodynamics) as well as absorption, distribution, metabolism, and excretion (pharmacokinetics) along with the adverse effects, clinical uses, interactions, doses, contraindications, and routes of administration of different classes of drugs.

Unit I: General Pharmacology

8 Hours

Introduction to Pharmacology- nature and source of drugs, essential drugs concept and routes of drug administration, Basic terminologies in pharmacology, types of agonists and antagonists. Overview of Pharmacodynamics: Principles and mechanisms of drug action, receptors pharmacology, drug receptors interactions signal transduction mechanisms, dose response relationship, therapeutic index, combined effects of drugs and factors modifying drug action. Pharmacokinetics- Membrane transport, absorption, distribution, metabolism, and excretion of drugs. Enzyme induction, enzyme inhibition. Adverse drug reactions. Overview of phases of clinical trials and pharmacovigilance.

Unit II: Pharmacology of drugs acting on Nervous system

8Hours

Neurohumoral transmission, co-transmission, and classification of neurotransmitters. Classification, mechanism, use and adverse effect of drugs in: Parasympathomimetic, Parasympatholytic, Sympathomimetics, sympatholytic., Neuromuscular blocking agents, skeletal muscle relaxants (peripheral), Local anesthetic agents, and glaucoma, General anesthetics and pre-anesthetics, Sedatives, hypnotics and centrally acting muscle relaxants, Anti-epileptics, Antipsychotics, antidepressants, anti-anxiety agents, anti-manics, Anti-parkinsonian Anti-Alzheimer's, and Opioid analgesics.

Unit III: Pharmacology of drugs acting on cardiovascular system and blood: 10 hours

Classification, mechanism, use and adverse effect of: Drugs used in congestive heart failure, Anti-hypertensive drugs, Anti-anginal drugs, Anti-arrhythmic drugs, Anti-hyperlipidemic drug, Hematinic, coagulants and anticoagulants. Fibrinolytics and anti-platelet drugs.

Unit IV: Drugs acting on Gastrointestinal tracts and Renal system:

11 hours

Drugs for Peptic ulcer, diarrhea, and constipation, Antiemetics and Prokinetics. Physiology of Urine formation, Pharmacology of diuretics and antidiuretics. Drugs acting on

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respiratory and NSAIDs: Expectorants, Antitussives, Drugs for Asthma, Pharmacology of Prostaglandins and NSAIDs. Chemotherapeutic agents and Immunopharmacology: General consideration of antimicrobial agents. Pharmacology of antifungal, antibacterial, antiviral, antimalarial and anticancer drugs. Pharmacology of Immunomodulators.

UNIT V. Toxicology

8 hours

Principles of Toxicology, Introduction to toxicology and toxicokinetic

Routes of exposure and factors influencing toxicity, Mechanisms of toxicity and types of toxicological hazards, Evaluation of Chemical Hazards, Risk assessment and risk management Carcinogenicity, mutagenicity, and teratogenicity, Environmental toxicology, and ecotoxicology

Total Periods: 45

Practical: Pharmacology and Toxicology Lab

1. Introduction to experimental pharmacology, commonly used instruments, and laboratory animals in experimental pharmacology.
2. Common laboratory techniques. Blood withdrawal, serum and plasma separation, anesthetics and euthanasia are used for animal studies.
3. Study of different routes of drugs administration in mice/rats.
4. Effect of drugs on ciliary motility of frog esophagus
5. Effect of drugs on rabbit eye.
6. Effects of skeletal muscle relaxants using rota-rod apparatus.
7. Study of anxiolytic activity of drugs using rats/mice (simulation based)
8. Effect of drugs on locomotor activity using actophotometer.
9. Anticonvulsant effect of drugs by MES and PTZ method (simulation based)
10. Effect of drugs on blood pressure and heart rate of dog.
11. Study of diuretic activity of drugs.
12. Effect of physostigmine and atropine on DRC of acetylcholine using frog rectus abdominis muscle and rat ileum
13. Bioassay Methods of drugs like acetylcholine, adrenaline, insulin, d-tubocurarine
14. Anti-inflammatory activity of drugs using carrageenan induced paw-edema model (simulation based).
15. Estimation of serum biochemical parameters by using semi- autoanalyzer.

Total Periods: 45

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Reference Books (latest edition)

1. Rang, H.P., Dale, M.M., Ritter, J.M. and Moore, P.K., 2003. Pharmacology, ChurchillLivingstone. New York, pp.3-4.
2. Goodman, L.S., 1996. Goodman and Gilman's the pharmacological basis of therapeutics (Vol.1549, pp. 1361-1373). New York: McGraw-Hill.
3. Goodman, L.S. and Gilman, A., 1955. The pharmacological basis of therapeutics. TheMacmillan.
4. Howland, R.D., and Mycek, M.J., 2006. Lippincotts Illustrated reviews, Pharmacology.Teaching Learning, 5, p.5.
5. Tripathi, K.D., 2003. Essentials of medical pharmacology, Jaypee Brothers. Med Pub Ltd NewDelhi Edn, 5, pp.93-94.

Course Outcomes After the completion of course, the students will be able to:

CO1: Explain targets and pharmacology of underlying diseases.

CO2: Understand the pharmacological actions of different categories of drugs.

CO3. Apply the basic pharmacological knowledge in the prevention and treatment of variousdiseases.

CO4: Analyse the effect of drugs on animals by simulated experiments.

CO5: Report the clinical applications, adverse effects and toxicities of drugs used as medicines.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Program Outcomes												
Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	-	-	-	-	1	1	-	1
CO 3	3	3	2	-	3	-	-	-	1	1	2	1
CO4	3	3	2	2	3	1	1	-	1	1	2	1
CO5	3	3	2	2	3	1	1	1	1	1	2	1
Average	3	3	2	2	3	1	1	1	1	1	2	1

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25CI302 DESIGN AND MANAGEMENT OF CLINICAL TRIALS

L-T-P-C: 2-1-1.5-4.5

COURSE OBJECTIVE

This course provides an overview of design, conduct, and management of clinical trials. This course describes the fundamental principles of clinical trial design, ethical considerations, data collection and analysis, regulatory requirements, and trial management strategies. It helps students in developing a strong foundation in planning and executing clinical trials, ensuring patient safety, and generating reliable and valid evidence for healthcare interventions.

UNIT 1. Clinical Trial Design **10 Hours**

Study objectives and research questions, Selection of study population and eligibility criteria, Randomization, blinding, and allocation concealment, Endpoints and Sample Size Calculation, Selection and measurement of primary and secondary endpoints, Power and sample size calculation, Considerations for non-inferiority and equivalence trials.

UNIT II. Data Collection and Management **10 Hours**

Case report form (CRF) development and data collection tools, Data validation and quality control, electronic data capture (EDC) systems and data management. Statistical Considerations in Clinical Trials, Overview of statistical concepts and hypothesis testing, Analysis plans and statistical analysis methods, Interim analysis, and adaptive trial design.

UNIT III. Participant Recruitment, Retention, and Adherence **12 Hours**

Strategies for participant recruitment and informed consent, Retention and follow-up of study participants, Adherence monitoring and strategies. Clinical Trial Monitoring and Safety, Role of clinical research monitors, Data safety monitoring boards (DSMBs) and interim analysis, Pharmacovigilance, and adverse event reporting

UNIT IV. Good Clinical Practice (GCP) and Trial Management **8 Hours**

Principles of GCP and regulatory compliance, Responsibilities of sponsors, investigators, and study coordinators, Study site management and coordination.

UNIT V. Virtual and decentralized trials **5 Hours**

Virtual and decentralized trials, Adaptive trial designs and precision medicine approaches, big data, and real-world evidence in clinical research

Total Periods: 45

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PRACTICAL

Hands on training on Protocol Development

1. Hands on training on Study Feasibility Assessment
2. Ethical Considerations and Informed Consent
3. Data Collection and Case Report Form (CRF) Design
4. Randomization and Blinding
5. Study Monitoring and Quality Assurance
6. Data Analysis and Interpretation
7. Safety Monitoring and Adverse Event Reporting
8. Regulatory Compliance and Reporting
9. Case Studies and Practical Scenarios.

Total Periods: 45

REFERENCES

1. Spriet, A., Dupin-Spriet, T. and Simon, P., 1994. Methodology of clinical drug trials. Karger Publishers.
2. Chow, S.C. and Liu, J.P., 2008. Design and analysis of clinical trials: concepts and methodologies (Vol. 507). John Wiley & Sons.
3. EDITION, T., 1988. Principles and Practice of Pharmaceutical Medicine.
4. Rees, J., 2004. The fundamentals of clinical discovery. Perspectives in Biology and Medicine, 47(4), pp.597-607.
5. Spriet, A., Dupin-Spriet, T. and Simon, P., 1994. Methodology of clinical drug trials. Karger Publishers.
6. Chow, S.C. and Liu, J.P., 2008. Design and analysis of clinical trials: concepts and methodologies (Vol. 507). John Wiley & Sons.
7. Karlberg, J.P.E. and Speers, M.A., 2010. Reviewing clinical trials: a guide for the ethics committee. Hong Kong.

COURSE OUTCOMES

On completion of this course, the students will be able to

- CO1.** Explain the purpose and importance of clinical trials in evaluating the safety and efficacy of healthcare interventions.
- CO2.** Describe the fundamental principles of clinical trial design, including study objectives, endpoints, sample size calculation, randomization, and blinding.
- CO3.** Identify the ethical considerations and regulatory requirements in

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clinical trial conduct, including informed consent, data privacy, and institutional review board (IRB) approval.

CO4. Develop skills in data collection, management, and analysis in clinical trials, including case report form (CRF) development and statistical considerations.

CO5. Describe the role of monitoring, quality assurance, and data safety monitoring boards(DSMBs) in ensuring trial integrity and patient safety.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	-	-	-	-	1	1	-	1
CO 3	3	2	2	-	3	-	-	-	1	1	2	1
CO4	3	2	2	2	3	1	2	-	1	1	2	
CO5	3	2	2	2	3	1	2	-	1	1	2	
Average	3	2	2	2	3	1	2	-	1	1	2	1

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**25CI303 CLINICAL SAMPLING AND ANALYTICAL TECHNIQUES
L-T-P-C:2-1-2-5**

COURSE OBJECTIVE

This course provides an overview of clinical sampling and analytical techniques used in healthcare and biomedical research. It covers the principles and procedures for sample collection, handling, and storage, as well as the analytical techniques used for measuring various clinical parameters. Learners will learn the importance of reliable and accurate hands on performing and analysing clinical samples.

UNIT I. Introduction to Clinical Sampling and Analysis 10 Hours

Importance of clinical sampling in healthcare and research, Ethical considerations and informed consent in sample collection, Overview of analytical methods used in clinical sample analysis. Types of Clinical Samples and Collection Techniques, blood, and serum samples: venipuncture, fingerstick, and arterial sampling, Urine, saliva, and other bodily fluid samples: collection methods and considerations, Tissue, and biopsy samples: collection techniques and processing.

UNIT II. Clinical Sample Handling 8 Hours

Clinical sample handling, storage, and transport, pre-analytical factors: sample collection tubes, anticoagulants, and additives, Sample handling procedures: centrifugation, aliquoting, and labeling, Storage conditions and sample stability, transport considerations and chain of custody. Analytical techniques in clinical sample analysis.

UNIT III. Quality Control and Assurance 8 Hours

Quality control and assurance in clinical sample analysis, calibration and standardization of analytical instruments used in clinical sample handling. Internal and external quality control measures, Validation and verification of analytical methods, Accreditation, and proficiency testing.

UNIT IV. Common Clinical Tests and Interpretation of Results 8 Hours

Complete blood count (CBC) and differential analysis, Basic metabolic panel (BMP) and liver function tests, Lipid profile, hormone assays, and therapeutic drug monitoring, Microbiological and molecular diagnostic tests.

UNIT V. Point-of-Care Testing (POCT) and Rapid Diagnostic 11 Hours

Overview of POCT devices and their applications, Selection and use of POCT devices in different clinical settings, Quality assurance and regulatory considerations for POCT. Emerging Technologies in Clinical Sample Analysis, Advances in molecular diagnostics: PCR, next-generation sequencing (NGS), Point-of-care molecular testing and biosensors, Omic technologies: genomics, proteomics, and metabolomics

Total Periods: 45

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Practical: Clinical Sampling and Analytical Lab

This lab course complements the theoretical knowledge gained in the Clinical Sampling and Analytical course. Students will gain hands-on experience in various clinical sampling techniques and analytical methods used in healthcare and biomedical research.

1. **Introduction to Clinical Sampling and Laboratory Safety:** To study Clinical sampling techniques, Laboratory safety protocols and practices.
2. **Introduction to laboratory instruments and equipment.**
3. **Blood Sample Collection and Analysis:** To study and understand Venipuncture technique and blood collection, Preparation and analysis of blood smears, Hematology tests: complete blood count (CBC) and differential analysis
4. **Urine Sample Collection and Analysis:** Study Collection methods and considerations for urine samples, Analysis of urine samples using dipstick tests, microscopic examination of urine sediment
5. **Biochemical Analysis of Clinical Samples:** Explain Principles of spectrophotometry and spectrometry, Analysis of clinical samples for biochemical parameters (e.g., glucose, cholesterol, liver enzymes), Use of automated analyzers for biochemical analysis
6. **Immunological and Serological Analysis:** Explain Enzyme-linked immunosorbent assays (ELISA), Rapid diagnostic tests for infectious diseases, Antibody, and antigen detection techniques.
7. **Molecular Diagnostics:** Understand DNA extraction techniques from clinical samples, Polymerase chain reaction (PCR) for molecular diagnostics, Next-generation sequencing (NGS) techniques
8. **Quality Control and Assurance in Clinical Sample Analysis:** Explain Calibration and standardization of instruments, Internal and external quality control measures, Validation of analytical methods
9. **Data Analysis and Interpretation:** Perform Statistical analysis of clinical data, Interpretation of laboratory test results, Reporting and documentation of laboratory findings.
10. **Emerging Technologies in Clinical Sample Analysis:** Introduction to advanced techniques (e.g., mass spectrometry, proteomics, genomics), Hands-on experience with emerging technologies (if available).

Total Periods: 60

REFERENCE and Textbooks

1. Harr, R.R., 2012. Medical laboratory science review. FA Davis.
2. Bishop, M.L., 2020. Clinical Chemistry: Principles, Techniques, and Correlations, Enhanced Edition: Principles, Techniques, and Correlations. Jones & Bartlett Learning.

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3. B McKenzie, S., 2015. Clinical laboratory hematology. Lynne Williams.
4. Grody, W.W., Nakamura, R.M., Kiechle, F.L. and Strom, C. eds., 2009. Molecular diagnostics: techniques and applications for the clinical laboratory. Academic Press.
5. Rifai, N., 2017. Tietz textbook of clinical chemistry and molecular diagnostics-e-book. Elsevier Health Sciences.

COURSE OUTCOME: Upon the completion of the course the student will be able to

CO1: Explain the principles and importance of clinical sampling in healthcare and biomedical research.

CO2: Describe different types of clinical samples and their collection techniques.

CO3: Demonstrate skills in performing common clinical tests, analytical techniques used for clinical sample analysis.

CO4: Explain quality control and quality assurance measures in clinical sample analysis.

CO5: Discuss the emerging technologies and advancements in clinical sample analysis.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	-	-	-	-	1	1	-	1
CO 3	3	3	2	-	3	-	-	-	1	1	2	1
CO4	3	3	2	2	3	1	1	-	1	1	2	1
CO5	3	-	2	2	3	1	1	1	1	1	2	1
Average	3	3	2	2	3	1	1	1	1	1	2	1

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25CI304 ENVIRONMENTAL SCIENCE

L-T-P-C:2-1-0-4

COURSE OBJECTIVES:

To introduce the basic concepts of environment, ecosystems and biodiversity and emphasize on the biodiversity of India and its conservation. To impart knowledge on the causes, effects and control or prevention measures of environmental pollution and natural disasters.

UNIT I ENVIRONMENT AND BIODIVERSITY

10Hours

Definition, scope and importance of environment – need for public awareness. Eco-system and Energy flow– ecological succession. Types of biodiversity: genetic, species and ecosystem diversity– values of biodiversity, India as a mega-diversity nation – hot-spots of biodiversity – threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – endangered and endemic species of India – conservation of biodiversity: In-situ and ex-situ.

UNIT II ENVIRONMENTAL POLLUTION

10Hours

Causes, Effects and Preventive measures of Water, Soil, Air and Noise Pollutions. Solid, Hazardous and E-Waste management. Case studies on Occupational Health and Safety Management system (OHASMS). Environmental protection, Environmental protection acts .

UNIT III RENEWABLE SOURCES OF ENERGY .

8Hours

Energy management and conservation, New Energy Sources: Need of new sources. Different types new energy sources. Applications of- Hydrogen energy, Ocean energy resources, Tidal energy conversion. Concept, origin and power plants of geothermal energy.

UNIT IV SUSTAINABILITY AND MANAGEMENT

9Hours

Development , GDP ,Sustainability- concept, needs and challenges-economic, social and aspects of sustainability-from unsustainability to sustainability-millennium development goals, and protocols-Sustainable Development Goals-targets, indicators and intervention areas Climate change- Global, Regional and local environmental issues and possible solutions-case studies. Concept of Carbon Credit, Carbon Footprint. Environmental management in industry-A case study.

UNIT V SUSTAINABILITY PRACTICES

8Hours

Zero waste and R concept, Circular economy, ISO 14000 Series, Material Life cycle assessment, Environmental Impact Assessment. Sustainable habitat: Green buildings, Green materials, Energy efficiency, Sustainable transports. Sustainable energy: Non-conventional Sources, Energy Cyclescarbon cycle, emission and sequestration, Green

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Engineering: Sustainable urbanization- Socioeconomical and technological change.

Total Periods: 45

TEXT BOOKS:

1. Anubha Kaushik and C. P. Kaushik's "Perspectives in Environmental Studies", 6th Edition, New Age International Publishers ,2018.
2. Benny Joseph, 'Environmental Science and Engineering', Tata McGraw-Hill, New Delhi, 2016.
3. Gilbert M.Masters, 'Introduction to Environmental Engineering and Science', 2nd edition, Pearson Education, 2004.
4. Allen, D. T. and Shonnard, D. R., Sustainability Engineering: Concepts, Design and Case Studies,1st Edition, Pearson, 2011.
5. Bradley. A.S; Adebayo, A.O., Maria, P. Engineering applications in sustainable design and development, Cengage learning.
6. Environment Impact Assessment Guidelines, Notification of Government of India, 2006.
7. Mackenthun, K.M., Basic Concepts in Environmental Management, Lewis Publication, London, 1998.

COURSE OUTCOME: Upon the completion of the course the student will be able to

CO1: To recognize and understand the functions of environment, ecosystems and biodiversity and their conservation.

CO2: • To identify the causes, effects of environmental pollution and natural disasters and contribute to the preventive measures in the society

CO3: To identify and apply the understanding of renewable and non-renewable resources and contribute to the sustainable measures to preserve them for future generations.

CO4: To recognize the different goals of sustainable development and apply them for suitable technological advancement and societal development.

CO5: To demonstrate the knowledge of sustainability practices and identify green materials,energy cycles and the role of sustainable urbanization

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3

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Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	-	-	-	-	1	1	-	1
CO 3	3	3	2	-	3	-	-	-	1	1	2	1
CO4	3	3	2	2	3	1	1	-	1	1	2	1
CO5	3	-	2	2	3	1	1	1	1	1	2	1
Average	3	3	2	2	3	1	1	1	1	1	2	1

25CI305

Ability Enhancement/Co-curricular

Credits-2

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SEMESTER IV

25CI401

PHARMACOVIGILANCE

L-T-P-C: 2-1-1.5-4.5

Course Objective: This course provides an overview of pharmacovigilance, the science and activities related to the detection, assessment, understanding, and prevention of adverse effects or any other drug-related problems. Students will learn about the importance of pharmacovigilance in ensuring drug safety, the global regulatory framework, and the methods used to collect, analyze, and report adverse drug reactions. They will also explore risk management strategies and pharmacovigilance in different populations.

UNIT I. Introduction to Pharmacovigilance

8 Hours

Definition, scope, and objectives of pharmacovigilance Historical background and milestones in pharmacovigilance, Roles, and responsibilities of stakeholders in pharmacovigilance, Pharmacovigilance databases and information systems, Global Regulatory Framework and Guidelines International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use (ICH) guidelines, World Health Organization (WHO) guidelines and programs, National regulatory authorities, and their roles in pharmacovigilance.

UNIT II. Adverse Drug Reaction (ADR) Reporting and Data Collection

8 Hours

ADR reporting methods, data collection and sources of ADR reporting: spontaneous reporting, clinical trials, literature, and electronic health records, Signal detection and data mining techniques, Adverse Event Assessment and Causality Assessment. Causality assessment methods: WHO-Uppsala Monitoring Centre (UMC) criteria, Naranjo algorithm, Severity assessment and seriousness criteria, Quality assessment of ICSR

UNIT III. Pharmacovigilance in Special Populations

8 Hours

Pediatric pharmacovigilance: challenges and considerations, Geriatric pharmacovigilance: age-related changes and medication safety, Pharmacovigilance in pregnancy and lactation, Pharmacovigilance in Drug Development and Clinical Trials, Pre-marketing safety assessment and clinical trial safety monitoring, Role of the clinical research team in pharmacovigilance, Reporting of serious adverse events (SAEs) in clinical trials.

UNIT IV. Pharmacovigilance in the Real-World Setting

10 Hours

Post-marketing surveillance and real-world evidence, Pharmacoepidemiology and observational studies in pharmacovigilance, Role of healthcare professionals and patients in pharmacovigilance, Pharmacovigilance Challenges and Future Trends, Challenges in signal detection and benefit-risk assessment. Pharmacovigilance in the era of big data and artificial intelligence, Emerging trends, and future directions in pharmacovigilance

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UNIT V. Vaccine safety surveillance and Materiovigilance

11 Hours

Vaccine pharmacovigilance, vaccination failure, adverse events following immunization. Pharmacovigilance methods passive surveillance – spontaneous reports and case series, stimulated reporting, active surveillance – sentinel sites, drug event monitoring and registries, comparative observational studies – cross sectional study, case control study and cohort study, targeted clinical investigations. Materiovigilance, Hemovigilance, Identification and reporting of medication errors, Communication in Pharmacovigilance

Total Periods: 45

Pharmacovigilance Practical

Practical pharmacovigilance activities aim to provide students with hands-on experience in the implementation of pharmacovigilance principles and processes.

- 1. Study and explain Adverse Drug Reaction (ADR): A Case Study Analysis**
- 2. Understand and explain pharmacovigilance databases** such as the WHO Global Individual Case Safety Report (ICSR) database or national databases. Report hypothetical or adverse drug reactions using standardized reporting forms.
- 3. Explain Signal Detection and Data Analysis:** Use pharmacovigilance data sets to perform spontaneous reporting. Apply signal detection techniques to identify potential safety concerns and generate signal reports.
- 4. Risk Management Planning: Develop** risk management plans for selected medications. Identify potential risks, propose risk minimization strategies, and design tools for risk communication to healthcare professionals and patients.
- 5. Pharmacovigilance Auditing and Inspection:** Simulation based pharmacovigilance audits or inspections.
- 6.** Prepare audit/inspection report, review relevant documentation, identify compliance gaps, and propose corrective actions.
- 7. Pharmacoepidemiology Study Design: Understand** pharmacoepidemiology study designs, such as cohort studies or case-control studies. Design Pharmacoepidemiologic studies to investigate potential associations between drug exposures and adverse events.
- 8. Adverse Event Reporting in Clinical Trials:** Demonstrate Monitoring and reporting serious adverse events (SAEs) occurring during clinical trials.
- 9. Role-Playing Exercises: Simulate** interactions between healthcare professionals, patients, and pharmacovigilance professionals.
- 10.** Practice counseling patients on ADR reporting, handling queries from healthcare professionals, or conducting medication safety presentations.

Total Periods: 45

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References and Textbooks

1. Mann, R.D., and Andrews, E.B. eds., 2007. Pharmacovigilance. John Wiley & Sons.
2. Waller, P. and Harrison-Woolrych, M., 2017. An introduction to pharmacovigilance. JohnWiley & Sons.
3. Andrews, E.B. and Moore, N. eds., 2014. Mann's pharmacovigilance. John Wiley & Sons.
4. Nour, S. and Plourde, G., 2018. Pharmacoepidemiology and pharmacovigilance: Synergistic tools to better investigate drug safety. Academic Press.
5. Plotkin, S.A., Orenstein, W. and Offit, P.A., 2012. Vaccines E-book. Elsevier HealthSciences.
6. Mal, D.K., Ehsan, I. and Mukherjee, B., MATERIOVIGILANCE and HAEMOVIGILANCE.

Course Outcome

After the completion of course, the students will be able to:

CO1. Explain the basic principle and importance of pharmacovigilance in drug safety monitoring.

CO2. Describe the global regulatory framework and guidelines for pharmacovigilance.

CO3. Explain the methods and tools used in collection, analysis, and reporting of ADR

CO4. Discuss the role of pharmacovigilance in signal detection and risk management.

CO5. Describe the challenges and future trends in pharmacovigilance and gain insights into pharmacovigilance in special populations.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	2	-	-
CO 2	3	-	2	-	-	-	-	-	1	2	-	2
CO 3	3	3	2	-	3	-	-	-	1	2	2	2

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CO4	3	3	2	2	3	1	1	-	1	2	2	2
CO5	3	3	2	2	3	1	1	1	1	2	2	2
Average	3	3	2	2	3	1	1	1	1	2	2	2

25CI402 CLINICAL PHARMACOKINETICS L-T-P-C:2-1-1.5-4.5

COURSE OBJECTIVES:

This course is intended to teach students the fundamental principles and applications of pharmacokinetics in designing the individualized dosage regimen, to interpret the plasma drug concentration profile in altered pharmacokinetics, drug interactions and in therapeutic drug monitoring processes to optimize the drug dosage regimen.

UNIT I: Introduction to clinical pharmacokinetics **10 Hours**

Compartmental and non-compartmental models, renal and non-renal clearance, models of hepatic clearance, estimation of bioavailability, multiple dosing, calculation of loading and maintenance doses. Designing of dosage regimens - determination of dose and dosing intervals, conversion from intravenous to oral dosing, nomograms, and tabulations in designing dosage regimen.

UNIT II: Pharmacokinetics of Drug Interaction **6 Hours**

Pharmacokinetic drug interactions, inhibition and induction of drug metabolism, inhibition of biliary excretion.

UNIT III Non-Linear Mixed Effects Modelling **10 Hours**

Structural or base model, modeling random effects, modeling covariate relationships, mixture model, estimation methods, model building techniques, covariate screening methods, testing the model assumptions, precision of the parameter estimates and confidence intervals, model misspecification and violation of the model assumptions, model validation, simulation of dosing regimens and dosing recommendations.

UNIT IV: Altered Pharmacokinetics **8 Hours**

Drug dosing in the elderly, pediatrics, obese patients, and during pregnancy and lactation, drug dosing in renal failure and extracorporeal removal of drugs, drug dosing in hepatic failure.

UNIT V: Therapeutic Drug Monitoring: **11 Hours**

Introduction, individualization of drug dosage regimen (variability – genetic, age, weight,

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disease and interacting drugs). TDM protocol, pharmacokinetic/pharmacodynamic correlation in drug therapy. TDM in Cardiovascular disease - lidocaine, digoxin, and amiodarone; Antibiotics - gentamicin, vancomycin, and meropenem; Seizure disorders - carbamazepine, phenytoin, and sodium valproate; Psychiatric conditions - fluoxetine, lithium, and amitriptyline; Organ transplantation - cyclosporine; Cytotoxic agents - 5-fluorouracil, methotrexate, and cisplatin.

Total Periods: 45

Practical: Clinical Pharmacokinetics Lab

1. To determine K_a , pK_a , and partition coefficient (PC) of Salicylic acid and study their relationship.
2. To study *in vitro* drug release of the given compressed tablet of Paracetamol.
3. To study *in vitro* drug release of the given fast release tablet.
4. To perform *in vitro* absorption-permeation study of marketed tablet.
5. To study the protein binding of salicylic acid by equilibrium dialysis method.
6. To determine various pharmacokinetic parameters from given blood data of IV bolus injection (one compartment model).
7. To determine the area under the curve for the given data by Trapezoidal method.
8. To determine the area under the curve for the given data by counting squares method and cutting & weighing method.
9. Determine the pharmacokinetic model from the given plasma concentration versus time data following 250 mg rapid intravenous bolus dose of a drug.

Total Periods: 30

REFERENCES

1. Gibaldi, M., 1991. Biopharmaceutics and clinical pharmacokinetics.
2. Shargel, L., Andrew, B.C. and Wu-Pong, S., 1999. Applied biopharmaceutics & pharmacokinetics (Vol. 264). Stamford: Appleton & Lange.
3. Huang, X.H. and Li, J., 2007. PETER L. BONATE. Pharmacokinetic-Pharmacodynamic Modeling and Simulation. New York: Springer, 2005. 387 ISBN: 038727197X.
4. Burton, M.E. ed., 2006. Applied pharmacokinetics & pharmacodynamics: principles of therapeutic drug monitoring. Lippincott Williams & Wilkins.
5. Wong, S.H. and Sunshine, I., 1996. Handbook of analytical therapeutic drug monitoring and toxicology. CRC Press.
6. Dhillon, S. and Kostrzewski, A. eds., 2006. Clinical pharmacokinetics. Pharmaceutical Press.
7. Notari, Robert E. Biopharmaceutics and clinical pharmacokinetics: an

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introduction. Routledge, 2017.

8. Doogue, M.P. and Polasek, T.M., 2013. The ABCD of clinical pharmacokinetics. Therapeutic advances in drug safety, 4(1), pp.5-7.

COURSE OUTCOMES:

Upon completion of this course, it is expected that students shall be able to:

CO1: Define pharmacokinetic drug interactions

CO2: Discuss pharmacokinetic parameters in clinical settings

CO3: Recommend dosage adjustment for pediatric and geriatric populations

CO4: Design the drug dosage regimen for individual patients; recommend dosage adjustment for patients with renal/hepatic impairment

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	-	-	-	-	1	1	-	1
CO 3	3	-	2	-	3	-	-	-	1	1	2	1
CO4	3	3	2	2	3	1	1	-	1	1	2	1
Average	3	3	2	2	3	1	1	1	1	1	2	1

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25CI403 BIOETHICS AND DRUG SAFETY L-T-P-C:2-1-1.5-4.5

Course Description: This course describes the ethical considerations and principles about the development, regulation, and use of pharmaceutical drugs. Students will understand the fundamentals of bioethics and drug safety, addressing ethical issues related to clinical trials, informed consent, access to medications, off-label use, adverse events, and post-marketing surveillance.

UNIT I. Introduction to Bioethics and Drug Safety **8 Hours**

Overview of bioethics and its relevance to drug safety, Ethical frameworks and principles in healthcare, Historical perspective on drug safety and regulation, Ethical Considerations in Clinical Trials, Ethical guidelines, and regulations for clinical trials.

UNIT II. Informed consent and participant autonomy **8 Hours**

Elements of informed consent, consent process and documentation, special considerations for vulnerable populations, Balancing risks and benefits in trial design and recruitment, Informed Consent in Drug Research and Use, Elements and challenges of informed consent, Ethics of placebo-controlled trials and equipoise. Participant Recruitment and Retention in clinical trials.

UNIT III. Access to Medications: Ethics and Equity **8 Hours**

Ethical issues in medication affordability and accessibility, Global health disparities and access to essential medicines, pharmaceutical industry practices and drug pricing. Off-label prescribing: benefits and risks, Compassionate use programs and expanded access, Informed decision-making and patient autonomy, Ethics of Adverse Event Reporting and Patient Safety

UNIT IV. Ethical Issues in Drug Safety **11 Hours**

Ethical considerations in monitoring drug safety post-approval, ethical case studies in drug safety, analysis of real-life case studies involving ethical issues in drug safety, Ethical reasoning, and decision-making in complex scenarios. Discussion and debate on different perspectives and ethical frameworks

UNIT V. Ethical Issues in Emerging Technologies and Innovative Therapies **10 Hours**

Ethical considerations in clinical trials for modern technologies, New Drugs in Clinical Trials 2019, Emerging Ethical Issues in Drug Safety, Exploration of emerging ethical challenges in drug safety, Ethical considerations in personalized medicine and gene therapy, Ethical implications of emerging technologies in drug development.

Total Periods: 45

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Practical: Bioethics and Drug Safety Lab

Practical activities in a course on Bioethics and Drug Safety provide students with opportunities to apply ethical principles and analyze real-world scenarios related to drug development, regulation, and use.

1. Perform Case Study Analysis: Analyze the case, identify the ethical issues involved, and propose ethical solutions or actions.
2. Evaluate the informed consent process, risk-benefit assessment, and protection of vulnerable populations.
3. Prepare ethical review reports highlighting any concerns or recommendations
Simulation based demonstration of informed consent process, including communication, comprehension, and documentation.
4. Demonstrates the roles of investigators, research participants, and ethics committee members (simulation based), discuss ethical issues and challenges encountered.
5. Ethics Committee Meeting Simulation: Mock research proposal involving ethical considerations.
6. Review the proposal, discuss the ethical aspects, and reach a consensus decision. Encourage thoughtful deliberation and ethical reasoning during the simulation.
7. Debate on Access to Medications: Discussion based activity on for and against different perspectives, such as affordability, intellectual property rights, and global health equity. Students should present evidence-based arguments and engage in constructive discussions.
8. Field Visit to Regulatory Agencies or Ethics Committees:
9. Visits to local regulatory agencies or ethics committees involved in drug safety and research oversight.
10. Guest Speakers from Bioethical Organizations: Discuss and learn about the practical challenges and ethical considerations in drug safety.

Total Periods: 45

REFERENCE and Textbooks

1. Rozovsky, F.A. and Adams, R.K., 2003. Clinical trials and human research: A practical guide to regulatory compliance.
2. Evans, B.J. and Meslin, E.M., 2006. Encouraging translational research through harmonization of FDA and common rule informed consent requirements for research with banked specimens. *The Journal of Legal Medicine*, 27(2), pp.119-166.
3. Gallin, J.I. and Ognibene, F.P. eds., 2012. Principles and practice of clinical research. Academic Press.
4. Karlberg, J.P.E. and Speers, M.A., 2010. Reviewing clinical trials: a guide for the

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ethics committee. Hong Kong.

5. Cartwright, A.C. and Matthews, B.R. eds., 2016. International pharmaceutical product registration (Vol. 200). CRC Press.
6. Beecher, H.K., 2001. Ethics and clinical research. Bulletin of the World Health Organization, 79, pp.367-372.
7. Angell, M., 1997. The ethics of clinical research in the Third World. New England journal of medicine, 337(12), pp.847-849.

Course Outcome: Upon the completion of the course the student will be able to

CO1: Explain the principles of bioethics and their application to drug safety.

CO2: Describe the ethical considerations in the design, conduct, and reporting of clinical trials.

CO3: Discuss the concept of informed consent and its importance in drug research and use.

CO4: Analyze the ethical dimensions of access to medications, including affordability and availability.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	3	-	-	-	2	1	-	2
CO 2	3	-	2	-	3	-	-	-	2	1	-	2
CO 3	3	-	2	-	3	-	-	-	2	1	2	2
CO4	3	3	2	2	3	1	1	-	2	1	2	2
Average	3	3	2	2	3	1	1	1	2	1	2	2

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COURSE OBJECTIVES

This course will cover on understanding the importance of Good Clinical Practices (GLP) aspects, regulations governing the conduct of Clinical Trials and the Intellectual Property Rights

UNIT-I Ethical Aspects

10 Hours

Ethical principles underlined research involving human subjects, legal authorities for Institutional Review Board (IRB), Health and Human Services regulations (HHS), FDA regulations, regulatory requirements, duties of IRBs, IRB membership, role of IRB in reviewing Clinical Drug Trials, Assessment of scientific design, competence of investigator, selection of subjects, balancing benefits, and risks. Compensation for research related injuries, special issues like role of Lay member of IRB, review of multi-institutional trials, duty to monitor, financial risks of clinical trial subject, compliance with new regulations.

UNIT-II Applicable GCP Guidelines

10 Hours

International Conference on Harmonization of technical requirements for registration of Pharmaceuticals for human use guidelines (ICH-GCP). Indian Council of Medical Research- Ethical Guidelines for Biomedical Research on Human participants (ICMR), Indian Good Clinical Practices. Regulatory guidelines for Biomaterials and Medical Devices

UNIT-III Council for International Organizations of Medical Science (CIOMS) guidelines

10 Hours

CIOMS International Ethical guidelines for biomedical research involving human subjects. Principles Of Medical Ethics Relevant to the Protection of Prisoners Against Torture (1983), Intellectual Property Rights: Terminology, Patent Laws, TRIPS (Trade Related Intellectual Property Rights) Agreement, Trademarks, copyrights, Clinical Trial Application Requirements:

UNIT-IV Investigational New drug (IND)

5 Hours

Classifications, IND application submission, check list, FDA IND review check list, IND application process, Information for sponsors-investigator submitting IND, IND forms and instructions

UNIT- V New Drug Application (NDA)

10 Hours

Pre NDA meeting, NDA submission Check list, FDA NDA review check list, Abbreviated New drug Application (ANDA): ANDA content, ANDA, Submission check list, FDA ANDA review check list, ANDA process for generic drugs, guidance documents for ANDAs, ANDA forms and electronic submissions, Orphan Drugs Application: Submission check list, FDA orphan drug review check list, FDA documents

Total Periods: 45

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Practical- Regulatory Aspects in Clinical Research Lab

1. Understanding the role of regulatory authorities and importance of regulatory guidelines.
2. Declaration of Helsinki and the Belmont Report
3. Informed Consent
4. Investigational New Drug/Device Application
5. Institutional Review Board/Ethics Committee
6. Good Clinical Practice (GCP)
7. Safety Reporting
8. Data Management and Quality Assurance
9. Regulatory Inspections
10. Reporting and Documentation

Total Periods: 45

Recommended Books:

1. The Fundamentals of Clinical Research: A Universal Guide for Implementing Good Clinical Practice, by P. Michael Dubinsky, Karen A. Henry Wiley; 1st edition (1 February 2022).
2. Clinical Trials Audit Preparation - A Guide for Good Clinical Practice (GCP) Inspections Illustrated, 18 June 2010, John Wiley & Sons Inc; 1st edition (18 June 2010).
3. Good Clinical Practice - Standard Operating Procedures for Clinical Researchers Paperback – 27 March 1998, John Wiley & Sons Inc; 1st edition (27 March 1998); 01149344934
4. Principles of Good Clinical Practice (Pharmacy Business Administration) Paperback – Pharmaceutical Press; 1st edition (3 August 2010)
5. Guide for Clinical Trial Staff: Implementing Good Clinical Practice Spiral-bound – Import, S Karger AG (14 October 2003)

COURSE OUTCOMES: Upon completion of this course the student will be able to

CO1. Explain about the conduct of Clinical Trials and understanding of ethical principles underlined research involving human subjects.

CO2. Improves ideas and updating information on the current trend in Good Clinical Practices

CO3. Explain the role of global as well as Indian regulatory authorities, guidelines while conducting clinical trials

CO4. Apply the Information of IPR and GCP guidelines for conducting clinical trials

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Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	3	-	-	-	1	1	-	-
CO 2	3	-	2	-	3	-	-	-	1	1	-	1
CO 3	3	-	2	-	3	-	-	-	1	1	2	1
CO4	3	3	2	2	3	1	1	-	1	1	2	1
Average	3	3	2	2	3	1	1	1	1	1	2	1

25CI405

Ability Enhancement/Co-curricular IV Credits-2

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SEMESTER V

25CI501 GLOBAL REGULATIONS OF CLINICAL TRIALS L-T-P-C:2-1-1.5-4.5

COURSE OBJECTIVES: This course is designed to impart the fundamental knowledge on the clinical development process of drugs, pharmaceuticals and medical devices, regulations and guidance governing the conduct of clinical research in India, USA, and EU.

UNIT I. Regulatory Authorities and Roles

8 Hours

International regulatory bodies: ICH, WHO, and their guidelines, Regional regulatory authorities: FDA, EMA, PMDA, and others, Roles, and responsibilities of regulatory agencies in clinical trial oversight.

UNIT II. Clinical Research Regulations

11 Hours

- a) Clinical research regulations in India – Schedule Y, New Drugs in Clinical Trials 2019 & Medical device guidelines-ISO14155. Guidance USA: Regulations to conduct drug studies in USA (FDA)
- b) NDA 505(b) (1) of the FD&C Act (Application for approval of a new drug)
- c) NDA 505(b) (2) of the FD&C Act (Application for approval of a new drug that relies, atleast in part, on data not developed by the applicant)
- d) ANDA 505(j) of the FD&C Act (Application for approval of a generic drug products)
- e) Biosimilar Guidelines

UNIT III. Clinical Research Related Guidelines

8 Hours

- a) E4 – Dose Response Information to support Drug Registration
- b) E7 – Studies in support of General Population: Geriatrics
- c) E8 – General Considerations of Clinical Trials
- d) E10 – Choice of Control Groups and Related Issues in Clinical Trials,
- e) E11 – Clinical Investigation of Medicinal Products in the Pediatric Population

UNIT IV. Code of federal regulations related to clinical trials.

10 Hours

- a) CFR 21Part 50: Protection of Human Subjects
- b) CFR 21Part 54: Financial Disclosure by Clinical Investigators
- c) CFR 21Part 312: IND Application
- d) CFR 21Part 314: Application for FDA Approval to Market a New Drug
- e) CFR 21Part 320: Bioavailability and bioequivalence requirements
- f) CFR 21Part 812: Investigational Device Exemptions

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g) CFR 21Part 822: post-market surveillance

UNIT V. Harmonization Efforts and Future Trends

8 Hours

Global harmonization initiatives in clinical trial regulations, international collaboration and mutual recognition agreements, Emerging trends, and future directions in global clinical trial regulations

Total Periods: 45

Practical: Global Regulations of Clinical Trials Lab

1. Regulatory Authority Case Studies: Prepare presentations or reports highlighting key aspects and differences in regulations.
2. Mock Clinical Trial Approval Process: Prepare an Investigational New Drug (IND) application or Clinical Trial Authorization (CTA) submission.
3. Mock ethics committee or regulatory review board, addressing ethical and regulatory considerations.
4. Ethical Review Board Simulation: Conduct a simulation where students take on the roles of ethics committee members or regulators.
5. Regulatory Compliance Audits: Mock clinical trial documentation, such as protocols, informed consent forms, and case report forms.
6. Reporting and Documentation Exercise: Reporting process, prepare adverse event reports, and document the necessary information following regulatory requirements.
7. Analyze the impact of their reporting on patient safety and trial conduct.
8. Data Management and Quality Assurance: Mock clinical trial datasets or case studies.
9. Perform data quality checks, assess data integrity, and ensure compliance with regulatory requirements for data collection and management.
10. Site Inspection Preparation
11. Prepare for a mock regulatory inspection by reviewing their site's documentation, ensuring compliance with regulations, and addressing potential compliance gaps.
12. Regulatory Updates and Emerging Trends.

Total Periods: 45

REFERENCES BOOKS

1. Chin, R. and Bairu, M. eds., 2011. Global clinical trials: effective implementation and management. Academic Press.
2. Gallin, J.I. and Ognibene, F.P. eds., 2012. Principles and practice of clinical research. Academic Press.
3. Gad, S.C. ed., 2009. Clinical trials handbook. John Wiley & Sons.
4. Plomer, A., 2013. The law and ethics of medical research: international

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bioethics and humanrights. Routledge.

5. Chow, S.C. and Liu, J.P., 2008. Design and analysis of clinical trials: concepts and methodologies(Vol. 507). John Wiley & Sons.
6. Lang, T. and Siribaddana, S., 2012. Clinical trials have gone global: is this a good thing? PLoSmedicine, 9(6), p.e1001228.
7. Weigmann, K., 2015. The ethics of global clinical trials: In developing countries, participation in clinical trials is sometimes the only way to access medical treatment. What should be done to avoid exploitation of disadvantaged populations? EMBO reports, 16(5), pp.566-570.

COURSE OUTCOME On completion of this course, the students will be able to

CO1: Discuss global regulations in clinical research, medical device development process and different types and phases of Clinical trials.

CO2: Describe different regulations governing clinical trials in India as well as USA and EU.

CO3: Explore the ethical considerations and principles underlying clinical trial regulations.

CO4: Examine the processes involved in clinical trial approval, monitoring, and reporting.

CO5. Analyze the challenges and harmonization efforts in global clinical trial regulations.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	3	-	-	-	1	1	-	2
CO 2	3	-	2	-	3	-	-	-	1	1	-	2
CO 3	3	-	2	-	3	-	-	-	1	1	2	2
CO4	3	3	2	2	3	1	1	-	1	1	2	2
CO5	3	3	2	2	3	1	1	-	1	1	2	2
Average	3	3	2	2	3	1	1	1	1	1	2	2

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25CI502 EPIDEMIOLOGY AND GLOBAL HEALTH L-T-P-C:2-1-1.5-4.5

Course Objectives

The course objective of this course helps student to describe the science of epidemiology and the etiology of specific diseases, interpret epidemiological information contained in scientific literature and design interventions for treating and preventing ill health.

Unit I: Epidemiology: Laying the Foundations

8 Hours

Definitions: Epidemiology, Public Health, and Health. Uses of Epidemiology, evolving patterns of morbidity and mortality-Demographics and disease patterns, mortality, and life expectancy trends. Roots of Epidemiology-John Graunt, Germ theory, The London Epidemiological Society, John Snow and William Farr. Global Health and Sustainable Development Goals.

Unit II: Epidemiological Study Designs

8 Hours

Etiologic research- Hypothesis statement, Epidemiological variables-Person, place, time, and data. Descriptive versus Analytical, Longitudinal versus cross-sectional. Epidemiological studies. Experimental versus observational, cohort versus case-control. Ethical conduct of studies involving human subjects.

Unit III: Outbreak Investigation and Causation

10 Hours

Background-Initial detection of Outbreaks, goals, and methods. CDC prescribed investigatory steps. Case study on the outbreak investigation of Covid 19 and Foodborne outbreak. Causation in Epidemiology-Natural history of the disease, Variability in the expression of disease, causal models, Concept of cause, Establishing the cause of a disease.

Unit IV: Health Services and Health Policy

8 Hours

Healthcare planning and evaluation, Measuring the quality of healthcare, the planning cycle, health public policy in practice, Planning a research project, and critical reading of published reports.

Unit V: Monitoring and Evaluation

11 Hours

Monitoring Health Care Systems, Organizations, and Programs Surveillance, Health services information and Evaluation. Dimensions of Evaluation, The Evaluation Process, Conceptual Framework for Specifying Evaluation Criteria, and Analytical Approaches to Evaluation.

Total Periods: 45

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Practical- Epidemiology and Global Health Lab

The course will also extend students' practical skills and ability to critically review the existing public health policies and Programs. Emphasis will be placed on epidemiological reasoning principles and findings interpretation. At the end of this course, students can apply quantitative and qualitative approaches to address population health challenges and interpret epidemiological information in scientific literature. It encompasses planning an epidemiological study and encouraging the publication of the results.

1. Write a critical review of any one theory of Disease.
2. Introduction to case study on Covid 19 (Region specific/Uttarakhand)
3. Give a summary of the recommended health strategy and an assessment of whether the health strategy was successful.
4. Project proposal on any one identified public health issue in a selected community.
5. Evaluation of the three articles and recommendation for an effective public health intervention
6. Community Screening: Framing open-ended and close-ended questionnaires/Probing.
7. Development of tools to assess knowledge, attitudes, and practices
8. Report on a Community Visit/Transect walk and social mapping.
9. Seminar
10. Peer reviews

Total Periods: 45

Practical hours include one seminar in which students will give presentations. The project proposal plan of the study will be presented in the seminar and discussed with other students. Students will also be engaged in peer reviews of one another's research plans.

Reference Books

1. Beaglehole, R. et al. (1993): *Basic Epidemiology*. Geneva, WHO.
2. Lilienfeld's Foundations of Epidemiology (Fourth Edition) Dona Schneider, David E. Lilienfeld
3. B. Burt Gerstman, Epidemiology Kept Simple: An Introduction to Traditional and Modern Epidemiology (3rd Edition)
4. Lilienfeld, Abraham M. (1994): *Foundations of Epidemiology*. New York, Oxford University Press, (Chapter 2)
5. Denise M. Oleske: Epidemiology and the Delivery of Health Care Services (Third edition): Methods and Applications, Springer.

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Course Outcomes: On completion of this course, the students will be able to

CO1-Describe the evolution of epidemiology.

CO2-Discuss various types of epidemiological studies and ethical considerations involving human subjects.

CO3-Evaluate the process of investigation of an outbreak and causation in Epidemiology.

CO4-Plan research project and critical reading of published reports.**CO5** Develop analytical approaches to monitoring and evaluation.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	-	-	-
CO 2	3	-	-	-	-	-	-	-	1	-	-	-
CO 3	-	-	-	-	3	-	-	-	1	-	-	-
CO4	-	3	-	2	-	1	1	-	1	-	-	-
CO5	-	3	2	2	3	1	1	3	1	-	2	-
Average	3	3	2	2	1	1	1	3	1	-	2	-

25CI503

IMMUNOLOGY

L-T-P-C: 2-1-1.5-4.5

COURSE OBJECTIVE

This course's main objective is to introduce the basic concepts of immunology and its application in disease diagnosis. This course addresses immunological disorders and discusses the techniques to be employed in diagnostic laboratories. This will also enable the students to know about the different cells which are associated with the body's defense mechanism against pathogens and cancer.

Unit I: History of immunology

8 Hours

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Introduction to Innate and Adaptive immunity; Contributions of scientists to the development of the field of immunology - Edward Jenner, Karl Landsteiner, Robert Koch, Paul Ehrlich, Elie Metchnikoff, Peter Medawar, MacFarlane Burnet, Neils K Jerne, Rodney Porter, and Susumu Tonegawa.

Unit II: Immunity: Cells, organs, and Antigen-Antibody Interaction 11 Hours

- (a) Structure, and Functions of: Immune Cells – Stem cell, T cell, B cell, NK cell, Macrophage, Neutrophil, Eosinophil, Basophil, Mast cell, Dendritic cell; and Immune Organs – Bone Marrow, Thymus, Lymph Node, Spleen, GALT, MALT, CALT.
- (b) Characteristics of antigens (Foreignness, Molecular size, and Heterogeneity); Haptens; Epitopes (T & B cell epitopes); T-dependent and T-independent antigens; Adjuvants; Structure, Types, Functions and Properties of antibodies; Antigenic determinants on antibodies (Isotypic, allotypic, idiotypic); VDJ rearrangements; Monoclonal and Chimeric antibodies.

Unit III: Immune responses

8 Hours

Primary and Secondary Immune Response; Generation of Humoral Immune Response (Plasma and Memory cells); Generation of Cell Mediated Immune Response (Self MHC restriction, T cell activation, Co-stimulatory signals).

Unit IV: Major Histocompatibility Complex and Complement System 8 Hours

Organization of MHC locus (Mice & Human); Structure and Functions of MHC I & II molecules; Antigen processing and presentation (Cytosolic and Endocytic pathways); Components of the Complement system; Activation pathways (Classical, Alternative and Lectin pathways).

Unit V: Immunological dysregulation and immunity in tumor

10 Hours

Types of Autoimmunity and Hypersensitivity with examples; Immunodeficiencies - SCID, DiGeorge syndrome, Chediak-Higashi syndrome, Leukocyte adhesion deficiency; Types of tumors, tumor Antigens, causes and therapy for cancers. Immune technology- Principles of Precipitation, Agglutination, Immunodiffusion, Immunoelectrophoretic, ELISA, Western blotting, Immunofluorescence, Flow cytometry, Immunoelectron microscopy.

Total Periods: 45

PRACTICALS

1. Identification of human blood groups.
2. Perform Total Leukocyte Count of the given blood sample.

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3. Perform Differential Leukocyte Count of the given blood sample.
4. Separate serum from the blood sample (demonstration).
5. Perform immunodiffusion by Ouchterlony method.
6. Perform DOT ELISA.
7. Perform immunoelectrophoretic.

Total Periods: 45

REFERENCES

1. Abbas AK, Lichtman AH, Pillai S. (2007). Cellular and Molecular Immunology. 6th edition Saunders Publication, Philadelphia.
2. Delves P, Martin S, Burton D, Roitt IM. (2006). Roitt's Essential Immunology. 11th edition Wiley Blackwell Scientific Publication, Oxford.
3. Goldsby RA, Kindt TJ, Osborne BA. (2007). Kuby's Immunology. 6th edition W.H. Freeman and Company, New York.
4. Murphy K, Travers P, Walport M. (2008). Janeway's Immunobiology. 7th edition Garland Science Publishers, New York.
5. Peakman M, and Vergani D. (2009). Basic and Clinical Immunology. 2nd edition Churchill Livingstone Publishers, Edinburgh.
6. Richard C and Geffrey S. (2009). Immunology. 6th edition. Wiley Blackwell Publication.

COURSE OUTCOMES

Upon completion of the course, the student should be able to:

THEORY

CO1: Outline the history of scientists who contributed to immunology.

CO2: Explain types of cells and organs that participate in defense mechanisms against pathogens and other foreign particles.

CO3: Discuss the defense mechanism through Antibodies, Major histocompatibility complex and Complement systems.

CO4: Apply advanced technologies in immunology to diagnose different diseases.

CO5: Demonstrate basic skills of total Leukocyte Count and differential Leukocyte Count of the given blood sample

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

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Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
CO 1	3	-	-	-	-	-	-	-	1	-	-	-
CO 2	3	-	-	-	-	-	-	-	1	-	-	-
CO 3	-	-	-	-	3	-	-	-	1	-	-	-
CO4	-	3	-	2	-	1	1	-	1	-	-	-
CO5	-	3	2	2	3	1	1	3	1	-	2	-
Average	3	3	2	2	1	1	1	3	1	-	2	-

25CI504

BIOINSTRUMENTATION

L-T-P-C:2-1-1.5-4.5

Course Objectives

This course is introduced with an idea to fill the gap between academia and research. This course introduces analytical techniques along with working principles, common instrumentation, and their applications. This course engenders students with the fundamental knowledge of analytical techniques required for research careers in allied health fields.

Unit-I Centrifugation

(6 Hours)

Basic principles of centrifugation, standard sedimentation coefficient, types of centrifuges based on speed, different types of rotors, principles and applications of differential, rate zonal and density gradient centrifuge.

Unit-II Chromatography

(10 Hours)

Introduction to chromatography, principle and applications of paper chromatography, thin layer chromatography, column chromatography (adsorption, gel filtration, ion exchange, and affinity) and high-performance liquid chromatography (HPLC).

Unit-III Electrophoresis

(8 Hours)

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Introduction to electrophoresis, polyacrylamide, and agarose gel electrophoresis SDS and Native-PAGE, isoelectric focusing (IEF) and 2-D gel electrophoresis.

Unit-IV Spectroscopy

(11 Hours)

Introduction to spectroscopy, electromagnetic spectrum, Jablonski's diagram, Lambert-Beer law, principle, instrumentation, and applications of UV-visible, fluorescence, Fourier-transform infrared (FT-IR) and nuclear magnetic resonance (NMR). Introduction, principle and applications of mass spectrometry, types of ionization methods (Electron impact, chemical ionization, ESI, MALDI).

Unit-V Microscopy

(10 Hours)

Principle of microscopy, resolving powers of different microscopes, magnification, principle, and applications of: Compound microscopy, dark field microscopy, fluorescent microscopy, phase contrast microscopy, confocal and electron microscopy (SEM & TEM), Fixation and staining.

Total Periods: 45

Practical

1. Separation of components of a given mixture using a laboratory scale centrifuge.
2. Separation of mixtures by paper / thin layer chromatography.
3. Separation of mixtures of molecules by any form of chromatography.
4. Separation of protein mixtures by sodium-dodecyl sulphate-polyacrylamide gel electrophoresis (SDS-PAGE).
5. Quantification of carbohydrates by DNS/Anthrone method using UV-Visible spectrophotometer.
6. Determination of fluorescence in bovine serum albumin.
7. Fluorescence spectroscopy-based confirmational analysis of protein molecules.
8. To perform simple direct staining to study the morphology of bacterial culture.

Total Periods: 45

Reference Books

1. Principles and Techniques of Biochemistry and Molecular Biology (2018) 8th ed., Wilson, K. and Walker, J. Cambridge University Press, ISBN 13: 978-1316614761.
2. Introduction to Practical Biochemistry (2009) Sawhney, S.K. and Singh R. Narosa Publishing House (New Delhi), ISBN-13: 978-8173193026.

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3. Biophysical Chemistry: Principles and Techniques (2016) Upadhyay A., Upadhyay K. and Nath N. Himalaya Publishing House, ISBN 13: 978-8183188654.
4. Principles of Fluorescence Spectroscopy (2010) 5th ed. Lakowicz J.R. Springer, ISBN 13: 978-0387312781.

Course Outcomes

After the completion of course, the students will be able to:

CO1: Outline different analytical techniques and their instrumentation, and operation.

CO2: Demonstrate skill in carrying out research projects by employing centrifugation, and chromatographic and electrophoresis-based separation techniques.

CO3: Describe the terms, principle, instrumentation, operation, and applications of molecular spectroscopic and microscopic techniques.

CO4: Apply appropriate bioanalytical techniques for identification, separation, isolation, and purification of biomolecules.

CO5: Apply principles of various analytical devices used in research and enhance problem solving techniques.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3- Substantial (high)

Program Outcomes												
Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
CO 1	3	-	-	-	-	-	-	-	1	-	-	-
CO 2	3	-	-	-	-	-	-	-	1	-	-	-
CO 3	3	3	2	-	3	-	-	-	1	-	2	-
CO4	3	3	2	2	-	1	1	-	1	-	2	-
CO5	3	3	2	2	3	1	1	3	1	-	2	-
Average	3	3	2	2	1	1	1	3	1	-	2	-

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25CI505	Ability Enhancement/Co-curricular V	Credits- 2
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Course Objectives

This course is designed to impart a thorough knowledge of the relevant aspects of pharmacotherapy of various conditions with reference to its pharmacological approaches and understanding of basic pathophysiological mechanisms. Hence, it will help study the syllabus of pharmacotherapeutic approaches and get baseline knowledge required to practice medicine safely, confidently, rationally, and effectively.

Unit I: Introduction to Pharmacotherapeutics

12 Hours

Pharmacotherapeutics – Introduction, scope, and objectives. Rational use of Medicines, Evidence Based Medicine, Essential Medicines List, Standard Treatment Guidelines (STGs). Basic principles of Cell injury and Adaptation: Introduction, definitions, Homeostasis, Components and Types of Feedback systems, Causes of cellular injury, Pathogenesis (Cell membrane damage, Mitochondrial damage, Ribosome damage, Nuclear damage), Morphology of cell injury – Adaptive changes (Atrophy, Hypertrophy, hyperplasia, Metaplasia, Dysplasia), Cell swelling, Intra cellular accumulation, Calcification, Enzyme leakage and Cell Death Acidosis & Alkalosis, Electrolyte imbalance

Unit II Basic mechanism involved in the process of inflammation

12 Hours

Basic mechanism involved in the process of inflammation and repair: Introduction, Clinical signs of inflammation, Different types of Inflammation, Mechanism of Inflammation – Alteration in vascular permeability and blood flow, migration of WBC's, Mediators of inflammation, Basic principles of wound healing in the skin, Pathophysiology of Atherosclerosis

Unit III Etiopathogenesis and treatment of cardiovascular disorders

13 Hours

Definition, etiopathogenesis, clinical manifestations, nonpharmacological and pharmacological management of the diseases associated with Cardiovascular System: Hypertension, congestive heart failure, ischemic heart disease (angina, myocardial infarction, atherosclerosis, and arteriosclerosis)

Unit IV Etiopathogenesis and treatment of respiratory, kidney and blood related disorders

13 Hours

Definition, etiopathogenesis, clinical manifestations, nonpharmacological and pharmacological management of the diseases associated with Respiratory system: Asthma, Chronic obstructive airways diseases. Renal system: Acute and chronic renal failure.

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Hematological Diseases: Iron deficiency, megaloblastic anemia (Vit B12 and folic acid), sickle cell anemia, thalassemia, hereditary acquired anemia, hemophilia.

Unit V Etiopathogenesis and treatment of nervous, endocrine, and gastrointestinal disorders **12 Hours**

Definition, etiopathogenesis, clinical manifestations, nonpharmacological and pharmacological management of the diseases associated with Endocrine system: Diabetes, thyroid diseases, disorders of sex hormones. Nervous system: Epilepsy, Parkinson's disease, stroke. Psychiatric disorders: depression, schizophrenia, and Alzheimer's disease. Gastrointestinal system: Peptic Ulcer.

Unit VI Etiopathogenesis and treatment of cancer, infectious diseases, bone, and joint disorders. **13 Hours**

Principles of Cancer: Classification, etiology, and pathogenesis of Cancer. Infectious diseases: Meningitis, Typhoid, Leprosy, Tuberculosis Urinary tract infections. Sexually transmitted diseases: AIDS, Syphilis, Gonorrhea. Definition, etiopathogenesis, clinical manifestations, nonpharmacological and pharmacological management of the diseases associated with diseases of bones and joints: Rheumatoid Arthritis, Osteoporosis, Gout.

Total Periods: 75

Practical: Pharmacotherapeutics

1. Preparation and discussion of SOAP (Subjective, Objective, Assessment and Plan) notes for at least SIX clinical cases (real / hypothetical) of the following disease conditions: Hypertension and Angina Pectoris
2. Preparation and discussion of SOAP (Subjective, Objective, Assessment and Plan) notes for at least SIX clinical cases (real / hypothetical) of the following disease conditions: Myocardial Infarction and Hyperlipidemia
3. Preparation and discussion of SOAP (Subjective, Objective, Assessment and Plan) notes for at least SIX clinical cases (real / hypothetical) of the following disease conditions: Rheumatoid arthritis and Asthma
4. Preparation and discussion of SOAP (Subjective, Objective, Assessment and Plan) notes for at least SIX clinical cases (real / hypothetical) of the following disease conditions: Diabetes and Epilepsy
5. Preparation and discussion of SOAP (Subjective, Objective, Assessment and Plan) notes for at least SIX clinical cases (real / hypothetical) of the following disease conditions: Stroke and Depression
6. Preparation and discussion of SOAP (Subjective, Objective, Assessment and Plan)

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notes for at least SIX clinical cases (real / hypothetical) of the following disease conditions: Tuberculosis and Anemia

7. Preparation and discussion of SOAP (Subjective, Objective, Assessment and Plan) notes for at least SIX clinical cases (real / hypothetical) of the following disease conditions: Dermatological conditions and Viral Infection
8. Patient counselling exercises using role plays based on the real / hypothetical clinical case scenarios
9. Simulated cases to enable dose calculation of selected drugs in pediatrics under various pathological conditions
10. Simulated cases to enable dose calculation of selected drugs in geriatrics under various pathological conditions.

Total Periods: 30

Reference Books

1. Rang, H.P., Dale, M.M., Ritter, J.M. and Moore, P.K., 2003. Pharmacology, Churchill Livingstone. New York, pp.3-4.
2. Katzung, B.G., 2012. Basic and clinical pharmacology.
3. Goodman, L.S., 1996. Goodman and Gilman's the pharmacological basis of therapeutics (Vol. 1549, pp. 1361-1373). New York: McGraw-Hill.
4. Goodman, L.S. and Gilman, A., 1955. The pharmacological basis of therapeutics. The Macmillan.
5. Howland, R.D., and Mycek, M.J., 2006. Lippincott's Illustrated reviews, Pharmacology. Teaching Learning, 5, p.5.
6. Tripathi, K.D., 2003. Essentials of medical pharmacology, Jaypee Brothers. Medical Publications New Delhi Edn, 5, pp.93-94.
7. Kulkarni SK. Handbook of experimental pharmacology. Vallabh Prakashan, Dawson, B. and Trapp, R.G. 2019.

Course outcomes: After the completion of course, the students will be able to:

CO1: Understand the etiology and pathogenesis of the disease states and basic mechanism of the disease.

CO2: Explain the functions of different drugs used in the mitigation of the diseased condition.

CO3: Identify the signs and symptoms of the diseases.

CO4: Describe various therapeutic approaches and the importance of individualized therapeutic plans based on diagnosis.

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CO5: Create a plan of rational medicine therapy for common diseases and design and deliver discharge counselling for patients.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	-	-	-	-	1	1	-	1
CO 3	3	3	2	-	3	-	-	-	1	1	2	1
CO4	3	3	2	2	3	1	1	-	1	1	2	1
CO5	3	3	2	2	3	1	1	1	1	1	2	1
Average	3	3	2	2	3	1	1	1	1	1	2	1

25CI602

BIOPHARMACEUTICS

L-T-P-C:4-1-1-6

Course Objective: This course provides an in-depth understanding of the principles and applications of biopharmaceutics, which is the study of the relationship between the physicochemical properties of drugs, their formulation, and their pharmacological and therapeutic effects. The course covers topics such as drug absorption, distribution, metabolism, and excretion, as well as drug delivery systems and bioavailability.

Unit I Introduction to Biopharmaceutics

12 Hours

Overview of biopharmaceutics and its significance in drug development and delivery. Basic concepts of drug absorption, distribution, metabolism, and elimination (ADME). Physiological, biochemical and cell biology-related background to the physiology of the gastrointestinal tract and the liver and different pathophysiological conditions associated with gastrointestinal tract and the liver.

Unit II Drug Delivery Systems

12 Hours

Routes of administration: oral, parenteral (intravenous, intramuscular, subcutaneous),

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topical, transdermal, inhalation, etc. Formulation considerations for different drug delivery systems. Drug release kinetics and mechanisms.

Unit III Drug Absorption and Distribution 13 Hours

Absorption: Mechanisms of drug absorption through GIT, factors influencing drug absorption through GIT, absorption of drug from Non per oral extra-vascular routes.

Tissue permeability of drugs, binding of drugs, apparent, volume of drug distribution, plasma and tissue protein binding of drugs, factors affecting protein-drug binding. Kinetics of protein binding, Clinical significance of protein binding of drugs.

Unit IV Drug metabolism and elimination

12 Hours

Drug metabolism and basic understanding metabolic pathways Microsomal oxidation. Phases of biotransformation of Drugs and endogenous toxic compounds. Microsomal oxidation reactions. II phase of biotransformation of medicines. Conjugation reactions in the liver. Renal excretion of drugs, factors affecting renal excretion of drugs, renal clearance, non-renal routes of drug excretion of drugs

Unit V Bioavailability and Bioequivalence

13 Hours

Definition and Objectives of bioavailability, absolute and relative bioavailability, measurement of bioavailability, in-vitro drug dissolution models, in-vitro-in-vivo correlations, bioequivalence studies. Regulatory aspects on the absorption, bioavailability, and bioequivalence of drug.

UNIT 6 Biopharmaceutics and Pharmacokinetics Modeling

13 Hours

Mathematical models for drug absorption, distribution, and elimination. PK/PD (pharmacokinetic/pharmacodynamic) modeling and simulation, Application of modeling in drug development and dosage regimen optimization.

Total Periods: 75

Practical – Biopharmaceutics Lab

Basic understanding of in silico, in vitro, in situ and in vivo methods that have been used to study dissolution, permeability, transit times, stability, metabolism, bioavailability, and bioequivalence. To use open-source platforms such as Swiss dock for prediction of ADMET parameters and their comparison with the practical values.

1. Use of Nomograms to adjust dose for persons having renal impairment.
2. Use of Nomograms to adjust dose for persons having hepatic impairment.
3. Study of Protein Binding of ciprofloxacin hydrochloride by Enhancing the concentration of Egg Albumin.

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4. To study the effect of solubility of drug on dissolution and absorption.
5. To study the effect of stability on absorption of the drugs.
6. To study the effect of pH on dissolution of marketed enteric coated tablets.
7. To study the effect of pH on dissolution of diclofenac sodium.
8. Calculation of bioequivalence of different dosage forms.
9. Effect of transit times on dissolution of tablets.

Total Periods: 30

Recommended books:

1. Shargel, L., Andrew, B.C. and Wu-Pong, S., 1999. Applied biopharmaceutics & pharmacokinetics (Vol. 264). Stamford: Appleton & Lange.
2. Notari, R.E., 1975. Biopharmaceutics and pharmacokinetics: an introduction. (No Title).
3. Gibaldi, M., 1971. Introduction to biopharmaceutics.
4. Wagne, J.G., 1970. Biopharmaceutics and relevant pharmacokinetics. Drug Information Bulletin, 4(2), pp.226-226.
5. Niazi, S., 1979. Textbook of biopharmaceutics and clinical pharmacokinetics.
6. Handbook of Bioequivalence Testing, by Niazi, S. K.
7. Macheras, P. and Iliadis, A., 2016. Modeling in biopharmaceutics, pharmacokinetics, and pharmacodynamics: homogeneous and heterogeneous approaches (Vol. 30). Springer.
8. Riviere, J.E. ed., 2011. Comparative pharmacokinetics: principles, techniques, and applications. John Wiley & Sons.

Course Outcome Upon the completion of the course the student will be able to

CO1 Understand biopharmaceutical, physiological, biochemical and cell biology-related aspects on the transport and metabolism of drugs in the gastrointestinal tract and in the liver.

CO2 Explain mechanisms behind the transport of drug and metabolism and how drugs can interact with other drugs and food and methods to study these interactions.

CO3 Describe the role of biopharmaceutics in clinical research within the pharmaceutical industry.

CO4 Apply regulatory requirements within the biopharmaceutical area.

CO5 Compile the findings of biopharmaceutical studies in clinical research.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

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Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	-	-	-	-	1	1	-	1
CO 3	3	3	2	-	3	-	-	-	1	1	2	1
CO4	3	3	2	2	3	1	1	-	1	1	2	1
CO5	3	3	2	2	3	1	1	1	1	1	2	1
Average	3	3	2	2	3	1	1	1	1	1	2	1

25CI603 DATA MANAGEMENT TECHNOLOGIES L-T-P-C:4-1-1-6

Course Objective: This course introduces data management technology focused on its applications in clinical research. Students will learn about the principles and techniques involved in managing clinical research data effectively and ensuring data quality and integrity. The course covers both theoretical concepts and practical skills necessary for data management in clinical research settings.

UNIT I Introduction to Data Management in Clinical Research 15 Hours

Overview of data management in the clinical research process, Role of data management in ensuring data integrity and regulatory compliance, Types of Clinical data including image data, tabular data, EM waves data, Bio signal data.

UNIT II Designing Databases for Clinical Research Studies 15 Hours

Study database architecture and design considerations, Electronic Case Report Forms (eCRFs) and data collection instruments, Data entry best practices and techniques, Data validation checks and edit checks, Double data entry and reconciliation. Identification and resolution of data discrepancies, Query management and resolution process, Data quality assessment and monitoring

UNIT III. Electronic Data Capture (EDC) Systems 15 Hours

Introduction to EDC systems and their features, EDC system implementation and user training, Data extraction and reporting from EDC systems, Data Security and Privacy,

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Protection of personal health information (PHI), Data encryption and anonymization techniques, Compliance with Health Insurance Portability and Accountability Act (HIPAA)

UNIT IV. Data Integration and Interoperability

15 Hours

Data standards in clinical research (e.g., CDISC standards), Data integration techniques for multi-site studies, Electronic Health Records (EHR) integration Risk Based Monitoring, statistical monitoring, central monitoring, involvement of real time data monitor, data visualization and monitoring.

UNIT V. Emerging Trends in Data Management for Clinical Research

15 Hours

Use of artificial intelligence and machine learning in data management, Mobile health (mHealth) technologies and their impact on data collection, AI based Healthcare applications, Challenges and Benefits of using AI in healthcare applications.

Total Periods: 75

Practical: - Data Management Technologies Lab

In a clinical research lab, effective data management is crucial to ensure the integrity, accuracy, and security of research data.

1. Electronic Data Capture (EDC) Systems: EDC systems are software applications designed to collect, manage, and store clinical research data electronically.
2. Clinical Trial Management Systems (CTMS): CTMS platforms are used to manage various aspects of clinical trials, including participant enrollment, study timelines, documentation, and communication.
3. Laboratory Information Management Systems (LIMS): LIMS software is specifically designed to manage laboratory workflows and data in research and healthcare settings.
4. Data Warehousing and Integration: Data warehousing and integration technologies play a crucial role in consolidating and integrating data from multiple sources.
5. Data Security and Privacy Measures: Data security is paramount in clinical research labs to protect participant confidentiality and comply with regulatory requirements.
6. Statistical Analysis Software: Statistical analysis software packages like SAS (Statistical Analysis System), R, or SPSS are commonly used in clinical research labs for data analysis and statistical modeling.
7. Electronic Medical Records (EMR) Systems: In some cases, clinical research labs may integrate with hospital or clinic EMR systems to access relevant patient data for research purposes.
8. Data Backup and Disaster Recovery: To ensure data integrity and prevent loss,

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clinicalresearch labs implement data backup and disaster recovery solutions.

Total Periods: 30

References and Textbooks

1. Rondel, R.K., Varley, S.A. and Webb, C.F. eds., 2000. Clinical data management. NewYork: Wiley.
2. Prokscha, S., 2011. Practical guide to clinical data management. CRC Press.
3. Ngari, M.M., Waithira, N., Chilengi, R., Njuguna, P., Lang, T. and Fegan, G., 2014. Experience of using an open-source clinical trials data management software system inKenya. BMC research notes, 7(1), pp.1-8.
4. Huberman, A.M., and Miles, M.B., 1994. Data management and analysis methods.
5. "Schadt, E.E., Linderman, M.D., Sorenson, J., Lee, L. and Nolan, G.P., 2010. Computational solutions to large-scale data management and analysis. Nature reviews genetics, 11(9), pp.647-657.
6. Callahan, S.P., Freire, J., Santos, E., Scheidegger, C.E., Silva, C.T. and Vo, H.T., 2006, June. VisTrails: visualization meets data management. In Proceedings of the 2006 ACM SIGMOD international conference on Management of data (pp. 745-747).

Course Outcome:

CO1: Explain and understand the importance of data management, regulatory requirements, and guidelines for data management in clinical research.

CO2: Develop skills in designing and implementing databases for clinical research studies.

CO3: Explain data entry, validation, and cleaning techniques specific to clinical research.

CO4: Apply knowledge and hands-on on electronic data capture (EDC) systems and understand the principles of data security and privacy in clinical research.

CO5: Explore emerging trends and technologies in data management for clinical research.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												

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CO 1	3	-	-	-	-	-	-	-	1	1	-	2
CO 2	3	-	2	-	-	-	-	-	1	1	-	2
CO 3	3	3	2	-	3	-	-	-	1	1	2	2
CO4	3	3	2	2	3	1	1	-	1	1	2	2
CO5	3	3	2	2	3	1	1	1	1	1	2	2
Average	3	3	2	2	3	1	1	1	1	1	2	2

25CI604	Ability Enhancement/Co-curricular	Credits- 2
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25CI605	Industrial Training/Survey/Project	Qualifying
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SEMESTER VII

25CI701 HEALTH ECONOMICS AND OUTCOME RESEARCH L-T-P-C:4-1-0-5

COURSE OBJECTIVES

This course provides an overview of the structure of health care markets and its economy. Students will understand the impact of the health care system on the competing goals of

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broad access, high quality, and affordability. Identify how consumers and providers respond to changes in incentives and develop an appreciation for opposing views on health care reform.

UNIT I Introduction to Health Economics

10Hours

Introduction to Health Economics, Demand; Supply; Market Equilibrium; Elasticity; Market Efficiency; Ethical principles in health economic analysis; Measuring status of health and cost of illness. Introduction to outcome research and its role in evaluating healthcare interventions.

UNIT II Economic Evaluation Methods

15 Hours

Cost-effectiveness analysis (CEA): principles, measures of effectiveness, and cost-effectiveness ratios. Cost-utility analysis (CUA): quality-adjusted life years (QALYs) and utility measures. Cost-benefit analysis (CBA): monetizing health outcomes and interpreting benefit-cost ratios. Budget impact analysis (BIA): assessing the financial implications of healthcare interventions.

UNIT III Health Technology Assessment and Health Outcomes Measurement 15 Hours

Overview of HTA and its role in evaluating the value of healthcare technologies. HTA agencies, guidelines, and processes in different countries. Conducting economic evaluations for HTA purposes. Health Outcomes Measurement: Patient-reported outcomes (PROs) and quality of life assessments. Health-related quality of life instruments and utility measurement. Preference-based valuation methods (e.g., time trade-off, standard gamble).

UNIT IV Economic Evaluation in Pharmaceutical Industry

12 Hours

Economic Evaluation in Pharmaceutical Industry: Economic considerations in pharmaceutical product development. Pricing and reimbursement strategies. Real-world evidence (RWE) and its role in economic evaluations.

UNIT V Health Policy and Health Economics

13Hours

Health policy analysis and its relationship with health economics. Economic evaluation in healthcare policy decision-making. The role of economic evaluations in healthcare resource allocation. Ethical and Methodological Issues: Ethical considerations in health economic research. Methodological challenges and limitations in conducting economic evaluations. Interpretation and reporting of health economic analyses.

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UNIT VI Advanced Topics in Health Economics and Outcome Research 10 Hours

Value-based pricing and value frameworks. Economic evaluations in specific disease areas. Emerging trends and methodologies in HEOR research.

Total Periods: 75

Recommended books/ References

1. Health Economics, Jay Battacharya, Timothy Hyde, and Peter Tu, 1st Edition, PalgraveMacmillan, 2014.
2. Health Care Reform: What It Is, Why It is Necessary, How It Works, Jonathan Gruber, 2012.
3. An Introduction to Health Economics (Pharmacoeconomics & outcomes research). Brookwood Medical Publications Ltd (1 January 1995). M.F. Drummond.
4. <https://www.econ.berkeley.edu/undergrad/home/learning-goals>

COURSE OUTCOME

- CO1.** Explain the use of central ethical principles that are considered when government makes decisions and influence public health.
- CO2.** Summarize which health indices can be used to combine different aspects of health and explain how to derive the cost of illnesses.
- CO3.** Describe how government insurance programs affect participants, medical providers, and private insurance markets.
- CO4.** Distinguish the different types of medical care data and how those measurements are driven by reimbursement and quality measurement needs.
- CO5.** Evaluate the impact of health care reform from both the free-market and government-interventionist perspectives.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3- Substantial (high)

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Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	-	-	-	-	1	1	-	1
CO 3	3	3	2	-	3	-	-	-	1	1	2	1
CO4	3	3	2	2	3	1	1	-	1	1	2	1
CO5	3	3	2	2	3	1	1	1	1	1	2	1
Average	3	3	2	2	3	1	1	1	1	1	2	1

25CI702 EMERGING TECHNOLOGIES IN CLINICAL TRIALS L-T-P-C:3-1-1-5

Course Objective The objective of the course on Emerging Technologies in Clinical Research is to provide students with an in-depth understanding of the latest technological advancements and their applications in the field of clinical research. The course aims to familiarize students with the potential benefits, challenges, and regulatory considerations associated with adopting emerging technologies in clinical trials.

UNIT I. Introduction to Emerging Technologies in Clinical Trials. 10 Hours

Overview of clinical trials and their importance in medical research, Introduction to emerging technologies and their impact on clinical trials, Ethical and regulatory considerations for adopting emerging technologies, Wearable Devices and Sensors in Clinical Trials, Introduction to wearable devices and sensors, Applications of wearables and sensors in data collection, Data integration and analysis from wearables and sensors, Challenges and considerations in using wearable devices in clinical trials

UNIT II Electronic Health Records (EHR) and Real-World Data (RWD) in Clinical Trials 10 Hours

Utilizing electronic health records in clinical trial design and recruitment; Leveraging real-world data for trial feasibility, patient selection, and outcomes assessment; Ensuring data quality and privacy in the use of EHR and RWD; Regulatory considerations and guidelines for utilizing EHR and RWD in clinical trials; Mobile Health (mHealth) Technologies in

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Clinical Trials

UNIT III . Introduction to mobile health technologies and applications 10 Hours

Mobile apps for patient recruitment, informed consent, and study participation, Remote monitoring and telemedicine in clinical trials, Data security and privacy considerations in mHealth technologies.

UNIT IV Artificial Intelligence (AI) and Machine Learning (ML) in Clinical Trials

10 Hours

Overview of AI and ML techniques in clinical trial design and analysis, Predictive modeling and patient stratification using AI and ML, Adverse event detection and signal identification using AI and ML, Ethical considerations and challenges in adopting AI and ML in clinical trials.

UNIT V .Blockchain Technology in Clinical Trials

10 Hours

Introduction to blockchain technology and its potential in clinical trials, Enhancing data integrity, transparency, and security with blockchain, Introduction to virtual and decentralized clinical trials, Remote patient monitoring and virtual visits, Data collection and management in virtual trials, Regulatory considerations and emerging best practices for virtual trials.

UNIT VI. Data Analytics and Big Data in Clinical Trials

10 Hours

Introduction to big data and its applications in clinical trials, Predictive analytics for patient recruitment and trial outcomes, Real-time monitoring and adaptive trial design using big data. Digital Biomarkers and Endpoint Assessment

Total Periods: 60

Practical: Emerging Technologies in Clinical Trials Lab

Introduction to the lab environment and tools used in emerging technologies in clinical trials

1. Familiarization with software applications and platforms relevant to data collection, analysis, and management
2. Hands-on experience with wearable devices and sensors commonly used in clinical trials
3. Data collection and integration from wearable devices and sensors
4. Accessing and working with electronic health records and real-world data sources
5. Data extraction, transformation, and integration from EHR and RWD platforms

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6. Analyzing and visualizing clinical data from EHR and RWD using statistical software or programming languages
7. Exploring mobile health applications and platforms for clinical trials
8. Hands-on experience with mHealth tools for patient recruitment, data collection, and remote monitoring
9. Integration and analysis of data obtained from mHealth technologies
10. Hands-on experience with blockchain platforms and tools used in clinical trials
11. Simulation of virtual and decentralized clinical trial scenarios
12. Hands-on experience with virtual trial platforms and remote monitoring tools
13. Working with large-scale clinical trial datasets. Exploratory data analysis and visualization using big data tools and platforms
14. Digital Biomarkers and Endpoint Assessment Lab
15. Identification and evaluation of digital biomarkers in clinical trial datasets and implementing algorithms for digital endpoint assessment

Total Periods: 30

Textbooks and References

1. Cortez, N., 2018. Digital Health: Scaling Healthcare to the World. Cham, Switzerland: Springer, Cham, pp.249-269.
2. Rosa, C., Campbell, A.N., Miele, G.M., Brunner, M. and Winstanley, E.L., 2015. Using e- technologies in clinical trials. Contemporary clinical trials, 45, pp.41-54.
3. Gresham, G., Schrack, J., Gresham, L.M., Shinde, A.M., Hendifar, A.E., Tuli, R., Rimel, B.J., Figlin, R., Meinert, C.L. and Piantadosi, S., 2018. Wearable activity monitors in oncology trials: current use of an emerging technology. Contemporary clinical trials, 64, pp.13-21.
4. "mHealth: Multidisciplinary Verticals" edited by Sasan Adibi
5. "Real-World Evidence Generation and Evaluation of Therapeutics: Proceedings of a Workshop" by the National Academies of Sciences, Engineering, and Medicine
6. "Blockchain in Healthcare: Innovations that Empower Patients, Connect Professionals and Improve Care" by Corey Todaro and John D. Halamka
7. "Artificial Intelligence in Healthcare: A Comprehensive Guide" edited by Shailendra Singh and Rajendra Akerkar
8. "Chang, H., 2015. Book review: Data-driven healthcare & analytics in a big data world. Healthcare informatics research, 21(1), pp.61-62.
9. Manogaran, G., Lopez, D., Thota, C., Abbas, K.M., Pyne, S. and Sundarasekar, R., 2017. Bigdata analytics in healthcare Internet of Things. Innovative healthcare systems for the 21st century, pp.263-284.

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Course Outcome Upon the completion of this course the student will be able to

CO1. Evaluate the impact of wearable devices, sensors, and other data collection technologies on clinical trial design and data quality.

CO2. Analyze the potential applications and benefits of electronic health records (EHR) and real-world data (RWD) in clinical research.

CO3. Assess the role of mobile health (mHealth) technologies in patient recruitment, remote monitoring, and data collection in clinical trials.

CO4. Examine the use of artificial intelligence (AI) and machine learning (ML) in clinical trial design, patient stratification, and adverse event detection.

CO5. Explain the principles and challenges of virtual and decentralized clinical trials and their impact on trial design and patient engagement.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	2
CO 2	3	-	2	-	-	-	-	-	1	1	-	2
CO 3	3	3	2	-	3	-	-	-	1	1	2	2
CO4	3	3	2	2	3	1	1	-	1	1	2	2
CO5	3	3	2	2	3	1	1	1	1	1	2	2
Average	3	3	2	2	3	1	1	1	1	1	2	2

25CI703

RESEARCH METHODOLOGY

L-T-P-C:3-1-1-5

Course Objectives

This course is designed to enable students to:

- understand basic concepts of research and its methodologies
- identify and define a research problem, state a hypothesis, select an appropriate research design, and implement research project
- discuss the concepts and procedures of sampling, ethical considerations, data

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collection, analysis, and reporting

- review literature, collect and analyze data, and write a report / dissertation

Unit I: Introduction to Research

14 hours

Definition of research, objectives of research, applications and types of research, research process and steps involved. Collecting and reviewing literature, types of literature review (survey, systematic, meta-analysis), conceptualization, and formulation of a research problem, constructing hypothesis, identifying variables, Synopsis.

Unit II: Design of Research and Sample Survey or Experiments

12 hours

Research Design-Selecting and defining a research problem, need for research design, features of a good research design, different research designs (exploratory, descriptive experimental and diagnostic research); Design of Sample Survey or Experiments: Census V/s Sample enumerations, objectives and principles of sampling, Types of sampling, Sampling and Non-sampling errors. Designing Questionnaires and interviews. Determination of the sample size.

Unit III: Measurement of Scaling Concepts

9 hours

Scales of measurements, nominal, ordinal, internal and ratio scales, Errors in measurements. Validity and Reliability in measurement, Scale Construction Techniques.

Unit IV: Data Collection & Analysis

16 hours

Primary & secondary data, Validity and Reliability of data collection procedures, data preparation, exploratory data analysis, parametric and nonparametric tests, correlation and regression analysis, ANOVA, Multivariate Techniques, Introduction to data representation methods/tools. Like SAS, SPSS, R, etc.

Unit V: Research Ethics & Scientific Communication

9 hours

Ethical conduct of research, Introduction to scientific misconduct, repeatability and reproducibility of research, Ethics, and plagiarism in publication, Art of Communicating Scientifically – formats for scientific presentation and writing, conclusion, referencing and, Bibliography; journal publications, Impact factor, Citation index, Research related Software - references management, Plagiarism detection etc.

Total Periods: 60

Recommended Books/ Resources:

1. Kothari C.R., “Research Methodology, Methods, and Techniques, Second edition, (2008), New Age International Publication.

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2. Krishna Swamy K.N., Siva Kumar A.I., Mathirajan M., “Management Research Methodology (2006), Pearson Education, New Delhi.
3. Ranjit Kumar: Research Methodology, A step by step guide for beginners, Pearson Education, Sixth Edition 2009.
4. Mark Saunders, Philip Lewis, Adrain Thornhiu: Research Methods for Business Students, Pearson Education.
5. Ram Ahuja, “Research Methods,” (2001), Rawat Publications, New Delhi. 6. Cooper D., Schindler P., Business research methods,” (2003) Tata Mc-Graw Hill, New Delhi
6. https://apps.who.int/iris/bitstream/handle/10665/206929/929061157X_eng.pdf?sequence=1&isAllowed=y

Practicals

1. Literature review

Detailed review of literature on the chosen topic

2. Reference management tools (Mendeley, EndNote, etc.)

Use of reference management tools and integration into MS office

3. Setting research question

Identify the research area/topic of interest, review the research trends, write about significance of the chosen topic/area, set research question(s), and develop hypothesis.

4. Setting objectives and protocols (methods)

Write specific objectives and design research methods for each objective.

5. Data collection, analysis, and representation

6. Report writing and communication

Finalize the report based on the collected and analyzed data, communicate for peer-review, etc.

Total Periods: 30

Course Outcomes

Upon completion of the course, the student should be able to:

CO1: Understand the basics of research, types, steps, and application.

CO2: Collect, review, and analyze literature and data

CO3: Identify and define a research problem, set hypothesis, and select an appropriate research design

CO4: Demonstrate the ability to choose methods appropriate to research aims and objectives

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CO5: Compile and Communicate research findings

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	1	-	-	-	1	1	-	1
CO 3	3	3	2	-	1	-	-	-	1	1	2	1
CO4	3	3	2	2	1	1	1	-	1	1	2	1
CO5	3	3	2	2	1	1	1	1	1	1	2	1
Average	3	3	2	2	1	1	1	1	1	1	2	1

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25CI704 GOOD LABORATORY AND MANUFACTURING PRACTICES

L-T-P-C: 3-1-1-5

COURSE OBJECTIVE

The main objective of this course is to introduce the basic concepts of Quality assurances in Good Laboratory Practices. This course addresses Federal Food and Drug Law and discusses the Trade and Company Standards Control by National, International. This will also enable the students to know about the Scope and importance of GLP and GMP.

Unit I: Good Laboratory Practices

10 Hours

Introduction to GLP, WHO guidelines on GLP and GMP, Quality assurances in Good Laboratory Practices and Quality Standards. Advantages and Disadvantages, Concept of Quality Control. Government and trade standards of quality Federal Food and Drug Law FDA Action BSTI Laws, BSTI action and activities other food laws (Legalization).

Unit II: Validation of Analytical Procedures

15 Hours

Implications of cGMP and Food plant sanitation. The regulations of cGMPs Planning of Plant Sanitation Programs and Construction factors Hygienic design of food plants and equipment Sanitation in warehousing, storage, shipping, receiving, containers and packaging materials Control of rats, rodents, birds, insects, and microbes. Cleaning and Disinfection: Physical and Microbiological Approach. Food Quality and Quality control including the HACCP system. (Critical quality control points in different stages of production including raw materials and processing materials)

Unit III: Good Manufacturing Practice

10 Hours

Good Manufacturing Practice: definitions, requirements, and historical background. Quality assurance, quality management, design of quality systems. Principles for documentation in GMP, Site Master File, SMF, Monographs, Protocols (production protocols, standard operating procedures, SOP).

Unit IV: Risk Analysis and Precautions

10 Hours

Risk analysis and risk assessment, Clinical trials, Qualification and validation, Microbiological test and quality control, Aseptic production, localities, clothing, Audit, monitoring, internal and external inspections

Unit V: Sampling

15 Hours

Sampling: Introduction, WHO guidelines, sampling plans and techniques, operating characteristics curves, maintenance of sampling records of finished product and packaging material. Ethics in manufacturing and control, Principles of quality by design (QBD). Introduction to the concept of Design of Experiment (DOE) Application of QBD principles in Biotech product development.

PRACTICALS

1. Basic procedures and precautions for working in a Laboratory
2. Pre-analysis preparations and management of materials.
3. Describe Sampling techniques and perform a sampling of raw materials
4. Accuracy and Precision of analysis method.
5. Validate the analysis metho
6. Basic procedures to produce hygiene food and drugs.
7. Procedure to produce a Food product.
8. Storage stability tests for Developed products.
9. Packaging and transportation requirements for a product.

Total Periods: 30

REFERENCES

1. cGMP starter guide: Principles in Good Manufacturing Practices for Beginners, Emmet P. Tobin, Createspace Independent Publishing Platform, April 2016.
2. Good Manufacturing Practices for Pharmaceuticals: GMP in Practice, B Cooper, CreatespaceIndependent Publishing Platform, July 2017.
3. Sarwar Beg and Md Saquib Hasnain, Pharmaceutical Quality by design: Principles and application, Academic press, March 2019.
4. Ron S. Kenett, Shelemياهو Zacks, Daniele Amberti, Modern Industrial Statistics: with applications in R, MINITAB and JMP, 2nd Edition, Wiley, January 2014.
5. N Politis S, Colombo P, Colombo G, M RekkasD.Design of experiments (DoE) in pharmaceutical development, Drug Dev Ind Pharm. 2017 Jun;43(6):889-901. doi: 10.1080/03639045.2017.1291672.
6. Andrew Teasdale, David Elder, Raymond W. Nims, ICH quality guidelines- An implementation guide, Dec 2017.
7. Singh, G., Agarwal, G. and Gupta, V. Drug regulatory affairs, CBS publication, 2005.
8. Marc P. Mathieu, New Drug Development: A regulatory overview, Nov 2000.
9. ICH guidelines available in the official website “<https://www.ich.org>”. Course Outcomes: Understand that the areas that come under the Good Laboratory Practices are personnel and organizational, testing facilities, equipment, testing and controls, records, reports, and protocolfor and conduct of non-clinical labs.

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COURSE OUTCOMES

Upon completion of the course, the student should be able to:

CO1: To know about the detailed guidelines on GLP and GMP.

CO2: To study the trade standards of quality Federal Food and Drug Law FDA.

CO3: Understand the concept of Regulatory requirements and approval procedures for NewDrugs and technologies.

CO4: Identify and understand the GLP and GMP

CO5: Describe the procedures and importance of GLP and GMP.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	1	-	-	-	1	1	-	1
CO 3	3	3	2	-	1	-	-	-	1	1	2	1
CO4	3	3	2	2	1	1	1	-	1	1	2	1
CO5	3	3	2	2	1	1	1	1	1	1	2	1
Average	3	3	2	2	1	1	1	1	1	1	2	1

Industrial Internship/Startup

Credits-15

**25CI801 FUNDAMENTALS OF COMPUTER APPLICATIONS & DATA BASE
MANAGEMENT**

L-T-P-C:6-2-0-8

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Course objectives

- ❖ Learning about the basic components of a computer system, the history of computers, and different types of computer systems.
- ❖ Learning the techniques and tools to design, build, and extract information from a database
- ❖ Learning how to perform basic database administration tasks, such as applying relational database design principles to build data models
- ❖ Learning how to organize, maintain, and retrieve information from a database management system

Unit- I Introduction to computer:

15 hours

Introduction, Characteristics of computer, block diagram of computer, Generations of computer, computer languages. Operating systems –concepts & types. Input output devices: Input devices (keyboard, point and draw devices, data scanning devices, digitizer, electronic card reader, voice recognition devices, vision-input devices), output devices (monitors, pointers, plotters, screen image projector, voice response systems). Processor and memory: The Central Processing Unit (CPU), main memory. Storage Devices: Sequential and direct access devices, magnetic tape, magnetic disk, optical disk, mass storage devices.

Unit II Microsoft office tools and Internet

15 hours

Introduction to MS-Word, Introduction to power-point: introduction, creating and manipulating presentation, views, formatting and enhancing text, slide with graphs. MS-PowerPoint, MS-Excel. Computer networks & and HTML: Introduction, types of networks (LAN, MAN, WAN, Internet, Intranet), network topologies (star, ring, bus, mesh, tree, hybrid), components of network. Internet and its Applications: definition, brief history, basic services (E-Mail, File Transfer Protocol, telnet, the World Wide Web (WWW)), www browsers, use of the internet. FTP TCP / IP, Electronic mail.

Unit III Operating Systems

15 hours

Identify operating systems and their attributes (i.e., DOS, Unix, Macintosh, Windows), Identify the advantages and disadvantages of the computer to individuals and business. Identify the roles and equipment used for input, processing, and output in an information system. Identify correct safety procedures Basic Data Processing- Input, update and store data into records in an existing database, Open stored spreadsheet, input and update data into spreadsheet, store revised spreadsheet and print revised spreadsheet.

Unit IV Database and Spreadsheet Operations

15 hours

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Plan and create database, input and update data into records, store database and print quick reports from database, Create spreadsheet, input data into spreadsheet, update data in spreadsheet and store spreadsheet. Introduction to Spread sheet packages- Data Modeling, Introduction to MS-Access, Forms, Filters and Queries, Charts and Reports, Command Buttons, Macros and Database Maintenance.

Unit V Visual Basics

15 hours

Introduction to VB, Programming fundamentals, Hospital Information System (HIS) with Electronic Medical Records (EMR) or Electronic Health Information Management System (HMIS)

Total Periods: 75

Reference Books

1. Database Fundamentals Neeraj Sharma, Liviu Perniu, Raul F. Chong, Abhishek Iyer, Chaitali Nandan, Adi-Cristina Mitea, Mallarswami Nonvinkere, Mirela Danubianu A book for the community by the community First Edition (November 2010). IBM Corporation 2010.
2. Fundamentals of Database Management Systems, Dalwinder Singh, RS Salaria ISBN: 978-93-92549-44-1

Course Outcome

After the completion of course, the students will be able to:

CO1. Explain the the Computer operating systems and languages

CO2. Describe the skills in using Microsoft Word, Power point, Excel

CO3. Apply the knowledge of computer networks.

CO4. Discuss the database processing

CO5. Design, develop and evaluate data base systems

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
CO 1	3	-	-	-	-	-	-	-	1	1	-	-

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CO 2	3	-	2	-	1	-	-	-	1	1	-	1
CO 3	3	3	2	-	1	-	-	-	1	1	2	1
CO4	3	3	2	2	1	1	1	-	1	1	2	1
CO5	3	3	2	2	1	1	1	1	1	1	2	1
Average	3	3	2	2	1	1	1	1	1	1	2	1

25CI802

HEALTH & BIO INFORMATICS

L-T-P-C:6-2-0-8

Course Objective: This course provides an overview of Manage health, medical, and bio-informatics information using best practices in data stewardship; data science and data analytics; and human-centered design and systems.

Syllabus

75 Hours

Unit-I Introduction to Health Care Informatics

15 hours

Overview of health care informatics, History and evolution of health care informatics, Health informatics workforce development, Future trends and directions in health care informatics

Unit-II Health Information Systems

15 hours

Electronic Health Records (EHR) - EHR implementation and adoption , EHR functionalities and benefits, EHR data management and security , Benefits and challenges of EHR Health Information Exchange (HIE) - Interoperability and data sharing in healthcare Standards and protocols for HIE , Health information exchange concepts and principles , HIE implementation challenges and solutions , Impact of HIE on healthcare coordination and continuity. Clinical Decision Support Systems (CDSS) - Clinical Decision Support Systems (CDSS), Applications of CDSS in healthcare , Evaluating CDSS effectiveness

Unit-III Health Data Management

15 hours

Data collection, storage, and retrieval , Data standards and interoperability, Health Data Standards and Interoperability , SNOMED CT, LOINC, and other terminologies , Health Level Seven International (HL7) standards , Health Analytics and Data Visualization, Introduction to data analytics in healthcare

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Techniques for data visualization

Unit-IV Health Analytics & Business Intelligence

15 hours

Data analysis and visualization, Predictive analytics in health care, Population health management , Data sources and data collection in healthcare ,Data mining and visualization techniques, Quality improvement using data analytics

Unit-V Introduction to Bioinformatics

15 hours

Definition and scope of bioinformatics, Importance of bioinformatics in biological research, Overview of bioinformatics databases and resources Bioinformatics tools and software, Introduction to sequence alignment, Pairwise sequence alignment algorithms (e.g., Needleman-Wunsch, Smith-Waterman), Multiple sequence alignment, Sequence motif analysis and identification, DNA and protein sequence databases, Sequence similarity and homology search tools, Introduction to next-generation sequencing data analysis

Total Periods: 75

References and Textbooks

- 1.Mann, R.D., and Andrews, E.B. eds., 2007. Pharmacovigilance. John Wiley & Sons.
- 2.Health informatics, Practical guide for health care and information technology professionals, VI Ed., Robert E Hoyt & Ann K. Yoshihashi, Ebook ISBN 978-0-9887529-2-4, <http://informaticseducation.org>
- 3.Health Informatics An interpersonal approach, III Ed., Lynda R hardy, ISBN 978-0-323-71196-8

Course Outcome

After the completion of course, the students will be able to:

CO1. Explain the basic principle of Health Care Informatics.

CO2. Describe the global Health Information Systems.

CO3. Explain the methods and tools used in Health Data Management

CO4. Discuss the role of Health Analytics & Business Intelligence.

CO5. Describe the challenges and basics of Bioinformatics

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

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Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	1	-	-	-	1	1	-	1
CO 3	3	3	2	-	1	-	-	-	1	1	2	1
CO4	3	3	2	2	1	1	1	-	1	1	2	1
CO5	3	3	2	2	1	1	1	1	1	1	2	1
Average	3	3	2	2	1	1	1	1	1	1	2	1

25CI803

DISEASE CLASSIFICATION SYSTEMS

L-T-P-C:6-2-0-8

Course Objectives

The aim of the course is to categorize diseases, health-related conditions, and external causes of disease and injury in order to be able to compile useful statistics in mortality and morbidity.

Unit I International Classification of Diseases

15hours

International Classification of Diseases (ICD-10) and Surgical Procedures (ICD-9CM) and SNOMED-CT, Nomenclatures and Classification Systems, Standard Nomenclatures of diseases (SNDO).

Unit II Current Medical Information Terminology

15hours

Current Medical Information Terminology, Systematized Nomenclature of Pathology (SNOP), Systematized Nomenclature of Medicine (SNOMED), Common Procedures Coding System (HCPCS).

Unit III Current Procedural Terminology

15hours

Current Procedural Terminology, International Classification of Functioning, Disability and Health (ICF), Case-Mix Classifications, Diagnosis Related Groups. ICD – 9 (CM),

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ICD – 11, ICD- Oncology (ICD - O). Disease and operation nomenclatures, International Classification of Disease 10, International Classification of Disease – 9CM indexing of patient care data.

Unit IV Introduction and usage of International Classification of Disease 15hours

Introduction and usage of International Classification of Disease in practical's
International Classification of Diseases - ICD-11, ICD-9 CM (Surgical Procedures) CPT
– Current Procedural Terminology (Introduction)

Unit V HCPCS 15hours

Healthcare Common Procedure Coding System (Introduction)- ICD-11 - Alpha-numeric coding ,Volume 1 – Tabular list , Volume 2 – Instruction manual , Volume 3–Alphabetical Index - Classification of Diseases according to Clinical Pertinence , ICD-9CM (Procedure) coding – International Classification of Diseases – Clinical modification, CPT – Introduction of CPT and HCPCS – 3 levels of codes ,SNOMED-CT

Total Periods: 75

Text book:

1. A Modern Approach to Disease Classification and Clinical Coding Folasayo Ayegbayo , 27 September 2018, ISBN-10 : 0359119492
2. Mony, Prem & C., Nagaraj. (2007). Health information management: An introduction to disease classification and coding. The National medical journal of India. 20. 307-10.

Course Outcome

After the completion of course, the students will be able to:

- CO1.** Identify standard clinical terminologies
- CO2.** Identify appropriate clinical classification system
- CO3.** Apply the disease classification system in Health information system
- CO4.** Explain the International Classification of Disease
- CO5.** Describe Healthcare Common Procedure Coding System

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

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Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	1	-	-	-	1	1	-	1
CO 3	3	3	2	-	1	-	-	-	1	1	2	1
CO4	3	3	2	2	1	1	1	-	1	1	2	1
CO5	3	3	2	2	1	1	1	1	1	1	2	1
Average	3	3	2	2	1	1	1	1	1	1	2	1

25CI804

BIO STATISTICS

L-T-P-C:6-2-0-8

Course objectives:

- ❖ Students learn the main concepts of biostatistics
- ❖ Students learn the basic principles of probability and how they relate to biostatistics
- ❖ Students learn how to use numeric and graphical techniques to display and summarize medical data
- ❖ Students learn how to design clinical trials that enable accurate data collection

UNIT I PROBABILITY AND RANDOM VARIABLES

15 Hours

Sample spaces - Events - Axiomatic approach to probability - Conditional probability – Additional theorem - Multiplication theorem - Baye's theorem – Random variables : Continuous and discrete random variables - Distribution function - Expectation with properties - Moments, mean, variance problems - Continuous and discrete distributions.

UNIT II STANDARD DISTRIBUTIONS

15 Hours

Bivariate distribution - Conditional and marginal distribution - Discrete distributions - Binomial, Poisson, Geometric distributions - Continuous distributions - Normal, Exponential and Negative exponential, Gamma distributions - Simple problems - Properties.

Unit III BIOSTATISTICS-INTRODUCTION, SAMPLING DISTRIBUTION 15 Hours

Introduction to Biostatistics, Types of variables & scales of measurements, Measures of central tendency and dispersion, Rate, Ratio, Proportion, Incidence & Prevalence, Concept

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of probability distribution- Normal, Poisson & Binomial distribution, Parameters & Applications- Concepts of sampling distribution -Standard Error & Confidence Interval
Skewness & Kurtosis

Unit IV TEST OF SIGNIFICANCE, CORRELATION & REGRESSION 20 Hours

Basics of Testing of Hypothesis- Null & Alternate Hypothesis, Level of significance (Parametric) & power of test-p Value, Tests of significance- test (Paired & Unpaired), Chi-Square test, Test of Proportion- One-way analysis of variance- Repeated measures of analysis of variance, Test of significance (non-parametric), Mann- Whitney U test, Wilcoxon test, Kruskal- Wallis analysis of variance, Friedmann's analysis of variance, Simple correlation- Pearson's & Spearman's, Testing the significance of correlation coefficient linear & multiple regression, Multivariate Analysis, Concept of multivariate analysis - Introduction to logistic regression & survival analysis. Reliability & Validity evaluation of diagnostic tests

Unit V Format of Scientific document

10 Hours

Structure of research protocol, Structure of thesis/ research report- Formats of reporting in scientific journals, Systematic review - Meta-analysis

Total Periods: 75

REFERENCES:

1. Devore, J. L., "Probability and Statistics for Engineering & Sciences", 8th Edition, Cengage Learning, 2014.
2. Gupta. S.C and Kapoor, V.K., "Fundamentals of Mathematical Statistics", 12th Edition, Sultan Chand and Sons, New Delhi, 2020.
3. Johnson, R.A., Miller, I and Freund J., "Miller and Freund's Probability and Statistics for Engineers", 9th Edition, Pearson Education, Asia, 2016.
4. Rice, J. A., "Mathematical Statistics and Data Analysis", 3rd Edition, Cengage Learning, 2013.
5. Ross, S. M., "Introduction to Probability and Statistics for Engineers and Scientists", 6th Edition, Elsevier, 2020.

Course Outcome

After the completion of course, the students will be able to:

- CO1: Identify Probability And Random Variables
- CO2 : Develop standard distribution
- CO3 : Conduct sampling research
- CO4: Analyze Test of Significance, Correlation
- CO5: Preparation of scientific document

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

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Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	1	-	-	-	1	1	-	1
CO 3	3	3	2	-	1	-	-	-	1	1	2	1
CO4	3	3	2	2	1	1	1	-	1	1	2	1
CO5	3	3	2	2	1	1	1	1	1	1	2	1
Average	3	3	2	2	1	1	1	1	1	1	2	1

25CI901 HEALTH CARE FINANCING

L-T-P-C:6-2-0-8

Course Objective: This course provides an overview of Manage health, medical, and financial information using best practices in data stewardship; data science and data analytics; and human-centered design and systems.

Unit I The Nature and purpose of Accounting, Accounting Concepts & Accounting records: 15hours

What is accounting information? Who needs it? What they need or expect? What do accountants do? Single Entry Book – keeping , Double Entry Book - keeping , What is an Account? Making entries. Five types of Accounts (Income, Expense, Asset, Liability, Capital) Book – keeping rules accounting books/ledgers (Nominal, Purchase, Sales, Journal etc) Dealing with cash, imprest system

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Unit II Preparation of various Financial Statements & Fixed assets and Depreciation:

15hours

Trial Balance , Receipts and Payments ,Income and Expenditure Account, Balance Sheet. What are fixed assets and why are they different? What is depreciation and why do we need it?. How do we calculate depreciation? (pros and cons of different methods) Accounting entries for depreciation

Unit III Costing and Pricing:

15hours

Financial accounting Vs. Cost accounting , Key terms: Direct/indirect, fixed/variable/semi-variable, Analysing results: Standard/budgeted/actual , Costing hospital services , Taken action: controllable /uncontrollable , Making decisions: Marginal/book/out –of pocket costs , Reporting costs: Cost Centres, allocation and apportionment of costs , Pricing methods and decisions.

Unit IV Accounting and Audit Procedures in Health Care Sector

15hours

Accounting System in hospital, Purpose of an audit and auditing principles, What the auditor does?, The audit report – “True and Fair View”, Legal requirements: layout, audit and filing of accounts. Definition and history of Health Insurance, Concepts in Health Insurance, Issues in Health Insurance, Effective Health Insurance, Good & Bad in Health Insurance

Unit V Types of health insurance

15hours

Social health insurance , Private health insurance, Community health insurance (CHI),Government-initiated health insurance schemes (GHI) , Differences in the four categories ,Health Insurance in India. Social Health insurance , Voluntary (commercial) health insurance , Daily hospitalization expenses – Royal Sundaram’s Hospital Cash , Critical illness cover – ICICI Prudential’s Crisis cover , Community health insurance (CHI) , Government-initiated health insurance schemes .

Total Periods: 75

Reference Books

1. Health care finance, Modern financial analysis for accelerating biomedical innovation, Andrew W Lo & Shomesh E Chaudhri, Princeton University Press
2. Health care finance, An introduction to accounting and financial management , VI Ed., Louis C Gapenski, Kristin L Reiter, Health administration press, ISBN **978-1567937411**

Course Outcome

After the completion of course, the students will be able to:

CO1. Explain the basic principle of accounts, finance and financial management.

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CO2.Describe the financial accounting and financial management principles

CO3.Explain the hospital management accounting and health insurance

CO4.Discuss the types of health insurance

CO5.Describe the Accounting and Audit Procedures in Health Care Sector

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes												
Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	1	-	-	-	1	1	-	1
CO 3	3	3	2	-	1	-	-	-	1	1	2	1
CO4	3	3	2	2	1	1	1	-	1	1	2	1
CO5	3	3	2	2	1	1	1	1	1	1	2	1
Average	3	3	2	2	1	1	1	1	1	1	2	1

25CI902 CLINICAL WORKFLOW AND PROCESS REDESIGNING

L-T-P-C:6-2-0-8

Overview of clinical workflow and its impact on healthcare delivery, Principles of process improvement and redesign, Ethical considerations in workflow redesign, Workflow Analysis Techniques, Workflow analysis methods and tools, Process mapping and value stream mapping, Time-motion studies and observation techniques

Identifying Inefficiencies and Bottlenecks- Identifying common bottlenecks and inefficiencies in clinical workflows, Analyzing root causes of workflow problems, Human factors and ergonomics in workflow redesign, Change Management in Workflow Redesign- Principles of change management in healthcare settings, Stakeholder analysis and engagement, Strategies for overcoming resistance to change

Technology and Workflow Optimization- Role of technology in clinical workflow optimization, Electronic health records (EHR) and clinical decision support systems, Mobile health technologies and their impact on workflow, Integration of technology with clinical processes, Implementation of Workflow Redesign- Developing an action plan for clinical process redesign, Pilot testing and evaluation of redesigned workflows, Monitoring and sustaining workflow improvements, Measuring outcomes and evaluating the impact of redesign efforts

Introduction to Lean and Six Sigma principles, DMAIC (Define, Measure, Analyze, Improve, Control) framework, Applying Lean and Six Sigma tools for process improvement, Change Management in Workflow Redesign, Understanding change management principles and frameworks, Strategies for engaging stakeholders and managing resistance to change, Communication and training in workflow redesign initiatives

Role of technology in clinical workflow optimization, Electronic health records (EHRs) and clinical decision support systems Impact of technology on patient safety and quality of care

Total Periods: 75

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Reference Book

1. Ozkaynak, M., Unertl, K., Johnson, S., Brixey, J., Haque, S.N. (2022). Clinical Workflow Analysis, Process Redesign, and Quality Improvement. In: Finnell, J.T., Dixon, B.E. (eds) Clinical Informatics Study Guide. Springer, Cham. https://doi.org/10.1007/978-3-030-93765-2_8.

Course Outcome

After the completion of course, the students will be able to:

CO1. Understand the concepts and importance of clinical workflow and process redesign.

CO2. Apply workflow analysis techniques to evaluate clinical processes

CO3. Create process maps to visualize and document clinical workflows Describe hospital administration and other services

CO4. Implement change management strategies to facilitate workflow redesign.

Maintain quality in health care

CO5. Evaluate the role of technology in enhancing clinical workflow and Develop and implement a plan for clinical process redesign.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes												
Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	1	-	-	-	1	1	-	1
CO 3	3	3	2	-	1	-	-	-	1	1	2	1
CO4	3	3	2	2	1	1	1	-	1	1	2	1
CO5	3	3	2	2	1	1	1	1	1	1	2	1
Average	3	3	2	2	1	1	1	1	1	1	2	1

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25CI903

HOSPITAL ADMINISTRATION

L-T-P-C:6-2-0-8

Course objectives of hospital administration include:

- ❖ Balancing cost-effective operations with the highest standards of patient care
- ❖ Managing, controlling, and securing patient health information
- ❖ Defining a vision for the future and identifying goals and objectives
- ❖ Establishing goals on productivity that health centers must maintain
- ❖ Promoting and advertising the healthcare services and facilities offered by a hospital

Unit- I Introduction to Management

15 hours

Importance of Management , Definition of Management -Characteristic features of Management , Roles of Management-Role of a Manager , Levels of Management and their functions- Process of Management.Manage rial skills, Management and Administration Principles of Organizational Management-Culture, Values and Mission , Organizational Structure , Planning, coordination, Decision making and controlling, organization, Hospital Organizational Structures.

Unit- II Organizational Behavior

15 hours

Personal Growth and Development Definition, characteristics, determinants, causes, Theories (Type, Trait, Intrapsychic, Social learning) Perception: Definition- Factors- Perceptual grouping and selectivity - Stimuli selection- Barriers - Honing perceptual skills.Attitudes and values: Definition, Characteristics, Functions and Formation of Attitudes-Definition, types, formation of values- Values and behavior- Values and ethics- Values and attitudes, Learning: Definition:– Components –Determinants- Theories (classical, operant, cognitive, social learning)- Principles of reinforcement- Punishment- Learning curves- Learning and behavior

Unit- III Managing People (Hospital Administration)

15 hours

Overview – scope and functions of HR dept., HR planning, Recruitment and Appointment, Training and Development, Goal setting, rewards systems and motivation, Performance Appraisal, Promotion, internal transfers, Problems and Legal issues, Leadership, Working in teams. Key administrators and their functions, overview of medical and para-medical specialties, main service departments: Overview of health services – government services: private & not for profit: primary, secondary & tertiary health care: types of hospital: community, super-specialty etc

Unit- IV Product-based services

Pharmacy purchasing and stores, Pharmacy dispensing, Prosthetics & Orthotics .Diagnostic Services (Radiology, Laboratories, Blood Bank etc.) Overview – main services and their functions, In-house services, Patient Services (non-medical), Reception, Welcome/Help Desk. Patient facilities, wheelchairs, Ambulances, Public Relations – objectives, functions, policies, different media, methodologies, networking- Managing Support Services, Overview of

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functions of all support services including Laundry, Catering, Cleaning, CSSD, Transport, Security, Materials (Purchase and Stores) etc. Functions of GS Office

Unit- V Hospital Infrastructure (Buildings and Plant)

15 hours

Civil Engineering – Planning and maintaining buildings, water & sewage, Electrical Engineering, Mechanical Engineering, Equipment Maintenance, Medical Gases, etc. Biomedical Engineering, Corporate Governance & legal matters, Relationships with other institutions and organizations. Healthcare Policies & Standards- health law, regulations, accreditations standards, and certification requirements. Compliance to organisational policy, Auditing methods and techniques □ ICT systems in compliance with applicable laws, regulations, standards and requirements Quality assurance program in the health system. Concepts of Quality of Care- Quality Improvement Approaches Standards and Norms Quality Improvement Tools.

Total Periods: 75

Reference books

1. Hospital Administration, 1st Edition, 2013, Sonu Dr Goel, , eBook ISBN: 9788131239032
2. Hospital Administration, CM Francis mario C De Souza, 2024, 3/e reprint, Jaypee Publisher, ISBN 97881717972192024

Course Outcome

After the completion of course, the students will be able to:

- CO1.** Explain organizational behavior
- CO2.** Describe principles of management
- CO3.** Describe hospital administration and other services
- CO4.** Maintain quality in health care
- CO5.** Adapt Medical ethics

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												

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CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	1	-	-	-	1	1	-	1
CO 3	3	3	2	-	1	-	-	-	1	1	2	1
CO4	3	3	2	2	1	1	1	-	1	1	2	1
CO5	3	3	2	2	1	1	1	1	1	1	2	1
Average	3	3	2	2	1	1	1	1	1	1	2	1

Elective Course

L-T-P-C: 3-2-0-5

25a606a

AYUSH AND LIFESTYLE

L-T-P-C:3-2-0-5

Course Objective: The objective of the course on Ayush and Lifestyle is to provide students with a comprehensive understanding of the principles, practices, and applications of Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homeopathy, along with the significance of lifestyle factors in promoting health and well-being. The course aims to equip students with the knowledge and skills to integrate Ayush principles and lifestyle practices into their personal lives and professional healthcare settings.

UNIT 1. Introduction to Ayush and Lifestyle

15 Hours

Overview of Ayush systems (Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homeopathy), Principles and philosophies underlying Ayush practices, Importance of lifestyle factors in maintaining health and well-being **Ayurveda:** Fundamental principles and concepts of Ayurveda, Doshas (Vata, Pitta, Kapha) and their role in health and disease, Ayurvedic diagnostic techniques and assessment of prakriti (individual constitution), Ayurvedic therapies and treatment modalities, Ayurvedic diet and nutrition. Concept of Dinacharya and how its related to modern times Circadian rhythm, Ritu Charya - Seasonal Cycles as per Ayurveda. **Yoga & Naturopathy:** Introduction to yoga and its philosophy, Asanas (physical postures) and pranayama (breathing techniques), Yogic relaxation techniques and meditation, Naturopathic principles and therapies for health promotion and disease prevention, Mind-body connection, and stress management

UNIT 2. Unani Medicine, Siddha and Homeopathy Medicine

15 Hours

Introduction to Unani medicine and its historical background, Concepts of temperament (Mizaj) and humors (Akhlat), Unani diagnostic methods and approaches to treatment, Unani

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pharmacology and herbal medicine, Unani dietetics and lifestyle recommendations. Siddha Medicine: Overview of Siddha medicine and its principles, Concepts of doshas (Vata, Pitta, Kapha) and seven humors (Dhatus), Siddha diagnostic methods and assessment of disease, Siddha treatment modalities, including herbal medicines and therapies, Siddha lifestyle practices for maintaining health and wellness. Homeopathy: Introduction to homeopathy and its fundamental principles, Law of similar and principles of potentization, Homeopathic material medica and remedy selection, Homeopathic case taking and individualization of treatment, Homeopathic management of common health conditions

UNIT 3. Lifestyle Medicine

13 Hours

Importance of lifestyle factors in health and disease prevention, Role of nutrition, physical activity, and stress management in promoting health, Lifestyle interventions for chronic diseases such as diabetes, cardiovascular diseases, and obesity, Integrating Ayush principles and practices into a healthy lifestyle, Evidence-based approaches to lifestyle medicine, Research and Evidence in Ayush and Lifestyle. Traditional food recipes from Ayush Systems, Ayurveda based diet and lifestyle guidelines.

UNIT 4. Introduction to research methodologies in Ayush and lifestyle medicine 12Hours

Critical appraisal of Ayush and lifestyle research studies, Evaluation of evidence-based practices in Ayush and lifestyle interventions, Ethical considerations in Ayush and lifestyle research, Public Health Perspectives in Ayush and Lifestyle, Ayush, and lifestyle interventions for community health promotion.

UNIT 5. Ayurveda as a curative and restorative science

10 Hours

Public health challenges and opportunities in integrating Ayush practices, Role of Ayush and lifestyle approaches in preventive healthcare, Health policy considerations in the Ayush and lifestyle sector, Emerging Trends and Future Directions in Ayush and Lifestyle.

UNIT 6. Innovations and advancements in Ayush practices

10 Hours

Integration of Ayush and lifestyle approaches into mainstream healthcare systems, Global perspectives and collaborations in Ayush and lifestyle medicine, Potential challenges, and opportunities in the field.

Total Periods: 75

Textbooks and References

1. "Ayurveda: The Science of Self-Healing" by Dr. Vasant Lad Lotus Press; 2nd edition (28 July 1993).

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2. "The Complete Book of Ayurvedic Home Remedies" by Dr. Vasant Lad, Little, Brown BookGroup (2 November 2006).
3. "The Yoga Bible" by Christina Brown, Krause Publications; 39820th edition (29 May 2003)
4. Naturopathic Medicine: Treating the Whole Person, Health Advisory Lectures & Literature; 3rd Revised edition (1 November 2000)
5. "Unani Medicine: Towards a Global Approach" by Hakeem Syed Zillur Rahman Betascript Publishing, 2011
6. Rasayana: Ayurvedic Herbs for Longevity and Rejuvenation, H.S. Puri · 2002
7. Tridosha Made Easy: The Basic Ayurvedic Principle, Janardhana V Hebbar, Raghuram YS, Manasa S · 2019
8. "Lifestyle Medicine: Lifestyle, the Environment, and Preventive Medicine in Health andDisease" edited by James M. Rippe.

Course Outcome: Upon the completion of this course the student will be able to

CO1. Explain the fundamental principles, philosophies, and concepts of Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homeopathy.

CO2. Analyze and apply diagnostic methods and assessment tools used in Ayush systems to evaluate individual constitution and health conditions.

CO3. Demonstrate the use of Ayush therapies, treatments, and lifestyle interventions to promote health, prevent diseases, and manage common health conditions.

CO4. Evaluate and apply principles of nutrition, dietetics, and herbal medicine in the context of Ayush and lifestyle practices.

CO5. Apply the principles of lifestyle medicine in addressing chronic diseases and health promotion.

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Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	-	-	-
CO 2	3	-	2	-	1	-	-	-	1	-	-	-
CO 3	3	3	2	-	1	-	-	-	1	-	2	-
CO4	3	3	2	2	1	1	1	-	1	-	2	-
CO5	3	3	2	2	1	1	1	1	1	-	2	-
Average	3	3	2	2	1	1	1	1	1	-	2	-

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Course Objective: This course will provide information on clinical research services, clinical research organization, clinical investigator, and pharmaceutical or Biotechnology Company, hospitals, or medical device company. An SMO specializes in managing clinical research sites.

UNIT1. Introduction to Site Management Operations 12 Hours

Overview of site management operations in clinical research, Roles and responsibilities of site managers and clinical research coordinators, Importance of good clinical practice (GCP) and regulatory compliance, Contract Research Organizations (CRO), their functions and importance in clinical trials, Difference between Site Management Organization (SMOs) and CROs, General comments on the site's status and GCP compliance, using company- or trial-specific checklists. Visit details – who was seen by whom, what was discussed, the date of the next planned visit.

UNIT 2. Site Documentation and Regulatory Compliance 12 Hours

Essential documents in clinical research and their management, Regulatory requirements and submissions for site approval, Compliance with local, national, and international regulations and guidelines

UNIT 3. Study Start-up and Site Initiation 13 Hours

Study start-up activities and timelines, Site initiation visits and investigator meetings, Study- specific training and site staff qualification, Subject Recruitment and Enrollment, Strategies for subject recruitment and retention, Informed consent process and documentation, Screening, eligibility criteria, and subject enrollment procedures

. UNIT 4. Site Monitoring and Quality Assurance 12 Hours

Role of site monitors and clinical research associates (CRAs), Monitoring visits, source data verification, and compliance checks, Quality assurance processes and corrective actions, Investigational Product Management, Handling and accountability of investigational products, Drug storage, dispensing, and reconciliation, Adherence to Good Manufacturing Practice (GMP) guidelines

UNIT 5 Global Site Management Organizations (SMO) Markets 12 Hours

Global Site Management Organizations Market (2021 to 2035) - by Therapeutic Areas, Trial Phases, Clinical Trial Components, Type of Interventions and Key Geographies. Site Management Operations in Specialized Settings, Considerations for site management operations in specialized trials (e.g., pediatric, oncology, global trials), Site management in decentralized and virtual trials, Ethical considerations in site management operations

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**UNIT 6: Study Close-out and Archiving, Study close-out activities and timelines.
12 Hours**

Documentation completion and final reporting, archiving and retention of study records
Emerging Trends and Future Directions in Site Management Operations, Advances in technology and their impact on site operations, Risk-based monitoring and remote monitoring approaches, Innovations, and challenges in site management operations.

Total Periods: 75

References

1. <https://www.insightaceanalytic.com/report/global-site-management-organizations-smo-market/1191>
2. <https://www.prnewswire.com/news-releases/global-site-management-organizations-market-2021-to-2035---by-therapeutic-areas-trial-phases-clinical-trial-components-type-of-interventions-and-key-geographies-301446900.html>

Text Book:

1. Swink, M., Melnyk, S.A., Cooper, M. and Hartley, J., 2014. Managing operations (Vol.1260547639). New York: McGraw-Hill/Irwin.
2. The CRA's Guide to Monitoring Clinical Research, Fourth Edition Perfect Paperback –1 June 2016.
3. "Site Management Handbook: An Operational Guide for Investigators, Sponsors, and Site Personnel" by Gregory K. Mislick and Laura C. Bonifacio
4. "Good Clinical Practice Guide" by Medicines and Healthcare products Regulatory Agency (MHRA)
5. "Clinical Trials Handbook: Design and Conduct" by Curtis L. Meinert
6. "Managing the Investigational Product Supply Chain: Ensuring Effective Distribution in Clinical Trials" by Nigel Rulewski
7. "Risk-Based Monitoring and Fraud Detection in Clinical Trials Using JMP and SAS" by Richard C. Zink and Christi Lalanne.

Course Outcome: Upon the completion of this course the student will be able to

CO1: Explain about the functions of monitor, investigator, sponsor in Clinical Research Organization

CO2: Demonstrate knowledge of good clinical practice (GCP) guidelines and regulatory requirements for conducting clinical research.

CO3. Apply site selection criteria and conduct feasibility assessments for clinical trial participation.

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CO4. Demonstrate Effective management of site documentation and ensure compliance with regulatory standards.

CO5. Describe study start-up activities, including site initiation visits and investigator meetings.

Co-Relationship Matrix

Relationship between the Course Outcomes (COs) and Program Outcomes (POs)

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3
Course Outcomes												
CO 1	3	-	-	-	-	-	-	-	1	1	-	-
CO 2	3	-	2	-	1	-	-	-	1	1	-	1
CO 3	3	3	2	-	1	-	-	-	1	1	2	1
CO4	3	3	2	2	1	1	1	-	1	1	2	1
CO5	3	3	2	2	1	1	1	1	1	1	2	1
Average	3	3	2	2	1	1	1	1	1	1	2	1

**ARUNAI UNIVERSITY
TIRUVANNAMALAI**

Program Name : B.A – தமிழ்க் கலாச்சாரம் மற்றும் நாகரிகம் (B.A – TAMIL CULTURE AND CIVILIZATION)

Program Duration : 3 years

(It Should be a Regional Language, If only Regional Language)

Philosophy&Significance of the Course

தமிழ் கலாச்சாரமும் நாகரிகமும் நம் முன்னோர்களின் வாழ்வியலையும், சிந்தனைகளையும், பண்பாடுகளையும் பிரதிபலிக்கின்றன. இவை தமிழர்களின் அடையாளமாக விளங்குகின்றன. இப்பாடத் திட்டம் மூலம், தமிழர்களின் கலாச்சார வரலாறு, சமூக அமைப்பு, பண்பாடு, ஆன்மீகம், கலை, இலக்கியம் மற்றும் வாழ்க்கை முறையின் தனித்துவங்கள் பற்றிய ஆழ்ந்த அறிவை மாணவர்கள் பெறுவது முக்கிய நோக்கமாகக் கொள்ளப்பட்டுள்ளது.

இந்த பாடத்திட்டம் தமிழர் பெருமையை ஆழமாக உணர்த்துவதுடன், மாணவர்களுக்கு அவர்களுடைய பண்பாட்டு அடையாளத்தை அறிவிக்கவும், அவர்களது சமூக பொறுப்புணர்வை மேம்படுத்தவும் உகந்ததாக அமைகிறது.

Distinctiveness and uniqueness of the program

1. ஆழமான வரலாற்றுப் பார்வை:

தமிழ் நாகரிகத்தின் தொடர்ச்சியான வளர்ச்சியையும் அதன் வரலாற்றுப் பெருமைகளையும் ஆராயும் வாய்ப்பை வழங்குகிறது.

2. கலை மற்றும் தத்துவ ஒருங்கிணைவு:

தமிழ் இசை, இலக்கியம், சிற்பம், மற்றும் தத்துவ சிந்தனைகளின் தனித்துவத்தையும் அதன் பங்களிப்பையும் விளக்குகிறது.

3. மெய்ப்பொருள் அடையாளம்:

தமிழர் பண்பாட்டை அறிவதன் மூலம் மாணவர்களுக்கு தங்கள் அடையாளத்தில் பெருமை வளர்க்கும் வகையில் வழிகாட்டுகிறது.

4. ஒப்பீட்டு ஆராய்ச்சி:

மற்ற நாகரிகங்களுடன் ஒப்பிடுவதன் மூலம் தமிழ் பண்பாட்டின் தனித்தன்மையையும் அதனுடைய உலகளாவிய முக்கியத்துவத்தையும் அடையாளம் காண உதவுகிறது.

5. அறமும் ஒழுக்கமும்:

தமிழர் சமூக ஒழுக்கங்களான அறம், சமத்துவம், மற்றும் பண்பாட்டுச் சிந்தனைகளின் இன்றைய சமூக வாழ்வில் உள்ள பயன்பாட்டையும் முக்கியத்துவத்தையும் விளக்குகிறது.

இந்தப் பாடத்திட்டம், மாணவர்களை தமிழ் பண்பாட்டு தூதர்களாக உருவாக்கி, தமிழர் பாரம்பரியத்தையும் நாகரிக வளர்ச்சியையும் உலகறியச் செய்யும் முக்கியமான தளமாக விளங்குகிறது.

**ARUNAI UNIVERSITY
TIRUVANNAMALAI**

**REGULATION-2025
(Choice Based Credit System)**

B.A – தமிழ்க் கலாச்சாரம் மற்றும் நாகரிகம்

(It Should be a Regional Language, If only Regional Language)

PROGRAMME OUTCOMES (POs):

Graduates of the Programme B.A – TAMIL Culture and Civilization will be able to

- PO1** தமிழ் பண்பாட்டு பாரம்பரியத்தைக் கருதல்: தமிழர் பாரம்பரியம், கலாச்சாரம், மற்றும் நாகரிகம் தொடர்பான உள்ளார்ந்த அடிப்படைகளைப் புரிந்துகொண்டு அவற்றின் முக்கியத்துவத்தை விளங்குதல்..
- PO2** திறன்களின் வளர்ச்சி: இலக்கியம், கலை, வரலாறு, மற்றும் சமூக அமைப்புகளில் இருக்கும் கலாச்சார அடையாளங்களை ஆய்வு செய்யும் திறன்களை மேம்படுத்துதல்..
- PO3** சமூக ஒற்றுமை உருவாக்கம்: தமிழர் கலாச்சாரத்தின்மூலம் சமூகத்தின் மனிதநேயத்தையும் ஒற்றுமையையும் வளர்க்குதல்.
- PO4** பாரம்பரியத்தை பாதுகாத்தல்: தமிழர் பாரம்பரிய கலாச்சாரங்களின் நவீனப் பயன்பாடுகளைப் புரிந்துகொண்டு அவற்றை பாதுகாத்தல் மற்றும் பரப்புதல்.
- PO5** தனிப்பட்ட ஆளுமை மேம்பாடு: கலாச்சாரம் சார்ந்த பாடங்கள் மூலம் மாணவர்களின் தனிநபர் திறன்களை மேம்படுத்துதல்.
- PO6** ஆய்வு மனப்பாங்கு: தமிழர் நாகரிகம் மற்றும் கலாச்சாரம் சார்ந்த பரிணாம வளர்ச்சிகளை ஆராயும் திறன்களை உருவாக்குதல்..
- PO7** கலை மற்றும் நாகரிக சமூகம்: தமிழ் கலை மற்றும் நாகரிகத்தின் சமூக பயன்பாடு தொடர்பான கருத்துகளை புரிந்துகொண்டு அவற்றை நடைமுறைப்படுத்துதல்.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs):

Graduates of the programme B.A – TAMIL Culture and Civilization will

- PEO1** கலை, கலாச்சார வளம் மற்றும் வரலாற்றுப் புரிதல்: தமிழர் பண்பாட்டின் பல்வேறு பரிமாணங்களையும் சமூகத்தில் அதன் தாக்கத்தையும் அறிய உதவுதல்..
- PEO2** ஆளுமை வளர்ச்சி: பண்பாட்டின் அடிப்படைகளைக் கற்றல் மூலம் மாணவர்களின் ஆளுமை வளர்ச்சி மற்றும் சமூக பொறுப்புணர்வை அதிகரித்தல்.
- PEO3** அறிவியல் திறன்கள்: பாரம்பரியத்தை நவீன தேவைமுறைகளுடன் இணைத்து புதிய துறைகளுக்கு வழிவகை செய்யக்கூடிய திறமைகளை உருவாக்குதல்.
- PEO4** சமூக முன்னேற்றம்: கலாச்சாரத்தைப் பயன்படுத்தி சமூக முன்னேற்றத்திற்கு பங்களிப்பை ஊக்குவித்தல்.

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- PEO5** மொழி மேம்பாடு: தமிழ் மொழியின் வளங்களையும் பன்முக அடையாளங்களையும் ஆய்வு செய்து பேச்சு மற்றும் எழுத்துத்திறன்களை மேம்படுத்துதல்.
- PEO6** சுயதிறன் மற்றும் சுயமுனைவு: பாரம்பரிய மற்றும் நவீன தமிழ் கலைகளின் தொழில்முனைவு வாய்ப்புகளை புரிந்துகொண்டு சுயமுனைவு உருவாக்குதல்.

PROGRAMME SPECIFIC OBJECTIVES (PSOs):

B.A – TAMIL Culture and Civilization program's specific goals are to create a virtue-based and a well-civilized society:

- PSO1** பாரம்பரியத்திற்கான ஆழ்ந்த புரிதல்: தமிழ் கலாச்சாரம் மற்றும் நாகரிகத்தின் முக்கியத்துவத்தை அறிந்து பாரம்பரியத்திற்கான தெளிவை ஏற்படுத்துதல்.
- PSO2** அறிவியல் மற்றும் கலாச்சார ஆய்வு திறன்: கலாச்சாரம் சார்ந்த ஆய்வுத் திறன்களை வளர்த்துக்கொண்டு, தற்கால சமூக தேவைகளுக்கு ஏற்ப புதிய யோசனைகளை உருவாக்குதல்.
- PSO3** தமிழ் கலாச்சாரத்தின் ஆவணப்படுத்தல்: தமிழ் பண்பாட்டு அடையாளங்களைச் சரியாக ஆவணப்படுத்தி உலகளாவிய பார்வைக்கு கொண்டு செல்லுதல்.
- PSO4** மாற்றுத்திறன்கள் மற்றும் வளர்ச்சி: மொழி, கலை மற்றும் பண்பாட்டு துறைகளில் சிறப்பு திறன்களை அடைய மாணவர்களை உதவுதல்.
- PSO5** சமூகப் பங்குபற்றல்: சமூகத்தின் கலாச்சார மற்றும் பண்பாட்டு சவால்களை அடிப்படையாகக் கொண்டு, உரிய தீர்வுகளைத் தயாரிக்க ஆர்வம் செலுத்துதல்.

**ARUNAI UNIVERSITY
TIRUVANNAMALAI**

B.A – TAMIL CULTURE AND CIVILIZATION (Full Time)

இளங்கலைப் பட்டப்படிப்பு

தமிழ்க் கலாச்சாரம் மற்றும் நாகரிகம் (முழுநேரம்)

பாடத்திட்டம்

ஆறு பருவங்கள்

SEMESTER I

S.N O.	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIODS
				L	T	P	C	
THEORY								
1		Tamil / Other Language	LAN	5	1	0	3	6
2		English -1	ENG	5	1	0	3	6
3	TAC25101	தமிழர் தொன்மையும் வரலாறும்	CC	5	0	0	5	5
4	TAC25102	தமிழர் கலை மற்றும் பண்பாடு	CC	5	1	0	5	6
5	TAE25103	தமிழர் வாழ்வியல்	EC	5	0	0	4	5
6	TASE25104	ஆளுமைத்திறன்	SEC	2	0	0	2	2
TOTAL				27	3	0	22	30

SMESTER II

S.NO .	COURSE CODE	COURSE TITLE	CATE GOR Y	PERIODS PER WEEK				TOTAL PERIODS
				L	T	P	C	
THEORY								
1		Tamil / Other Language	LAN	5	1	0	3	6
2		English -2	ENG	5	1	0	3	6
3	TAC25201	தமிழர் சமய பாரம்பரியம்	CC	5	0	0	5	5
4	TAC25202	தமிழர் பொருளாதாரம்	CC	5	1	0	5	6

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5	TAE25203	தமிழர் சமுதாயப் பண்பாடு மற்றும் அரசியல் சிந்தனை	EC	5	0	0	4	5
6	TASE25204	கைவினைத்திறன்	SEC	2	0	0	2	2
TOTAL				27	3	0	22	30

SEMESTER III

S.NO .	COURSE CODE	COURSE TITLE	CAT EGO RY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
THEORY								
1		Tamil / Other Language	LAN	5	1	0	3	6
2		English -3	ENG	5	1	0	3	6
3	TAC25301	தமிழர் கல்வி வளர்ச்சி	CC	5	0	0	5	5
4	TAC25302	தமிழர் பண்பாட்டு மற்றும் கலாச்சார ஆய்வுகள்	CC	5	1	0	5	6
5	TAE25303	தமிழர் சிந்தனை மற்றும் தத்துவ மரபுகள்	EC	5	0	0	4	5
6	TASE25304	கூத்தும் இசையும்	SEC	2	0	0	2	2
TOTAL				27	3	0	22	30

SEMESTER IV

S.NO .	COURSE CODE	COURSE TITLE	CAT EGO RY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
THEORY								
1		Tamil / Other Language	LAN	5	1	0	3	6
2		English -3	ENG	5	1	0	3	6
3	TAC2540 1	தமிழர் பண்பாட்டு இலக்கியங்கள்	CC	5	0	0	5	5
4	TAC2540 2	தமிழர் நவீன துறை மற்றும் சமூக செயல்பாடுகள்	CC	5	1	0	5	6

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5	TAE2540 3	வாய்மொழி இலக்கியங்களும் தமிழ்க் கலாச்சாரமும்	EC	5	0	0	4	5
6	TASE254 04	படைப்பிலக்கியம்	SEC	2	0	0	2	2
TOTAL				27	3	0	22	30

SEMESTER V

S.NO .	COURSE CODE	COURSE TITLE	CAT EGO RY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
THEORY								
1		தமிழர் சமூக மாற்றங்கள்	CC	5	1	0	4	5
2		தமிழர் பண்பாட்டு ஆதார நூல்கள் மற்றும் ஆய்வுகள்	CC	5	1	0	4	5
3		தமிழர் பண்பாட்டுக் கொள்கைகள்	CC	5	0	0	4	5
4		தமிழ் மரபு மருத்துவம்	CC	5	1	0	4	5
5		தமிழரின் தற்சார்பு வாழ்வியல்	EC	4	0	0	3	4
6		தமிழர் வானியல் அறிவு	EC	4	0	0	3	4
TOTAL				28	3	0	22	28

SEMESTER VI

S.NO .	COURSE CODE	COURSE TITLE	CAT EGO RY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
THEORY								
1		வள்ளுவ வாழ்வியல்	CC	4	0	0	3	5
2		நவீன தமிழர் சமூகப் பண்பாடு	CC	4	0	0	3	5
3		தமிழர் பண்பாட்டு மறுவளர்ச்சி	CC	4	1	0	3	5
PRACTICAL / EXTENSION ACTIVITIES								

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5		FIELD WORK	EDC	6	1	0	5	5
6		PROJECT	EDC	7	3	0	6	6
TOTAL				27	5	0	20	26

TOTAL CREDITS: 130

CORE COURSES (CC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	TAC2510 1	தமிழர் தொன்மையும் வரலாறும்	CC	5	0	0	5	5
2	TAC2510 2	தமிழர் கலை மற்றும் பண்பாடு	CC	5	1	0	5	6
3	TAC2520 1	தமிழர் சமய பாரம்பரியம்	CC	5	0	0	5	5
4	TAC2420 2	தமிழர் பொருளாதாரம்	CC	5	1	0	5	6
5	TAC2530 1	தமிழர் கல்வி வளர்ச்சி	CC	5	0	0	5	5
6	TAC2530 2	தமிழர் பண்பாட்டு மற்றும் கலாச்சார ஆய்வுகள்	CC	5	1	0	5	6
7	TAC2540 1	தமிழர் பண்பாட்டு இலக்கியங்கள்	CC	5	0	0	5	5
8	TAC2540 2	தமிழர் நவீன துறை மற்றும் சமூக செயல்பாடுகள்	CC	5	1	0	5	6
9		தமிழர் சமூக மாற்றங்கள்	CC	5	1	0	4	5
10		தமிழர் பண்பாட்டு ஆதார நூல்கள் மற்றும் ஆய்வுகள்	CC	5	1	0	4	5

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11		தமிழர் பண்பாட்டுக் கொள்கைகள்	CC	5	0	0	4	5
12		தமிழ் மரபு மருத்துவம்	CC	5	1	0	4	5
13		வள்ளுவ வாழ்வியல்	CC	4	0	0	3	5
14		நவீன தமிழர் சமூகப் பண்பாடு	CC	4	0	0	3	5
15		தமிழர் பண்பாட்டு மறுவளர்ச்சி	CC	4	1	0	3	5
TOTAL CREDITS								65

LIST OF ELECTIVE COURSES (EC)

SEMESTER I, ELECTIVE I

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	TAE2510 3	தமிழர் வாழ்வியல்	EC	5	0	0	4	5
TOTAL CREDITS								4

SEMESTER II, ELECTIVE II

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	TAE2520 3	தமிழர் சமுதாயப் பண்பாடு மற்றும் அரசியல் சிந்தனை	EC	5	0	0	4	5
TOTAL CREDITS								4

SEMESTER III, ELECTIVE III

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	

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1	TAE2530 3	தமிழர் சிந்தனை மற்றும் தத்துவ மரபுகள்	EC	5	0	0	4	5
TOTAL CREDITS								4

SEMESTER IV, ELECTIVE IV

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	TAE2540 3	வாய்மொழி இலக்கியங்களும் தமிழ்க் கலாச்சாரமும்	EC	5	0	0	4	5
TOTAL CREDITS								4

SEMESTER IV, ELECTIVE V

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1		தமிழரின் தற்சார்பு வாழ்வியல்	EC	4	0	0	3	4
2		தமிழர் வானியல் அறிவு	EC	4	0	0	3	4
TOTAL CREDITS								6

LIST OF SKILL ENHANCEMENT COURSE (SEC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	TASE251 04	ஆளுமைத்திறன்	SEC	2	0	0	2	2
	TASE242 04	கைவினைத்திறன்	SEC	2	0	0	2	2

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	TASE253 04	கூத்தும் இசையும்	SEC	2	0	0	2	2
	TASE254 04	படைப்பிலக்கியம்	SEC	2	0	0	2	2
TOTAL CREDITS							8	

EMPLOYMENT DEVELOPMENT COURSES (EDC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
2		FIELD WORK	EDC	6	1	0	5	5
3		PROJECT	EDC	7	3	0	6	6
TOTAL CREDITS							11	

TAC25101

தாள் 1: தமிழர் தொன்மையும் வரலாறும்

L T P C

5 0 0 5

OBJECTIVES:

- To learn about statistical methods and their applications in engineering, including estimation theory, hypothesis testing, correlation, regression, randomized variables & probability distributions.

அலகு 1.தமிழர் தொன்மையின் அறிமுகம் 10

தமிழர்களின் தொன்மை வரலாறு - தமிழ் மொழியின் வளர்ச்சி - தொன்மை அடையாளங்கள்

அலகு 2.சங்க இலக்கியத்தில் தமிழர் வாழ்க்கை 10

சங்க இலக்கியத்தின் வகைகள் - தொல்காப்பியம், புறநானூறு - சங்க கால வாழ்க்கைத் தலைப்புகள்

அலகு 3.தொல்காப்பிய கால சமூக அமைப்பு 10

தொல்காப்பிய காலச் சமூக அமைப்புகள் - கல்வி, குடும்ப அமைப்பு - பொருளாதார அடிப்படைகள்

அலகு 4.தமிழர் அரசியல் வரலாறு 10

தமிழர் அரசியல் அமைப்புகள் - முத்தமிழ் சங்கங்கள் - கலிங்கப் போர், சோழ, பாண்டியர்கள்

அலகு 5.உலக தமிழ் மரபுகள் 10

உலகமெங்கும் தமிழின் பரவல் - புலம்பெயர் தமிழர்கள் - தமிழ் மரபுகளை பாதுகாக்கும் முயற்சிகள்

Total Periods: 50

OUTCOMES:

After completing this course, students should demonstrate competency in the following topics:

CO1 தமிழர் தொன்மை, பரம்பரை, மற்றும் கலாச்சார அடிப்படைகளை ஆராய்கிறது.

CO2 சங்க இலக்கியம் மூலம் வாழ்வியல் பண்புகளை உணருதல்.

CO3 தொல்காப்பியத்தில் கூறப்பட்ட சமூக அமைப்பை புரிதல்.

CO4 தமிழர் அரசியல் அமைப்புகள் மற்றும் முறைமைகளை ஆராய்கிறது

CO5 உலகளாவிய பார்வையில் தமிழ் மரபுகளை அறிதல்.

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பாடநூல் :

தமிழின் பண்டைய வரலாறு மற்றும் கலாச்சாரம், ஆசிரியர்: டாக்டர். ந. கிருஷ்ணமூர்த்தி பதிப்பகம்:
பாரதி பதிப்பகம்

பார்வைநூல்கள்:

தமிழர் கலாச்சாரம் ஒரு அறிமுகம், ஆசிரியர்: திரு. ம. நா. சு. கா. சாந்தா , பதிப்பகம்: சிந்தனைப்
பதிப்பகம்

தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, ஆசிரியர்: திரு. ச. ஜெயராமன் பதிப்பகம்:
ஆழமான் பதிப்பகம்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	2	2	3	3
CO2	3	3	3	3	2	2	2	2	2
CO3	3	3	3	3	2	2	2	2	2
CO4	3	3	3	3	2	2	2	3	3
CO5	3	3	3	3	2	2	2	3	3
Avg	3	3	3	3	2	2	2	2	2

*** 3- High, 2- Medium, 1- Low**

TAC25102

தாள் 2: தமிழர் கலை மற்றும் பண்பாடு

L T P C

5 1 0 5

OBJECTIVES:

- ☐ To study and understand the properties of modern construction materials used in construction.
- ☐ To understand special concretes, metals, composites, water proofing compounds, non weathering materials, and smart materials.

அலகு 1.பாரம்பரிய கலைகள்

10

தமிழின் பாரம்பரிய கலை வடிவங்கள் - வெண்ணெய் ஓவியங்கள், வெப்ப பூசல், உடையாள் ஓவியம்

அலகு 2.நாட்டுப்புற மரபுகள் 10

நாட்டுப்புறக் கலை மற்றும் கலாச்சாரம் - நாட்டுப்புற இசை, நாடகம் - கிராமப்புற பண்பாட்டுப் பணிகள்

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அலகு 3.சிற்ப மற்றும் ஓவிய கலை 10

சிற்பக் கலை வகைகள் - ஓவிய கலை: விளக்கம் மற்றும் பாரம்பரியம் - அயலவர்களின் கலைத்தொகைகள்

அலகு 4.இசை மற்றும் நாடக மரபுகள் 10

திருக்கோயில்களில் இசையின் பங்கு - நாட்டுப்புற இசை மற்றும் நாடகக் கலைகள் - கலைஞர்களின் பங்கு

அலகு 5.தமிழர் திருவிழா கலாச்சாரம் 10

திருவிழாக்களின் முக்கியத்துவம் - சமய ரீதியான திருவிழாக்கள் - பரம்பரிய நாள்களில் சமூக பங்கேற்பு

Total Periods: 50

OUTCOMES:

After completing this course, students should demonstrate competency in the following topics:

- CO1** தமிழர் பாரம்பரிய கலை வடிவங்களை ஆராய்கிறது.
- CO2** நாட்டுப்புற மரபுகளின் சமூக மற்றும் கலாச்சார தாக்கங்களை உணருதல்.
- CO3** தமிழர் சிற்ப மற்றும் ஓவிய பாரம்பரியத்தை ஆராய்கிறது.
- CO4** தமிழர் இசை மற்றும் நாடக மரபுகளை அறிதல்.
- CO5** திருவிழாக்களில் தமிழர் பங்கு மற்றும் அதன் சமூகதன்மையை உணர்தல்.

பாடநூல் :

1.தமிழின் கலைகள் மற்றும் பண்பாட்டு மரபுகள், ஆசிரியர்: டாக்டர். ரா. செல்லப்பா, பதிப்பகம்: தமிழ்சாரா பதிப்பகம்

பார்வைநூல்கள்:

- 1.தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, ஆசிரியர்: திரு. ச. ஜெயராமன், பதிப்பகம்: ஆழமான் பதிப்பகம்
- 2.தமிழின் கலாச்சாரம் மற்றும் அது உலகத்திற்கு வழங்கிய பங்களிப்புகள், ஆசிரியர்: முனைவர். எல். சுந்தரமணி , பதிப்பகம்: சிந்தனைப் பதிப்பகம்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	2	3	3	3	2	2	2	3
CO2	2	1	2	2	2	2	2	2	3

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CO3	2	2	2	3	3	2	1	2	3
CO4	2	2	2	3	3	2	2	1	1
CO5	2	2	3	3	3	2	1	2	2
Avg	2	2	2	3	3	2	2	1	2

*** 3- High, 2- Medium, 1- Low**

TAC25103

தாள் 3: தமிழர் வாழ்வியல்

L T P C

5 0 0 5

OBJECTIVES:

- ☐ Building services are the essential services provided in the buildings for improving functioning of the buildings in efficient manner for the desired use of the building.
- ☐ The electrical services, mechanical services such as air conditioning, lighting, ventilation, fire protection, acoustics and sound insulations, elevators, escalators, as well as civil engineering services such as water supply, sanitary services, etc. have become most essential services for residential, industrial, high rise, hotels, motels, monumental buildings.

அலகு 1.குடும்ப மற்றும் சமூக அமைப்பு

தமிழர் குடும்ப அமைப்புகள் - சமூக நடத்தை விதிகள் - குடும்ப அமைப்பின் வகைகள்

அலகு 2.விவசாய மரபுகள்

தமிழின் விவசாய மரபுகள் - ஆறு மற்றும் வடிகால் விவசாயம் - பொதுவான விவசாய பழக்கங்கள்

அலகு 3.தொழில்கள் மற்றும் வர்த்தகம்

பண்டைய தொழில்கள் - வர்த்தக பரிமாற்றங்கள் - கைத்தொழில்கள்

அலகு 4.திருமணம் மற்றும் சடங்குகள்

தமிழர் திருமண வழக்குகள் - திருமண பரிசுகளும் சடங்குகளும் - சங்க கால திருமண வழிமுறைகள்

அலகு 5.பாரம்பரிய நெறிகள்

பண்டைய தமிழில் உரிமைகள் - சமூக ஒற்றுமையை உருவாக்கும் நெறிகள் - சமூக கட்டமைப்பில் நெறிகளின் பங்கு

OUTCOMES:

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On the successful completion of this course, the students will be able:

- CO1** தமிழர் குடும்ப அமைப்பு மற்றும் அதன் மாறுபாடுகளை அறிதல்.
- CO2** விவசாய மரபுகளைப் புரிந்துகொள்வது.
- CO3** தொழில்களும் வர்த்தக முறைகளும் பண்டைய சமூகத்திற்கான பங்களிப்புகளை விளக்குகிறது.
- CO4** திருமண மரபுகள் மற்றும் நிகழ்வுகளை புரிந்துகொள்கிறது.
- CO5** பாரம்பரிய நெறிகளின் சமூக முக்கியத்துவம்.

பாடநூல் :

- 1.தமிழ் சமுதாயம் மற்றும் வாழ்க்கை, டாக்டர். ப. குமரசாமி, சிந்தனைப் பதிப்பகம்

பார்வைநூல்கள்:

- 1.தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, திரு. ச. ஜெயராமன், ஆழமான் பதிப்பகம்
- 2.தமிழின் கலாச்சாரம் ஓர் பார்வை, முனைவர். செ. குமரன், தமிழ்ச்சங்கம் பதிப்பகம்
- 3.தமிழின் கலாச்சாரம் மற்றும் அது உலகத்திற்கு வழங்கிய பங்களிப்புகள், முனைவர். எல். சுந்தரமணி, சிந்தனைப் பதிப்பகம்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	2	2	2	2
CO2	2	2	3	2	3	2	2	3	2
CO3	2	3	1	2	3	2	2	3	2
CO4	3	1	1	2	3	2	2	3	2
CO5	3	2	3	2	3	2	2	3	2
Avg	3	2	3	2	3	2	2	3	2

*** 3- High, 2- Medium, 1- Low**

இரண்டாம் பருவம்

TAC24201

தாள் 1: தமிழர் சமய பாரம்பரியம்

L T P C

OBJECTIVES:

- Building services are the essential services provided in the buildings for improving functioning of the buildings in efficient manner for the desired use of the building.
- The electrical services, mechanical services such as air conditioning, lighting, ventilation, fire protection, acoustics and sound insulations, elevators, escalators, as well as civil engineering services such as water supply, sanitary services, etc. have become most essential services for residential, industrial, high rise, hotels, motels, monumental buildings.

அலகு 1.தமிழர் சமய அடிப்படை 10

தமிழரின் ஆதிசமயத் தோற்றம் - பெரும்பான்மையான சமய வழிமுறைகள் - சமய விளக்கங்கள் மற்றும் ஆழமான மத தத்துவங்கள்

அலகு 2.தமிழர் கோயில்கள் மற்றும் மதகோட்பாடுகள் 10

பிரபலமான தமிழ் கோயில்கள் - கோயில்களின் பங்கு மற்றும் மத வழிமுறைகள் - கோயில் திருவிழாக்கள்

அலகு 3.தமிழ் சமய இலக்கியங்கள் 10

திருவாசகம், திருமந்திரம் - சமய இலக்கியங்கள், கவிதைகள் - சமயச் செய்திகளை சமூகத்தில் அடையாளப்படுத்துதல்

அலகு 4.மத ஒழுக்கங்கள் 10

தமிழின் தர்ம ஒழுக்கங்கள் - மத ஒழுக்க வாழ்க்கை முறைகள் - சமய அமைப்பின் பண்புகள்

அலகு 5.நவீன சமய கோட்பாடுகள் 10

சமூகவியல் பார்வையில் சமய மாற்றங்கள் - பழமையான சமயகோட்பாடுகளின் தாக்கங்கள் -- சமயம் மற்றும் சமூக ஒற்றுமை

Total Periods: 50

OUTCOMES:

On the successful completion of this course, the students will be able:

- CO1** தமிழர் சமய மரபுகளின் முக்கியத்துவத்தை அறிதல்.
- CO2** கோயில்கள் மற்றும் மத கோட்பாடுகளைப் புரிதல்.
- CO3** சமய இலக்கியங்களின் வரலாற்று பங்களிப்புகளை விளக்குகிறது.
- CO4** மத ஒழுக்கங்களின் சமூக முக்கியத்துவம்.

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CO5 சமய கோட்பாடுகளின் பரிணாம மாற்றத்தை புரிதல்

பாடநூல் :

1.தமிழின் கலை, கலாச்சாரம் மற்றும் சமூக அமைப்புகள், ஆசிரியர்: டாக்டர். ந. விஸ்வநாதன், பதிப்பகம்: பாரதி பதிப்பகம்

பார்வைநூல்கள்:

1.தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, ஆசிரியர்: திரு. ச. ஜெயராமன், பதிப்பகம்: ஆழமான் பதிப்பகம்

2.தமிழின் கலாச்சாரம்: ஓர் பார்வை, ஆசிரியர்: முனைவர். செ. குமரன், பதிப்பகம்: தமிழ்ச்சங்கம் பதிப்பகம்

3.தமிழ் சமுதாயம் மற்றும் வாழ்க்கை, ஆசிரியர்: டாக்டர். ப. குமரசாமி, பதிப்பகம்: சிந்தனைப் பதிப்பகம்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	2	2	2	2
CO2	2	2	3	2	3	2	2	3	2
CO3	2	3	1	2	3	2	2	3	2
CO4	3	1	1	2	3	2	2	3	2
CO5	3	2	3	2	3	2	2	3	2
Avg	3	2	3	2	3	2	2	3	2

*** 3- High, 2- Medium, 1- Low**

TAC25202

தாள் 2: தமிழர் பொருளாதாரம்

L T P C

5 1 0 5

OBJECTIVES:

- ☐ To learn about the maintenance objectives, functions, systems & overhaul & repair. To acquire knowledge in inspection decisions and NUCREC method.

அலகு 1.பண்டைய பொருளாதார அமைப்பு 10

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சமுதாய வளர்ச்சியில் பொருளாதார அமைப்பின் பங்கு - பொருளாதார தொழில்நுட்பங்கள் - இந்திய சமய பொருளாதாரம்

அலகு 2.வர்த்தக வளர்ச்சியில் தமிழர் பங்களிப்பு 10

தமிழரின் வணிகப் பண்புகள் - வர்த்தகப் பொருளாதார வளர்ச்சி - கடலோர வர்த்தகம் மற்றும் அதன் பங்கு

அலகு 3.கடலோர வர்த்தகம் 10

தமிழர்களின் கடலோர வர்த்தக வழிமுறைகள் - வானிலை மாற்றங்களும் பண்டைய வர்த்தக சுழற்சிகளும் - உலக பொருளாதார சந்தைகளில் தமிழரின் பங்கு

அலகு 4.தொழில்துறை மேம்பாடு 10

தமிழர்களின் தொழில்துறை வளர்ச்சி - நவீன தொழில்நுட்ப மாற்றங்கள் - தொழில்துறை நிலையான வளர்ச்சி

அலகு 5.நவீன பொருளாதாரம் 10

நவீன தொழில்நுட்ப பொருளாதாரம் - பொருளாதார கட்டமைப்புகள் மற்றும் வளர்ச்சி - தமிழர் பொருளாதார துறைகளில் பங்கு

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Total Periods: 50

OUTCOMES:

At the end of the course the students shall have the abilities to:

- CO1** தமிழர் பண்டைய பொருளாதார அமைப்பின் அடிப்படைகளை ஆராய்தல்.
- CO2** தமிழர் வர்த்தக வரலாற்றை விளக்குகிறது.
- CO3** தமிழர் கடலோர வர்த்தகத்தின் தாக்கத்தை அறிதல்.
- CO4** தொழில்துறையில் தமிழர் பங்களிப்பை விளக்குகிறது.
- CO5** நவீன பொருளாதாரத்தில் தமிழர் பங்களிப்பை ஆராய்தல்.

பாடநூல் :

1.தமிழர் பொருளாதாரம் மற்றும் வர்த்தகம், டாக்டர். ப. விக்னேஷ், தமிழ்சாரா பதிப்பகம்

பார்வைநூல்கள்:

1.தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, திரு. ச. ஜெயராமன், ஆழமான் பதிப்பகம்

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COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	3	3	2	2	2	3	2	2
CO2	2	3	3	2	3	3	2	2	1
CO3	3	3	3	2	3	3	2	3	1
CO4	3	3	3	2	2	2	3	2	2
CO5	3	3	3	2	3	2	2	2	1
Avg	3	3	3	2	3	2	2	2	1

*** 3- High, 2- Medium, 1- Low**

BM24104

தாள் 3: தமிழர் சமுதாயப் பண்பாடு மற்றும் அரசியல் சிந்தனை

L T P C

3 0 0 3

OBJECTIVE:

- ☐ To study the properties of concrete making materials, tests, mix design, special concretes, and various methods for making concrete.

அலகு 1.தமிழர் சமூக ஒழுங்கு 10

அலகு 2.தமிழர் அரசியல் வரலாறு 10

அலகு 3.நவீன அரசியல் சிந்தனைகள் 10

அலகு 4.காவிய அரசியல் தத்துவங்கள் 10

அலகு 5.தமிழர் மற்றும் உலக அரசியல் 10

Total Periods: 50

OUTCOMES:

On completion of the course, the student is expected to be able to:

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- CO1** தமிழர் சமூக ஒழுங்கு மற்றும் அதன் மாற்றங்களை ஆராய்தல்.
- CO2** பண்டைய அரசியல் அமைப்புகளின் தமிழர் பங்களிப்பை அறிதல்.
- CO3** தமிழர் நவீன அரசியல் சிந்தனைகளின் தாக்கத்தை ஆராய்தல்.
- CO4** காவியங்களில் தமிழர் அரசியல் தத்துவத்தை புரிதல்.
- CO5** தமிழர் அரசியல் பார்வையின் உலகளாவிய தாக்கத்தை அறிதல்.

பாடநூல் :

தமிழ் இலக்கியம் மற்றும் அதன் சமூக பங்குகள், முனைவர். வ. ராமநாதன், தமிழ்ச்சங்கம் பதிப்பகம்

பார்வைநூல்கள்:

தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, திரு. ச. ஜெயராமன், ஆழமான் பதிப்பகம்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	2	3	3	2	2	2	2	1	1
CO2	2	3	3	2	3	3	2	2	2
CO3	3	3	3	2	3	3	3	1	2
CO4	3	3	3	2	2	2	2	2	1
CO5	3	3	3	2	3	2	2	2	2
Avg	3	3	3	2	3	2	2	2	2

*** 3- High, 2- Medium, 1- Low**

இரண்டாம் ஆண்டு

மூன்றாம் பருவம்

TAC24301

தாள் 1: தமிழர் கல்வி வளர்ச்சி

L T P C

5 0 0 5

OBJECTIVES:

- ☐ On completion of the course, the student is expected to be able to

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அலகு 1: தமிழர் கல்வியின் வரலாற்றுப் பின்னணி 10

பண்டைய கல்வி முறை (குருகுலம் மற்றும் கலசங்கள்) - தமிழர் கல்வியில் சங்க காலத்தின் பங்கு - கல்வி மையங்கள்: மதுரை, காஞ்சி மற்றும் நாலந்தா - கல்வியில் தமிழ் மொழியின் நிலை - கல்வியின் சமூக மற்றும் கலாச்சார தாக்கம்

அலகு 2: கல்வியில் ஏற்பட்ட மாற்றங்கள் 10

தமிழ் கல்வியின் சீரழிவு -காரணங்கள் - பாண்டியர், சோழர் காலங்களில் கல்வி - மராத்தியர், மொகலாயர் ஆட்சியில் கல்வி அமைப்புகள் - ஆங்கிலேயரின் கல்வி திட்டத்தின் தாக்கம் - தமிழர் பாரம்பரிய கல்வியின் புதுப்பிப்பு

அலகு 3: நவீன கல்வி மற்றும் தமிழர் பங்கு 10

ஜெ.அம்பேத்கர் மற்றும் கல்வி புரட்சிகள் - தமிழர் சமூகத்தின் கல்விச் சீர்திருத்தங்கள் - பெண்கள் கல்வியில் தமிழரின் முன்னோடி பணி - தமிழர் கல்வியில் நூலகங்களின் பங்கு - கல்வி துறையில் தமிழரின் சாதனைகள்

அலகு 4: தமிழ் மொழி மற்றும் கல்வி 10

தமிழர் கல்வியில் மொழியின் பங்களிப்பு - தமிழ் இலக்கியங்கள் கல்வியின் ஆதாரம் - தமிழ் கல்வி மற்றும் அரசியல் மாற்றங்கள் - தற்கால கல்வியில் தமிழ் மொழியின் நிலை - தமிழ் கல்வியில் தொழில்நுட்பத்தின் பங்கு

அலகு 5: தமிழர் கல்வியின் எதிர்காலம் 10

கல்வியில் தமிழர் பாரம்பரியத்தை பாதுகாப்பது எப்படி? - நவீன தமிழ் கல்வி சவால்கள்
தொழில்நுட்ப ஆவணங்களில் தமிழ் மொழியின் இடம் - உலகளாவிய கல்வி அளவில் தமிழர் பங்குகளை உயர்த்துதல் - புதிய தலைமுறையில் தமிழர் கல்வி வளர்ச்சி

Total Periods: 50

OUTCOMES:

On completion of the course, the student is expected to be able to:

- CO1** தமிழர் கல்வியின் வரலாறு மற்றும் முக்கியத்துவம்
- CO2** கல்வியில் ஏற்பட்ட மாற்றங்கள் மற்றும் நவீன தமிழ் கல்வி வளர்ச்சி
- CO3** தமிழ் மொழியின் கல்வித் துறையில் பங்களிப்பு
- CO4** எதிர்கால கல்வியில் தமிழர் பங்கு
- CO5** மாணவர்கள் தமிழர் கல்வியின் வரலாற்றுப் பின்னணி மற்றும் அதில் ஏற்பட்ட மாற்றங்களை புரிந்து கொள்வார்கள்.

பாடநூல் :

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1.தமிழர் கல்வியின் வரலாறு மற்றும் வளர்ச்சி, முனைவர் இரா. சண்முகம், தமிழ்நாடு கல்வி மற்றும் கலாச்சார நிறுவனம், சென்னை

2.தமிழர் கல்வி வரலாற்றுப் பரிமாணங்கள், முனைவர் சி. சுதாகர், உயிர்மை பதிப்பகம், சென்னை

பார்வைநூல்கள்:

1.தமிழர் கல்வி: பாரம்பரியம் முதல் நவீனத்திற்கே, புலவர் மா. கணேசன், தமிழ்நாடு நூலக கல்வி நிறுவனம், சென்னை

2.தமிழர் கல்வி: பாரம்பரியமும் புதுமையும், முனைவர் எம். இளங்கோ, கிழக்கு பதிப்பகம், மதுரை

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	3
CO2	1	1	3	3	1	2	2	2	2
CO3	1	1	3	3	1	2	3	2	3
CO4	1	1	3	3	1	2	3	2	3
CO5	1	1	3	3	1	2	2	2	2
Avg	1	1	3	3	1	2	3	2	3

* 3- High, 2- Medium, 1- Low

TAC24301

தாள் 2: தமிழர் பண்பாட்டு மற்றும் கலாச்சார ஆய்வுகள்

L T P C

5 0 0 5

OBJECTIVES:

On completion of the course, the student is expected to be able to

அலகு 1.தமிழர் மரபு மற்றும் பண்பாட்டின் அடிப்படைகள் 10

அலகு 2.தமிழர் பண்பாட்டு தொல்லியல் 10

அலகு 3.உலகளாவிய கலாச்சார ஒப்பீடுகள் 10

அலகு 4.தமிழ் மொழியின் வளர்ச்சியியல் 10

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அலகு 5.நவீன உலகில் தமிழர் கலாச்சாரம் 10

Total Periods: 50

OUTCOMES:

On completion of the course, the student is expected to be able to:

- CO1** தமிழர் மரபுகளை அழியாதவாறு பாதுகாக்கும் உத்திகள்.
- CO2** தொல்லியல் மூலம் தமிழர் பண்பாட்டை ஆராய்தல்.
- CO3** உலகளாவிய அளவில் தமிழர் கலாச்சார ஒப்பீடுகளை புரிதல்.
- CO4** தமிழ் மொழியின் பரிணாம மாற்றங்களை ஆராய்தல்.
- CO5** நவீன உலகில் தமிழர் பண்பாட்டின் தாக்கத்தை அறிதல்.

பாடநூல் :

தமிழர் கலாச்சாரம் ஒரு அறிமுகம், திரு. ம. நா. சு. கா. சாந்தா, சிந்தனைப் பதிப்பகம்

பார்வைநூல்கள்:

தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, திரு. ச. ஜெயராமன், ஆழமான் பதிப்பகம்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	3	3	2	3
CO2	2	3	3	3	2	3	3	2	3
CO3	2	3	3	3	2	3	3	2	3
CO4	3	3	3	3	2	3	3	2	3
CO5	3	3	3	3	2	3	3	2	3
Avg	3	3	3	3	2	3	3	2	3

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*** 3- High, 2- Medium, 1- Low**

TAC24301

தாள் 3: தமிழர் சிந்தனை மற்றும் தத்துவ மரபுகள்

L T P C

5 0 0 5

OBJECTIVES:

On completion of the course, the student is expected to be able to

அலகு 1.தமிழர் தத்துவ பார்வைகள் 10

அலகு 2.ஞானக் கோட்பாடுகள் 10

அலகு 3.மெய்யியல் கோட்பாடுகள் 10

அலகு 4.தமிழர் சமய தத்துவங்கள் 10

அலகு 5.தற்கால தத்துவ சிந்தனைகள் 10

Total Periods: 50

OUTCOMES:

On completion of the course, the student is expected to be able to:

CO1 தமிழர் தத்துவ அடிப்படைகளை அறிதல்.

CO2 தமிழர் ஞானம் மற்றும் அதன் சமூக பங்களிப்பை ஆராய்தல்.

CO3 மெய்யியலில் தமிழர் பங்களிப்பு.

CO4 தமிழர் சமய தத்துவ பார்வைகளின் தாக்கத்தை அறிதல்.

CO5 தமிழர் தத்துவங்களில் சமகால மாற்றங்கள்.

பாடநூல் :

தமிழர் கலாச்சாரம் ஒரு அறிமுகம், திரு. ம. நா. சு. கா. சாந்தா, சிந்தனைப் பதிப்பகம்

பார்வைநூல்கள்:

தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, திரு. ச. ஜெயராமன், ஆழமான் பதிப்பகம்

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COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	3	3	2	3
CO2	2	3	3	3	2	3	3	2	3
CO3	2	3	3	3	2	3	3	2	3
CO4	3	3	3	3	2	3	3	2	3
CO5	3	3	3	3	2	3	3	2	3
Avg	3	3	3	3	2	3	3	2	3

* 3- High, 2- Medium, 1- Low

நான்காம் பருவம்

TAC24401

தாள் 1: தமிழர் பண்பாட்டு இலக்கியங்கள்

**L T P C
5 0 0 5**

OBJECTIVES:

On completion of the course, the student is expected to be able to

அலகு 1.இலக்கியங்களில் வாழ்வியலும் பண்பாடும் 10

அலகு 2.புராணங்களிலும் காவியங்களிலும் பண்பாட்டு கூறுகள் 10

அலகு 3.சமகால இலக்கியங்கள் மற்றும் சமூக மாறுதல்கள் 10

அலகு 4.குடியரசு இலக்கியப் பங்களிப்புகள் 10

அலகு 5.தமிழ் இலக்கியங்களில் பண்பாட்டு ஆய்வுகள் 10

Total Periods: 50

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OUTCOMES:

On completion of the course, the student is expected to be able to:

- CO1** இலக்கியங்களில் தமிழர் வாழ்வியல் பண்பாட்டின் தாக்கத்தை அறிதல்.
- CO2** புராணங்களில் பண்பாட்டு கூறுகளை ஆராய்தல்.
- CO3** சமகால இலக்கியங்களில் சமூக மாற்றங்களை அறிதல்.
- CO4** குடியரசு கால இலக்கியங்களில் தமிழர் பங்களிப்பை புரிதல்.
- CO5** தமிழ் இலக்கியங்களின் பண்பாட்டு அடிப்படையை ஆராய்தல்

பாடநூல் :

தமிழின் கலைகள் மற்றும் பண்பாட்டு மரபுகள், டாக்டர். ரா. செல்லப்பா, தமிழ்சாரா பதிப்பகம்

பார்வைநூல்கள்:

தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, திரு. ச. ஜெயராமன், ஆழமான் பதிப்பகம்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	3
CO2	1	1	3	3	1	2	2	2	2
CO3	1	1	3	3	1	2	3	2	3
CO4	1	1	3	3	1	2	3	2	3
CO5	1	1	3	3	1	2	2	2	2
Avg	1	1	3	3	1	2	3	2	3

*** 3- High, 2- Medium, 1- Low**

TAC24402

தாள் 2: தமிழர் நவீன துறை மற்றும் சமூக செயல்பாடுகள்

L T P C

5 0 0 5

**ARUNAI UNIVERSITY
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OBJECTIVES:

On completion of the course, the student is expected to be able to

அலகு 1.இலக்கியங்களில் வாழ்வியலும் பண்பாடும் 10

அலகு 2.புராணங்களிலும் காவியங்களிலும் பண்பாட்டு கூறுகள் 10

அலகு 3.சமகால இலக்கியங்கள் மற்றும் சமூக மாறுதல்கள் 10

அலகு 4.குடியரசு இலக்கியப் பங்களிப்புகள் 10

அலகு 5.தமிழ் இலக்கியங்களில் பண்பாட்டு ஆய்வுகள் 10

Total Periods: 50

OUTCOMES:

On completion of the course, the student is expected to be able to:

CO1 தமிழர் கல்வி வளர்ச்சியின் தாக்கத்தை அறிதல்.

CO2 தொழில்நுட்ப வளர்ச்சியில் தமிழர் பங்களிப்பை ஆராய்தல்.

CO3 மருத்துவத்துறையில் தமிழர் பங்களிப்பை அறிதல்.

CO4 சமூக ஆராய்ச்சியில் தமிழர் பங்களிப்பை விளக்குதல்.

CO5 பொதுநல திட்டங்களில் தமிழர் பங்குபற்றிய விழிப்புணர்வு.

பாடநூல் :

தமிழர் கலாச்சாரம் ஒரு அறிமுகம், திரு. ம. நா. சு. கா. சாந்தா, சிந்தனைப் பதிப்பகம்

பார்வைநூல்கள்:

தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, திரு. ச. ஜெயராமன், ஆழமான் பதிப்பகம்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	3

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CO2	1	1	3	3	1	2	2	2	2
CO3	1	1	3	3	1	2	3	2	3
CO4	1	1	3	3	1	2	3	2	3
CO5	1	1	3	3	1	2	2	2	2
Avg	1	1	3	3	1	2	3	2	3

*** 3- High, 2- Medium, 1- Low**

TAC24402

தாள் 3: வாய்மொழி இலக்கியங்களும் தமிழ்க் கலாச்சாரமும்

L T P C

5 0 0 5

OBJECTIVES:

On completion of the course, the student is expected to be able to

அலகு 1: வாய்மொழி இலக்கியங்களின் வரலாறு மற்றும் முக்கியத்துவம் 10

வாய்மொழி இலக்கியங்களின் தோற்றம் - பழமையான வாய்மொழி இலக்கியங்களின் அடிப்படைகள் - சமூகவியல் ஆய்வுகளின் மூலம் வாய்மொழியின் பங்கு

அலகு 2: சங்க இலக்கியங்களின் பங்களிப்பு 10

சங்க கால இலக்கியங்களின் தரம் - சங்க இலக்கியங்களின் கலாச்சார பக்கங்கள் - சங்க இலக்கியங்கள் தமிழின் கலாச்சாரத்தில்

அலகு 3: தமிழ் நவீன இலக்கியங்கள் மற்றும் கலாச்சாரம் 10

நவீன தமிழ் இலக்கியங்களின் தோற்றம் - கலாச்சார மாற்றங்கள் - சமீபத்திய தமிழ் இலக்கியங்கள்

அலகு 4: தமிழின் கலை மற்றும் இலக்கியமான பாரம்பரியம் 10

தமிழின் கலைவினைகள் - பாடல்கள், இசை மற்றும் நாடகக் கலை - தமிழில் கலைத்துறைகளின் பங்களிப்பு

அலகு 5: தமிழின் உலகளாவிய பங்களிப்புகள் 10

தமிழின் உலக அளவில் வளர்ச்சி - தமிழின் பங்களிப்பு மற்றும் சமூக மறுமொழி - உலகளாவிய தமிழரின் தாக்கம்

Total Periods: 50

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OUTCOMES:

On completion of the course, the student is expected to be able to:

- CO1** வாய்மொழி இலக்கியங்களின் வரலாற்றுப் புள்ளிவிவரம்
CO2 சங்க இலக்கியங்கள் மற்றும் தமிழ் பண்பாட்டின் முன்னேற்றம்
CO3 தமிழ் நவீன இலக்கியங்கள் மற்றும் அதன் எதிர்கால தாக்கங்கள்
CO4 தமிழ் கலை மற்றும் இலக்கியத்தின் பரிமாணங்கள்
CO5 தமிழ் மொழியின் உலகளாவிய பங்கு

பாடநூல்:

வாய்மொழி இலக்கியங்கள் மற்றும் தமிழ்க் கலாச்சாரம், வெ. இராசபெருமாள், தமிழ் எழுத்து பதிப்பகம்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	3
CO2	1	1	3	3	1	2	2	2	2
CO3	1	1	3	3	1	2	3	2	3
CO4	1	1	3	3	1	2	3	2	3
CO5	1	1	3	3	1	2	2	2	2
Avg	1	1	3	3	1	2	3	2	3

* 3- High, 2- Medium, 1- Low

மூன்றாம் ஆண்டு

ஐந்தாம் பருவம்

TAC24501

தாள் 1: தமிழர் சமூக மாற்றங்கள்

L T P C

5 0 0 5

OBJECTIVES:

ARUNAI UNIVERSITY
TIRUVANNAMALAI

On completion of the course, the student is expected to be able to

அலகு 1.குடியரசு காலத்தில் சமூக மாற்றங்கள் 10

அலகு 2.தமிழர் பெண்கள் முன்னேற்றம் 10

அலகு 3.பிரபல தமிழர் சமூகச் செயல்பாடுகள் 10

அலகு 4.பன்னாட்டு தமிழ் சமூகங்கள் 10

அலகு 5.இணையதள தமிழர் சமூகங்கள் 10

Total Periods: 50

OUTCOMES:

On completion of the course, the student is expected to be able to:

CO1 தமிழர் கல்வியின் வரலாறு மற்றும் முக்கியத்துவம்

CO2 கல்வியில் ஏற்பட்ட மாற்றங்கள் மற்றும் நவீன தமிழ் கல்வி வளர்ச்சி

CO3 தமிழ் மொழியின் கல்வித் துறையில் பங்களிப்பு

CO4 எதிர்கால கல்வியில் தமிழர் பங்கு

CO5 மாணவர்கள் தமிழர் கல்வியின் வரலாற்றுப் பின்னணி மற்றும் அதில் ஏற்பட்ட மாற்றங்களை புரிந்து கொள்வார்கள்.

பாடநூல் :

தமிழர் கலாச்சாரம் ஒரு அறிமுகம், திரு. ம. நா. சு. கா. சாந்தா, சிந்தனைப் பதிப்பகம்

பார்வைநூல்கள்:

தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, திரு. ச. ஜெயராமன், ஆழமான் பதிப்பகம்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	3

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CO2	1	1	3	3	1	2	2	2	2
CO3	1	1	3	3	1	2	3	2	3
CO4	1	1	3	3	1	2	3	2	3
CO5	1	1	3	3	1	2	2	2	2
Avg	1	1	3	3	1	2	3	2	3

*** 3- High, 2- Medium, 1- Low**

TAC24501

தாள் 2: தமிழர் பண்பாட்டு ஆதார நூல்கள் மற்றும் ஆய்வுகள்

L T P C

5 0 0 5

OBJECTIVES:

On completion of the course, the student is expected to be able to

அலகு 1.சங்க இலக்கியங்களில் பண்பாடு 10

அலகு 2.இலக்கிய ஆய்வுகள் 10

அலகு 3.நவீன தமிழ் ஆய்வுகள் 10

அலகு 4.தொலைநோக்கு பார்வையில் தமிழ் பண்பாடு 10

அலகு 5.பல்வேறு துறைகளில் தமிழ் அறிவியல் 10

Total Periods: 50

OUTCOMES:

On completion of the course, the student is expected to be able to:

CO1 சங்க இலக்கியங்கள் மூலம் பண்பாட்டு அம்சங்களை ஆராய்தல்.

CO2 இலக்கியங்களில் தமிழர் கலாச்சார பங்களிப்பை விளக்குதல்.

CO3 நவீன தமிழ் பண்பாட்டு ஆய்வுகளை அறிதல்.

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CO4 தமிழர் பண்பாட்டின் நீண்டகால தாக்கங்களை புரிதல்.

CO5 பல்வேறு துறைகளில் தமிழர் பங்களிப்பை அறிதல்.

பாடநூல் :

1.தமிழர் கலாச்சாரம் ஒரு அறிமுகம், திரு. ம. நா. சு. கா. சாந்தா, சிந்தனைப் பதிப்பகம்

பார்வைநூல்கள்:

1.தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, திரு. ச. ஜெயராமன், ஆழமான் பதிப்பகம்

2.தமிழின் கலாச்சாரம் மற்றும் அது உலகத்திற்கு வழங்கிய பங்களிப்புகள், முனைவர். எல். சுந்தரமணி, சிந்தனைப் பதிப்பகம்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	3
CO2	1	1	3	3	1	2	2	2	2
CO3	1	1	3	3	1	2	3	2	3
CO4	1	1	3	3	1	2	3	2	3
CO5	1	1	3	3	1	2	2	2	2
Avg	1	1	3	3	1	2	3	2	3

*** 3- High, 2- Medium, 1- Low**

TAC24501

தாள் 2: தமிழர் பண்பாட்டு ஆதார நூல்கள் மற்றும் ஆய்வுகள்

L T P C

5 0 0 5

OBJECTIVES:

On completion of the course, the student is expected to be able to

அலகு 1.சங்க இலக்கியங்களில் பண்பாடு 10

அலகு 2.இலக்கிய ஆய்வுகள் 10

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அலகு 3.நவீன தமிழ் ஆய்வுகள் 10

அலகு 4.தொலைநோக்கு பார்வையில் தமிழ் பண்பாடு 10

அலகு 5.பல்வேறு துறைகளில் தமிழ் அறிவியல் 10

Total Periods: 50

OUTCOMES:

On completion of the course, the student is expected to be able to:

- CO1** சங்க இலக்கியங்கள் மூலம் பண்பாட்டு அம்சங்களை ஆராய்தல்.
- CO2** இலக்கியங்களில் தமிழர் கலாச்சார பங்களிப்பை விளக்குதல்.
- CO3** நவீன தமிழ் பண்பாட்டு ஆய்வுகளை அறிதல்.
- CO4** தமிழர் பண்பாட்டின் நீண்டகால தாக்கங்களை புரிதல்.
- CO5** பல்வேறு துறைகளில் தமிழர் பங்களிப்பை அறிதல்.

பாடநூல் :

1.தமிழர் கலாச்சாரம் ஒரு அறிமுகம், திரு. ம. நா. சு. கா. சாந்தா, சிந்தனைப் பதிப்பகம்

பார்வைநூல்கள்:

- 1.தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, திரு. ச. ஜெயராமன், ஆழமான் பதிப்பகம்
- 2.தமிழின் கலாச்சாரம் மற்றும் அது உலகத்திற்கு வழங்கிய பங்களிப்புகள், முனைவர். எல். சுந்தரமணி, சிந்தனைப் பதிப்பகம்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	3
CO2	1	1	3	3	1	2	2	2	2
CO3	1	1	3	3	1	2	3	2	3
CO4	1	1	3	3	1	2	3	2	3

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CO5	1	1	3	3	1	2	2	2	2
Avg	1	1	3	3	1	2	3	2	3

* 3- High, 2- Medium, 1- Low

TAC24501

தாள் 3: தமிழர் பண்பாட்டுக் கொள்கைகள்

L T P C

5 0 0 5

OBJECTIVES:

On completion of the course, the student is expected to be able to

அலகு 1.பாரம்பரிய கொள்கைகளின் அடிப்படைகள் 10

அலகு 2.நீதி மற்றும் பண்பாட்டு கொள்கைகள் 10

அலகு 3.தொழில்முறை நெறிமுறைகள் 10

அலகு 4.சமுதாய நலனுக்கான கொள்கைகள் 10

அலகு 5.அரசியல் மற்றும் பொருளாதார கொள்கைகள் 10

Total Periods: 50

OUTCOMES:

On completion of the course, the student is expected to be able to:

CO1 தமிழர் பாரம்பரிய கொள்கைகளின் சமூக தாக்கங்களை அறிதல்.

CO2 தமிழர் நீதி முறை மற்றும் அதன் பண்பாட்டு அடிப்படைகளை ஆராய்தல்.

CO3 தமிழர் தொழில்முறை நெறிமுறைகளின் தாக்கத்தை புரிதல்.

CO4 தமிழர் சமூக நலனுக்கான கொள்கைகளை வெளிப்படுத்தல்.

CO5 தமிழர் அரசியல் மற்றும் பொருளாதாரக் கொள்கைகளின் வளர்ச்சியை ஆராய்தல்.

பாடநூல் :

ARUNAI UNIVERSITY
TIRUVANNAMALAI

1.தமிழர் கலாச்சாரம் ஒரு அறிமுகம், திரு. ம. நா. சு. கா. சாந்தா, சிந்தனைப் பதிப்பகம்

பார்வைநூல்கள்:

1.தமிழர் சமுதாயத்தின் அமைப்புகள் மற்றும் பண்பாடு, திரு. ச. ஜெயராமன், ஆழமான் பதிப்பகம்

2.தமிழின் கலாச்சாரம் மற்றும் அது உலகத்திற்கு வழங்கிய பங்களிப்புகள், முனைவர். எல். சுந்தரமணி, சிந்தனைப் பதிப்பகம்

COs-POs & PSO's MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	3
CO2	1	1	3	3	1	2	2	2	2
CO3	1	1	3	3	1	2	3	2	3
CO4	1	1	3	3	1	2	3	2	3
CO5	1	1	3	3	1	2	2	2	2
Avg	1	1	3	3	1	2	3	2	3

* 3- High, 2- Medium, 1- Low

TAC24501

தமிழ் மரபு மருத்துவம்

L T P C

5 0 0 5

OBJECTIVES:

On completion of the course, the student is expected to be able to

அலகு 1: தமிழ் மரபு மருத்துவத்தின் வரலாறு மற்றும் அடிப்படைகள் 10

தமிழ் மரபு மருத்துவம் வரலாற்று அமைப்பு - பிராரம்பிக காலத்தில் மரபு மருத்துவத்தின் பயன்பாடு - மரபு மருத்துவம் மற்றும் அதன் சமூக பண்பாடு

அலகு 2: தமிழ் மரபு மருத்துவம் மற்றும் குணங்கள் 10

பசும்பொன் மற்றும் அங்கவினியோகங்கள் - மரபு மருத்துவம் மூலம் குணப்படுத்தும் முறைகள் - பிரசித்தி பெற்ற தமிழ் மருத்துவ நூல்கள்

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அலகு 3: தமிழ் மரபு மருத்துவத்தின் உபயோகப்பொருட்கள் 10

மஞ்சள், பெருங்காயம் மற்றும் எலுமிச்சம் - மூலிகைகள் மற்றும் அவற்றின் மருத்துவ பங்கு - குவாமரம், சித்தார்த்தி போன்ற மருந்துகளின் பயன்கள்

அலகு 4: தமிழ் மரபு மருத்துவத்தில் ஊடகவியல் முறைகள்

உடல் ஆரோக்கியத்திற்கு ஊடகவியல் முறைகள், பரிசோதனைகள் மற்றும் பராமரிப்பின் ஆராய்ச்சி, மரபு மருத்துவத்தில் ஊடகவியல் தத்துவம்

அலகு 5: தமிழ் மரபு மருத்துவத்தின் எதிர்காலம்

தமிழில் மரபு மருத்துவத்தின் பரவல், மரபு மருத்துவம் மற்றும் புதிய சித்தாந்தங்கள், சித்த மருத்துவம் உலகளாவிய நிலை

Total Periods: 50

OUTCOMES:

On completion of the course, the student is expected to be able to:

- CO1** தமிழ் மரபு மருத்துவத்தின் வரலாற்று பண்புகள்
- CO2** தமிழ் மரபு மருத்துவத்தில் குணப்படுத்தும் திறன்கள்
- CO3** தமிழ் மரபு மருத்துவத்திலும் மூலிகைகளின் பங்கு
- CO4** ஊடகவியல் முறைகளின் பங்கு மரபு மருத்துவத்தில்
- CO5** தமிழ் மரபு மருத்துவத்தின் எதிர்கால சவால்கள்

பாடநூல்:

- 1.தமிழ் மரபு மருத்துவம், டாக்டர். ரா. மஞ்சளா, தமிழ்க் கல்வி பதிப்பகம்
- 2.தமிழ் மரபின் ஆயுர்வேதம் - மு. ராமநாதன், பழமையான மருத்துவ நடைமுறைகள் - க. சுரேஷ்

பார்வை நூல்கள்:

- 1.அயுர்வேத மரபு - சா. மகேஷ்வரி
- 2.களஞ்சிய மருத்துவம் - அ. கிருஷ்ணமூர்த்தி
- 3.மூலிகைகள் மருத்துவம் - வ. விஜயா
- 4.தமிழில் மருந்துகள் - ப. சந்திரன்
- 5.சித்த மருத்துவம் - ர. விஜயசாரதி
- 6.உணர்வு மருத்துவம் - ஜி. காமாட்சி
7. பழமையான மருத்துவத்துக்கான புதுமை - ம. துரைமுருகன்
- 8.அறிவியல் மற்றும் மரபு மருத்துவம் - ப. செல்வராஜ்

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COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	3
CO2	1	1	3	3	1	2	2	2	2
CO3	1	1	3	3	1	2	3	2	3
CO4	1	1	3	3	1	2	3	2	3
CO5	1	1	3	3	1	2	2	2	2
Avg	1	1	3	3	1	2	3	2	3

*** 3- High, 2- Medium, 1- Low**

TAC24501

தமிழரின் தற்சார்பு வாழ்வியல்

L T P C

5 0 0 5

OBJECTIVES:

On completion of the course, the student is expected to be able to

அலகு 1: தமிழ் மரபு மருத்துவத்தின் வரலாறு மற்றும் அடிப்படைகள் 10

தமிழ் மரபு மருத்துவம் வரலாற்று அமைப்பு - பிராரம்பிக காலத்தில் மரபு மருத்துவத்தின் பயன்பாடு - மரபு மருத்துவம் மற்றும் அதன் சமூக பண்பாடு

அலகு 2: தமிழ் மரபு மருத்துவம் மற்றும் குணங்கள் 10

பசும்பொன் மற்றும் அங்கவினியோகங்கள் - மரபு மருத்துவம் மூலம் குணப்படுத்தும் முறைகள் - பிரசித்தி பெற்ற தமிழ் மருத்துவ நூல்கள்

அலகு 3: தமிழ் மரபு மருத்துவத்தின் உபயோகப்பொருட்கள் 10

மஞ்சள், பெருங்காயம் மற்றும் எலுமிச்சம் - மூலிகைகள் மற்றும் அவற்றின் மருத்துவ பங்கு - குவாமரம், சித்தார்த்தி போன்ற மருந்துகளின் பயன்கள்

அலகு 4: தமிழ் மரபு மருத்துவத்தில் ஊடகவியல் முறைகள்

உடல் ஆரோக்கியத்திற்கு ஊடகவியல் முறைகள், பரிசோதனைகள் மற்றும் பராமரிப்பின் ஆராய்ச்சி, மரபு மருத்துவத்தில் ஊடகவியல் தத்துவம்

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அலகு 5: தமிழ் மரபு மருத்துவத்தின் எதிர்காலம்

தமிழில் மரபு மருத்துவத்தின் பரவல், மரபு மருத்துவம் மற்றும் புதிய சித்தாந்தங்கள், சித்த மருத்துவம் உலகளாவிய நிலை

Total Periods: 50

OUTCOMES:

On completion of the course, the student is expected to be able to:

- CO1** தமிழ் மரபு மருத்துவத்தின் வரலாற்று பண்புகள்
- CO2** தமிழ் மரபு மருத்துவத்தில் குணப்படுத்தும் திறன்கள்
- CO3** தமிழ் மரபு மருத்துவத்திலும் மூலிகைகளின் பங்கு
- CO4** ஊடகவியல் முறைகளின் பங்கு மரபு மருத்துவத்தில்
- CO5** தமிழ் மரபு மருத்துவத்தின் எதிர்கால சவால்கள்

பாடநூல்:

- 1.தமிழ் மரபு மருத்துவம், டாக்டர். ரா. மஞ்சளா, தமிழ்க் கல்வி பதிப்பகம்
- 2.தமிழ் மரபின் ஆயுர்வேதம் - மு. ராமநாதன், பழமையான மருத்துவ நடைமுறைகள் - க. சுரேஷ்

பார்வை நூல்கள்:

- 1.அயுர்வேத மரபு - சா. மகேஷ்வரி
- 2.களஞ்சிய மருத்துவம் - அ. கிருஷ்ணமூர்த்தி
- 3.மூலிகைகள் மருத்துவம் - வ. விஜயா
- 4.தமிழில் மருந்துகள் - ப. சந்திரன்
- 5.சித்த மருத்துவம் - ர. விஜயசாரதி
- 6.உணர்வு மருத்துவம் - ஜி. காமாட்சி
7. பழமையான மருத்துவத்துக்கான புதுமை - ம. துரைமுருகன்
- 8.அறிவியல் மற்றும் மரபு மருத்துவம் - ப. செல்வராஜ்

COs-POs & PSOs MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	1	1	2	2	2	2	3	2	3

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CO2	1	1	3	3	1	2	2	2	2
CO3	1	1	3	3	1	2	3	2	3
CO4	1	1	3	3	1	2	3	2	3
CO5	1	1	3	3	1	2	2	2	2
Avg	1	1	3	3	1	2	3	2	3

*** 3- High, 2- Medium, 1- Low**

ஆறாம் பருவம்

தாள் 1: வள்ளுவ வாழ்வியல்

**L T P C
5 0 0 5**

அலகு 1: திருக்குறளின் அறத்துப்பால் வாழ்வியலின் அடிப்படை

10

அறத்துப்பாலின் சாராம்சம்

திருக்குறளின் அறம்: தனிநபர் ஒழுக்கம்

குடும்பம் மற்றும் சமூக உறவுகளில் அறம்

அறத்துப்பாலின் மெய்ப்பொருள் உதவி, நன்றி, பொறுமை

அறத்தின் பாதிப்புகள்: நவீன வாழ்க்கை சூழலில் அதன் பங்கு

அலகு 2: பொருட்பால் பொருளாதார வாழ்வியல்

10

பொருட்பாலின் அடிப்படை கொள்கைகள்

பொருளாதார வாழ்க்கை: வேலை, செல்வம், நீதி

நல்லாட்சிக்கான திருக்குறளின் வழிகாட்டல்

அரசியல் வாழ்வியல்: ஒரு தலைவரின் பணிகள்

பொருளாதார சமநிலையின் அடிப்படைத் தத்துவங்கள்

அலகு 3: காமத்துப்பால் காதலின் வாழ்வியல் தத்துவம்

10

காமத்துப்பாலின் பொதுவாய்த்த கோட்பாடுகள்

காதல் வாழ்வின் தர்மங்கள்

காதல் வாழ்க்கையின் அழகு மற்றும் சிக்கல்கள்

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உறவுகள்: காதலிலும் அறத்திலும் இணைந்த வாழ்க்கை
காமத்துப்பாலின் உளவியல் தத்துவங்கள்

அலகு 4: வாழ்வியல் அறம் திருக்குறளின் முழுமையான பின்புலம்

10

வாழ்வியல் அறம் தனிநபரின் நற்பண்புகள்
சமூகத்தில் ஒருங்கிணைந்த வாழ்வியல்
ஒழுக்கம் மற்றும் நேர்மை
அன்பு, பொறுமை, மனிதநேயம்
அறத்துப்பாலும் பொருட்பாலும் சமநிலையான வாழ்க்கை

அலகு 5: திருக்குறள் வாழ்வியலின் நவீன பயன்பாடு

10

திருக்குறள் மற்றும் நவீன வாழ்வியல்: ஒழுக்கத்தின் அவசியம்
பொருளாதாரம், சமூக நீதி மற்றும் மனிதநேயம்
நவீன குடும்ப அமைப்புகளில் திருக்குறளின் பங்கு
திருக்குறளின் தத்துவங்கள் நேரம், பண்பாடு மற்றும் பாரம்பரியம்
உலகளாவிய வாழ்வியல் தத்துவமாக திருக்குறள்

- ஊழுளு1 : அறம், பொருள், காமம் என்ற மூன்று பகுதிகளின் முழுமையான விளக்கம்.
ஊழுளு2 : திருக்குறளின் வாழ்வியல் கொள்கைகள் மற்றும் அவற்றின் நடைமுறைப் பயன்கள்.
ஊழுளு3 : ஒழுக்கம், காதல், பொருளாதாரம், சமூக அமைப்புகள் பற்றிய தத்துவங்கள்.
ஊழுளு4 : திருக்குறளின் வரலாற்று பின்னணி மற்றும் இலக்கியத்துடன் சேர்க்கை.
ஊழுளு5 : திருக்குறளின் தத்துவங்களை ஆய்வு செய்து அறத்தின் முக்கியத்துவத்தை விளக்குதல்.
ஊழுளு6 : வாழ்வியல் சிக்கல்களை திருக்குறளின் அடிப்படையில் அணுகுதல்.

Total Periods: 50

பாடநூல்கள் : திருக்குறள் வாழ்வியல் தத்துவங்கள்

டாக்டர் முனைவர் இரா. கணேசன்
தமிழ்நாடு மொழி வளர்ச்சி நிறுவனம், சென்னை

திருக்குறளின் வாழ்வியல் மூலாதாரம்
முனைவர் ப. திருநாவுக்கரசு
சாகித்திய அகாதமி வெளியீடு

பார்வைநூல்கள்: திருக்குறள்: வாழ்வியல் தத்துவ நெறிகள்
முனைவர் சி. ஜெயராமன்
உயிர்மை பதிப்பகம், சென்னை

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திருக்குறள்: வாழ்வியல் முன்மாதிரிகள்
புலவர் க. மணிவண்ணன்
கிழக்கு பதிப்பகம், மதுரை

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	3	3	3	3
COS 2:	3	3	3	3	3
COS 3:	3	3	3	3	3
COS 4:	3	3	3	3	3
COS 5:	3	3	3	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (ர்ப்பா) 2 (ஆநனரைஅ) 1 (டுழற) ”-” (ழே ஆயிபெ):

தாள் 2: நவீன தமிழர் சமூகப் பண்பாடு

L T P C

5 0 0 5

ஊழுளு1 : தமிழர் பண்பாட்டு வரலாறு மற்றும் சமூக வளர்ச்சியில் ஏற்பட்ட மாற்றங்கள்.

ஊழுளு2 : நவீன வாழ்க்கைமுறைகளில் தமிழர் பண்பாட்டின் இடம் மற்றும் தாக்கம்.

ஊழுளு3 : உலகமயமாக்கலின் பின்னணியில் தமிழர் சமூகத்தின் நிலை.

ஊழுளு4 : மாணவர்கள் நவீன தமிழர் சமூகப் பண்பாட்டை நன்கு புரிந்து கொள்வார்கள்.

அலகு 1: தமிழர் சமூகப் பண்பாட்டின் வரலாற்றுப் பின்னணி

10

பழந்தமிழர் சமூக அமைப்புகள்

பண்பாட்டின் பரிமாணங்கள் (குடும்பம், மதம், கல்வி)

சமூக ஒழுக்கங்கள் மற்றும் மரபுகள்

காலப்போக்கில் ஏற்பட்ட மாற்றங்கள்

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பண்டைய சமுதாய பண்பாட்டின் தொடர்ச்சிகள்

அலகு 2: சமூக மாற்றங்கள் மற்றும் நவீன தமிழர் வாழ்க்கை 10

நகர்ப்புற வாழ்க்கை மற்றும் தொழில்நுட்பத்தின் தாக்கம்
கல்வி மற்றும் வேலைவாய்ப்பில் ஏற்பட்ட மாற்றங்கள்
மதம் மற்றும் சமூகத்தின் நிலை
தமிழர் வாழ்வியலில் புதுமையான அம்சங்கள்
நவீன வாழ்க்கை முறையில் தமிழ் மரபுகளின் இடம்

அலகு 3: பண்பாட்டு அடையாளங்கள் மற்றும் நவீன சவால்கள் 10

தமிழர் மொழி மற்றும் பண்பாட்டின் அடையாளம்
உலகமயமாக்கலின் தாக்கம்
மரபு மற்றும் நவீனத்தின் இடைவெளி
தமிழர் இசை, நடனம், இலக்கியங்களில் நவீன மாற்றங்கள்
பண்பாட்டு அடையாளங்களை பாதுகாக்கும் முயற்சிகள்

அலகு 4: சமூக முன்னேற்றத்திற்கான தமிழர் பங்களிப்பு 10

சமூக சேவையில் தமிழர் பங்கு
அரசியல், கலை, கல்வி துறைகளில் தமிழர்களின் முன்னேற்றம்
பெண்கள் கல்வி மற்றும் சமுதாயத்தில் ஏற்பட்ட வளர்ச்சி
தமிழர் தொழில் நுட்ப முன்னேற்றம்
திரை உலகம் மற்றும் செய்தி ஊடகங்களில் தமிழர்களின் தாக்கம்

அலகு 5: நவீன தமிழ் சமூகத்தின் எதிர்காலம் 10

பண்பாட்டு ஒருங்கிணைவு மற்றும் வளர்ச்சி
சமூகப் பிணைப்பு மற்றும் இணைவு
தமிழர் பண்பாட்டு விழிப்புணர்வு
புதிய தலைமுறை மற்றும் மரபு
தமிழர் பண்பாட்டின் உலகளாவிய நிலை

* Total Periods: 50

பாடநூல் : நவீன தமிழர் சமூகப் பண்பாடு
ஆசிரியர்: வெ.பெருமாள்
பதிப்பகம்: உயிர்மை பதிப்பகம், சென்னை.

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COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	3	3	3	3
COS 2:	3	3	3	3	3
COS 3:	3	3	3	3	3
COS 4:	3	3	3	3	3
COS 5:	3	3	3	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (ர்பை) 2 (ஆநனரைஅ) 1 (டுழற) ”-” (ழே ஆயிபெ):

தாள் 3: தமிழர் பண்பாட்டு மறுவளர்ச்சி`

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ஊழுளு : தமிழர் பண்பாட்டின் பண்டைய பெருமை மற்றும் மறுவளர்ச்சியின் தேவை.

ஊழுளு :இலக்கியம், கலை மற்றும் கல்வியில் ஏற்பட்ட மறுவளர்ச்சி.

ஊழுளு : தமிழர் சமூகத்திற்கான பண்பாட்டு மீள்பிறப்பின் பாதைகள்.

ஊழுளு : மாணவர்கள் தமிழர் பண்பாட்டை மீட்டெடுக்கும் முயற்சிகளை அறிந்து கொள்ளுதல்

அலகு 1: தமிழர் பண்பாட்டின் சீரழிவும் மறுவளர்ச்சியின் தேவை

10

பண்டைய தமிழர் பண்பாட்டு முறை

ஒழுங்குநிலை சீரழிவு மற்றும் அதன் காரணங்கள்

தமிழ் மரபுகளை மீட்டெடுக்கும் முயற்சிகள்

கொழுந்து மரபுகளை மீட்டெடுத்த தமிழக ஆளுமைகள்

மறுவளர்ச்சியின் முக்கியத்துவம்

அலகு 2: தமிழ் இலக்கியத்தில் மறுவளர்ச்சி

10

சங்க கால இலக்கியத்தின் மறுமலர்ச்சி

பாண்டிய, சோழ, சேர மன்னரின் பங்களிப்பு

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பக்தி இலக்கியங்கள் மற்றும் தற்கால பண்பாட்டு தாக்கம்
மறுவளர்ச்சியில் இலக்கிய இயக்கங்கள்
தமிழ் மொழி காக்கும் அரசியல் மற்றும் சமூக இயக்கங்கள்

அலகு 3: கலைகள் மற்றும் பண்பாட்டு பாரம்பரியம்

10

தமிழர் நடனம், இசை மற்றும் கலைமுறை
கோயில் கட்டிடக்கலை மற்றும் சிற்பங்கள்
மக்கள் கலைகள் மற்றும் தற்கால முன்னேற்றம்
திரை உலகம் மற்றும் புதுக்கவிதை
கலைஞர்களின் பங்கு மற்றும் பண்பாட்டின் மறு எழுச்சி

அலகு 4: கல்வி, அறிவியல் மற்றும் தொழில்நுட்பத்தின் பங்களிப்பு

10

தமிழர் கல்வியில் ஏற்பட்ட மறுவளர்ச்சி
தொழில்நுட்ப வளர்ச்சியில் தமிழரின் பங்கு
தமிழ் அறிவியல் பண்பாட்டின் பரிமாணங்கள்
பண்பாட்டு முன்னேற்றத்தில் கல்வியின் இடம்
இன்றைய சமூகத்தில் அறிவியல் பண்பாட்டு தத்துவங்கள்

அலகு 5: தமிழர் பண்பாட்டு மறுவளர்ச்சியின் எதிர்காலம்

10

தமிழர் மரபுகளின் மீட்பு முயற்சிகள்
தமிழ் மொழி பாதுகாப்பில் அரசின் பங்கு
புதிய தலைமுறையிடம் தமிழர் பண்பாட்டை அழியாமல் செய்யும் முயற்சிகள்
உலகளாவிய பண்பாட்டு ஒருங்கிணைவு
தமிழர் பண்பாட்டின் எதிர்கால வளர்ச்சி பாதை

Total Periods: 50

பாடநூல் : தமிழர் பண்பாட்டு மறுவளர்ச்சி

மா.கார்த்திகேசு

தமிழ்நாடு கலை இலக்கிய வளர்ச்சி அமைப்பு, சென்னை.

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	3	3	3	3
COS 2:	3	3	3	3	3

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COS 3:	3	3	3	3	3
COS 4:	3	3	3	3	3
COS 5:	3	3	3	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High) 2 (Medium) 1 (Low) "-" (No Mapping):

**விரிவாக்க செயல்பாடுகள்
ஆய்வேடு**

ஆய்வேடு பணிக்க மேற்கொள்ள வேண்டிய நெறிமுறைகள்:

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- ❖ ஆய்வு பற்றிய விளக்கம் அறிதல்
- ❖ ஆய்வுத் தலைப்பு தேர்வு செய்தல்
- ❖ முதன்மைத் துணை ஆதாரங்களைத் திரட்டுதல்.
- ❖ ஆய்வு நெறிமுறைகளை அறிந்து ஆய்வேடு எழுதுதல்.
- ❖ மேற்கோள்களைத் தேர்ந்தெடுத்தல்,
- ❖ ஆய்வுப் பயனை வெளிப்படுத்துதல்,
- ❖ களஆய்வினை மேற்கொள்ளுதல்
- ❖ துணை நூற்பட்டியல் தயாரித்தல்.

Total Periods: 30

Program Name : முதுகலை – தொழில்சார் கலைகள் (MA – VOCATIONAL ARTS)

Program Duration: 2 years

Philosophy/Significance of the Course

தத்துவம்:

இன்றைய தொழில்நுட்ப மையப்படுத்தப்பட்ட உலகில், தொழில்சார் கலைகளின் மகத்துவம் அதிகரித்து வருகின்றது. இப்பாடத் திட்டத்தின் தத்துவம், பாரம்பரியமும் நவீனத்துவமும் இணைந்த தொழில்சார் திறன்களை மாணவர்களுக்கு வழங்குவதே ஆகும். தொழில்களுக்கான அடிப்படை திறன்களையும், தொழில்முறை மனப்பாங்கையும் உருவாக்கி, மாணவர்களை சுயமாகவும் பொருளாதார ரீதியிலும் வலுவாக உருவாக்குவதே இதன் நோக்கமாகும்.

முக்கியத்துவம்:

1. **தொழில்நுட்ப திறன் வளர்ச்சி:** மாணவர்களுக்கு தொழில்நுட்ப-மையப்பட்ட கலைகளில் திறன்களை அடையச் செய்தல்.
2. **தொழில் சந்தையில் போட்டித்திறன்:** உலக சந்தையில் தொழில்துறையின் மாறும் தேவைவரிசைகளுக்கேற்ப திறமைகளை உருவாக்குதல்.
3. **திறன் மேம்பாடு:** மாணவர்களின் தனிநபர் திறன்களை மேம்படுத்தி, அவர்களின் பொருளாதார சுயமுன்னேற்றத்தை உறுதி செய்தல்
4. **சுயதொழில் ஊக்குவிப்பு:** தனி தொழில் தொடங்குவதற்கான கலைகள் மற்றும் கருத்துக்கள் மூலம் மாணவர்களுக்கு உதவுதல்.

இந்த பாடத்திட்டம், தொழில்சார் திறன்களையும் தொழில் சார்ந்த மனப்பாங்கையும் மாணவர்களுக்கு வழங்கி, அவர்களை சுயமுனைவு மிக்க, பொருளாதார ரீதியிலான முன்னேற்றத்திற்கு வழிவகுக்கும் ஒருவகை சாதனமாக இருக்கும்.

Distinctiveness and uniqueness of the program

தனித்துவமும் சிறப்புமிகுதியும் (Distinctiveness and Uniqueness):

1. நடைமுறை கற்கை:

இது முழுக்க தொழில்நுட்பத்தின் அடிப்படையில் நடைமுறையில் பயன்படக்கூடிய கலைகளைப் பயிற்சி அளிக்கிறது.

2. தொழில்முறை திறன்கள்:

மாணவர்களின் தொழில்நுட்ப அறிவையும் செயல்முறை திறன்களையும் மேம்படுத்துவதற்கான சிறப்பான பயிற்சியை வழங்குகிறது.

3. சந்தை சார்ந்த கற்பித்தல்:

தொழில்சந்தையின் தேவைப்படி பாடத்திட்டம் வடிவமைக்கப்பட்டுள்ளதால், மாணவர்கள் உடனடி வேலை வாய்ப்பை அடைவதற்கு முடிகிறது.

4. சொந்த தொழில் உருவாக்கம்:

மாணவர்கள் வேலைக்காக மட்டுமல்லாமல், தாங்களே தொழில்முனைவர்களாக செயல்பட அவர்களின் திறமைகளை உருவாக்குகிறது.

5. சமூக மற்றும் பொருளாதார முன்னேற்றம்:

தொழில்முறை கலைகள் சமூக முன்னேற்றத்திற்கும் பொருளாதார மேம்பாட்டிற்கும் முக்கிய பங்காற்றக்கூடிய கல்வி துறையாக உருவாகியுள்ளது.

இப்பாடம், மாணவர்களின் தனிப்பட்ட திறன்களையும் தொழில்முறை தேவைகளையும் ஒருங்கிணைத்து, அவர்கள் வாழ்வில் சிறந்த முன்னேற்றத்திற்கான பாதையை அமைக்கிறது

REGULATION-2025
(Choice Based Credit System)
B.A. VOCATIONAL ARTS

இளங்கலை – தொழில்சார் கலைகள்

(It Should be a Regional Language, If only Regional Language)

PROGRAMME OUTCOMES (POs):

Graduates of the Programme M.A. VOCATIONAL ARTS will be able to

- PO1** தமிழ் பண்பாட்டு பாரம்பரியத்தைக் கருதல்: தமிழர் பாரம்பரியம், கலாச்சாரம், மற்றும் நாகரிகம் தொடர்பான உள்ளார்ந்த அடிப்படைகளைப் புரிந்துகொண்டு அவற்றின் முக்கியத்துவத்தை விளங்குதல்.
- PO2** திறன்களின் வளர்ச்சி: இலக்கியம், கலை, வரலாறு, மற்றும் சமூக அமைப்புகளில் இருக்கும் கலாச்சார அடையாளங்களை ஆய்வு செய்யும் திறன்களை மேம்படுத்துதல்.
- PO3** சமூக ஒற்றுமை உருவாக்கம்: தமிழர் கலாச்சாரத்தின்மூலம் சமூகத்தின் மனிதநேயத்தையும் ஒற்றுமையையும் வளர்க்குதல்.
- PO4** பாரம்பரியத்தை பாதுகாத்தல்: தமிழர் பாரம்பரிய கலாச்சாரங்களின் நவீனப் பயன்பாடுகளைப் புரிந்துகொண்டு அவற்றை பாதுகாத்தல் மற்றும் பரப்புதல்.
- PO5** தனிப்பட்ட ஆளுமை மேம்பாடு: கலாச்சாரம் சார்ந்த பாடங்கள் மூலம் மாணவர்களின் தனிநபர் திறன்களை மேம்படுத்துதல்.
- PO6** ஆய்வு மனப்பாங்கு: தமிழர் நாகரிகம் மற்றும் கலாச்சாரம் சார்ந்த பரிணாம வளர்ச்சிகளை ஆராயும் திறன்களை உருவாக்குதல்.
- PO7** கலை மற்றும் நாகரிக சமூகம்: தமிழ் கலை மற்றும் நாகரிகத்தின் சமூக பயன்பாடு தொடர்பான கருத்துகளை புரிந்துகொண்டு அவற்றை நடைமுறைப்படுத்துதல்.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs):

Graduates of the programme B.A. VOCATIONAL ARTS will

- PEO1** கலை, கலாச்சார வளம் மற்றும் வரலாற்றுப் புரிதல்: தமிழர் பண்பாட்டின் பல்வேறு பரிமாணங்களையும் சமூகத்தில் அதன் தாக்கத்தையும் அறிய உதவுதல்.
- PEO2** ஆளுமை வளர்ச்சி: பண்பாட்டின் அடிப்படைகளைக் கற்றல் மூலம் மாணவர்களின் ஆளுமை வளர்ச்சி மற்றும் சமூக பொறுப்புணர்வை அதிகரித்தல்.
- PEO3** அறிவியல் திறன்கள்: பாரம்பரியத்தை நவீன தேவைமுறைகளுடன் இணைத்து புதிய துறைகளுக்கு வழிவகை செய்யக்கூடிய திறமைகளை உருவாக்குதல்.

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- PEO4** சமூக முன்னேற்றம்: கலாச்சாரத்தைப் பயன்படுத்தி சமூக முன்னேற்றத்திற்கு பங்களிப்பை ஊக்குவித்தல்.
- PEO5** மொழி மேம்பாடு: தமிழ் மொழியின் வளங்களையும் பன்முக அடையாளங்களையும் ஆய்வு செய்து பேச்சு மற்றும் எழுத்துத்திறன்களை மேம்படுத்துதல்.
- PEO6** சுயதிறன் மற்றும் சுயமுனைவு: பாரம்பரிய மற்றும் நவீன தமிழ் கலைகளின் தொழில்முனைவு வாய்ப்புகளை புரிந்துகொண்டு சுயமுனைவு உருவாக்குதல்.

PROGRAMME SPECIFIC OBJECTIVES (PSOs):

B.A. VOCATIONAL ARTS program's specific goals are to create a virtue-based and a well-civilized society:

- PSO1** பாரம்பரியத்திற்கான ஆழ்ந்த புரிதல்: தமிழ் கலாச்சாரம் மற்றும் நாகரிகத்தின் முக்கியத்துவத்தை அறிந்து பாரம்பரியத்திற்கான தெளிவை ஏற்படுத்துதல்.
- PSO2** அறிவியல் மற்றும் கலாச்சார ஆய்வு திறன்: கலாச்சாரம் சார்ந்த ஆய்வுத் திறன்களை வளர்த்துக்கொண்டு, தற்கால சமூக தேவைகளுக்கு ஏற்ப புதிய யோசனைகளை உருவாக்குதல்.
- PSO3** தமிழ் கலாச்சாரத்தின் ஆவணப்படுத்தல்: தமிழ் பண்பாட்டு அடையாளங்களைச் சரியாக ஆவணப்படுத்தி உலகளாவிய பார்வைக்கு கொண்டு செல்லுதல்.
- PSO4** மாற்றுத்திறன்கள் மற்றும் வளர்ச்சி: மொழி, கலை மற்றும் பண்பாட்டு துறைகளில் சிறப்பு திறன்களை அடைய மாணவர்களை உதவுதல்.
- PSO5** சமூகப் பங்குபற்றல்: சமூகத்தின் கலாச்சார மற்றும் பண்பாட்டு சவால்களை அடிப்படையாகக் கொண்டு, உரிய தீர்வுகளைத் தயாரிக்க ஆர்வம் செலுத்துதல்.

B.A – B.A. VOCATIONAL ARTS (Full Time)

இளங்கலைப் பட்டப்படிப்பு
தொழில்சார் கலைகள் (முழுநேரம்)
பாடத்திட்டம்
நான்கு பருவங்கள்

SEMESTER I

S.N O.	COURSE CODE	COURSE TITLE	CAT EG OR Y	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	

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THEORY								
1	PTSF25101	தமிழக கலாச்சாரம் மற்றும் பண்பாடு	FC	5	0	0	3	5
2	PTSC25102	தொழில்நுட்ப எழுத்துத்திறன்	PCC	5	0	0	3	5
3	PTSC25103	நூலிழைக் கலைகள் (Weaving Arts)	PCC	5	1	0	5	6
4	PTSC25104	தொழில்முனைவோர் வளர்ச்சி (Entrepreneurial Development)	PCC	4	1	0	4	5
5	PTSC25105	சுற்றுச்சூழல் மேலாண்மை மற்றும் தீர்வுகள்	PCC	5	0	0	4	5
6	PTSEC25106	திறந்தேடல் மற்றும் சாதனை வளமாதிரி (Skill Enhancement - Practical)	EC	4	0	0	4	4
TOTAL				28	2	0	22	30

SEMESTER II

S.N O.	COURSE CODE	COURSE TITLE	CAT EG OR Y	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
THEORY								
1	PTSF25201	அறம் மற்றும் சமுதாய ஒழுக்கம்	FC	5	0	0	3	5
2	PTSC25202	மின்னணு ஆவண மேலாண்மை	PCC	5	0	0	3	5
3	PTSC25203	தொடர்புக்கலைகள் (Communication Arts)	PCC	5	1	0	5	6
4	PTSC25204	கைவினைக் கலைகள் (Handicraft Design)	PCC	4	1	0	4	5
5	PTSC25205	தொழில்சார் திட்ட மேலாண்மை	PCC	5	0	0	4	5
6	PTSEC25206	கிராமம் மற்றும் நகரம் – தொழில்சார் கலைகள்	EC	4	0	0	4	4
TOTAL				28	2	0	22	30

SEMESTER III

S.N O.	COURSE CODE	COURSE TITLE	CAT EG OR Y	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	

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THEORY								
1	PTSF25301	சர்வதேச சந்தை உற்பத்தி அறிமுகம்	PCC	5	0	0	3	5
2	PTSC25302	இலக்கியத்திற்கான தொழில்நுட்ப அணுகுமுறை	PCC	5	0	0	3	5
3	PTSC25303	தொழில்முனைவு உற்பத்தி பயிற்சி (Industrial Training)	PCC	5	1	0	5	6
4	PTSC25304	மின் கலைகள் (Digital Arts)	PCC	4	1	0	4	5
5	PTSC25305	மனித வள மேலாண்மை (HR Management)	EC	5	0	0	4	5
PRACTICAL / EXTENSION ACTIVITIES								
6	PTSEC25306	விரிவான ஆய்வு - பகுதி I (Research Project - Part I)	EDC	4	0	0	4	4
TOTAL				28	2	0	22	30

SEMESTER IV

S.N O.	COURSE CODE	COURSE TITLE	CAT EG OR Y	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
THEORY								
1	PTSF25401	தொழில்சார் சந்தை அறிவியல்	PCC	4	1	0	3	5
2	PTSC25402	தொழில்சார் தகவல் தொடர்பு மேலாண்மை	PCC	5	1	0	3	6
3	PTSC25403	சமூக ஊடகங்களுக்கான கலை வடிவமைப்பு	PCC	5	1	0	3	6
PRACTICAL / EXTENSION ACTIVITIES								
4	PTSC25404	FIELD WORK	EDC	5	1	0	5	6
5	PTSC25405	விரிவான ஆய்வு - பகுதி II (Research Project - Part II)	EDC	5	2	0	6	7
TOTAL				24	6	0	20	30

FOUNDATION COURSES (FC)

S.N O.	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	PTSF25101	தமிழக கலாச்சாரம் மற்றும் பண்பாடு	FC	5	0	0	3	5
2	PTSF25201	அறம் மற்றும் சமுதாய ஒழுக்கம்	FC	5	0	0	3	5
TOTAL CREDITS							6	10

PRINCIPAL CORE COURSES (PCC)

S.N O.	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	PTSC25102	தொழில்நுட்ப எழுத்துத்திறன்	PCC	5	0	0	3	5
2	PTSC25103	நூலிழைக் கலைகள் (Weaving Arts)	PCC	5	1	0	5	6
3	PTSC25104	தொழில்முனைவோர் வளர்ச்சி (Entrepreneurial Development)	PCC	4	1	0	4	5
4	PTSC25105	சுற்றுச்சூழல் மேலாண்மை மற்றும் தீர்வுகள்	PCC	5	0	0	4	5
5	PTSC25202	மின்னணு ஆவண மேலாண்மை	PCC	5	0	0	3	5
6	PTSC25203	சம்பக்கலைகள் (Communication Arts)	PCC	5	1	0	5	6
7	PTSC25204	கைவினைக் கலைகள் (Handicraft Design)	PCC	4	1	0	4	5
8	PTSC25205	தொழில்சார் திட்ட மேலாண்மை	PCC	5	0	0	4	5
9	PTSF25301	சர்வதேச சந்தை உற்பத்தி அறிமுகம்	PCC	5	0	0	3	5
10	PTSC25302	இலக்கியத்திற்கான தொழில்நுட்ப அணுகுமுறை	PCC	5	0	0	3	5
11	PTSC25303	தொழில்முனைவு உற்பத்தி பயிற்சி (Industrial Training)	PCC	5	1	0	5	6
12	PTSC25304	மின் கலைகள் (Digital Arts)	PCC	4	1	0	4	5
13	PTSF25401	தொழில்சார் சந்தை அறிவியல்	PCC	4	1	0	3	5

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14	PTSC25402	தொழில்சார் தகவல் தொடர்பு மேலாண்மை	PCC	5	1	0	3	6
15	PTSC25403	சமூக ஊடகங்களுக்கான கலை வடிவமைப்பு	PCC	5	1	0	3	6
வுழுவயுடு ஊசுநுனுஐவுளு								65

LIST OF ELECTIVE COURSES (EC)
SEMESTER I, ELECTIVE I

S.N O.	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	PTSEC251 06	திறன்தேடல் மற்றும் சாதனை வளமாதிரி (Skill Enhancement - Practical)	EC	4	0	0	4	4
2	PTSEC252 06	கிராமம் மற்றும் நகரம் – தொழில்சார் கலைகள்	EC	4	0	0	4	4
3	PTSC25305	மனித வள மேலாண்மை (HR Management)	EC	5	0	0	4	5
TOTAL CREDITS								22

EMPLOYMENT DEVELOPMENT COURSES (EDC)

S.NO .	COURSE CODE	COURSE TITLE	CATE GORY	PERIODS PER WEEK				TOTAL PERIOD S
				L	T	P	C	
1	PTSEC25 306	விரிவான ஆய்வு - பகுதி I (Research Project - Part I)	EDC	4	0	0	4	4
2	PTSC254 04	FIELD WORK	EDC	5	1	0	5	6
3	PTSC254 05	விரிவான ஆய்வு - பகுதி II (Research Project - Part II)	EDC	5	2	0	6	7
TOTAL CREDITS								11

முதல் ஆண்டு

முதல் பருவம்

PTSF25101

தாள் - 1 தமிழகக் கலாச்சாரம் மற்றும் பண்பாடு

L T P C

5 0 0 3

CO1 தமிழர் கலாச்சாரத்தின் பின்புலம் மற்றும் தொன்மையை விளக்குதல்.

CO2 தமிழர் பாரம்பரிய நடனம் மற்றும் இசைக் கலைகளைப் புரிந்துகொள்வது.

CO3 தமிழர் கட்டிடக்கலை மற்றும் அதன் தொழில்திறன்களை ஆராய்தல்.

CO4 பாரம்பரிய தமிழ் கைவினை கலைகளை புரிந்து தொழில் வாய்ப்புகளை காணுதல்.

CO5 தமிழர் சமூக ஒழுக்கம் மற்றும் பண்பாட்டு மரபுகளை அனுசரித்தல்.

அலகு 1: தமிழர் தொன்மை மற்றும் வரலாறு

10

அலகு 2: தமிழ் நடனம் மற்றும் இசை

10

அலகு 3: கட்டிடக்கலை

10

அலகு 4: தமிழ் கைவினைக் கலைகள்

10

அலகு 5: பண்பாட்டு மரபுகள்

10

Total Periods: 50

பாடநூல்கள்:

க. ராசமணி – தமிழர் பண்பாட்டுச் சாரம், கம்பன் பத்திரிகைகள்

ஆ. சண்முகம் – தமிழகத்தின் கலாச்சாரம் மற்றும் வரலாறு, தமிழ் புத்தகபதி

பார்வை நூல்கள்:

மு. வரதராசன் – தமிழ் இலக்கிய வரலாறு (முதல் பாகம்)

பதிப்பகம்: கம்பன் பத்திரிகைகள்

இரா. நாகசாமி – தமிழர் பண்பாட்டு வரலாற்றுச் சிற்பங்கள்

பதிப்பகம்: தமிழரசு பதிப்பகம்

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
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COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25102

தாள் - 2 தொழில்நுட்ப எழுத்துத்திறன்

L T P C

5 0 0 5

- CO1 தொழில்நுட்பம் சார்ந்த உரை வடிவங்களை அறிதல்.
- CO2 தொழில்நுட்பத்திற்கேற்ப நேர்மையான உரை தயாரித்தல்.
- CO3 தொழில்நுட்ப ஆவணங்களை தரவுகளின் அடிப்படையில் உருவாக்குதல்.
- CO4 துல்லியமான ஆய்வறிக்கைகள் உருவாக்கும் திறனை வளர்த்தல்.
- CO5 தொழில்முனைவோர் வளர்ச்சிக்கான தகுந்த சொற்பொழிவுகளை தயாரித்தல்.

அலகு 1: தொழில்நுட்ப எழுத்தின் அடிப்படைகள்	10
அலகு 2: எளிய உரையாடல் தயாரித்தல்	10
அலகு 3: தொழில்நுட்ப ஆவண தயாரித்தல்	10
அலகு 4: ஆய்வறிக்கைகள் எழுதுதல்	10
அலகு 5: தொழில்முனைவர் எண்ணங்களை பதிவு செய்யும் திறன்	10

Total Periods: 50

பாடநூல்கள்:

கல்கி நரசிம்மன் – தொழில்நுட்பத் தமிழ்

பதிப்பகம்: தமிழ் பதிப்பகம்

எஸ். ஆறுமுகம் – தமிழ் எழுத்துத்திறன் முறைமைகள்

பதிப்பகம்: வடிவமைப்பு பதிப்பகம்

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பார்வை நூல்கள்:

பாரதி தாசன் – இலக்கியத் திறனாய்வுச் சிந்தனைகள்
பதிப்பகம்: சங்கதிப்பு பதிப்பகம்
டேவிட் எபனேசர் – தொழில்நுட்பத் தமிழ் பயன்பாடுகள்
பதிப்பகம்: நாராயணா பதிப்பகம்

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25103
P C

நூலிழைக் கலைகள் (Weaving Arts)

L T

5 0 0 5

CO1: நூலிழை கலைகளின் வரலாற்றை ஆராய்தல்.

CO2: நூலிழை வடிவமைப்பு செயல்முறைகளை புரிந்துகொள்வது.

CO3: நவீன தொழில்நுட்பங்களை நூலிழையில் பயன்படுத்துதல்.

CO4: கையால் நூல் உற்பத்தி முறைகளின் முக்கியத்துவம் அறிதல்.

CO5: மின்னணு கருவிகளை பயன்படுத்தி நூல் வடிவமைப்பு செய்வது.

அலகு 1: நூலிழையின் வரலாறு

10

அலகு 2 : நூலிழை வடிவமைப்பு

10

அலகு 3: நவீன நூல் தொழில்நுட்பம்

10

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அலகு 4: கையால் நூல் உருவாக்குதல்

10

அலகு 5: மின்னணு நூலிழை

10

Total Periods: 50

பாடநூல்கள்:

மு. அய்யனார் – தமிழக நெசவாளர் வரலாறு

பதிப்பகம்: தமிழ்த்தேச பத்திரிகைகள்

பெரியசாமி திருநாவுக்கரசு – தமிழகக் கைவினைக் கலைகள்

பதிப்பகம்: கம்பன் பதிப்பகம்

பார்வை நூல்கள்:

க. முத்துசாமி – நூலிழைக் கலை மற்றும் பாரம்பரிய வளர்ச்சி

பதிப்பகம்: தமிழ் பதிப்பகம்

Dr. K.R. Nandhini – Textile Design and Weaving Techniques (தமிழில் மொழிபெயர்ப்பு)

பதிப்பகம்: வடிவமைப்பு பதிப்பகம்

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25104

தொழில்முனைவோர் வளர்ச்சி

L T P C

5 0 0 5

CO1: தொழில்முனைவரின் அடிப்படைத் திறன்களை அடைவது.

CO2: சந்தையின் தேவைகளை ஆய்வு செய்யும் திறன்களைப் பெறுதல்.

CO3: தொழில்நுட்ப நிதி மேலாண்மை முறைகளை அறிதல்.

CO4: புது தொழில் தொடங்கத் தேவையான திட்டங்களை வடிவமைத்தல்.

CO5: தொழில்முனைவர் செயல்திறனை மேம்படுத்துதல்.

அலகு 1: தொழில்முனைவரின் அடிப்படைகள்	10
அலகு 2: சந்தை ஆய்வு	10
அலகு 3: நிதி மேலாண்மை	10
அலகு 4: தொழில்முனைவர் திட்டம் தயாரித்தல்	10
அலகு 5: செயல்திறனில் வளர்ச்சி	10

Total Periods: 50

பாடநூல்கள்:

வசந்த தேசாய் – தொழில்முனைவு வளர்ச்சி மற்றும் மேலாண்மை
பதிப்பகம்: வார்த்தை பதிப்பகம்
எஸ். கோபாலகிருஷ்ணன் – தமிழகத் தொழில்முனைப்பு
பதிப்பகம்: கம்பன் பதிப்பகம்

பார்வை நூல்கள்:

சி.பி. குப்தா – தொழில்முனைப்பு வளர்ச்சி: தமிழ் அறிமுகம்
பதிப்பகம்: தமிழ்நாடு பதிப்பகம்
பி. கணேஷ் – தொழில்நுட்பமும் தொழில்முனைவர்களும்
பதிப்பகம்: தமிழர் பதிப்பகம்

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25105 திறன்தேடல் மற்றும் சாதனை வளமாதிரி (Skill Enhancement - Practical) L T P C

5 0 0 5

- CO1: தனிநபர் வளர்ச்சிக்கான அடிப்படைகளை நிர்ணயித்தல்.
 CO2: குழு செயல்பாடுகளில் ஒத்துழைப்பு ஆற்றல் வளர்த்தல்.
 CO3: தொழில்நுட்ப சாதனங்களைப் பயன்படுத்தி வேலையைச் செய்வது.
 CO4: புதிய வேலை வாய்ப்புகளை அடையத் தகுந்த திறனை வளர்த்தல்.
 CO5: தொழில்நுட்பத்துடன் கூடிய கலைகளை அறிதல்.

அலகு 1: தனிநபர் திறன்கள்	10
அலகு 2: குழு வேலைப்பாடுகள்	10
அலகு 3: தொழில்நுட்பத்தைப் பயன்படுத்துதல்	10
அலகு 4: வேலைதிறனை மேம்படுத்துதல்	10
அலகு 5: தொழில்நுட்ப கலையை கற்றல்	10

Total Periods: 50

பாடநூல்கள்:

மெய்யப்பன் ராஜா – திறன்தேடல் மற்றும் ஆளுமை மேம்பாடு
 பதிப்பகம்: தமிழ் வலை பதிப்பகம்
 மைதிலி ராமகிருஷ்ணன் – தொழில்நுட்பத் திறன்கள்
 பதிப்பகம்: தமிழ்நாடு பதிப்பகம்

பார்வை நூல்கள்:

ஸ்டீபன் கோவி – சிறந்த நெறிமுறைகள் (தமிழில் மொழிபெயர்ப்பு)
 பதிப்பகம்: மாபெரும் பதிப்பகம்
 டேல் கார்னெகி – மனித உறவுகள் மற்றும் வெற்றித் திறன் (தமிழில்)
 பதிப்பகம்: தமிழர் பதிப்பகம்

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2

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COS 5:	3	2	2	3	3
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விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

இரண்டாம் பருவம்

PTSF25201

அறம் மற்றும் சமுதாய ஒழுக்கம்

L T P C

5 0 0 3

CO1: அறத்தின் தத்துவ அடிப்படைகளையும் அதன் சமூக தேவையையும் விளக்குதல்.

CO2: ஒழுக்கநெறிகள் சமூகத்தின் மேம்பாட்டில் உறுதியான பங்கு வகிக்கின்றன என்பதை ஆராய்தல்.

CO3: தமிழ் இலக்கியங்களில் அறத்தின் வடிவங்களைப் புரிந்துகொள்வது.

CO4: தொழில்நுட்பம் மற்றும் அறம் இணைந்த சமூகத்தைக் கட்டமைத்தல்.

CO5: பாரம்பரிய சமூக ஒழுக்கங்களின் தேவையை அறிதல்.

அலகு 1: அறத்தின் அடிப்படைகள்	10
அலகு 2: சமுதாய ஒழுக்கநெறிகள்	10
அலகு 3: தமிழ் இலக்கியத்தில் அறம்	10
அலகு 4: அறத்தின் உற்பத்தி நெறிகள்	10
அலகு 5: சமுதாய பரம்பரைகள்	10

Total Periods: 50

பாடநூல்கள்:

திருக்குறள் – அறத்துப்பால் (பாரிமேலழகர் உரை), தமிழ் பதிப்பகம்

ஸ்வாமி சித்பவானந்தா – திருக்குறள் விளக்கம் (அறம் பாகம்), தமிழ்நாடு பதிப்பகம்

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பார்வை நூல்கள்:

நன்னூல் – ஒழுக்கநெறிகள் மற்றும் அறவழிகள், சுதாரிகா பதிப்பகம்

எம்.ஜி. கோவிந்தன் – சமுதாய ஒழுக்கம் மற்றும் ஒழுக்கநெறி, தமிழ் வலை பதிப்பகம்

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25202

மின்னணு ஆவண மேலாண்மை

L T P C

5 0 0 3

மயின் அடிப்படைத் தொழில்நுட்பங்களை கற்றல்.

CO2 ஆவணங்கள் மற்றும் தரவுகளின் பாதுகாப்பை உறுதிப்படுத்துதல்.

CO3 ஆவணங்களை கட்டமைக்க ஆவண அடுக்குமுறையை பயன்படுத்துதல்.

CO4 மின்னணு சேமிப்பு முறைகளை ஆய்வு செய்வது.

CO5 ஆவண மேலாண்மையில் உள்ள சிக்கல்களுக்கான தீர்வுகளை வழங்குதல்.

அலகு 1: ஆவண மேலாண்மையின் அடிப்படைகள்	10
அலகு 2: மின்னணு தகவல் பாதுகாப்பு	10
அலகு 3: ஆவண அடுக்குமுறை	10
அலகு 4: மின்னணு சேமிப்பு தொழில்நுட்பங்கள்	10
அலகு 5: ஆவண மேலாண்மை சிக்கல்களை தீர்ப்பது	10

Total Periods: 50

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COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25203

தொடர்பு கலைகள் (Communication Arts)

L

T P C

5 0 0 3

CO1 தகவல்தொடர்பின் அடிப்படைகளை விளக்குதல்.

CO2 வார்த்தைத் தேர்வு மற்றும் எழுத்துப் பிரதி திறன்களை மேம்படுத்துதல்.

CO3 மின்னஞ்சல் மற்றும் சமூக ஊடக தகவல்தொடர்பு முறைகளின் பயன்பாட்டை அறிதல்.

CO4 தொழில்சார் தகவல்தொடர்புக்கான சிறந்த நடைமுறைகளை கற்றல்.

CO5 தகவல்தொடர்பின் திறனை தொழில்துறையின் தேவைகளுக்கு ஏற்ப மேம்படுத்துதல்.

அலகு 1: தகவல்தொடர்பு முறைமைகள் 10

அலகு 2: வார்த்தை மற்றும் எழுத்துத் திறன் 10

அலகு 3: மின்னணு தகவல்தொடர்பு 10

அலகு 4: தொழில்துறையின் தகவல்தொடர்பு முறைகள் 10

அலகு 5: தகவல்தொடர்பில் எதிர்பார்ப்புகள் 10

Total Periods: 50

பாடநூல்கள்:

பி. இளங்கோவன் – தகவல் தொடர்பு துறைமைகள், கம்பன் பதிப்பகம்

கே. கிருஷ்ணமூர்த்தி – தொழில்நுட்பத் தகவல்தொடர்பு முறைமைகள், தமிழரசு பதிப்பகம்

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பார்வை நூல்கள்:

அரங்கநாதன் – தமிழக தகவல்தொடர்பு வரலாறு, தமிழ் புத்தகபதி

லெஸிகார் மற்றும் .:பிளாட்லி – தொழில்துறை தகவல் தொடர்பு (தமிழில் மொழிபெயர்ப்பு), சுதாரிகா பதிப்பகம்

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25204

கைவினைக் கலைகள் (Handicraft Design)

L T P C

5 0 0 3

CO1: கைவினைக் கலைகளின் துவக்கத்தின் வரலாற்றையும் அதன் வளர்ச்சியையும் அறிதல்.

CO2: கைவினைக் கலைகளுக்கான நவீன தொழில்நுட்பங்களை உள்வாங்குதல்.

CO3: பாரம்பரிய கைவினைகளையும் நவீன கைவினைகளையும் ஒப்பிடுதல்.

CO4: சந்தையை முன்னிட்டு கைவினை பொருட்களை வடிவமைத்தல்.

CO5: கைவினைகளின் மூலம் தொழில்முனைவின் வாய்ப்புகளை ஆராய்தல்.

அலகு 1: கைவினைக் கலைகளின் வரலாறு 10

அலகு 2: கைவினை வடிவமைப்பு தொழில்நுட்பங்கள் 10

அலகு 3: பாரம்பரிய மற்றும் நவீன கைவினை 10

அலகு 4: விற்பனைக்கான கைவினை தயாரிப்பு 10

அலகு 5: கைவினை மற்றும் தொழில்முனைவு 10

Total Periods: 50

பாடநூல்கள்:

பழனி செல்வன் – தமிழகக் கைவினைகள், தமிழ் பதிப்பகம்

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சுந்தரபாண்டியன் – கைவினைக் கலைகளின் வரலாறு, வணிக பத்திரிகைகள்

பார்வை நூல்கள்:

எஸ். கே. ஜெயின் – கைவினைக் கலைகள் மற்றும் பாரம்பரியம், தமிழரசு பதிப்பகம்
ராணி குமாரி – தமிழ் நாட்டின் கைவினைக் கலைகள் (தமிழ் மொழிபெயர்ப்பு), தமிழர் பதிப்பகம்

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25205

தொழில்சார் திட்ட மேலாண்மை

L

T P C

5 0 0 3

CO1: திட்ட மேலாண்மையின் அடிப்படைகளை விளக்குதல்.

CO2: திட்டத்திற்கு முன் செய்ய வேண்டிய ஆய்வுகளை அமைத்தல்.

CO3: திட்டத்தை செயல்படுத்த தொழில்நுட்ப முறைமைகளை அறிதல்.

CO4: திட்ட வெற்றியை அளவீடு செய்யும் முறைமைகளை கற்றல்.

CO5: திட்ட மேலாண்மையில் புதுமைகளை மேற்கொள்ளுதல்.

அலகு 1: திட்ட மேலாண்மையின் அடிப்படைகள்

10

அலகு 2: திட்டத்திற்கான முன்னோட்ட ஆய்வு

10

அலகு 3: திட்ட செயலாக்கம்

10

அலகு 4: திட்டத்தின் அளவீடு

10

அலகு 5: மேம்படுத்தப்பட்ட திட்டங்கள்

10

Total Periods: 50

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பாடநூல்கள்:

தொழில்துறை மேலாண்மை, ஆர். ராமசாமி, சுலோகா பப்ளிகேஷன்ஸ்இ சென்னை
திட்ட மேலாண்மை முறைகள், பி. சந்திரசேகரன், விகாஸ் பப்ளிகேஷன்ஸ்
திட்ட வளர்ச்சி மற்றும் நிர்வாகம், என். சுப்ரமணியம், நியூ செஞ்சுரி புக் ஹவுஸ்

பார்வை நூல்கள்:

தொழில்முறை மேலாண்மை, கந்தசாமி, எவ்ரேஸ்ட் பப்ளிகேஷன்ஸ்இ கோயம்புத்தூர்
திட்ட செயல்திறன் மதிப்பீடு, ஜி. முத்துவேல், தமிழ்நாடு படிக்காசு பதிப்பகம்

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25206

கிராமம் மற்றும் நகரம் – தொழில்சார் கலைகள்

L T P C

5 0 0 3

அலகு 1: கிராமத்து தொழில்நுட்பங்கள்

10

அலகு 2: நகரத்து கலை மற்றும் தொழில்நுட்ப வளர்ச்சி

10

அலகு 3: நகரப் பயன்பாடுகள் மற்றும் பரிபாலன வணிகம்

10

அலகு 4: கிராம தொழில்கள் மற்றும் ஊராட்சி உற்பத்தி

10

அலகு 5: சமூகமானது கிராமத் தொழில்கள்

10

Total Periods: 50

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
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COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

இரண்டாம் ஆண்டு

மூன்றாம் பருவம்

PTSF25301 சர்வதேச சந்தை உற்பத்தி அறிமுகம் (Introduction to Global Market Production) L T P C

5 0 0 3

CO1: சர்வதேச சந்தை அமைப்பு மற்றும் அதன் அடிப்படைகளை விளக்குதல்.

CO2: சர்வதேச சந்தையில் ஏற்றுமதி மற்றும் இறக்குமதியின் முக்கியத்துவத்தை புரிந்துகொள்வது.

CO3: வாடிக்கையாளர் தேவைகளைத் தீர்க்கும் தொழில்முறை அணுகுமுறையை கற்றல்.

CO4: பன்னாட்டு வணிகத்தின் மேல் இந்திய பொருளாதாரத்தின் தாக்கத்தை ஆராய்தல்.

CO5: உலக சந்தையில் உற்பத்தி திறனை மேம்படுத்துதல்.

அலகு 1: சர்வதேச சந்தை அடிப்படைகள் 10

அலகு 2: ஏற்றுமதி மற்றும் இறக்குமதி முறைமைகள் 10

அலகு 3: வாடிக்கையாளர் சேவை மேலாண்மை 10

அலகு 4: பன்னாட்டு வணிகம் மற்றும் பொருளாதார ஆதிக்கம் 10

அலகு 5: உலக சந்தை உற்பத்தி நடைமுறைகள் 10

Total Periods: 50

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THIRUVANNAMALAI

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25302

இலக்கியத்திற்கான தொழில்நுட்ப அணுகுமுறை

L T P C

5 0 0 3

CO1: இலக்கிய நவீன படைப்புகளில் தொழில்நுட்பத்தின் தாக்கத்தை அறிதல்.

CO2: மின்னணு வடிவங்களில் இலக்கியங்களை அடையாளம் காணுதல்.

CO3: இலக்கிய அனுமதிக்கான தொழில்நுட்ப வழிகளை விளக்குதல்.

CO4: மொழிபெயர்ப்பு கருவிகளை இலக்கியத்துடன் இணைத்தல்.

CO5: தொழில்நுட்பத்தினை பயன்படுத்தி இலக்கிய ஆய்வுகளை மேற்கொள்வது.

அலகு 1: இலக்கியத்தில் தொழில்நுட்பம் 10

அலகு 2: மின்னணு இலக்கிய அறிமுகம் 10

அலகு 3: இலக்கிய அனுமதி குறியீடுகள் 10

அலகு 4: இலக்கிய மொழிபெயர்ப்பு தொழில்நுட்பம் 10

அலகு 5: இலக்கிய ஆய்வுக்கான கலைவழிகள் 10

Total Periods: 50

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COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25303

தொழில்முனைவு உற்பத்தி பயிற்சி (Industrial Training)

L T P C

5 0 0 3

CO1: தொழில்முனைவு உற்பத்தி முறைகளை அறிதல்.

CO2: உற்பத்தி திட்டங்களை வடிவமைக்கும் திறனை மேம்படுத்துதல்.

CO3: தொழில்துறையில் செயல்படும் நேரடி குழல்களுடன் பழகுவதல்.

CO4: தொழில்நுட்ப கருவிகளை உற்பத்தியில் பயன்படுத்துதல்.

CO5: தொழில்முனைவு பயிற்சியின் மூலம் தொழில் நுட்பங்களை முறைப்படுத்துதல்.

அலகு 1: தொழில்முனைவு உற்பத்தி அடிப்படைகள்

10

அலகு 2: உற்பத்தி திட்ட கட்டமைப்பு

10

அலகு 3: தொழில்துறை உற்பத்தி குழல்கள்

10

அலகு 4: தொழில்நுட்ப உற்பத்தி பணிகள்

10

அலகு 5: தொழில்துறையுடன் ஒருங்கிணைந்து செயல்படுதல்

10

Total Periods: 50

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COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25304

மின் கலைகள் (Digital Arts)

L T P C

5 0 0 3

ஊழி1: மின்னணு கலைகளின் துவக்கம் மற்றும் வளர்ச்சி பற்றிய அடிப்படைகளை அறிதல்.

ஊழி2: நவீன டிஜிட்டல் கருவிகளை கற்றல் மற்றும் பயன்படுத்தல்.

ஊழி3: டிஜிட்டல் கலை வடிவமைப்பில் பயன்படுத்தப்படும் மென்பொருட்களை ஆராய்தல்.

ஊழி4: கலை வடிவமைப்பிற்கான அசைவூட்டல் தொழில்நுட்பத்தை கற்றல்.

ஊழி5: மின்னணு கலைகளை தொழில்முறை பயன்பாட்டிற்கு தகுந்தவாறு உருவாக்குதல்

அலகு 1: மின்னணு கலைகளின் வரலாறு 10

அலகு 2: டிஜிட்டல் வடிவமைப்பு கருவிகள் 10

அலகு 3: மின்னணு கலைகளுக்கான மென்பொருட்கள் 10

அலகு 4: அசைவூட்டல் மற்றும் கிராபிக்ஸ் 10

அலகு 5: தொழில்சார் மின்னணு கலைகள் 10

Total Periods: 50

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COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25305

மனித வள மேலாண்மை (HR Management)

L T P C

5 0 0 3

அலகு 1: மனித வள அடிப்படைகள்

10

அலகு 2: ஊழியர் தொடர்புகள்

10

அலகு 3: உந்துதல் மற்றும் மதிப்பீடு

10

அலகு 4: வேலைகளின் மாறுபாடு

10

அலகு 5: திறமையான மனித வளம் உருவாக்கம்

10

Total Periods: 50

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

ARUNAI UNIVERSITY
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அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

நான்காம் பருவம்

PTSF253401

தொழில்சார் சந்தை அறிவியல் (Market Science for Professionals)

L T P C

5 0 0 3

CO1: சந்தை ஆராய்ச்சிக்கான தொழில்நுட்ப கருவிகளை அறிதல்.

CO2: வாடிக்கையாளர் தேவைகளை அடையாளம் காணுதல்.

CO3: சந்தையின் புள்ளிவிவரங்களை கணிக்கல் மற்றும் மதிப்பீடு செய்தல்.

CO4: மாற்றிய கலந்தாய்வு முறைகளைச் செயல்படுத்துதல்.

CO5: புது சந்தை உருவாக்க முறைமைகளை கற்றல்.

அலகு 1: சந்தை ஆராய்ச்சி தொழில்நுட்பம்

10

அலகு 2: வாடிக்கையாளர் பார்வைகள்

10

அலகு 3: சந்தை புள்ளிவிவரங்கள்

10

அலகு 4: சுழல்நிலை விற்பனை நுட்பங்கள்

10

அலகு 5: புது சந்தை உருவாக்கம்

10

Total Periods: 50

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

ARUNAI UNIVERSITY
THIRUVANNAMALAI

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF253401

தொழில்சார் தகவல் தொடர்பு மேலாண்மை

L T P C

5 0 0 3

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

PTSF25204

சுற்றுச்சூழல் மேலாண்மை மற்றும் தீர்வுகள்

L T P C

5 0 0 3

அலகு 1: சுற்றுச்சூழல் மேலாண்மையின் அடிப்படைகள்	10
அலகு 2: பசுமை தொழில்துறை	10
அலகு 3: புவிசார் வளங்கள் பயன்பாடு	10
அலகு 4: சுற்றுச்சூழலுக்கு பொருத்தமான தொழில்நுட்பங்கள்	10
அலகு 5: பசுமை மற்றும் புதுமை செயல்முறை	10

COS/POS	POS 1	POS 2	POS 3	POS 4	POS 5
COS 1:	3	2	2	2	2
COS 2:	3	2	3	3	3
COS 3:	2	3	3	3	2
COS 4:	3	3	2	3	2
COS 5:	3	2	2	3	3

விளக்கம்:

அளவீட்டு அளவுகள்:

3 (High), 2 (Medium), 1 (Low), "" (No Mapping).

விரிவாக்க செயல்பாடுகள்

ஆய்வேடு

ஆய்வேடு பணிக்க மேற்கொள்ள வேண்டிய நெறிமுறைகள்:

- ❖ ஆய்வு பற்றிய விளக்கம் அறிதல்
- ❖ ஆய்வுத் தலைப்பு தேர்வு செய்தல்
- ❖ முதன்மைத் துணை ஆதாரங்களைத் திரட்டுதல்.
- ❖ ஆய்வு நெறிமுறைகளை அறிந்து ஆய்வேடு எழுதுதல்.
- ❖ மேற்கோள்களைத் தேர்ந்தெடுத்தல்,
- ❖ ஆய்வுப் பயனை வெளிப்படுத்துதல்,
- ❖ களஆய்வினை மேற்கொள்ளுதல்
- ❖ துணை நூற்பட்டியல் தயாரித்தல்.

B.COM – Financial Risk Management and Taxation

PHILOSOPHICAL TOPICS:

1. Ethics and Morality

- The Moral Obligation of Auditors: Is it morally justifiable for auditors to prioritize shareholder interests over those of other stakeholders?
- Tax Evasion vs. Tax Avoidance: Is tax evasion morally equivalent to tax avoidance? What are the implications for taxation policy?

2. Theoretical Foundations

- The Social Contract Theory of Accounting: How does the social contract theory inform our understanding of accounting and its role in society?
- The Stakeholder Theory of Accounting: Who are the stakeholders in accounting, and how do their interests shape accounting practices?

3. Taxation and Social Justice

- The Ethics of Taxation: Is taxation morally justifiable? What are the implications for social justice?
- Taxation and Income Inequality: How does taxation impact income inequality? What are the moral implications of taxation policy?

4. Auditing and Accountability

- The Ethics of Auditing: Is auditing morally justifiable? What are the implications for accountability?
- Auditing and Corporate Governance: How does auditing impact corporate governance? What are the moral implications of auditing practices?

5. Financial Reporting and Transparency

- The Ethics of Financial Reporting: Is financial reporting morally justifiable? What are the implications for transparency?
- Financial Reporting and Stakeholder Needs: How does financial reporting impact stakeholder needs? What are the moral implications of financial reporting practices?

SIGNIFICANCE OF AUDIT:

- ❖ Ensures Accuracy and Reliability: Auditing ensures that financial statements are accurate, reliable, and free from material misstatements.
- ❖ Provides Assurance: Auditing provides stakeholders with assurance that financial statements are presented fairly and in accordance with accounting standards.

- ❖ **Helps in Decision-Making:** Audited financial statements help stakeholders make informed decisions about investments, lending, and other business activities.
- ❖ **Promotes Transparency and Accountability:** Auditing promotes transparency and accountability in financial reporting, which helps to build trust and confidence in the organization.
- ❖ **Identifies and Mitigates Risks:** Auditing helps to identify and mitigate risks associated with financial reporting, such as fraud, error, and non-compliance.

SIGNIFICANCE OF TAXATION:

- ❖ **Revenue Generation:** Taxation is a significant source of revenue for governments, which is used to fund public goods and services.
- ❖ **Redistribution of Wealth:** Taxation helps to redistribute wealth from the rich to the poor, promoting social justice and equality.
- ❖ **Economic Stabilization:** Taxation helps to stabilize the economy by influencing aggregate demand and supply.
- ❖ **Encourages Compliance:** Taxation encourages individuals and businesses to comply with tax laws and regulations, promoting a culture of tax compliance.
- ❖ **Fosters Economic Growth:** Taxation can foster economic growth by providing incentives for investment, innovation, and entrepreneurship.

Interrelationship between Audit and Taxation:

- ❖ **Tax Compliance:** Auditing helps to ensure tax compliance, which is critical for taxation revenue generation.
- ❖ **Financial Reporting:** Taxation relies on accurate financial reporting, which is ensured through auditing.
- ❖ **Risk Management:** Auditing helps to identify and mitigate risks associated with taxation, such as tax evasion and avoidance.
- ❖ **Transparency and Accountability:** Auditing promotes transparency and accountability in financial reporting, which is essential for taxation.

Course Structure

Course Title: Audit and Taxation

Course Description: This undergraduate program in Audit and Taxation is designed to equip students with a comprehensive understanding of auditing and taxation principles, practices, and regulations. The program aims to develop professionals who can provide expert advice on taxation and auditing matters, and who can help organizations maintain transparency, accountability, and compliance with regulatory requirements.

Course Objectives:

1. To develop a comprehensive understanding of auditing and taxation principles, practices, and regulations.
2. To equip students with the skills and knowledge required to provide expert advice on taxation and auditing matters.
3. To develop professionals who can help organizations maintain transparency, accountability, and compliance with regulatory requirements.
4. To prepare students for professional certifications in auditing and taxation, such as CA, CMA, CIA, and CTP.

Subjects / Papers to be included:

Paper 1: Financial Accounting

Paper 2: Principals of Auditing

Paper 3: Business Law

Paper 4: Fundamentals of Commerce

Paper 5: Auditing Theory and Practice

Paper 6: Financial Management

Paper 7: Goods and service Tax

Paper 8: Business Communication

Paper 9: Corporate Accounting

Paper 10: Management Accounting

Paper 11: Business Statistics - I

Paper 12: Auditing Standard and Procedure

Paper 13: EVS

Paper 14: International Finance

Paper 15: Advanced Taxation

Paper 16: Business Statistics – II

Paper 17: Advanced Auditing

Paper 18: Value Education

Paper 19: Cost Accounting

Paper 20: Income Tax Law

Paper 21: Auditing and Assurance

Paper 22: Entrepreneurial Development

Paper 23: International Financial Reporting Standard (IFRS)

Paper 24: Corporate Governance and Ethics

Paper 25: Financial Analysis and Interpretation

Paper 26: Advanced Financial Management

Paper 27: Taxation of Business Income

Paper 28: Auditing and Financial Reporting

Paper 29: Group / Individual Project

Paper 30: Tally & Enterprise resource planning

Pedagogy of the course selected – Assessment methods

1. Lecture Method: Deliver lectures to introduce key concepts, principles, and practices in auditing and taxation.
2. Case Study Method: Use real-life case studies to illustrate auditing and taxation concepts and practices.
3. Group Discussions: Encourage group discussions to develop critical thinking, problem-solving, and communication skills.
4. Practical Assignments: Provide practical assignments to develop hands-on skills in auditing and taxation.
5. Guest Lectures: Invite industry experts to deliver guest lectures and share their experiences.
6. Role-Playing: Use role-playing to simulate real-life auditing and taxation scenarios.
7. Online Resources: Utilize online resources, such as videos, podcasts, and online tutorials, to supplement course materials.

Assessment Methods

1. Quizzes and assignments (20%)
2. Mid-term exams (30%)
3. Final exams (30%)
4. Project work and presentations (20%)

BACHELOROF COMMERCE

B.COM - Financial Risk Management and Taxation

DEGREECOURSE

The Course of Study and the Scheme of Examinations

S. No.	Part	Study Components		Ins. Hrs/ week	Credit	Title of the Paper	Maximum Marks		
		Course Title					CIA	Uni. Exam	Total
		SEMESTER I							
1.	I	Language	Paper-1	6	4	Tamil/Other Languages	25	75	100
2.	II	English	Paper-1	6	4	English I	25	75	100
3.	III	Core Theory	Paper-1	6	5	Financial Accounting	25	75	100
4.	III	Core Theory	Paper- 2	5	4	Principles of Auditing	25	75	100
5.	III	Elective I	Paper-1	4	3	Business Law	25	75	100
6.	IV	Skill	Paper-1	3	2	Fundamentals of Commerce	25	75	100
		Sem. Total		30	22		150	450	600
		SEMESTER II					CIA	Uni. Exam	Total
7.	I	Language	Paper-2	6	4	Tamil II/Other Languages	25	75	100
8.	II	English	Paper-2	6	4	English II	25	75	100
9.	III	Core Theory	Paper-3	6	5	Auditing Theory and Practice	25	75	100
10.	III	Core Theory	Paper-4	5	4	Financial management	25	75	100
11.	III	Elective II	Paper-2	4	3	Goods and service Tax	25	75	100
12.	IV	Skill	Paper-2	3	2	Business Communications	25	75	100
		Sem. Total		30	22		150	450	600
		SEMESTER III					CIA	Uni. Exam	Total
13.	I	Language	Paper-3	6	4	Tamil III/Other Languages	25	75	100
14.	II	English	Paper-3	5	4	English III	25	75	100
15.	III	Core Theory	Paper-5	5	4	Corporate Accounting	25	75	100

16.	III	Core Theory	Paper-6	5	4	Management Accounting	25	75	100
17.	III	Elective III	Paper-3	5	3	Business Statistics-I	25	75	100
18.	IV	Skill Based Subject	Paper-3	2	2	Auditing Standard and Procedure	25	75	100
19.	IV	EVS	Paper-1	2	1	EVS	25	75	100
		Sem. Total		30	22		175	525	700
		SEMESTER IV							
20.	I	Language	Paper-4	6	4	Tamil IV/Other Languages	25	75	100
21.	II	English	Paper-4	6	4	English –IV	25	75	100
22.	III	Core Theory	Paper-7	5	4	International Finance	25	75	100
23.	III	Core Theory	Paper-8	4	3	Advanced Taxation	25	75	100
24.	III	Elective IV	Paper-4	4	3	Business Statistics-II	25	75	100
25.	IV	Skill Based Subject	Paper-4	3	2	Advanced Auditing	25	75	100
26.	IV	Value Education	Paper-2	2	2	Value Education	25	75	100
		Sem. Total		30	22		175	525	700
		SEMESTER V							
27.	III	Core Theory	Paper-9	6	5	Cost Accounting	25	75	100
28.	III	Core Theory	Paper-10	6	5	Income Tax Law	25	75	100
29.	III	Core Theory	Paper-11	5	4	Auditing and Assurance	25	75	100
30.	III	Elective IV	Paper-5	4	2	Entrepreneurial Development	25	75	100
31.	III	Elective IV	Paper-6	4	2	International Financial Reporting Standards (IFRS)	25	75	100
32.	III	Internal Elective	Paper-1	3	2	Corporate Governance and Ethics	25	75	100
33.	IV	Skill Based Subject	Paper-5	2	2	Financial Analysis and interpretation	25	75	100
		Sem. Total		30	22		175	525	700
		SEMESTER VI							
34.	III	Core Theory	Paper-12	7	5	Advanced Financial Management	25	75	100
35.	III	Core Theory	Paper-13	7	5	Taxation of Business Income	25	75	100
36.	III	Core Theory	Paper-14	6	4	Auditing and Financial Reporting	25	75	100
37.	III	Compulsory Project	Paper-1	6	5	Group/Individual Project	25	75	100

38.	IV	Skill based Subject	Paper-6	4	3	Tally & Enterprise Resource Planning	25	75	100
		Sem. Total		30	22		125	375	500
					132				3800

FIRST YEAR – SEMESTER – I

CORE PAPER – 1 FINANCIAL ACCOUNTING

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	5				5	6	25	75	100
Learning Objectives									
LO1	To understand the basic accounting concepts and standards.								
LO2	To know the basis for calculating business profits.								
LO3	To familiarize with the accounting treatment of depreciation.								
LO4	To learn the methods of calculating profit for single entry system.								
LO5	To gain knowledge on the accounting treatment of insurance claims.								
Prerequisites: Should have studied Accountancy in XII Std									
Unit	Contents								No. of Hours
I	Fundamentals of Financial Accounting Financial Accounting – Meaning, Definition, Objectives, Basic Accounting Concepts and Conventions - Journal, Ledger Accounts– Subsidiary Books — Trial Balance - Classification of Errors – Rectification of Errors – Preparation of Suspense Account – Bank Reconciliation Statement - Need and Preparation								15
II	Final Accounts Final Accounts of Sole Trading Concern- Capital and Revenue Expenditure and Receipts – Preparation of Trading, Profit and Loss Account and Balance Sheet with Adjustments.								15
III	Depreciation and Bills of Exchange Depreciation - Meaning – Objectives – Accounting Treatments - Types - Straight Line Method – Diminishing Balance method – Conversion method. Units of Production Method – Cost Model vs Revaluation Bills of Exchange – Definition – Specimens – Discounting of Bills – Endorsement of Bill – Collection – Noting – Renewal – Retirement of Bill under rebate								15
IV	Accounting from Incomplete Records – Single Entry System Incomplete Records -Meaning and Features - Limitations - Difference between Incomplete Records and Double Entry System - Methods of Calculation of Profit - Statement of Affairs Method – Preparation of final statements by Conversion method.								15
V	Royalty and Insurance Claims Meaning – Minimum Rent – Short Working – Recoupment of Short Working – Lessor and Lessee – Sublease – Accounting Treatment. Insurance Claims –Calculation of Claim Amount-Average clause (Loss of Stock only)								15
TOTAL								75	
THEORY 20% & PROBLEM 80%									
CO	Course Outcomes								

CO1	Remember the concept of rectification of errors and Bank reconciliation Statements
CO2	Apply the knowledge in preparing detailed accounts of sole trading concerns
CO3	Analyse the various methods of providing depreciation
CO4	Evaluate the methods of calculation of profit
CO5	Determine the royalty accounting treatment and claims from insurance companies in case of loss of stock.
Textbooks	
1.	S. P. Jain and K. L. Narang Financial Accounting- I, Kalyani Publishers, New Delhi.
2.	S.N. Maheshwari, Financial Accounting, Vikas Publications, Noida.
3.	Shukla Grewal and Gupta, “Advanced Accounts”, volume 1, S.Chand and Sons, New Delhi.
4.	Radhaswamy and R.L. Gupta: Advanced Accounting, Sultan Chand, New Delhi.
5.	R.L. Gupta and V.K. Gupta, “Financial Accounting”, Sultan Chand, New Delhi.
Reference Books	
1.	Dr. Arulanandan and Raman: Advanced Accountancy, Himalaya Publications, Mumbai.
2.	Tulsian , Advanced Accounting, Tata McGraw Hills, Noida.
3.	Charumathi and Vinayagam, Financial Accounting, S.Chand and Sons, New Delhi.
4.	Goyal and Tiwari, Financial Accounting, Taxmann Publications, New Delhi.
5.	Robert N Anthony, David Hawkins, Kenneth A. Merchant, Accounting: Text and Cases. McGraw-Hill Education, Noida.
NOTE: Latest Edition of Textbooks May be Used	
Web Resources	
1.	https://www.slideshare.net/mcsharma1/accounting-for-depreciation-1
2.	https://www.slideshare.net/ramusakha/basics-of-financial-accounting
3.	https://www.accountingtools.com/articles/what-is-a-single-entry-system.html

**MAPPING WITH PROGRAMME OUTCOMES AND
PROGRAMME SPECIFIC OUTCOMES**

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	3	3	2	3	2	2	3	2	2
CO2	3	2	3	3	3	2	2	2	3	2	2
CO3	3	2	3	3	3	2	2	2	3	2	2
CO4	3	2	3	3	2	2	2	2	3	2	2
CO5	3	2	3	3	3	2	2	2	3	2	2
TOTAL	15	10	15	15	13	11	10	10	15	10	10
AVERAGE	3	2	3	3	2.6	2.2	2	2	3	2	2

3 – Strong, 2- Medium, 1- Low

FIRST YEAR – SEMESTER – I

CORE PAPER– 2 PRINCIPLES OF AUDITING

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	5				4	5	25	75	100
Learning Objectives									
LO1	To understand the basic accounting concepts and standards.								
LO2	To know the basis for calculating business profits.								
LO3	To familiarize with the accounting treatment of depreciation.								
LO4	To learn the methods of calculating profit for single entry system.								
LO5	To gain knowledge on the accounting treatment of insurance claims.								
Prerequisites: Should have studied Accountancy in XII Std									
Unit	Contents								No. of Hours
I	INTRODUCTION Meaning and Definition of Auditing - Nature and Scope of Auditing - Accountancy and auditing, Auditing and Investigation - Objectives of auditing - Limitations of audit - Advantages of audit - classification of audit.								15
II	AUDITPROGRAMMEANDINTERNALCONTROL Meaning and definition of audit program - Advantage and disadvantage - audit file, audit note book, audit working papers - purposes and importance of working papers - Internal check - meaning, objectives of Internal check – features of good Internal check system – Internal Control - meaning, objectives and features of good Internal control .								15
III	VOUCHING Vouching - meaning of vouching - Importance - objects - Vouching of cash transactions - Verification of assets and liabilities - meaning of verification - objectives - Distinction between vouching and verification - distinction between Valuation and Verification.								15
IV	COMPANYAUDITORS Company auditors - Qualification and Disqualification of an auditor - Appointment and Removal of an auditor - Powers and Duties of auditors - Liabilities of an auditor.								15
V	AUDITOR’SREPORT Auditor’s Report-Importance of auditor’s report-contents of audit report-Kinds of reports.								15
TOTAL								75	
CO	Course Outcomes								
CO1	Define auditing and its process.								

CO2	Compare and contrast essence of internal check and internal control.
CO3	Identify the role of audit or sin companies.
CO4	Define the concept of Corporate Governance.
CO5	Appraise the implications of Corporate Social Responsibility

Text Books

S.NO	AUTHORS	TITLE	PUBLISHERS
1.	B.N.Tandon	A handbook of practical Auditing.	S.Chand
2.	T.R.Sharma	Auditing	Sahitya Bhavan, Agra.
3.	B.N.Tandon Sudharsanam, Sundharababu	Practical Auditing	S.Chand,.
4.	Dr.K.Sundar and K.Parri	Practical Auditing	Vijay Nicole Imprints Pvt., Ltd.,
5.	S.K.Basu	Auditing and Principles and Techniques	Pearson
6	B.N.Tandon	Practical Auditing	S. Chand.
7	Dr.Premavathy	Auditing	Vishnu Publications
8	Dinkar pagare	Principles and practice of auditing	Sultan Chand & Sons
9	Raymond J.Noss	Practical auditing techniques	Univers
10	B.N.Tandon	The handbook of practical auditing	S.chand

Reference Books:

S.NO	AUTHORS	TITLE	PUBLISHERS
1.	Dr.L.Natarajan	Practical Auditing	Margham Publications
2.	Kamal Gupta and Ashok Arora	Fundamentals of Auditing	Tata Mc., Graw Hill.
3.	R.G.Saxena.	Auditing	Himalaya Publishing House
4.	S.N.Maheshwari	Banking Theory, law and practice	Kalyani Publications.

5	S. Vengadaman	Practical Auditing	Margham Publication.
6	B.N.Tandon	Practical auditing for ug courses for madras university	S.chand
7	Dr.Natarajan	practical auditing	Margam
8	A.Jesenth Rani	Practical auditing	Charulatha
9	George Benton Renn	Practical auditing	Kessinger
10	Dr.R.N Sengupta	Practical guide to auditing	New central book house

Mapping with Programme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	S	S	S	S	S	S	S	S	S
CO2	S	S	S	S	S	S	S	S	S	S
CO3	S	S	S	S	S	S	S	S	S	S
CO4	S	S	S	S	S	S	S	S	S	S
CO5	S	S	S	S	S	S	S	S	S	S

PO–Programme Outcome, CO –Course outcome
S – Strong, M – Medium, L–Low (may be avoided)

FIRST YEAR – SEMESTER – I

ELECTIVE-1: BUSINESS LAW

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	5				3	4	25	75	100
Learning Objectives									
LO1	To know the nature and objectives of Mercantile law and the essentials of valid contract								
LO2	To gain knowledge on performance contracts								
LO3	To be acquainted with the rules of Indemnity and Guarantee								
LO4	To make aware of the essentials of Bailment and pledge								
LO5	To understand the provisions relating to sale of goods								

Prerequisites: Should have studied Commerce in XII Std		
Unit	Contents	No. of Hours
I	Elements of Contract Indian Contract Act 1872: Definition of Contract, Essentials of Valid Contract, Classification of Contract, Offer and Acceptance – Consideration – Capacity to Contract – Free Consent - Legality of Object – Contingent Contracts – Void Contract	15
II	Performance of Contract Meaning of Performance, Offer to Perform, Devolution of Joint liabilities & Rights, Time and Place of Performance - Remedies for Breach of Contract - Termination and Discharge of Contract.	15
III	Contract of Indemnity and Guarantee Contract of Indemnity and Contract of Guarantee- Extent of Surety's Liability, Kinds of Guarantee, Rights of Surety, Discharge of Surety.	15
IV	Bailment and Pledge Bailment and Pledge – Bailment – Concept – Essentials – Classification of Bailment's, Duties and Rights of Bailor and Bailee – Law of Pledge – Meaning – Essentials of Valid Pledge, Pledge and Lien, Rights of Pawner and Pawnee.	15
V	Sale of Goods Act 1930: Definition of Contract of Sale – Formation – Essentials of Contract of Sale – Conditions and Warranties - Transfer of Property - Rights and duties of buyer - Rights of an Unpaid Seller	15
	TOTAL	75
Course Outcome		
CO1	Explain the Objectives and significance of Mercantile law	
CO2	Understand the clauses and exceptions of Indian Contract Act.	
CO3	Outline the contract of indemnity and guarantee	
CO4	Familiar with the provision relating to Bailment and Pledge	
CO5	Explain the various provisions of Sale of Goods Act 1930	

Textbooks	
1	N.D.Kapoor,BusinessLaws-SultanChandandSons,NewDelhi.
2	R.S.N.Pillai–BusinessLaw,S.Chand,NewDelhi.
3	MCKuchhal&VivekKuchhal,Businesslaw,SChandPublishing,NewDelhi
4	M.V.Dhandapani,BusinessLaws,SultanChandandSons,NewDelhi.
5	ShusmaAurora,BusinessLaw,Taxmann,NewDelhi.
ReferenceBooks	
1	PreethiAgarwal,BusinessLaw,CAfoundationstudymaterial,Chennai.
2	BusinessLawbySaravanel,Sumathi,Anu,HimalayaPublications, Mumbai.
3	KavyaandVidhyasagar,BusinessLaw,NithyaPublication,NewDelhi.
4	D.Geet,BusinessLawNiraliPrakashanPublication,Pune.
5	M.R.Sreenivasan,BusinessLaws,MarghamPublications,Chennai.
NOTE:LatestEditionofTextbooksMaybeUsed	
WebResources	
1	www.cramerz.comwww.digitalbusinesslawgroup.com
2	http://swcu.libguides.com/buslaw
3	http://libguides.slu.edu/businesslaw

MAPPINGWITHPROGRAMMEOUTCOMESANDPROGRAMMESPECIFIC OUTCOMES

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PSO 1	PSO 2	PSO 3
CO1	3	2	2	3	2	2	2	2	2	2	2
CO2	3	2	3	3	2	2	2	2	2	2	2
CO3	3	2	2	3	2	2	2	2	2	2	2
CO4	3	2	3	3	2	2	2	2	2	2	2
CO5	3	2	3	3	2	2	2	2	2	2	2
TOTAL	15	10	13	15	10	10	10	10	10	10	10

3–Strong, 2-Medium, 1-Low

FIRST YEAR – SEMESTER – I**SKILL – 1: FUNDAMENTALS OF COMMERCE**

Subject Code	L	T	p	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	3				2	3	25	75	100
Learning Objectives									
LO1	Understand the meaning of Commerce and Industry.								
LO2	Familiarize with Various Accounting methods.								
LO3	Explore about Market and Marketing								
LO4	Understand the various Acts prevailing in India.								
LO5	Gain knowledge about Taxation and Filing of Income Tax.								
Unit	Contents								No of Hours
I	Commerce- Introduction: Definition of Commerce - Importance – Meaning of Barter system - Business–Industry- Trade–Hindrances of Trade- Branches of Commerce.								6
II	Accounting–Introduction: Bookkeeping – Meaning - Definition- Objectives - Accounting – Meaning- Definition – objectives – Branches of Accounting - Financial Accounting – Cost Accounting - Management Accounting - its features and Differences.								6
III	Introduction to Marketing: Definition of Market –Classification of Markets –Marketing –Meaning and Definition- Characteristics - Difference Between Market and Marketing – Approaches to Study of Marketing.								6
IV	Introduction to Legal aspects of Business– Meaning of: Indian Contract Act 1872- Negotiable Instruments Act 1881 - Sale of Goods Act 1930-Partnership Act 1932 - Banking Regulation Act 1948 - Income Tax Act 1961 – Insolvency and Bankruptcy Code 2016– GST Act 2017 - Anti Money Laundering Act 2020.								6
V	Tax Return Filing: Meaning and Types of Taxation - Types of Returns - Filing of Income Tax Return- Filing of GST return - Slab rates.								6
	TOTAL								30
Course Outcomes									
CO1	To make the students familiar with the concepts of Commerce and Industry.								
CO2	To encourage and motivate the students for the Accounting Education.								
CO3	To Analyze the Various classification of Markets and Marketing.								
CO4	To make the students aware towards the various commercial Laws.								
CO5	To aware the types of Taxation and slab rates.								

Text Books	
1	S.P.JainandK.LNarang2023,FinancialAccounting-I,KalyaniPublishers, New Delhi
2	N.D.Kapoor, MercantileLaw,SultanChand&Sons,NewDelhi.
3	Dr.L.Natarajan,MarghamPublications,Chennai.
Reference Books	
1	Hariharan N, Income Tax Law & Practice, Vijay Nicole Imprints Pvt. Ltd. Chennai.
2	R.S.N.PillaiAndBagavathi,BusinessLaw,S.ChandPublishing.
3	T.Srinivasan–IncomeTax&Practice–VijayNicole ImprintsPvt.Limited,Chennai.
4	T.S.Reddy&DrY.HariprasadReddy,ManagementAccounting.Margham Publications, Chennai.
NOTE: Latest Edition of Text Books May be Used	
Web Resources	
1	https://www.incometaxmanagement.com/Direct-Taxes/AY-2021-22/assessment/1-assessment-of-an-individual.html
2	https://dea.gov.in/sites/default/files/moneylaunderingact.pdf
3	https://www.mca.gov.in/Ministry/pdf/TheInsolvencyandBankruptcyofIndia.pdf

MAPPINGWITHPROGRAMMEOUTCOMESAND PROGRAMME SPECIFICOUTCOMES

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	3	2	3	2	3	3	3	2	2
CO2	3	2	2	3	2	2	2	2	3	2	3
CO3	3	2	3	3	3	2	3	3	3	2	2
CO4	2	2	2	2	2	2	2	2	3	2	2
CO5	3	2	3	3	3	2	3	3	3	2	3
Total	14	10	13	13	13	10	13	13	15	10	12
Average	2.8	2	2.6	2.6	2.6	2	2.6	2.6	3	2	2.4

FIRST YEAR – SEMESTER – II

CORE PAPER – 3: AUDITING THEORY AND PRACTICE

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	7				5	6	25	75	100
Learning Objectives									
LO1	To enable students to understand process of auditing and its classification.								
LO2	To impart knowledge on internal check and internal control.								
LO3	To illustrate the role of auditors in company.								
LO4	To help students understand the framework, theories and models of Corporate Governance.								
LO5	To provide insights into the concept of Corporate Social Responsibility								
Prerequisite: Should have studied Commerce in XII Std									
Unit	Contents								No. of Hours

I	Introduction to Auditing Meaning and Definition of Auditing – Distinction between Auditing and Accounting – Objectives – Advantages and Limitations of Audit – Scope of Audit – Classifications of Audits – Audit of For Profit enterprises and Non-profit Organizations	15
II	Audit Procedures and Documentation Audit Planning – Audit Programme – Procedures - Internal Audit - Internal Control – Internal Check System – Vouching – Cash and Trade Transactions - Verification of Assets and Liabilities and its Valuation	15
III	Company Auditor Appointment and Removal of Auditors – Rights, Duties and Liabilities of Auditor – Audit Report - Recent Trends in Auditing - Information Systems Audit (ISA) – Auditing around the computer – Auditing through the computer- e-audit tools.	15
IV	Introduction to Corporate Governance Conceptual Framework of Corporate Governance: Theories & Models, Broad Committees - Corporate Governance Reforms. Major Corporate Scandals in India and Abroad: Common Governance Problems Noticed in various Corporate Failures.	15
V	Corporate Social Responsibility Concept of CSR, Strategic Relationship of CSR with Corporate Sustainability - CSR and Business Ethics, CSR and Corporate Governance - CSR Provisions under the Companies Act, 2013 (Section 135 schedule – VII).	15
	TOTAL	75

Course Outcomes	
CO1	Define auditing and its process.
CO2	Compare and contrast essence of internal check and internal control.
CO3	Identify the role of auditors in companies.
CO4	Define the concept of Corporate Governance.
CO5	Appraise the implications of Corporate Social Responsibility
Textbooks	
1	Dinkar Pagare, Principles and Practice of Auditing, Sultan Chand & Sons, New Delhi
2	B. N. Tandon, S. Sudharsanam & S. Sundharabahu, Practical Auditing, S. Chand & Sons New Delhi.
3	Dr. T. R. Sharma, Dr. Gaurav Sankalp, Auditing & Corporate Governance, Sahitya Bhawan Publications, Agra
4	Aruna Jha, Auditing & Corporate Governance, Taxmann Publication Pvt. Ltd, New Delhi.
Reference Books	
1	Kevin Keasey, Steve Thompson & Mike Wright, Governance & Auditing, Emerald Group Publishing Limited, Bingley
2	Dr. T. R. Sharma, Auditing, Sahitya Bhawan Publications, Agra
3	C. B. Gupta, Neha Singhal, Auditing & Corporate Governance, Scholar Tech Press, New Delhi.
4	Shri. Vengadamani, Practical Auditing, Margham Publication, Chennai.
NOTE: Latest Edition of Textbooks May be Used	
Web Resources	
1	https://www.wallstreetmojo.com/audit-procedures/
2	https://theinvestorsbook.com/company-auditor.html
3	https://www.investopedia.com/terms/c/corp-social-responsibility.asp

Mapping with Programme Outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PSO 1	PSO 2	PSO 3
CO1	3	2	3	2	3	2	3	3	3	3	2
CO2	3	2	2	2	2	2	2	2	3	2	2
CO3	3	3	3	2	3	2	3	3	3	3	2
CO4	3	2	2	2	2	2	2	2	3	3	2
CO5	3	3	3	2	3	2	3	3	3	2	2
TOTAL	15	12	13	10	13	10	13	13	15	13	10
AVERAG E	3	2.2	2.6	2	2.6	2	2.6	2.6	3	2.6	2

3 –Strong,2-Medium,1-Low

FIRST YEAR – SEMESTER – II**CORE PAPER – 4 FINANCIAL MANAGEMENT**

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	5				4	5	25	75	100
Learning Objectives									
LO1	To provide expert knowledge on setting financial objectives & goals.								
LO2	To manage Financial Resource, financial risk management and through understanding of investment portfolios and financial instruments.								
LO3	To Maximize the Cost of Capital by Developing a Sound and Economical combinations of Corporate Securities, Proper Estimation and Requirement for Expansion and Growth								
LO4	To Ensure adequate return on Investment								
LO5	To Maintain Proper Cash Flow Creating Reserves and Goodwill								
Prerequisites: Should have studied Accountancy in XII Std									
Unit	Contents								No. of Hours
I	Nature and Importance of Finance Functions - Organizing Finance Functions - Functions of Finance Manager - Objectives of Finance Function - Methods and Sources of Raising Finance - Critical Appraisal of the Various Sources of Finance.								10
II	Goals of Finance Function - Financing Decisions - Financial Planning - Financial Forecasting - Capital Structure Decisions - Net Income Approach, NOI Approach and MM Approach- Capitalization - Cost of Capital - Computation of Cost of Capital-Dividend Policy-Factors Determining Dividend Policy.								13
III	Investment Decisions - Estimation of Cash Flows - Evaluation of Alternative Investment Proposals like NPV, ARR, IRR Methods - Decision Making Under Risk and Uncertainty - Inflation and Investment Decisions								13
IV	Working Capital - Meaning, Concept, Types and Significance-Gross and Net Working Capital - Determinants of Working Capital - Sources of WC - Credit and Collection Policies.								12
V	Security Analysis and Portfolio Management - Leverages -Meaning, Types of Leverage. Degree of Operating and Financial Leverage - Financial Ratio Analysis.								12
TOTAL								60	

CO1	The student will be able to understand the basic Principles and practices of Financial management.
CO2	The student will be able to determine the amount of Capital, Organization and Structure. Reduce cost of Capital and Operating Risks
CO3	The student will be able to have the knowledge and practice of arriving financial Decision makings
CO4	The student will be able to acquire practical knowledge on Calculation of working capital
CO5	The student will be able to gain knowledge on leverage and portfolio management

TEXT BOOKS:

S.No	Authors	Title	Publishers
1.	Dr.S.N.Maheswari	Financial Management	Sultan Chand & Sons, New Delhi
2.	Dr.A.Murthy	Financial Management	Margham Publications, Chennai.

3.	Dr.J.Srinivasan,Sridharand Ramalingam	FinancialManagement	VijayNicole Imprints Pvt .Ltd.Chennai
4.	R.K.Sharma	FinancialManagement	KalyaniPublishers,New Delhi
5.	I.M.Pandey	FinancialManagement	VikasPublishingHouse Pvt Ltd, 01-Nov-2009
6.	P.c.Kulkarni	FinancialManagement	B.G. Sathyaprasad, HimalayaPublications, 2004
7.	Dr..V.R.Palanivelu,	FinancialManagement	S.ChandPublication,2010
8.	I.MPanday	FinancialManagement	Pearson
9.	Berk	FinancialManagement	Pearson
10.	PrasanaChandra	Financial Managementtheoryand practice	McGrawhilleducation Pvt. Ltd India

REFERENCE BOOKS

S.NO	AUTHORS	TITLE	PUBLISHERS
1.	I.M.Pandey	FinancialManagement	VikasPublisionhousePvt Ltd. Noida
2.	PrasannaChandra	FinancialManagement10ed.	McGrawhilleducation Pvt. Ltd India
3.	Subirkumar Banerjee	FinancialManagement	PHILearningPvtLtd
4.	VyuptakeshSharan	FundamentalsofFinancial Management	Pearson Education
5.	Dr.N. Premavathy	FinancialManagement	SriVishnuPublications, Chennai.
6.	S.C. Kuchhal	FinancialManagement	Chaitanya
7.	P.V.Kulkarni&B.G.Sa tyaprasad	FinancialManagement	HimalayaPublishingHouse
8	M.YKhanP.K Jain	Financialmanagementtext problem and cases	McGraw hill education Pvt. Ltd India

9	PrasannaChandra	Fundamentals of financial management	Taxmann
10	Dr.S.p. Gupta	Advanced Financial management	Sathiyabhavan

Reference: Journals

1. Indian Journal of Business Finance and Accounting
2. Journal of Financial Reporting and Accounting
3. Asian Academy Management Journal of Accounting and Finance
4. Review of Accounting and Finance

E-Materials

1. EconomicTimes.Com
2. FinancialExpress
3. Reserchgate.net
4. Entrepreneur.com
5. The Hindu business line

Mapping with Programme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	S	S	S	S	S	S	S	S	S
CO2	S	S	S	S	S	S	S	S	S	S
CO3	S	S	S	S	S	S	S	S	S	S
CO4	S	S	S	S	S	S	S	S	S	S
CO5	S	S	S	S	S	S	S	S	S	S

PO–Programme Outcome, CO –Course outcome
S –Strong, M – Medium, L–Low (may be avoided)

FIRST YEAR – SEMESTER – II

ELECTIVE PAPER - 2 GOOD AND SERVICE TAX

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	4				3	4	25	75	100
Learning Objectives									
LO1	To Understand the Basics of GST.								
LO2	To Study the Registration and Computation of GST.								
LO3	To Acquaint the Students with Filing of Returns.								
LO4	To understand the concepts of GST technology.								
LO5	To explore the knowledge relating to the penalties and offences relating to GST								
Prerequisite: Should have studied Commerce in XII Std									
Unit	Contents								No. of Hours
I	Overview of Goods and Service Tax-Introduction - Meaning of GST - IGST - Scope of GST - Present/Old Tax Structure V/S GST - GST In Other Countries- Existing Taxes Proposed to be Subsumed Under GST-Principles Adopted for Subsuming the Taxes-Dual GST-Benefits of GST-GST Council-GST Network (GSTN) And GST Regime-Integrated Goods and Services Tax Act- 2017- Title and Definitions- Administration.								12
II	Registration Under GST -Rules-Procedure of Registration-Exempted Goods And Services Under GST - Rates of GST– GST At 5 % - GST At 12 % - GST At 18 % - GST At 28% - Procedure Relating To Levy (CGST &SGST) -Various Schedules Related To Supply - Computation of Taxable Value and Tax Liability [CSGT &SGST] -Procedure Relating To Levy (IGST)								12
III	Input Tax Credit (ITC)- Eligibility and Conditions for Taking Input Tax Credit Apportionment of ITC and Blocked Credit -Various Documents Under GST- Tax Invoice Bill for Supply- Debit Note- Credit Note- Payment Voucher- Receipt Voucher- E-Way Bill HSN Code and SAC Code - Simple Problems on Utilization Of Input Tax Credit.								12
IV	Assessment and Administration of GST - Types of GST Returns- Types of Assessment &Assessment Procedures- Role and Functions of GST Council-Tax Authorities and Their Powers; Tax Deduction at Source &Tax Collection at Source- Refund of Tax-Offence and Penalties.								12
V	GST and technology -Introduction to GSTN – Power and Functions of GSTN – Design and Implementation Framework – Design &Implementation Framework GSTN. Goods And Service Tax Suvidha Provider (GSP) - Concept.								12
TOTAL								60	

CourseOutcomes	
CO1	The students will get a general understanding of the GST law in the country
CO2	The students will provide an Insight into practical aspects of GST
CO3	The students will equip themselves to become tax practitioners.
CO4	The students will be able to calculate GST at different level.
References	
1	V.S Datey – Taxman’s: GST.
2	Step By Step Guide to GST – Compliances - Avinash Poddar.
3	GST Handbook to students – CA Vivek KR Agarwal, 2022, Neelam Book House.
4	A Complete Guide To Goods and Services Tax – Sanjiv Agarwal.
5	GST for laymen, CA Apeshika Solanki, 2021, Bloomsbury India.
6	Madhykar N Hiregange: GST
Web Resources	
1	https://www.taxmann.com/academy/
2	https://gstcouncil.gov.in/

MAPPING WITH PROGRAMME OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	3	3	2	3	2	2	3	2	2
CO2	3	2	3	3	3	2	2	2	3	2	2
CO3	3	2	3	3	3	2	2	2	3	2	2
CO4	3	2	3	3	2	2	2	2	3	2	2
CO5	3	2	3	3	3	2	2	2	3	2	2
TOTAL	15	10	15	15	13	11	10	10	15	10	10
AVERAGE	3	2	3	3	2.6	2.2	2	2	3	2	2

3 – Strong, 2- Medium, 1- Low

FIRST YEAR – SEMESTER – II

SKILL PAPER – 2 BUSINESS COMMUNICATIONS

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	3				2	3	25	75	100
LearningObjectives									
LO1	Toenablethestudentsto knowabouttheprinciples,objectivesand importanceof communication in commerce and trade.								
LO2	Todevelopthestudentsto understandabouttradeenquiries								
LO3	Tomakethestudentsawareabout varioustypesofbusinesscorrespondence.								
LO4	Todevelopthestudentstowritebusinessreports.								
LO5	Toenablethelearnerstoupdatewithvarioustypesofinterviews								
Prerequisites: Should have studied Commerce in XII Std									
Unit	Contents								No. of Hours
I	IntroductiontoBusinessCommunication Definition – Meaning – Importance of Effective Communication – Modern CommunicationMethods–BarrierstoCommunication–ECommunication -BusinessLetters:Need-Functions–EssentialsofEffectiveBusiness Letters – Layout								12
II	TradeEnquiries TradeEnquiries–Ordersandtheir Execution–Credit andStatusEnquiries– Complaints andAdjustments – Collection Letters – Sales Letters – Circulars								12
III	Banking&InsuranceCorrespondence Banking Correspondence – Types – Structure of Banking Correspondence–ElementsofaGoodBankingCorrespondence– Insurance – Meaning and Types – Insurance Correspondence – Difference between Life and General Insurance – Meaning of Fire Insurance – Kinds – Correspondence Relating to Marine Insurance – AgencyCorrespondence–Introduction–Kinds–StagesofAgent Correspondence – Terms of Agency Correspondence								12
IV	SecretarialCorrespondence Company Secretarial Correspondence – Introduction – Duties of Secretary – Classification of Secretarial Correspondence – Specimen letters – Agenda andMinutesofReportwriting–Introduction–TypesofReports– PreparationofReportWriting								12

V	Interview Preparation Application Letters – Preparation of Resume – Interview: Meaning – Objectives and Techniques of Various Types of Interviews –Creating & maintaining Digital Profile	12
	TOTAL	60

Course Outcomes	
CO1	Acquire the basic concept of business communication.
CO2	Exposed to effective business letter
CO3	Paraphrase the concept of various correspondences.
CO4	Prepare Secretarial Correspondence like agenda, minutes and various business reports.
CO5	Acquire the skill of preparing an effective resume
Textbooks	
1	Rajendra Pal & J.S. Korlahalli, Essentials of Business Communication-Sultan Chand & Sons- New Delhi.
2	Gupta and Jain, Business Communication, Sahitya Bahvan Publication, New Delhi.
3	K.P. Singha, Business Communication, Taxmann, New Delhi.
4	R.S.N. Pillai and Bhagavathi. S, Commercial Correspondence, Chand Publications, New Delhi.
5	M.S. Ramesh and R. Pattenshetty, Effective Business English and Correspondence, S. Chand & Co, Publishers, New Delhi.
Reference Books	
1	V.K. Jain and Om Prakash, Business communication, S. Chand, New Delhi.
2	Rithika Motwani, Business communication, Taxmann, New Delhi.
3	Shirley Taylor, Communication for Business-Pearson Publications-New Delhi.
4	Bovee, Thill, Schatzman, Business Communication Today-Pearson Education, Private Ltd-New Delhi.
5	Penrose, Rasbery, Myers, Advanced Business Communication, Bangalore.
Web Resources	
1	https://accountingseekho.com/
2	https://www.testpreptraining.com/business-communications-practice-examquestions
3	https://bachelors.online.nmims.edu/degree-programs

MAPPING WITH PROGRAMME OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PSO 1	PSO 2	PSO 3
CO1	3	3	2	3	2	2	2	2	2	2	2
CO2	3	3	2	3	2	2	2	3	2	2	2
CO3	3	3	2	3	2	2	2	2	2	2	2
CO4	3	3	2	3	2	2	2	2	2	2	2
CO5	3	3	2	3	2	2	2	2	2	2	2
TOTAL	15	15	15	15	10	10	10	11	10	10	10
AVERAGE	3	3	3	3	2	2	2	2.2	2	2	2

3 – Strong, 2-Medium, 1-Low

SECOND YEAR – SEMESTER – III

CORE PAPER –5: CORPORATE ACCOUNTING

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	5				4	5	25	75	100
Learning Objectives									
LO1	To understand about the issue of shares and pro-rata allotment.								
LO2	To understand the Underwriting of Shares and Redemption of Preference Shares								
LO3	To know about Issue and Redemption of Debentures.								
LO4	To learn the form and contents of financial statements as per Schedule III of Companies Act 2013								
LO5	To examine the various methods of valuation of Goodwill and shares								
Prerequisite: Should have studied Financial Accounting in I Year									
Unit	Contents								No. of Hours
I	Issue of Shares Issue of Shares–Premium-Discussion -Forfeiture-Reissue–Pro-rata Allotment.								15
II	Issue & Redemption of Preference Shares Underwriting of Shares and Debentures – Underwriting Commission - Types of Underwriting. Redemption of Preference Shares–Provisions of Companies Act– Capital Redemption Reserve – Minimum Fresh Issue – Redemption at Par, Premium and Discount.								15
III	Issue and Redemption of Debentures Debentures: Issue and Redemption–Meaning–Methods–In One lot–in Installment – Purchase in the Open Market includes Ex Interest and Cum Interest-Sinking Fund Investment Method.								15

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IV	Final Accounts Introduction–Final Accounts–Form and Contents of Financial Statements as Per Schedule III of Companies Act 2013 – Part I Form of Balance Sheet – Part II Form of Statement of Profit and Loss.	15
V	Valuation of Goodwill – Meaning – Need for Valuation of Goodwill – Methods of Valuing Goodwill– Average Profit –Super Profit –Annuity and Capitalization Method. Valuation of Shares – Need for Valuation of Shares – Methods of Valuation of Shares – Net Assets Method – Yield and Fair Value Methods.	15
	TOTAL	75
THEORY 20% & PROBLEMS 80%		
Course Outcomes		

CO1	Prepare and account for various entries to be passed in case of issue of shares
CO2	To understand the Underwriting of shares and Redemption of preference shares .
CO3	To have thorough knowledge about issue and redemption of Debentures.
CO4	Construct Financial Statements applying relevant accounting treatments
CO5	Compute the value of goodwill and shares under different methods and assess its applicability
Textbooks	
1	S.P.Jain and N.L.Narang, Advanced Accounting Vol I, Kalyani Publication, New Delhi.
2	R.L.Gupta and M.Radhaswamy, Advanced Accounts Vol I, Sultan Chand, New Delhi.
3	Broman, Corporate Accounting, Taxmann, New Delhi.
4	Shukla, Grewal and Gupta- Advanced Accounts Vol I, S.Chand, New Delhi.
5	M.C.Shukla, Advanced accounting Vol I, S.Chand, New Delhi.
Reference Books	
1	T.S.Reddy, A.Murthy–Corporate Accounting–Margham Publication, Chennai.
2	D.S.Rawat & Nozer Shroff, Students Guide To Accounting Standards, Taxmann, New Delhi
3	Prof.Mukesh Bramhbut, Devi, Corporate Accounting I, Ahilya Publication, Madhya Pradesh
4	Anil Kumar, Rajesh Kumar, Corporate accounting I, Himalaya Publishing house, Mumbai.
5	Prasanth Athma, Corporate Accounting I, Himalaya Publishing house, Mumbai.

NOTE: Latest Edition of Textbooks May be Used	
Web Resources	
1	https://www.tickertape.in/blog/issue-of-shares/
2	https://www.taxmann.com/bookstore/bookshop/bookfiles/chapter12valuationofgoodwillandshares.pdf
3	https://www.mca.gov.in/content/mca/global/en/acts-rules/ebooks/accountingstandards.html

MAPPING WITH PROGRAMME OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	2	2	2	3	2	2
CO2	3	2	3	2	2	2	2	2	3	2	2
CO3	3	2	3	2	3	2	2	2	3	2	2
CO4	3	1	3	2	3	2	2	2	3	2	2
CO5	3	3	3	2	3	2	2	2	3	2	2
TOTAL	15	11	15	10	13	10	10	10	15	10	10
AVERAGE	3	2.2	3	2	2.6	2	2	2	3	2	2

3 – Strong, 2-Medium, 1-Low

SECOND YEAR – SEMESTER – III

CORE PAPER – 6 MANAGEMENT ACCOUNTING

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	5				4	5	25	75	100
Learning Objectives									
LO1	To understand basics management accounting								
LO2	To know the aspects of Financial Statement Analysis								
LO3	To familiarize with fund flow and cash flow analysis								
LO4	To learn about budgetary control								
LO5	To gain insights into marginal costing.								
Unit	Contents								No. of Hours
I	Introduction to Management Accounting Management Accounting – Meaning – Scope – Importance- Limitations - Management Accounting Vs Cost Accounting – Management Accounting Vs Financial Accounting. Analysis and Interpretation of Financial Statements– Nature and Significance –Types of Financial Analysis–Tools of Analysis– Comparative Statements– Common Size Statement–Trend Analysis.								15
II	Ratio Analysis Ratio Analysis: Meaning – Advantages – Limitations – Types of Ratios – Liquidity Ratios – Profitability Ratios -Turnover Ratios – Solvency Ratios – Leverage Ratios-Preparation of Financial Statements from Ratios.								15
III	Funds Flow & Cash Flow Analysis Introduction, Meaning of Funds Flow Statement-Ascertainment of Flow of Funds - Schedule of Changes in Working Capital-Adjusted Profit and Loss Account-Preparation of Funds Flow Statement. Cash Flow Statement: Meaning–Advantages–Limitations– Preparation of Cash Flow Statement as per AS 3– Cash Flow from Operating, Financing and Investing activities								15
IV	Budget and Budgetary Control Meaning–Preparation of Various Budgets–Cash Budget – Flexible Budget– Production Budget–Sales Budget– Master Budget–Budgetary Control –Benefits								15

V	Marginal Costing: Meaning - Features – Marginal Costing vs Absorption Costing - Fixed Cost, Variable Cost and Semi Variable Cost- Contribution- Marginal Cost Equation- P/V Ratio - Break Even Point - Margin of Safety – Cost-Volume Profits Analysis.	15
	TOTAL	75
THEORY20%&PROBLEMS80%		
CO	Course Outcomes	
CO1	Rememberandrecallbasicsinmanagementaccounting	
CO2	ApplytheknowledgeofpreparationofFinancialStatements	
CO3	Analysetheconceptsrelatingtofundflowandcashflow	
CO4	Evaluatetechniquesof budgetarycontrol	
CO5	Formulatecriteriafordecisionmakingusingprinciplesofmarginalcosting.	
Textbooks		
1	JainS.P.&NarangK.L.(2018)CostandManagementAccounting,KalyaniPublications,	
2	Rds.Maheswari,CostandManagementAccounting,SultanChandSonsPublications,NewDelhi.	
3	SharmaandShashiK.Gupta,ManagementAccounting,KalyaniPublishers,Chennai.	
4	JenitraLMervin,DasltonLCecil,ManagementAccounting,LerantecPress,Chennai.	
5	T.S.Reddy&Y.HariPrasadReddy,ManagementAccounting,MarhamPublications,Chennai.	
ReferenceBooks		
1	Chadwick–TheEssenceofManagementAccounting,FinancialTimesPublications,England.	
2	CharlesT.HorngrenandGaryN.Sundem–IntroductiontoManagementAccounting,Pearson,Chennai.	
3	MurthyAandGurusamyS,ManagementAccounting-Theory&Practice,VijayNicoleImprintsPvt.Ltd .Chennai.	
4	Hansen-Mowen,CostManagementAccountingandControl,SouthWesternCollege,India.	
5	N.P.Srinivasan,ManagementAccounting,NewAgepublishers,Chennai.	
NOTE:LatestEditionofTextbooksMaybeUsed		
WebResources		
1	https://www.accountingnotes.net/companies/fund-flow-analysis/fund-flowanalysis-accounting/13300	
2	https://accountingshare.com/budgetary-control/	

MAPPING WITH PROGRAMME OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PSO 1	PSO 2	PSO 3
CO1	3	2	3	2	3	2	3	3	3	2	2
CO2	3	2	2	2	3	2	2	2	3	2	3
CO3	3	2	3	2	3	2	3	3	3	2	2
CO4	3	2	2	2	3	2	2	2	3	2	2
CO5	3	3	3	2	3	2	3	3	3	2	3
TOTAL	15	11	13	10	15	10	13	13	15	10	12
AVERAGE	3	2.1	2.6	2	2	2	2.6	2.6	3	2	2.4

3–Strong,2-Medium,1-Low

BUSINESS STATISTICS - I

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	5				3	5	25	75	100
Learning Objective									
LO1	To familiarize students with the basic concepts in Business Mathematics and Statistics.								
LO2	To make students understand various Measure of central tendency.								
LO3	To know principles of construction of Dispersion.								
LO4	To know the principles of Correlation and Regression								
LO5	To be able to choose rational options in practical decision								
UNIT	Contents								No.of Hours
I	Statistics Statistics – Definition – scope and Limitation – Presentation of data- Simple Bar Diagram, Multiple Bar Diagram ,Component Bar Diagram ,Percentage Bar Diagram ,Pictogram Diagrammatic and graphical Representation of Data- Frequency Polygon, Frequency Curve, Cumulative Frequency Curve- Quartile,percentiles and Deciles								16
II	Measure of central tendency Arithmetic Mean ,Weighted Arithmetic Mean –Frequency Distribution, Properties of AM Combination Mean ,Geometric Mean ,Harmonic Mean - Median and Mode ,								19
III	Measure of Dispersion Range, Merit and Demerit - Mean Deviation – Quartile Deviation - Standard Deviation – Relative Measure- Coefficient Variation								19
IV	Correlation Correlation -Karl Pearson’s Correlation ,Positive Correlation ,Negative Correlation ,No Correlation ,Simple Correlation – Scatter Diagram – Numerical Value of the Correlation Coefficient - Concurrent Deviation method – Rank Correlation – Properties of Correlation Coefficient ,Limitation -Uses of Correlation in Business Regression – Regression Lines – Regression coefficients – Uses of Regression in Business Problems.								19
V	Annuity Mathematics for Finance – Simple and compound Interest, Effective rate of interest – Annuities, Leasehold estate, Free Hold Estate, Amortization, Immediate Annuity, Present value of an immediate annuity - Discounts and mathematics present values.								17
	Total								90
Weightage of Marks: Theory 20% and Problems 80%									
CourseOutcomes								ProgrammeOutcome	
CO	Upon completion of the course the students would be Able to:								
1	After the study of unit-1, the student will be able to apply basic terms of statistical data solving practical problems in the field of business.								
2	After the study of unit-2, the student will be able to explain basic methods of Measure of central tendency.								
3	After the study of unit-3, the student will be able to solve problems in the areas of simple and compound interest								

	account, use of compound interest.	
4	After the study of unit-4, the student will be able to apply Correlation & regression.	
5	After the study of unit-4, the student will be able to discuss the effects of various types and methods of interest account..	

Text Book

1. Dr. P.R. Vittal, Business Mathematics and Statistics ,Margham Publications,Chennai,2013
2. S P Rajagopalan, R Sattanathan - Business Mathematics,Vijay Nicole Imprints (p) Ltd,2013
3. Prof. A. V. Rayarikar , P. G. Dixit,Business Mathematics and Statistics, Nirali Prakshan Publishers,2019
4. J.K. Sharma Business Statistics,Pearson Publication.
5. P.A.Naviitham , Business Mathematics and Statistics, Jai Publishers, Trichy, April 2013
6. D.Joseph Anbarasu, Business Statistics,Vijay Nicole Imprints Private Limited, Chennai,2015.
7. B L Agarwal Basic Statistics, New Age International Private Limited,New Delhi,4th Edition ,2006.
8. T. Veerarajan, Fundamentals of Mathematical Statistics,Yes Dee Publishing Pvt Ltd,Chennai,2017.
9. R.S.N Pillai, Bagavathi, Statistics Theory and Practice, S. Chand Company LimitedNew Delhi,7th Edition 2008.
10. P. A.Naviitham , Business Statistics & Operations Research, Jai Publishers, Trichy.

Reference Books

1. Agarwal B.M., Business Mathematics & Statistics ,Ane Books Pvt Ltd, 2009
2. Andre Francis Business Mathematics and Statistics, Six Edition
3. A Francis; Ben Mousley,Business Mathematics and Statistics, Andover, United Kingdom Cengage Learning, 2014
4. Prof. A. V. Rayarikar , P. G. Dixit Business Mathematics And Statistics Kindle Edition
5. M.Hajameeral, M.Ravinthammal University Statistical Methods and their applications I, Science Press New Delhi 2013
6. S.P.Gupta, Statistical Methods, Sultan Chand & Sons, New Delhi, 36th edition 2008
7. S.P.Gupta, Statistical Methods, Sultan Chand & Sons,New Delhi, 45th edition 2017
8. S.C. Gupta, V.K.Kapoor Fundamentals of Mathematical Statistics , Sultan Chand & Sons, New Delhi,12th Edition 2020
9. D.C. Sancheti , V.K.Kapoor, Business Mathematics,Sultan Chand & Sons, New Delhi,11th Edition 1993
10. Dr. P.R. Vittal, Business Statistics & Operations Research , Margham Publications,Chennai ,2013

Web Resources

- 1 <https://theintactone.com/2019/09/01/ccsubba-204-business-statistics/>
- 2 <https://ug.its.edu.in/sites/default/files/Business%20Statistics.pdf>
- 3 <http://www.statisticshowto.com>
- 4 <https://statisticsbyjim.com/basics/measures-central-tendency-mean-median-mode/>
- 5 <https://www.toppr.com/guides/business-mathematics-and-statistics/index-numbers/>

Mapping with Programme Outcomes:

CO/PSO	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6
CO 1	3	3	3	3	3	3
CO 2	3	3	2	3	3	3
CO 3	3	2	2	3	2	3
CO 4	3	3	3	2	2	3
CO 5	3	2	3	2	3	2

Weightage of course contributed to each PSO	15	13	13	13	13	14
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S-Strong-3M-Mediu

SECOND YEAR – SEMESTER – III

SKILL BASED SUBJECT-3: AUDITING STANDARD AND PROCEDURE

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst.Hours	Marks		
									CIA	External	Total
	Auditing Standard and Procedure			-	-	-	2	2	25	75	100
Learning Objectives											
LO1	To educate the concept of auditing and audit programmes.										
LO2	To provide insight on internal auditing and vouching of trading transactions										
LO3	To provide the procedures to be followed for the verification and valuation of assets and liabilities.										
LO4	To enhance the practical knowledge relating the procedures of auditing practices of Joint stock companies										
LO5	To enrich knowledge about the provisions of investigation under companies act.										
Unit	Contents									No. of Hours	
I	Auditing– Origin – Definition – Objectives – Types – Advantages and Limitations – Qualities of an Auditor – Audit Programmes									15	
II	Internal Control – Internal Check and Internal Audit –Audit Note Book – Working Papers. Vouching – Voucher – Vouching of Cash Book – Vouching of Trading Transactions – Vouching of Impersonal Ledger.									15	
III	Verification and Valuation of Assets and Liabilities – Auditor’s position regarding the valuation and verifications of Assets and Liabilities – Depreciation – Reserves and Provisions – Secret Reserves.									15	
IV	Audit of Joint Stock Companies – Qualification – Dis-qualifications – Various modes of Appointment of Company Auditor – Rights and Duties – Liabilities of a Company Auditor – Share Capital and Share Transfer Audit – Audit Report – Contents and Types.									15	
V	Investigation – Objectives of Investigation – Audit of Computerised Accounts – Electronic Auditing – Investigation under the provisions of Companies Act.									15	
TOTAL HOURS									75		

CourseObjective	
CO1	The main objectives of this course are to: 1. To understand the various concepts of auditing and the procedure for the conduct of internal audit
CO2	2. To familiarize with the process of valuing assets and liabilities
CO3	3. To understand the process of auditing the joint stock companies and investigation mechanism
TextBook	
1	Practical Auditing -B.N. Tandon Namara,WilliamRab, PearsonPublication
2	Auditing Principles & Practices Pradeep Kumar, Baldev Suchdeva Kalyani Publishers 8 th edition Reprint 2014.
3.	Principles of Auditing Dinkar Pagare Sultan Chand & Sons, New Delhi. 11 th Edition 2007.
References:	
1	Study material of Institute of Chartered Accountants of India.
2	R.M De Paula, “Auditing-the English language Society and Sir Isaac Pitman and Sons Ltd,London
3	Spicer and Pegler, “Auditing: Khatalia’s Auditing” 4. Kamal Gupta, “Auditing “ , Tata Mcgriall Publications
WebResources	
1.	MOOC, SWAYAM, NPTEL, Websites,etc.
2.	https://www.himpub.com/documents/Chapter3442.pdf
3.	http://ebooks.lpude.in/commerce/bcom/term_3/DCOM204_AUDITING_THEORY.pdf

MAPPINGWITHPROGRAMMEOUTCOMESANDPROGRAMMESPECIFIC OUTCOMES

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	3	3	2	3	2	2	3	2	2
CO2	3	2	3	3	3	2	2	2	3	2	2
CO3	3	2	3	3	3	2	2	2	3	2	2
CO4	3	2	3	3	2	2	2	2	3	2	2
CO5	3	2	3	3	3	2	2	2	3	2	2
TOTAL	15	10	15	15	13	11	10	10	15	10	10
AVERAGE	3	2	3	3	2.6	2.2	2	2	3	2	2

3 –Strong,2-Medium, 1-Lo

SECOND YEAR – SEMESTER – IV
CORE PAPER -7 INTERNATIONAL FINANCE

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	5				4	5	25	75	100

Learning Objective

LO1	To get acquainted with the concept of foreign exchange markets	
LO2	To understand the actual problems of Multinational Corporation within an environment	
LO3	To gain knowledge of foreign exchange market techniques	
LO4	To understand the flow of foreign capitals	
LO5	To develop an understanding of floating exchange rate	
Unit	Contents	No. of Hours
I	International Financial Environment: Overview, Nature and Scope of International Finance Evolution of international financial system gold standard, Breton woods standard, floating exchange rate; International Finance Management VS Domestic Financial Management.	15
II	International Financial Markets: Euro currency market, international bond market, international equity market, international money market	15
III	International Financial Institutions: IMF, Bank for International Settlements; international banking-euro bank, types of banking offices-correspondent bank, representative office, foreign branch, subsidiary bank, offshore bank.	15
IV	International Financial Instruments: Introduction to International Financial Instruments Types of International Financial -Euro CP, Eurobonds, foreign bonds, global bonds, euro equity, ADR, GDRs	15
V	Multinational Corporate Decisions in Global Market: Foreign investment decision- Foreign direct investment (FDI)- motives, FDI theories-theory of comparative advantage, OLI paradigm of FDI in India. FII's Definition, role of FII's, Different between FDI & FII.	15
TOTAL HOURS		75

Course Objective	
CO1	To revise the Concept of International Financial Management
CO2	To discuss the Concept of International Financial Markets and International Financial Instruments and FDI
CO3	To identify with the Concept of International Financial Institutions and multinational corporate decisions in Global Markets
Text Book	
1	O.P. Agarwal International Financial Management, 3 rd Edition 2014 HP
2	Gupta Shashi K., Rangi Praneet International Finance 2 nd Edition 2017, Kalyani Publishers
References:	
1	Eun C.S., Resnick B.G., "International Financial Management", 2010, Tata McGraw Hill Education Pvt. Ltd., 4th Ed. Special Indian Edition
2	Shailaja G., "International Finance", 2010, 2nd Ed. Orient Black 'swan.
3	Hendrik VandenBerg, "International Finance and Open Economy Macroeconomics", 2009, 1st Ed. Cambridge.
Web Resources	
1.	https://josephscollege.ac.in
2.	https://rccindore.com

MAPPING WITH PROGRAMME OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	2	3	2	2	2	3	3	2	2
CO2	3	2	3	3	3	2	3	2	3	2	2
CO3	3	2	3	3	3	2	2	2	3	2	2
CO4	2	2	3	3	2	2	3	2	3	2	2
CO5	3	2	3	3	3	2	2	3	3	2	2
TOTAL	14	10	14	15	13	10	12	12	15	10	10
AVERAGE	3	2	3	3	2.6	2.2	2	2	3	2	2

3 –Strong, 2-Medium, 1-Low

SECOND YEAR – SEMESTER – IV

CORE PAPER – 8: ADVANCE TAXATION

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst.Hours	Marks		
									CIA	External	Total
	ADVANCE TAXATION			-	-	-	3	4	25	75	100
Learning Objectives											
LO1	To know about new taxes, concept of Income tax, exemptions and deductions related to incomes.										
LO2	To describe how to arrive taxable income from salary.										
LO3	To find out the taxable income from house property.										
LO4	To calculate the taxable income from Business and Profession.										
LO5	To ascertain the capital gains and income from other sources.										
Unit	Contents									No.of Hours	
I	INCOME TAX ACT 1961 AND RESIDENTIAL STATUS: Income Tax Act, 1961 – Definitions – Basis of charge different types of assessee – Rates of Tax- previous year and Assessment year – capital and revenue income, expenditure and loss – incomes exempted under section10 – Residential status – Scope of Total Income.									15	
II	INCOME FROM SALARY: Definition - Basis of charge – Different forms of salary – Allowances –Perquisites – Gratuity – Pension – Computation of Pension – Leave Salary – Profits in lieu of Salary- Provident Funds -Computation of taxable salary – deductions from salary.									15	
III	INCOME FROM HOUSE PROPERTY: Income from House Property: Basis of charge – determination of annual value – GAV, NAV – income from let-out property – self occupied property – deductions – computation of taxable income.									15	
IV	INCOME FROM BUSINESS OR PROFESSION: Income from Business or Profession: Basis of charge – methods of accounting – deductions – disallowances, computation of taxable income – profit and gains of business and profession.									15	
V	INCOME FROM CAPITAL GAINS: Income from Capital Gains: Basis of charge – short and long term capital gains – indexed cost of acquisition and improvement – exemptions – computation of taxable capital gains – Computation of Income from other sources.(Problem 80 %, Theory 20 %)									15	
TOTAL HOURS									75		

Course Objective	
CO1	Demonstrate the understanding of the basic concepts and definitions under the Income Tax Act.
CO2	Assess the residential status of an assessee & the incidence of tax.
CO3	Compute income of an individual under the head salaries.
CO4	Ability to compute income from house property.
CO5	Evaluate income from a business carried on or from the practice of a Profession.
TextBook	
1	GaurandNarang, "IncomeTax Law and Practice" KalyaniPublishers, NewDelhi.
2	Dr.HC.Mehrotra, "IncomeTax Law and Accounts" SahithyaBhavan publishers.
3	T. Srinivasan – Income Tax & Practice – Vijay Nicole Imprints Private Limited , Chennai.
4	Vinod, K.Singhania, Students Guide to IncomeTax, Taxman Publications Pvt. Ltd.
References	
1	Mehrotra- Income Tax Law&Accounts, Goyal, SathiyaBhavan Publications.
2	A. Murthy " Income Tax Law & Practice" Vijay Nichole, Imprints (P) Ltd.
3	Dr. T.S. Reddy & Hariprasad Reddy ' Income tax' – MarghamPublications, Chennai
Web Resources	
1.	www.incometaxindia.gov.in
2.	www.incometaxindiaefiling.gov.in
3.	www.onlineservices.tin.egov-nsdl.com

**MAPPING WITH PROGRAMME
OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES**

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	2	2	2	2	2	2
CO2	3	2	3	2	2	2	2	2	2	2	2
CO3	3	2	3	2	2	2	2	2	2	2	2
CO4	3	2	3	2	2	2	2	2	2	2	2
CO5	3	2	3	2	2	2	2	2	2	2	2
TOTAL	15	10	15	10	10	10	10	10	10	10	10

AVERAGE	3	2	3	2	2	2	2	2	2	2	2
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3–Strong,2-Medium,1- Low

SECOND YEAR – SEMESTER – IV

ELECTIVE PAPER 4 : BUSINESS STTISTICS - II

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst.Hours	Marks		
									CIA	Internal	Total
	Business Statistics - II			-	-	-	3	4	25	75	100
Learning Objectives											
LO1	To familiarize students with the basic concepts in operation research.										
LO2	To be able to choose rational options in practical decision making problems using standard mathematical models of operation research.										
LO3	To know principles of construction of mathematical models situation and mathematical analysis methods of operation research.										
LO4	To be able to choose rational options in Time Series.										
LO5	To develop skills in analysis of Index Number and weighted Index Numbers.										
UNIT	Contents										No.of Hours
I	Operation Research Operation Research-origin-Definition-various model and modeling-Application and Scope- Merits and De-merits-Linear Programming characteristics-Formulation graphical method solution to Graphical method Alternative method of solving LPP-(simple problems).										15
II	Sequencing Problem Sequencing-Introduction-Sequencing problems-processing n jobs to two machines- processing n jobs to three machines- processing two jobs through m machines- processing n jobs through m machines.Replacement models-Introduction-individual replacement policy-Group replacement policy-Miscellaneous replacement problems(basic problems).										19
III	Game theory Introduction-Terminologies of game theory-game with mixed and pure strategies-value of game-optimum strategy-with saddle point-without saddle point-Dominance property-Graphical solution 2Xn and mX2 games.										18
IV	Time series Time series-component of time series-secular trend-seasonal variation-cyclical variation-irregular variation-measurement of trend-Graphic method-semi Average method-Moving average method-Method of least squares-Measurement of seasonal variations-Simple average method-Ratio to moving average method.										19
V	Index Number Index number-Weighted and UN weighted Index Numbers – Cost of Living Index Number–Average of Relative Price Indices-Quality Index										19

	Number-Test on index Numbers- Time reversal test, Factors reversal test-circular test.	
	Total	90
Weightage of Marks: Theory 20% and Problems 80%		
CourseOutcomes		ProgrammeOutcome
CO	Upon completion of the course the students would be Able to:	
1	After studied unit -1, the students will be able to identify and develop operation research models from the verbal description of the real system.	
2	After studied unit -2, the students will be able use mathematical tools to solve the proposed model.	
3	After studied unit -3, the students will be able to understand the mathematical tools that are needed to solve optimization problems.	
4	After studying unit-4, the student will be able to develop Time Series	
5	After studying unit-5, Students can Use Index Number , Weighted and UN weighted Index Numbers in practical application .	
Text Book		
1. Fundamental of Mathematical Statistics - S.C. Gupta & V.K. Kapoor - Sultan Chand 2 2 Dr. P.R. Vittal Business Mathematics and Statistic, Margham Publications, Chennai , 2013 3. S P Rajagopalan, R Sattanathan - Business Mathematics, Vijay Nicole Imprints (P) Ltd, 2013 4. Dr. P.R. Vittal Operations research - Margham Publications, Chennai. 5. Gurusamy S - Elements of operation Research - Vijay Nochole Imprints (P) Ltd. 6. Prem kumar Gupta & D.S.Hira, Operations research - S. Chand & Company, New Delhi. 7. R. Paneerselvam, Operations research - PHI Learning Pvt. Ltd. 8. J.K. Sharma, Operations research - Laxmi Publications Pvt.Ltd. 9. Dr. P.R. Vittal & V.Malini - Operations research - Margham Publications - Chennai 2017 10. Er Prem Kumar Gupta & Dr.D.S.Hira, Problems in Operation Research, S.Chand & Company Limited, New Delhi - 2020		
Reference Books		
1. Agarwal B.M. Business Mathematics & Statistics Ane Books Pvt Ltd, 2009 2. S P Rajagopalan, R Sattanathan - Business Mathematics, Tata McGraw- Hill Publishing company Ltd, New Delhi, 2009 3. A Francis; Ben Mousley Business Mathematics and Statistics, United Kingdom Cengage Learning, 2014 4. Prof. A. V. Rayarikar , P. G. Dixit Business Mathematics And Statistics, Nirali Prakshan Publishers, 2019 5. B M Aggarwal, Business Mathematics and Statistics, Ane Books Pvt Ltd, 2010 6. Dr.A. Thilagaraj & Dr.N.Soundarraj, Business Mathematics, Learntech Press , Trichy, 2010 7. Dr. P.R. Vittal Business Statistics , Margham Publications, 2009, Chennai 8. J.P.Singh , Business Mathematics , Ane Books Pvt.Ltd New Delhi 2016. 9. S.P.Gupta- Statistical Methods, Sultan Chand & Sons - New Delhi- 36th Edition 2008 10. S.P.Gupta- Statistical Methods, Sultan Chand & Sons - New Delhi- 45th Edition 2017 1. Hamdy A.Taha, Operations Research, Prentice Hall of India, New Delhi, 2007. 2. Kanti Swarup, P.K.Gupta, Manmohan, Operations Research, Sultan Chand & Sons, New Delhi, 2008. 3. Sasieni, Arthur Yaspan, Lawrence Friedman, Operations Research Methods and Problems, Wiley International Edition, 1959. 4. S.D. Sharma, Operations Research, Kedarnath Ram Nath & Co Publishers, 15th Edition 2007. 5. Gurusamy S - Operations Research - Vijay Nichole Imprints (P) Ltd.		

6. Prof. A. Chandasekaran , B.Kalpna, A Text book of Operations Research , Dhanam Publications , 1st Edition,2016
7. J.K.Sharma, Operations Research Theory And Applications , Macmillan Publisher India Limited, New Delhi,2009,4th Edition
8. Dr.S.P.Gupta, Dr.P.K.Gupta & Dr.Manmohan,Business Statistics and Opearations Research,Sultan Chand and Sons , New Delhi,2007
9. S.P.Rajagopalan & R.Sattanatha, Business Statistics and Operations Research 3rd Edition,Vijay Nicole Imprints Pvt Ltd -2014.
10. Dr. P.R. Vittal Business Statistics & Operations Research ,Margham Publications,Chennai ,2013

Web Resources

- 1 chromeextension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.rccmindore.com/w p-content/uploads/2021/04/Operations-Research.pdf
- 2 chromeextension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.bbau.ac.in/dept/UIET/EMER601%20Operation%20Research%20Queuing%20theory.pdf
- 3 https://www.onlinemathlearning.com › linear-programming-example
- 4 https://www.kellogg.northwestern.edu › weber › Notes_6_Decision_trees
- 5 www.pondiuni.edu.in › sites › default › files

Mappingwith Programme Outcomes:

CO/PSO	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6
CO 1	3	3	3	3	3	3
CO 2	3	3	2	3	3	3
CO 3	3	2	2	3	2	3
CO 4	3	3	3	2	2	3
CO 5	3	2	3	2	3	2
Weightageofcourse contributed to each PSO	15	13	13	13	13	14

S-Strong-3 M-Medium-2L-Low-1

SECOND YEAR – SEMESTER – IV
SKILL BASED SUBJECT – 4 : ADVANCED AUDITING

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst.Hours	Marks		
									CIA	External	Total
	Advanced Auditing			-	-	-	2	3	25	75	100
Learning Objectives											
LO1	To gain basic knowledge of the auditing standards, principles and procedures regarding preparation and verification of financial accounting statement										
LO2	To gain an awareness regarding verification of documentary evidence and its importance in auditing.										
LO3	To identify the different ways frauds and errors found in financial statements.										
Unit	Contents										No. of Hours
I	Meaning and Definition of Auditing – Distinction between Auditing and Accounting - Objectives - Advantages and Limitations of Audit - Scope of Audit - Classifications of Audit – Audit Planning - Meaning. Audit programme – Meaning – Objectives and Contents. Audit Note Book, contents, Usefulness of Audit Note Book - Audit working papers - meaning. Ownership and Custody – Test checking and Routine checking - Meaning. Internal control – Meaning – Definition – Objectives – Technique for evaluation of Internal Control System. Internal check – Meaning - Objectives difference between Internal control, Internal check and Internal Audit.										15
II	Vouching and Verification Vouching – Meaning and Definitions - Objectives. Trading Transactions – Audit of Ledger - Scrutinizing of ledgers – Vouching of cash Receipts and Payments - Vouching of outstanding Assets and Liabilities – Verification – Meaning - Objectives and Process – Valuation of Assets and liabilities – Distinction between Verification and Valuation.										15
III	Audit and Accounting Standards Types of Audit – Statutory Audit – Concurrent Audit – Stock Audit – Cost Audit – Secretarial Audit – CAG Audit – Management Audit. Accounting Standards – Standards on Auditing Standards on Internal Audit – Penal Provisions – Role of National Financial Reporting Authority (NFRA)										15
IV	Auditors and Audit Report Appointment – Procedures – Eligibility and Qualifications – Powers and Duties – Rotation and Removal of Auditors – Resignation of Auditors – Remuneration of Auditors - Audit report – Preparation and presentation. Auditor's Responsibilities and liabilities towards Shareholders, Board and Audit Committee. Restriction on other Services.										15
V	Recent Trends in Auditing EDP Audit – Meaning – Division of auditing in EDP environment. Impact of Computerization on Audit Approach – Online Computer System Audit – Types of Online Computer System Audit – Audit around with the										15

	Computers – Procedure of Audit under EDP system- Green Audit- Introduction	
TOTAL HOURS		75

CourseObjective	
CO1	To make the students to understand the concept of present day Auditing Practices.
CO2	To enable the students to gain knowledge of various techniques of Auditing.
Text Books	
1	Auditing, D.P. Jain Konark Publishers Pvt. Ltd.
2	Auditing, Principles and Practice, Ravinder Kumar and Virender Sharma, Eastern Economy Edition..
References Books	
1	Practical Auditing, B.N.Tandon, Sultan Chand and Co., NewDelhi.
2	Contemporary Auditing, Kamal Gupta,Tata McGraw Hill.
3	Practical Auditing, Dinkar Pagare, Sultan Chand & Sons
4	Sundar.K&Paari.K,PracticalAuditing,VijayNicoleImprintsPvt.Ltd.Chennai
WebResources	
1	http://www.osbornebooksshop.co.uk/p/auditing_tutorial
2	www.mu.ac.in
3	www.learnthat.com

Mappingwithprogrammeoutcomesandprogrammespecificoutcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	3	3	2	3	2	2	3	2	2
CO2	3	2	3	3	3	2	2	2	3	2	2
CO3	3	2	3	3	3	2	2	2	3	2	2
CO4	3	2	3	3	2	2	2	2	3	2	2
CO5	3	2	3	3	3	2	2	2	3	2	2
TOTAL	15	10	15	15	13	11	10	10	15	10	10
AVERAGE	3	2	3	3	2.6	2.2	2	2	3	2	2

3 –Strong,2-Medium, 1-Low

THIRDYEAR - SEMESTER - V
CORE PAPER - 9 COST ACCOUNTING

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst.Hours	Marks		
									CIA	External	Total
	COST ACCOUNTING			-	-	-	5	6	25	75	100
Learning Objectives											
LO1	To gain basic knowledge of the auditing standards, principles and procedures regarding preparation and verification of financial accounting statement.										
LO2	To gain an awareness regarding verification of documentary evidence and its importance in auditing.										
LO3	To identify the different ways frauds and errors found in financial statements.										
Unit	Contents									No. of Hours	
I	Nature and scope of cost Accounting Cost Accounting: Nature and Scope-Objectives, Advantages and Limitations-Financial Vs. Cost Accounting-CostingSystem-TypesofCostingandCostClassification-CostSheetandTenders-Cost Unit -Cost Centre and Profit Centre.									10	
II	Material Purchase and Control Purchase Department and its Objectives - Purchase Procedure - Classification and Codification of Materials, Material Control: Levels of Stock and EOQ - Perpetual Inventory System, ABC and VED Analysis - Accounting of Material Losses.									15	
III	Methods of pricing of Material Issues Cost Price Methods: FIFO, LIFO, Average Price Methods: Simple and Weighted Average Price Methods, Notional Price Methods: Standards Price, and Market Price Methods									20	
IV	Labour Cost Control Labour Turnover: Causes, Methods of Measurement and Reduction of Labour Turnover - Idle and Over Time - Remuneration and Incentive: Time and Piece Rate - Taylor’s, Merricks and Gantt’s Task-Premium Bonus System-Halsey ,Rowan and Emerson’s Plans- Calculation of Earnings of Workers.									25	
V	Overheads Classification of overheads cots- departmentalization of overheads – allocation absorption and appointment of overheads costs- primary and secondary distribution of overheads – computation of machine hour rate and labour hour rate.									20	
TOTAL HOURS									90		

Note: Questions in Sec. A, B & C shall be in the proportion of 20: 80 between Theory and Problems.

TEXT BOOKS:

S.no	Authors	Title	Publishers
1	S.P.JainandNarang	Cost Accounting	KalyaniPublishers,NewDelhi
2.	T.S.Reddy&HariPrasad Reddy	Cost Accounting	MarghamPublications, Chennai.
3.	S.P. Iyengar	Cost Accounting	SultanChand&Sons, New Delhi.
4	ManoshDutta	Cost Accounting	DorlingKindersley(India)Pvt. Ltd, 2010
5	A.Murthyand S. Gurusamy,	Cost Accounting	Vijay Nicole Imprints Private Ltd., Chennai.
6	KhannaB.S.PandeyI.M., Ahuja G.K., and Arora M.N	PracticalCosting	S. Chand &Sons
7	AroraM.N	Cost Accounting	S. Chand &Sons
8	R.S.N.Pillai &Bhagavati	Cost Accounting	S. Chand &Sons
9	Bhabatosh Banerjee	Cost Accounting – Theory & Practices	SultanChand &Sons
10	V.KSaxena,C.D Vashist,	Cost Accounting problemsandsolutions	SultanChand &Sons

REFERENCEBOOKS:

S.No	Authors	Title	Publishers
1.	Tulsian	Cost Accounting	TataMcGraw Hills.
2.	S.N.Maheswari	Principles of Cost Accounting	Sultan Chand & sons, New Delhi
3.	ManashDutta,	Cost Accounting	Pearson Education (Singapore) Pvt. Ltd, Second Edition Print, 2005
4	M.C. Shukla, T.S. Grewal, Dr.M.P.Gupta,	Cost Accounting	S.Chand&CompanyLtd, 2010.
5	ReddyandMurthy	Cost Accounting	MarghamPublications
6	Inamdar, S. M. (Satish Inamdar)	Cost & Management Accounting	EverestPublishingHouse
7	Kishore,R.M.	Cost & Management Accounting	TaxmanAllied Service
8	V.KSaxena,C.D Vashist,	Advanced Cost & ManagementAccounting	SultanChand & Sons
9	Jawaharlal	Cost Accounting	MCGrawHill
10	M.E.ThukaramRao	Cost and Management Accounting	NewAgeInternational

Reference Journal

1. BusinessandEconomicsJournal,
2. GlobalEconomics,
3. Accounting& Marketing,
4. AccountingResearch Journal,
5. AsianReviewof Accounting,
6. Asia-PacificJournalofAccountingandEconomics,
7. JournalofAccountingandOrganizationalChange,
8. JournalofContemporaryAccountingand Economics

E-Materials

1. www.icwai.org
2. www.nasbaregistry.org.

MappingwithProgramme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	S	S	S	S	S	S	S	S	S
CO2	S	S	S	S	S	S	S	S	S	S
CO3	S	S	S	S	S	S	S	S	S	S
CO4	S	S	S	S	S	S	S	S	S	S
CO5	S	S	S	S	S	S	S	S	S	S

PO–Programme Outcome, CO –Course outcome

S – Strong, M – Medium, L–Low(may be avoided)

THIRD YEAR - SEMESTER - V
CORE PAPER - 10 INCOME TAX LAW

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst.Hours	Marks		
									CIA	External	Total
	INCOME TAX LAW			-	-	-	5	6	25	75	100
Learning Objectives											
LO1	To acquire Knowledge of Different Income Tax Concepts										
LO2	The Main Objective of Taxation is Economic Development										
LO3	To Overcome the Scarcity of Capital, Taxes are regarded as effective means to Control Inflation										
LO4	To Control Cyclic Fluctuations										
LO5	Reduction of Balance of Payments Difficulties										
Unit	Contents									No. of Hours	
I	Introduction- Income Tax Act 1961- Basic Concepts - Assessment Year - Previous Year - Person – Assessee - Income - Agricultural Income - Capital and Revenue Receipts - Capital and Revenue Expenditures - Exempted Incomes u/s 10. Residential status of an individual-Residential status of a HUF - Residential status of a firm and association of persons - Residential status of a company - incidence of tax liability.									15	

II	INCOME FROM HOUSE PROPERTY Annual value - Determination of annual value- Income from let out house property - Income from self-occupied house property - Deductions allowed from Income from house property u/s 24.	15
III	SALARIES Meaning and features of Salary - Allowances - Perquisites - Profits in lieu of Salary - Provident Fund and its types - payments exempted u/s 10: Leave travel concession; gratuity; pension; leave encashment; retrenchment compensation; VRS - Deductions from salary: EA and professional tax- deduction u/s 80C- taxable salary	30
IV	PROFIT AND GAINS OF BUSINESS OR PROFESSION AND DEPRECIATION Meaning of business and profession - deductions expressly allowed - expenses expressly disallowed - treatment/admissibility of certain expenses and incomes - income from business- income from profession-Meaning of depreciation - conditions for depreciation - actual cost - written down value-computation of allowable depreciation.	20
V	INCOME TAX AUTHORITIES CBDT - powers - Director General of income tax - Chief commissioner of income tax - Assessing officer - appointment - Jurisdiction - powers relating to search and seizure.	10
TOTAL HOURS		90
THEORY 20% & PROBLEM 80%		

TEXT BOOKS:

S.NO	AUTHORS	TITLE	PUBLISHERS
1.	Gaur & Narang	Income Tax Law & Practice	Kalyani Publishers
2.	Dr. A. Murthy	Income Tax Law & Practice	Vijay Nicole Imprints Pvt. Ltd. Chennai
3.	Reddy, T.S. & Hari Prasad Reddy,	Income Tax Theory, Law & Practice	Margham Publications, Chennai.
4.	V.B. Gaur & Narang	Income Tax Law And Practice	Kalayani Publishers, 2001
5.	Dr. Vinod K. Singhania	Income Tax Law And Practice	Taxmann Publications Pvt. Limited, 2005.
6.	T.N. Manoharan & H. R. Hari	Taxation	Ankit Thakkar for snow white publication pvt ltd
7.	A. Murthy	Income Tax Law And Practice	Vijay Nicole

8.	N.Hariharan	IncomeTaxLawAnd Practice	Mcgrewhill
9.	T .Srinivasan	IncomeTaxLawAnd Practice	VijayNicole
10.	Rajavelu	IncomeTaxLawAnd Practice	S.V.P publications

REFERENCEBOOKS:

S.NO	AUTHORS	TITLE	PUBLISHERS
1.	Mehrotra	IncomeTaxLaw&Accounts	SahithiyaBhavan Publications
2.	Vinod,K.Singhania	StudentsGuidetoIncomeTax	TaxmanPublicationsPvt. Ltd
3.	Anita Raman	IncomeTaxLaw&Practice	McGrawHill
4	V.BalaChandran,S. Thothadri,	TaxationLawand Practice	Published by Asoke K. Ghosh, PHI Learning PrivateLimited, Volume1, 2003
5	V.P Gaur	Incometax law& practice	Kalyani
6	Dr.H.CMehrotra	Taxationlawand practice	Sathiyabhawan
7	M.Jeevarathinam	Incometax law& practice	Winnerswisdom
8	Expert Teacher	Taxationlawand practice	Sathishandbrothers
9	G.S.Mitra	Incometax law& practice	Mahaveer publication
10	Dr.R.K.Jain	Taxationlawand practice	SPBD publication

Reference Journal

1. Indian Journal of Tax Law
- 2.Taxman.com/Journal
3. Vision Journal of Indian Taxation
4. Income Tax Reports, Company Law, Institute of India Pvt Ltd, Chennai

E-Materials

1. GSTand IncomeTaxFortnightlyEMagazine
2. CapitalGainClearTax
3. Indiafiling.com
4. ClearTax.in
5. IncomeTaxManagement.com

MappingwithProgramme Outcomes

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	S	S	S	S	S	S	S	S	S
CO2	S	S	S	S	S	S	S	S	S	S
CO3	S	S	S	S	S	S	S	S	S	S
CO4	S	S	S	S	S	S	S	S	S	S
CO5	S	S	S	S	S	S	S	S	S	S

PO–Programme Outcome, CO –Course outcome
S –Strong, M – Medium,L–Low(maybeavoided)

THIRDYEAR - SEMESTER - V

CORE PAPER – 11 AUDITING AND ASSURANCE

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst.Hours	Marks		
									CIA	External	Total
	Auditing And Assurance			-	-	-	4	5	25	75	100
Learning Objectives											
LO1	Students are able to study an audit report and analyze the company performance.										
LO2	Students are able to do differentiate between internal audit and external audit.										
LO3	Students are made aware of the regulations and rules under various acts regarding Auditing.										
LO4	Collect the information about types of audit conducted in any one organization.										
LO5	Visit an audit firm; write about the procedure followed by them in auditing the books of accounts of a firm.										

UNIT	Contents	No. of Hours
I	INTRODUCTION OF AUDITING [STANDARDS OF AUDITING (SA 200-299)] Auditing – Meaning, Definition, Objectives of an audit – Primary & Secondary objective, Case Laws on Audit Objectives, Types of Audit – Statutory & Independent Audit, Meaning of errors, Classification of errors, Its detection by an auditor, Frauds – meaning, intention, classification & detection by auditor and prevention of frauds by an auditor, Window dressing of financial statements, An overview of Auditing and Assurance Standards issued by ICAI.	15
II	AUDIT PLANNING [STANDARDS OF AUDITING (SA 200 – 599)] Commencing an Audit – Audit Engagement letter, Commencement procedures – (SA 210) Documentation – Documentation as under SA 230 – Audit working papers, Audit files: Permanent and current audit files, Ownership and custody of working papers, Materiality (SA 320), Audit Evidence (SA 500) and documentation.	15
III	INTERNAL CONTROL (SA 265) Concept of Internal Control, Internal check & Internal Audit, objectives, Features of a good Internal Control System, Methods of recording, Existing Internal Control Systems followed by an auditor – Questionnaire, Check list & flow chart methods, Role of the management, Evaluation of Internal Control System, Reporting to clients on Internal Control Weakness (SA 265) . Vouching & Verification in specific areas—receipt / payments, sales and debtors, purchase and creditors, cash and bank balance, fixed assets and investments.	15
IV	THE COMPANY AUDITOR Qualification & Disqualification of an Auditor, Appointment of Auditor, Remuneration, Duties of an Auditor, Rights & Liabilities of an Auditor as per Companies Act 2013.	15
V	AUDIT REPORT (SA 700 - 799) Auditors Opinion, Audit Report- Characteristics, Importance and Significance of Audit Report, Types of Audit Report, Qualifications in the Auditors Report - Audit examination, Information & explanation from the management, Statement of facts v/s. Expression of opinion, date of report & signing,	15
TOTAL HOURS		75

Course Objective	
CO1	To gain basic knowledge of the auditing standards, principles and procedures regarding preparation and verification of financial accounting statements.
CO2	To gain an awareness regarding verification of documentary evidence and its importance in auditing.
CO3	To identify the different ways frauds and errors found in financial statements.
Text Book	
1	B. N. Tandon, S. SudharsanamandS. Sundharabahu- A Handbook of PracticalAuditing -S. Chand & Co. Delhi
2	B. N. Tandon-Auditing - S. Chandra & Co. Ltd., Delhi.
3.	D. P. Jain-Auditing - Konark Publishers Pvt. Ltd, Delhi
References:	
1	DinakarPagare-Practice of Auditing
2	Dr. T. R. SharmaAuditing - SahityaBhavan, Agra.
3	Kamal Gupta & Ashok Arora-Fundamentals of Auditing - Tata McGraw Hills
Web Resources	
1.	Jagadeesh Prakash-Auditing

MAPPING WITH PROGRAMME OUTCOMESAND PROGRAMME SPECIFIC OUTCOMES

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	3	3	2	3	2	2	3	2	2
CO2	3	2	3	3	3	2	2	2	3	2	2
CO3	3	2	3	3	3	2	2	2	3	2	2
CO4	3	2	3	3	2	2	2	2	3	2	2
CO5	3	2	3	3	3	2	2	2	3	2	2
TOTAL	15	10	15	15	13	11	10	10	15	10	10
AVERAGE	3	2	3	3	2.6	2.2	2	2	3	2	2

3 – Strong, 2- Medium, 1- Low

THIRDYEAR - SEMESTER - V

ELECTIVE PAPER : 5 – ENTREPRENEURIAL DEVELOPMENT

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
	4				2	4	25	75	100
Learning Objectives									
LO1	To know the meaning and characteristics of entrepreneurship								
LO2	To identify the various business opportunities								
LO3	To understand the Process of setting up an enterprise								
LO4	To gain knowledge in the aspects of legal Compliance of setting up of an enterprise								
LO5	To develop an understanding of the role of MSME in economic growth								
Prerequisite: Should have studied Commerce in XII Std									
Unit	Contents								No. of Hours
I	Introduction to Entrepreneur Meaning of Entrepreneurship–Characteristics of Entrepreneurship – Types of Entrepreneurships–Self Employment– Difference between Entrepreneurship and Employment– Meaning of Entrepreneur– Classification.								12
II	Design Thinking Idea Generation–Identification of Business Opportunities– Design Thinking Process–Creativity–Invention–Innovation– Differences–Value Addition–Concept and Types–Tools and Techniques of Generating an idea–Turning Idea in to Business Opportunity.								12
III	Setting up of an Enterprise Process of Setting Up an Enterprise–Forms of an Enterprise–Sole Proprietorship – Partnership – Limited Liability Partnership Firm – Joint Stock Company – One Man partnership – Choice of Form of an Enterprise–Marketing, Technical, Financial, Commercial and Economical.								12
IV	Business Model Canvas and Formulation of Project Report Introduction – Contents of Project Report–Project Description– Market Survey–Fund Requirement–Registration–Source of Funds– Modern Sources of Funds.								12
V	MSME's and Support Institutions Government Schemes and Women Entrepreneurship– Importance of MSME for Economic Growth–MSME–Definition– Role of Government Organizations in Entrepreneurship Development –MSME DI – DIC – Khadi and Village Industries Commission – NSIC– NABARD, SICVI, SFC.								12

TOTAL		60
Course Outcomes		
CO1	Identify the various trait so fan entrepreneur	
CO2	Turnideasintobusinessopportunities	
CO3	Dofeasibilitystudybeforestartingaproject	
CO4	Identifythesourcesoffundsforfundingaproject	
CO5	DevelopanunderstandingabouttheGovernmentschemesavailablefor womenentrepreneurs	
Textbooks		
1	JayashreeSuresh,(Reprint2017)EntrepreneurialDevelopment,MarghamPu blications.Chennai.	
2	Dr.C.B.Gupta&Dr.S.S.Khanka(Reprint2014).EntrepreneurshipAndSmallBusinessM anagement, Sultan Chand & Sons, NewDelhi.	
3	CharantimathPoornima,(Reprint2014.),Entrepreneurshipdevelopment- Small,PearsonEducation,India.	
4	RajShankar,(Reprint2016),EntrepreneurshipTheoryandPractice,VijayNicoleandImpr intsPvt.Ltd, Chennai.	
5	Vasant Desai, (Reprint 2017).Dynamics ofEntrepreneurialDevelopment &ManagementTwentyFourthEdition.HimalayaPublishingHouse.Mumbai.	
ReferenceBooks		
1	Anilkumar,Poornima,PrinciplesofEntrepreneurialdevelopment,Newagepub lication,Chennai.	
2	Dr.A.K.singh,Entreprenuerialdevelopmentandmanagement,Laxmipublications,Chenn ai.	
3	Dr.R.K.Singal,Entreprenuerialdevelopmentandmanagement,S.K.Katariapublishers,Ne wDelhi.	
4	Dr.M.C.Garg,EntrepreneurialDevelopment,NewDelhi.	
5	E.Gordon,K.Natrajan,Entreprenuerialdevelopment,Himalayapublishing,Mumbai.	
NOTE:LatestEditionofTextbooksMaybeUsed		

Web Resources	
1.	https://www.interaction-design.org/literature/topics/design-thinking
2.	https://www.bms.co.in/steps-involved-in-setting-up-of-an-enterprise/
3.	http://www.msme.gov.in/

MAPPING WITH PROGRAMME OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PSO 1	PSO 2	PSO 3
CO1	3	2	3	2	3	2	3	3	3	2	2
CO2	3	2	2	2	2	2	2	2	3	2	3
CO3	3	3	3	2	3	2	3	3	3	2	2
CO4	3	2	2	2	2	2	2	2	3	2	2
CO5	3	3	3	2	3	2	3	3	3	2	3
TOTAL	15	12	13	10	13	10	13	13	15	10	12
AVERAGE	3	2.2	2.6	2	2.6	2	2.6	2.6	3	2	2.4

3–Strong,2-Medium,1-Low

THIRD YEAR - SEMESTER - V

ELECTIVE PAPER: 6 INTERNATIONAL FINANCIAL REPORTING STANDARDS

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst. Hours	Marks		
									CIA	External	Total
	International Financial Reporting Standards			-	-	-	2	4	25	75	100
Learning Objectives											
LO1	Students will be able to apply conceptual and regulatory frameworks for financial reporting										
LO2	Students will gain knowledge on Account for transactions in accordance with International accounting standards										
LO3	Students can analyze and interpret financial statements										
LO4	Students will be equipped with concepts of business combinations in accordance with International accounting standards										
LO5	Preparation of consolidated financial statements										

Unit	Contents	No. of Hours
I	International Financial Reporting Standards Meaning of IFRS - relevance of IFRS to India; merits and limitations of IFRS; process of setting IFRS- Practical challenges in implementing IFRS.	15
II	Accounting for Transactions Accounting for transactions in financial statements: Tangible non-current assets - Intangible assets - Impairment of assets - Financial instruments - Leasing - Provisions and events after the reporting period.	15
III	Analysis and Interpretation of Financial Statements Analyzing and interpreting financial statements :Limitations of financial statements - Calculation and interpretation of accounting ratios and trends to address users' and stakeholders' needs - Limitations of interpretation techniques - Specialized, not-for-profit, and public sector entities	15
IV	Financial Statements Preparation of financial statements: Preparation of single entity financial statements: Statement of Financial position- statement of profit and loss account and other comprehensive income- statement of changes in equity- Statement of Cash flows.	15
V	Consolidated Financial Statements Preparation of consolidated financial statements: Consolidated statement of financial position Consolidated statement profit and loss account and other comprehensive income	15
	TOTAL HOURS	75

Course Objective	
The objective of the subject is to enable the students to understand the need and method of presentation of financial statements in accordance with IFRS.	
To develop knowledge and skills in understanding and applying accounting standards and the theoretical framework	
References	
Financial Reporting, BPP Learning Media LTD, 2016	
Advanced Financial Accounting, David Pendrill and Richard Lewis, Pearson Education	
Web Resources	
1	Financial Reporting, Becker Educational Development Corp., 2016
2	Financial Reporting, Kaplan Publishing, 2016

**MAPPING WITH PROGRAMME OUTCOMES AND PROGRAMME
SPECIFIC OUTCOMES**

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PSO 1	PSO 2	PSO 3
CO1	3	2	3	3	2	3	2	2	3	2	2
CO2	3	2	3	3	3	2	2	2	3	2	2
CO3	3	2	3	3	3	2	2	2	3	2	2
CO4	3	2	3	3	2	2	2	2	3	2	2
CO5	3	2	3	3	3	2	2	2	3	2	2
TOTAL	15	10	15	15	13	11	10	10	15	10	10
AVERAGE	3	2	3	3	2.6	2.2	2	2	3	2	2

3 – Strong, 2- Medium, 1- Low

THIRDYEAR - SEMESTER - V

INTERNAL ELECTIVE: 1 CORPORATE GOVERNANCE AND BUSINESS ETHICS

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst. Hours	Marks		
									CIA	External	Total
	Corporate Governance And Business Ethics			-	-	-	2	3	25	75	100
Learning Objectives											
LO1	To get acquainted with the concept corporate governance and business ethics.										
LO2	To develop an understanding of the various sources of business ethics and evaluating business ethics.										
LO3	To gain knowledge of corporate bodies and forums.										
LO4	To understand the corporate governance committee.										
LO5	To develop an understanding of social responsibilities and business ethics.										
Unit	Contents									No. of Hours	
I	Corporate Governance - meaning - objectives - need - importance - principles - corporate - governance and organisation success. Corporate governance in India.									15	
II	Levels of governance structure - Corporate governance and role, responsibilities, and powers - Board of directors, Corporate Management committee and Divisional Management Committee.									15	
III	Corporate Governance Forums - CII Code on corporate governance - features - various Corporate Governance forums - CACG, OECD, ICGN, AND NFCG.									15	
IV	Corporate Social Responsibility - Definition - nature - levels - phases and approaches, principles, Indian models - dimensions. Corporate social reporting - Objectives of corporate Social Reporting and case studies.									15	
V	Business Ethics - meaning, significance, scope -factors responsible for ethical and unethical business decisions. Unethical practices in business - Business ethics in India - Ethics training programme.									15	
TOTAL HOURS									75		

Course Objective	
C01	To gain knowledge and an insight into the corporates and business ethics.

C02	To learn techniques of various business ethics.
C03	To build capability for applying such learning to address of corporate bodies and related business ethics programmes.
Text Book	
1	Dr. Neeru Vasishth and Dr.Namita Rajput - Corporate Government values and ethics, Taxman Publications Pvt Ltd, New Delhi.
2	S.Sankaran - International Business and environment, Margam Publications, Chennai.
3.	Dr.S.S. Khanka - Business Ethics and Corporate Government , S.Chand Publications
References:	
1	Sunder. K. Business Ethics and values, vijay nichole Prints, Chennai.
2	Taxman - Corporate Governance, Indian Institute of Corporate Affairs.
3	A.C. Fernando, K.P. Muralidharan and E.K. Satheesh - Corporate Governance, Principles, policies and practices, Pearson Education.
Web Resources	
1.	http://elearningindustry.com
2.	http://essentialskillz.com
3.	http://journal of corporate governance research - macrothink institute.

**MAPPING WITH PROGRAMME OUTCOMES AND
PROGRAMME SPECIFIC OUTCOMES**

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	3	3	2	3	2	2	3	2	2
CO2	3	2	3	3	3	2	2	2	3	2	2
CO3	3	2	3	3	3	2	2	2	3	2	2
CO4	3	2	3	3	2	2	2	2	3	2	2
CO5	3	2	3	3	3	2	2	2	3	2	2
TOTAL	15	10	15	15	13	11	10	10	15	10	10
AVERAGE	3	2	3	3	2.6	2.2	2	2	3	2	2

3 – Strong, 2- Medium, 1- Low

THIRDYEAR - SEMESTER - V
SKILL BASED SUBJECT: 5 FINANCIAL ANALYSIS AND INTERPRETATION

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst. Hours	Marks		
									CIA	External	Total
	Financial analysis and interpretation			-	-	-	2	2	25	75	100
Learning Objectives											
LO1	To get acquainted with the concept of financial analysis and interpretation.										
LO2	To develop and understanding of the various sources of financial interpretations.										
LO3	To gain knowledge of financial analysis techniques.										
LO4	To understand the risk about financial growth.										
LO5	To develop an understanding financial profitability growth.										
Unit	Contents										No. of Hours
I	Introduction to financial Analysis - Meaning - Objectives - importance - functions - Types of financial statements - Tools and techniques of Financial analysis.										15
II	Financial Statement analysis - Analysis and interpretation of financial statements - Nature significance - Types of Financial Analysis - Tools of Analysis - Comparative Statement - Common size Statement - Trend analysis. Cash flow and Fund flow analysis - Concepts and importance - preparation of cash flow statement (as per accounting standards) - Fund flow statement.										15
III	Analysis and Profitability and Growth - Analysis of Operating efficiency - marginal costing - Break even Analysis - Valuation of analysis - valuation of firms and equity shares - Earnings per share (EPS) - Price to Earnings Ratio (P/E).										15
IV	Financial planning and forecasting - Techniques of Financial Forecasting - Pro Forma Financial Statements - Risk and return analysis - Understanding Risk in Financial Decisions - Expected Returns and Variance - Risk, return Trade off.										15
V	Interpretation of financial Data - Interpretation for decision making - Case studies and real life examples										15
	TOTAL HOURS										75

Course Objective	
CO1	To gain knowledge and an insight into the financial analysis.
CO2	To learn financial techniques.
CO3	To build a capability for applying such learning of financial interpretations.
References	
1	Hansen - Mowen, Cost Management Accounting and control, south western college.
2	Financial statement analysis and security valuation - stephen H.Penman
3.	Principles of corporate finance-Richar A.Brealey, Stewart C.Myers, and Franklin Allen.
4.	Financial management theory and practice- EugeneF.Brigham and Michale Ehrhardt.
5.	Fundamentals of financial management - James C.Van Horne and John M.Wachowicz
6.	Analysis for financial management - Robert C. Higgins.
Web Resources	
1	WWW.accountingcoach.com
2	WWW.financialtimes.com

MAPPING WITH PROGRAMME OUTCOMESAND PROGRAMME SPECIFIC OUTCOMES

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	3	3	2	3	2	2	3	2	2
CO2	3	2	3	3	3	2	2	2	3	2	2
CO3	3	2	3	3	3	2	2	2	3	2	2
CO4	3	2	3	3	2	2	2	2	3	2	2
CO5	3	2	3	3	3	2	2	2	3	2	2
TOTAL	15	10	15	15	13	11	10	10	15	10	10
AVERAGE	3	2	3	3	2.6	2.2	2	2	3	2	2

3 – Strong, 2- Medium, 1- Lo

THIRDYEAR - SEMESTER - VI
CORE PAPER - 12 ADVANCED FINANCIAL MANAGEMENT

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst. Hours	Marks		
									CIA	External	Total
	Advanced Financial Management			-	-	-	5	7	25	75	100
Learning Objectives											
LO1	To have the understanding of the functions of Finance Management.										
LO2	To expand the awareness of Long-term Sources of Funds.										
LO3	To facilitate the students to the understanding of Capital Structure and Leverage										
LO4	To bring subject knowledge about capital investment decision among the students.										
LO5	To let students to be acquainted with the subject of working capital management.										
Unit	Contents										No. of Hours
I	Financial Management - Financial Management - Functions - Goals of Financial Management - Maximization Vs. optimizations - Risk-return trade off.										18
II	Management of Funds - Management of funds – Long-term sources - Shares and Debentures - Convertible Securities and Term Loans - Working Capital financing - Sources and approaches- Bank Credit-Basic principles and Methods of assessment- Other sources of short-term finance Operating environment of working capital										18
III	Capital Structure of Planning - Capital structure planning: Concepts of cost of capital - cost of equity, debt, retained earning- Weighted average cost of capital - Capital structure theories - Net income, Net operating income, MM and Traditional Theories - Leverage - Types and significance. Dividend policy and practices - Dividend policies - Factors affecting dividend decision - Dividend theories - Graham, Gordon, Walter and MM Theories.										18
IV	Management of Fixed Assets - Management of fixed assets - Evaluation of capital investment decision: Payback period - ARR - IRR - NPV - CAPM.										18
V	Working Capital Management - Working capital management-working capital cycle-forecasting of working capital requirements-Factors influencing working capital-Management of inventory, cash and accounts receivables-payables management-credit and collection policies.										18
	TOTAL HOURS										90

Text Books

1. I M Pandey, Financial Management, Vikas Publishing House Pvt Ltd.
2. John H Hampton, Financial Decision Making, Prentice Hall of India Ltd.
3. Rustagi, R. P. Galgotia, Financial Management: Theory, Concepts & Problems
4. Inamdar, Financial Management, S.M. Everest
5. A. Murthy, Financial Management, Margham Publications

Reference Books

1. Prasanna Chandra, Financial Management, Tata McGraw Hill Publishing Company Limited.
2. M.Y.Khan and P.K.Jain, Financial Management, Tata McGraw Hill Publishing Company Limited.
3. P.V.Ratnam, Financial Management Theory, Problems and Solutions, Kitab Mahal.
4. Eugene F. Brigham and Michael C. Ehrhardt, Financial Management: Theory & Practice,
5. Kishore R.M, Financial Management, Taxman Allied Service

Course Materi Mapping with Programme Outcomes

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	M	M	S	S	S	M	M	S	M
CO2	S	M	S	S	S	M	M	S	S	S
CO3	S	S	S	S	M	S	M	S	M	S
CO4	S	S	S	S	M	M	S	S	M	M
CO5	S	S	S	S	M	S	S	M	S	M

PO – Programme Outcome, CO – Course outcome S – Strong, M – Medium, L – Low
(may be avoided)

THIRDYEAR - SEMESTER - VI
CORE PAPER - 13 TAXATION OF BUSINESS INCOME

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst.Hours	Marks		
									CIA	External	Total
	Taxation Of Business Income			-	-	-	5	7	25	75	100
Learning Objectives											
LO1	Students who complete this course will be able to computation of total income tax liability.										
LO2	To understand the basic concepts and definitions of Income from Business and Profession										
LO3	To Assess companies provisions and rules. To develop ability to calculate taxable income of firms, cooperative societies and charitable trust.										
LO4	To Study the Registration and Computation of GST.										
LO5	To understand the provisions Relating to appeals, Revision, Penalties and prosecutions, Tax Administration.										
Unit	Contents									No. of Hours	
I	Definitions (Theory only) Person, Assessee, Assessment Year, Previous Year, Casual income, Agricultural income, Dividend, Total Gross income & Total Taxable income. Residential Status (Theory only) Residential Status of and Incidence of tax (for an individual assessee only) General deduction available under section 80 C, 80 D, 80 E, 80 G and 80 U Exempted Incomes									15	
II	Income from Business and Profession: Concept of Income from Business and Profession – Computation of Income on estimated basis of particular business. (U/S 44AD, 44AE, and 44ADA.)									15	
III	Assessment of Firm, Co-operative Societies, HUF, Company: Assessment of Firm: Provision and rules – Computation of total income tax liability. Assessment of non-resident of India. Assessment of Hindu undivided family: Provisions and rules – Computation of total income and tax liability. Assessment of Co-operative societies: Provisions and rules – Computation of total income and tax liability. Assessment of Company: Provisions and rules, Computation of total income and tax liability. Assessment of charitable and other trust.									15	
IV	GST: Meaning of GST – IGST – Scope of GST – Benefits of GST – Registration under GST – Rules – Procedure of Registration - Computation of Taxable Value and Tax Liability [CSGT &SGST] - Procedure Relating To Levy (IGST) - Input Tax Credit (ITC) - Simple Problems on Utilization Of Input Tax Credit.									15	

V	Procedure of Assessment: Tax Deduction and collection Number (TAN), Permanent Account Number (PAN), e-Fileing of return, Deduction of Tax at source – Advance payment of Tax – Recovery and refund of tax – penalties and prosecution – Appeal and Revision, Tax Administration.	15
TOTAL HOURS		75
THEORY 20% & PROBLEMS 80%		

Course Objective	
CO1	Demonstrate the understanding of the basic concepts and definitions under the Income Tax Act.
CO2	Evaluate income from a business carried on or from the practice of a Profession.
CO3	Assess the residential status of an assessee& the incidence of tax.
CO4	The students will get a general understanding of the GST law in the country.
TextBook	
1	Reddy,T.S. &HariprasadReddy,Y,Income Tax Theory, Law & Practice,Margham Publications,Chennai.
2	AhujaGirish and Ravi Gupta, “Systematic Approach to Income Tax”, Bharat Law House, Delhi.
3	Singhaniavinod k. and Singhaniamonica “Students guide to Income Tax” ,Axmann publicationPvt ltd, New Delhi.
4	Gaur &Narang, Income Tax Law & Practice, Kalyani Publishers.
References	
1	Hariharan N, Income Tax Law & Practice, Vijay Nicole Imprints Pvt. Ltd.Chennai.
2	T.Srinivasan – Income Tax & Practice –Vijay Nicole Imprints Pvt. Limited,Chennai.
3	Dr.A.Murthy, Income Tax Law & Practice, Vijay Nicole Imprints Pvt. Ltd.Chennai
Web Resources	
1.	https://cleartax.in/s/residential-status/
2.	https://www.legalraasta.com/itr/income-from-salary/
3.	https://www.incometaxmanagement.com/Direct-Taxes/AY-2021-22/assessment/1-assessment-of-an-individual.html .

MAPPING WITH PROGRAMME OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	2	3	2	3	2	2
CO2	3	2	3	2	3	2	3	2	3	2	2
CO3	3	2	3	2	3	2	3	2	3	2	2
CO4	3	2	3	2	2	2	3	2	3	2	2
CO5	3	2	3	2	2	2	3	2	3	2	2
TOTAL	15	10	15	10	12	10	15	10	15	10	10
AVERAGE	3	2	3	2	2.4	2	3	2	3	2	2

3–Strong,2-Medium,1- Low

THIRD YEAR - SEMESTER - VI
CORE PAPER - 14 AUDITING AND FINANCIAL REPORTING

Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst.Hours	Marks		
									CIA	External	Total
	Auditing and Financial Reporting			-	-	-	4	6	25	75	100
Learning Objectives											
LO1	To make the students to understand the concept of present day Auditing Practices.										
LO2	To enable the students to gain knowledge of various techniques of Auditing.										
LO3	To get knowledge of the financial accounting and reporting frameworks of the business.										
LO4	To achieve the knowledge and skill of accounting principal										
LO5	To get knowledge how to perform in corporate finance professionals.										
Unit	Contents										No. of Hours
I	Audit programme – Meaning – Objectives and Contents. Audit Note Book, contents, Usefulness of Audit Note Book - Audit working papers - meaning. Routine checking - Meaning. Internal check – Meaning - Objectives difference between Internal control, Internal check and Internal Audit.										15

II	Types of Audit – Statutory Audit – Concurrent Audit – Stock Audit – Cost Audit – Secretarial Audit – CAG Audit – Management Audit. Accounting Standards – Standards on Auditing Standards on Internal Audit – Penal Provisions – Role of National Financial Reporting	15
III	Balance sheet - income statement - Statement of Comprehensive Income - Statement of changes in equity - Statement of cash flows - Integrated reporting	15
IV	Acquisition of Fixed Assets - Capitalization of Interest - Costs Incurred After Acquisition - Depreciation - Impairment - Asset Retirement Obligation - Disposal & Involuntary Conversions - Knowledge-based intangibles (R&D, software) - Legal rights based intangibles (patent, copyright, trademark, franchise, license, leasehold	15
V	Paid-in capital - Retained earnings - Accumulated other comprehensive income - Stock dividends and stock splits - Stock options - Business Combinations & Consolidations	15
	TOTAL HOURS	75
	REFERENCES:	
1	Intermediate Accounting; Donald E. Kieso, Jerry J. Weygandt, Terry D. Warfield- Wiley	
2	Advanced Accounting; Joe Ben Hoyle, Thomas Schaefer, Timothy Doupnik; Mc Graw Hill	
3	Practical Auditing, B.N. Tandon, Sultan Chand and Co., New Delhi.	
	WEB RESOURCES:	
1	http://www.osbornebooksshop.co.uk/p/auditing_tutorial	
2	www.learnthat.com	

MAPPING WITH PROGRAMME OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3
CO1	3	2	3	3	2	3	2	2	3	2	2
CO2	3	2	3	3	3	2	2	2	3	2	2
CO3	3	2	3	3	3	2	2	2	3	2	2
CO4	3	2	3	3	2	2	2	2	3	2	2
CO5	3	2	3	3	3	2	2	2	3	2	2
TOTAL	15	10	15	15	13	11	10	10	15	10	10
AVERAGE	3	2	3	3	2.6	2.2	2	2	3	2	2

3 –Strong, 2-Medium, 1-Low

THIRDYEAR - SEMESTER - VI**SKILL BASED SUBJECT 6 : TALLY&ENTERPRISE RESOURCEPLANNING**

Subject Code	L	T	P	S	Credits	Inst. Hours	Marks		
							CIA	External	Total
					3	4	25	75	100
Learning Objectives									
LO1	To explore to the aspects relating a creation of Company.								
LO2	To learn about stock and vouchers.								
LO3	To be acquainted with Taxation and the concept BRS.								
LO4	To learn about implement of ERP.								
LO5	To assimilate knowledge on ERP packages.								
Prerequisite: Should have studied Commerce in XII Std									

Unit	Contents	No. of Hours
I	INTRODUCTION OF TALLY Introduction to Tally– Company Creation–Select a company– Alter–Display – Shut a company– Delete a company– Accounts info – Ledger Creation – single ledger –multiple ledger– Altering and Displaying ledger – Delete a ledger– Group creation – Single group creation – Multiple group creation – Alteration & Delete a group– Business Transactions – Types of Business Transactions–Bookkeeping.	12
II	INVENTORY&VOUCHERS Inventory info – Creation of Stock Group –Creation of Stock Category– Creation of Unit of Measures – Creation of Godowns – Reports- Inventory vouchers–Re-order level and Re-order Quantity–Batch-Bill wise Details– Bill of Material– Accounting vouchers	12
III	TAXATION Statutory and Taxation–Value Added Tax(VAT) –Tax Deducted at Source (TDS)–Tax Collected at Source(TCS) –Service Tax– Security Control and Tally audit – Export and Import – Backup and Restore – Payroll features in ERP – Cheque Management – Bank Reconciliation System.	12
IV	ERP INTRODUCTION&IMPLEMENTATION ERP: An Overview, Benefits of ERP, ERP and Related Technologies, Business Process Reengineering (BPR), Data Warehousing, Data Mining, OLAP, SCM. ERP Implementation Life cycle, Implementation Methodology, Hidden Costs, Organizing the Implementation, Vendors, Consultants and Users, Contract with Vendors.	12
V	THE BUSINESS MODULES Business modules in an ERP Package, Finance, Manufacturing, Human Resources, Plant Maintenance, Materials Management, Quality Management, Sales and Distribution	12

	TOTAL	60
CO	Course Outcomes	
CO1	Understanding the creation of company and Ledger in Tally	
CO2	Understanding the creation of stock categories in Tally	
CO3	Understanding the importance of GST in Tally	
CO4	Understanding the Characteristics the ERP implementation procedures	
CO5	Understanding the elements of ERP and available of ERP packages.	
Textbooks		
1	AlexisLeon,“ERPDemystified”,TataMcGrawHill,NewDelhi,2000	
2	MSOffice2000forEveryOne - VikasPublishingHousePvt.Ltd.,SanjaySaxena.	
3	MSOffice2000ProfessionalFast&Easy–JuneSlton,BPBpublications.	
4	TheABC&MS OfficeProfessionalEdition-GuyHart-Davis..	
ReferenceBooks		
1	JosephABrady,EllenFMonk,BretWagner, “ConceptsinEnterprise Resource Planning”,ThompsonCourseTechnology,USA,2001.	
2	VinodKumarGargandVenkatakrishnanNK, “EnterpriseResourcePlanning–Co nceptsandPractice”,PHI,NewDelhi,2003	
3	AakashBusinessTools:SpokenTutorialProjectIITBombay	
4	MasteringTally:DineshMaidasani,FirewalMedia	
5	ImplementingTallyERP9:A.KNadhaniandK.KNadhani,BPBPublishations	
NOTE:LatestEditionofTextbooksMaybeUsed		
WebResources		
1	https://help.tallysolutions.com https://elements.envat	

MAPPING WITH PROGRAMME OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PSO 1	PSO 2	PSO 3
CO1	3	2	3	2	3	2	3	3	3	2	2
CO2	3	2	2	2	2	2	2	2	3	2	3
CO3	3	3	3	2	3	2	3	3	3	2	2
CO4	3	2	2	2	2	2	2	2	3	2	2
CO5	3	3	3	2	3	2	3	3	3	2	3
TOTAL	15	12	13	10	13	10	13	13	15	10	12
AVERAGE	3	2.2	2.6	2	2.6	2	2.6	2.6	3	2	2.2

3–Strong,2-Medium,1-LoW

ARUNAI UNIVERSITY

REGULATION - 2025

P.G DIPLOMA IN Solar PV and Battery Management Systems

Program Duration: 1 year (2 semesters)

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs):

- Find employment in Core solar and battery management service sectors.
- Get elevated to technical lead position and lead the organization competitively.
- Become consultant and provide solutions to the practical problems of core organization.
- Become an entrepreneur and be part of solar and battery designing product and service industries

PROGRAM SPECIFIC OUTCOMES (PSOs):

On completion of PG diploma in solar and battery management systems program, the student will gain the following Program Specific Outcomes.

1. **Foundation of solar system design and development:** Ability to understand the principles and working of PV components, inverter design, control techniques that are forming a part of power generation, Grid interfacing, and energy storage. Students can assess the power management, auditing, crisis and energy saving aspects.
2. **Foundation of battery management system Concepts:** Ability to apply to design and develop mathematical methodologies to solve problems related with PV arrays and structure, thermal aspects using appropriate engineering tools and algorithms.
3. **Computing and Research Ability:** Ability to use knowledge in various domains to identify research gaps and hence to provide solution which leads to new ideas and innovations.

PROGRAM DESCRIPTION:

The **PG Diploma in Solar and Battery Management Systems** aims to address the growing demand for skilled professionals in the renewable energy and energy storage sectors. Graduates of this program will emerge as **skilled engineers, designers, and energy consultants**, ready to contribute to the global transition toward renewable energy and sustainable power solutions. The objectives of this program are as follows:

- ❖ Provide in-depth understanding of **solar energy systems** and **battery storage technologies**, including their design, operation, and maintenance.
- ❖ Equip students with the ability to analyze and solve technical challenges in integrating solar energy and battery systems into existing power grids and standalone setups.

- ❖ Train professionals in **design optimization, efficiency improvement, and cost analysis** of solar and battery systems for residential, commercial, and industrial applications.
- ❖ Encourage the adoption of energy-efficient and eco-friendly technologies to reduce carbon footprints and promote environmental conservation.
- ❖ Inspire students to become **entrepreneurs** in the renewable energy sector by offering knowledge in project management, financial modelling, and energy economics.

CURRICULUM AND SYLLABI FOR SEMESTERS I & II

SEMESTER I

S. NO.	COURSE CODE	COURSE TITLE	TOTAL CONTACT PERIODS	CREDITS
1.	PGDSB101	Principles of Solar Energy Systems	4	4
2.	PGDSB102	Basics of Battery Technology	4	4
3.	PGDSB103	Solar Energy Circuitry and Power Distribution	3	3
4.	PGDSB104	Energy Storage and Power Management	3	3
5.	PGDSB105	Renewable Energy Policies and Economics	3	3
6.	PGDSB106	PV Cell and Battery Testing Laboratory	3	3
TOTAL			20	20

SEMESTER II

S. NO.	COURSE CODE	COURSE TITLE	TOTAL CONTACT PERIODS	CREDITS
1.	PGDSB201	Advanced Solar System Design	4	4
2.	PGDSB202	Power Electronics in Solar and Storage Systems	3	3
3.	PGDSB203	Electric Vehicle Technology	3	3
4.	PGDSB204	Design of Battery Management Systems	4	4
5.	PGDSB205	Project Work	6	6
TOTAL			20	20

PGDSB101

PRINCIPLES OF SOLAR ENERGY SYSTEMS

L T P C
4 0 0 4

OBJECTIVES:

- ❖ To study the concepts of solar photovoltaics
- ❖ To impart knowledge on solar cells
- ❖ To learn the operation of solar cell technologies
- ❖ To provide knowledge emerging solar cell technologies
- ❖ To educate on the applications of solar photovoltaic.

UNIT – 1 SOLAR PHOTOVOLTAICS 12

Solar Energy: Sun and Earth-Basic Characteristics of solar radiation- angle of sunrays on solar collector-Estimating Solar Radiation Empirically - Equivalent circuit of PV Cell- Photovoltaic cell characteristics: P-V and I-V curve of cell-Impact of Temperature and Insolation on I-V characteristics-Shading Impacts on I-V characteristics-Bypass diode - Blocking diode.

UNIT – 2 AN INTRODUCTION TO SOLAR CELLS 12

PN junction diode – equilibrium condition and non equilibrium conditions – PN junction under illumination – design of solar cells – losses in solar cells – solar cell design – analytical techniques – solar simulator – quantum efficiency measurement – minority carrier life time and diffusion length measurement.

UNIT – 3 SOLAR CELL TECHNOLOGIES 12

Production of Si – growth of solar PV industry and Si requirements- Preparation of Si wafers- Si Sheets- Si feedstock for solar cell industry- Solar grade silicon – Si usage in solar PV- process flow of commercial Si cell technology – High efficiency Si Solar cells – chalcopyrite solar cell technologies – Microcrystalline Si thin film technology – Thin film Epitaxial Si solar cell.

UNIT – 4 EMERGING SOLAR CELL TECHNOLOGIES AND CONCEPTS 12

Organic solar cells – solar cell structure – materials – dye sensitized solar cell – GaAs solar cells – thermo photovoltaic (TPV) – current cell technologies – crystalline Si Multi junction solar cells – intermediate band gap, impurity PV and quantum well solar cells – spectrum modification approaches – hot carrier solar cells.

UNIT – 5 SOLAR PHOTOVOLTAIC APPLICATIONS 12

Solar radiation – sun and earth movement – angle of sunrays on solar collector – sun tracking – estimating solar radiation empirically – solar radiation on tilted surface – measurement of solar radiation.

TOTAL: 60 PERIODS

OUTCOMES:

CO1: To understand the standards the concept of solar photovoltaics

CO2: To understand solar cells and workings.

CO3: To analyse and design procedure of solar cell technologies.

CO4: To Acquire knowledge on emerging solar cells and its concepts.

CO5: To Understand the concepts related with photovoltaic applications.

TEXT BOOKS

1. Chetan Singh So “Solar Photovoltaics: Fundamentals, Technologies And Applications” PHI learning Private limited,.
2. Yogi Goswami, Frank Kreith, Jan F. Kreider Principles of Solar Engineering, taylor & francis 2nd edition.
3. Dr. Sundaravadivelu S, Mr. Suresh R. Norman, Dr. Johns Stella I, Dr. Suresh Kumar “A Solar Photovoltaic Power Systems Principles, Design and Applications” notion press.

PGDSB102

BASICS OF BATTERY TECHNOLOGY

L	T	P	C
4	0	0	4

OBJECTIVES:

- ❖ To study the concepts battery technologies
- ❖ To impart knowledge on battery thermal management systems
- ❖ To learn the operation and simulation of batteries
- ❖ To provide knowledge in various types of batteries.
- ❖ To educate on the applications of integrated batteries.

UNIT – 1 BATTERY TECHNOLOGIES 12

Battery technologies – terms and definitions – specific values versus density – state of charge and depth of discharge – charge efficiency – side reaction – battery main challenges – conventional battery technologies – future technologies – life cycle assessment – environmental impact assessment of battery technology.

UNIT – 2	BATTERY THERMAL MANAGEMENT SYSTEMS	12
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Introduction – design of battery thermal management systems – categories – air based – liquid based – phase change material (PCM) – liquid vapour phase change – recent scenario in battery thermal management systems – electrical design of battery pack – mechanical design of battery pack – thermal design of battery pack- design integration.

UNIT – 3	SIMULATION OF BATTERIES	12
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Spatial dimension – simulation domain – governing equations – initial value and guesses – boundary conditions – mesh generation – ill posedness and compatibility equation – famous numerical methods- problems.

UNIT – 4	TYPES OF BATTERIES	12
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Lead acid battery – zinc silver oxide - lithium based batteries - components – types – governing equations – thermal runaway problems – simulation.

UNIT – 5	INTEGRATED BATTERY BASED SYSTEMS	12
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Introduction – integrated battery based system in transportation – case study – PEM fuel cell assisted lithium ion battery electric vehicle – hybrid electric aircraft propulsion systems.

TOTAL: 60 PERIODS

OUTCOMES:

CO1: To understand the standards the concept of battery technologies

CO2: To understand battery behaviour and thermal management systems.

CO3: To analyse and simulation procedure of batteries.

CO4: To Acquire knowledge on types of batteries.

CO5: To understand the concepts integrated battery based systems.

TEXT BOOKS

1. Marc A Rosen, Aida Fa “ Battery Technology: From Fundamentals to Thermal Behaviour and Management” Academic press
2. Farschad Torabi “Simulation of Battery Systems: Fundamentals and Applications” Academic press
3. Slobodan Petrovic “Battery Technology Crash Course: A Concise Introduction” Springer

PGDSB103 SOLAR ENERGY CIRCUITRY AND POWER L T P C
DISTRIBUTION

$$\begin{array}{cccc} 3 & 0 & 0 & 3 \end{array}$$

OBJECTIVES:

- ❖ To study the concepts solar wiring and fundamentals.
- ❖ To impart knowledge on installation procedures.
- ❖ To learn the operation and working of rooftop solar.
- ❖ To provide knowledge in installation of grid connected solar systems.
- ❖ To educate on the applications and maintenance of PV systems.

UNIT – 1 SOLAR WIRING CIRCUIT FUNDAMENTALS 9

Series and parallel wiring – DC electrical circuit rules – physical array layout – series string size – module support structure – electrical wiring diagram - string inverter – micro inverter – wire types and characteristics.

UNIT – 2 COMPONENTS INSTALLATION GUIDE AND RULE 9

Batteries C ratings – charge controller – combiner box ACDB and DCDB – PV Configuration – calculation of payback period – residential, commercial an utility scale PV market – policies and regulation – over current protection – wiring methods – standard test conditions.

UNIT – 3 ROOFTOP AND BIPV SOLAR PV SYSTEMS 9

Introduction – rooftop solar PV systems – components of grid tied rooftop systems – building integrated PV (BIPV) systems – design consideration BIPV – reliability of rooftop systems – lifecycle – electrical arcing -cost analysis.

UNIT – 4 GRID CONNECTED SOLAR PV POWER SYSTEMS 9

Introduction – components – pv module types – array junction box – dc string combiner box- module mounting structure and types – ac cable – grounding cable connection – inverter selection guide – tilt angle – dc to ac ratio – energy yield estimation – lightning arrester – isolator and insulators – simulation tools.

UNIT – 5 GRID INTEGRATION PERFORMANCE AND MAINTENANCE 9

ACDB – transformer requirements – instrument transformer – switchgear – transmission line requirement – protection requirement for solar pv power system – reliability of grid connected solar system – inverter care – pv module care – transformer care – switchyard, control panel and relays – infrared thermography.

TOTAL: 45 PERIODS

OUTCOMES:

- CO1: To understand the standards the concept solar wiring
CO2: To understand installation and rule guidelines.
CO3: To design and develop rooftop solar system.
CO4: To Acquire knowledge on grid connected PV systems.
CO5: To understand the concepts in grid tie inverter maintenance.

TEXT BOOKS

1. Joseph Burdick, Philip Schmidt, “Install Your Own Solar Panels: Designing and Installing a Photovoltaic System to Power Your Home” advanced energy systems LLC.
2. Tara Cunningham, Duane Cunningham, “One Sun: A Guide Into Solar: A Comprehensive Classroom Guide for the PV Associate Exam” Vicious Creations LLC
3. Rabindra Kumar Satpathy, Venkateswarlu Pamuru, “Solar PV Power: Design, Manufacturing and Applications from Sand to Systems” Academic press

PGDSB104 ENERGY STORAGE AND POWER MANAGEMENT L T P C

3 0 0 3

OBJECTIVES:

- ❖ To study the concepts energy storage technologies.
- ❖ To impart knowledge on electrical energy storage system.
- ❖ To learn the operation and working of rooftop solar.
- ❖ To provide knowledge in installation of grid connected solar systems.
- ❖ To educate on the applications PV storage for small island.

UNIT – 1 INTRODUCTION TO ENERGY STORAGE TECHNOLOGIES 9

Energy storage overview – history of energy storage systems – types of energy storage technology – global energy storage capacity.

UNIT – 2 SOLAR ELECTRICAL ENERGY STORAGE 9

Solar electrical energy storage technologies – distributed scale storage technologies – economics of solar electrical energy storage technologies.

UNIT – 3 BATTERY MANAGEMENT SYSTEMS 9

Battery management systems parts – power module – dc/dc converter – example of BMS – comparison - battery design – battery characteristics.

UNIT – 4 BATTERY CHARGING ALGORITHMS 9

Charging algorithms for li-ion batteries – fast charging – influence of charge voltage/ current on charging process - simulation for fast charging of li-ion battery.

UNIT – 5 PV STORAGE SYSTEMS FOR REMOTE SMALL ISLAND 9

Introduction – need for energy storage in remote islands –main components of the PV ESS - ESS sizing - PV ESS characteristics – ESS cost analysis - case study –PV PHS system components and operational modes.

TOTAL: 45 PERIODS

OUTCOMES:

- CO1: To understand the concept of Energy storage
- CO2: To understand energy storage technologies.
- CO3: To design and develop battery management systems.
- CO4: To Acquire knowledge on battery charging algorithms.
- CO5: To understand the concepts energy storage in small island.

TEXT BOOKS

1. Paul Breeze “Power System Energy Storage Technologies” Academic press
2. Bent Sorensen “Solar Energy Storage” Academic press
3. H.J. Bergveld, W.S. K “Battery Management Systems: Design by Modelling” Springer science business media

OBJECTIVES:

- ❖ To study the concepts energy policies.
- ❖ To impart knowledge on renewable energy policies.
- ❖ To learn the energy policies in power sector.
- ❖ To provide knowledge economic analysis of energy investment.
- ❖ To educate on the policies and economics in renewable energy sector

UNIT – 1 ENERGY EFFICIENCY POLICIES IN POWER SECTOR 9

Codes and standards – direct financial incentives – demand based incentives – regulations and strategic plans – technical assistance – comparative analysis based on sectors.

UNIT – 2 RENEWABLE ENERGY POLICIES 9

Incentives, education and research – capacity building – carbon tax – energy tax – energy production payment – grant – green labelling – rebate – research and development – tax credit – tax reduction and tax exemption – public finance – regulations – choosing the most suitable renewable energy policy.

UNIT – 3 RENEWABLE ENERGY POLICIES IN POWER SECTOR 9

Incentives, education and research – public finance – regulations – case study.

UNIT – 4 ECONOMIC ANALYSIS OF ENERGY INVESTMENTS 9

Introduction – main characteristic of energy projects – essentials of the economic analysis of projects – identification of costs and benefits – indication of cost benefits comparison – method without time value and employing time value.

UNIT – 5 ECONOMICS OF RENEWABLE ENERGY SUPPLY 9

Introduction – renewable and alternative energy background – status of renewable energies – renewable electricity generation – drivers of renewable energy – support mechanism.

TOTAL: 45 PERIODS

OUTCOMES:

CO1: To understand the concept of energy policies

CO2: To understand renewable energy policies.

CO3: To understand energy investment policies.

CO4: To Acquire knowledge on energy policies in power sector.

CO5: To understand the concepts economic analysis of energy investment.

TEXT BOOKS

1. Neyre Tekbiyık Ersoy “Energy Efficiency and Renewable Energy Policies” Synthesis Lectures on Renewable Energy Technologies - Springer
2. Subhes C. Bhattacharyya, “Energy Economics: Concepts, Issues, Markets and Governance” springer 2nd edition.

COURSE OBJECTIVES:

- ❖ To enable the students to understand the behaviour of semiconductor device based on experimentation
- ❖ Be exposed to converter and inverter simulation studies.
- ❖ Familiarize the operation and characteristics of MPPT algorithm.
- ❖ Explore the characteristics and working of on grid and off grid inverters.
- ❖ Learn the synchronization methods of grid tie inverters.

List of experiments

1. Study of switching characteristics of Power MOSFET , IGBT and SCR
2. Simulations of AC-DC-AC converters, PWM inverters and Matrix Converters with Resistive and dynamic loads
3. Simulation on modelling of Solar PV System- V I Characteristics
4. Circuit Simulation of MPPT algorithm
5. Circuit Simulation of Charge controller testing
6. MPPT algorithm testing for an stand-alone system
7. MPPT algorithm testing for an grid tie inverter
8. Synchronization process for single phase and three phase solar Grid tied PV system

TOTAL: 45 PERIODS

COURSE OUTCOMES:

Upon successful completion of the course, the students will be able to:

- CO1: Analyse the characteristics of Power MOSFET, IGBT and SCR configurations experimentally
- CO2: Analyse the characteristics of converters, PWM inverters and Matrix Converters by simulation.
- CO3: Analyse the working of MPPT algorithm
- CO4: Analyse the Synchronization process grid tie inverters.
- CO5: Analyse the characteristics I V Characteristics

OBJECTIVES:

- ❖ To study the concepts advanced photovoltaic system design.
- ❖ To impart knowledge on solar design criteria.
- ❖ To learn the photovoltaic array configuration and sizing.
- ❖ To provide knowledge energy storage devices.
- ❖ To learn on the charge control and MPPT.

UNIT – 1 ADVANCED PHOTOVOLTAIC SYSTEM DESIGN AND DESIGN CRITERIA 12

Grades of pv systems – client requirements – goals- objectives – solar resource – location – climate – altitude – shading – airflow – wind – space available – structural challenges – orientation – mounting – maintenance. – system life.

UNIT – 2 EVALUATION AND DESIGN CRITERIA 12

Monitoring – operations and system maintenance – cleaning panels, filters and equipment cavities – system performance modelling – quality of installation – testing schedule and procedure – quality assurance – service entrance sections – national and local electrical codes – panels – UL panel certification and other certifications.

UNIT – 3 PHOTOVOLTAIC ARRAY CONFIGURATION AND SIZING 12

PV module – Panel fundamentals – monocrystalline silicon – multicrystalline silicon – amorphous silicon- ribbon drawn or pulled silicon – thin film technology – module selection – number of modules – photovoltaic array sizing- sizing PV array – orientation – central inverter concept – module inverter concept.

UNIT – 4 ENERGY STORAGE DEVICES (ESD) SYSTEM DESIGN 12

Batteries in pv system – battery design and construction – battery types – lead acid battery classifications – types of lead acid batteries – gel cell batteries – value regulated lead acid (VRLA) batteries – nickel cadmium batteries – nickel iron batteries – lithium ion batteries – battery maintenance – battery safety considerations.

UNIT – 5 CHARGE CONTROL AND MAXIMUM POWER POINT TRACKING 12

Battery charge controllers in PV systems – charge controller terminology and definitions – charge controller designs – shunt controllers – single stage series controllers – diversion controllers- pulse width modulations controller PWM – MPPT – PV controller system design – voltage regulation set point selection – charge controller selection – sizing charge controller.

TOTAL: 60 PERIODS**OUTCOMES:**

- CO1: To understand the concept of advanced photovoltaic system design.
- CO2: To understand solar design criteria.
- CO3: To understand photovoltaic array configuration and sizing. .
- CO4: To Acquire knowledge on energy storage devices.
- CO5: To understand the concepts charge control and MPPT.

TEXT BOOKS

1. John R. Balfour, Mic “Advanced Photovoltaic System Design” the art and science of photovoltaics, jones & Bartlett learning.
2. L. Ashok Kumar, K. Mohana Sundaram "Solar PV System Design, Installation, Operation and Maintenance" Nova Science Publishers 2021

OBJECTIVES:

- ❖ To study the concepts inverter topologies for solar PV.
- ❖ To impart knowledge on multilevel inverter topologies for solar PV.
- ❖ To learn the Dc-Dc converter topologies for solar PV .
- ❖ To provide knowledge emerging DC – DC converter topologies.
- ❖ To learn on control techniques of solar inverter and converters.

UNIT – 1 INVERTER TOPOLOGIES FOR SOLAR PV 9

Introduction – single stage dc-ac converter – line commutated photovoltaic inverter – self commutated photovoltaic inverter – grid tie inverters – inverter with high frequency with line frequency transformer – half bridge zero voltage state converters – H bridge inverter.

UNIT – 2 MULTILEVEL INVERTER TOPOLOGIES FOR SOLAR PV 9

Introduction – comparison of multilevel inverter – reduced order multilevel inverter – advanced multilevel inverter topologies – switched battery boost multilevel inverter – switched capacitor multilevel inverter – string inverter – multistring inverter - transformer less inverter topologies – low frequency inverter – HERIC inverter.

UNIT – 3 DC-DC CONVERTER TOPOLOGIES FOR SOLAR PV 9

Introduction – topologies of dc-dc converter – unidirectional dc-dc converter – bidirectional dc-dc converter – double input pulse width modulation dc –dc converter – single input multiple output dc – dc converter.

UNIT – 4 EMERGING DC – DC CONVERTER TOPOLOGIES 9

SEPIC converter – LUO converter – Integrated SEPIC – CUK converter – Flyback converter – zeta converter – self lift CUK converter – push pull converter – advantages and disadvantages of converters.

UNIT – 5 CONTROL TECHNIQUES OF SOLAR INVERTER AND CONVERTERS 9

Closed loop control – buck boost converter – boost converter – buck converter – interleaved boost converter – LLC resonant converter – forward converter- two level inverter – three level inverter.

TOTAL: 45 PERIODS**OUTCOMES:**

- CO1: To understand the concept of inverter topologies for solar PV.
 CO2: To understand multilevel inverter topologies for solar PV.
 CO3: To understand the Dc-Dc converter topologies for solar PV.
 CO4: To Acquire knowledge on emerging DC – DC converter topologies.
 CO5: To understand the concepts control techniques of solar inverter and converters.

TEXT BOOKS

1. L. Ashok Kumar, S.Albert Alexander, Madhuvanthani Rajendran “Power Electronic Converters for Solar Photovoltaic Systems” Academic Press.
2. Frede Blaabje “Applications of Power Electronics” Volume 1 MDPI

OBJECTIVES:

- ❖ To study the concepts need for electric vehicles.
- ❖ To impart knowledge on electric vehicles architecture.
- ❖ To learn the energy storage.
- ❖ To provide knowledge electric drives and control.
- ❖ To learn on control electric vehicles.

UNIT I	NEED FOR ELECTRIC VEHICLES	9
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History and need for electric and hybrid vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies, comparison of diesel, petrol, electric and hybrid vehicles, limitations, technical challenges

UNIT II	ELECTRIC VEHICLE ARCHITECHTURE	9
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Electric vehicle types, layout and power delivery, performance – traction motor characteristics, tractive effort, transmission requirements, vehicle performance, energy consumption, Concepts of hybrid electric drive train, architecture of series and parallel hybrid electric drive train, merits and demerits, mild and full hybrids, plug-in hybrid electric vehicles and range extended hybrid electric vehicles, Fuel cell vehicles.

UNIT III	ENERGY STORAGE	9
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Batteries – types – lead acid batteries, nickel based batteries, and lithium based batteries, electrochemical reactions, thermodynamic voltage, specific energy, specific power, energy efficiency, Battery modeling and equivalent circuit, battery charging and types, battery cooling, Ultra-capacitors, Flywheel technology, Hydrogen fuel cell, Thermal Management of the PEM fuel cell

UNIT IV	ELECTRIC DRIVES AND CONTROL	9
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Types of electric motors – working principle of AC and DC motors, advantages and limitations, DC motor drives and control, Induction motor drives and control, PMSM and brushless DC motor - drives and control , AC and Switch reluctance motor drives and control – Drive system efficiency – Inverters – DC and AC motor speed controllers

UNIT V	DESIGN OF ELECTRIC VEHICLES	9
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Materials and types of production, Chassis skate board design, motor sizing, power pack sizing, component matching, Ideal gear box – Gear ratio, torque–speed characteristics, Dynamic equation of vehicle motion, Maximum tractive effort – Power train tractive effort Acceleration performance, rated vehicle velocity – maximum grad ability, Brake performance, Electronic control system, safety and challenges in electric vehicles. Case study of Nissan leaf, Toyota Prius, tesla model 3, and Renault Zoe cars

TOTAL: 45 PERIODS

OUTCOMES:

- CO1: To understand the concept of need for electric vehicles.
CO2: To understand electric vehicles architecture.
CO3: To understand the energy storage.
CO4: To Acquire knowledge on electric drives and control.
CO5: To understand the concepts of electric vehicles.

TEXT BOOKS

1. Iqbal Hussein, *Electric and Hybrid Vehicles: Design Fundamentals*, 2nd edition CRC Press, 2011.
2. Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, *Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design*, CRC Press, 2004.
3. James Larminie, John Lowry, *Electric Vehicle Technology Explained* - Wiley, 2003.
4. Ehsani, M, "Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design", CRC Press, 2005

PGDSB204	DESIGN OF BATTERY MANAGEMENT SYSTEMS	L	T	P	C
		4	0	0	4

OBJECTIVES:

- ❖ To study the concepts of battery management systems.
- ❖ To impart knowledge on BMS options.
- ❖ To learn the BMS functions.
- ❖ To provide knowledge custom BMS design.
- ❖ To learn on control techniques deploying a BMS.

UNIT – 1 BATTERY MANAGEMENT SYSTEMS 12

Battery management systems parts – power module – dc/dc converter – example of BMS – comparison - battery design – battery characteristics.

UNIT – 2 BMS OPTIONS 12

Functionality – CCCV charges – regulators – meters – balancers – protectors – functionality comparison – technology – simple analog – sophisticated digital – technology comparison – topology.

UNIT – 3 BMS FUNCTIONS 12

Measurement – voltage – temperature – current – protection – thermal management – balancing – redistribution – distributed charging – evaluation – external communications – logging and telemetry.

UNIT – 4 CUSTOM BMS DESIGN 12

BMS ASIC – selection – Analog BMS design – ready made digital BMS design – custom digital BMS design – cell interface- distributed charging.

UNIT – 5 DEPLOYING A BMS 12

Installing – battery pack design – BMS connection to pack – BMS connections to system – configuring – cell configuration – pack configuration – system configuration – testing – trouble shooting.

TOTAL: 60 PERIODS

OUTCOMES:

- CO1: To understand the concept of battery management systems.
CO2: To understand BMS options.
CO3: To understand the BMS functions.
CO4: To Acquire knowledge on BMS design.
CO5: To understand the concepts of control techniques deploying a BMS.

TEXT BOOKS

1. Davide Andrea “Battery Management Systems for Large Lithium Ion Battery Packs” Artech house 2010.
2. Henk Jan Bergveld, W.S. Kruijt, P.H.L Notten “Battery Management Systems: Design by Modelling” Kluwer academic publishers.

COURSE OBJECTIVE:

To develop the ability to solve a specific problem right from its identification and literature review till the successful solution of the same. To train the students in preparing project reports and to face reviews and viva voce examination.

P.G DIPLOMA AIML in Health Care Management

PHILOSOPHY / SIGNIFICANCE OF

THE COURSE :

- ❖ Patient-centered care: The course emphasizes the importance of patient-centered care, where healthcare managers prioritize patients' needs, preferences, and values.
- ❖ Evidence-based management: The course promotes evidence-based management practices, where healthcare managers make informed decisions based on research, data, and best practices.
- ❖ Collaboration and teamwork: The course highlights the importance of collaboration and teamwork among healthcare professionals, managers, and stakeholders to achieve common goals.
- ❖ Quality improvement and safety: The course emphasizes the importance of quality improvement and safety in healthcare management, where managers strive to reduce errors, improve patient outcomes, and enhance the overall quality of care.
- ❖ Addressing healthcare challenges: The course addresses pressing healthcare challenges, such as access, affordability, quality, and safety, by equipping healthcare managers with the necessary knowledge, skills, and competencies.
- ❖ Developing healthcare leaders: The course aims to develop healthcare leaders who can navigate complex healthcare systems, lead change initiatives, and improve healthcare outcomes.
- ❖ Improving healthcare systems: The course focuses on improving healthcare systems, services, and processes, by applying management principles, theories, and models to real-world healthcare scenarios.
- ❖ Enhancing patient care and experience: The course emphasizes the importance of patient-centered care, patient safety, and patient satisfaction, by equipping healthcare managers with the necessary knowledge and skills to improve patient care and experience.

Subjects / Papers to be included:

Paper 1: Hospital Management

Paper 2: Healthcare Management

Paper 3: Healthcare Finance

Paper 4: Research methods in Healthcare

Paper 5: Project

Paper 6: Healthcare Operations Management

Paper 7: Human Resources in Healthcare management

Paper 8: Healthcare policy & Ethics

Paper 9: Quality Management in Hospital

Paper 10: Internship program

PEDAGOGY OF THE COURSE SELECTED – ASSESSMENT METHODS

1. Lectures: Traditional lectures will be used to introduce key concepts, theories, and frameworks in

healthcare management.

2. Case Studies: Real-life case studies will be used to illustrate key concepts and challenges in healthcare

management.

3. Group Discussions: Group discussions will be used to encourage critical thinking, problem-solving, and

collaboration among students.

4. Role-Playing: Role-playing exercises will be used to simulate real-life scenarios in healthcare

management, such as patient interactions, team meetings, and conflict resolution.

5. Guest Lectures: Guest lectures from healthcare professionals and industry experts will be used to provide

students with real-world insights and perspectives.

6. Project-Based Learning: Students will work on projects that apply theoretical concepts to real-world

healthcare management challenges.

ASSESSMENT METHODS:

1. Quizzes and Exams: Quizzes and exams will be used to assess students' knowledge and understanding of

key concepts and theories.

2. Case Study Analyses: Students will be required to analyze case studies and provide written reports that

demonstrate their critical thinking and problem-solving skills.

3. Group Projects: Group projects will be used to assess students' ability to work collaboratively and apply

theoretical concepts to real-world healthcare management challenges.

4. Reflective Journaling: Students will be required to maintain a reflective journal throughout the course,

which will be used to assess their critical thinking and reflection skills.

5. Class Participation: Class participation will be used to assess students' engagement and contribution to

class discussions and activities.

LEARNING OUTCOMES :

Upon completing this course, students will be able to:

1. Analyze healthcare management concepts and theories: Students will be able to analyze and apply key

concepts and theories in healthcare management.

2. Apply critical thinking and problem-solving skills: Students will be able to apply critical thinking and

problem-solving skills to real-world healthcare management challenges.

3. Demonstrate effective communication and collaboration skills: Students will be able to demonstrate

effective communication and collaboration skills in a healthcare management context.

4. Evaluate healthcare management practices and policies: Students will be able to evaluate healthcare

management practices and policies and provide recommendations for improvement.

P.G DIPLOMA AIML in Health Care Management

DIPLOMA COURSE

The Course of Study and the Scheme of Examinations

S. No.	Part	Study Components		Ins. Hrs/ week	Credit	Title of the Paper	Maximum Marks		
		Course Title					CIA	Uni. Exam	Total
		SEMESTER I							
1.	III	Core Theory	Paper-1	6	4	Hospital Management	25	75	100
2.	III	Core Theory	Paper- 2	6	4	Healthcare Management	25	75	100
3.	III	Core Theory	Paper- 3	6	4	Healthcare Finance	25	75	100
4.	III	Elective I	Paper- 1	6	4	Research Methods in Health care	25	75	100
5.	III	Elective II	Paper-2	6	4	Project	25	75	100
		Sem. Total		30	20		150	450	600
		SEMESTER II					CIA	Uni. Exam	Total
7.	III	Core Theory	Paper- 4	6	4	Healthcare Operation Management	25	75	100
8.	III	Core Theory	Paper- 5	6	4	Human Resource in Healthcare Management	25	75	100
9.	III	Core Theory	Paper- 6	6	4	Quality Management in Hospital	25	75	100

10.	III	Elective I	Paper- 3	6	4	Healthcare policy & Ethics	25	75	100
11.	III	Elective II	Paper-4	6	4	: Internship program	25	75	100
		Sem. Total		30	20		150	450	600

PGDHM SYLLABUS 2025

SEMESTER - I

HEALTHCARE FINANCE

UNIT – I

Introduction to Healthcare Finance - The Financial Environment - Financial Accounting Basics and the Income Statement - The Balance Sheet and Statement of Cash Flows.

UNIT – II

Managerial Accounting Basics, Cost Behavior, and Profit Analysis - Cost Allocation - Pricing and Service Decisions - Planning and Budgeting.

UNIT – III

Time Value Analysis - Financial Risk and Required Return - Long-Term Debt Financing - Equity Financing - Capital Structure and the Cost of Capital

UNIT – IV

The Basics of Capital Budgeting - Project Risk Assessment and Incorporation.

UNIT – V

Current Asset Management and Financing -Financial Condition Analysis - Lease Financing and Business Valuation.

Reference Book: -

Health Finance –L.E.Feldman.

HEALTHCARE MANAGEMENT

UNIT – I

International Health Care: A Twelve-Country Comparison - Organized Delivery System - Legal Implications of Business Arrangements in the Healthcare Industry.

UNIT –II

Financial Management of Organized Healthcare Delivery Systems - Labor and Employment Laws Applicable to Organized Delivery Systems.

UNIT –III

The Management of Nursing Services - Marketing Healthcare Services - Healthcare Information Technologies in an Era of Healthcare Reform: A Complex Adaptive System Perspective.

UNIT - IV

Management Engineering - Ambulatory Care -Bioterrorism Preparedness - The Hospital in an Organized Delivery System.

UNIT –V

The History of Hospitals – Laboratories – Pharmacy -Material and Resource Management.

Reference Book: -

Health Care Administration Managing Organized Delivery Systems Fifth Edition
- Lawrence F. Wolper

RESEARCH METHODS IN HEALTHCARE

UNIT – I

Introduction - Genetic Data Available- Methodology - Structural Equation Model - Score Function Formulation -Two-Stage Learning - Further Issues -Search Algorithm -Global Optimal Solution Search -Heuristic Algorithm for a Local Optimal Solution

Unit-II

Data - Tools -Algorithms - Literature Review - Methodology -
Morph proteomics - Biomedical Analytics - Integrating Morph proteomics and Biomedical Analytics -Case Studies Clinical: Therapeutic Recommendations for Pancreatic Aden carcinoma Clinical: Biology Underlying Exceptional Responder
in Refractory Hodgkin's Lymphoma Research: Role of the Hypoxia Pathway in Both Ontogenesis and Embryogenesis

UNIT – III

Sensor-Based - Characterization and Modeling of Nonlinear Dynamics -Multiracial
Spectrum Analysis of Nonlinear Time Series - Recurrence Quantification Analysis – Multi scale
Recurrence Quantification Analysis - Healthcare Applications - Nonlinear Characterization of
Heart Rate Variability – Multi scale Recurrence Analysis of Space–Time Physiological Signals.

UNIT – IV

Statistical Modeling of ECG for Disease Diagnosis - ECG Signal Denoising Waveform
Detection Feature Extraction Disease Classification and Diagnosis An Example: Detection of
Obstructive Sleep Apnea from a Single ECG Lead Introduction to Obstructive Sleep Apnea
Materials and Methods Database QRS Detection and RR Correction R Wave Amplitudes and
EDR Signal Feature Set Classifier Training with Feature Selection.

UNIT- V

Measurement Uncertainty: Background and Analytical - Estimation - Uncertainty in
Clinical Laboratories - Uncertainty in Clinical Laboratories: A System- Approach Model
Development Guidelines - System Description and Process Phases - Modeling Guidelines -
Implementation of Guidelines: Enzyme Assay Uncertainty Model - Calibration Phase Sample
Analysis Phase - Results and Analysis.

Reference Book: -

Healthcare analytic – Hui yang, Eva K.Lee , Wiley.

HOSPITAL MANAGEMENT

UNIT- I

Introduction to Hospital Administration: Definition – Objectives and functions Meaning-
Importance – Implication – Need–Types Scope, Features Management of change

UNIT-II

Role and structure of Human Resource Function in organizations Challenges in Hospital Administration

UNIT- III

Approaches to Hospital Administration: Meaning- Importance – Implication – Need– Types- Scope, Features Management of change

UNIT- IV

Human Resource Planning: Personnel Policy - Characteristics Meaning- Importance – Implication – Need–Types- Scope, Features Management of change

UNIT-V:

HR Planning: Need, Scope and Process – Job analysis – Job description – Job specification- Succession Planning

Reference Book: -

1)HOSPITAL ADMINISTRATION -Dr Jaspreet Kaur, Associate Professor, Trinity Institute of Management (GGSIPU) Units (1-3)

2)HOSPITAL ADMINISTRATION -PG Aquinas, Professor and Dean, Aloysius Institute of Business Administration, St. Aloysius College, Mangalore(units-4,5)

HEALTHCARE OPERATION MANAGEMENT

UNIT –I

The Challenge and the Opportunity- History of Performance Improvement- Evidence-Based Medicine and Pay-for-Performance.

UNIT –II

Strategy and the Balanced Scorecard- Operations Management in Action- Definition of a Project- Project Selection and Chartering- Project Scope and Work Breakdown- Project Control- Quality Management, Procurement, the Project Management Office, and Project Closure- The Project Manager and Project Team.

UNIT –III

Tools for Problem Solving and Decision Making - Using Data and Statistical Tools for Operations Improvement -Quality Management: Focus on Six Sigma -The Lean Enterprise – Simulation.

UNIT –IV

Process Improvement and Patient Flow –Scheduling and Capacity Management - Supply Chain Management.

UNIT –V

Human Resource Planning -Managerial Accounting -Control - Which Tools to Use: A General Algorithm . - Operational Excellence -VVH Strives for Operational Excellence - The Healthcare Organization of the Future.

Reference Book: -

Healthcare operations management- Daniel B. McLaughlin Julie M. Hays

HUMAN RESOURCES IN HEALTHCARE MANAGEMENT

UNIT –I

Introduction to Health Care Human Resources Management - Defining Elements of Progressive Health Care Organizations - Current Perceptions of Health Care Organizations - Spheres of Influence Model- Five Significant Change Dynamics of Modern Health Care - Profile

of a Progressive Health Care Organization - Composition of a Progressive Health Care Human Resources Management Department.

UNIT –II

The Changing Role of Human Resources Management - Human Resources Outsourcing - Strategic Human Resources Management and Human Resources Planning - Anticipating Future Needs - Evaluating the Effectiveness of Strategic Human Resources Management - Problems and Implications of Strategic Human Resources Management.

UNIT –III

Performance Matrix of Superstars, Steadies, and Nonplayers - PACT Formula - Strategic Requirements for a Progressive Health Care Human Resources Management Department - Federal Equal Employment Opportunity Laws 84 Proving Employment Discrimination - Contents Affirmative Action: Executive Orders and Other Federal Laws - Constitutional Rights- Additional Protections for Employees.

UNIT –IV

Cultural Factors Relevant to Health Care Human Resources - Practical Application: A Tale of Two Jersey Cultures -Selection and Placement Strategy Considerations for Volunteers - Maximizing Health Care Volunteer Performance - Agenda Topics for Volunteer Summits - Ten Essential Rules for Volunteer Placement.

UNIT –V

Legal Significance of Job Analysis - Job Analysis Information - Designing a Job Analysis Program - Job Descriptions and Job Specifications - Competency Modeling - Job Analysis Techniques - Contents - Preselection Process - Critical Dimensions of External

Recruitment - Preparing and Planning for the Interview - Conducting the Interview 199

Evaluating the Interview.

Reference Book: -

Human Resources Management for Health Care Organizations -JOAN E. PYNES
DONALD N. LOMBARDI.

HEALTHCARE POLICY AND ETHICS

UNIT –I

An introduction to healthcare ethics-Unpacking vulnerability in healthcare ethics

UNIT –II

The difference between right and wrong -The quality of healthcare: A care ethics
approach

UNIT -III

The Influence of culture, values and faith on end-of-life decisions -ethics and human
rights in healthcare

UNIT –IV

Euthanasia, do we dare? -Community-based participatory research in healthcare:
ethical issues and impact on healthcare practitioners

UNIT -V

Ethical decision making processes-Spirituality and healthcare

Reference Book: -

HEALTHCARE ETHICS FOR HEALTHCARE PRACTITIONERS- LAETUS O.K.
LATEGAN & GERT J. VAN ZY

QUALITY MANAGEMENT IN HOSPITAL

UNIT –I

Quality-Dimensions of Quality in Health Care-Evolution of the Concept of Quality-Present International Scenario -International Society for Quality in Health Care (ISQua) -Accrediting the Accreditors-Indian Scenario-Cost of Quality: Economic Feasibility

UNIT II

How to Improve the Quality of Services in Hospitals-Essentials of Quality Improvement in Health Care Service-Improvement of Quality in Health Care: Different Approaches-Implementation of Quality Management Program in an Organisation.

UNIT –III

ISO Certification-Benefits of ISO Certification-Structure of ISO 9001: 2000 Standards-Quality Manual-NABH Accreditation-Joint Commission on Accreditation of Health Care-Organizations (JCAHO)-JCI Accreditation Program-Ideal Approach to Quality

UNIT –IV

Patient Friendly Hospital-Front Office-Out-patient Department-Emergency Service-Department of Dental Surgery-Pathology Department-Radiology Department.

UNIT –V

Operation Theatre (OT) Department-Blood Bank-Physiotherapy and Rehabilitation Department-Maternity Services- In-patient Areas (Ward Units)-Intensive Care Unit (ICU)-Neonatal Intensive Care Unit-Dialysis Service-Pharmacy Service-Medical Records Department-Central Sterile Supply Department (CSSD)-Dietary Service-Laundry Service-House Keeping Service-Hospital Engineering Service

Reference Book: -

Quality Management in Hospitals -S K Joshi